

A Survey of Electron Swarm Data

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An electron swarm consists of a small number density n of electrons in a gas of much higher number density N . The mean energy and energy distribution of such a swarm are determined by the value of E/N , where E is the electric field. At any given value of E/N the swarm may be characterized by the values of eight parameters, viz; drift velocity, diffusion coefficient, (diffusion coefficient)/mobility, excitation coefficient, electron attachment coefficient, electron detachment coefficient, ionization coefficient, recombination coefficient. In this survey, data on these parameters obtained by a variety of experimental techniques are collected, discussed, and compared graphically. Also included on the graphs are computed values of the parameters obtained in many cases from cross sections and energy distributions chosen to give the best fit with the swarm data. Selected tabulations of the data are also given except in cases for which the accuracy of the data is not sufficient to warrant numerical presentation. The mean energy of the electron swarms ranges from thermal to several electron volts and the gases for which data are given are the rare gases, the common molecular gases (H_2 , N_2 , O_2 , CO , NO , CO_2 , NO_2) and air. The survey also contains an extensive bibliography which includes references (i) to publications on electron swarms in a much wider range of gases than those for which data are given and (ii) to papers concerned with energy distributions, conductivity, and ionization coefficients in crossed electric and magnetic fields in addition to those relating to the eight parameters listed above.

Key words: Electrical breakdown of gases; electrical discharges; electron attachment coefficient; electron detachment coefficient; electron diffusion coefficient; electron drift velocity; electron excitation coefficient; electron-ion recombination coefficient; electron ionization coefficient; electron swarm; electron transport coefficients; ionized gases.

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1. Introduction

In order to understand many types of plasma and ionized gases, a knowledge is required of the behavior of mixed swarms of charged and neutral particles under various conditions of pressure (particle density) and temperature in the presence of electric and magnetic fields.

The behavior of electron swarms is characterized by a number of parameters that are determined experimentally in the laboratory. These parameters may, for convenience, be listed as follows:

- (1) the drift velocity W of the electrons in an electric field E ;
- (2) the ratio of the diffusion coefficient D to the drift velocity, usually expressed as D/μ where $\mu = W/E$ is the mobility; this ratio is a measure of the mean energy of the electrons;
- (3) the diffusion coefficient of electrons D ;
- (4) the electron attachment coefficient η ;
- (5) the detachment coefficient of electrons from negative ions δ (strictly a property of ion swarms but often measured in electron swarm experiments);
- (6) the electron excitation coefficient ϵ ;
- (7) the primary ionization coefficient α ;
- (8) the coefficient of recombination of electrons with positive ions r .

Values of the above coefficients are usually determined for gas pressures in the range from about 1 to 1000 Torr (number density N from about 3.5×10^{16} to 3.5×10^{19} molecules/cm³) at neutral gas temperatures in the range from about 50 to 500 K. By definition, the number density of electrons in a swarm is much less than that of the gas in which it moves, and the dominant collisions are consequently those between electrons and gas atoms. Thus, since the determination of most of the above properties requires the application of an electric field, the mean energy of the electrons is in general higher than that of the neutral gas (up to typically a factor of 50 times greater at $E/N \sim 150 \times 10^{-17}$ V. cm²) and the electron energy distribution is not Maxwellian.

In some experiments, however, especially in most of those on recombination and on the measurement of coefficients at very low values of E/N , the electrons have a Maxwellian distribution. In these cases, rate coefficients determined in the laboratory may be directly applicable, for example, to processes occurring in the ionosphere.

Among the interactions of importance in determining the behavior of swarms are those of electrons with neutral particles, electrons with ions, and negative ions with neutral particles. Thus, in addition to their intrinsic value, the swarm data can be used to supplement information obtained from direct measurements of cross sections. The use of swarm data is of particular value both for the determination of cross sections at low electron energies, where direct measurements of

cross sections are notoriously difficult, and for obtaining information about three-body processes which are insignificant at the particle densities usually used in beam experiments.

In order to obtain cross sections from swarm data it is, however, necessary to know the energy distribution under the conditions of the experiment, and this presents considerable difficulties. Little experimental evidence on electron energy distributions under swarm conditions is available, although there have been a few recent measurements [1651].¹ There have been considerable recent theoretical advances in the treatment of the problem [181, 218, 260, 1062, 2553],¹ in which self-consistent sets of cross sections for momentum transfer and for some inelastic processes have been obtained. The method used was to assume reasonable, energy-dependent cross sections, to integrate the Boltzmann equation numerically in order to obtain an energy distribution, and then to use this distribution together with the assumed cross sections to determine certain swarm coefficients such as W and D/μ . The assumed cross sections were then adjusted in order to give the best fit between calculated and observed swarm data.

Eventually, it is conceivable that the stage may be reached at which the cross sections of all electron collision processes are known, together with the energy distribution of an electron swarm under any given set of conditions. At present, however, the coefficients representing the behavior of the swarm form a necessary "half-way house," as it were, giving information that is useful in the solution of a wide range of fundamental and applied problems, from the determination of cross sections and energy distributions on the one hand, to the understanding of the behavior of ionized gases in laboratory devices and in astrophysical phenomena on the other. There is thus considerable need for an up-to-date compilation of available data, and towards this end a bibliography of electron swarm data was started in 1966. An interim version of this bibliography appeared as JILA Information Center Report No. 4 in October 1967. This bibliography has been completed and brought more up-to-date and is given below in sections 4 and 5. In addition, a data survey for the inert and the more common molecular gases has been carried out and is given in section 3. The effective terminal date for the comprehensive literature search for the bibliography was December 1968, but a more selective search, related to the data survey but including other bibliographic material, was continued until the end of 1972.

2. Selection and Arrangement of Data

2.1. Symbols and Units

As mentioned in section 1, the mean energy of an electron swarm in a given gas depends on the value of E/N . Thus the coefficients which characterize the

¹ Numbers in brackets refer to references listed in section 5.2.

behavior of swarms are usually presented as functions of E/N . Until recently, however, it has been common practice to use E/p_0 (where p_0 is the gas pressure reduced to 0 °C) rather than E/N and in view of this both E/p_0 and E/N are given in all graphs in order to help familiarize the conversion from E/p_0 to E/N . An increasingly widely used unit in which to express the value of E/N is the Townsend (1 Td = 10^{-17} V cm²), first suggested by Huxley, Crompton, and Elford (Brit. J. Appl. Phys. **17**, 1237, 1966).

With the exception of the drift velocity and the diffusion coefficient, the coefficients with which we are concerned can be expressed either (a) as rate coefficients or (b) as coefficients per unit drift distance in the direction of the electric field. There appear to be no generally agreed symbols for the full range of coefficients discussed, so in an attempt to achieve at least internal consistency, the following nomenclature is used here:

a) *Rate coefficients* are denoted by the appropriate lowercase type symbol with a suffix when appropriate to denote whether it is a two-body or a three-body reaction, e.g., the symbol representing three-body attachment rate coefficient is a_3 ;

b) *Coefficients* (per unit drift distance) are denoted by a lowercase Greek letter, e.g., the symbol for the attachment coefficient per unit drift distance is η . If the nature of the reaction is known, a subscript is added to denote whether it is a two- or three-body reaction, e.g., η_3 denotes the three-body attachment coefficient.

The relationship of the rate coefficient per unit drift distance to the collision frequency depends on the nature of the process. For a two-body process (such as ionization or radiative attachment, for example) the *collision frequency* for a given process ν_2 , say, is related to the *coefficient* Π_2 and the *rate coefficient* p_2 for this process by

$$dn = n\Pi_2 dx = n\Pi_2 W dt = n\nu_2 dt = np_2 N dt,$$

where dn is the number of new electrons formed by a concentration n electrons in a drift distance dx or in a time dt and W is the electron drift velocity.

Thus $\Pi_2 = \nu_2/W = p_2 N/W$, Π_2 being in units of cm⁻¹, ν_2 in s⁻¹ and p_2 in cm³ s⁻¹.

For three-body processes (such as three-body attachment or recombination) on the other hand, the collision frequency ν_3 , coefficient per unit drift distance Π_3 , and rate coefficient p_3 for the process are related by

$$dn = n\Pi_3 dx = n\Pi_3 W dt = n\nu_3 dt = np_3 N^2 dt,$$

so that $\Pi_3 = \nu_3/W = p_3 N^2/W$, Π_3 being in units of cm⁻¹, ν_3 in s⁻¹ as before, but p_3 in cm⁶ s⁻¹.

A complete list of the symbols assigned to each of the coefficients and other parameters together with the units in which they are measured is given in section 6.

2.2. Scope and Arrangement of Data

An arbitrary choice has been made to limit the data presented to those gases that are of most general interest, viz: the monatomic gases, the common molecular gases H₂, N₂, O₂, CO, NO, CO₂, and air.

The aim is to provide a collection of the most reliable data on these gases and in order to avoid confusion, much of the earlier data, which has been superseded by values obtained with higher purity gas samples, is omitted. Since the coefficients are dependent on E/N rather than on E/p , results from papers that do not give the gas temperature to which the gas pressure corresponds are excluded unless there is a dearth of other data available. In these cases, the values of E/N are calculated assuming a stated value of the temperature and if they are the only data available the values of E/N are designated by $(E/N)_{Est}$ in graphs and tables. Other considerations relevant to specific coefficients are given in the introduction to the appropriate section and in the text accompanying the diagrams.

The collected data are presented in both graphical and tabular form. Except for data excluded for the reasons given above, an attempt has been made to include all published data for the coefficient involved on any given graph. Thus the graphs provide an indication of the amount and range of data available in each case and of the degree of agreement which exists.

On the other hand, the tables which are grouped at the end of each section have mostly been prepared with a view to providing usable accurate sets of numerical values (where these are available) rather than for comparative purposes. Thus, for cases in which there are a number of sets of experimental and theoretical results in good agreement, if there is an experimental set of data which seems more accurate than the others, this has often been the only set tabulated. Where there are no published experimental values, theoretically computed values have been used in some cases.

For the coefficients discussed in sections 3.1 to 3.7, however, the theoretical results have, in most cases, been obtained, as indicated in the text, by using cross sections chosen to give the best fit with the experimental swarm data. The theoretical values on the graphs in these sections thus do not provide an independent confirmation of the experimental data, themselves. They are included on the figures to show the cases in which such calculations have been made and the degree of agreement which has been achieved using this procedure. In section 3.8, on the other hand, the theoretical values for the recombination coefficient do not depend on the experimental values for r and thus, where they exist, provide an independent comparison.

Where there is a considerable spread in the available data and no clear reason to prefer one set of data over the others, no table has been included because values, accurate to within the spread, can readily be obtained

from the graphs. Theoretical data are often presented as continuous curves in the original papers and the theoretical points on many of the graphs given in this review represent no more than an arbitrary selection of points, chosen at sufficiently frequent intervals on the original curves to indicate their trend. There may thus well be more points given on the graphs than were originally calculated. Occasionally, in figures on which there are a large number of results in close agreement, a few of the intermediate points in some sets of data have been omitted for clarity. Unless otherwise stated the data given were obtained at ambient temperatures. Temperature is mentioned for specific cases only when data are given for both ambient and other temperatures.

2.3. Method of Preparation of Figures and Tables

Where the original data were published in tabular form, figures were prepared directly from the tables. In most cases, however, the data were published in the form of graphs and in these cases the graphs were enlarged and the co-ordinates of the data points obtained in digital form using a Gerber Digital Reduction System connected to an IBM card punch. The least count of the Gerber reader is 0.1 percent of full scale. The digital data were stored on magnetic tapes and the figures then photocomposed from these tapes on a DD.280 Cathode ray tube at the NOAA Boulder Laboratories computing facility. The plotting error amounts to less than 0.5 percent near the largest value on any particular graph and is somewhat larger where the coefficient is small (< 10 percent of the largest value).

For cases in which the original data were in tabular form, the original tables are reproduced. Where the data were given graphically, the co-ordinates of the data points obtained using the Gerber Digital Reduction System were used to prepare the tables. Since this system gives values of each co-ordinate to four significant figures, a decision on how many of these figures to retain was required. In the published paper, the errors in the coefficient are often not estimated and, even more frequently, no estimate is given in the experimental papers of the errors in the measured parameters that give the values of E/N . However, the value of E/N is seldom determined more accurately than to within about 0.1 percent and the coefficients themselves are usually not determined to better than within about 1 percent. In general, in these circumstances, therefore, as far as the coefficients are concerned, three significant figures are retained for values lying between 1.00 and 2.99 and two significant figures for values lying between 3.0 and 9.9, while in the case of E/N , four significant figures are retained for values lying between 1.000 and 2.999 and three significant figures for those between 3.00 and 9.99. Particular cases in which this procedure is not appropriate are indicated where the tables occur and the procedure adopted is also given there.

3. Data Survey

3.1. Drift Velocity

An electron swarm in the presence of a steady electric field both diffuses and moves as a whole in the direction of the field. In the steady state, the center of mass of the swarm attains a velocity W , which may be expressed [1432] (see also W. P. Allis, in *Handbuch der Physik*, S. Flugge, ed., Springer, Berlin, 1956, Vol. 21, p. 383), in terms of the ratio E/N and of atomic parameters by the relationship²

$$W = - (4\pi e/3Nm) \int_0^\infty (c^2/q_m) (df/dc) dc, \quad (1)$$

where e and m are the electronic charge and mass respectively, c is the speed of an electron, q_m the momentum transfer cross section which is a function of the speed and f the speed distribution function normalized through the equation

$$4\pi \int_0^\infty f(c) c^2 dc = 1.$$

Using pulse techniques, W may be determined experimentally by direct measurement of the time taken for the swarm to move through a measured distance. At relatively high values of E/N , where the short transit times make precise direct measurements of this sort difficult, many of the data available were determined using a less direct method originally devised by Townsend and Tizard [2104] before the advent of pulse techniques. In this method, the deflection, in a direction perpendicular to both the electric and magnetic fields, of a swarm travelling a measured distance under the influence of crossed uniform electric and magnetic fields is determined. From measurements of this sort, a quantity W_M , which is related to W by

$$W_M = \psi W, \quad (2)$$

is determined. The constant ψ is sometimes known as the magnetic deflection coefficient [181], and often has a value near unity (see for example Townsend, Phil. Mag. **23**, 880, 1937, and Huxley, Australian J. Phys. **13**, 718, 1960). It can be shown that

$$\psi = \frac{3 \int_0^\infty (c/q_m^2) (df/dc) dc}{4\pi \left[\int_0^\infty (c^2/q_m) (df/dc) dc \right]^2}, \quad (3)$$

for small magnetic fields B such that $\omega = eB/m \ll Nq_m c$. The value of ψ thus depends on the distribution function

²This relationship is strictly true only for the case of elastic collisions but in fact has a much wider range of applicability (see R. W. Crompton, Adv. in Electronics & Electron Physics **27**, 1, 1969).

and on the variation of q_m with c , being unity if cq_m is a constant. Where they are available, the values of both W and W_M are given in the following graphs and tables for the gases listed in section 2.2.

3.1.a. Drift Velocity—Helium

Much attention has been paid to the region of low values of $E/N < 13 \times 10^{-17}$ V cm², because this is a particularly favorable one in which to obtain values of q_m from data on drift velocities. The experimental values obtained in this region are given in figure 1.1. Consideration has to be given to the effect of diffusion which leads to a correction of the measured value of the drift velocity by a factor of the form $[1 + (c_1/d)(D/W)]$ where d is the drift distance, D the diffusion coefficient and c_1 is a dimensionless constant. It has however been shown [1434] that under usual experimental conditions this correction is small (< 10 percent).

The two most recent sets of data [530, 2433] are in good agreement. The error in the data of Crompton, Elford, and Jory [2433] is estimated as less than 1 percent over the whole range investigated and less than 0.5 percent for $E/N < 1.82 \times 10^{-17}$ V cm². Accordingly, this complete set of data is tabulated in table 1.1(a) together with the results of Pack and Phelps [530] at values of E/N below those investigated by Crompton et al. (Crompton et al. indicate that the data of Bowe [1623] have been revised since they were published and now agree within 2 percent of their data, but no details are available.)

There have been a number of recent investigations of the variation of W with gas pressure in helium both at room temperature [4055, 4058] and at 77 K [4913, 4355]. No significant variation of W with pressure was observed for N up to about 3×10^{20} cm⁻³ and E/N down to about 0.008×10^{-17} V cm². At higher pressures, however, the decrease in W with increasing N shown in table 1.1(b) was reported [4055], the accuracy of most of the measurements being estimated to be within 1 or 2 percent.

The results of a number of theoretical computations of W in this region of low values of E/N are given in figure 1.2. It can be seen that the different theoretical treatments give values in general agreement with each other and with experiment, even though they are based on different assumptions and different variations of q_m with energy. Two of the recent sets of data [1062, 2433] in figure 1.2 were obtained by an iterative process involving numerical integration of the Boltzmann equation and are in detailed agreement within the experimental error with the two most recent sets of accurate experimental data. The difference in the values of q_m required to give this detailed agreement with the two different sets of experimental data is small, being about 15 percent at 3 eV but less than 3 percent below 0.6 eV. In both cases a useful cross-check on the values of q_m obtained was provided by using them to calculate values of other

swarm parameters which were compared with experimental values (see W_M and sec. 3.2.a). Also given in figure 1.2 are the theoretical data resulting from a stochastic treatment [3463] of the problem.

At higher values of E/N , values of W for electrons in glow discharges in helium have been obtained experimentally from measurements on the positive columns of such discharges. The results which are shown in figure 1.3 and table 1.2 are in good agreement with each other.

There are no completely satisfactory theoretically computed values of W for helium for $E/N > 1.3 \times 10^{-16}$ V cm². The most reliable to date are those obtained by Monte Carlo methods [3897, 5490]. Both these sets of data are in good agreement with experiment and one is shown in figure 1.3. The treatment, however, effectively ignored the effect of electron diffusion and the production of slow electrons by ionization on the energy distribution [see 3752].

Results [530, 4355] illustrating how the drift velocity at very low values of E/N varies with temperature in the range from room temperature down to 77 K are shown in figure 1.4 and the values obtained at 77 K tabulated in table 1.1(c).

Other measurements [1714, 2266] at temperatures near 4 K and at pressures up to the saturated vapor pressure show that the mobility in that region lies well below the kinetic theory values, being a factor of about 10^4 smaller at the saturated vapor pressure.

3.1.b. Values of W_M —Helium

Values of W_M accurate to within about ± 2 percent obtained recently by Crompton, Elford, and Jory [2433] are compared in figure 1.5 with the earlier values of Townsend and Bailey [201]. The values of q_m obtained by Crompton et al. from considerations of their experimentally measured values of W (see previous section) were used to calculate W_M and these computed values are given together with their experimental data for W_M in table 1.3.

3.1.c. Drift Velocity—Neon

A number of direct measurements of drift velocities have been made for neon for values of E/N below 4×10^{-17} V cm². Robertson [4862] has shown that low levels of N₂ impurity (~ 20 ppm) can significantly affect the values of W in neon and the most reliable sets of data seem to be those of Nielsen [1387], of Pack and Phelps [530] and of Robertson [4862], the latter being the most accurately determined with an estimated error of ± 1 percent. These results which were obtained at ambient temperatures are given in table 1.4(a) and compared with the other results in figure 1.6.

Robertson was also able to find values of the momentum transfer cross section which gave calculated values of the drift velocity in agreement with experiment and these values are also shown in figure 1.6.

Values obtained [4862] at a temperature of 77 K are also given in figure 1.6 and table 1.4(b).

There are no experimental values of W in neon for $E/N > 4 \times 10^{-17}$ V cm² measured under swarm conditions. Values of W for glow discharge in neon have however been obtained [2164, 5780] by measuring the current and electron concentration. These values are compared in figure 1.7 with the most reliable theoretically computed swarm values available, which are those of Thomas and Thomas [3752] who obtained good agreement between results calculated using a numerical solution of the Boltzmann equation [3753] and using Monte Carlo methods [3752].

3.1.d. Values of W_M —Neon

The only values of W_M obtained for neon using the Townsend method are those of Bailey [2280] for neon known to contain 1 percent helium. (See fig. 1.8.) Robertson [4862] has suggested that the gas may well have also contained small amounts of molecular impurities (see 3.2.b).

3.1.e. Drift Velocity—Argon

There have been a large number of investigations of the drift velocity of electrons in argon giving data mainly in the range $3 \times 10^{-18} < E/N < 4 \times 10^{-16}$ V cm². During the course of this work, it has become clear [732, 2042] that small amounts of impurity (<1 percent) particularly of molecular gases such as N₂ give rise to very different values (typically a factor of two higher) than those obtained for pure argon. Relatively low values characteristic of pure argon have been obtained in several investigations and these results are shown in figure 1.9 and given in table 1.5 from which it can be seen that there is a fair measure of agreement. Theoretical values in the same range of values of E/N are shown in figure 1.10 from which it can be seen that there is a general agreement both between them and with the experimental results shown in figure 1.9, although a variety of methods and assumptions have been used.

The theoretical values calculated by Englehardt and Phelps [292] were obtained on the basis of values of q_m (as a function of energy) determined by comparing experimental and theoretical values of both W and D/μ over the range $6 \times 10^{-21} < E/N < 10^{-15}$ V cm². (The values of q_m have, however, been questioned, see Golden, Phys. Rev. **151**, 48, 1966.) The calculated values of W for the lower end of this range [1062] are in detailed agreement with the experimental data at temperatures of 300 K and 77 K as shown in figure 1.11. The experimental values are given in table 1.6 together with one theoretical value at the lowest value of E/N .

There are fewer experimental data available for $E/N > 4 \times 10^{-16}$ V cm², those of Jager and Otto [2153] of Wagner [4971] and of Brambring [4944] being shown in figure 1.12 and in table 1.7. These data are in good agreement with each other and fit smoothly with the

values of W obtained at low values of E/N shown in figure 1.9. Also shown in figure 1.12 are the theoretical values computed by Golant [881] on the basis of smoothed experimental collision cross sections and the energy distribution which he calculated and which was found to be in good agreement with that of Englehardt and Phelps [292] at $E/N = 3 \times 10^{-16}$ V cm².

Measurements [4058, 5188] covering a total range of pressure from about 750 to 76,000 Torr show no variation of drift velocity with pressure outside the experimental error.

3.1.f. Values of W_M —Argon

The only published experimental data for W_M for argon are the early results of Townsend and Bailey [197, 199]. The results from [199] which were for less contaminated gas samples are shown in figure 1.13 and table 1.8 together with the theoretical values by Englehardt and Phelps [292] calculated using their values of q_m obtained from consideration of data on W and D/μ . A possible explanation of the difference is that the theoretical data refer to the limit of vanishingly small magnetic fields and Chen's calculations (J. Appl. Phys. **37**, 2205, 1966) of the Hall effect show that W_M is lower at higher fields.

3.1.g. Drift Velocity—Krypton and Xenon

Few experimental data are available for xenon and krypton, the most reliable at low values of $E/N < 4 \times 10^{-17}$ V cm² being those of Pack, Voshall, and Phelps [439] whose results are shown in figure 1.14(a) and given in tables 1.9 and 1.10.

Frost and Phelps [1062] were able to determine values of q_m as a function of energy which gave good agreement with the above experimental data, although in the case of krypton the values of q_m required lie well below the measured values (Ramsauer, Ann. Physik **12**, 529, 1932) of total cross section and other calculated values (O'Malley, Phys. Rev. **130**, 1020, 1963) of q_m ; there are differences also in the case of xenon when compared with other results.

At higher values of E/N the only data available are those of Wagner [4971] for xenon for $112 \times 10^{-17} < E/N < 255 \times 10^{-17}$ V cm² which are shown in figure 1.14(b) and given in table 1.10(a).

There are no published values of W_M for either krypton or xenon.

3.1.h. Drift Velocity—Hydrogen

A particularly thorough investigation of W for hydrogen over the range $2.8 \times 10^{-19} < E/N < 5.7 \times 10^{-16}$ V cm² was carried out by Lowke [1155], who showed that using modern experimental techniques and making the necessary corrections for diffusion [see also 1434], measurements of W can be made to within 1 percent. His results for $E/N < 2.82 \times 10^{-17}$ V cm² are shown in figure 1.15 together with those of Pack and Phelps [530] and Bradbury and Nielsen [1381] with which they agree

within the experimental error. The two recent sets of data [530, 1381] are given in table 1.11(a).

In this region of low E/N , the values of W depend on the gas temperature, and the results given in figure 1.16(a) (for clarity of presentation the figure has been limited to values of $E/N < 2.82 \times 10^{-18}$ V cm² and the data at a temperature of 195 K in [530] omitted) show that the two published sets of experimental data [530, 1155] for this temperature dependence agree well with each other and with the theoretically computed values [181, 3463]. The experimental data at 77 K are given in table 1.11(b).

The majority of the results of the relatively large number of investigations which have been made of W for hydrogen lie within the range $2.8 \times 10^{-17} < E/N < 113 \times 10^{-17}$ V cm², and most of the published data (which it is possible to express unambiguously in terms of E/N) are shown in figure 1.17 and selected values given in table 1.12. It can be seen that, bearing in mind that the experimental error increases to about 2 or 3 percent at the upper end of this range, there is good agreement between the various sets of data.

Theoretical values of W were obtained by Frost and Phelps [181] on the basis of a numerical solution of the Boltzmann equation and as can be seen from figures 1.16(a) and 1.17 these values were in good agreement with experiment for a determined set of values of the appropriate momentum transfer, rotational and vibrational cross sections involved. The theoretical values of W obtained for a temperature of 77 K by Bell and Kostin [3463] using a stochastic treatment also agree well with experiment.

In a recent interesting series of experiments [4450, 4913, 4916, 4484] aimed at investigating the pressure dependence of the drift velocity [4913] and determining cross sections for rotational and vibrational transitions [4450, 4484] in hydrogen, measurements have been made of the drift velocity for both normal and para-hydrogen at 77 K and for normal hydrogen at 293 K. These results are collated together in [4916] from which the data shown in figure 1.16(b) were obtained. The data for normal hydrogen in figure 1.16(b) are the most accurate available and comparison with figures 1.16(a) and 1.15 shows that they are in good agreement with the earlier experimental data. The data for para-hydrogen agree to within 0.5 percent with the earlier results of [3659] and [2267].

Gibson [4450] obtained a set of cross sections that led to calculated values of W in agreement within ± 0.7 percent with these experimental results for hydrogen. In view of the experimental accuracy of the data these cross sections which are considerably different from those obtained earlier [181] are the most accurate available to date. In the above series of experiments the variation of drift velocity with gas pressure was also investigated for $6 \times 10^{18} < N < 10^{20}$ cm⁻³ for both normal and para-hydrogen at 77 K and some of the results for normal hydrogen at the higher pressures

are given in table 1.14(a). The linear decrease with increasing pressure observed in both normal and para-hydrogen is consistent with the hypothesis [3665] that temporary negative ion formation associated with rotational excitation occurs.

Results of experiments at higher pressures in normal hydrogen at room temperature [3382, 4058] which are shown in table 1.14(b) seem to require a different explanation (as do other results [5103] at 77.8 K over a similar range of values of N) and a multiple scattering theory has been proposed (W. Legler, Phys. Lett. A29, 719, 1969) for this range.

In the region above $E/N = 112 \times 10^{-17}$ V cm², only one extensive set of data exists. This was obtained by Schlumbohm [1625] from a study of avalanche transit times, and the results are shown in figure 1.18 and table 1.13, which also contain three values obtained by Blevin and Hasan [3459] at the lower end of this range.

3.1.i. Values of W_M —Hydrogen

Creaser [3384] recently obtained values of W_M , with an error of about 1 percent as shown in figure 1.19 and in table 1.15, where they are compared with the early values of Townsend and Bailey [195].

3.1.j. Drift Velocity—Nitrogen

There have been well over thirty experimental investigations of W for nitrogen. It is convenient to consider these in two groups depending on whether the value of E/N used was larger or smaller than about 5.6×10^{-16} V cm². For the lower values of E/N , electrical shutter methods first employed by Bradbury and Nielsen [1381] have been widely used [530, 1155, 1381, 2285] and as shown by Lowke [1155] can give results accurate to within 1 percent when appropriate precautions are taken. For the most part, the results obtained at higher values of E/N are from studies of electron pulses and avalanches [1314, 1716, 1763, 2136, 2154].

Those results for $E/N < 5.6 \times 10^{-16}$ V cm² which can be expressed as functions of E/N are shown in figure 1.20. (For clarity of presentation, this is given in two parts (a) and (b) with one set of results common to both to aid comparison.) Results in the range $5.6 \times 10^{-16} < E/N < 5.6 \times 10^{-15}$ V cm² are shown in figure 1.21 from which it can be seen that there is more scatter in the results in this region than at the lower values of E/N covered by figure 1.20, where there is good agreement between the various sets of data obtained. Numerical values in the two different E/N ranges are given in tables 1.16 and 1.17.

There is only one set of measurements published for $E/N > 5.6 \times 10^{-15}$ V cm². These are shown in figure 1.22 and given in table 1.18.

At low values of E/N , the drift velocity is dependent on the temperature and the experimental values of W at the extreme temperatures used both below and above room temperature are shown in figure 1.23 where they are compared with some of the results at room tempera-

ture. Values at temperatures of 77 K, 195 K and 473 K are given in table 1.19. Also shown in figure 1.23 are the values calculated [218] on the basis of a numerical solution of the Boltzmann equation and appropriate choice of the collision cross sections involved.

For highly compressed gases at room temperature, Grunberg [3382] has shown that the drift velocity depends not only on E/N but also on N and this dependence is shown in table 1.20. This table also includes the results of Allen and Prew [5188] which also show a significant decrease of W with increasing N at the lower values of E/N used.

3.1.k. Values of W_M —Nitrogen

An extensive investigation of the values of W_M in nitrogen, which extended and substantially confirmed the early work of Townsend and Bailey [195], was recently carried out by Jory [1429]. The results of these experimental investigations are shown in figure 1.24 and in table 1.21.

Values of the magnetic deflection coefficient ψ obtained using the value of W_M shown in figure 1.24 together with the values of W obtained by Lowke [1155], are shown in figure 1.25, where they are compared with the values of ψ obtained theoretically by Englehart, Risk, and Phelps [218] with which there is good agreement.

3.1.l. Drift Velocity—Oxygen

A complication in oxygen and other electronegative gases is the formation of negative ions, with the result that fewer measurements have been made in these gases than in hydrogen and nitrogen.

The majority of the data [727, 1336, 1412, 2553, 5226, 4901] available has been obtained using drift-tube measurements and the results obtained for $2.8 \times 10^{-17} < E/N < 57 \times 10^{-17}$ V cm² are shown in figure 1.26(a). These results include one set [4901] for oxygen containing 2 percent hydrogen in which the dissociative detachment reaction between O⁻ and H₂ rapidly removes the O⁻, so avoiding some of the complications arising from the presence of negative ions. Recently measurements have been extended [5135] to very low values of E/N using the drift-dwell-drift method (see sec. 3.3.a). The values obtained are compared with the earlier drift tube studies, which extend into the region of E/N below 2.8×10^{-17} , in figure 1.26(b). As can be seen from figure 1.26(a) and (b) the self-consistent set of cross sections chosen to give agreement with the earlier drift velocity data and other measured swarm parameters seem to give values of W that are too low at very low values of E/N . Numerical data for the region $E/N < 57 \times 10^{-17}$ V cm² are given in table 1.22.

For values of $E/N > 57 \times 10^{-17}$ V cm² most of the experimental data have been obtained from studies of pulsed avalanches [1314, 2370] and these are given in figure 1.27 together with a few experimental results of drift tube measurements [5226] and theoretically com-

puted values [2553] at the lower end of the range. The experimental data in this range are given in table 1.23.

3.1.m. Values of W_M —Oxygen

The only values of W_M available for oxygen are those obtained in the early work of Townsend and Bailey [195] and of Brose [2089] working in the same laboratory. Since the results of Townsend and Bailey show a dependence of W on N as well as on E/N which was removed by the improved techniques used by Brose, only Brose's results are given in figure 1.28 and in table 1.24.

3.1.n. Drift Velocity—Carbon Monoxide

The only data for W for carbon monoxide that can be expressed unambiguously in terms of E/N are those of Pack, Voshall, and Phelps [439] which were obtained for three different temperatures. These are shown in figure 1.29 and table 1.25. The authors state that there is scatter in their data for $T = 300$ K at low values of E/N ($< 8.5 \times 10^{-19}$ V cm²) because of attachment, which did not, however, occur excessively under other conditions.

3.1.o. Values of W_M —Carbon Monoxide

The only values of W_M available for carbon monoxide are those of Skinker [200]. The value of the temperature to which the values of E/p given in the paper correspond is not given and a temperature of 15° C was assumed to obtain the data given in figure 1.30 and table 1.26.

3.1.p. Drift Velocity—Nitric Oxide

The only published values of drift velocities for nitric oxide are the recent results of Parkes and Sugden [4943], obtained using a pulsed drift tube. These are shown in figure 1.31. The results were obtained at temperatures of 294 and 459 K and for a range of values of N from 2.1 to 7.1×10^{17} cm⁻³ but no systematic change with temperature or with gas number density was found within these ranges.

3.1.q. Values of W_M —Nitric Oxide

There have been two experimental investigations of W_M for nitric oxide. Skinker and White [200] reported a variation of W_M with N at a given E/N attributed to the formation of negative ions, but gave a graph of W , E/N which is in fair agreement with later results of Bailey and Somerville [2385] using a method [see 2278] designed to eliminate errors due to the presence of negative ions. Bailey and Somerville's results are given in figure 1.32 and table 1.27.

3.1.r. Drift Velocity—Carbon Dioxide

Several sets of experimental values of W for CO₂ have recently been obtained for values of $E/N < 2.82 \times 10^{-16}$ V cm² both from studies of electron pulses [725, 1833, 2920, 1184] and by using the electrical shutter method [439, 2141]. The latter method is capable of high accuracy provided adequate consideration is given to

possible sources of error. Elford [2041], using this method, obtained results with an estimated error of less than 0.5 percent for values of $E/N < 9.1 \times 10^{-17} \text{ V cm}^2$ and of less than 1 percent for $9.1 \times 10^{-17} < E/N < 2.12 \times 10^{-16} \text{ V cm}^2$. These results, together with the others obtained for $E/N < 2.82 \times 10^{-16} \text{ V cm}^2$, are shown in figure 1.33 in which the theoretical values (calculated [2553] using a self-consistent set of cross sections chosen to give agreement with a range of data on electron swarms) are also given. A set of numerical values is given in table 1.28.

There is a marked dependence of W on pressure in carbon dioxide as shown by the results of Allen and Prew [5188] which are given in figure 1.34. This agrees with the earlier results obtained by Lehning [4203] at $E/N < 2.8 \times 10^{-16} \text{ V cm}^2$ for the ratio of the drift velocity at various pressures in the range 8000 to 25,000 Torr to that at 500 Torr. The ratio decreased rapidly throughout the range, becoming less than 1/100 at pressures of 20,000 Torr and above.

There are no published experimental data for W in CO_2 in the range $2.82 \times 10^{-16} < E/N < 10^{-15} \text{ V cm}^2$ but there are values in the region above $E/N = 10^{-15} \text{ V cm}^2$ obtained by Frommhold [2154] and by Schlumbohm [1314]. Frommhold's values are given in figure 1.35. Schlumbohm does not give his experimental data, but states that they can be represented by the equation $W = 1.58 \times 10^6 (E/p_{20})^{0.591}$ for E/p_{20} from 150 to 2000 $\text{V cm}^{-1} \text{ Torr}^{-1}$ and values calculated using this equation are also shown in figure 1.35. The data of figure 1.35 are given numerically in table 1.29.

3.1.s. Values of W_M —Carbon Dioxide

The only experimental values of W_M available for CO_2 are the early results of Skinker [198]. These results are shown in figure 1.36 from which it can be seen that the

experimental values are in agreement with the theoretical values of Hake and Phelps (also shown in the figure) up to values of E/N about $4 \times 10^{-16} \text{ V cm}^2$ but become smaller than the theoretical values at higher values of E/N .

3.1.t. Drift Velocity—Air

There have been few measurements of W for air. There are two sets of experimental data available for $E/N < 10^{-16} \text{ V cm}^2$, one [3165] based on measurements of electron pulses and the other on the electrical shutter method [1412]. The results are given in figure 1.37 and table 1.30.

The only theoretical values for air are those of Heylen [1447] which were obtained using a Maxwellian energy distribution and an assumed form for the variation of the cross section with energy. These theoretical values for $E/N < 10^{-16} \text{ V cm}^2$ are also shown in figure 1.37. (It is not clear from the original paper [1447] to what temperature the values of E/p used correspond. The theoretical values in figure 1.37 have been obtained on the assumption that it is 0°C .)

For $E/N > 9.9 \times 10^{-17} \text{ V cm}^2$ there are also two sets of experimental data available [2305, 2370], both obtained from studies of electron avalanches. These data are shown in figure 1.38 and are given in table 1.31. The available theoretical values [1447] in this region are also shown in figure 1.38, and the theoretical values in the region for which there are no experimental data are included in table 1.31.

3.1.u. Values of W_M —Air

The two sets of experimental data available for W_M for air are shown in figure 1.39 and tabulated in table 1.32.

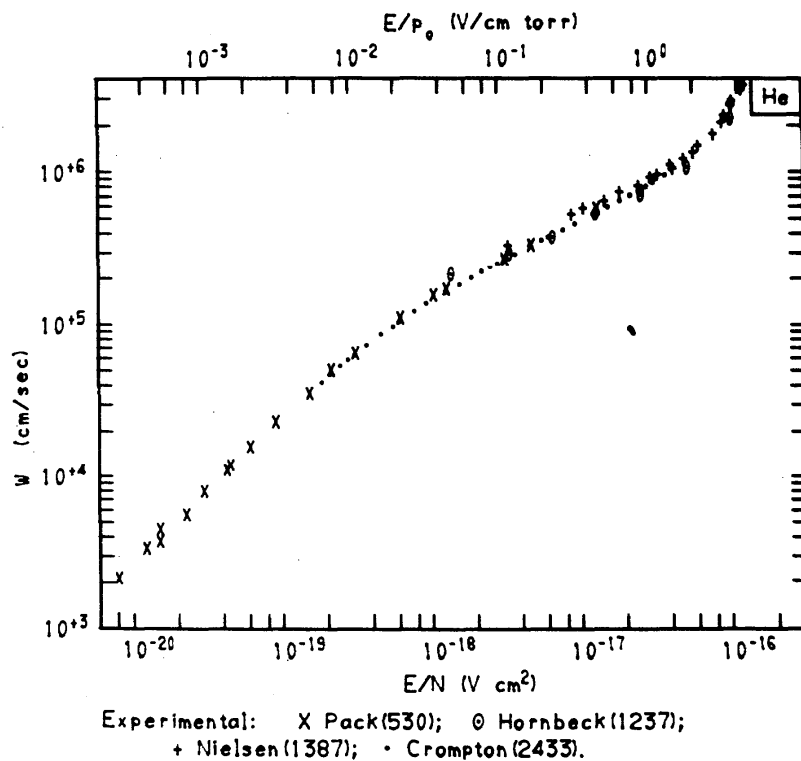


FIGURE 1.1. Experimental values of W as a function of E/N for values of $E/N < 1.3 \times 10^{-16}$ V cm² in helium.

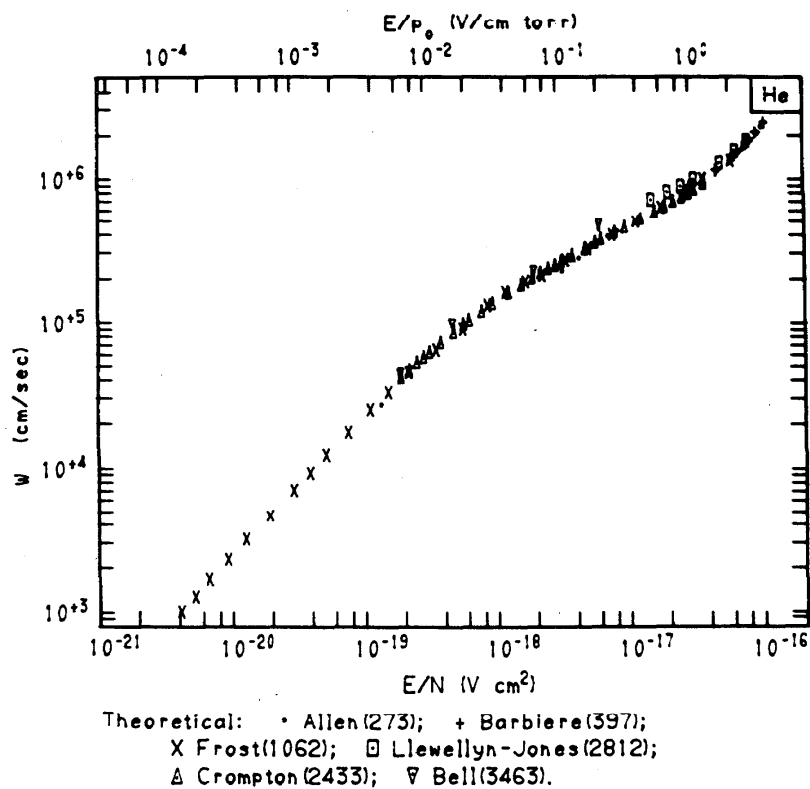


FIGURE 1.2. Theoretically computed values of W as a function of E/N for values of $E/N < 1.3 \times 10^{-16}$ V cm² in helium.

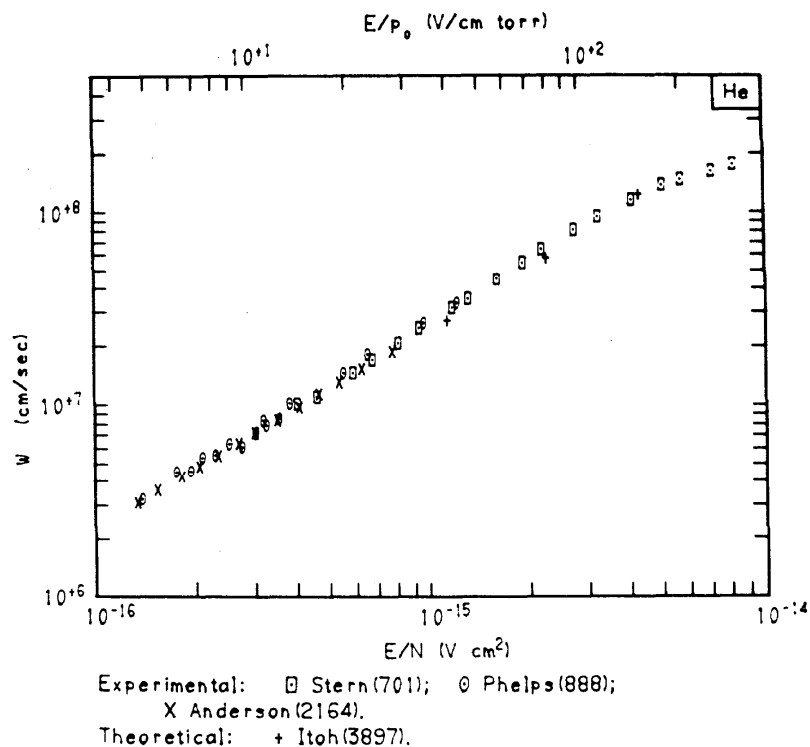


FIGURE 1.3. Experimental values of W for electrons in glow discharges together with computed values for $E/N > 1.3 \times 10^{-16} \text{ V cm}^2$ in helium. (Stern's data have been smoothed to avoid a confusing number of points.)

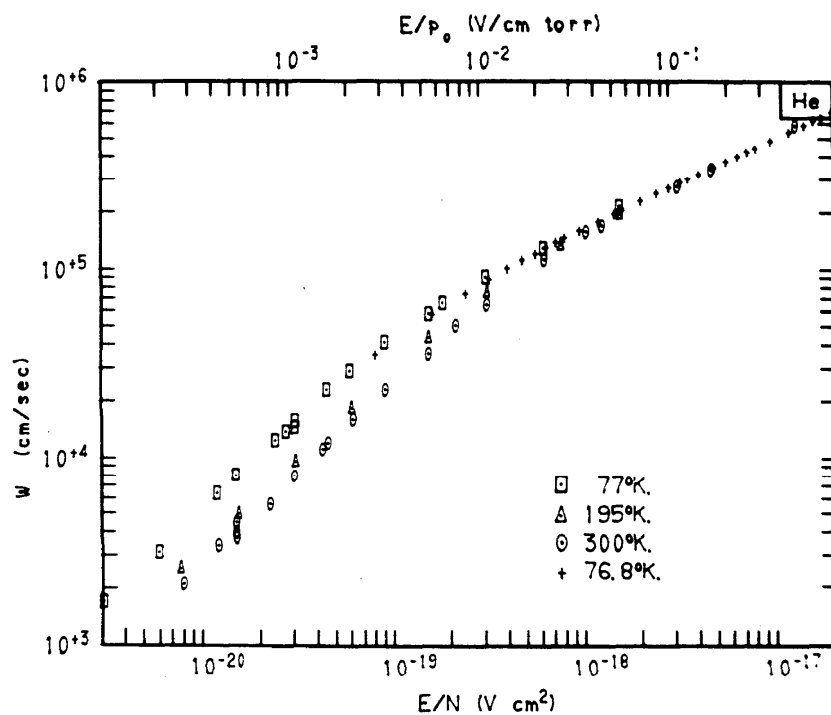
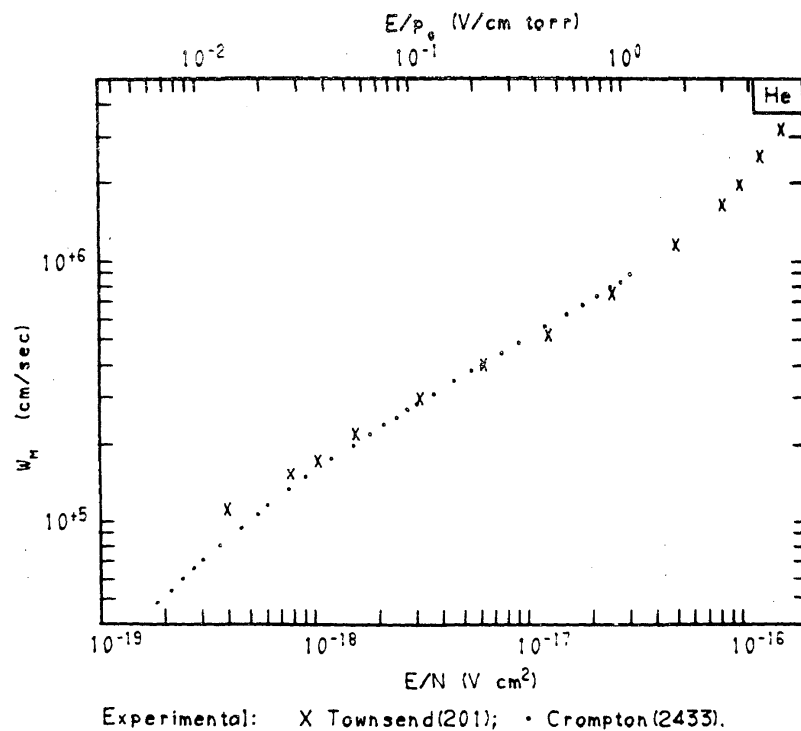
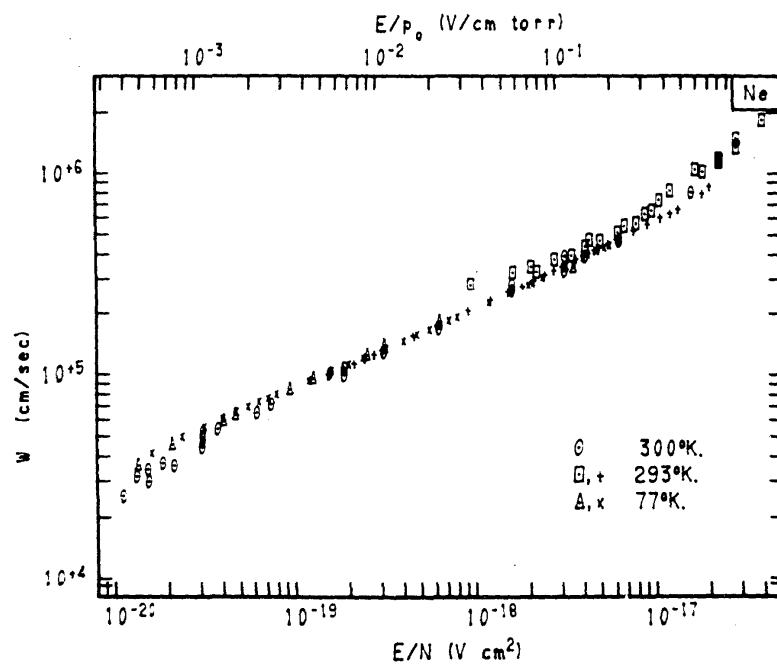


FIGURE 1.4. W as a function of E/N in helium at different temperatures.

FIGURE 1.5. W_N , E/N for helium.FIGURE 1.6. W , E/N for neon for $E/N < 4 \times 10^{-17}$ V cm².

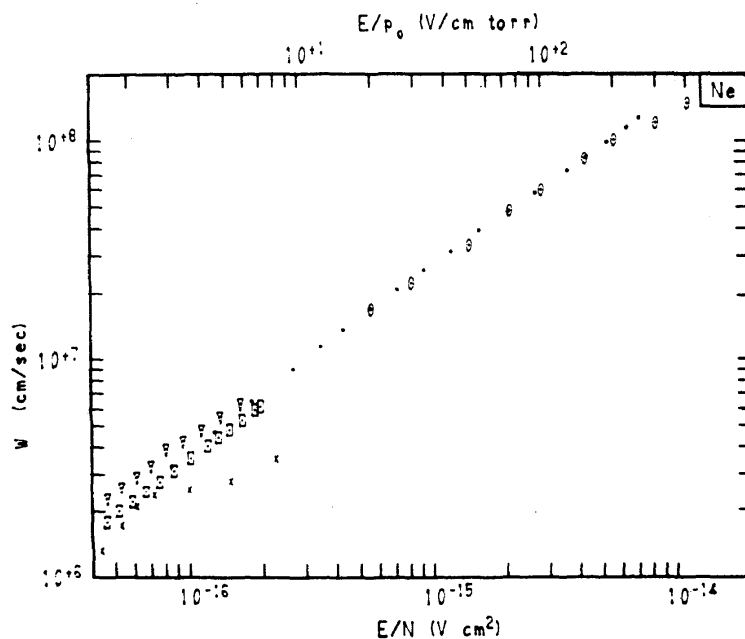


FIGURE 1.7. Values of W as a function of E/N for neon for $E/N > 4 \times 10^{-17}$ V cm².

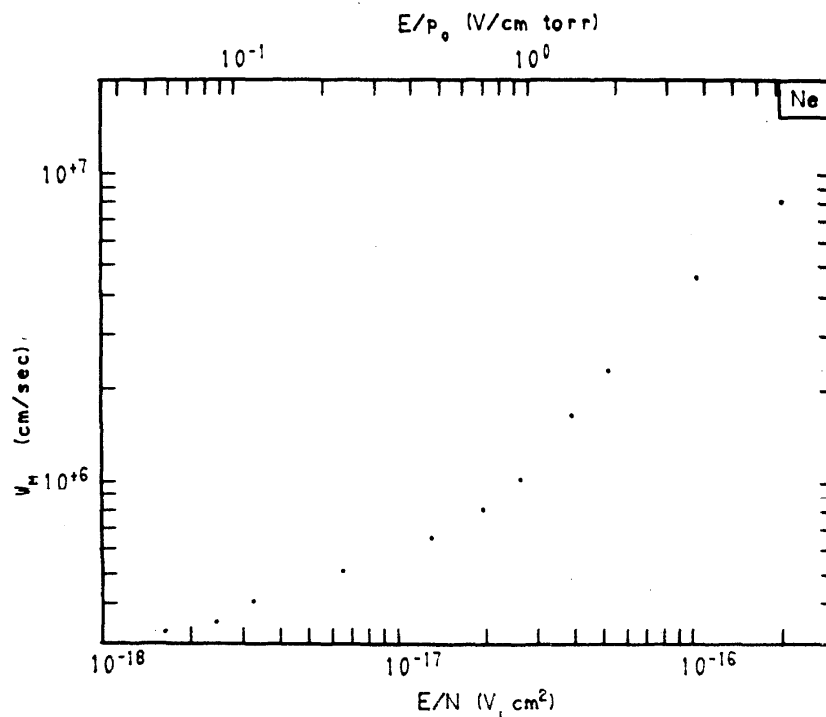


FIGURE 1.8. W_M , E/N for neon.

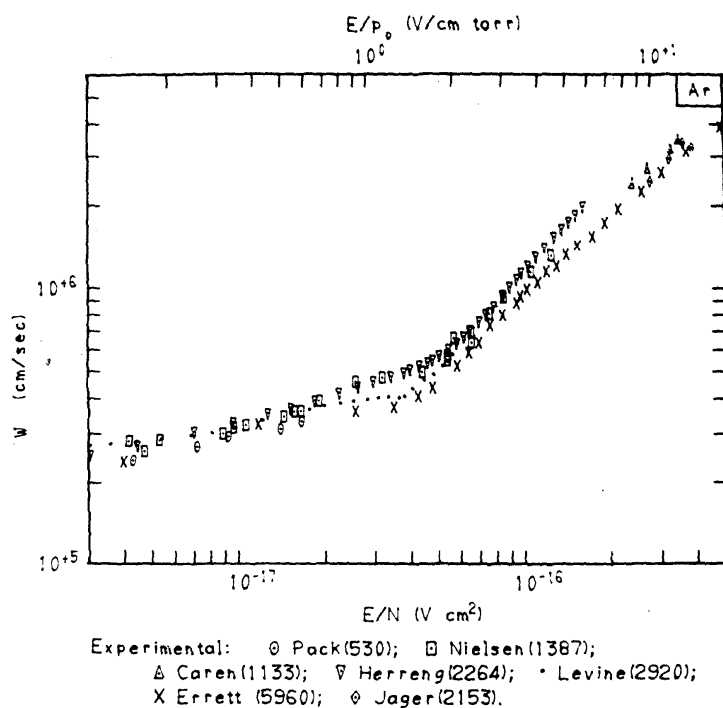


FIGURE 1.9. Experimental values of W as a function of E/N for argon in the range $3 \times 10^{-18} < E/N < 4 \times 10^{-16} \text{ V cm}^2$.

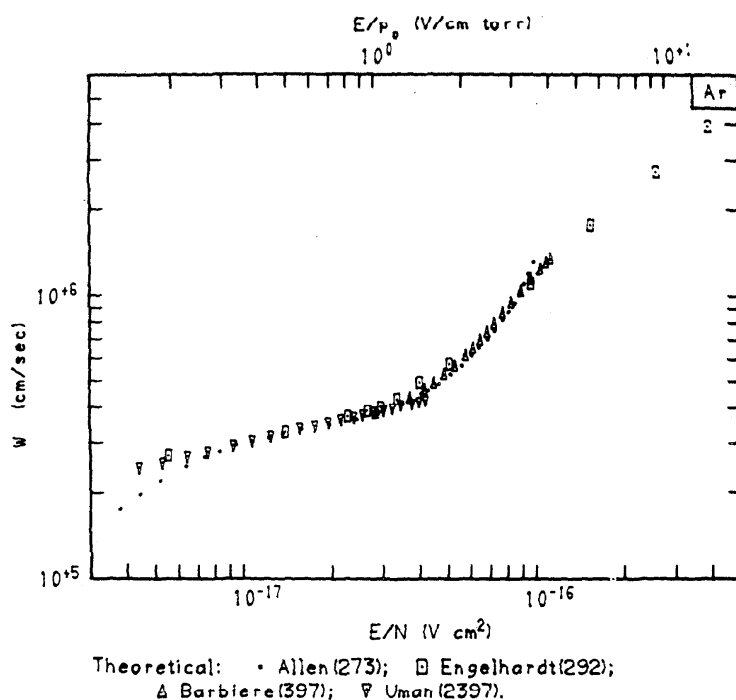
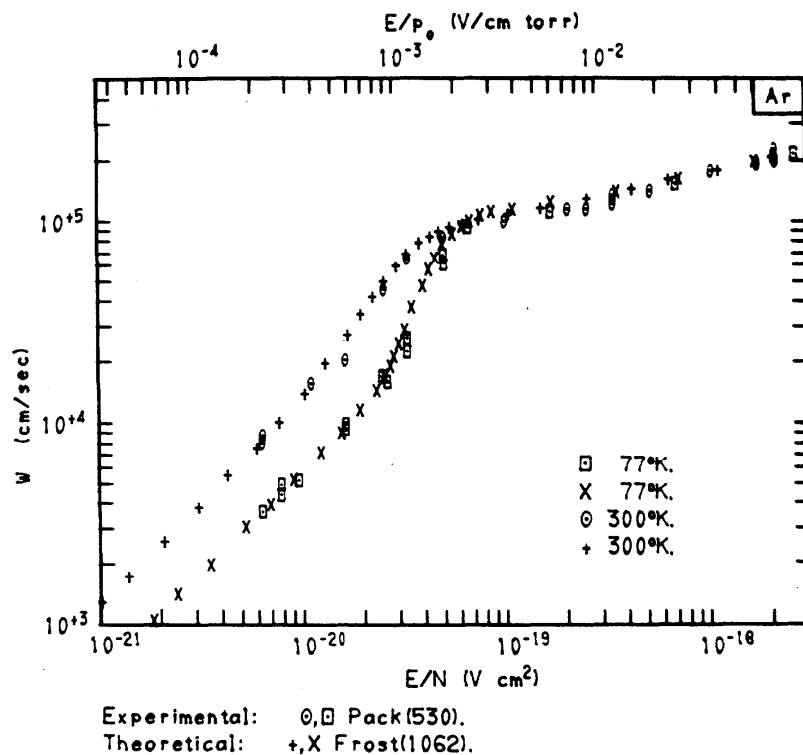
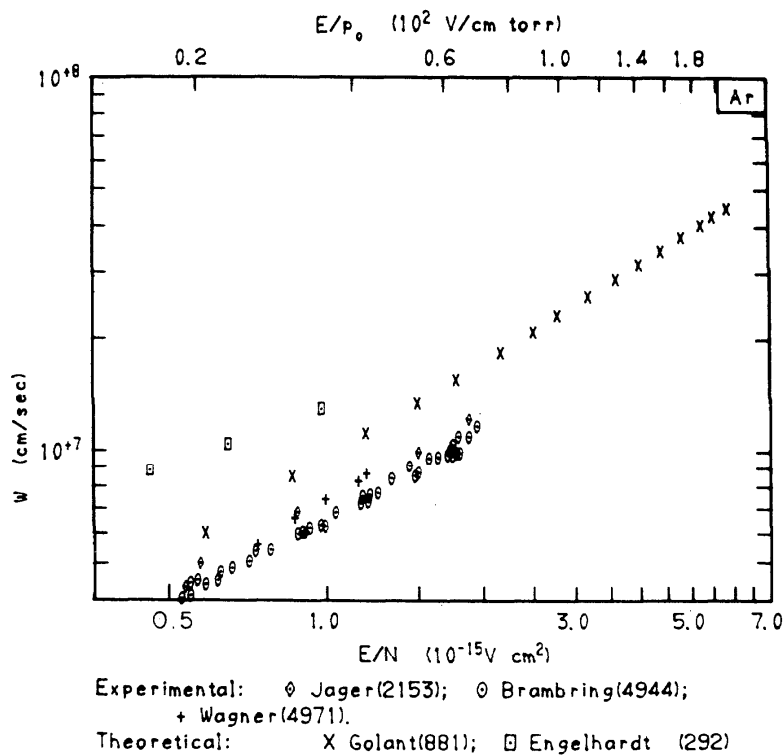
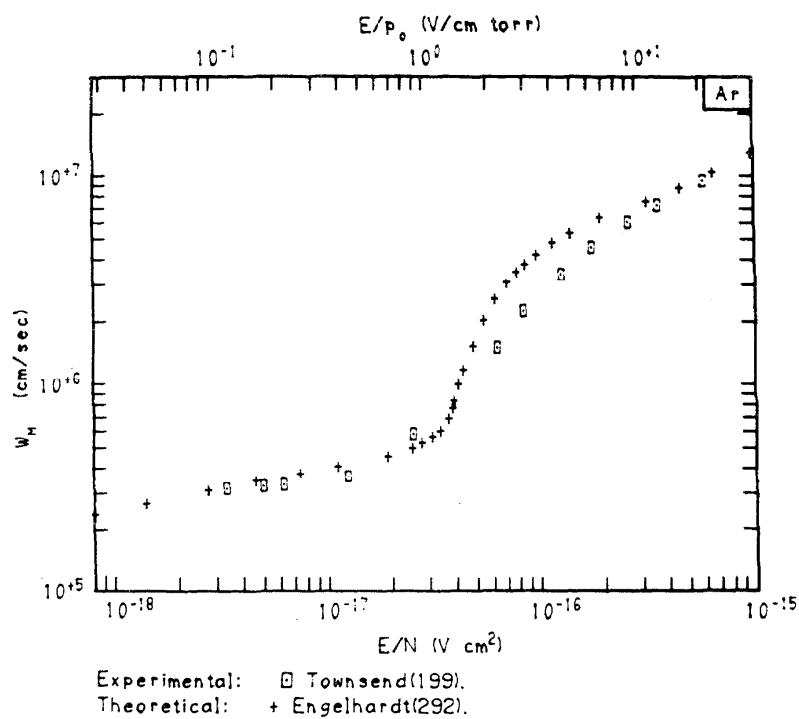
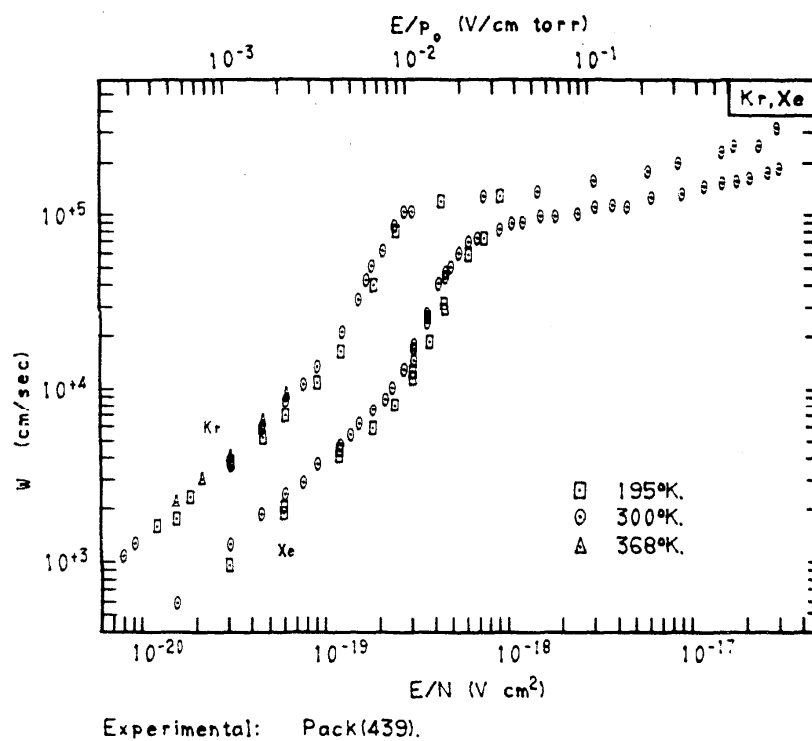


FIGURE 1.10. Theoretical values of W as a function of E/N for argon in the range $3 \times 10^{-18} < E/N < 4 \times 10^{-16} \text{ V cm}^2$.

FIGURE 1.11. W , E/N for argon for $E/N < 3 \times 10^{-18} \text{ V cm}^2$.FIGURE 1.12. W , E/N for argon for $E/N > 4 \times 10^{-16} \text{ V cm}^2$.

FIGURE 1.13. W_M , E/N for argon.FIGURE 1.14(a). Experimental values of W as a function of E/N for krypton and xenon for $E/N < 4 \times 10^{-17} \text{ V cm}^2$.

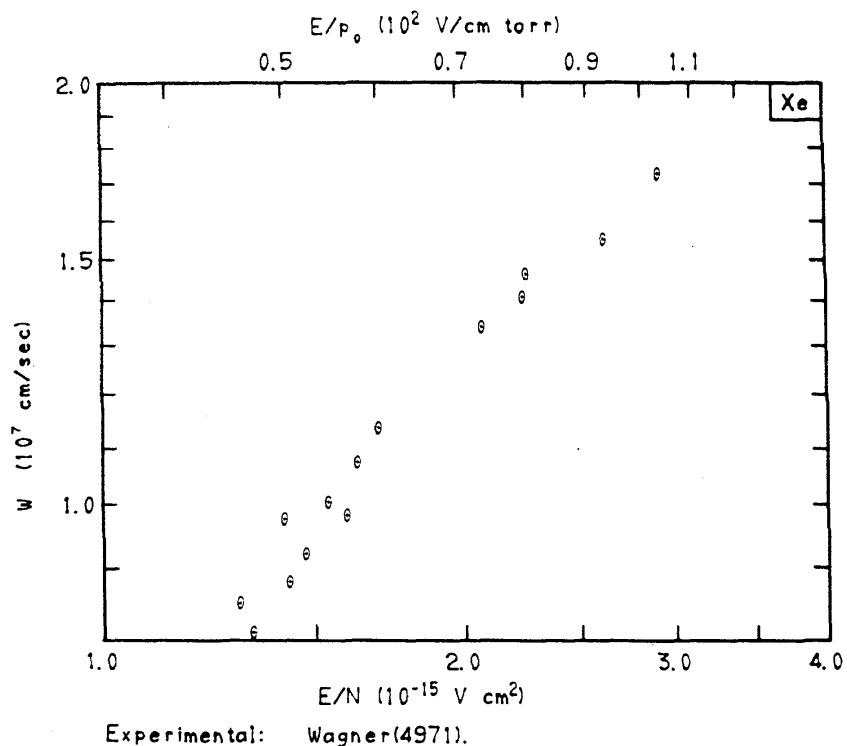


FIGURE 1.14(b). Experimental values of W as a function of E/N for xenon for $E/N > 112 \times 10^{-17} \text{ V cm}^2$.

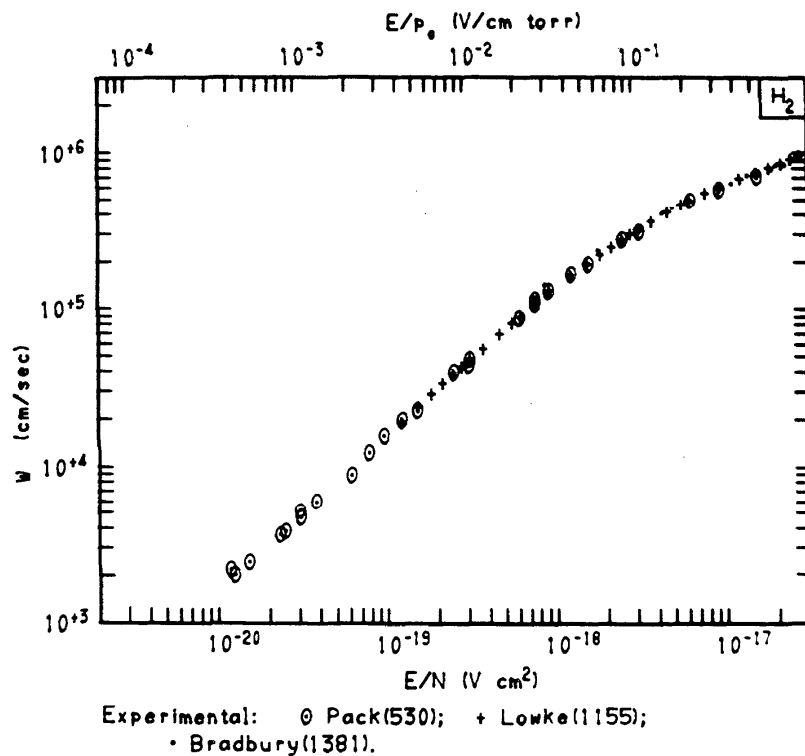
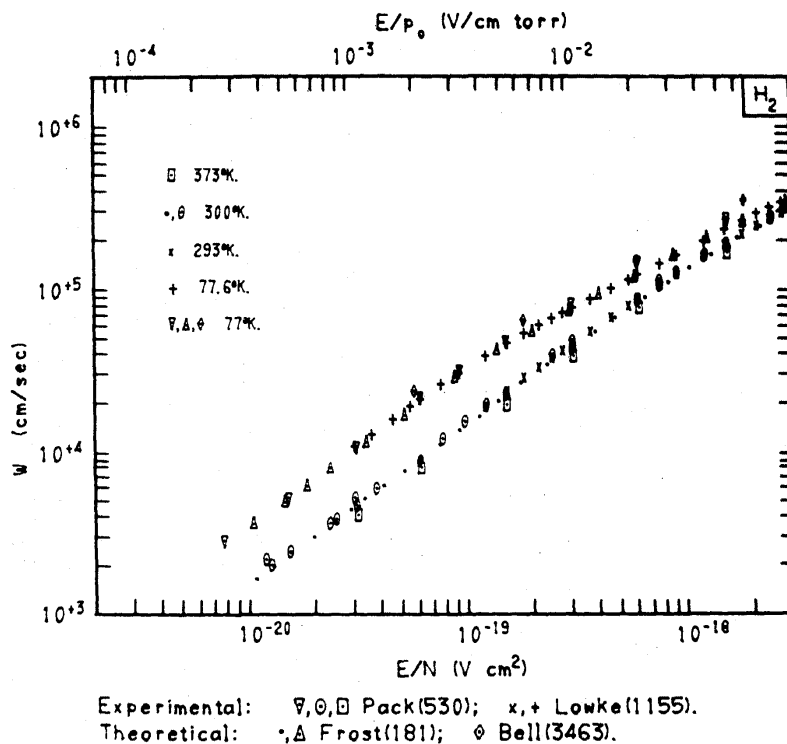
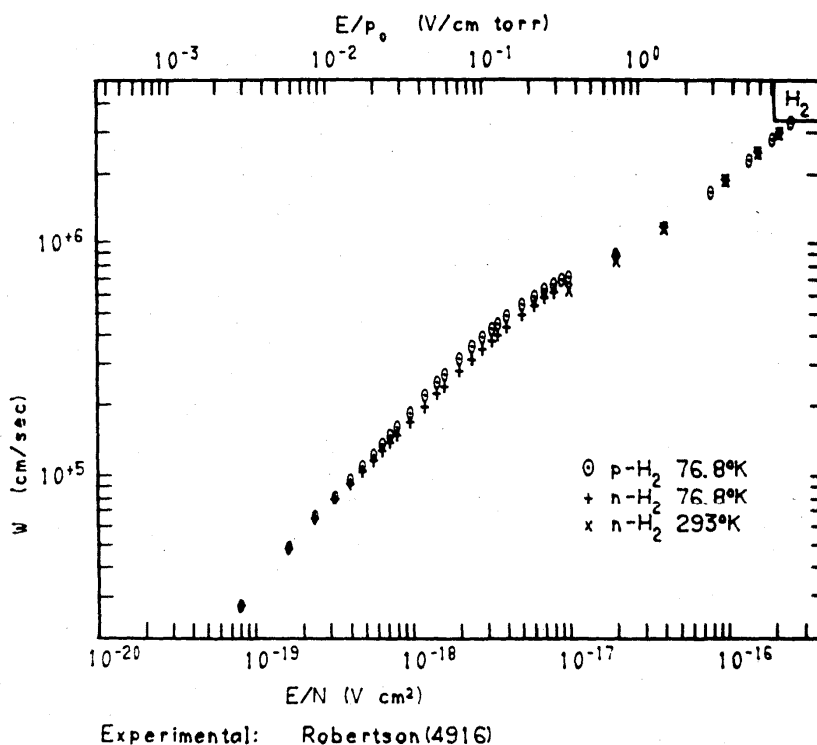


FIGURE 1.15. Experimental values of W as a function of E/N for hydrogen for $E/N < 2.82 \times 10^{-17} \text{ V cm}^2$.

FIGURE 1.16(a). W , E/N for hydrogen for various temperatures at low values of E/N .FIGURE 1.16(b). Experimental values of W in para-hydrogen at 76.8 K and normal hydrogen at 76.8 and 293 K.

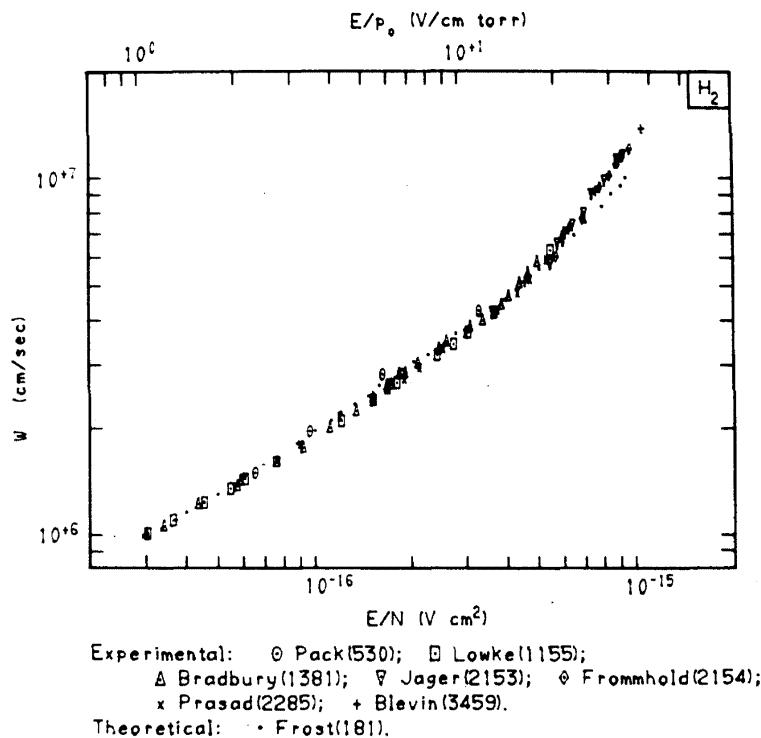


FIGURE 1.17. W , E/N for hydrogen in the range $2.8 \times 10^{-17} < E/N < 113 \times 10^{-17} \text{ V cm}^2$.

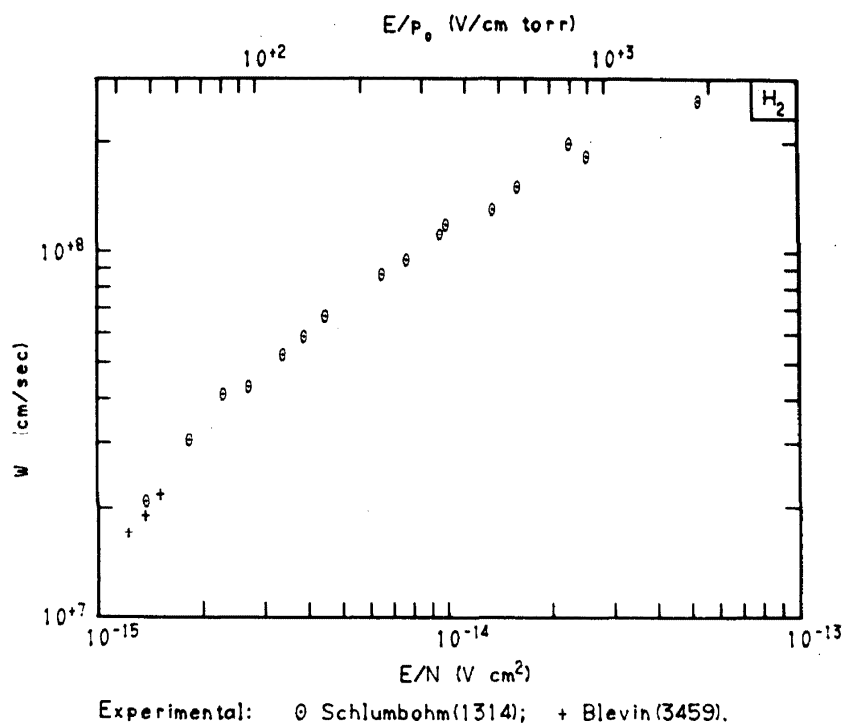
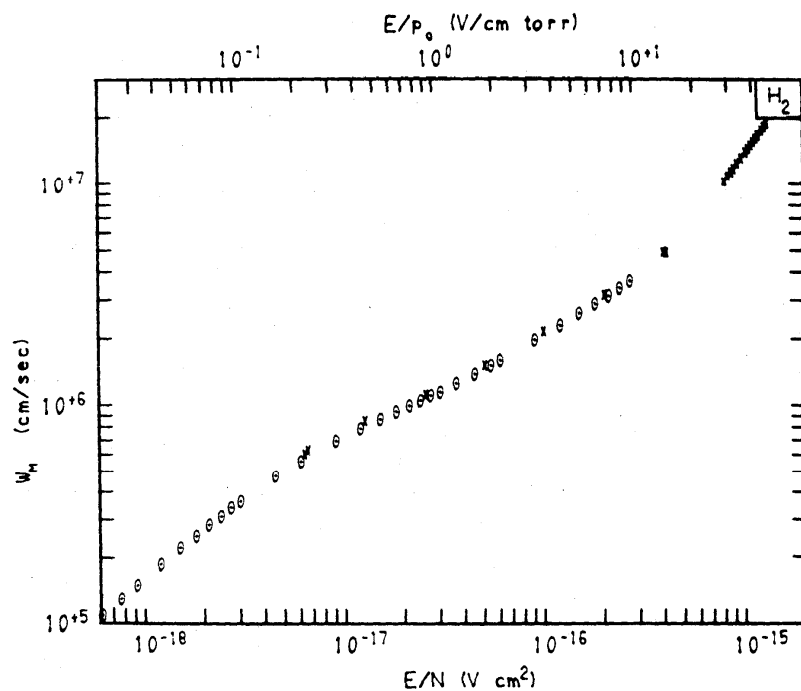
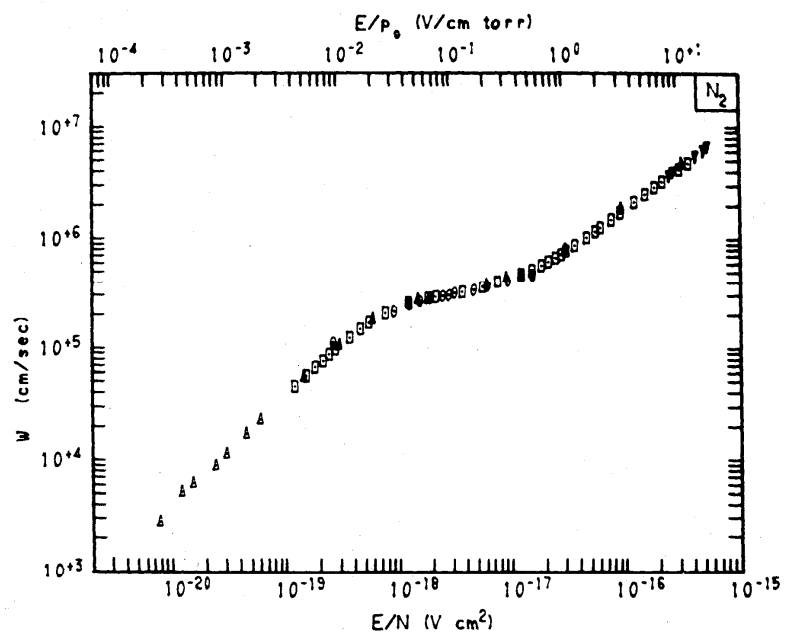


FIGURE 1.18. W , E/N for hydrogen for $E/N > 112 \times 10^{-17} \text{ V cm}^2$.



Experimental: $\circ$ Creaser(3384); $\times$ Townsend(195).

FIGURE 1.19. Experimental W_H , E/N for hydrogen.



Experimental: Δ Pack(530); $\circ$ Christophorou(725);
 $\square$ Lowke(1155); ∇ Prasad(2285).

FIGURE 1.20(a). Experimental values of W as a function of E/N for nitrogen for $E/N < 5.6 \times 10^{-16}$ V cm².

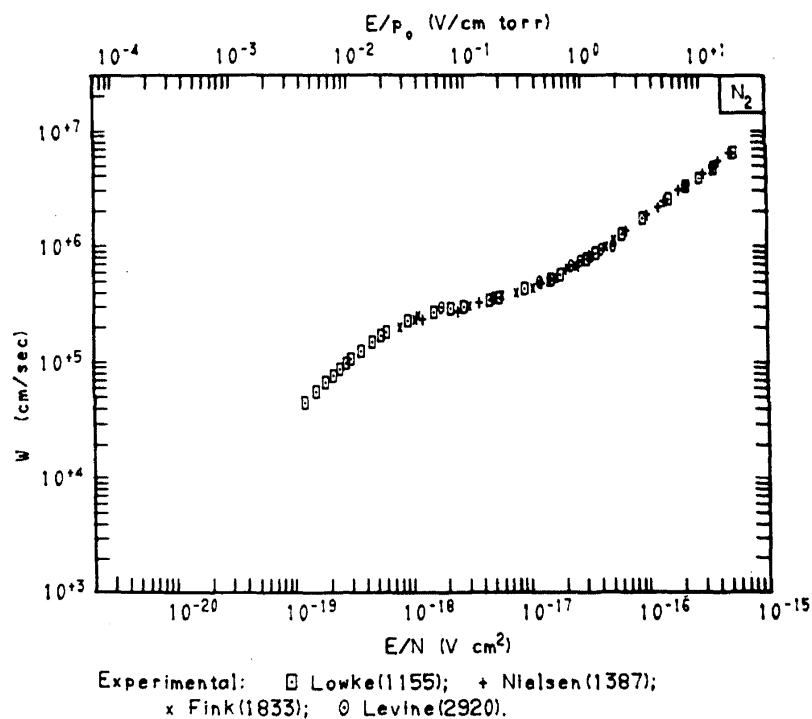


FIGURE 1.20(b). Experimental values of W as a function of E/N for nitrogen for $E/N < 5.6 \times 10^{-16}$ V cm².

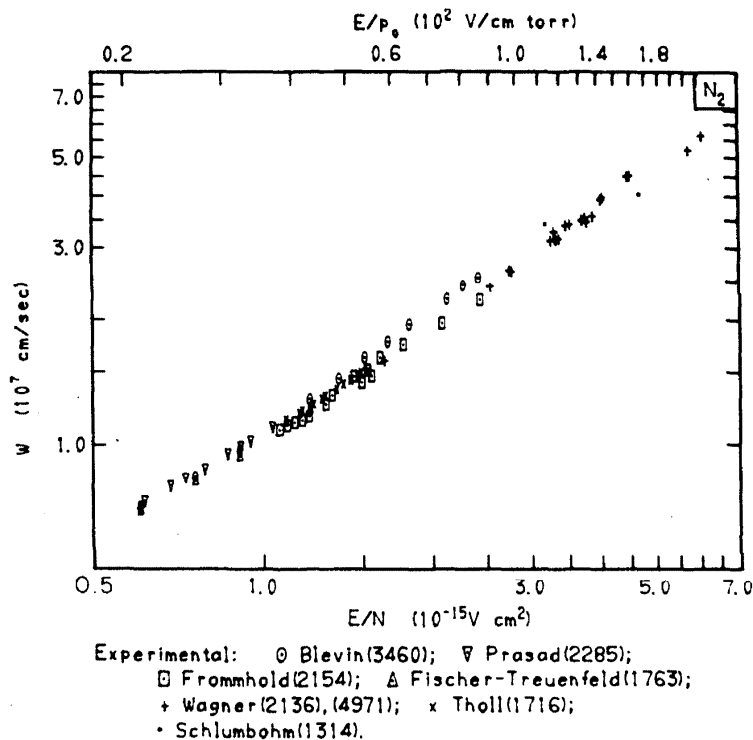
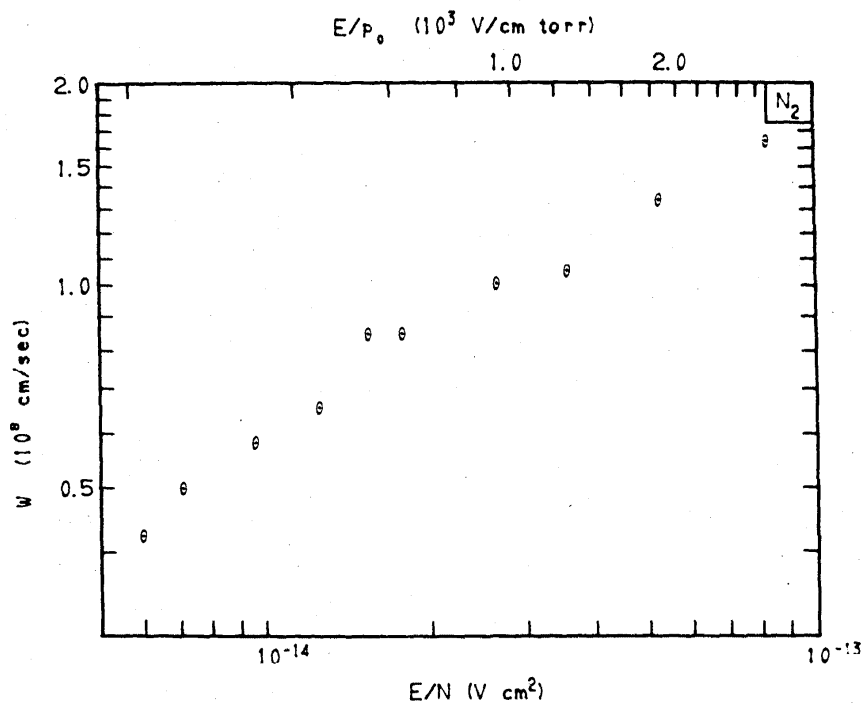
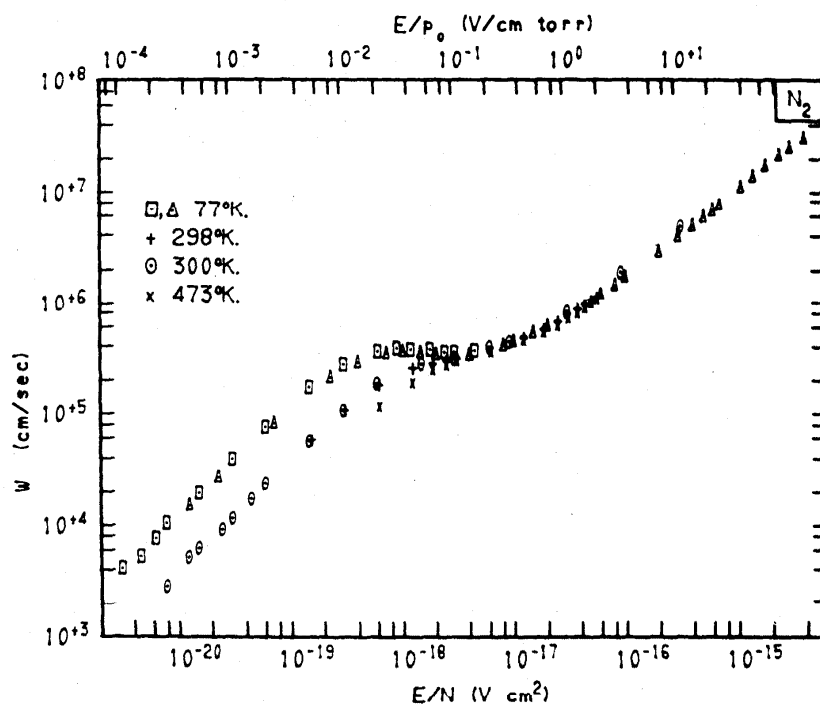


FIGURE 1.21. Experimental values of W as a function of E/N for nitrogen for $5.6 \times 10^{-16} < E/N < 6.2 \times 10^{-15}$ V cm².



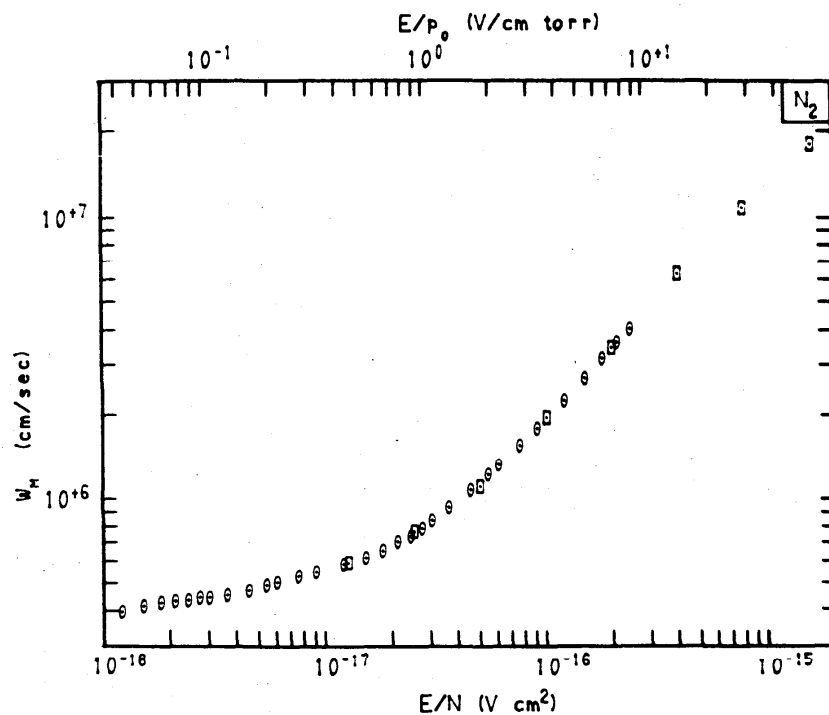
Experimental: O Schlumbohm(1314).

FIGURE 1.22. Experimental values of W as a function of E/N for nitrogen for $E/N > 5.6 \times 10^{-15}$ V cm².



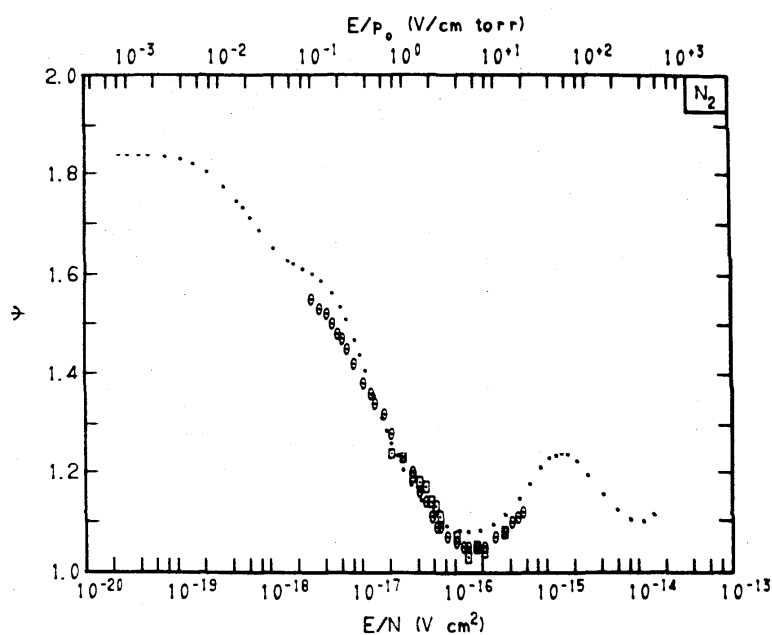
Experimental: □, ○ Pack(530); +, x Hendrick(3662).
Theoretical: Δ Engelhardt(218).

FIGURE 1.23. Values of W as a function of E/N for nitrogen at various temperatures.



Experimental: $\circ$ Jory(1429); $\square$ Townsend(195).

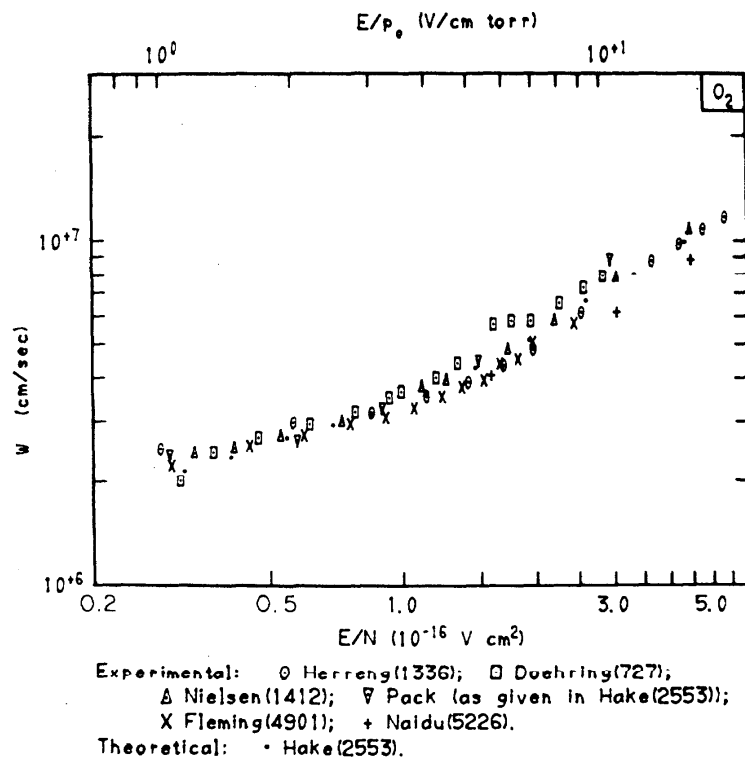
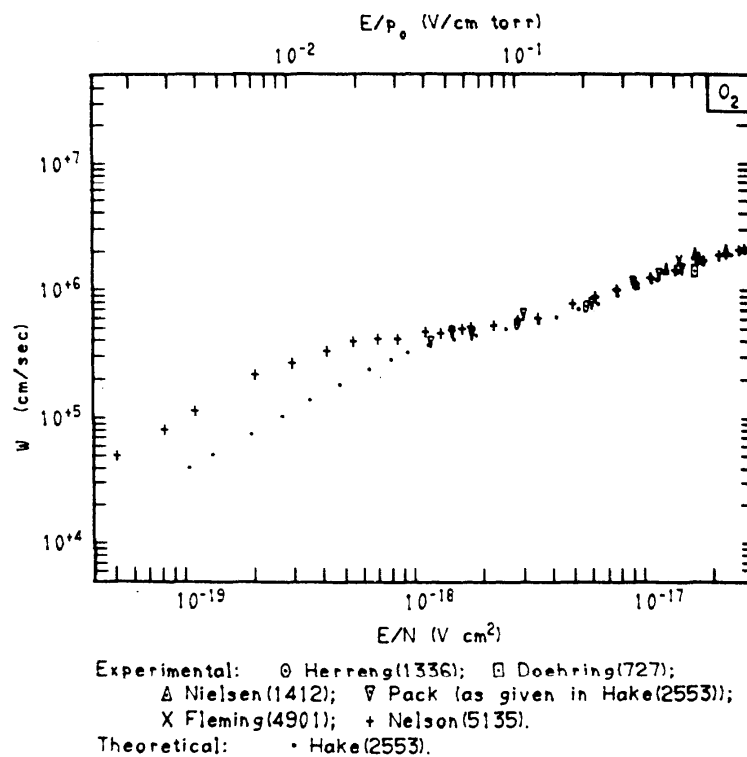
FIGURE 1.24. Experimental values of W_M as a function of E/N for nitrogen.

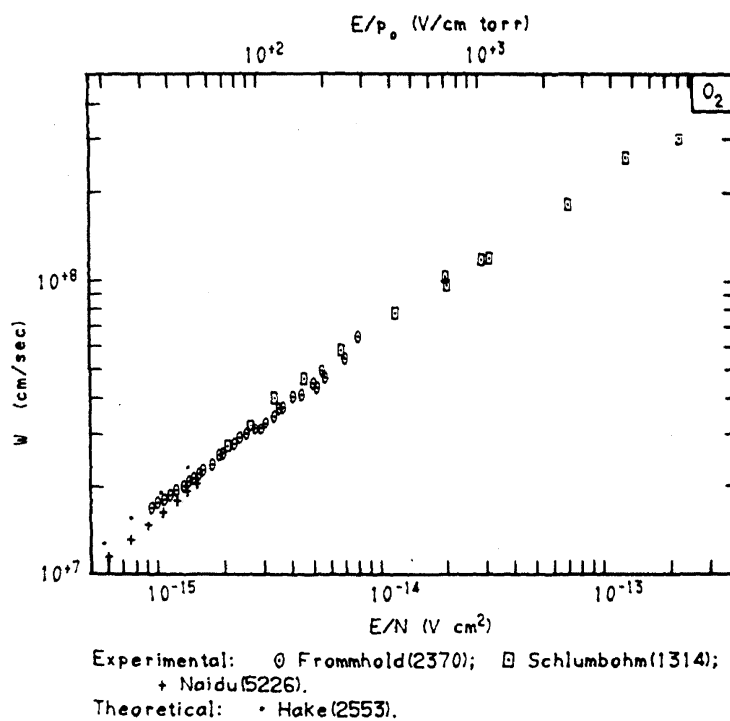
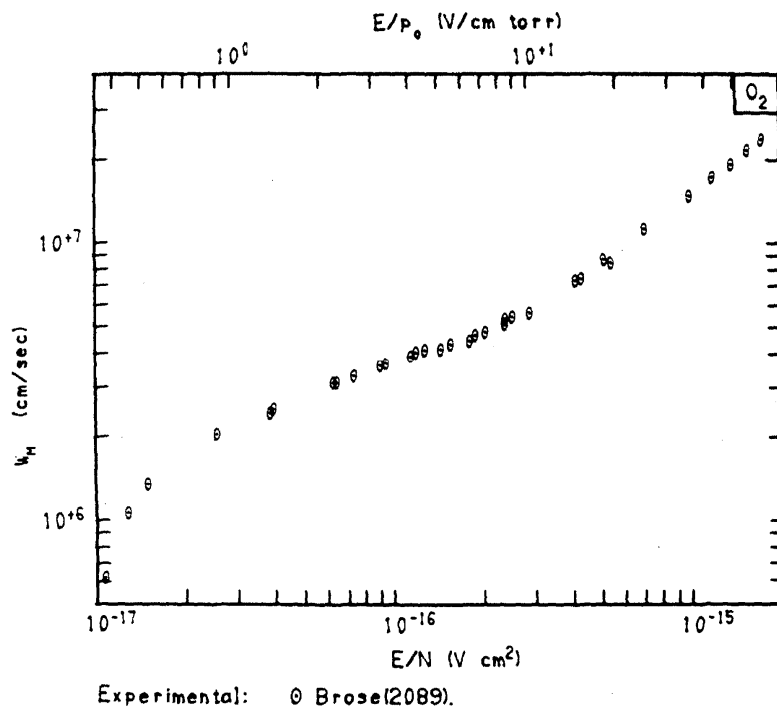


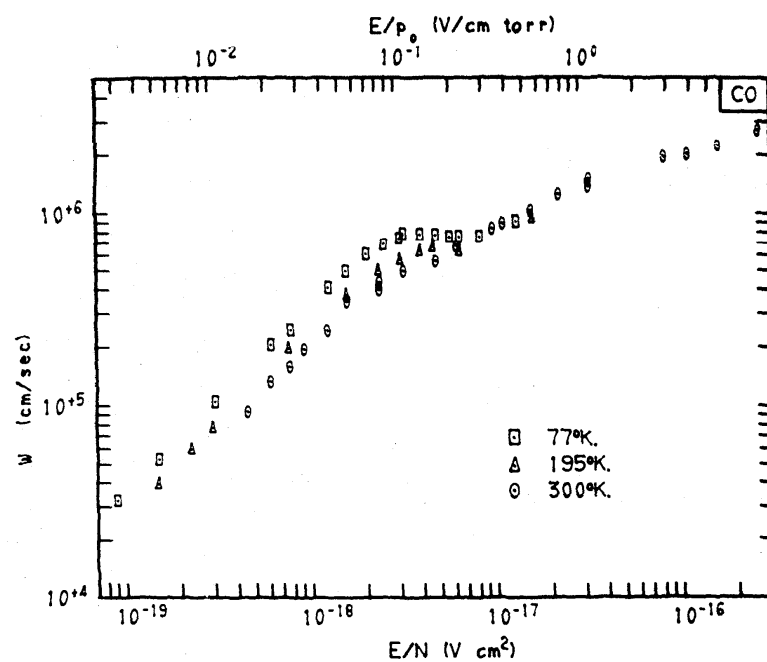
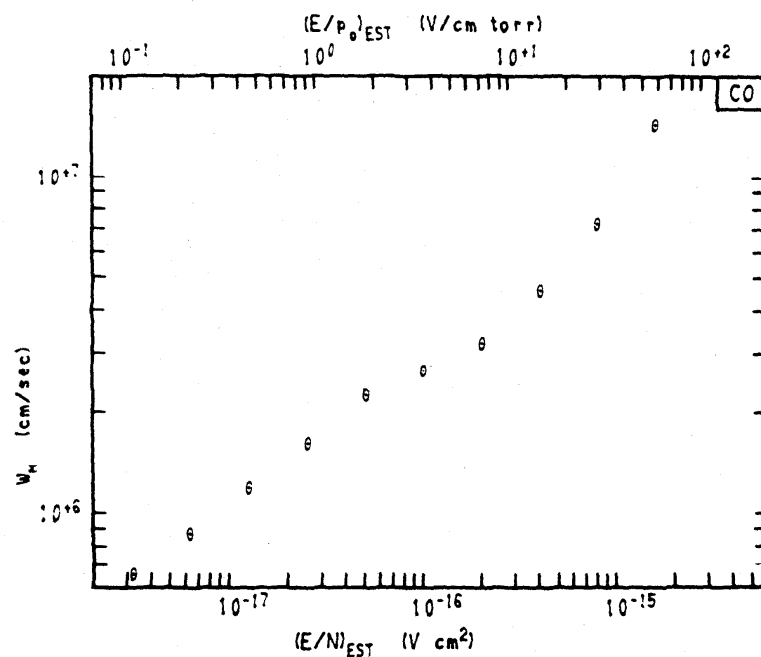
Experimental: $\circ$ Using W_M from Jory(1429) and W from Lowke(1155); $\square$ Using W_M from Townsend(195) and W from Lowke(1155).

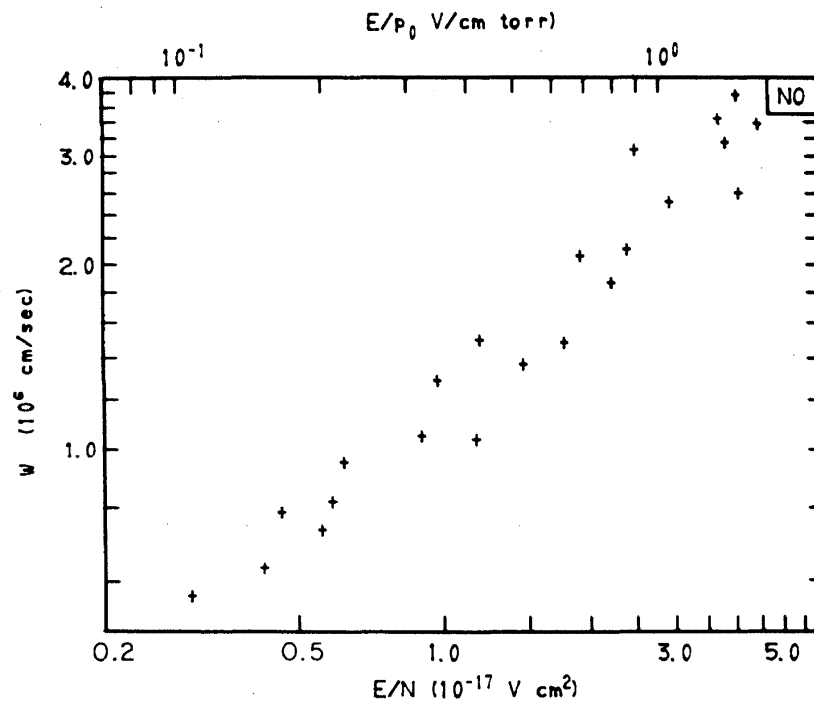
Theoretical: $\cdot$ Engelhardt(218).

FIGURE 1.25. Values of ψ as a function of E/N for nitrogen.

FIGURE 1.26(a). W , E/N for oxygen for $2.8 \times 10^{-17} < E/N < 57 \times 10^{-17} \text{ V cm}^2$.FIGURE 1.26(b). W , E/N for oxygen for $E/N < 2.8 \times 10^{-17} \text{ V cm}^2$.

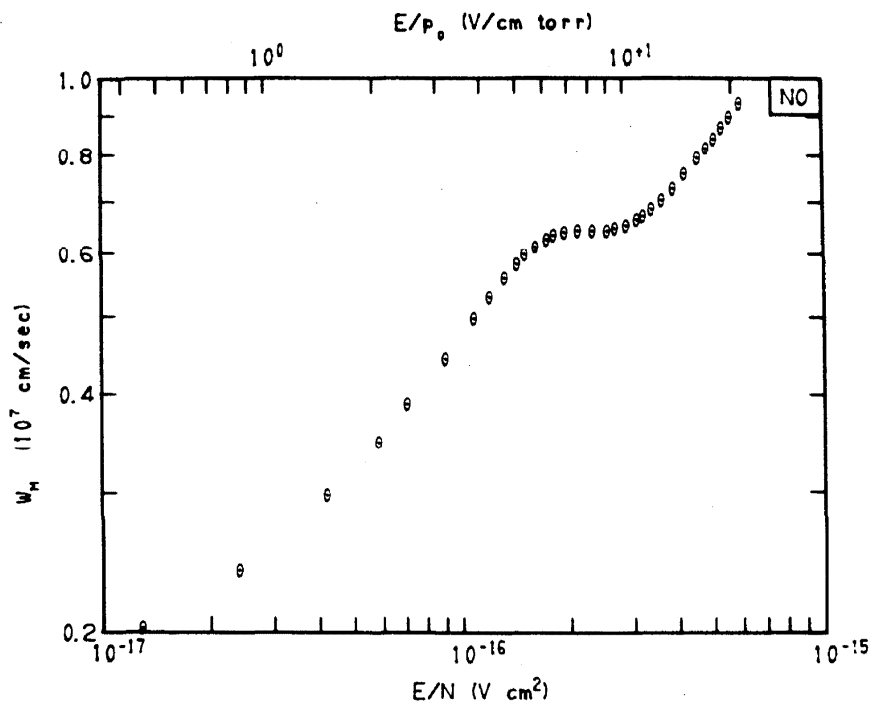
FIGURE 1.27. W , E/N for oxygen for $E/N > 5.7 \times 10^{-16} \text{ V cm}^2$.FIGURE 1.28. Experimental values of W_H , E/N for oxygen.

FIGURE 1.29. Experimental W , E/N for carbon monoxide.FIGURE 1.30. Experimental W_M , $(E/N)_{EST}$ for carbon monoxide.



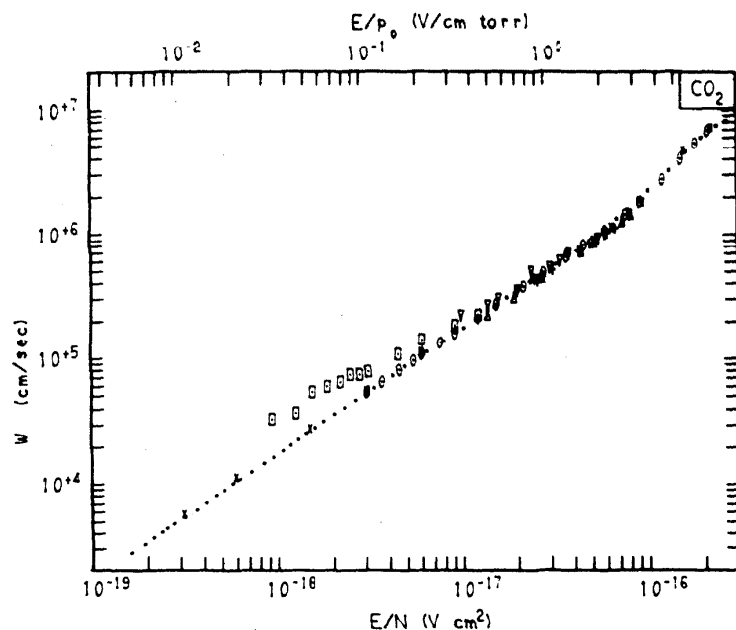
Experimental: Parkes(4943).

FIGURE 1.31. Experimental W , E/N for nitric oxide.



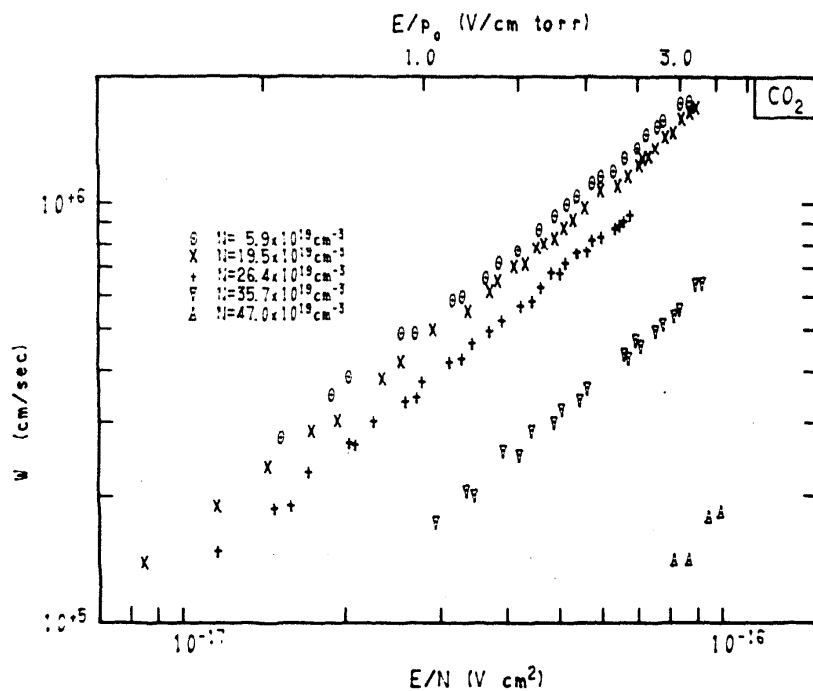
Experimental: o Bailey(2385).

FIGURE 1.32. Experimental W_M , E/N for nitric oxide.



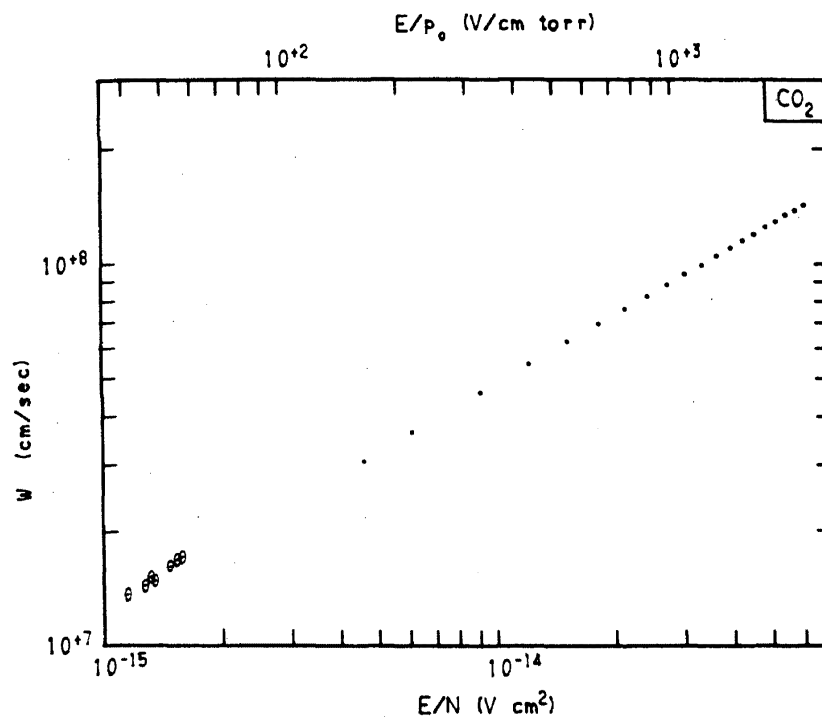
Experimental: $\circ$ Elford(2041); $\square$ Christophorou(725);
 Δ Fink(1833); ∇ Levine(2920); $+$ Hurst(1184);
 $\times$ Pack(439).
 Theoretical: Hake(2553).

FIGURE 1.33. W , E/N for carbon dioxide for $E/N < 2.82 \times 10^{-16}$ V cm².



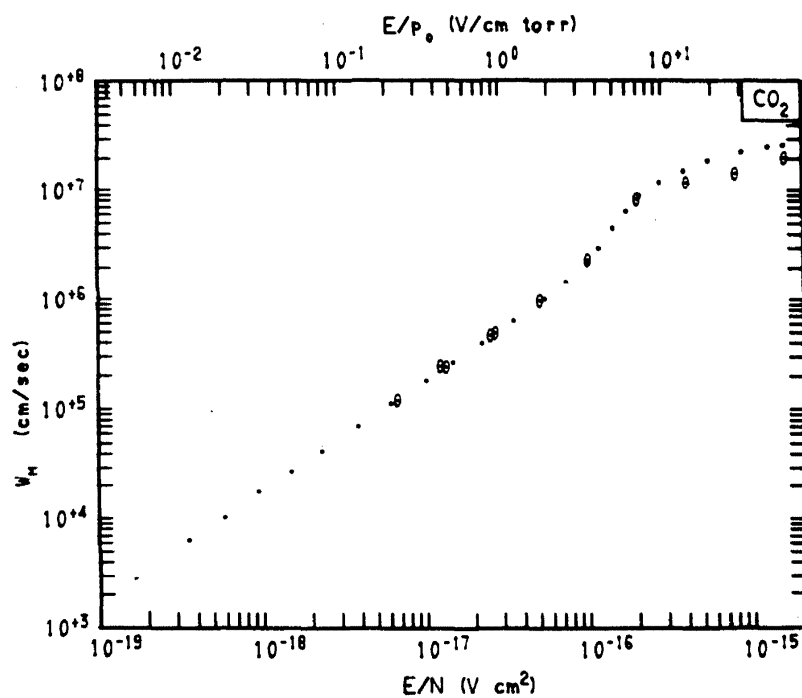
Experimental: all data from Allen(5188).

FIGURE 1.34. Variation of W with gas number density for carbon dioxide.



Experimental: • Schlumbohm(1963); ○ Frommhold(1954).

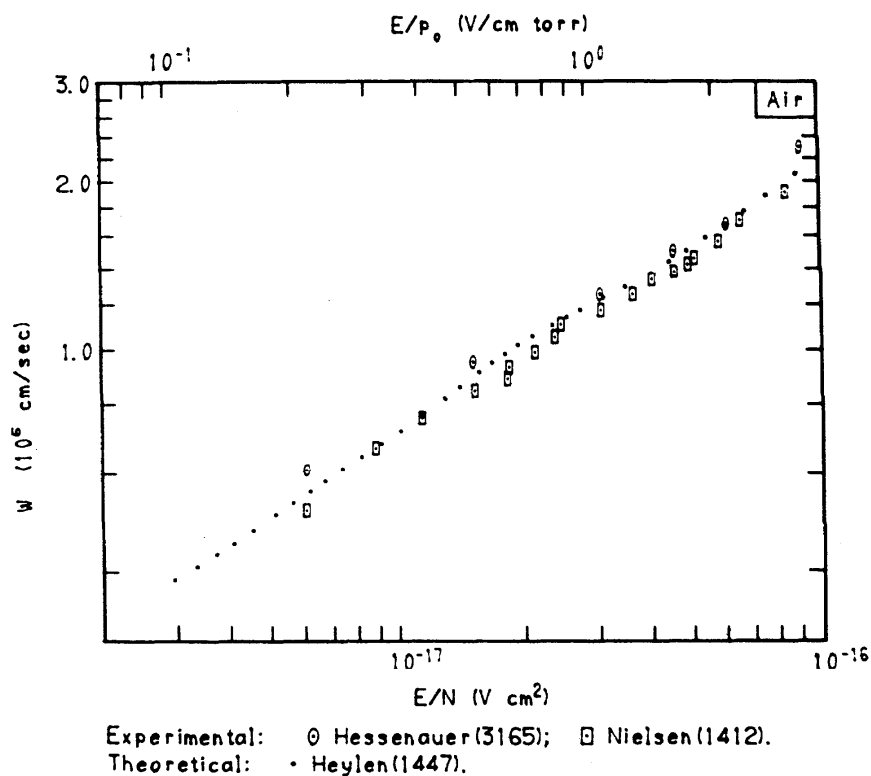
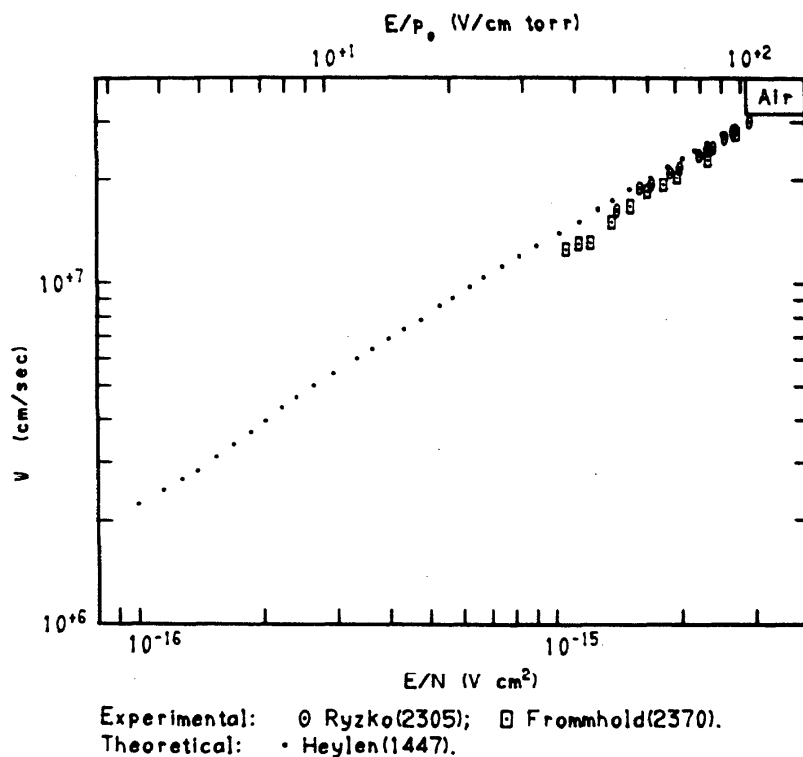
FIGURE 1.35. Experimental W , E/N for carbon dioxide for $E/N > 10^{-15}$ V cm².

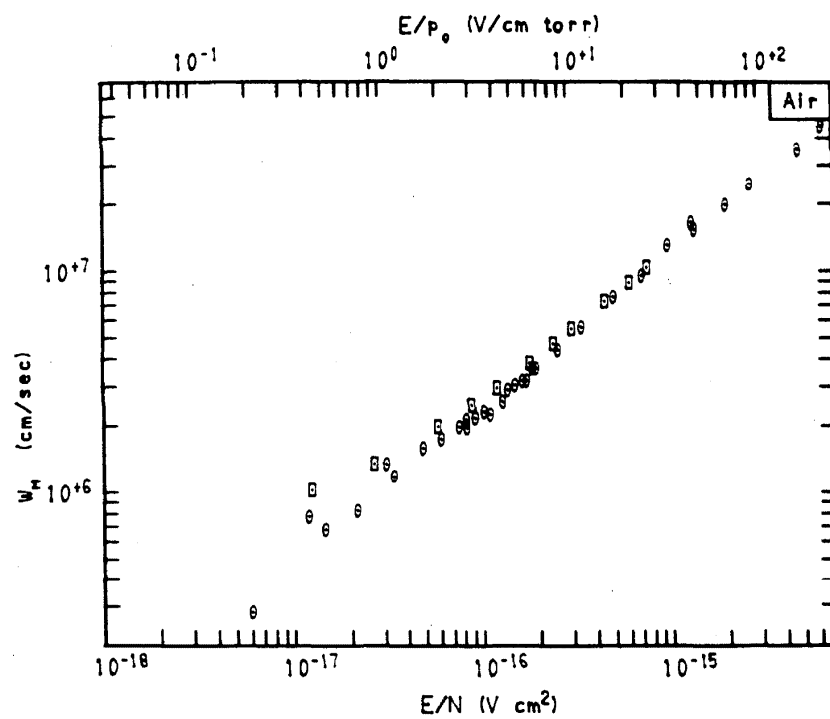


Experimental: ○ Skinker(1968).

Theoretical: • Hake(1953).

FIGURE 1.36. W_M , E/N for carbon dioxide.

FIGURE 1.37. W , E/N for air for $E/N < 9.87 \times 10^{-17} \text{ V cm}^2$.FIGURE 1.38. W , E/N for air for $E/N > 9.87 \times 10^{-17} \text{ V cm}^2$.



Experimental: $\circ$ Townsend(2104); $\square$ Huxley(2054).

Figure 1.39. Experimental W_M , E/N for air.

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Table 1.1(a)

Values of W for helium for $E/N < 4 \times 10^{-17} \text{Vcm}^2$ at ambient temperature

E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W (10^5cm/sec)
0.000795	0.0210 (a)	0.02730	0.586 (b)	0.364	2.93 (b)
0.001219	0.034 (a)	0.0303	0.637 (b)	0.455	3.28 (b)
	0.037 (a)	0.0364	0.733 (b)	0.546	3.60 (b)
0.001518	0.045 (a)	0.0455	0.863 (b)	0.607	3.79 (b)
	0.040 (a)	0.0546	0.980 (b)	0.759	4.23 (b)
0.002281	0.056 (a)	0.0607	1.052 (b)	0.910	4.63 (b)
0.00303	0.080 (a)	0.0759	1.22 (b)	1.214	5.33 (b)
0.00427	0.111 (a)	0.0910	1.37 (b)	1.517	5.97 (b)
0.00455	0.121 (a)	0.1214	1.62 (b)	1.820	6.55 (b)
0.00609	0.161 (a)	0.1517	1.84 (b)	2.124	7.07 (b)
0.00905	0.230 (a)	0.1820	2.04 (b)	2.430	7.57 (b)
0.01526	0.36 (a)	0.2124	2.21 (b)	2.730	8.07 (b)
0.01820	0.418 (b)	0.2430	2.37 (b)	3.03	8.57 (b)
0.02124	0.477 (b)	0.2730	2.53 (b)	3.64	9.47 (b)
0.02430	0.533 (b)	0.303	2.67 (b)		

Experimental:

(a) Pack, *et al.*, Phys. Rev. **121**, 798 (1961).

(b) Crompton, *et al.*, Austr. J. Phys. **20**, 369 (1967).

Table 1.1(b)

Values of W for helium showing the variation of W with N at high values of N at ambient temperature

$N(\text{cm}^{-3})$	1.074×10^{21}	7.892×10^{20}	5.732×10^{20}	2.783×10^{20}
E/N (10^{-17}Vcm^2)	W (10^4cm/sec)			
.00909	2.03	2.08	2.12	2.19
.0121	2.66	2.74	2.78	2.91
.0151	3.3	3.4	3.4	3.6
.0182	3.9	4.0	4.1	4.2
.0242	5.1	5.2	5.3	5.4
.0303	6.2	6.2	6.3	6.4
.0455	8.6	8.6	8.7	8.7
.0606	10.5	10.6	10.6	10.6

Experimental:

All data taken from Grunberg, Z. Naturforsch. **24a**, 1838 (1969).

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Table 1.1(c)
Values of W for helium at 77°K

E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)	
.000277	.0168	(a)	.0278	.91	(a)	.278	2.67	(b)
.000545	.031	(a)	.0318	.874	(b)	.318	2.84	(b)
.00110	.065	(a)	.0398	.991	(b)	.350	2.97	(b)
.00136	.081	(a)	.0477	1.10	(b)	.398	3.16	(b)
.00218	.124	(a)	.0557	1.19	(b)	.477	3.45	(b)
.00247	.138	(a)	.0559	1.29	(a)	.557	3.71	(b)
.00275	.146	(a)	.0636	1.28	(b)	.636	3.96	(b)
.00275	.157	(a)	.0716	1.36	(b)	.716	4.19	(b)
.00410	.230	(a)	.0795	1.44	(b)	.795	4.40	(b)
.00540	.290	(a)	.0954	1.56	(b)	.954	4.80	(b)
.00795	.345	(b)	.119	1.76	(b)	1.19	5.34	(b)
.00814	.41	(a)	.140	2.13	(a)	1.43	5.85	(b)
.0138	.59	(a)	.143	1.93	(a)	1.59	6.16	(b)
.0159	.570	(b)	.159	2.03	(a)	1.75	6.46	(b)
.0165	.66	(a)	.199	2.27	(a)	1.99	6.89	(b)
.0239	.737	(b)	.239	2.47	(a)			

Experimental:

- (a) Pack, *et al.*, Phys. Rev. **121**, 798 (1961).
 (b) Crompton, *et al.*, Austr. J. Phys. **20**, 369 (1967).

Table 1.2

Values of W for electrons in helium glow discharges for $E/N > 13 \times 10^{-17}$ Vcm² at ambient temperatures

E/N (10^{-17} Vcm ²)	W (10^8 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^8 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^8 cm/sec)	
13.40	0.031	(a)	32.4	0.079	(b)	93.6	0.246	(c)
13.74	0.032	(b)	35.0	0.084	(a)	96.4	0.262	(b)
15.40	0.036	(a)	35.4	0.085	(b)	117.2	0.31	(c)
17.44	0.046	(b)	38.0	0.102	(b)	121.5	0.33	(b)
18.17	0.043	(a)	40.0	0.101	(c)	130.5	0.35	(c)
19.34	0.046	(b)	40.7	0.097	(a)	160.5	0.44	(c)
20.60	0.048	(a)	46.2	0.110	(c)	192.4	0.54	(c)
20.94	0.054	(b)	46.8	0.113	(a)	218.9	0.64	(c)
22.86	0.055	(b)	53.8	0.131	(a)	272.8	0.81	(c)
23.36	0.055	(a)	55.3	0.146	(b)	323	0.95	(c)
25.15	0.064	(b)	59.0	0.146	(c)	407	1.15	(c)
26.84	0.064	(a)	63.0	0.153	(a)	503	1.37	(c)
27.44	0.061	(b)	65.3	0.182	(b)	573	1.47	(c)
30.0	0.072	(a)	67.5	0.170	(c)	704	1.63	(c)
30.2	0.072	(b)	77.4	0.187	(a)	823	1.75	(c)
31.7	0.084	(b)	80.5	0.208	(c)			

Experimental:

- (a) Anderson, Phys. Fluids **7**, 1517 (1964).
 (b) Phelps, *et al.*, Phys. Rev. **117**, 470 (1960).
 (c) Stern, in *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, eds. (Serma, Paris, 1963), Vol. I, p. 331.

Table 1.3
Values of W_M for helium

E/N (10^{-17}Vcm^2)	W_M (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^5cm/sec)
0.01820	{0.480 (a) 0.486 (1)	0.1214	{1.77 (a) 1.77 (1)	0.607	{4.00 (a) 4.08 (1)
0.02124	{0.542 (a) 0.551 (1)	0.1517	{1.98 (a) 2.01 (1)	0.759	{4.45 (a) 4.55 (1)
0.0243	{0.602 (a) 0.613 (1)	0.1820	{2.19 (a) 2.21 (1)	0.910	{4.86 (a) 4.98 (1)
0.0273	{0.660 (a) 0.671 (1)	0.2124	{2.37 (a) 2.40 (1)	1.214	{5.61 (a) 5.73 (1)
0.0303	{0.715 (a) 0.727 (1)	0.243	{2.53 (a) 2.58 (1)	1.517	{6.24 (a) 6.38 (1)
0.0364	{0.812 (a) 0.830 (1)	0.273	{2.71 (a) 2.74 (1)	1.820	{6.80 (a) 6.98 (1)
0.0455	{0.947 (a) 0.970 (1)	0.303	{2.84 (a) 2.89 (1)	2.124	{7.32 (a) 7.53 (1)
0.0546	{1.07 (a) 1.10 (1)	0.364	{3.11 (a) 3.17 (1)	2.43	{7.92 (a) 8.05 (1)
0.0607	{1.15 (a) 1.17 (1)	0.455	{3.49 (a) 3.54 (1)	2.73	{8.31 (a) 8.55 (1)
0.0759	{1.33 (a) 1.35 (1)	0.546	{3.82 (a) 3.88 (1)	3.03	{8.91 (a) 9.03 (1)
0.0910	{1.48 (a) 1.50 (1)				

Experimental:

(a) Crompton, *et al.*, Austr. J. Phys. 20, 369 (1967).

Theoretical:

(1) Crompton, *et al.*, Austr. J. Phys. 20, 369 (1967).

Table 1.4(a)
Values of W for neon at ambient temperatures

E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W (10^5cm/sec)
0.001107	0.251 (a)	0.02732	1.24 (b)	0.486	4.2 (b)
0.001318	0.32 (a)	0.0304	1.30 (b)	0.546	4.5 (b)
0.001538	0.34 (a)	0.0455	1.53 (b)	0.607	4.7 (b)
0.001556	0.30 (a)	0.0607	1.72 (b)	0.759	5.2 (b)
0.001854	0.37 (a)	0.0911	2.04 (b)	0.911	5.6 (b)
0.002143	0.36 (a)	0.1214	2.3 (b)	1.062	5.9 (b)
0.00308	0.44 (a)	0.1518	2.5 (b)	1.214	6.3 (b)
0.00311	0.51 (a)	0.1821	2.7 (b)	1.336	6.5 (b)
0.00374	0.54 (a)	0.2125	2.9 (b)	1.821	7.8 (b)
0.00619	0.65 (a)	0.2428	3.1 (b)	2.003	8.5 (b)
0.00740	0.71 (a)	0.2732	3.3 (b)	2.270	11.6 (c)
0.01518	0.98 (b)	0.304	3.4 (b)	2.846	14.1 (c)
0.01821	1.05 (b)	0.364	3.7 (b)	3.94	18.4 (c)
0.02125	1.12 (b)	0.425	4.0 (b)		
0.02428	1.19 (b)	0.455	4.1 (b)		

Experimental:

(a) Pack, *et al.*, Phys. Rev. 121, 798 (1961), data taken at 300 °K.

(b) Robertson, J. Phys. B 5, 648 (1972), data taken at 293 °K.

(c) Nielsen, Phys. Rev. 50, 950 (1936), data taken at 293 °K.

Table 1.4(b)

Values of W for neon at 77 °K

E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)	E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)	E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)
0.001594	0.42	0.01594	1.06	0.1195	2.32
0.002390	0.51	0.01992	1.15	0.1594	2.62
0.00319	0.57	0.02390	1.24	0.1992	2.88
0.00398	0.62	0.0319	1.38	0.2104	2.95
0.00478	0.67	0.0398	1.50	0.2390	3.1
0.00558	0.71	0.0478	1.61	0.319	3.6
0.00637	0.75	0.0558	1.71	0.398	3.9
0.00717	0.78	0.0637	1.80	0.478	4.2
0.00797	0.82	0.0717	1.88	0.526	4.4
0.01195	0.95	0.0797	1.96	0.558	4.6
				0.637	4.8

Experimental:All data taken from Robertson, J. Phys. B 5, 648 (1972).

Table 1.5

Values of W for argon in the range $0.3 \times 10^{-17} < E/N < 40 \times 10^{-17} \text{ Vcm}^2$

E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)		E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)		E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)	
0.304	0.246	(a)	2.526	0.39	(b)	7.56	0.81	(d)
0.307	0.271	(b)	2.572	0.46	(d)	7.61	0.72	(c)
0.362	0.276	(b)	2.615	0.44	(a)	7.82	0.85	(a)
0.379	0.242	(c)	2.825	0.40	(b)	8.32	0.79	(c)
0.412	0.280	(d)	2.965	0.45	(a)	8.41	0.93	(a)
0.428	0.239	(e)	3.03	0.37	(c)	8.48	0.92	(d)
0.441	0.281	(b)	3.08	0.40	(b)	8.90	0.99	(a)
0.444	0.269	(a)	3.18	0.47	(d)	9.15	0.87	(c)
0.471	0.259	(d)	3.33	0.41	(b)	9.39	1.06	(a)
0.488	0.257	(c)	3.41	0.47	(a)	9.69	0.93	(c)
0.530	0.285	(d)	3.63	0.39	(c)	9.78	1.12	(a)
0.543	0.289	(b)	3.78	0.40	(b)	10.27	0.98	(c)
0.627	0.270	(c)	3.79	0.49	(a)	10.30	1.19	(a)
0.685	0.296	(b)	3.99	0.41	(b)	10.58	1.15	(d)
0.700	0.30	(a)	4.06	0.50	(a)	11.07	1.29	(a)
0.719	0.269	(e)	4.11	0.43	(b)	11.08	1.05	(c)
0.779	0.282	(c)	4.30	0.41	(c)	11.18	1.05	(c)
0.811	0.30	(b)	4.40	0.52	(a)	11.83	1.12	(c)
0.883	0.30	(d)	4.44	0.50	(d)	11.91	1.40	(a)
0.903	0.292	(c)	4.57	0.46	(b)	11.97	1.12	(c)
0.918	0.292	(e)	4.60	0.44	(c)	12.49	1.31	(d)
0.957	0.32	(a)	4.76	0.53	(a)	12.84	1.52	(a)
0.962	0.32	(d)	4.82	0.54	(a)	13.00	1.19	(c)
0.976	0.31	(b)	5.01	0.49	(b)	13.63	1.63	(a)
1.060	0.32	(d)	5.04	0.46	(c)	14.09	1.32	(c)
1.119	0.31	(c)	5.19	0.57	(a)	14.40	1.73	(a)
1.220	0.33	(b)	5.34	0.52	(b)	15.17	1.83	(a)
1.261	0.35	(a)	5.38	0.49	(c)	15.49	1.43	(c)
1.334	0.32	(c)	5.40	0.55	(d)	16.16	1.97	(a)
1.403	0.31	(e)	5.41	0.57	(d)	17.40	1.55	(c)
1.433	0.35	(d)	5.42	0.49	(c)	19.25	1.73	(c)
1.517	0.37	(a)	5.63	0.54	(b)	21.55	1.94	(c)
1.519	0.35	(b)	5.65	0.59	(a)	24.22	2.42	(f)
1.571	0.36	(d)	5.72	0.57	(b)	25.86	2.23	(c)
1.594	0.33	(c)	5.81	0.66	(d)	27.38	2.71	(f)
1.649	0.36	(d)	6.14	0.52	(c)	27.91	2.43	(g)
1.660	0.33	(e)	6.46	0.63	(a)	30.6	2.61	(c)
1.763	0.36	(b)	6.47	0.53	(c)	32.4	2.91	(g)
1.844	0.39	(a)	6.52	0.66	(a)	32.9	3.2	(f)
1.904	0.39	(d)	6.93	0.70	(a)	34.9	3.5	(f)
1.976	0.35	(c)	7.11	0.61	(c)	36.4	3.4	(g)
1.991	0.38	(b)	7.35	0.66	(d)	37.3	3.1	(c)
2.241	0.41	(a)		0.75	(a)	38.9	3.2	(g)
2.274	0.39	(b)		0.66	(c)			
2.452	0.36	(c)		0.79	(a)			

Experimental:

- (a) Herreng, Compt. Rend. 217, 75 (1943).
- (b) Levine, J. Appl. Phys. 35, 2618 (1964).
- (c) Errett, Ph.D. Thesis, Purdue University, 1951.
- (d) Nielsen, Phys. Rev. 50, 950 (1936).
- (e) Pack, et al., Phys. Rev. 121, 798 (1961).
- (f) Garen, Phys. Rev. 131, 1904 (1963).
- (g) Jager, et al., Z. Physik 169, 517 (1962).

Table 1.6(a)

Values of W for argon for $E/N < 0.3 \times 10^{-17} \text{ Vcm}^2$ at 300 K

E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)
0.0001	0.0122 (1)	0.00330	0.65 (a)	0.03436	1.29 (a)
0.000625	0.080 (a)	0.00499	0.84 (a)	0.0525	1.42 (a)
0.000631	0.087 (a)	0.01005	1.01 (a)	0.1045	1.78 (a)
0.001107	0.157 (a)	0.02042	1.15 (a)	0.1758	1.91 (a)
0.001622	0.205 (a)	0.02523	1.15 (a)	0.2123	2.27 (a)
0.002506	0.46 (a)			0.2148	2.00 (a)

Experimental:

(a) Pack, et al., Phys. Rev. 121, 798 (1961).

Theoretical:

(1) Frost, et al., Phys. Rev. 136, A1538 (1964).

Table 1.6(b)

Values of W for argon for $E/N < 0.3 \times 10^{-17} \text{ Vcm}^2$ at 77 K

E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)
0.0001	0.0056 (1)	0.001622	0.095 (a)	0.00499	0.65 (a)
0.000625	0.036 (a)		0.099 (a)	0.00664	0.94 (a)
0.000781	0.044 (a)	0.002478	0.169 (a)	0.01688	1.12 (a)
	0.050 (a)	0.002613	0.161 (a)	0.0698	1.53 (a)
0.000955	0.052 (a)	0.00330	0.243 (a)	0.2655	2.15 (a)

Experimental:

(a) Pack, et al., Phys. Rev. 121, 798 (1961).

Theoretical:

(1) Frost, et al., Phys. Rev. 136, A1538 (1964).

Table 1.7

Values of W for argon for $E/N > 50 \times 10^{-17} \text{ Vcm}^2$

E/N (10^{-17} Vcm^2)	W (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W (10^7 cm/sec)	
52.8	0.40	(b)	93.0	0.62	(b)	170.9	0.98	(b)
53.6	0.43	(a)	97.6	0.63	(b)	172.8	1.01	(b)
54.9	0.41	(b)	99.5	0.63	(b)	174.3	1.04	(b)
	0.44	(b)	99.7	0.74	(c)	174.5	0.98	(b)
56.7	0.45	(b)	104.3	0.68	(b)	177.7	1.55	(1)
57.3	0.50	(a)	115.3	0.83	(c)	177.8	0.99	(b)
58.6	0.44	(b)	116.5	0.72	(b)	179.1	1.10	(b)
58.8	0.60	(1)	117.3	0.76	(b)	180.6	0.99	(b)
62.0	0.45	(b)	118.9	1.12	(1)	187.2	1.22	(a)
62.6	0.48	(b)	119.5	0.87	(c)	187.3	1.10	(b)
66.0	0.49	(b)	120.2	0.73	(b)	194.6	1.17	(b)
71.0	0.51	(b)	121.3	0.77	(b)	216.1	1.84	(1)
73.1	0.54	(b)	125.5	0.77	(b)	250.0	2.10	(1)
73.6	0.56	(c)	133.4	0.84	(b)	277.4	2.31	(1)
77.9	0.54	(b)	144.5	0.91	(b)	318	2.60	(1)
86.2	0.85	(1)	148.3	0.85	(b)	359	2.88	(1)
87.2	0.66	(c)	149.4	1.34	(1)	398	3.2	(1)
88.0	0.60	(b)	150.0	1.00	(a)	438	3.4	(1)
	0.69	(a)	150.4	0.87	(b)	480	3.7	(1)
90.1	0.60	(b)	157.5	0.96	(b)	523	4.0	(1)
90.8	0.60	(a)	163.8	0.97	(b)	550	4.2	(1)
						585	4.5	(1)

Experimental:

(a) Jager, *et al.*, Z. Physik **169**, 517 (1962).(b) Brambring, Z. Physik **179**, 539 (1964).(c) Wagner, Z. Physik **178**, 64 (1964).

Theoretical:

(1) Golant, Sov. Phys.-Tech. Phys. **4**, 680 (1959).

Table 1.8

Values of W_M for argon

E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)	
0.0001	0.000057	(1)	0.01	0.0150	(1)	1.252	0.036	(a)
0.0007	0.00040	(1)	0.02	0.0178	(1)	2.518	0.059	(a)
0.0015	0.00087	(1)	0.0329	0.0196	(1)	3.0	0.050	(1)
0.002	0.00133	(1)	0.104	0.0240	(1)	6.32	0.152	(a)
0.0025	0.00228	(1)	0.329	0.030	(1)	8.43	0.226	(a)
0.003	0.0041	(1)	0.334	0.031	(a)	12.67	0.34	(a)
0.0035	0.0063	(1)	0.501	0.032	(a)	17.58	0.46	(a)
0.0045	0.0097	(1)	0.626	0.033	(a)	26.11	0.60	(a)
0.0065	0.0128	(1)	1.04	0.039	(1)	36.1	0.73	(a)
						59.0	0.95	(a)

Experimental:

(a) Townsend, *et al.*, Phil. Mag. **44**, 1033 (1922).

Theoretical:

(1) Engelhardt, *et al.*, Phys. Rev. **133**, A375 (1964).

Table 1.9(a)

Values of W for krypton at ambient temperatures and above

E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)	
0.000791	0.0108	(a)	0.00774	0.106	(a)	0.0306	1.04	(a)
0.000924	0.0128	(a)	0.00918	0.133	(a)	0.0758	1.27	(a)
0.001525	0.0226	(b)	0.01254	0.210	(a)	0.1496	1.34	(a)
0.002144	0.031	(b)	0.01550	0.33	(a)	0.304	1.57	(a)
0.00306	0.042	(b)	0.01712	0.42	(a)	0.609	1.78	(a)
0.00308	0.036	(a)	0.01838	0.51	(a)	0.892	2.00	(a)
0.00458	0.058	(a)	0.02118	0.62	(a)	1.53	2.30	(a)
0.00461	0.066	(b)	0.02440	0.86	(a)	1.786	2.51	(a)
0.00617	0.085	(a)	0.02772	1.04	(a)	2.440	2.51	(a)
0.00622	0.095	(b)				3.06	3.2	(a)

Experimental:

(a) Pack, *et al.*, Phys. Rev. 127, 2084 (1962), data taken at 300 K.(b) *Ibid.*, data taken at 368 K.

Table 1.9(b)

Values of W for krypton at 195 K

E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)		E/N (10^{-17} Vcm ²)	W (10^5 cm/sec)	
0.001210	0.0158		0.00465	0.052		0.01891	0.40	
0.001561	0.0174		0.00617	0.069		0.02475	0.80	
0.001851	0.0238		0.00918	0.109		0.0442	1.19	
0.00308	0.038		0.01236	0.162		0.0924	1.27	

Experimental:

Pack, *et al.*, Phys. Rev. 127, 2084 (1962).

Table 1.10(a)

Values of W for xenon at 300 K

E/N (10^{-17} V cm^2)	W (10^5 cm/sec)	E/N (10^{-17} V cm^2)	W (10^5 cm/sec)	E/N (10^{-17} V cm^2)	W (10^5 cm/sec)
0.001542	0.0058 (a)	0.0499	0.50 (a)	1.837	1.57 (a)
0.00305	0.0126 (a)	0.0552	0.60 (a)	2.178	1.64 (a)
0.00454	0.0188 (a)	0.0618	0.70 (a)	2.735	1.76 (a)
0.00611	0.0250 (a)	0.0692	0.73 (a)	3.15	1.86 (a)
0.00767	0.0292 (a)	0.0918	0.82 (a)	129.5	85 (b)
0.00923	0.037 (a)	0.1074	0.89 (a)	132.8	81 (b)
0.01225	0.047 (a)	0.1239	0.90 (a)	141.4	98 (b)
0.01393	0.054 (a)	0.1556	0.97 (a)	142.5	88 (b)
0.01560	0.062 (a)	0.1871	0.97 (a)	147.1	92 (b)
0.01849	0.059 (a)	0.2483	1.01 (a)	153.9	101 (b)
0.02163	0.087 (a)	0.307	1.10 (a)	159.5	98 (b)
0.02355	0.101 (a)	0.386	1.13 (a)	162.7	108 (b)
0.02754	0.128 (a)	0.463	1.10 (a)	169.3	114 (b)
0.0308	0.163 (a)	0.652	1.25 (a)	206.9	134 (b)
0.0366	0.256 (a)	0.916	1.31 (a)	224.1	141 (b)
0.0428	0.41 (a)	1.216	1.46 (a)	225.9	147 (b)
0.0466	0.45 (a)	1.528	1.53 (a)	261.9	155 (b)
				291.3	173 (b)

Experimental:

(a) Pack, *et al.*, Phys. Rev. 127, 2084 (1962).(b) Wagner, Z. Physik 178, 64 (1964).

Table 1.10(b)

Values of W for xenon at 195 K

E/N (10^{-17} V cm^2)	W (10^4 cm/sec)	E/N (10^{-17} V cm^2)	W (10^4 cm/sec)	E/N (10^{-17} V cm^2)	W (10^4 cm/sec)
0.00301	0.095	0.01849	0.75	0.0376	1.84
0.00603	0.199	0.02422	0.80	0.0459	2.98
0.01208	0.428	0.0304	1.21	0.0618	5.9
				0.07537	7.3

Experimental:

Pack, *et al.*, Phys. Rev. 127, 2084 (1962).

Table 1.11(a)

Values of W for hydrogen for $E/N < 2.8 \times 10^{-17} \text{Vcm}^2$ at ambient temperatures

E/N (10^{-17}Vcm^2)	W (10^5cm/sec)		E/N (10^{-17}Vcm^2)	W (10^5cm/sec)		E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	
0.001174	0.0217	(a)	0.0303	0.46	(b)	0.303	3.2	(b)
0.001254	0.0203	(a)	0.0364	0.55	(b)	0.364	3.6	(b)
0.001528	0.0245	(a)	0.0455	0.68	(b)	0.455	4.1	(b)
0.002330	0.037	(a)	0.0546	0.80	(b)	0.546	4.6	(b)
0.002488	0.039	(a)	0.0606	0.88	(b)	0.606	4.9	(b)
0.00303	0.052	(a)	0.0758	1.07	(b)	0.758	5.5	(b)
0.00307	0.047	(a)	0.0909	1.26	(b)	0.909	6.0	(b)
0.01212	0.188	(b)	0.1212	1.59	(b)	1.212	6.8	(b)
0.01516	0.235	(b)	0.1516	1.89	(b)	1.516	7.5	(b)
0.01819	0.283	(b)	0.1819	2.18	(b)	1.819	8.0	(b)
0.02122	0.33	(b)	0.2122	2.45	(b)	2.122	8.6	(b)
0.02425	0.37	(b)	0.2425	2.70	(b)	2.425	9.1	(b)
0.02728	0.42	(b)	0.2728	2.94	(b)	2.728	9.6	(b)

Experimental:

(a) Pack, *et al.*, Phys. Rev. **121**, 798 (1961), data taken at 300 K.(b) Lowke, Austr. J. Phys. **16**, 115 (1963), data taken at 293 K.

Table 1.11(b)

Values of W for hydrogen for $E/N < 0.3 \times 10^{-17} \text{Vcm}^2$ at 77 K

E/N (10^{-17}Vcm^2)	W (10^5cm/sec)		E/N (10^{-17}Vcm^2)	W (10^5cm/sec)		E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	
0.00303	0.108	(b)	0.02425	0.66	(b)	0.0795	1.47	(a)
0.00364	0.129	(b)	0.02728	0.72	(b)	0.0909	1.62	(b)
0.00455	0.158	(b)	0.0303	0.77	(b)	0.0954	1.67	(a)
0.00546	0.188	(b)	0.0318	0.79	(a)	0.1193	1.95	(a)
0.00606	0.207	(b)	0.0364	0.87	(b)	0.1212	1.97	(b)
0.00758	0.254	(b)	0.0398	0.92	(a)	0.1431	2.21	(a)
0.00795	0.273	(a)	0.0455	1.01	(b)	0.1516	2.32	(b)
0.00909	0.297	(b)	0.0477	1.04	(a)	0.1590	2.38	(a)
0.01212	0.383	(b)	0.0546	1.14	(b)	0.1819	2.62	(b)
0.01516	0.46	(b)	0.0557	1.15	(a)	0.1988	2.77	(a)
0.01590	0.48	(a)	0.0606	1.23	(b)	0.2122	2.90	(b)
0.01819	0.53	(b)	0.0636	1.26	(a)	0.2386	3.12	(a)
0.02122	0.59	(b)	0.0716	1.37	(a)	0.2425	3.2	(b)
0.02386	0.64	(a)	0.0758	1.43	(b)	0.2728	3.4	(b)
						0.2783	3.4	(a)

Experimental:

(a) Robertson, Austr. J. Phys. **24**, 445 (1971).(b) Lowke, Austr. J. Phys. **16**, 115 (1963).

Table 1.12

Values of W for hydrogen for $2.8 \times 10^{-17} < E/N < 110 \times 10^{-17} \text{Vcm}^2$ at ambient temperatures

E/N (10^{-17}Vcm^2)	W (10^7cm/sec)		E/N (10^{-17}Vcm^2)	W (10^7cm/sec)		E/N (10^{-17}Vcm^2)	W (10^7cm/sec)	
3.03	0.101	(a)	20.	0.281	(b)	58.1	0.65	(e)
3.64	0.110	(a)	21.22	0.295	(a)	60.6	0.70	(c)
4.	0.115	(b)	22.	0.297	(b)	62.3	0.72	(e)
4.55	0.123	(a)	24.	0.31	(b)	64.2	0.74	(d)
5.46	0.135	(a)	24.25	0.32	(a)	68.9	0.77	(e)
6.	0.141	(b)	26.	0.33	(b)	69.6	0.80	(d)
6.06	0.143	(a)	27.28	0.34	(a)	73.8	0.90	(d)
7.58	0.161	(a)	30.3	0.37	(a)	75.8	0.92	(c)
8.	0.165	(b)	36.4	0.42	(a)	78.1	0.94	(e)
9.09	0.179	(a)		0.42	(c)	80.8	0.98	(d)
10.	0.187	(b)		0.51	(a)	84.0	1.01	(e)
12.	0.207	(b)	45.5	0.51	(c)	88.3	1.10	(e)
12.12	0.210	(a)		0.63	(a)		1.14	(d)
14.	0.227	(b)	54.6	0.62	(c)	90.9	1.15	(c)
15.16	0.240	(a)	54.7	0.57	(d)	92.1	1.17	(d)
16.	0.246	(b)	56.7	0.60	(e)	97.0	1.22	(e)
18.	0.264	(b)	57.5	0.66	(d)	106.1	1.38	(c)
18.19	0.267	(a)		0.63	(e)			

Experimental:

- (a) Lowke, Austr. J. Phys. 16, 115 (1963).
- (b) Robertson, Austr. J. Phys. 24, 445 (1971).
- (c) Blevin, *et al.*, Austr. J. Phys. 20, 735 (1967).
- (d) Jager, *et al.*, Z. Physik 169, 517 (1962).
- (e) Frommhold, Z. Physik 160, 554 (1960).

Table 1.13

Values of W for hydrogen for $E/N > 120 \times 10^{-17} \text{Vcm}^2$ at ambient temperatures

E/N (10^{-17}Vcm^2)	W (10^8cm/sec)		E/N (10^{-17}Vcm^2)	W (10^8cm/sec)		E/N (10^{-17}Vcm^2)	W (10^8cm/sec)	
121.2	0.168	(a)	270.0	0.43	(b)	952	1.12	(b)
136.4	0.189	(a)	338	0.53	(b)	992	1.20	(b)
136.6	0.207	(b)	387	0.59	(b)	1347	1.32	(b)
151.6	0.216	(a)	444	0.67	(b)	1587	1.51	(b)
183.1	0.31	(b)	646	0.86	(b)	2231	1.98	(b)
227.7	0.41	(b)	760	0.95	(b)	2505	1.83	(b)
						5260	2.62	(b)

Experimental:

- (a) Blevin, *et al.*, Austr. J. Phys. 20, 735 (1967).
- (b) Schlumbohm, Z. Physik 182, 317 (1965).

Table 1.14(a)

Values of W showing the dependence of W on N at various values of E/N in hydrogen at 77 K

N(cm ⁻³)	1.257×10 ¹⁹	2.515×10 ¹⁹	3.772×10 ¹⁹	5.030×10 ¹⁹	6.287×10 ¹⁹	7.544×10 ¹⁹	8.802×10 ¹⁹
E/N(10 ⁻¹⁷ Vcm ²)	W(10 ⁵ cm/sec)						
0.01590		0.478	0.477	0.476	0.476	0.475	0.474
0.02386	0.648	0.645	0.644	0.644	0.644	0.644	0.644
0.0318		0.789	0.788	0.788	0.789	0.789	0.789
0.0398	0.921	0.918	0.917	0.917	0.917	0.918	0.918
0.0477	1.04	1.04	1.04	1.04	1.04	1.04	1.04
0.0557	1.16	1.15	1.15	1.15	1.15	1.15	1.15
0.0636	1.27	1.26	1.26	1.26	1.26	1.26	1.26
0.0716	1.37	1.37	1.37	1.36	1.36	1.36	1.36
0.0795	1.48	1.47	1.47	1.47	1.47	1.46	1.46
0.0954	1.68	1.67	1.66	1.66	1.66	1.66	1.66
0.1193	1.95	1.95	1.94	1.94	1.94	1.94	1.93
0.1431	2.21	2.21	2.20	2.20	2.20	2.19	2.19
0.1590	2.38	2.37	2.37	2.36	2.36	2.36	
0.1988	2.76	2.76	2.75	2.75	2.74		

Experimental:

All data taken from Robertson, Austr. J. Phys. 24, 445 (1971).

Table 1.14(b)

Values of W showing the dependence of W on N at various values of E/N in hydrogen at ambient temperatures

N(cm ⁻³)	1.110×10 ²¹	8.380×10 ²⁰	6.000×10 ²⁰	1.300×10 ²⁰	2.744×10 ¹⁹
E/N(10 ⁻¹⁷ Vcm ²)	W(10 ⁶ cm/sec)				
0.006			0.0084		
0.008	0.0111	0.0118	0.0124		
0.011	0.0147	0.0155	0.0164		
0.014	0.0182	0.0194	0.0207		
0.017	0.0220	0.0234	0.0246		
0.020	0.0255	0.0275	0.0287		
0.023	0.0290	0.0310	--		
0.025	0.0326	0.0351	0.0368		
0.028	0.036	0.039	0.0405		
0.034	0.043	0.046	--		
0.042	0.054	0.057	--		
0.051	0.064	0.068	--		
0.057	0.072	0.075	0.078		
0.071	0.088	0.092	0.095		
0.085	0.103	0.107	0.111		
0.0909	0.101		0.110	0.126	0.129
0.1212	0.133		0.145	0.159	0.164
0.1819	0.185		0.198	0.215	--
0.2425	0.235		0.246	0.266	0.275
0.364	0.305		0.317	0.334	--
0.485	0.373		0.399	0.421	0.430
0.728	0.488		0.510	0.518	0.533
0.970	0.563		0.584	0.598	0.612
2.425	0.842		0.867	0.886	0.890
4.85	1.17		1.20	1.22	1.24
9.09	1.65		1.71	1.715	1.73
12.12	1.98		2.05	2.035	2.04
18.19			2.54	2.55	2.60
24.25				3.08	3.06
36.4				4.02	4.07

Experimental:

(a) Grünberg, Z. Naturforsch. 23A, 1994 (1968).

(b) Grünberg, Z. Physik 204, 12 (1967).

Table 1.15
Values of W_M for hydrogen

E/N (10^{-17}Vcm^2)	W_M (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^5cm/sec)
0.0610	1.07 (a)	1.820	9.33 (a)	24.28	33.7 (a)
0.0760	1.29 (a)	2.130	9.93 (a)	27.30	36.3 (a)
0.0920	1.49 (a)	2.430	10.5 (a)	40.2	49 (b)
0.1210	1.87 (a)	2.569	11.2 (b)	41.4	49 (b)
0.1520	2.22 (a)	2.730	11.0 (a)	81.7	103 (b)
0.1820	2.53 (a)	3.03	11.5 (a)	84.8	109 (b)
0.2120	2.83 (a)	3.64	12.5 (a)	87.8	114 (b)
0.2430	3.11 (a)	4.55	13.8 (a)	90.8	118 (b)
0.2730	3.39 (a)	5.10	15.2 (b)	94.5	125 (b)
0.303	3.63 (a)	5.46	15.1 (a)	99.1	132 (b)
0.455	4.74 (a)	6.07	16.0 (a)	103.9	140 (b)
0.607	5.59 (a)	9.11	19.7 (a)	106.9	145 (b)
0.632	6.0 (b)	10.10	21.6 (b)	110.7	150 (b)
0.659	6.3 (b)	12.14	23.0 (a)	113.6	156 (b)
0.911	6.90 (a)	15.18	25.9 (a)	117.1	162 (b)
1.210	7.88 (a)	18.21	28.6 (a)	120.9	168 (b)
1.267	8.5 (b)	20.24	32 (b)	125.0	175 (b)
1.520	8.63 (a)	21.25	31.2 (a)	129.0	182 (b)
				133.6	189 (b)

Experimental:

(a) Creaser, Austr. J. Phys. 20, 547 (1967).

(b) Townsend, *et al.*, Phil. Mag. 42, 874 (1921).

Table 1.16

Values of W for nitrogen for $E/N < 56 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	E/N (10^{-17}Vcm^2)	W (10^5cm/sec)
0.01212	0.46	0.2122	2.90	3.03	7.7
0.01516	0.56	0.2425	2.96	3.64	8.7
0.01819	0.67	0.2728	3.03	4.55	10.2
0.02122	0.77	0.303	3.09	5.46	11.7
0.02425	0.88	0.364	3.22	6.06	12.7
0.02728	0.97	0.455	3.43	7.58	14.9
0.0303	1.07	0.546	3.63	9.09	17.1
0.0364	1.25	0.606	3.76	12.12	21.1
0.0455	1.49	0.758	4.02	15.16	25.0
0.0546	1.71	0.909	4.28	18.19	28.8
0.0606	1.82	1.212	4.76	21.22	32.3
0.0758	2.08	1.516	5.2	24.25	35.7
0.0909	2.28	1.819	5.7	27.28	39.0
0.1212	2.55	2.122	6.2	30.31	42.0
0.1516	2.71	2.425	6.7	36.4	48.2
0.1819	2.81	2.728	7.2	45.5	56.8
				54.6	65.1

Experimental:

All results taken from Lowke, Austr. J. Phys. 16, 115 (1963).

Table 1.17

Values of W for nitrogen for $60 \times 10^{-17} < E/N < 500 \times 10^{-17} \text{ Vcm}^2$

E/N (10^{-17} Vcm^2)	W (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W (10^7 cm/sec)	
60.6	0.69 (a)		118.8	1.22 (e)		151.6	1.62 (b)	
	0.70 (b)		119.2	1.23 (e)		151.7	1.51 (d)	
61.6	0.72 (c)		120.5	1.25 (c)		153.5	1.51 (f)	
68.6	0.79 (c)		120.6	1.17 (f)		155.6	1.46 (f)	
73.1	0.83 (c)			1.23 (e)		161.6	1.62 (f)	
75.8	0.82 (a)		121.0	1.25 (c)		164.2	1.58 (d)	
	0.84 (b)			1.23 (e)		166.7	1.77 (b)	
79.1	0.87 (c)			1.23 (a)		177.5	1.74 (f)	
86.7	0.95 (c)		121.2	1.29 (b)		181.9	1.95 (b)	
86.9	0.94 (c)		122.2	1.25 (e)		208.0	1.96 (f)	
90.9	0.95 (a)		122.8	1.25 (e)		212.2	2.24 (b)	
	0.97 (b)		123.9	1.26 (e)		227.3	2.43 (b)	
91.0	0.99 (c)		124.9	1.27 (e)		242.5	2.53 (b)	
95.1	1.02 (c)		125.4	1.28 (e)		243.0	2.23 (f)	
104.0	1.10 (c)		126.9	1.29 (e)		254.8	2.40 (d)	
104.4	1.09 (e)		127.4	1.29 (e)		275.5	2.63 (d)	
106.0	1.12 (e)		128.8	1.31 (e)		278.6	2.62 (g)	
	1.12 (a)		128.9	1.25 (f)		319	3.4 (h)	
106.1	1.11 (b)		129.3	1.31 (e)		326	3.1 (g)	
106.5	1.09 (f)		130.9	1.32 (e)		330	3.3 (d)	
106.9	1.11 (c)		132.3	1.31 (f)		333	3.1 (g)	
107.8	1.13 (e)		132.9	1.34 (e)		337	3.1 (d)	
109.3	1.15 (e)		135.0	1.36 (e)		347	3.4 (g)	
109.9	1.11 (f)		136.4	1.44 (b)		352	3.4 (d)	
110.8	1.15 (e)		136.9	1.37 (e)		371	3.5 (g)	
110.9	1.13 (f)		138.9	1.40 (e)		375	3.5 (d)	
112.6	1.17 (e)		141.0	1.41 (e)		378	3.4 (g)	
113.3	1.17 (c)		143.0	1.43 (e)		387	3.5 (d)	
113.8	1.13 (f)		144.6	1.46 (f)		400	3.9 (g)	
114.1	1.18 (e)		145.0	1.45 (e)		403	3.9 (d)	
115.8	1.19 (e)		147.1	1.46 (e)		446	4.5 (g)	
117.4	1.21 (e)		149.1	1.49 (e)		450	4.5 (d)	
117.5	1.15 (f)		149.6	1.41 (f)		467	4.0 (h)	

Experimental:

- (a) Fischer-Treuenfeld, Z. Physik 185, 336 (1965).
- (b) Blevin, *et al.*, Austr. J. Phys. 20, 741 (1967).
- (c) Prasad, *et al.*, Brit. J. Appl. Phys. 18, 371 (1967).
- (d) Wagner, Z. Physik 178, 64 (1964).
- (e) Tholl, Z. Physik 178, 183 (1964).
- (f) Frommhold, Z. Physik 160, 554 (1960).
- (g) Wagner, *et al.*, Z. Physik 170, 540 (1962).
- (h) Schlumbohm, Z. Physik 182, 317 (1965).

Table 1.18

Values of W for nitrogen for $E/N > 500 \times 10^{-17} \text{ Vcm}^2$

E/N (10^{-17} Vcm^2)	W (10^8 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^8 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^8 cm/sec)
597	0.42	1253	0.66	2647	1.01
706	0.50	1540	0.85	3560	1.05
955	0.58	1775	0.85	5240	1.34
				8240	1.64

Experimental:

All data taken from Schlumbohm, Z. Physik 182, 317 (1965).

Table 1.19(a)

Values of W for nitrogen at 473 K

E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)
0.0617	1.17	0.617	3.6	3.70	8.1
0.1234	1.90	1.234	4.6	4.32	8.9
0.1851	2.48	1.851	5.4	4.96	9.9
0.2468	2.75	2.468	6.3	5.55	10.7
0.309	2.99	3.09	7.2		

Experimental:

All data taken from Hendrick, *et al.*, ORNL-TM-1444, Oak Ridge National Laboratory, Oak Ridge, Tennessee (1968).

Table 1.19(b)

Values of W for nitrogen at 77 K

E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)
0.000306	0.041	0.00303	0.40	0.1188	3.8
0.000455	0.053	0.00600	0.76	0.1449	3.8
0.000615	0.077	0.0147	1.77	0.1762	3.8
0.000760	0.106	0.02944	2.78	0.2388	3.6
0.001219	0.147	0.0600	3.7	0.2944	3.6
0.001507	0.200	0.0889	3.9	0.443	3.7

Experimental:

All data taken from Pack, *et al.*, Phys. Rev. 121, 798 (1961).

Table 1.19(c)

Values of W for nitrogen at 195 K

E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^5 cm/sec)
0.000760	0.045	0.00607	0.34	0.0592	2.40
0.001524	0.087	0.0149	0.82	0.0889	2.82
0.00306	0.180	0.02944	1.55	0.1205	2.93
				0.1786	3.1

Experimental:

All data taken from Pack, *et al.*, Phys. Rev. 121, 798 (1961).

Table 1.20

Values of W showing the dependence of W on N at various values of E/N in nitrogen

N(cm ⁻³)	1.039 × 10 ²¹	1.84 × 10 ²¹	5.233 × 10 ²⁰	1.036 × 10 ²⁰	2.690 × 10 ¹⁹	7.82 × 10 ¹⁹
E/N (10 ⁻¹⁷ Vcm ²)	W (10 ⁶ cm/sec)					
0.1212	0.224 (a)		0.242 (a)	0.256 (a)	0.260 (a)	
0.1819	0.264 (a)		0.275 (a)	0.284 (a)	0.284 (a)	
0.2425	0.285 (a)		0.292 (a)	0.298 (a)	0.298 (a)	
0.303	0.307 (a)		0.306 (a)	0.312 (a)	0.312 (a)	
0.455	0.327 (a)		0.336 (a)	0.343 (a)	0.345 (a)	
0.606	0.353 (a)		0.363 (a)	0.368 (a)	0.371 (a)	
0.909	0.404 (a)		0.415 (a)	0.418 (a)	0.422 (a)	
1.212						0.558 (b)
1.263						0.523 (b)
1.516	0.484 (a)		0.493 (a)	0.505 (a)	0.500 (a)	
1.942						0.645 (b)
2.132		0.603 (b)				
2.421		0.623 (b)				
2.425	0.62 (a)		0.637 (a)	0.649 (a)	0.649 (a)	
2.502						0.750 (b)
2.553						0.769 (b)
2.721		0.745 (b)				
3.01		0.781 (b)				
3.24						0.855 (b)
3.33		0.800 (b)				
3.56						0.886 (b)
3.64		0.892 (b)				
3.92		0.909 (b)				0.949 (b)
4.38						1.030 (b)
6.06	1.15 (a)		1.205 (a)	1.225 (a)	1.225 (a)	
9.09	1.60 (a)		1.63 (a)	1.62 (a)	1.64 (a)	
24.25			3.40 (a)	3.33 (a)	3.36 (a)	
36.4				4.68 (a)	4.55 (a)	

Experimental:

(a) Grunberg, Z. Physik 204, 12 (1967).

(b) Allen, *et al.*, J. Phys. B 3, 1113 (1972).

Table 1.21

Values of W_M for nitrogen

E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)		E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)	
0.1212	0.0394	(a)	1.212	0.059	(a)	6.06	0.133	(a)
0.1516	0.0415	(a)	1.267	0.059	(b)	7.58	0.156	(a)
0.1819	0.0427	(a)	1.516	0.062	(a)	9.09	0.179	(a)
0.2122	0.0435	(a)	1.819	0.066	(a)	10.04	0.195	(b)
0.2425	0.0438	(a)	2.122	0.071	(.)	12.12	0.225	(a)
0.2728	0.0445	(a)	2.425	0.074	(a)	15.16	0.270	(a)
0.303	0.045	(a)	2.527	0.077	(b)	18.19	0.316	(a)
0.364	0.046	(a)	2.728	0.079	(a)	20.12	0.34	(b)
0.455	0.047	(a)	3.03	0.084	(a)	21.22	0.36	(a)
0.546	0.049	(a)	3.64	0.094	(a)	24.25	0.401	(a)
0.606	0.050	(a)	4.55	0.108	(a)	40.2	0.62	(b)
0.758	0.053	(a)	5.01	0.111	(b)	80.5	1.08	(b)
0.909	0.055	(a)	5.46	0.123	(a)	161.8	1.81	(b)

Experimental:

(a) Jory, Austr. J. Phys. 18, 237 (1965).(b) Townsend, et al., Phil. Mag. 42, 874 (1921).

Table 1.22

Values of W for oxygen for $E/N < 57 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^6cm/sec)		E/N (10^{-17}Vcm^2)	W (10^6cm/sec)		E/N (10^{-17}Vcm^2)	W (10^6cm/sec)	
0.00503	0.050	(a)	2.002	1.77	(a)	9.32	3.5	(f)
0.00810	0.080	(a)	2.156	1.85	(a)	9.76	3.6	(d)
0.01118	0.111	(a)	2.310	1.91	(a)	9.92	3.7	(f)
0.02043	0.216	(a)	2.318	2.09	(d)	10.59	3.3	(e)
0.02968	0.26	(a)	2.464	1.99	(a)	10.90	3.9	(f)
0.0420	0.33	(a)	2.618	2.05	(a)	11.03	3.8	(d)
0.0543	0.39	(a)	2.772	2.10	(a)	11.30	3.5	(c)
0.0697	0.41	(a)	2.825	2.50	(c)	11.89	4.0	(f)
0.0851	0.41	(a)	2.880	2.43	(d)	12.24	3.5	(e)
0.1128	0.47	(a)	2.926	2.16	(a)	12.57	4.0	(d)
0.1186	0.38	(b)	2.965	2.38	(b)	13.31	4.4	(f)
0.1313	0.45	(a)	3.00	2.23	(e)	13.67	3.8	(e)
0.1466	0.47	(b)	3.080	2.22	(a)	14.10	4.6	(f)
0.1467	0.48	(a)	3.14	2.00	(f)	14.12	3.9	(c)
0.1621	0.49	(a)	3.37	2.45	(d)	14.55	4.4	(d)
0.1775	0.51	(a)	3.38	2.20	(f)	14.86	4.4	(b)
0.1786	0.45	(b)	3.73	2.45	(f)	15.28	3.9	(e)
0.2237	0.52	(a)	4.14	2.54	(d)	15.82	5.0	(f)
0.2825	0.55	(c)	4.27	2.67	(d)	15.88	4.0	(g)
0.2853	0.58	(a)	4.49	2.70	(f)	16.10	4.8	(d)
0.301	0.64	(b)	4.50	2.56	(e)		5.7	(f)
0.347	0.60	(a)	4.55	2.70	(f)	16.60	4.4	(e)
0.493	0.77	(a)	4.64	2.70	(d)	16.95	4.3	(c)
0.565	0.74	(c)	4.72	2.70	(f)	17.35	4.9	(d)
0.597	0.77	(b)	5.27	2.76	(d)	17.66	5.8	(f)
						18.24	4.5	(e)
0.616	0.88	(a)	5.34	2.80	(f)			
	0.87	(a)	5.65	2.99	(c)	18.36	5.5	(f)
0.770	1.00	(a)	5.77	2.63	(b)	18.76	5.3	(d)
	0.99	(a)	5.91	2.74	(d)	18.82	5.3	(d)
0.906	1.14	(b)	5.97	2.74	(e)	19.49	5.8	(f)
						19.63	5.1	(e)
0.924	1.12	(a)	6.16	2.95	(f)			
1.078	1.23	(a)	6.24	2.94	(d)	19.77	4.8	(c)
1.186	1.33	(b)	6.50	3.00	(f)	21.07	6.2	(f)
1.232	1.34	(a)	6.81	3.05	(f)	22.07	6.0	(d)
1.264	1.50	(d)	6.94	3.0	(d)	22.60	5.5	(c)
						22.63	6.6	(f)
1.386	1.43	(a)	7.06	3.1	(f)			
1.412	1.51	(c)	7.24	3.0	(d)	24.42	5.7	(e)
1.44	1.69	(e)	7.58	2.95	(e)	25.42	6.1	(c)
1.486	1.46	(b)	7.77	3.2	(f)	25.71	7.3	(f)
1.540	1.52	(a)	8.37	3.3	(d)	28.04	7.5	(d)
						28.25	6.8	(c)
1.686	1.93	(d)	8.42	3.3	(f)			
1.694	1.61	(a)	8.48	3.2	(c)	28.33	7.9	(f)
1.695	1.40	(f)	8.50	3.4	(f)	29.65	8.8	(b)
1.756	1.79	(d)	8.93	3.3	(b)	30.4	7.9	(d)
1.848	1.69	(a)		3.5	(f)	30.5	6.1	(g)
			9.12	3.1	(e)	31.1	7.5	(c)

Table 1.22 (continued)

Values of W for oxygen for $E/N < 57 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^6cm/sec)		E/N (10^{-17}Vcm^2)	W (10^6cm/sec)		E/N (10^{-17}Vcm^2)	W (10^6cm/sec)	
33.9	8.2	(c)	42.4	9.8	(c)	48.0	10.8	(c)
36.4	9.2	(d)	44.6	10.7	(d)	50.8	11.3	(c)
36.5	8.8	(d)	44.8	10.9	(d)	52.9	11.6	(d)
36.7	8.7	(c)	45.0	8.8	(g)	53.7	11.7	(c)
39.6	9.3	(c)	45.2	10.3	(c)	56.5	12.0	(c)

Experimental:

- (a) Nelson, *et al.*, J. Chem. Phys. **57**, 4079 (1972).
- (b) Pack, as given in Hake, *et al.*, Phys. Rev. **158**, 70 (1967).
- (c) Herreng, Cahiers Phys. **38**, 6 (1952).
- (d) Nielsen, *et al.*, Phys. Rev. **51**, 69 (1937).
- (e) Fleming, *et al.*, J. Phys. D **5**, 291 (1972).
- (f) Doebling, Z. Naturforsch. **7A**, 253 (1952).
- (g) Naidu, *et al.*, J. Phys. D **3**, 957 (1970).

Table 1.23

Values of W for oxygen for $E/N > 57 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^8cm/sec)		E/N (10^{-17}Vcm^2)	W (10^8cm/sec)		E/N (10^{-17}Vcm^2)	W (10^8cm/sec)	
60.7	0.113	(c)	159.1	0.227	(a)	404	0.41	(a)
75.5	0.130	(c)	175.2	0.237	(a)	439	0.41	(a)
90.9	0.145	(c)	189.1	0.256	(a)	453	0.46	(b)
92.8	0.169	(a)	197.0	0.259	(a)	499	0.45	(a)
98.8	0.176	(a)	204.3	0.272	(b)	516	0.43	(a)
105.9	0.162	(c)	204.9	0.275	(a)	547	0.49	(a)
106.7	0.181	(a)	219.8	0.278	(a)	561	0.47	(a)
113.7	0.187	(a)	224.9	0.288	(a)	672	0.58	(b)
121.5	0.194	(a)	233.7	0.292	(a)	698	0.55	(a)
121.7	0.178	(c)	249.5	0.30	(a)	803	0.64	(a)
126.1	0.198	(a)	259.6	0.32	(b)	1180	0.78	(b)
131.2	0.201	(a)	273.1	0.31	(a)	1973	1.03	(b)
135.6	0.192	(c)	288.9	0.31	(a)	2001	0.97	(b)
137.3	0.208	(a)	303	0.33	(a)	2885	1.18	(b)
144.3	0.213	(a)	332	0.40	(b)	3140	1.20	(b)
149.7	0.216	(a)	333	0.35	(a)	7106	1.82	(b)
150.3	0.203	(c)	349	0.37	(a)	1293	2.60	(b)
153.1	0.220	(a)	364	0.38	(a)	2239	3.01	(b)

Experimental:

- (a) Frommhold, Fortschr. Physik **12**, 597 (1964).
- (b) Schlumbohm, Z. Physik **182**, 317 (1965).
- (c) Naidu, *et al.*, J. Phys. D **3**, 957 (1970).

Table 1.24

Values of W_M for oxygen

E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)	E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)	E/N (10^{-17} Vcm^2)	W_M (10^7 cm/sec)
1.067	0.062	12.70	0.41	28.73	0.56
1.281	0.106	14.36	0.42	40.8	0.73
1.502	0.134	14.78	0.43	42.0	0.75
2.533	0.203	15.14	0.43	42.6	0.75
3.84	0.242	15.38	0.43	51.0	0.87
5.93	0.250	15.53	0.43	53.6	0.85
6.26	0.31	18.03	0.44	70.0	1.12
6.44	0.31	18.69	0.46	99.2	1.48
7.33	0.33	18.84	0.47	118.9	1.72
9.00	0.36	20.33	0.48	119.6	1.76
9.45	0.37	23.60	0.51	137.7	1.92
11.41	0.39	23.69	0.53	156.5	1.15
11.86	0.40	25.09	0.54	174.6	2.35

Experimental:

All data taken from Brose, Phil. Mag. 50, 536 (1925).

Table 1.25(a)

Values of W in carbon monoxide at 300 K

E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)
0.00901	0.032	0.1529	0.50	0.472	0.77
0.01506	0.053	0.1990	0.61	0.556	0.76
0.0304	0.105	0.2462	0.69	0.630	0.76
0.0613	0.207	0.2971	0.74	0.810	0.76
0.0777	0.246	0.312	0.79	1.287	0.91
0.1236	0.41	0.387	0.78		

Experimental:

All data taken from Pack, et al., Phys. Rev. 127, 2084 (1962).

Table 1.25(b)

Values of W in carbon monoxide at 77 K

E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)	E/N (10^{-17} Vcm^2)	W (10^6 cm/sec)
0.0450	0.093	0.2314	0.40	1.566	1.04
0.0600	0.133		0.43	2.195	1.249
0.0761	0.158	0.312	0.50	3.15	1.44
0.0918	0.196	0.472	0.56	8.05	1.945
0.1224	0.244	0.614	0.67	10.73	1.99
0.1571	0.35	0.950	0.83	16.01	2.22
		1.090	0.88	26.06	2.71

Experimental:

All data taken from Pack, et al., Phys. Rev. 127, 2084 (1962).

Table 1.25(c)

Values of W in carbon monoxide at 195 K

E/N (10^{-17}Vcm^2)	W (10^6cm/sec)	E/N (10^{-17}Vcm^2)	W (10^6cm/sec)	E/N (10^{-17}Vcm^2)	W (10^6cm/sec)
0.01487	0.040	0.1549	0.38	0.455	0.67
0.02248	0.061	0.2284	0.51	0.638	0.64
0.02962	0.078	0.301	0.58	1.573	0.96
0.0758	0.201	0.387	0.64		

Experimental:

All data taken from Pack, *et al.*, Phys. Rev. 127, 2084 (1962).

Table 1.26

Values of W_M for carbon monoxide

E/N (10^{-17}Vcm^2)	W_M (10^7cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^7cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^7cm/sec)
0.319	0.065	2.551	0.162	20.41	0.32
0.638	0.087	5.10	0.225	40.8	0.46
1.276	0.119	10.19	0.265	81.6	0.72
				163.3	1.44

Experimental:

All data taken from Skinner, *et al.*, Phil. Mag. 46, 630 (1923).

Table 1.27

Values of W_M for nitric oxide

E/N (10^{-17}Vcm^2)	W_M (10^6cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^6cm/sec)	E/N (10^{-17}Vcm^2)	W_M (10^6cm/sec)
1.280	2.02	14.99	6.0	31.9	6.7
2.377	2.39	16.09	6.1	33.7	6.8
4.21	2.98	17.28	6.3	36.1	7.0
5.85	3.5	18.10	6.3	38.7	7.2
7.04	3.9	19.29	6.4	41.7	
8.96	4.4	21.03	6.4	45.2	7.9
10.79	5.0	23.13	6.4	47.7	8.2
11.88	5.3	25.32	6.4	50.2	8.4
13.07	5.6	26.60	6.5	52.8	8.7
14.17	5.8	28.71	6.5	55.6	9.0
		30.7	6.6	59.5	9.3

Experimental:

All data taken from Bailey, *et al.*, Phil. Mag. 17, 1169 (1934).

Table 1.28

Values of W for carbon dioxide for $E/N < 25 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^5cm/sec)		E/N (10^{-17}Vcm^2)	W (10^5cm/sec)		E/N (10^{-17}Vcm^2)	W (10^5cm/sec)	
0.0311	0.056	(a)	0.759	1.35	(b)	3.64	6.5	(b)
0.0594	0.115	(a)	0.910	1.62	(b)	4.55	8.2	(b)
0.1506	0.283	(a)	1.214	2.16	(b)	6.07	11.3	(b)
0.301	0.56	(a)	1.517	2.70	(b)	7.59	14.5	(b)
0.303	0.54	(b)	1.820	3.2	(b)	9.10	18.2	(b)
0.364	0.65	(b)	2.124	3.8	(b)	12.14	27.3	(b)
0.455	0.81	(b)	2.43	4.3	(b)	15.17	40	(b)
0.546	0.97	(b)	2.730	4.9	(b)	18.20	55	(b)
0.607	1.08	(b)	3.03	5.4	(b)	21.24	68	(b)

Experimental:

- (a) Pack, *et al.*, Phys. Rev. 127, 2084 (1962).
 (b) Elford, Austr. J. Phys. 19, 629 (1966).

Table 1.29

Values of W for carbon dioxide for $E/N > 100 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^8cm/sec)		E/N (10^{-17}Vcm^2)	W (10^8cm/sec)		E/N (10^{-17}Vcm^2)	W (10^8cm/sec)	
114.4	0.134	(a)	909	0.46	(b)	3940	1.09	(b)
126.8	0.143	(a)	1212	0.55	(b)	4240	1.14	(b)
131.5	0.150	(a)	1516	0.62	(b)	4550	1.19	(b)
135.1	0.147	(a)	1819	0.69	(b)	4850	1.24	(b)
147.7	0.162	(a)	2122	0.76	(b)	5150	1.28	(b)
151.5	0.166	(b)	2425	0.82	(b)	5460	1.33	(b)
153.5	0.169	(a)	2728	0.88	(b)	5760	1.37	(b)
158.7	0.172	(a)	3030	0.94	(b)	6062	1.41	(b)
455	0.31	(b)	3330	0.99	(b)			
606	0.36	(b)	3640	1.04	(b)			

Experimental:

- (a) Frommhold, Z. Physik 160, 554 (1960).
 (b) Schlumbohm, Z. Physik 182, 317 (1965).

Table 1.30

Values of W for air for $E/N < 9.5 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^6cm/sec)	E/N (10^{-17}Vcm^2)	W (10^6cm/sec)	E/N (10^{-17}Vcm^2)	W (10^6cm/sec)
0.292	0.39 (1)	1.409	0.86 (1)	3.48	1.29 (1)
0.333	0.41 (1)	1.516	0.95 (b)	3.64	1.25 (a)
0.370	0.43 (1)	1.528	0.84 (a)	4.00	1.37 (1)
0.406	0.45 (1)	1.567	0.91 (1)	4.03	1.33 (a)
0.452	0.47 (1)	1.679	0.94 (1)	4.45	1.43 (1)
0.514	0.51 (1)	1.803	0.98 (1)	4.55	1.50 (b)
0.565	0.53 (1)	1.828	0.89 (a)		1.37 (a)
	0.51 (a)	1.849	0.93 (a)	4.89	1.50 (1)
0.606	0.61 (b)	1.932	1.02 (1)	4.92	1.42 (a)
0.619	0.56 (1)	2.099	1.05 (1)	5.10	1.46 (a)
0.673	0.58 (1)	2.125	0.99 (a)	5.43	1.58 (1)
0.740	0.61 (1)	2.333	1.10 (1)	5.83	1.56 (a)
0.820	0.65 (1)	2.370	1.05 (a)	6.03	1.67 (1)
0.882	0.67 (a)	2.449	1.10 (a)	6.06	1.68 (b)
0.912	0.68 (1)	2.529	1.14 (1)	6.55	1.70 (a)
1.014	0.71 (1)	2.716	1.17 (1)	6.70	1.77 (1)
1.140	0.76 (1)	3.30	1.25 (b)	7.53	1.88 (1)
1.143	0.76 (a)	3.05	1.17 (a)	8.43	1.91 (a)
1.297	0.82 (1)	3.090	1.23 (1)	8.87	2.06 (1)
				9.09	2.3 (b)

Experimental:

(a) Nielsen, *et al.*, Phys. Rev. **51**, 69 (1937).(b) Hessenauer, Z. Physik **204**, 142 (1967).

Theoretical:

(1) Heylen, Proc. Phys. Soc. (London) **79**, 284 (1962).

Table 1.31

Values of W for air for $E/N > 9.5 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	W (10^7cm/sec)	E/N (10^{-17}Vcm^2)	W (10^7cm/sec)	E/N (10^{-17}Vcm^2)	W (10^7cm/sec)
9.98	0.222 (1)	43.3	0.73 (1)	152.2	1.67 (a)
11.51	0.244 (1)	47.5	0.78 (1)	160.6	1.89 (b)
12.76	0.264 (1)	52.8	0.86 (1)	167.2	1.85 (a)
13.87	0.283 (1)	56.6	0.90 (1)	171.6	1.94 (b)
15.42	0.31 (1)	62.2	0.97 (1)	183.1	1.94 (a)
16.94	0.34 (1)	67.6	1.04 (1)	189.7	2.11 (b)
18.58	0.37 (1)	75.0	1.12 (1)	197.2	2.04 (a)
20.18	0.40 (1)	82.4	1.20 (1)	199.8	2.19 (b)
22.18	0.43 (1)	90.6	1.28 (1)	223.0	2.39 (b)
24.04	0.46 (1)	103.0	1.40 (1)	233.5	2.48 (b)
26.42	0.50 (1)	106.3	1.25 (a)	234.2	2.32 (a)
29.38	0.54 (1)	114.2	1.30 (a)	241.6	2.52 (b)
33.4	0.60 (1)	122.1	1.31 (a)	258.0	2.69 (b)
36.3	0.64 (1)	137.2	1.51 (a)	270.6	2.81 (b)
39.8	0.69 (1)	140.8	1.62 (b)	274.9	2.77 (a)
				295.9	2.99 (b)

Experimental:

(a) Frommhold, Fortschr. Physik **12**, 597 (1964).(b) Ryzko, Proc. Phys. Soc. (London) **85**, 1283 (1965).

Theoretical:

(1) Heylen, Proc. Phys. Soc. (London) **79**, 284 (1962).

Table 1.32
Values of W_M for air

E/N (10^{-17} Vcm 2)	W_M (10^7 cm/sec)	E/N (10^{-17} Vcm 2)	W_M (10^7 cm/sec)	E/N (10^{-17} Vcm 2)	W_M (10^7 cm/sec)
0.589	0.0277 (a)	10.02	0.231 (a)	33.2	0.56 (a)
1.178	0.078 (a)	10.80	0.227 (a)	44.3	0.73 (b)
1.218	0.102 (b)	11.81	0.295 (b)	49.1	0.77 (a)
1.473	0.068 (a)	12.67	0.256 (a)	49.5	0.76 (a)
2.160	0.083 (a)	13.45	0.288 (a)	59.2	0.89 (b)
2.624	0.136 (b)	14.51	0.302 (a)	60.5	0.95 (a)
3.04	0.135 (a)	16.20	0.32 (a)	62.6	0.94 (a)
3.34	0.119 (a)	16.99	0.32 (a)	65.3	0.88 (a)
4.81	0.158 (a)	17.81	0.38 (b)	65.6	0.95 (a)
5.72	0.198 (b)	18.36	0.36 (a)	68.9	0.95 (a)
5.96	0.175 (a)	19.15	0.36 (a)	74.0	1.04 (b)
5.99	0.174 (a)	23.52	0.47 (b)	95.5	1.30 (a)
7.37	0.197 (a)	24.75	0.44 (a)	126.9	1.64 (a)
8.05	0.212 (a)	29.52	0.55 (b)	131.8	1.53 (a)
8.15	0.196 (a)	30.1	0.53 (a)	193.4	1.99 (a)
8.72	0.246 (b)	31.2	0.54 (a)	256.3	2.46 (a)
8.94	0.217 (a)	32.1	0.55 (a)	468	3.55 (a)
				619	4.49 (a)

Experimental:

- (a) Townsend, *et al.*, Proc. Roy. Soc. (London) Ser. A 88, 336 (1913).
(b) Huxley, *et al.*, Proc. Roy. Soc. (London) Ser. A 196, 402 (1949).

3.2. (Diffusion Coefficient)/Mobility

As an electron swarm drifts with a velocity W under the influence of an electric field E , it also diffuses due to the agitational velocity of the electrons. The diffusion is characterized by a diffusion coefficient D defined by the equation

$$j = -D \nabla n, \quad (4)$$

where j is the number of electrons flowing normally across unit area per second due to the density gradient of the electrons. At very low electric field strengths, the diffusion is symmetric and D is a constant independent of direction. As the electric field strength increases from very low values, the diffusion becomes asymmetric and the diffusion coefficient becomes a tensor, with components D_T and D_L transverse to and along the direction of E , respectively.

A useful and widely measured parameter of an electron swarm is the ratio W/D_T , because

- it is directly related to the mean energy of the swarm;
- it can be related to the collision cross section (see eq (8)), and
- when taken together with measured values of W , it gives values of D_T , which is a difficult parameter to measure directly.

The equation relating W/D_T to mean energy may be written as

$$W/D_T = (2F/3k) (Ee/kT), \quad (5)$$

where k is Boltzmann's constant, T is the absolute temperature and, k is the so-called Townsend energy factor which is defined as the ratio of the mean energy of the electrons to that of gas atoms at 15 °C, and F is a dimensionless constant that depends on the energy distribution (see L. G. H. Huxley and R. W. Crompton, in *Atomic and Molecular Processes*, D. R. Bates, ed., Academic Press, New York, 1962, pp. 336-73); F takes the value 1.5 for a Maxwellian distribution and 1.312 for a Druyvesteyn distribution.

If E is in V cm $^{-1}$, the other quantities in cgs units and $T=288$ K is adopted as a standard temperature, eq (5) becomes

$$W/D_T = 40.3 (E/k_1), \quad (6)$$

where

$$k_1 = 3k/2F.$$

In order to relate W/D_T to the collision cross section, the expression

$$D_T = (4\pi/3N) \int_0^\infty (c^3/q_m) f dc, \quad (7)$$

which relates the diffusion coefficient to atomic parameters, may be used in combination with eq (1) to give

$$D_T E/W = - (m/e) \frac{\int_0^\infty (c^3/q_m) f dc}{\int_0^\infty (c^2/q_m) (df/dc) dc}, \quad (8)$$

which has been used [see, e.g., 1062, 2433] to obtain information about q_m from data on electron swarms.

Experimentally, W/D_T is determined by means of the Townsend method (see J. S. Townsend, Proc. Roy. Soc. **A80**, 207, 1908; **A81**, 464, 1908) in which the distribution of the electron current over a given plane perpendicular to the direction of drift of an electron swarm originating at a point or line source is measured. There are considerable difficulties in the analysis of such measurements, but recent discussion (Crompton, Australian J. Phys. **25**, 409, 1972) has shown that despite these difficulties the values obtained are valid in most cases.

The parameter determined is W/D_T but as can be seen from eq (6) this quantity depends on N as well as on E/N . Thus the results of the experiments are usually given in terms of the parameter

$$D_T E/W = D_T/\mu,$$

where $\mu = W/E$ is the electron mobility. Alternatively, the results may be expressed in terms of

$$\epsilon_k = e(D_T/\mu),$$

which is usually known [260] as the "characteristic energy," because, as can be seen from eq (5), it is related to the mean energy. Both ϵ_k and D_T/μ are functions only of E/N ; the value of ϵ_k in electron volts is, of course, the same as that of D_T/μ in volts.

Recently, measurements have also been made [see, e.g., 3313] of W/D_L (where D_L is the longitudinal diffusion coefficient in a direction parallel to the electric field) using time-of-flight methods.

Values of D_T/μ for all the gases with which this survey is concerned and of D_L/μ where available are given in the following sub-sections.

3.2.a. Values of D_T/μ and of D_L/μ —Helium

A critical examination of the Townsend method by Crompton, Elford, and Jory [2433; see also 1622 and 1438] has shown that it is possible, provided sufficient care is taken particularly in relation to field uniformity,

to obtain values of D_T/μ with an error of less than 1 percent. Their results for helium [2433] are shown in figure 2.1(a). Also shown are the earlier experimental data [1036, 201], some of which [1036] required an empirical correction for geometrical distortion. Both Crompton et al. [2433] and Frost and Phelps [1062] obtained theoretical values by using the value of q_m discussed in section 3.1.a and the agreement obtained with their experimental values is shown in figure 2.1(b). As shown in figure 2.2, there is also good agreement between experimental [1036] and theoretical [1062] values of D_T/μ obtained at a temperature of 77 K. Sets of values of D_T/μ both at ambient temperature and 77 K are given in table 2.1.

There have also been recent measurements [3313] of D_L/μ for helium using a time-of-flight method and the values obtained are given in figure 2.1(a) and table 2.2. (The temperature to which the values of E/p given in this paper correspond is not clear, but later work [see 5135] shows it was probably 27 °C so this temperature was assumed here and wherever it is referred to in the rest of this section.) Also shown in figure 2.1(a) are the theoretical values obtained by Lowke and Parker [4052] using the values of q_m obtained in [2433]. There is agreement to within about 15 percent between theory and experiment.

3.2.b. Values of D_T/μ —Neon

There are very few data available for D_T/μ for neon. The only experimental measurements are those of Bailey [2280] on neon containing 1 percent helium. These results are shown in figure 2.3 together with the available theoretical values [273, 2250] which were obtained using the energy distribution of [1388] and various assumed cross sections. Using values of the momentum transfer cross section obtained from drift velocity measurements, Robertson [4862] calculated values of D_T/μ which are stated to be about 20 percent below Bailey's values. Since Robertson finds the values of D_T/μ very sensitive to nitrogen impurities he suggests that molecular impurities in Bailey's gas samples might account for the difference.

There are no experimental or theoretical values of D_L/μ available for neon.

3.2.c. Values of D_T/μ and D_L/μ —Argon

There have been two experimental investigations of D_T/μ for argon, the early results of Townsend and Bailey [199] covering the range $3 \times 10^{-18} < E/N < 6 \times 10^{-16}$ V cm² at room temperature, and the more recent data of Warren and Parker [1036] in the range $4.5 \times 10^{-21} < E/N < 1.3 \times 10^{-18}$ V cm² at temperatures of 77 K and 87 K. Both sets of data are shown in figure 2.4 and tabulated in table 2.3. The theoretical results of Engelhardt and Phelps [292] calculated using the values of q_m discussed in section 3.1.e are also shown in figure 2.4. There is good agreement between theory and experi-

ment except in the range of E/N from about 3×10^{-18} to 10^{-17} V cm².

Recent theoretical values of D_L/μ for argon at a temperature of 77 K, calculated [4052] using the above mentioned cross sections [292] and also the cross sections obtained by Golden (Phys. Rev. **151**, 48, 1966), are shown in figure 2.5. It can be seen that the results differ for values of E/N between about 10^{-20} and 10^{-19} V cm², but agree at higher values of E/N where they lie below the recent experimental results [3313] obtained at ambient temperatures. These experimental and theoretical data are given in table 2.4.

At higher values of E/N , Wagner [4971], from observation of the drift and diffusion of pulsed electron avalanches, gave values for $D_T/\mu \approx 6.3 \pm 1.1$ for $72 \times 10^{-17} < E/N < 136 \times 10^{-17}$ V cm².

3.2.d. Values of D_T/μ and D_L/μ —Krypton and Xenon

The only available experimental result for D/μ for krypton and xenon is a value of about 3.1 ± 0.9 V for D_T/μ for E/N in the range from 120×10^{-17} to 275×10^{-17} V cm² obtained [4971] from observations on pulsed avalanches.

At lower values of E/N ($< 3 \times 10^{-17}$ V cm²) theoretical values of D_T/μ were obtained by Frost and Phelps [1062] and of D_L/μ by Lowke and Parker [4052] using the cross sections mentioned in section 3.1.g. These theoretical values are shown in figure 2.6 and given in tables 2.5 and 2.6.

3.2.e. Values of D_T/μ and D_L/μ —Hydrogen

There have been a relatively large number of investigations of D/μ for hydrogen and the results at room temperature may most conveniently be considered in two regions of E/N , above and below $E/N = 10^{-16}$ V cm². The results in the region of E/N below 10^{-16} V cm² are shown in figure 2.7. The most recent data [2992] for D_T/μ are accurate to within ± 1 percent; at the higher values of E/N , these data agree with the earlier data [1438, 689, 352] obtained in the same laboratory and at the lower values of E/N , are regarded as more accurate. Thus to avoid confusion they are the only results of Crompton's group shown on the figure. It can be seen that all the experimental results [2992, 736, 195] for D_T/μ shown in figure 2.7 are in good agreement at the higher values of E/N , but that the results of [736] increasingly diverge from the others as E/N decreases. Theoretical values [181] calculated using the cross sections referred to in section 3.1.h are in good agreement with the recent experimental data and are shown in figure 2.7. The numerical values given in table 2.7 for $E/N < 7 \times 10^{-17}$ V cm² are those of [2992].

The recent experimental values [3313] of D_L/μ for hydrogen are shown in figure 2.7, together with the recent theoretical values [4052] obtained using the same cross sections as those used for the calculation of D_T/μ . The experimental values of D_L/μ are also given in table 2.8.

Results for D_T/μ and D_L/μ at values of $E/N > 10^{-16}$ V cm² are shown in figure 2.8. Ionization processes begin to become important in hydrogen when $E/N > 5.6 \times 10^{-16}$ V cm². Some of the results [1437, 1778, 5230, 5234] were obtained using an analysis taking ionization into account, whereas no account was taken of ionization in [195] and an unspecified correction for ionization was made in [3303]. The cross sections which gave theoretical ionization coefficients in good agreement with experiment (see sec. 3.7.b(i)) give theoretical values [260] of D_T/μ much higher (up to 40 percent greater) than the experimental values and are not shown. The experimental values of D_T/μ obtained taking ionization into account are included in table 2.7.

Schlumbohm [1625] determined the pulse shape of electron avalanches in a uniform field electrode system, so that his values of D/μ , which are also shown in figure 2.8, represent values of D_L/μ .

The two sets [2992, 1036] of available experimental data for D_T/μ at 77 K are in good agreement with one another, although that given in [1036] is based on an empirical calibration of the apparatus used. The values are shown, together with the theoretical values [260] for this temperature, in figure 2.9. A set of numerical values is given in table 2.9.

3.2.f. Values of D_T/μ and D_L/μ —Nitrogen

It is convenient to consider the results for D_T/μ obtained for nitrogen in two ranges, above and below a value of E/N of about 14×10^{-17} V cm². The results for $E/N < 14 \times 10^{-17}$ V cm² at ambient temperatures are shown in figure 2.10. The most recent results [689 and 1429] have an estimated error of less than 1 percent and these are tabulated in table 2.10. Small differences between these results and some of the earlier data [736 and 352] are ascribable to differences in purity of the gas. Commercial tank nitrogen with an impurity level of about 5 in 10^3 was used in the earlier work compared with samples containing impurities of a few ppm in the later experiments.

Also shown in figure 2.10 are the theoretical values obtained [181, 4052] by numerical solution of the Boltzmann equation.

The most recent experimental data [3303] for D_T/μ at values of $E/N > 14 \times 10^{-17}$ V cm² show that the results for spectroscopically pure nitrogen agree with those for tank nitrogen in this range. These results are shown, together with earlier experimental data and theoretically computed values [4052] in figure 2.11. The experimental data are also given in table 2.10.

There are also both experimental [1036] and theoretical [218] results available at 77 K for the low E/N range. These are shown in figure 2.12 and a set of numerical values tabulated in table 2.11.

There are two sets of experimental data [1833, 3313] available for D_L/μ for values of $E/N < 14 \times 10^{-17}$ V cm² at room temperature and these are given in figure 2.13 where they are compared with theoretical results [4052]

obtained by numerical integration of the Boltzmann equation.

For higher values of E/N there is only one set of experimental data [1625] available and these results are compared with theoretical values [5233] obtained by Monte Carlo methods in figure 2.14 (a temperature of 0 °C has been assumed in reducing the values of E/p given in this theoretical paper to E/N). The theoretical values obtained from integration of the Boltzmann equation [4052] for values of $E/N > 14 \times 10^{-17} \text{ V cm}^2$ are also shown in figure 2.14.

Table 2.12 gives a set of numerical values for D_L/μ for the whole range investigated.

3.2.g. Values of D_T/μ and D_L/μ —Oxygen

The available data for D_T/μ for oxygen are given in figure 2.15 and show that the more recent results [649, 1269, 5226] which have a quoted error of less than about 3 percent lie considerably above the early values [195, 2089] for $E/N > 5 \times 10^{-17} \text{ V cm}^2$. In an attempt to overcome some of the difficulties experienced in obtaining values of D_T/μ for electrons in the presence of negative ions, measurements have also recently been made [4901] in oxygen containing 1.7 percent hydrogen. The presence of the hydrogen rapidly removes O^- ions by an associative detachment reaction. These results for D_T/μ are also shown in figure 2.15 and can be seen to be in good agreement with the recent determinations in oxygen alone.

It has also been shown [2553] that a self-consistent set of values of the cross sections involved can be found that gives theoretical values of D_T/μ in good agreement with the recent experimental values. These theoretical values are also shown in figure 2.15.

A set of values of D_T/μ throughout the range of E/N investigated is given in table 2.13.

Experimental results for D_L/μ at very low values of E/N have recently been obtained using the drift-dwell-drift method (see section 3.3.a). These values are shown in figure 2.16 where they are compared with theoretical values [4052] which have been obtained over the range $10^{-18} < E/N < 10^{-15} \text{ V cm}^2$ using numerical integration of the Boltzmann equation. The disagreement between experimental and theoretical values suggests that some revision of the cross sections used in the theoretical calculations is necessary.

Schlumbohm [1625] obtained values of D/μ from investigations of avalanche shape at relatively high values of $E/N > 10^{-15} \text{ V cm}^2$. This type of experiment also gives values of D_L/μ and the results are shown in figure 2.17.

A composite set of tabulated values of D_L/μ for oxygen is given in table 2.14.

3.2.h. Values of D_T/μ and of D_L/μ —Carbon Monoxide

The only data at room temperature (unspecified) available for carbon monoxide are those of Skinker and White [200] for D_T/μ shown in figure 2.18. Also shown

in figure 2.18 are the more recent results of Warren and Parker [1036] which were obtained at a temperature of 77 K using an apparatus that required empirical calibration because of slight field distortions. The theoretical values obtained for 77 K by Hake and Phelps [2553] and for 293 K by Lowke [4052] using the cross sections of Hake and Phelps are also given in figure 2.18 and can be seen to be in good agreement with the experimental values.

Sets of values of D_T/μ at room temperature and at 77 K are given in tables 2.15(a) and (b), respectively.

There is one set of experimental data [3313] available for D_L/μ . This was obtained at 293 K and is shown together with theoretical values [4052], obtained at both 293 K and 77 K, in figure 2.19.

Sets of values of D_L/μ are tabulated in tables 2.16(a) and (b) for temperatures of 293 K and 77 K, respectively.

3.2.i. Values of D_T/μ —Nitric Oxide

The only published data on D/μ available for nitric oxide are the early experimental results of Skinker and White [200] and of Bailey and Somerville [2385] for D_T/μ . Although, as can be seen from figure 2.20, there is fair agreement between the two sets of data, it should be noted that the results of Skinker and White showed a marked dependence of D_T/μ on pressure, especially at the low values of E/N , probably because of the formation of negative ions.

3.2.j. Values of D_T/μ and D_L/μ —Carbon Dioxide

Measurements of D_T/μ for carbon dioxide have been made over the range $5.6 \times 10^{-19} < E/N < 1.4 \times 10^{-15} \text{ V cm}^2$. Recent work [2140] has shown experimentally that below values of E/N of about $14 \times 10^{-17} \text{ V cm}^2$ attachment has negligible influence on the results obtained. Several sets of data [159, 198, 736, 1036, and 2140] have been obtained in this range, but only those of Rees [2140] and of Warren and Parker [1036] extrapolated approximately to the expected value of D_{th}/μ (see 3.3) at low values of E/N and these results are shown in figure 2.21 and given in table 2.17.

At higher values of E/N , it was shown that under the experimental conditions used in [2140] the effect of attachment continued to be negligible up to a value of E/N of about $56 \times 10^{-17} \text{ V cm}^2$, and the various sets of results [2140, 1036, 198, 159] (the data for D_T/μ ascribed to [159] are actually given in Healey and Reed, *The Behavior of Slow Electrons in Gases*, 1941, p. 98) are seen in figure 2.22 to be in good agreement in this region.

At values of E/N greater than about $56 \times 10^{-17} \text{ V cm}^2$, the results obtained in [2140] showed that processes of both attachment and ionization become significant and that values of ionization coefficients and attachment coefficients are required for the analysis of the results. As a result, the values of D_T/μ are not accurately established in this region. The results of [2140] given in figure 2.22 were obtained using an expression that took

ionization but not attachment into account. The magnitude of the further correction required for attachment is uncertain, because values for the attachment coefficient are not accurately known. The values of D_T/μ at the higher values of E/N obtained in [2140] and given in figure 2.22 are probably too low by an amount varying between 4 percent and 9 percent depending on the values used for the attachment coefficient. As can be seen from figure 2.22, the data of [198], in which there is no discussion of ionization and attachment processes, are somewhat lower at these higher values of E/N .

The variation of D_T/μ with temperature was investigated experimentally by Warren and Parker [1036] and the results are shown in figure 2.23 where it can be seen that they are in good agreement with results calculated by Hake and Phelps [2553] using a self-consistent set of collision cross sections.

Sets of numerical values of D_T/μ at ambient temperatures and 195 K are given in tables 2.17(a) and (b), respectively.

At relatively low values of $E/N < 10^{-15}$ V cm² there exists one published set of theoretical values [4052] of D_L/μ for CO₂ and these are compared in figure 2.24 with the experimental data [1833, 3313] which are available over a more limited range.

The only values of D_L/μ at high values of $E/N > 10^{-15}$ V cm² in carbon dioxide are those obtained by Schlum-

bohm [1625] from the analysis of the arrival times of electrons in an avalanche and these values are shown in figure 2.25. Table 2.18 gives a set of values of D_L/μ in CO₂.

3.2.k. Values of D_T/μ —Air

Recent values [2813, 2097] of D_T/μ with an estimated error of ± 2 percent, obtained for dry air, from which CO₂ had been removed, are given in figure 2.26 and table 2.19. (In the region of overlap both sets of results agree and for clarity only those of [2097] are included in the figure.) These results are in good agreement with the later results of Rao [5247] and the early work of Bailey [1332] (not shown) if the temperature in the latter work is assumed to be 15 °C. As can be seen from figure 2.26, Townsend and Tizard's [2104] early results for dry air lie considerably below the recent values, possibly because the CO₂, which is expected to reduce the mean energy of the electrons, was not removed in this case. Above E/N of about 10^{-16} V cm² ionization becomes significant and may have affected the results of [2104].

There are no published theoretical values of D_T/μ available. Neither are there any published data for D_L/μ .

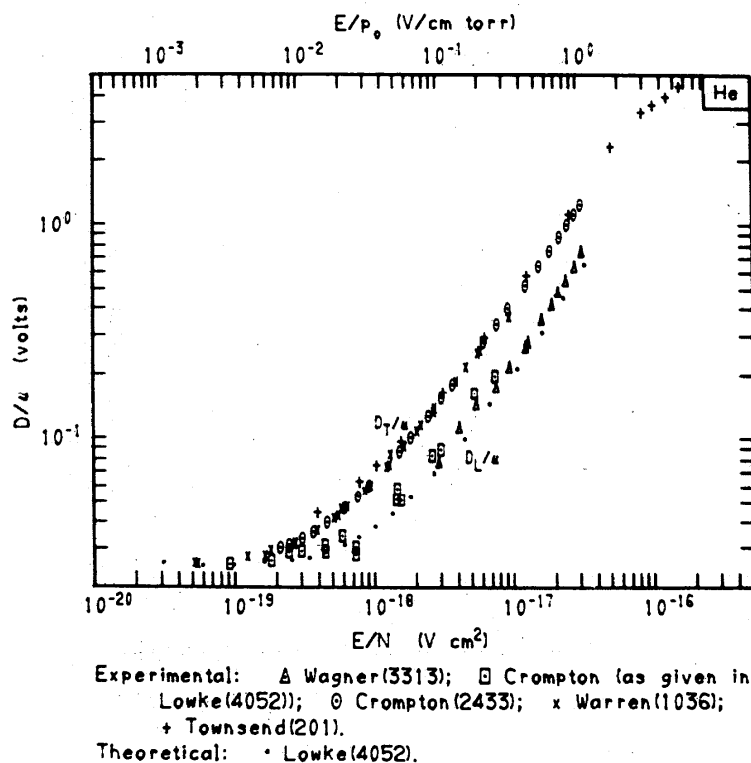


FIGURE 2.1(a). D_T/μ and D_L/μ as functions of E/N for helium at ambient temperature.

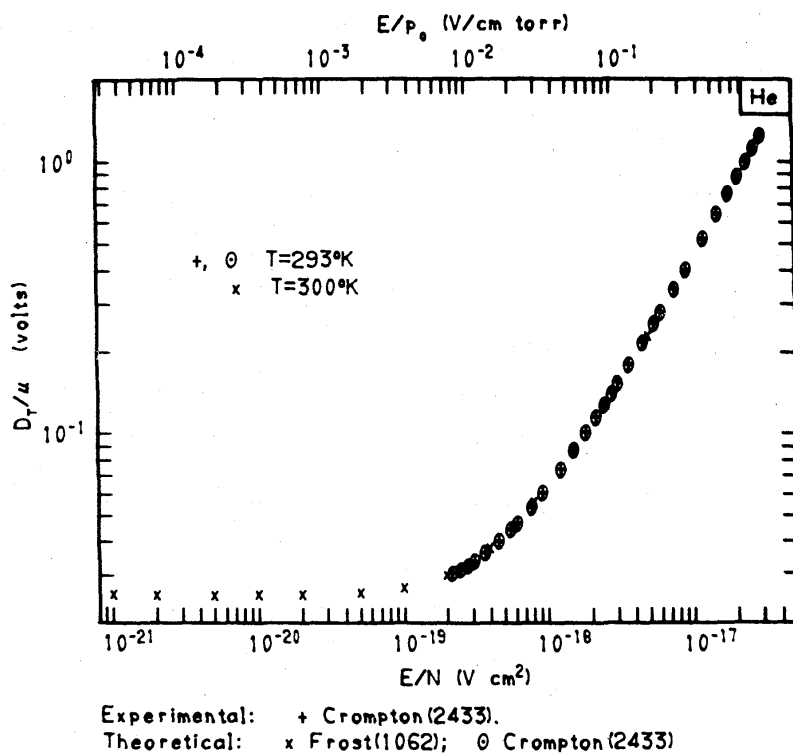


FIGURE 2.1(b). Comparison of experimental and theoretical values of D_T/μ for helium at ambient temperatures.

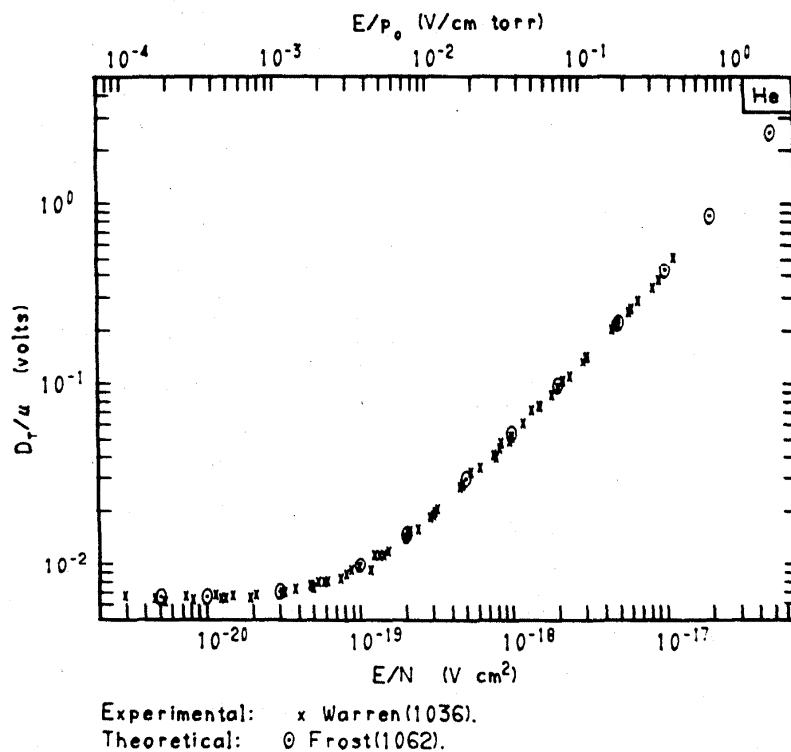


FIGURE 2.2. D_T/μ as a function of E/N in helium at 77 K.

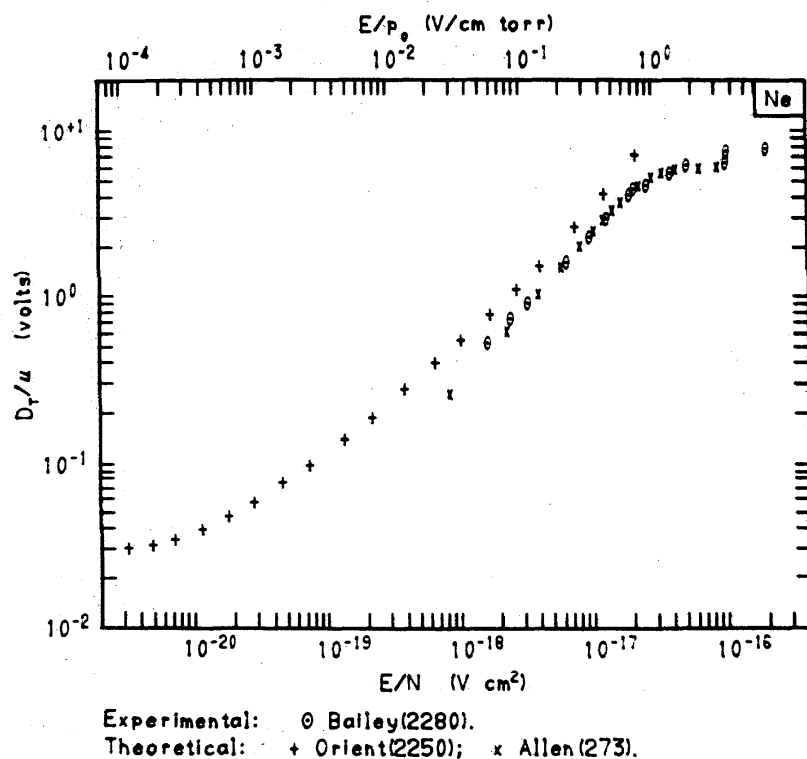


FIGURE 2.3. D_T/μ , E/N for neon. (The experimental results are for neon containing 1 percent helium. The temperature to which the values of E/p correspond in [273] is assumed to be 14°C.)

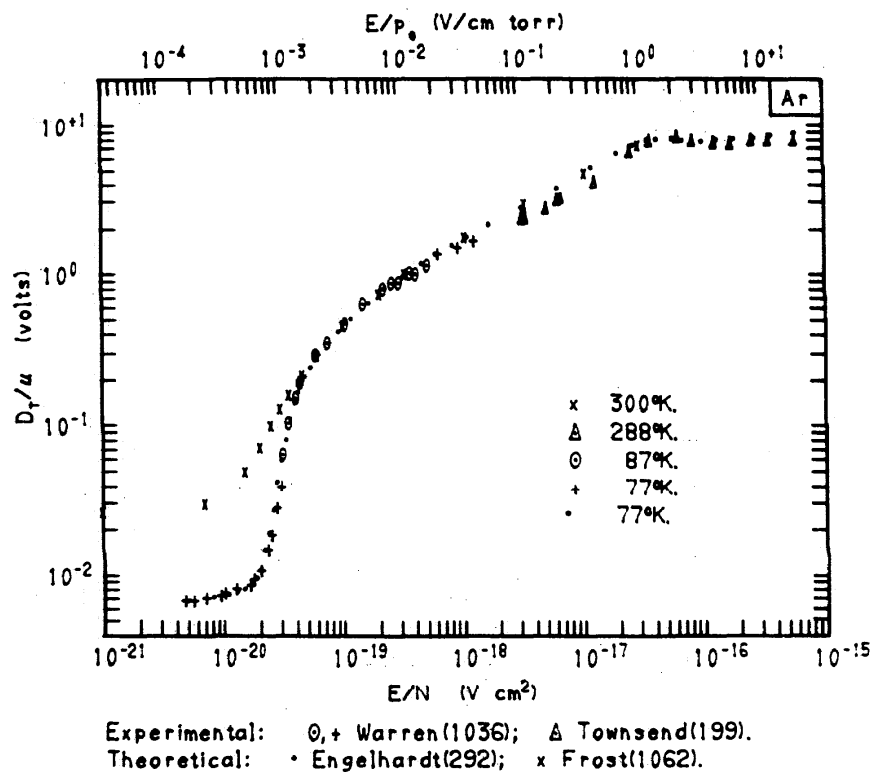
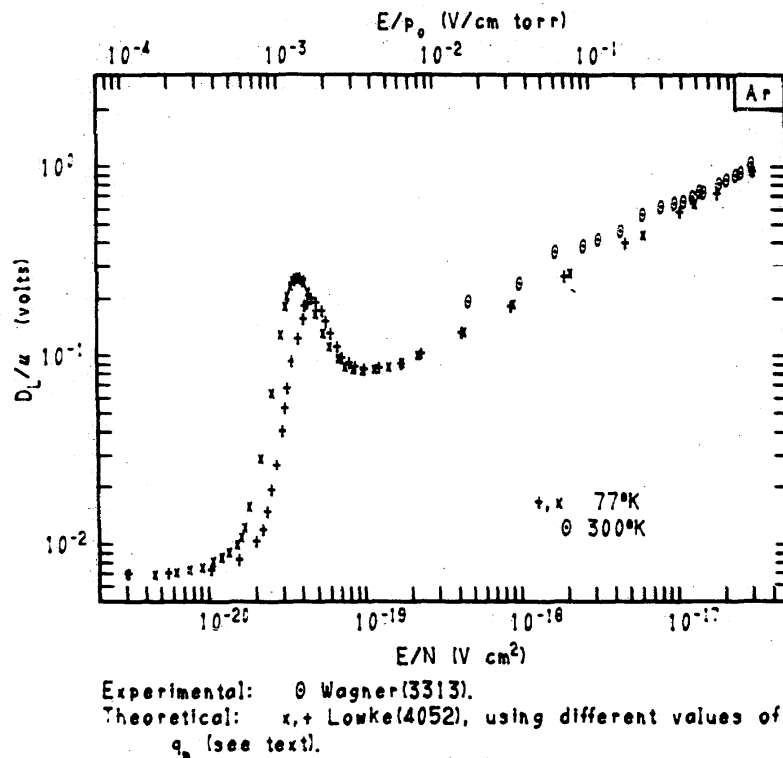
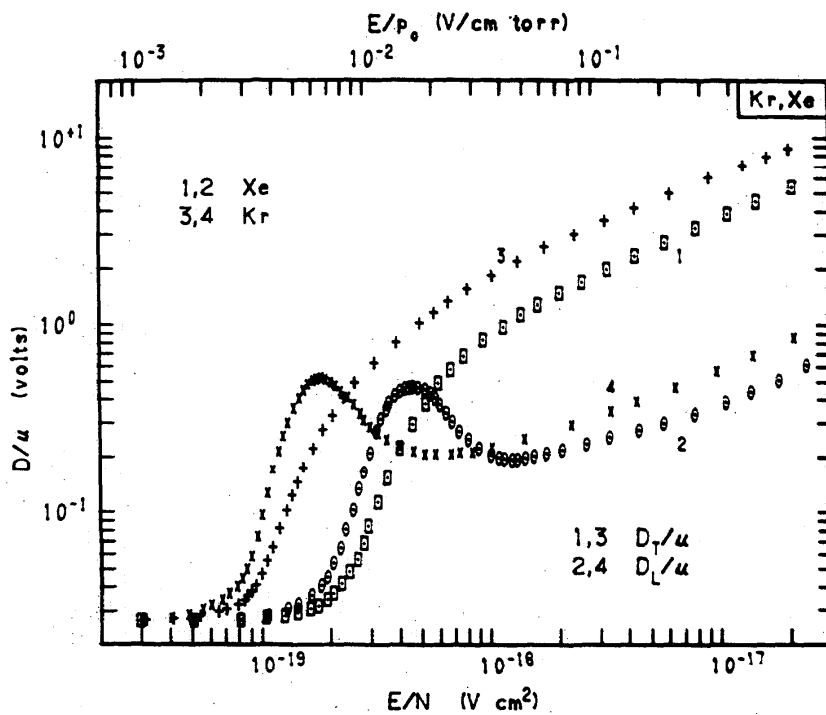


FIGURE 2.4. D_T/μ , E/N for argon at various temperatures.

FIGURE 2.5. D_L/μ , E/N for argon.FIGURE 2.6. Theoretical values of D_T/μ and D_L/μ for krypton and xenon.

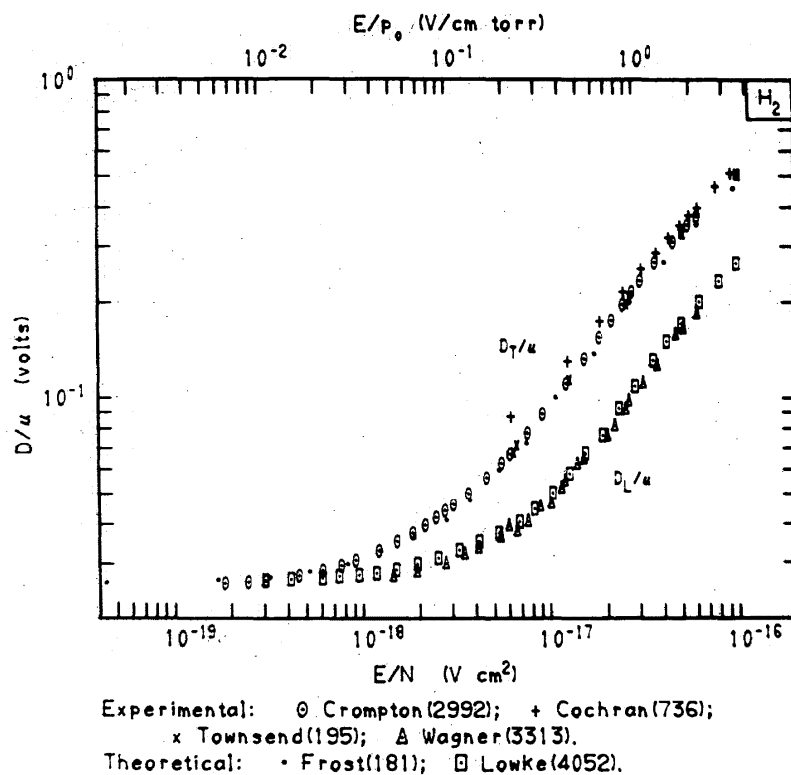


FIGURE 2.7. D_T/μ and D_L/μ as functions of E/N for hydrogen at room temperature for $E/N < 10^{-16}$ V cm^2 .

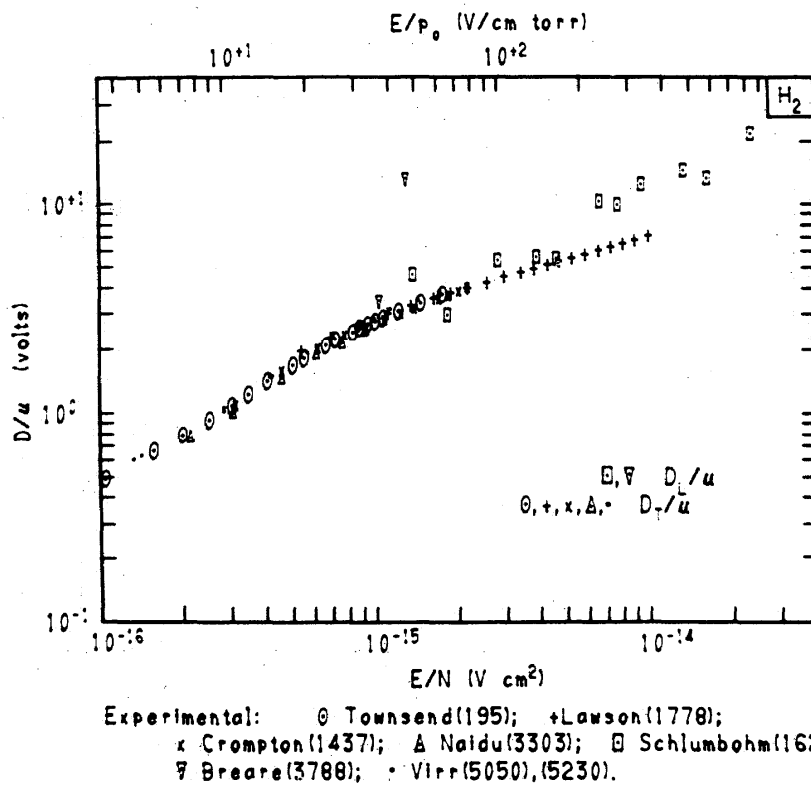
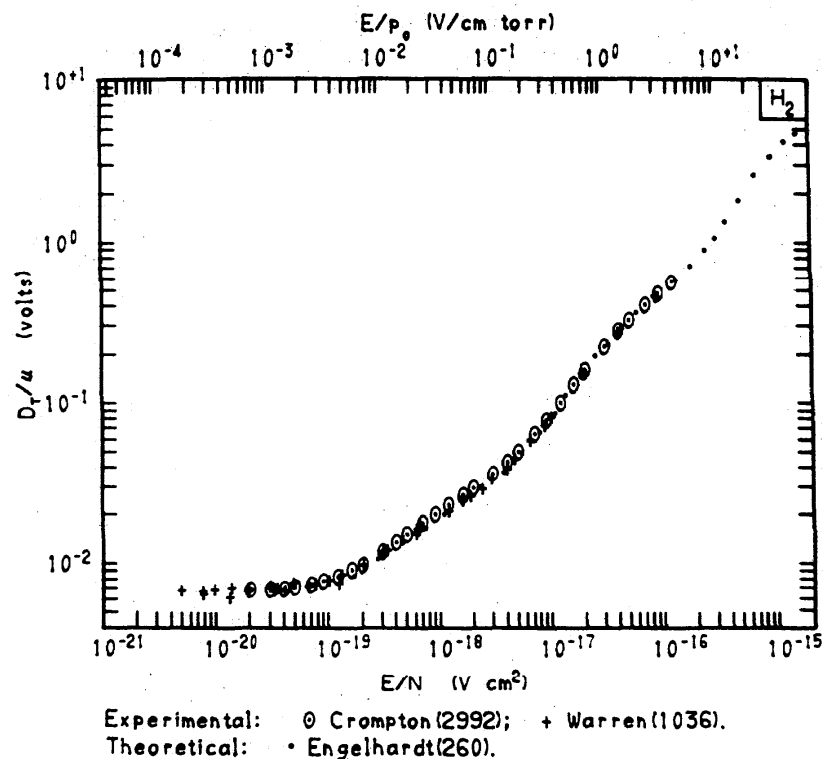
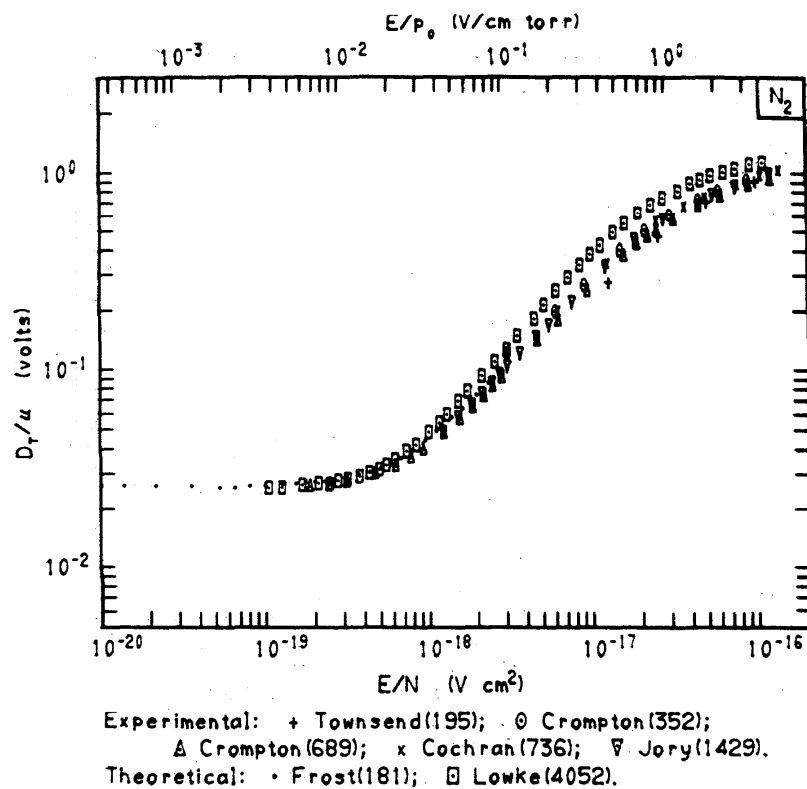
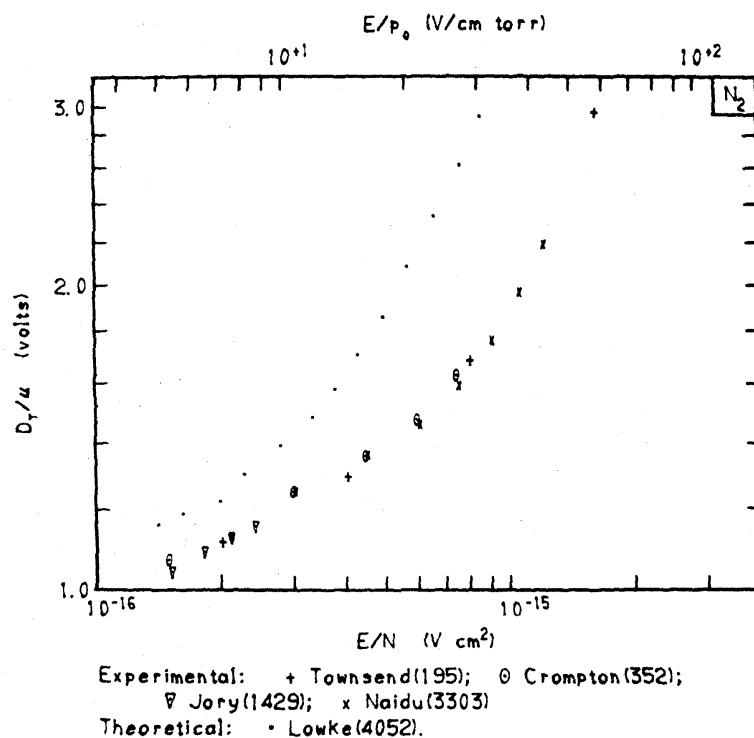
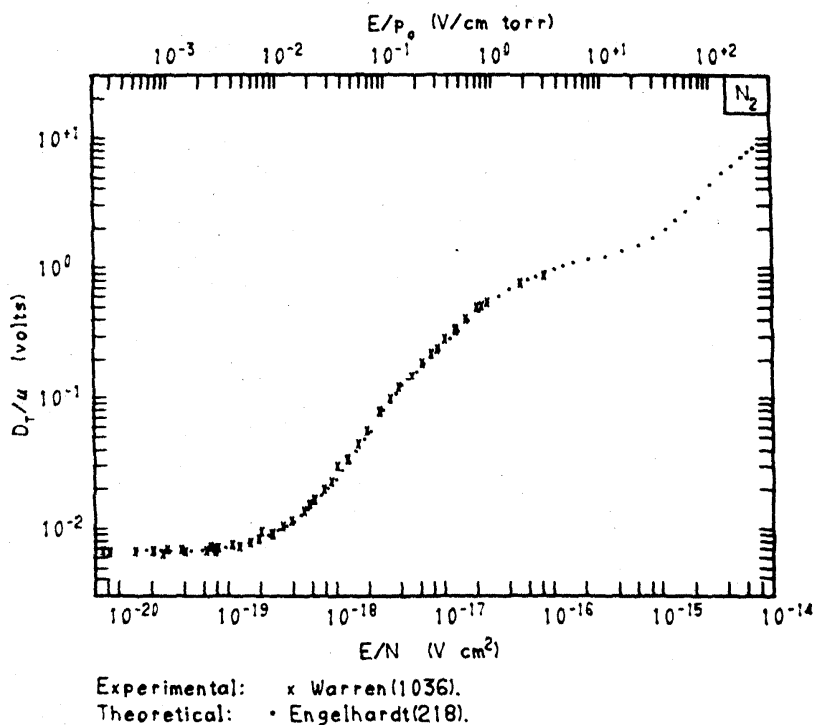
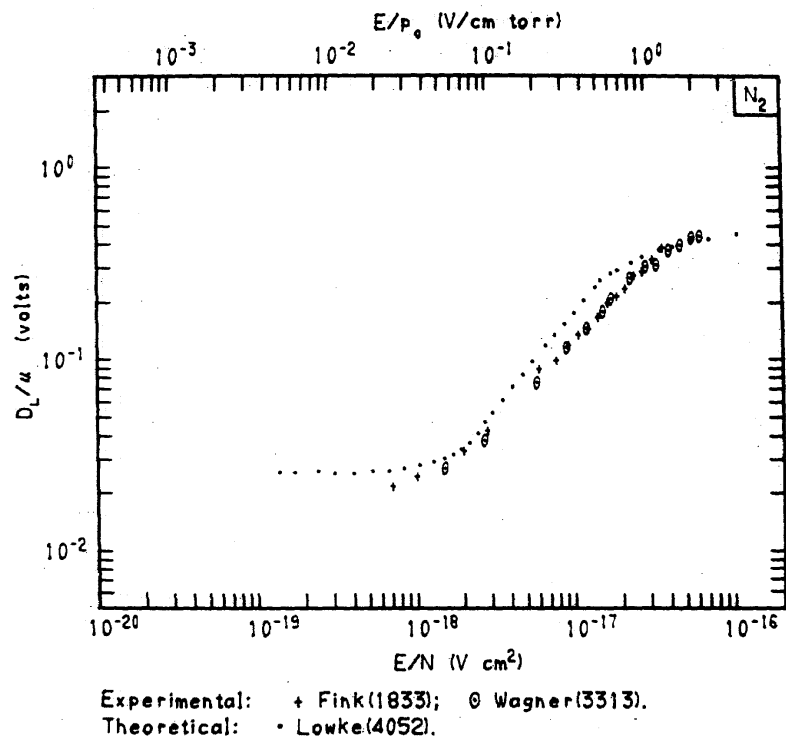
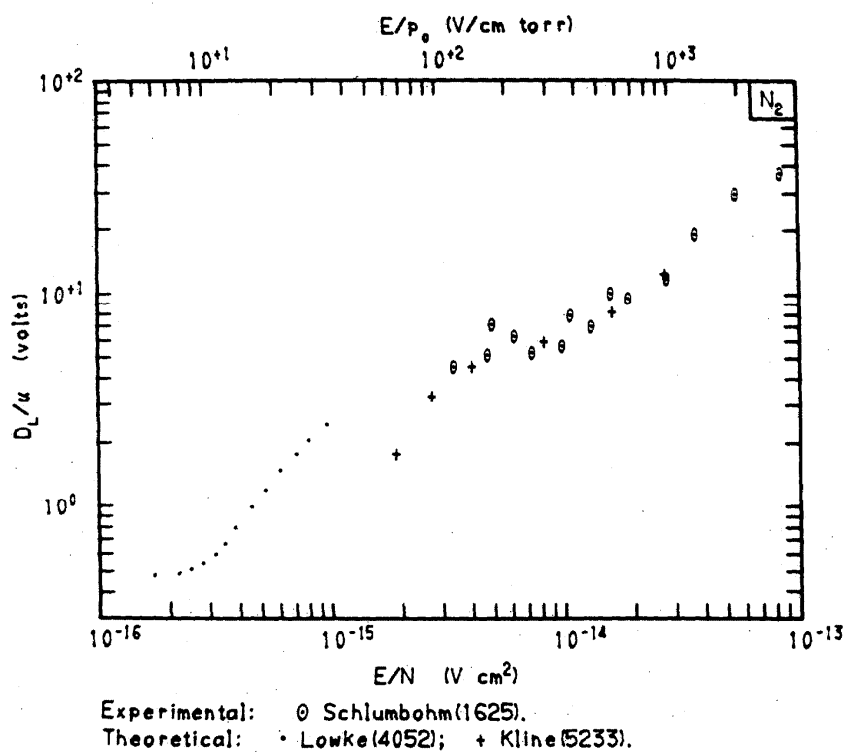
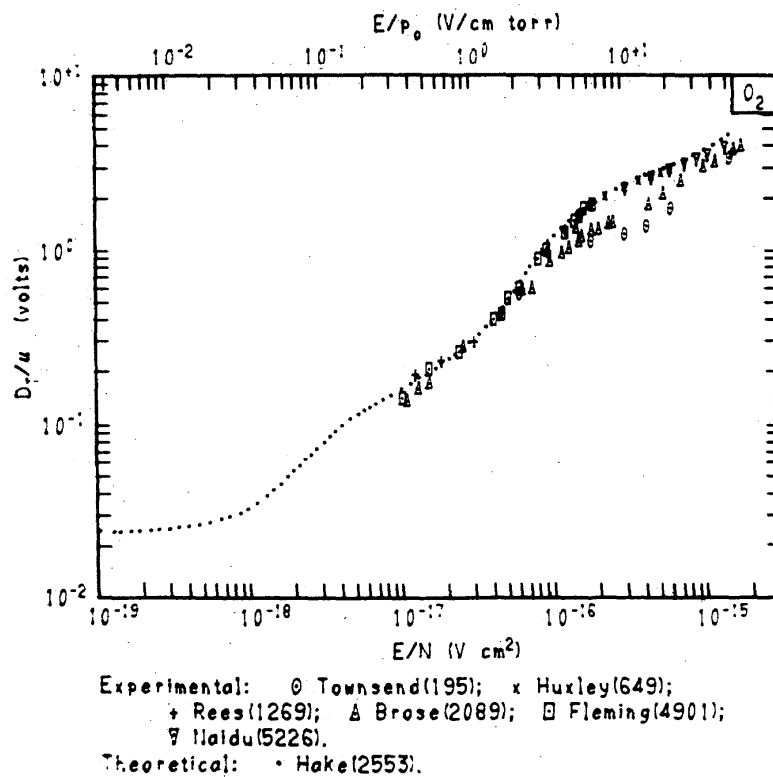
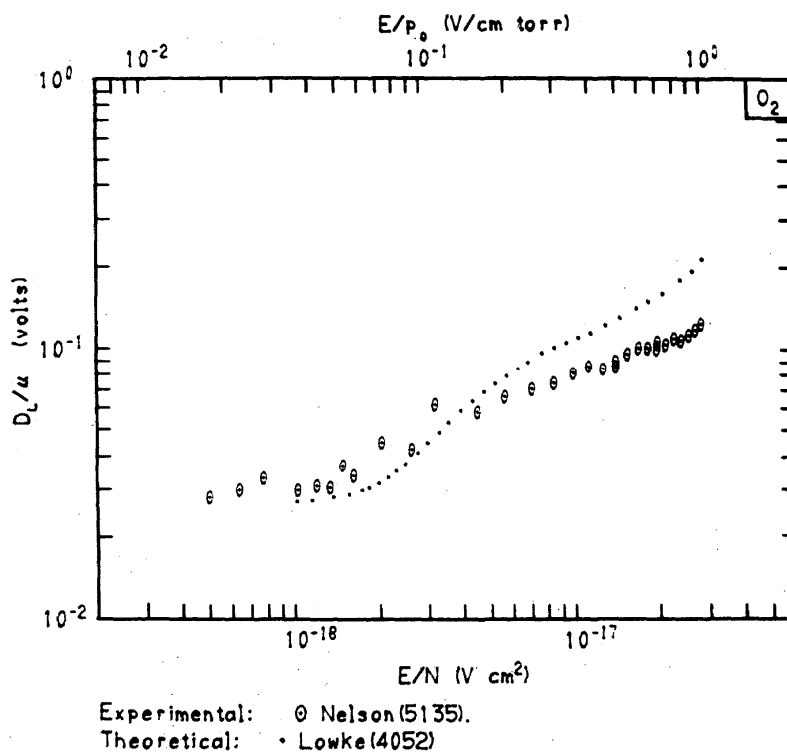


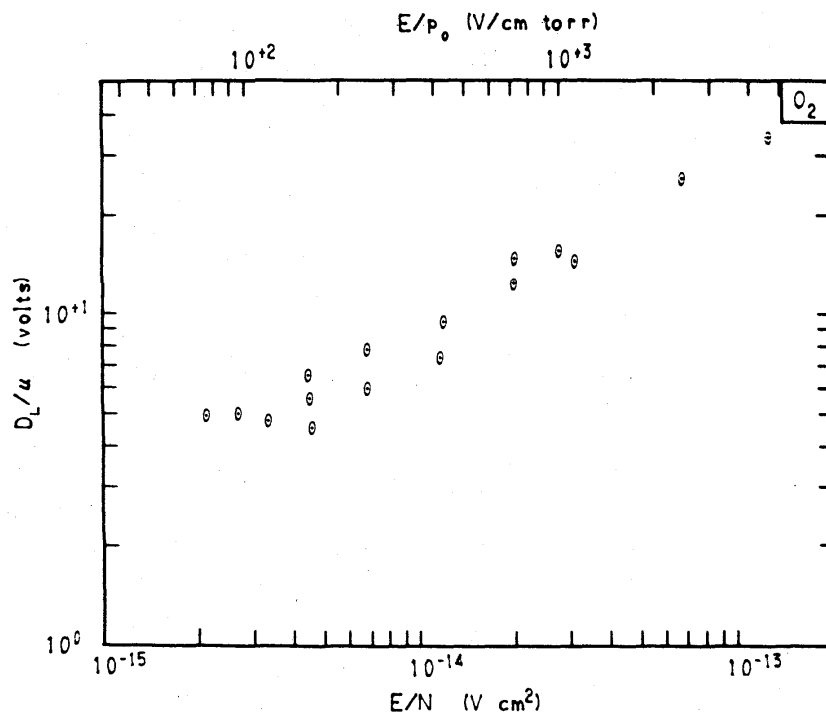
FIGURE 2.8. D_T/μ and D_L/μ as functions of E/N for hydrogen at room temperature for $E/N > 10^{-16}$ V cm^2 .

FIGURE 2.9. D_T/μ as a function of E/N for hydrogen at a temperature of 77 K.FIGURE 2.10. D_T/μ , E/N for nitrogen for $E/N < 14.1 \times 10^{-17}$ V cm² at ambient temperature.

FIGURE 2.11. D_T/μ , E/N for N_2 for $E/N > 14 \times 10^{-17} \text{ V cm}^2$ at ambient temperature.FIGURE 2.12. D_T/μ , E/N for nitrogen at a temperature of 77 K.

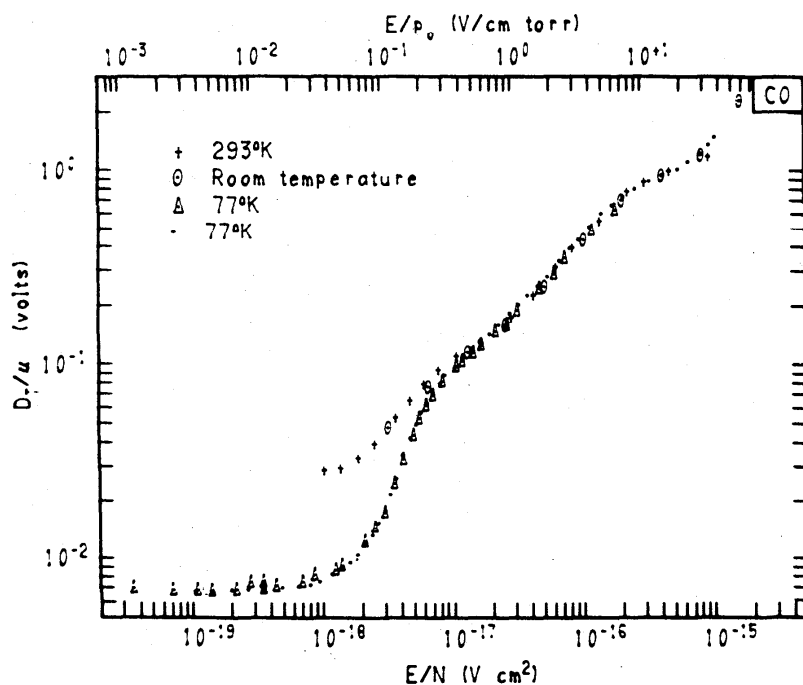
FIGURE 2.13. D_L/μ , E/N for nitrogen for $E/N < 14 \times 10^{-17}$ V cm².FIGURE 2.14. D_L/μ , E/N for nitrogen for $E/N > 14 \times 10^{-17}$ V cm².

FIGURE 2.15. D_T/μ , E/N for oxygen.FIGURE 2.16. D_L/μ , E/N for oxygen for $E/N < 3 \times 10^{-17}$ V cm².



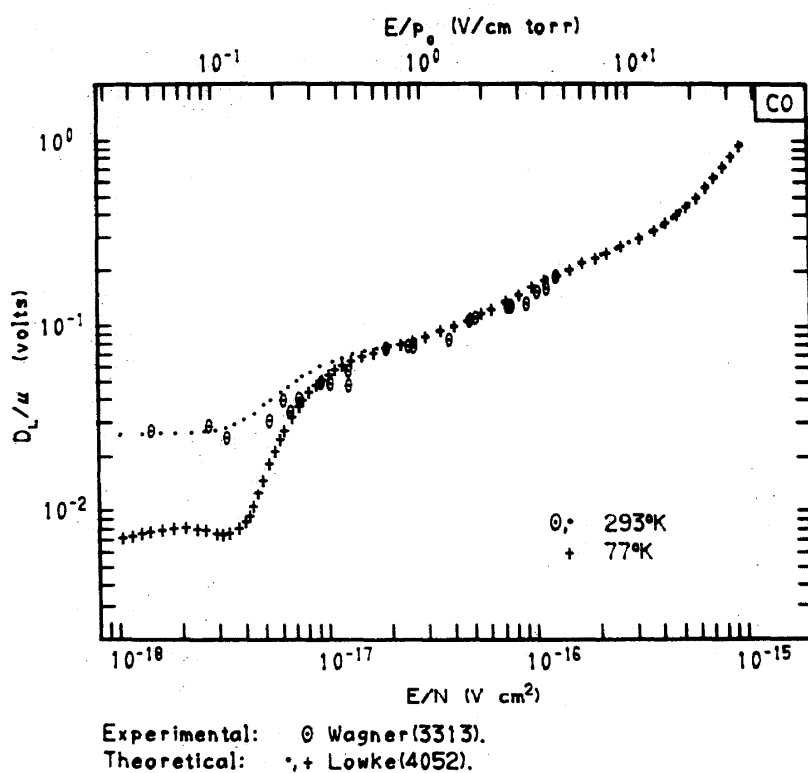
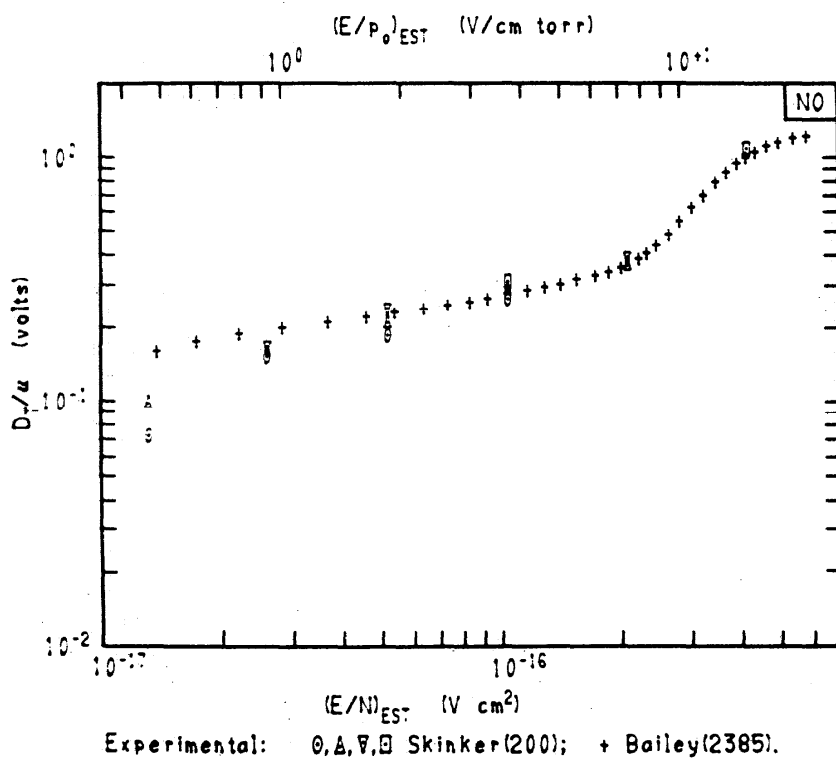
Experimental: O Schlumbohm(1625).

FIGURE 2.17. Experimental D_L/μ , E/N for oxygen for $E/N > 10^{-15}$ V cm².



Experimental: O Skinker(200); Δ Warren(1036).
Theoretical: + Hake(2553); - Lowke(4052).

FIGURE 2.18. D_T/μ , E/N for carbon monoxide at ambient temperatures and 77 K. (A temperature of 15 °C was assumed for the conversion of the values of E/p given in [200] to values of E/N .)

FIGURE 2.19. D_L/μ , E/N for carbon monoxide at 293 K and 77 K.FIGURE 2.20. Experimental D_T/μ , $(E/N)_{EST}$ for nitric oxide. (A temperature of 15 °C was assumed in both cases to convert the given values of E/p to values of E/N .)

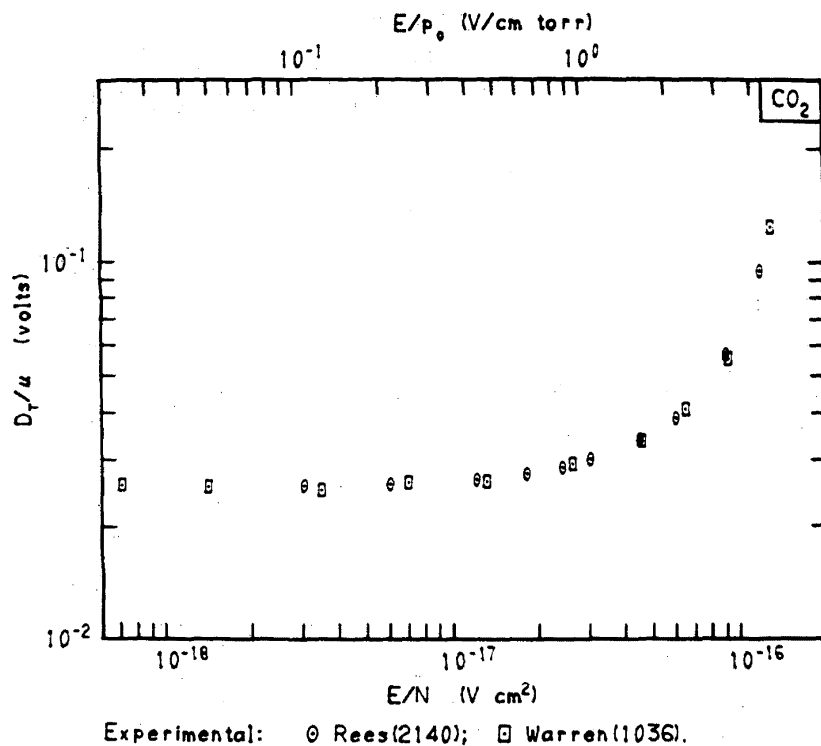


FIGURE 2.21. Experimental D_T/μ , E/N for carbon dioxide for $E/N < 14 \times 10^{-17} V\text{ cm}^2$.

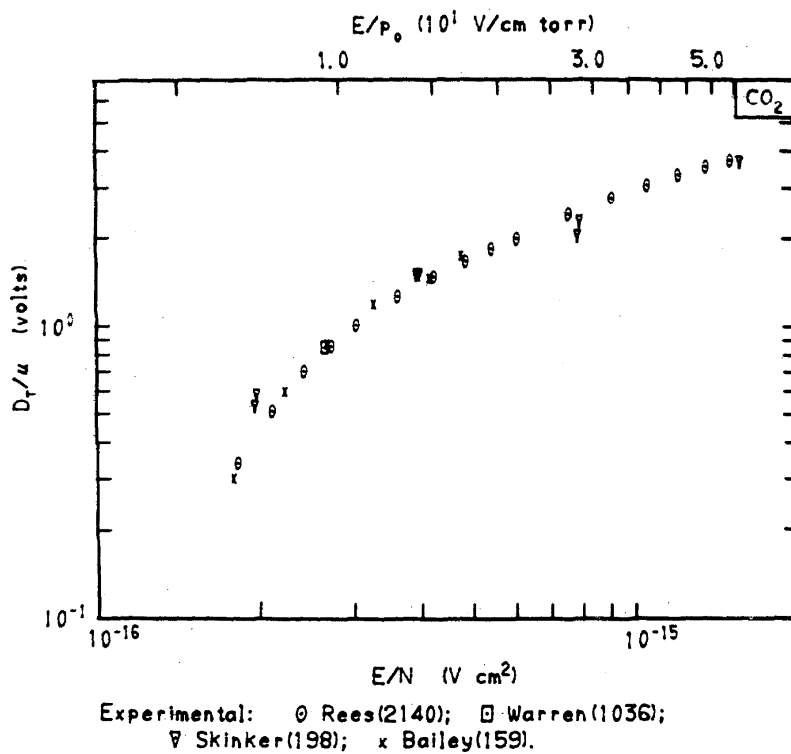
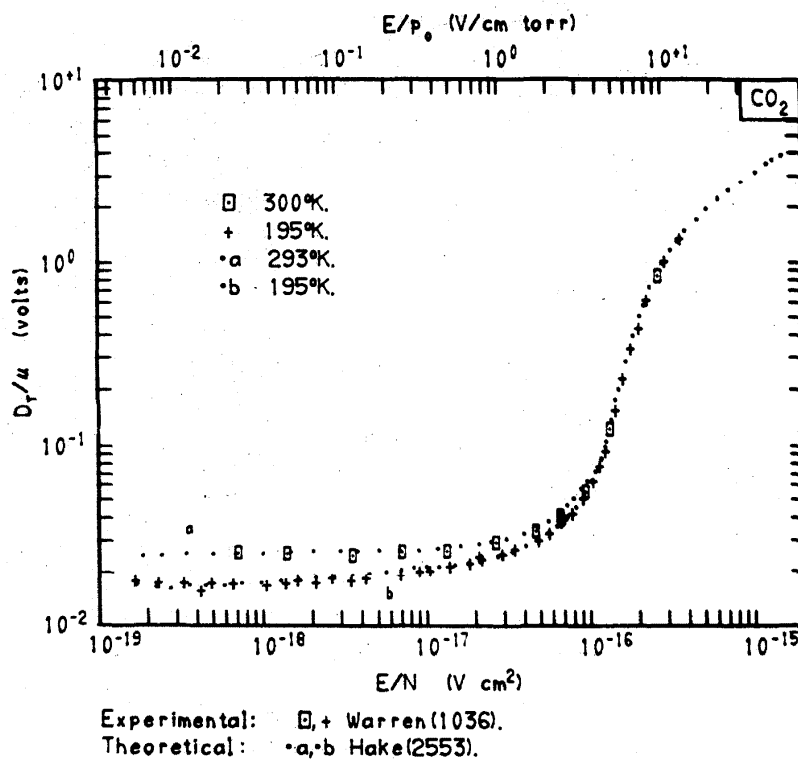
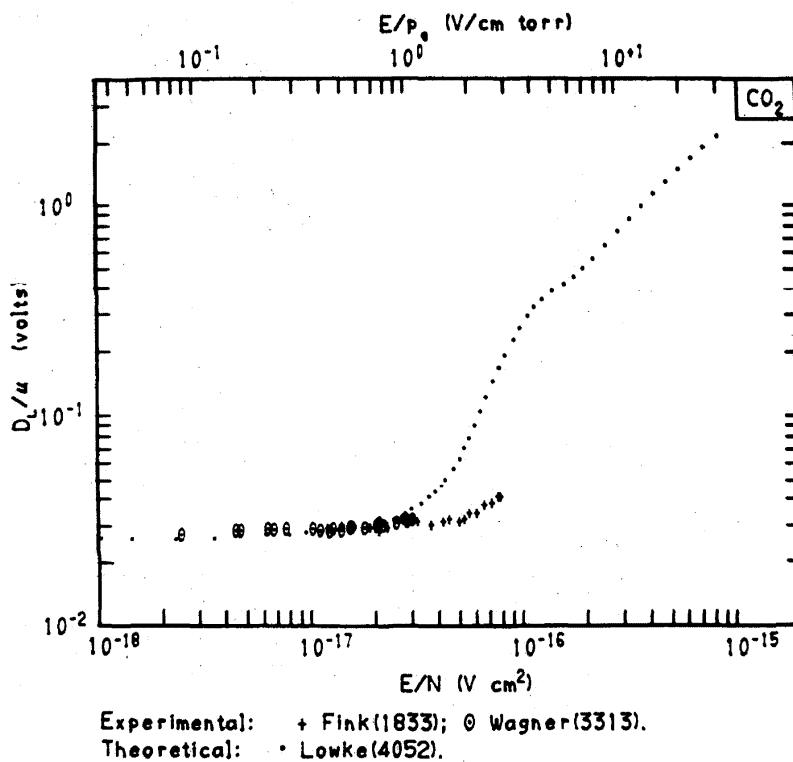
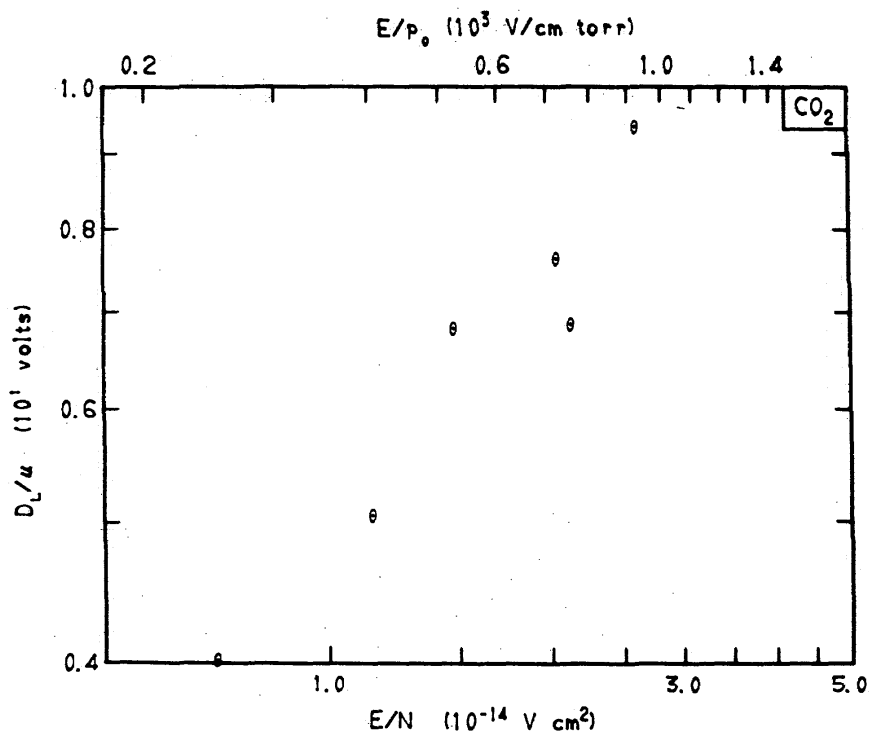


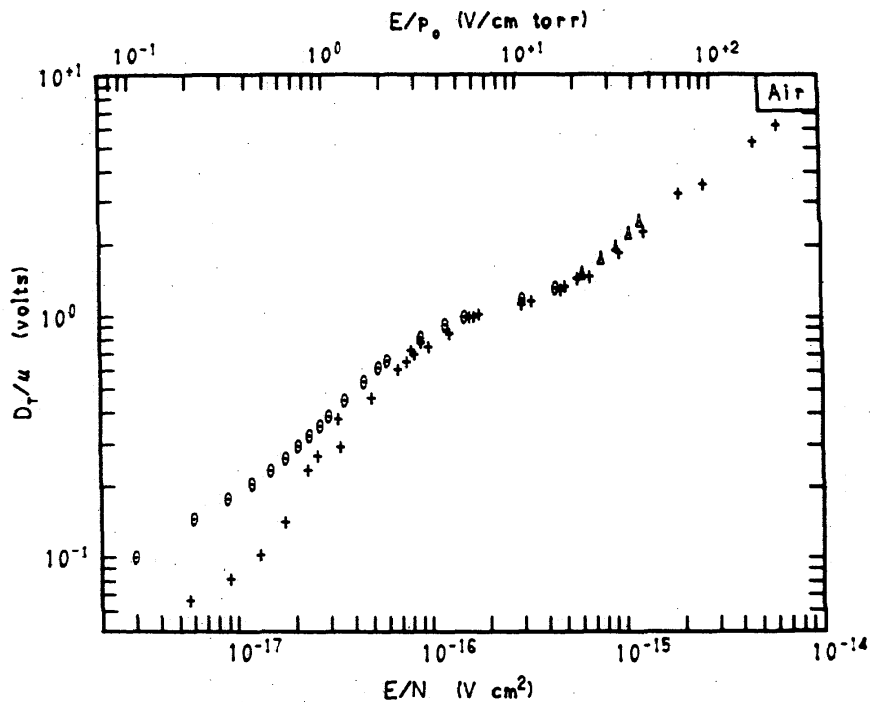
FIGURE 2.22. Experimental D_T/μ , E/N for carbon dioxide for $E/N > 14 \times 10^{-17} V\text{ cm}^2$.

FIGURE 2.23. D_T/μ , E/N for carbon dioxide at various temperatures.FIGURE 2.24. D_L/μ , E/N for carbon dioxide for $E/N < 10^{-15}$ V cm².



Experimental: θ Schlumbohm(1625).

FIGURE 2.25. Experimental D_T/μ , E/N for carbon dioxide for $E/N > 10^{-15}$ V cm².



Experimental: θ Crompton(2097); Δ Rees(2813);
+ Townsend(2104).

FIGURE 2.26. Experimental D_T/μ , E/N for dry air. (The data of [2097] and [2813] refer to dry air with CO₂ removed, whereas the data of [2104] refer to dry air containing CO₂. The temperature assumed for the conversion of E/p to E/N in [2104] is 15 °C.)

Table 2.1(a)

Values of D_T/μ for helium at ambient temperatures

E/N (10^{-17} vcm^2)	D_T/μ (10^{-1} v)	E/N (10^{-17} vcm^2)	D_T/μ (10^{-1} v)	E/N (10^{-17} vcm^2)	D_T/μ (10^{-1} v)
0.001133	0.203 (a)	0.02124	0.302 (b)	0.455	2.170 (b)
0.001804	0.234 (a)	0.02430	0.313 (b)	0.546	2.545 (b)
0.002834	0.256 (a)	0.02730	0.325 (b)	0.607	2.789 (b)
0.00360	0.255 (a)	0.0303	0.337 (b)	0.759	3.400 (b)
0.00515	0.254 (a)	0.0364	0.362 (b)	0.910	3.990 (b)
0.00543	0.254 (a)	0.0455	0.400 (b)	1.214	5.20 (b)
0.00720	0.260 (a)	0.0546	0.441 (b)	1.517	6.40 (b)
0.00776	0.260 (a)	0.0607	0.468 (b)	1.820	7.55 (b)
0.00823	0.262 (a)	0.0759	0.534 (b)	2.124	8.76 (b)
0.01133	0.271 (a)	0.0910	0.604 (b)	2.43	9.96 (b)
0.01177	0.266 (a)	0.1214	0.741 (b)	2.730	11.17 (b)
0.01242	0.275 (a)	0.1517	0.874 (b)	3.03	12.41 (b)
0.01645	0.279 (a)	0.1820	1.01 (b)	5.02	22.8 (c)
0.01707	0.284 (a)	0.2124	1.141 (b)	8.35	33. (c)
0.01773	0.280 (a)	0.243	1.271 (b)	10.07	36. (c)
0.01804	0.295 (a)	0.273	1.405 (b)	12.52	39. (c)
0.01878	0.295 (a)	0.303	1.536 (b)	15.80	44. (c)
		0.364	1.792 (b)		

Experimental:

- (a) Warren, *et al.*, Phys. Rev. **128**, 2661 (1962), data taken at 300 K.
 (b) Crompton, *et al.*, Austr. J. Phys. **20**, 369 (1967), data taken at 293 K.
 (c) Townsend, *et al.*, Phil. Mag. **46**, 657 (1923), data taken at 288 K.

Table 2.1(b)

Values of D_T/μ for helium at 77 K

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)
0.0002943	0.066	0.00882	0.094	0.0615	0.34
0.000463	0.065	0.00959	0.096	0.0751	0.41
0.000537	0.063	0.00990	0.098	0.0779	0.40
0.000736	0.067	0.01158	0.113	0.0820	0.45
0.000820	0.064	0.01161	0.093	0.0829	0.44
0.000844	0.063	0.01183	0.104	0.0844	0.48
0.001158	0.069	0.01236	0.115	0.0882	0.47
0.001236	0.066	0.01288	0.112	0.0959	0.49
0.001288	0.065	0.01344	0.116	0.0990	0.52
0.001344	0.065	0.01416	0.116	0.1161	0.58
0.001475	0.067	0.01475	0.129	0.1183	0.61
0.001940	0.066	0.01506	0.118	0.1236	0.635
0.001984	0.066	0.01524	0.120	0.1288	0.67
0.002052	0.066	0.01822	0.133	0.1344	0.73
0.002105	0.070	0.01984	0.144	0.1416	0.71
0.002114	0.069	0.02052	0.150	0.1506	0.77
0.002689	0.069	0.02114	0.156	0.1524	0.75
0.002943	0.071	0.02151	0.150	0.1822	0.88
0.00301	0.070	0.02397	0.160	0.1940	0.95
0.00309	0.068	0.02689	0.185	0.1984	0.96
0.00311	0.070	0.02903	0.187	0.2052	1.02
0.00323	0.071	0.0301	0.194	0.2151	1.05
0.00385	0.074	0.0305	0.190	0.2397	1.11
0.00410	0.074	0.0309	0.203	0.2903	1.35
0.00463	0.077	0.0311	0.199	0.301	1.42
0.00475	0.077	0.0323	0.206	0.305	1.38
0.00484	0.077	0.0385	0.248	0.309	1.44
0.00497	0.074	0.0410	0.250	0.311	1.45
0.00537	0.080	0.0435	0.254	0.456	2.03
0.00587	0.080	0.0456	0.270	0.472	2.12
0.00608	0.094	0.0475	0.279	0.497	2.19
0.00615	0.081	0.0484	0.286	0.581	2.5
0.00736	0.090	0.0497	0.286	0.609	2.6
0.00751	0.084	0.0537	0.32	0.668	2.9
0.00779	0.088	0.0581	0.33	0.829	3.4
0.00820	0.089	0.0587	0.33	0.916	3.8
0.00844	0.092	0.0608	0.33	1.161	5.2

Experimental:

All data taken from Warren, *et al.*, Phys. Rev. 128, 2661 (1962).

Table 2.2

Values of D_L/μ for helium at ambient temperatures

E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	
0.00914	0.253	(a)	0.1563	0.51	(a)	1.184	2.69	(b)
0.01826	0.265	(a)	0.2607	0.83	(a)	1.234	2.83	(b)
0.02442	0.291	(a)	0.2819	0.78	(b)	1.540	3.6	(b)
0.0302	0.293	(a)	0.300	0.89	(a)	1.816	4.2	(b)
0.0441	0.31	(a)	0.397	1.13	(b)	2.031	4.9	(b)
0.0446	0.29	(a)	0.526	1.61	(a)	2.322	5.5	(b)
0.0585	0.34	(a)	0.530	1.45	(b)	2.667	6.4	(b)
0.0736	0.29	(a)	0.728	1.76	(b)	3.00	7.6	(b)
0.1442	0.51	(a)	0.740	1.95	(a)			
0.1457	0.58	(a)	0.915	2.17	(b)			

Experimental:

- (a) Crompton, as given in Lowke, *et al.*, Phys. Rev. **181**, 302 (1969).
 (b) Wagner, *et al.*, J. Chem. Phys. **47**, 3138 (1967).

Table 2.3(a)

Values of D_T/μ for argon at 288 K

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	
0.313	23.6		1.252	42.		12.67	77.	
0.334	23.8		2.518	68.		17.58	76.	
0.501	27.5		3.73	80.		26.11	80.	
0.626	31.		6.32	84.		36.1	81.	
0.668	32.		8.43	79.		59.0	80.	

Experimental:

All data taken from Townsend, *et al.*, Phil. Mag. **44**, 1033 (1922).

Table 2.3(b)

Values of D_T/μ for argon at 77 K

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	
0.000466	0.066		0.001257	0.080		0.002524	0.182	
0.000553	0.0663		0.001630	0.085		0.002794	0.277	
0.000699	0.069		0.001760	0.093		0.00302	0.39	
0.000931	0.072		0.002012	0.106		0.0627	13.6	
0.001006	0.075		0.002328	0.144		0.0931	15.1	
						0.1257	16.4	

Experimental:

All data taken from Warren, *et al.*, Phys. Rev. **128**, 2661 (1962).

Table 2.3(c)

Values of D_T/μ for argon at 88 K

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)
0.00311	0.65	0.00584	2.91	0.02561	8.8
0.00348	1.03	0.00730	3.5	0.02924	8.8
0.00394	1.50	0.01024	4.7	0.0363	10.2
0.00438	1.91	0.01462	6.4	0.0410	10.0
		0.02189	8.0	0.0512	11.5

Experimental:
All data taken from Warren, *et al.*, Phys. Rev. **128**, 2661, (1962).

Table 2.4(a)

Values of D_L/μ for argon at ambient temperature

E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)
0.0445	1.91	0.607	5.6	1.473	7.4
0.0971	2.38	0.797	6.1	1.866	8.1
0.1659	3.6	0.963	6.4	2.080	8.6
0.2509	3.8	1.101	6.6	2.384	9.0
0.312	4.2	1.271	6.9	2.578	9.2
0.437	4.6	1.384	7.4	3.02	10.4

Experimental:
All data taken from Wagner, *et al.*, J. Chem. Phys. **47**, 3138, (1967).

Table 2.4(b)

Values of D_L/μ for argon at 77 K

E/N (10^{-17} Vcm^2)	D_L/μ (10^{-1} V)	E/N (10^{-17} Vcm^2)	D_L/μ (10^{-1} V)	E/N (10^{-17} Vcm^2)	D_L/μ (10^{-1} V)
0.000300	0.068 (2)	0.00364	2.49 (1)	0.00991	0.84 (2)
0.000303	0.067 (1)	0.00378	1.23 (2)	0.01214	0.84 (1)
0.000455	0.067 (1)	0.00394	2.58 (1)	0.01242	0.86 (2)
0.000552	0.070 (2)	0.00406	1.55 (2)	0.01721	0.91 (2)
0.000607	0.068 (1)	0.00418	1.81 (2)	0.0182	0.93 (1)
0.001043	0.073 (2)	0.00425	2.43 (1)	0.02349	1.02 (2)
0.001214	0.076 (1)	0.00436	1.86 (2)	0.0243	1.03 (1)
0.001551	0.082 (2)	0.00455	2.11 (1)	0.0303	1.12 (1)
0.00182	0.120 (1)	0.00461	1.98 (2)	0.0420	1.32 (2)
0.002001	0.103 (2)	0.00495	1.89 (2)	0.0455	1.33 (1)
0.002209	0.119 (2)	0.00539	1.71 (2)	0.0607	1.53 (1)
0.002371	0.148 (2)	0.00546	1.41 (1)	0.0876	1.80 (2)
0.00243	0.455 (1)	0.00570	1.51 (2)	0.1909	2.59 (2)
0.002545	0.192 (2)	0.00607	1.09 (1)	0.4790	3.9 (2)
0.002732	0.268 (2)	0.00612	1.30 (2)	1.073	5.7 (2)
0.002974	0.40 (2)	0.00676	1.11 (2)	1.864	7.2 (2)
0.00303	1.49 (1)	0.00715	0.97 (2)	3.15	9.4 (2)
0.00306	0.53 (2)	0.00801	0.91 (2)		
0.00324	0.68 (2)	0.00872	0.87 (2)		
0.00348	0.94 (2)	0.00910	0.82 (1)		

Theoretical:

- (1) Lowke, *et al.*, Phys. Rev. **181**, 302 (1969), using q_m given in Engelhardt, *et al.*, Phys. Rev. **133**, A375 (1964).
 (2) *Ibid.*, using values of q_m given by Golden, Phys. Rev. **151**, 48 (1966).

Table 2.5(a)

Values of D_T/μ for krypton

E/N (10^{-17} Vcm^2)	D_T/μ (V)	E/N (10^{-17} Vcm^2)	D_T/μ (V)	E/N (10^{-17} Vcm^2)	D_T/μ (V)
0.002083	0.0261	0.01715	0.229	0.0417	0.85
0.00400	0.0268	0.019	0.292	0.0659	1.26
0.00659	0.0291	0.02083	0.35	0.2083	2.78
0.00932	0.039	0.02553	0.49	0.659	5.2
0.01318	0.106	0.02946	0.59	2.083	8.3

Theoretical:

All data taken from Frost, *et al.*, Phys. Rev. **136**, A1538 (1964).

Table 2.5(b)

Values of D_L/μ for krypton

E/N (10^{-17}Vcm^2)	D_L/μ (V)	E/N (10^{-17}Vcm^2)	D_L/μ (V)	E/N (10^{-17}Vcm^2)	D_L/μ (V)
0.002083	0.0263	0.01715	0.515	0.0417	0.221
0.00400	0.0279	0.019	0.538	0.0659	0.214
0.00659	0.0337	0.02083	0.512	0.2083	0.293
0.00932	0.0673	0.02553	0.381	0.659	0.467
0.01318	0.281	0.02946	0.299	2.083	0.821

Theoretical:

All data taken from Lowke, *et al.*, Phys. Rev. **181**, 302 (1969).

Table 2.6(a)

Values of D_T/μ for xenon

E/N (10^{-17}Vcm^2)	D_T/μ (V)	E/N (10^{-17}Vcm^2)	D_T/μ (V)	E/N (10^{-17}Vcm^2)	D_T/μ (V)
0.002494	0.0259	0.0400	0.210	0.1576	1.21
0.0100	0.0268	0.0499	0.35	0.2494	1.65
0.01576	0.0289	0.066	0.55	0.788	3.2
0.02494	0.052	0.0788	0.67	2.494	5.4
0.0353	0.146	0.1114	0.93		

Theoretical:

All data taken from Frost, *et al.*, Phys. Rev. **136**, A1538 (1964).

Table 2.6(b)

Values of D_L/μ for xenon

E/N (10^{-17}Vcm^2)	D_L/μ (V)	E/N (10^{-17}Vcm^2)	D_L/μ (V)	E/N (10^{-17}Vcm^2)	D_L/μ (V)
0.002494	0.0260	0.0400	0.44	0.1576	0.208
0.0100	0.0281	0.0499	0.47	0.2494	0.233
0.01576	0.034	0.066	0.32	0.788	0.33
0.02494	0.108	0.0788	0.243	2.494	0.61
0.0353	0.34	0.1114	0.194		

Theoretical:

All data taken from Lowke, *et al.*, Phys. Rev. **181**, 302 (1969).

Table 2.7

Values of D_T/μ for hydrogen at ambient temperatures

E/N (10^{-17}Vcm^2)	D_T/μ (V)	E/N (10^{-17}Vcm^2)	D_T/μ (V)	E/N (10^{-17}Vcm^2)	D_T/μ (V)
0.01821	0.0258 (a)	3.64	0.267 (a)	134.7	3.2 (c)
0.02428	0.0262 (a)	4.55	0.310 (a)	138.5	3.2 (b)
0.0304	0.0265 (a)	5.46	0.350 (a)	142.6	3.5 (d)
0.0455	0.0275 (a)	6.07	0.374 (a)	153.6	3.3 (b)
0.0607	0.0286 (a)	13.24	0.60 (d)	157.2	3.7 (d)
0.0759	0.0297 (a)	18.24	0.84 (c)	162.8	3.5 (c)
0.0911	0.0308 (a)	26.66	1.16 (c)	169.5	3.5 (b)
0.1214	0.0330 (a)	28.20	1.02 (d)	171.5	3.8 (d)
0.1518	0.0353 (a)	31.0	1.08 (b)	184.5	3.6 (b)
0.1821	0.0375 (a)	39.3	1.54 (c)	188.0	3.7 (c)
0.2125	0.0396 (a)	41.4	1.47 (d)	200.0	3.8 (b)
0.2428	0.0418 (a)	45.6	1.56 (b)	215.9	4.0 (b)
0.2732	0.0439 (a)	53.3	1.92 (c)	216.1	4.0 (c)
0.304	0.0459 (a)	56.4	1.95 (d)	254.0	4.2 (c)
0.364	0.0500 (a)	61.9	2.00 (b)	294.6	4.5 (c)
0.455	0.0562 (a)	68.8	2.27 (c)	340	4.7 (c)
0.546	0.0625 (a)	70.3	2.40 (d)	379	4.9 (c)
0.607	0.0668 (a)	77.4	2.32 (b)	425	5.1 (c)
0.759	0.0779 (a)	84.9	2.65 (d)	469	5.3 (c)
0.911	0.0888 (a)	87.0	2.62 (c)	521	5.5 (c)
1.214	0.1112 (a)	92.5	2.55 (b)	580	5.7 (c)
1.518	0.1334 (a)	98.8	2.84 (d)	651	6.0 (c)
1.821	0.1551 (a)	107.9	2.78 (b)	721	6.2 (c)
2.125	0.1761 (a)	110.8	2.98 (c)	798	6.5 (c)
2.428	0.1962 (a)	113.4	3.2 (d)	884	6.7 (c)
2.732	0.215 (a)	123.4	2.97 (b)	995	7.1 (c)
3.04	0.233 (a)	127.3	3.2 (d)		

Experimental:

- (a) Crompton, *et al.*, Austr. J. Phys. **21**, 43 (1968).
 (b) Crompton, *et al.*, in Proc. of the Seventh International Conf. Phen. Ion. Gases (Belgrade, 22-27 August 1965) B. Perovic and D. Tosic, eds., Gradevinska Knjiga Publishing House, Belgrade **1**, 86 (1966).
 (c) Lawson, *et al.*, Brit. J. Appl. Phys. **16**, 1813 (1965).
 (d) Virr, *et al.*, 1st Int. Conf. Gas Discharges, London 530 (1970).

Table 2.8

Values of D_L/μ for hydrogen at ambient temperature

E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)
0.1394	0.275	0.737	0.41	2.143	0.83
0.1879	0.286	0.858	0.46	2.455	0.93
0.2698	0.30	0.982	0.47	2.561	0.99
0.336	0.33	1.112	0.53	3.05	1.12
0.400	0.34	1.155	0.56	3.64	1.28
0.524	0.37	1.358	0.63	4.56	1.60
0.582	0.40	1.464	0.65	5.02	1.67
0.646	0.39	1.955	0.77	5.86	1.87

Experimental:

All data taken from Wagner, *et al.*, J. Chem. Phys. **47**, 3138 (1967).

Table 2.9

Values of D_T/μ for hydrogen at 77 K

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	
0.000484	0.066	(a)	0.0300	0.118	(b)	1.600	1.31	(b)
0.000491	0.066	(a)	0.0350	0.127	(b)	1.800	1.45	(b)
0.000761	0.066	(a)	0.0400	0.135	(b)	2.000	1.60	(b)
0.000770	0.064	(a)	0.0450	0.143	(b)	2.50	1.93	(b)
0.000969	0.068	(a)	0.0500	0.151	(b)	3.00	2.24	(b)
0.001232	0.069	(a)	0.0600	0.165	(b)	3.50	2.52	(b)
0.001326	0.061	(a)	0.0700	0.178	(b)	4.00	2.78	(b)
0.001385	0.070	(a)	0.0800	0.189	(b)	4.50	3.02	(b)
0.001919	0.067	(a)	0.0900	0.200	(b)	5.00	3.25	(b)
0.001937	0.068	(a)	0.1000	0.210	(b)	6.00	3.66	(b)
0.002000	0.0676	(b)	0.1200	0.229	(b)	7.00	4.05	(b)
0.002500	0.0681	(b)	0.1400	0.247	(b)	8.00	4.40	(b)
0.00300	0.0685	(b)	0.1600	0.263	(b)	9.00	4.74	(b)
0.00350	0.0690	(b)	0.1800	0.278	(b)	10.00	5.06	(b)
0.00400	0.0696	(b)	0.2000	0.294	(b)	12.00	5.65	(b)
0.00450	0.0702	(b)	0.2500	0.329	(b)	14	6.11	(1)
0.00500	0.0709	(b)	0.300	0.364	(b)	17	6.94	(1)
0.00600	0.0723	(b)	0.350	0.398	(b)	20	7.24	(1)
0.00700	0.0738	(b)	0.400	0.433	(b)	30	10.71	(1)
0.00800	0.0755	(b)	0.450	0.467	(b)	40	14.73	(1)
0.00900	0.0772	(b)	0.500	0.502	(b)	50	18.68	(1)
0.01000	0.0790	(b)	0.600	0.572	(b)	60	22.9	(1)
0.01200	0.0827	(b)	0.700	0.642	(b)	70	26.3	(1)
0.01400	0.0867	(b)	0.800	0.713	(b)	80	29.6	(1)
0.01600	0.0907	(b)	0.900	0.786	(b)	90	32.4	(1)
0.01800	0.0946	(b)	1.000	0.860	(b)	100	35.0	(1)
0.02000	0.0986	(b)	1.200	1.01	(b)	125	39.9	(1)
0.02500	0.108	(b)	1.400	1.16	(b)	150	44.5	(1)

Experimental:

(a) Warren, *et al.*, Phys. Rev. **128**, 2661 (1962).(b) Crompton, *et al.*, Austr. J. Phys. **21**, 43 (1968).

Theoretical:

(1) Engelhardt, *et al.*, Phys. Rev. **131**, 2115 (1963).

Table 2.10

Values of D_T/μ for nitrogen at ambient temperatures

E/N (10^{-17} Vcm^2)	D_T/μ (V)		E/N (10^{-17} Vcm^2)	D_T/μ (V)		E/N (10^{-17} Vcm^2)	D_T/μ (V)	
0.01833	0.0262	(a)	0.620	0.180	(a)	9.17	0.88	(a)
0.02451	0.0270	(a)	0.758	0.219	(b)	12.12	0.98	(b)
0.0310	0.0279	(a)	0.909	0.256	(b)	12.26	0.94	(a)
0.0469	0.0305	(a)	0.937	0.255	(a)	14.90	1.07	(c)
0.0620	0.033	(a)	1.212	0.33	(b)	15.16	1.04	(b)
0.0775	0.037	(a)	1.516	0.39	(b)	18.19	1.09	(b)
0.0927	0.040	(a)	1.567	0.38	(a)	20.13	1.11	(d)
0.1212	0.048	(b)	1.819	0.45	(b)	21.22	1.12	(b)
0.1240	0.048	(a)	1.874	0.44	(a)		1.13	(e)
0.1516	0.057	(b)	2.122	0.50	(b)	24.25	1.15	(b)
0.1567	0.057	(a)	2.192	0.48	(a)	29.80	1.25	(c)
0.1819	0.066	(b)	2.425	0.54	(b)	30.3	1.25	(e)
0.1874	0.066	(a)	2.479	0.51	(a)	40.4	1.29	(d)
0.2122	0.075	(b)	2.728	0.57	(b)	44.7	1.36	(c)
0.2192	0.075	(a)	2.77	0.55	(a)	45.5	1.36	(e)
0.2425	0.085	(b)	3.03	0.61	(b)	59.6	1.48	(c)
0.2479	0.083	(a)	3.13	0.58	(a)	60.6	1.46	(e)
0.2728	0.094	(b)	3.64	0.66	(b)	74.5	1.63	(c)
0.2803	0.093	(a)	4.53	0.69	(a)	75.8	1.59	(e)
0.303	0.103	(b)	4.55	0.72	(b)	80.5	1.68	(d)
0.364	0.120	(b)	5.46	0.77	(b)	90.9	1.76	(e)
0.455	0.145	(b)	6.06	0.80	(b)	106.1	1.97	(e)
0.464	0.142	(a)	6.13	0.77	(a)	121.2	2.19	(e)
0.546	0.168	(b)	7.58	0.85	(b)	163.8	2.95	(d)
0.606	0.182	(b)	9.09	0.90	(b)			

Experimental:

- (a) Crompton, *et al.*, in Proceedings of the Sixth International Conference on Ionization Phenomena in Gases, Paris 8-13 July 1963, (P. Hubert and E. Cremieu-Alcan, Eds., Serma, Paris, 1963), Vol. 1, p. 337.
- (b) Jory, *Austr. J. Phys.* **18**, 237 (1965).
- (c) Crompton, *et al.*, *Proc. Roy. Soc. London, Ser A* **215**, 467 (1952).
- (d) Townsend, *et al.*, *Phil. Mag.* **42**, 874 (1921).
- (e) Naidu, *et al.*, *Brit. J. Appl. Phys. (J. Phys. D 2)* **1**, 763 (1968).

Table 2.11
Values of D_T/μ for nitrogen at 77 K

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	
0.000699	0.067	(a)	0.0320	0.104	(a)	1.009	2.87	(a)
0.000813	0.065	(a)	0.0385	0.115	(a)	1.267	3.3	(a)
0.000832	0.066	(a)	0.0506	0.136	(a)	1.279	3.4	(a)
0.001397	0.067	(a)	0.0574	0.155	(a)	1.599	4.1	(a)
0.002033	0.068	(a)	0.0633	0.165	(a)	2.018	5.0	(a)
0.002080	0.068	(a)	0.0640	0.168	(a)	2.238	5.2	(a)
0.002555	0.065	(a)	0.0770	0.202	(a)	2.530	5.4	(a)
0.002794	0.070	(a)	0.0894	0.230	(a)	5.06	7.6	(a)
0.00376	0.069	(a)	0.1009	0.30	(a)	8.41	8.8	(a)
0.00407	0.068	(a)	0.1267	0.34	(a)	10	9.9	(1)
0.00640	0.068	(a)	0.1276	0.34	(a)	20	1.16	(1)
0.00699	0.074	(a)	0.1605	0.45	(a)	30	1.26	(1)
0.00751	0.072	(a)	0.1925	0.57	(a)	40	1.34	(1)
0.00770	0.068	(a)	0.2018	0.57	(a)	50	1.42	(1)
0.00813	0.073	(a)	0.2530	0.79	(a)	60	1.51	(1)
0.01087	0.076	(a)	0.2555	0.78	(a)	70	1.60	(1)
0.01276	0.074	(a)	0.320	0.99	(a)	110	2.10	(1)
0.01599	0.078	(a)	0.385	1.21	(a)	140	2.53	(1)
0.01925	0.083	(a)	0.506	1.50	(a)	180	3.1	(1)
0.02033	0.095	(a)	0.633	1.89	(a)	240	3.9	(1)
0.02530	0.091	(a)	0.770	2.22	(a)	300	4.7	(1)
0.02555	0.093	(a)	0.863	2.43	(a)	400	5.9	(1)

Experimental:

(a) Warren, *et al.*, Phys. Rev. **128**, 2661 (1962).

Theoretical:

(1) Engelhardt, *et al.*, Phys. Rev. **135**, A1566 (1964).

Table 2.12
Values of D_L/μ for nitrogen at ambient temperatures

E/N (10^{-17}Vcm^2)	D_L/μ (V)		E/N (10^{-17}Vcm^2)	D_L/μ (V)		E/N (10^{-17}Vcm^2)	D_L/μ (V)	
0.0104	.026	(1)	1.164	0.146	(a)	200	2.47	(1)
0.0136	.026	(1)	1.476	0.178	(a)	334	4.5	(b)
0.0171	.026	(1)	1.664	0.207	(a)	469	5.1	(b)
0.0240	.026	(1)	2.209	0.266	(a)	491	7.2	(b)
0.0304	.025	(1)	2.761	0.31	(a)	616	6.3	(b)
0.0402	.025	(1)	3.21	0.31	(a)	729	5.3	(b)
0.0526	.026	(1)	3.85	0.37	(a)	977	5.7	(b)
0.0671	.026	(1)	4.56	0.39	(a)	1064	7.9	(b)
0.0828	.027	(1)	5.37	0.43	(a)	1310	7.0	(b)
0.105	.028	(1)	6.05	0.44	(a)	1592	10.1	(b)
0.128	.029	(1)	10.0	0.42	(1)	1898	9.6	(b)
0.1455	.027	(a)	20.0	0.45	(1)	2749	11.8	(b)
0.2606	.038	(a)	40.0	0.50	(1)	3660	19.1	(b)
0.561	.075	(a)	60.0	0.63	(1)	5440	29.6	(b)
0.867	.115	(a)	100	1.22	(1)	8470	37	(b)

Experimental:

(a) Wagner, *et al.*, J. Chem. Phys. **47**, 3138 (1967).

(b) Schlumbohm, Z. Physik **184**, 492 (1965).

Theoretical:

(1) Lowke, *et al.*, Phys. Rev. **181**, 302 (1969).

Table 2.13

Values of D_T/μ for oxygen

E/N (10^{-17}Vcm^2)	D_T/μ (V)		E/N (10^{-17}Vcm^2)	D_T/μ (V)		E/N (10^{-17}Vcm^2)	D_T/μ (V)	
0.01029	0.0241	(1)	0.457	0.108	(1)	9.0	1.01	(a)
0.01271	0.0241	(1)	0.512	0.115	(1)	9.07	0.99	(a)
0.01403	0.0241	(1)	0.565	0.121	(1)	9.09	0.98	(b)
0.01662	0.0244	(1)	0.615	0.126	(1)	10.0	1.08	(a)
0.01886	0.0244	(1)	0.669	0.131	(1)	12.0	1.27	(a)
0.02234	0.0248	(1)	0.749	0.137	(1)	12.12	1.31	(b)
0.02537	0.0249	(1)	0.850	0.145	(1)	12.22	1.29	(a)
0.02963	0.0251	(1)	0.993	0.159	(1)	14.0	1.47	(a)
0.0341	0.0257	(1)	1.0	0.14	(a)	14.90	1.57	(c)
0.0398	0.0260	(1)	1.189	0.185	(a)	15.0	1.55	(a)
0.0465	0.0266	(1)	1.212	0.188	(b)	15.16	1.60	(b)
0.0551	0.0276	(1)	1.489	0.205	(a)	15.20	1.57	(a)
0.0653	0.0288	(1)	1.516	0.207	(b)	15.40	1.57	(d)
0.0741	0.0296	(1)	1.785	0.225	(a)	16.0	1.73	(a)
0.0865	0.031	(1)	1.819	0.225	(b)	17.88	1.81	(c)
0.0969	0.033	(1)	2.0	0.21	(a)	18.19	1.84	(b)
0.1116	0.035	(1)	2.382	0.256	(a)	18.32	1.84	(a)
0.1284	0.039	(1)	2.425	0.258	(b)	22.35	2.05	(c)
0.1418	0.042	(1)	2.980	0.293	(a)	29.80	2.32	(c)
0.1587	0.046	(1)	3.0	0.31	(a)	30.0	2.23	(d)
0.1777	0.050	(1)	3.03	0.293	(b)	37.3	2.51	(c)
0.1961	0.054	(1)	4.0	0.40	(a)	44.7	2.68	(c)
0.2134	0.057	(1)	4.54	0.43	(a)	45.0	2.58	(d)
0.2323	0.063	(1)	4.55	0.43	(b)	52.2	2.80	(c)
0.2528	0.068	(1)	5.0	0.53	(a)	59.6	2.93	(c)
0.2751	0.073	(1)	5.99	0.60	(a)	60.1	2.87	(d)
0.308	0.080	(1)	6.0	0.65	(a)	75.2	3.1	(d)
0.335	0.087	(1)	6.06	0.60	(b)	90.2	3.3	(d)
0.365	0.092	(1)	7.0	0.76	(a)	105.9	3.5	(d)
0.403	0.099	(1)	8.0	0.89	(a)	121.2	3.6	(d)
						136.9	3.8	(d)
						150.7	4.0	(d)

Experimental:

- (a) Fleming, *et al.*, J. Phys. D 5, 291 (1972).
 (b) Rees, Austr. J. Phys. 18, 41 (1965).
 (c) Huxley, *et al.*, Austr. J. Phys. 12, 303 (1959).
 (d) Naidu, *et al.*, J. Phys. D 3, 957 (1970).

Theoretical:

- (1) Hake, *et al.*, Phys. Rev. 158, 70 (1967).

Table 2.14

Values of D_L/μ for oxygen

E/N (10^{-17}Vcm^2)	D_L/μ (V)		E/N (10^{-17}Vcm^2)	D_L/μ (V)		E/N (10^{-17}Vcm^2)	D_L/μ (V)	
0.0543	0.0279	(a)	1.232	0.086	(a)	20.0	1.36	(1)
0.0697	0.0296	(a)	1.386	0.084	(a)	40.0	1.80	(1)
0.0851	0.033	(a)		0.090	(a)	80.0	2.22	(1)
0.1128	0.0297	(a)	1.540	0.085	(a)	150	2.69	(1)
0.1313	0.031	(a)	1.694	0.096	(a)	211	4.9	(b)
0.1467	0.030	(a)	1.848	0.101	(a)	267	5.0	(b)
0.1621	0.037	(a)	2.002	0.101	(a)	333	4.8	(b)
0.1775	0.034	(a)	2.156	0.106	(a)	446	6.5	(b)
0.2237	0.045	(a)		0.100	(a)	452	5.5	(b)
0.2853	0.042	(a)	2.310	0.104	(a)	457	4.5	(b)
0.347	0.062	(a)	2.464	0.109	(a)		7.8	(b)
0.493	0.058	(a)	2.618	0.108	(a)	685	5.9	(b)
0.616	0.066	(a)	2.772	0.115	(a)	1165	7.4	(b)
	0.065	(a)		0.113	(a)	1192	9.4	(b)
0.770	0.071	(a)	2.926	0.118	(a)	1994	12.4	(b)
	0.069	(a)						
	0.074	(a)	3.080	0.123	(a)	2017	14.8	(b)
0.924	0.075	(a)	3.5	0.193	(1)	2804	15.6	(b)
	0.073	(a)	5.00	0.285	(1)	3130	14.5	(b)
	0.079	(a)	7.00	0.47	(1)	6870	25.6	(b)
1.078	0.081	(a)	10.0	0.76	(1)	12890	34	(b)

Experimental:

(a) Nelson, *et al.*, J. Chem. Phys. 57, 4079 (1972).(b) Schlumbohm, Z. Physik 184, 492 (1965).

Theoretical:

(1) Lowke, *et al.*, Phys. Rev. 181, 302 (1969).

Table 2.15(a)

Values of D_T/μ for carbon monoxide at ambient temperatures

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	
0.100	0.273	(1)	0.638	0.76	(a)	10.19	4.3	(a)
0.200	0.340	(1)	1.276	1.15	(a)	20.41	7.0	(a)
0.300	0.448	(1)	2.551	1.59	(a)	40.8	9.5	(a)
0.319	0.47	(a)	5.10	2.50	(a)	81.6	12.0	(a)
						163.3	22.8	(a)

Experimental:

(a) Skinker, *et al.*, Phil. Mag. 46, 630 (1923).

Theoretical:

(1) Lowke, *et al.*, Phys. Rev. 181, 302 (1969).

Table 2.15(b)

Values of D_T/μ for carbon monoxide at 77 K

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	
0.00354	0.070	(a)	0.2080	0.124	(a)	1.416	1.17	(a)
0.00708	0.069	(a)	0.2477	0.149	(a)	1.664	1.28	(a)
0.01099	0.070	(a)	0.2934	0.176	(a)	2.123	1.50	(a)
0.01416	0.069	(a)	0.354	0.250	(a)	2.642	1.61	(a)
0.02198	0.071	(a)	0.416	0.33	(a)	3.14	1.91	(a)
0.02831	0.076	(a)	0.497	0.44	(a)	4.69	2.48	(a)
0.0354	0.074	(a)	0.556	0.54	(a)	6.15	2.96	(a)
0.0441	0.073	(a)	0.624	0.62	(a)	7.30	3.5	(a)
0.0708	0.077	(a)	0.708	0.70	(a)	11.74	4.9	(a)
0.0879	0.083	(a)	0.832	0.83	(a)	17.60	6.4	(a)
0.1248	0.087	(a)	1.062	0.99	(a)	30	8.1	(1)
0.1416	0.093	(a)	1.174	1.05	(a)	70	10.8	(1)
						100	13.9	(1)

Experimental:

(a) Warren, *et al.*, Phys. Rev. **128**, 2661 (1962).

Theoretical:

(1) Hake, *et al.*, Phys. Rev. **158**, 70 (1967).

Table 2.16(a)

Values of D_L/μ for carbon monoxide at ambient temperature

E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)		E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	
0.0407	0.251		0.931	0.49		4.79	1.06	
0.1413	0.270		1.031	0.49		5.13	1.10	
0.2706	0.289		1.261	0.57		7.33	1.27	
0.329	0.252		1.262	0.48		7.66	1.28	
0.529	0.31		1.907	0.75		9.05	1.31	
0.615	0.40		2.439	0.78		10.19	1.52	
0.659	0.35		2.568	0.77		11.28	1.59	
0.730	0.41		3.82	0.84		12.54	1.85	

Experimental:

All data taken from Wagner, *et al.*, J. Chem. Phys. **47**, 3138 (1967).

Table 2.16(b)

Values of D_L/μ for carbon monoxide at 77 K

E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_L/μ (10^{-1}V)
0.100	0.0712	0.400	0.0913	2.5	0.848
0.200	0.0796	0.55	0.215	6.00	1.22
0.25	0.0787	0.700	0.357	10.0	1.73
0.31	0.0741	1.00	0.543	30.0	2.92
0.35	0.0764	1.7	0.732	70.0	6.22
				100	10.2

Theoretical:

All data taken from Lowke, *et al.*, Phys. Rev. **181**, 302 (1969).

Table 2.17(a)

Values of D_T/μ for carbon dioxide at ambient temperatures

E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)	E/N (10^{-17}Vcm^2)	D_T/μ (10^{-1}V)
0.0702	0.255 (a)	6.58	0.41 (a)	22.51	6.2 (a)
0.1403	0.254 (a)	9.09	0.57 (b)	24.25	7.0 (b)
0.303	0.255 (b)	9.25	0.55 (a)	26.39	8.4 (a)
0.351	0.249 (a)	12.12	0.95 (b)	26.82	8.7 (c)
0.606	0.258 (b)	13.19	1.23 (a)	27.28	8.5 (b)
0.702	0.261 (a)	14.37	1.53 (a)	28.87	10.0 (a)
1.212	0.265 (b)	16.21	2.3 (a)	30.3	10.1 (b)
1.319	0.263 (a)	17.88	2.98 (c)	32.8	11.9 (c)
1.819	0.275 (b)	18.01	3.3 (a)	36.0	13.2 (a)
2.425	0.286 (b)	18.19	3.4 (b)	36.4	12.7 (b)
2.639	0.293 (a)	19.64	5.2 (d)	39.3	14.9 (d)
3.03	0.301 (b)	19.79	5.7 (d)	39.9	15.0 (d)
4.55	0.34 (b)	20.24	4.3 (a)	41.7	14.6 (c)
4.63	0.34 (a)	21.22	5.1 (b)	42.4	14.8 (b)
6.06	0.39 (b)	22.35	6.0 (c)	47.7	17.4 (c)
				48.5	16.7 (b)
				54.6	18.4 (b)

Experimental:

- (a) Warren, *et al.*, Phys. Rev. **128**, 2661 (1962).
- (b) Rees, Austr. J. Phys. **17**, 462 (1962).
- (c) Bailey, *et al.*, Phil. Mag. **14**, 1033 (1932).
- (d) Skinner, Phil. Mag. **44**, 994 (1922).

Table 2.17(b)

Values of D_T/μ for carbon dioxide at 195 K

E/N (10^{-17} V cm^2)	D_T/μ (10^{-2} V)	E/N (10^{-17} V cm^2)	D_T/μ (10^{-2} V)	E/N (10^{-17} V cm^2)	D_T/μ (10^{-2} V)
0.01639	1.74	0.2620	1.82	2.887	2.50
0.02297	1.70	0.342	1.76	3.45	2.65
0.0329	1.70	0.419	1.82	4.84	2.97
0.0419	1.52	0.683	1.90	5.62	3.3
0.0491	1.70	0.888	2.00	6.64	3.7
0.0655	1.68	1.043	2.02	6.89	3.9
0.1043	1.64	1.366	2.13	7.79	4.1
0.1366	1.69	1.801	2.21	9.00	5.0
0.1636	1.75	2.089	2.40	10.34	6.3
0.2089	1.70	2.113	2.33	11.33	7.7
				12.32	9.2
				13.78	13.8

Experimental:

All data taken from Warren *et al.*, Phys. Rev. **128**, 2661 (1962).

Table 2.18

Values of D_L/μ for carbon dioxide

E/N (10^{-17} V cm^2)	D_L/μ (V)	E/N (10^{-17} V cm^2)	D_L/μ (V)	E/N (10^{-17} V cm^2)	D_L/μ (V)
0.2476	0.0271 (a)	1.397	0.0285 (a)	3.03	0.032 (a)
0.443	0.0284 (a)	1.534	0.0296 (a)	12.14	0.139 (1)
0.469	0.0284 (a)	1.560	0.0296 (a)	15	0.242 (1)
0.632	0.0291 (a)	1.789	0.0293 (a)	30.3	0.431 (1)
0.673	0.0287 (a)	2.044	0.030 (a)	60.7	0.903 (1)
0.769	0.0291 (a)	2.095	0.031 (a)	182	2.36 (1)
1.024	0.0291 (a)	2.199	0.030 (a)	704	4.0 (b)
1.112	0.0284 (a)	2.554	0.031 (a)	1145	5.1 (b)
1.234	0.0282 (a)	2.794	0.033 (a)	1472	6.8 (b)
1.282	0.0290 (a)	2.853	0.031 (a)	2036	7.6 (b)
				2124	6.9 (b)
				2604	9.4 (b)

Experimental:

(a) Wagner, *et al.*, J. Chem. Phys. **47**, 3138 (1967).(b) Schlumbohm, Z. Physik **184**, 492 (1965).

Theoretical:

(1) Lowke, *et al.*, Phys. Rev. **181**, 302 (1969).

Table 2.19

Values of D_T/μ for dry air from which the carbon dioxide has been removed

E/N (10^{-17}Vcm^2)	D_T/μ (V)	E/N (10^{-17}Vcm^2)	D_T/μ (V)	E/N (10^{-17}Vcm^2)	D_T/μ (V)
0.2980	0.099 (a)	2.682	0.35 (a)	14.90	1.00 (a)
0.600	0.146 (a)	2.980	0.39 (a)	29.80	1.18 (a)
0.894	0.179 (a)	3.58	0.45 (a)	44.7	1.32 (a)
1.192	0.206 (a)	4.47	0.54 (a)	59.6	1.48 (a)
1.490	0.233 (a)	5.36	0.61 (a)	60.6	1.50 (b)
1.788	0.261 (a)	5.96	0.66 (a)	75.8	1.73 (b)
2.086	0.293 (a)	8.94	0.82 (a)	90.9	1.97 (b)
2.384	0.32 (a)	11.92	0.93 (a)	106.1	2.24 (b)
				121.2	2.53 (b)

Experimental:

- (a) Crompton, *et al.*, Proc. Roy. Soc. London, Ser A, **218**, 507 (1953).
 (b) Rees, Austr. J. Phys. **17**, 307 (1964).

3.3. Diffusion Coefficient

For an electron swarm in thermal equilibrium with a gas, the diffusion is symmetric and can be characterized by a coefficient of diffusion, which is a constant independent of direction and which will be designated here as the thermal equilibrium diffusion coefficient D_{th} . When equilibrium is disturbed by the application of an electric field E , the diffusion becomes asymmetric and, as mentioned in section 3.2, the diffusion coefficient becomes a tensor, having components D_T and D_L , respectively, in directions perpendicular to and along the direction of the electric field. As $E \rightarrow 0$, both D_T and $D_L \rightarrow D_{th}$.

The only values of $D_T N$ and $D_L N$ given in sections 3.3.b and 3.3.c are those measured directly. Other values for $D_L N$ and $D_T N$ can, of course, be obtained by multiplying the values of D_L/μ and D_T/μ given in section 3.2 by values of $\mu N = W/(E/N)$ obtainable from the results given in section 3.1. Values of $D_{th} N$ can be obtained by extrapolating the data given in previous sections to zero field or by using previously determined values of the cross section in eq (7). Since this involves procedures other than purely arithmetical manipulation of the data given previously, theoretical values of D_{th} are included in section 3.3.a for comparison with the experimental values.

All the coefficients of diffusion are inversely proportional to the number density of the gas, so that it is convenient to present the data as values of $D_{th} N$ and as graphs of $D_T N$ as functions of E/N , as is done in the following sections.

3.3.a. Thermal Equilibrium Diffusion Coefficient D_{th}

The only experimental data for D_{th} are those obtained recently by Cavalleri [4049, 4050] and by Nelson and

Davis [4048, 5135]. In Cavalleri's so-called sampling diffusion chamber (SDC) method, the decay of the number of electrons in a gas, initially weakly ionized by a burst of X-rays, was measured. In Nelson and Davis' so-called drift-dwell-drift (DDD) method, an electron swarm produced by a pulsed light source was allowed to drift under the influence of an applied field which was then reduced to zero and subsequently re-applied after a measured period of time which could be varied. The distribution of the time of arrival of the electrons at the detector, which was measured, gave a measure of D_{th} . The experimental values obtained are given in the following tables, where the experimental methods are designated by SDC and DDD, respectively.

Several methods of calculating theoretical values for D_{th} were used by Nelson and Davis [4048], and the values obtained using the two methods which would be expected to give the most reliable results are given in tables 3.1 to 3.8. These methods were:

i) to calculate D_{th} from eq (7) by using values of q_m from various sources and the Maxwellian velocity distribution, and

ii) to calculate D_{th} by extrapolating measured values of W to zero field ($W_{E \rightarrow 0} = W_0$, say) and inserting W_0 in the equation $W_0/D_{th} = eE/kT$. The methods are designated q_m and W_0 , respectively, in the tables and references to the source of the data on q_m or W_0 used are also included with the method.

The following references to sources of data in addition to those in the bibliography (section 5) are designated in tables 3.1 to 3.7 by the following letters:

- a) D. E. Golden, Phys. Rev. **151**, 48 (1966)
 b) T. F. O'Malley, Phys. Rev. **130**, 1025 (1963)
 c) R. W. Crompton (unpublished).

In tables 3.1 to 3.8 values of D_{th} are given for helium, neon, argon, hydrogen, nitrogen, oxygen, carbon

monoxide, and carbon dioxide. There are no published values available for krypton, xenon, nitric oxide, or air.

3.3.b. Longitudinal Diffusion Coefficient D_L

There are very few directly measured values of D_L available. For helium, argon, and carbon monoxide the only experimental data are those of Wagner, Davis, and Hurst [3313] which are shown in figure 3.1 and given in table 3.9.

At relatively low values of $E/N < 2 \times 10^{-16} \text{ V cm}^2$ two sets of values [1833, 3313] have been obtained for $D_L N$ for nitrogen and carbon dioxide. These are shown in figure 3.2, together with values for hydrogen obtained by Wagner et al. [3313] and values extending to very low values of E/N for oxygen obtained by Nelson and Davis [5135].

The only sets of data over an extensive range of values E/N for $E/N > 10^{-15} \text{ V cm}^2$ are those of Schlumbohm [1625] for hydrogen, nitrogen, oxygen, and carbon dioxide. These are shown in figure 3.3, together with two values obtained for hydrogen at $E/N \sim 10^{-15} \text{ V cm}^2$ obtained by Breare and von Engel [3788].

The data for $D_L N$ in the various gases are tabulated in table 3.9.

3.3.c. Transverse Diffusion Coefficient D_T

Values of the transverse diffusion coefficient are usually obtained by combining the data of sections 3.1 and 3.2, but direct measurements of D_T have recently been made for helium by Cavalleri [4049] and these are shown in figure 3.4 and table 3.10. The results are in good agreement with those obtained from the values of W and D_T/μ obtained in [2433].

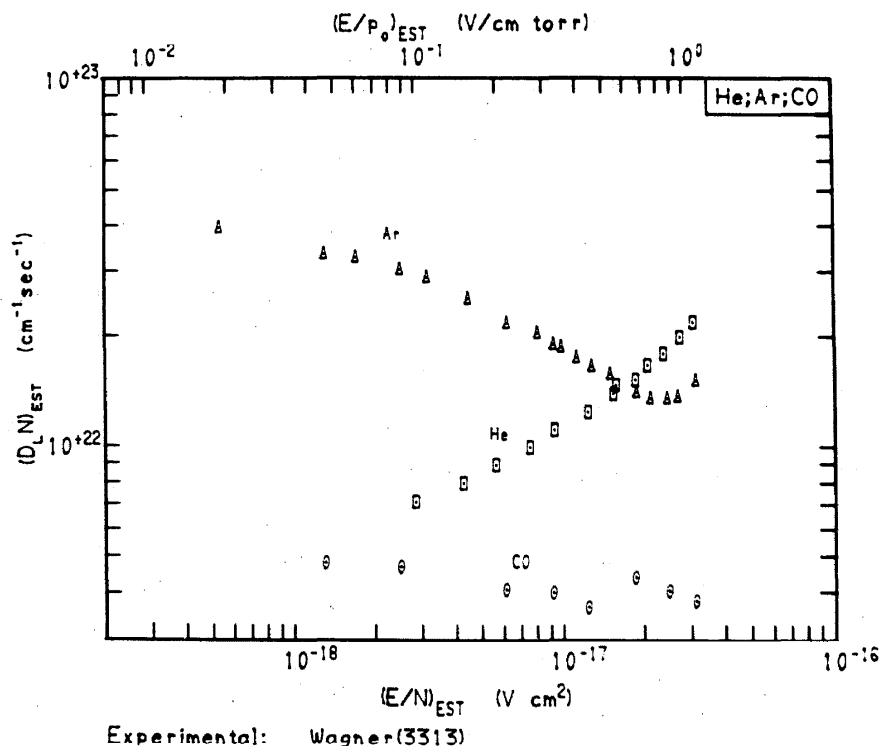


FIGURE 3.1. Experimental values of $(D_L N)_{\text{Est}}$ as a function of $(E/N)_{\text{Est}}$ for helium, argon, and carbon monoxide. (A temperature of 27 °C has been assumed to convert the values of D_p and E/p given in [3313].)

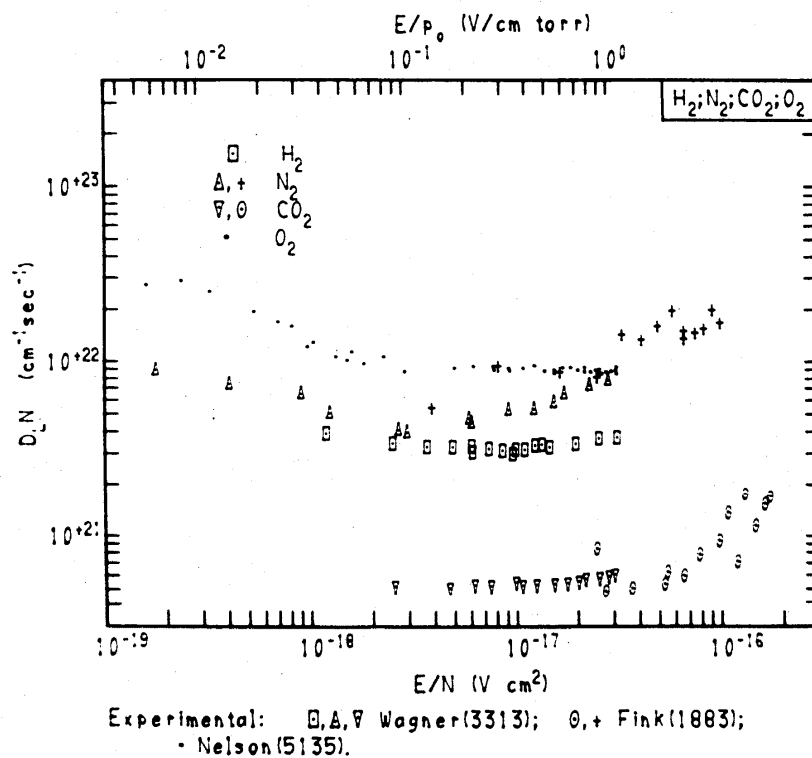


FIGURE 3.2. Experimental values of D_1N for hydrogen, nitrogen, oxygen, and carbon dioxide for $E/N < 2 \times 10^{-16} \text{ V cm}^2$. (A temperature of 27 °C has been assumed to convert the values of Dp and E/p given in [3313].)

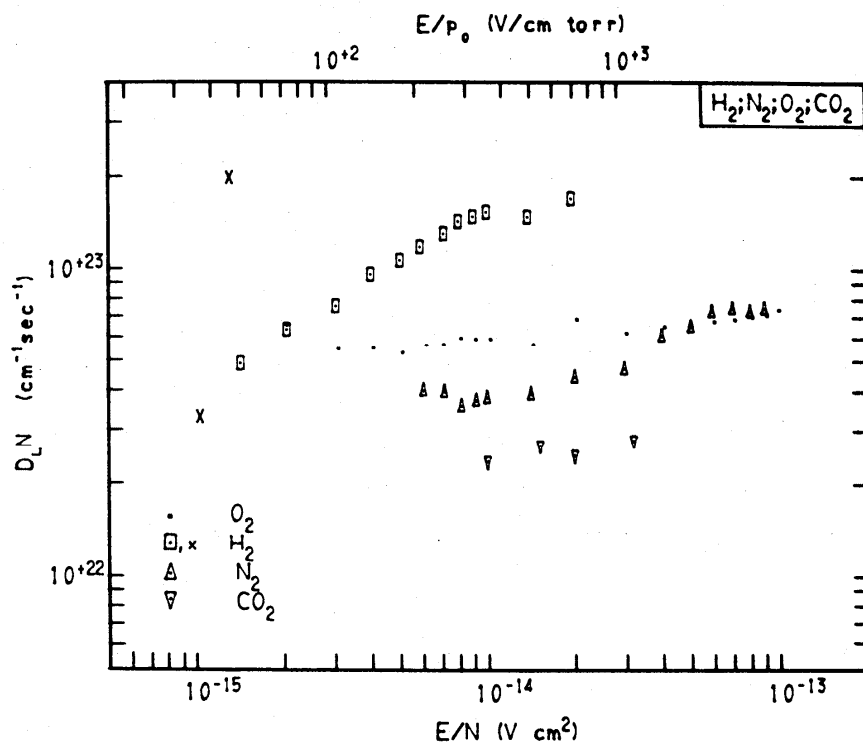
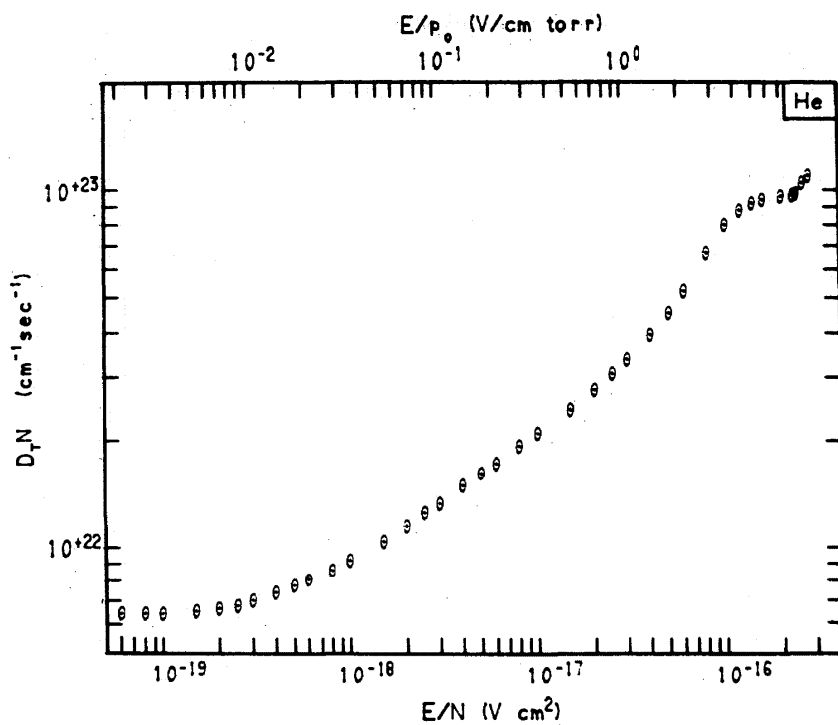


FIGURE 3.3. Experimental values of D_1N for hydrogen, nitrogen, oxygen and carbon dioxide for $E/N > 10^{-15} \text{ V cm}^2$. (A temperature of 20 °C has been assumed to convert the values of Dp and E/p given in [3788].)



Experimental: $\circ$ Cavalleri(4049).

FIGURE 3.4. Experimental values of $D_T N$ as a function of E/N for helium.

Table 3.1*

Values of $D_{th}N$ for helium

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
6.9		DDD	Nelson (4048)
6.4		SDC	Cavalleri (4049)
	6.6	$W_o(530)$	Nelson (4048)
	6.6	$q_m(1062)$	
	6.4	$q_m(2433)$	
	6.9	$q_m(a)$	
	6.4	$q_m(b)$	

Table 3.2 *

Values of $D_{th}N$ for neon

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
65		DDD	Nelson (4048)
73		SDC	Cavalleri (4050)
44			Lloyd (4063)
	71	$q_m(c)$	Nelson (4048)
	58	$q_m(b)$	

Table 3.3 *

Values of $D_{th}N$ for argon

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
21		DDD	Nelson (4048)
	37	$W_o(530)$	
	32	$q_m(1062)$	
	21	$q_m(b)$	
	27	$q_m(a)$	

Table 3.4 *

Values of $D_{th}N$ for hydrogen

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
4.3		DDD	Nelson (4048)
	3.5	$W_o(530)$	
	3.9	$q_m(1062)$	

*For an outline of the methods used and the meaning of the symbols in the method column see section 3.3.a of the text.

Table 3.5*

Values of $D_{th}N$ for nitrogen

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
9.5	8.2 9.8	DDD $W_o(530)$ $q_m(1062)$	Nelson (4048)

Table 3.6*

Values of $D_{th}N$ for oxygen

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
39		DDD	Nelson (5135)

Table 3.7*

Values of $D_{th}N$ for carbon monoxide

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
5.7	5.4	DDD $W_o(439)$	Nelson (4048)

Table 3.8*

Values of $D_{th}N$ for carbon dioxide

$D_{th}N(10^{21} \text{ cm}^{-1} \text{ sec}^{-1})$		Method	Reference
Experimental	Theoretical		
0.49	0.47	DDD $W_o(439)$	Nelson (4048)

*For an outline of the methods used and the meaning of the symbols in the method column see section 3.3.a. of the text.

Table 3.9(a)

Values of $D_L N$ for helium

E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)
0.2832	0.74	0.933	1.16	1.852	1.59
0.425	0.83	1.233	1.29	2.065	1.75
0.560	0.93	1.534	1.45	2.354	1.87
0.756	1.04	1.570	1.54	2.715	2.09
				3.033	2.29

Experimental:

All data taken from Wagner, *et al.*, J. Chem. Phys. 47, 3138 (1967).

Table 3.9(b)

Values of $D_L N$ for argon

E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)
0.0531	4.2	0.620	2.29	1.487	1.66
0.1298	3.5	0.809	2.16	1.871	1.47
0.1711	3.5	0.927	2.01	2.106	1.42
0.2478	3.2	0.985	1.99	2.425	1.42
0.312	3.1	1.121	1.85	2.655	1.43
0.442	2.7	1.275	1.75	3.08	1.60

Experimental:

All data taken from Wagner, *et al.*, J. Chem. Phys. 47, 3138 (1967).

Table 3.9(c)

Values of $D_L N$ for hydrogen

E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)
0.1180	0.40 (a)	1.104	0.33 (a)	204.2	6.3 (c)
0.2478	0.36 (a)	1.239	0.35 (a)	302	7.6 (c)
0.366	0.34 (a)	1.346	0.35 (a)	398	9.5 (c)
0.489	0.34 (a)	1.457	0.34 (a)	501	10.6 (c)
0.608	0.35 (a)	1.959	0.36 (a)	589	11.7 (c)
0.613	0.32 (a)	2.537	0.38 (a)	708	13.0 (c)
0.737	0.33 (a)	3.10	0.39 (a)	794	14.2 (c)
0.856	0.33 (a)	103.1	3.3 (b)	891	14.7 (c)
0.968	0.31 (a)	130.3	20. (b)	1000	15.3 (c)
0.991	0.33 (a)	141.3	4.9 (c)	1380	14.7 (c)
				1950	16.9 (c)

Experimental:

(a) Wagner, *et al.*, J. Chem. Phys. 47, 3138 (1967).(b) Breare, *et al.*, Proc. Roy. Soc. (London) Ser. A 282, 390 (1964).

(c) Schlumbohm, Z. Physik, 184, 492 (1965).

Table 3.9(d)

Values of $D_L N$ for nitrogen

E/N (10^{-17} Vcm^2)	$D_L N$ ($10^{22} \text{ cm}^{-1} \text{ sec}^{-1}$)		E/N (10^{-17} Vcm^2)	$D_L N$ ($10^{22} \text{ cm}^{-1} \text{ sec}^{-1}$)		E/N (10^{-17} Vcm^2)	$D_L N$ ($10^{22} \text{ cm}^{-1} \text{ sec}^{-1}$)	
0.01769	0.96	(a)	1.716	0.70	(a)	603	4.0	(c)
0.0403	0.79	(a)	2.278	0.79	(a)	708	4.0	(c)
0.0900	0.70	(a)	2.491	0.81	(b)	813	3.6	(c)
0.1241	0.54	(a)	2.809	0.84	(a)	912	3.8	(c)
0.2670	0.43	(a)	3.28	1.4	(b)	1000	3.8	(c)
0.2949	0.42	(a)	4.13	1.3	(b)	1413	4.0	(c)
0.385	0.53	(b)	4.93	1.6	(b)	1995	4.5	(c)
0.590	0.50	(a)	5.82	2.0	(b)	2951	4.7	(c)
0.608	0.47	(a)	6.59	1.3	(b)	3981	6.1	(c)
0.819	0.94	(b)	6.62	1.5	(b)	5012	6.6	(c)
0.922	0.56	(a)	7.46	1.5	(b)	5888	7.4	(c)
1.226	0.57	(a)	8.26	1.5	(b)	6918	7.6	(c)
1.536	0.62	(a)	9.11	2.0	(b)	7943	7.4	(c)
1.639	0.86	(b)	9.87	1.7	(b)	8913	7.6	(c)

Experimental:

(a) Wagner, *et al.*, J. Chem. Phys. **47**, 3138 (1967).(b) Fink, *et al.*, Helv. Phys. Acta. **38**, 717 (1965).(c) Schlumbohm, Z. Physik **184**, 492 (1965).

Table 3.9(c)

Values of $D_L N$ for oxygen

E/N (10^{-17} Vcm^2)	$D_L N$ ($10^{22} \text{ cm}^{-1} \text{ sec}^{-1}$)		E/N (10^{-17} Vcm^2)	$D_L N$ ($10^{22} \text{ cm}^{-1} \text{ sec}^{-1}$)		E/N (10^{-17} Vcm^2)	$D_L N$ ($10^{22} \text{ cm}^{-1} \text{ sec}^{-1}$)	
0.00400	3.6		0.2278	1.06		1.855	0.93	
0.01606	2.7		0.2889	0.87		2.011	0.90	
0.02351	2.9		0.502	0.91		2.168	0.87	
0.0329	2.5		0.621	0.94		2.318	0.92	
0.0533	1.95		0.778	0.93			0.88	
0.0700	1.68		0.784	0.90		2.462	0.89	
0.0818	1.60		0.928	0.91		2.550	0.90	
0.0975	1.21		0.934	0.88			0.85	
0.1044	1.29		1.091	0.92		2.625	0.85	
0.1338	1.06		1.241	0.95		2.775	0.87	
0.1522	1.01		1.397	0.88		2.919	0.88	
0.1595	1.14		1.548	0.90		3.08	0.90	
0.1832	0.96		1.554	0.86		3.10	0.94	
			1.705	0.93			0.86	

Experimental:

All data taken from Nelson, *et al.*, J. Chem. Phys. **57**, 4079 (1972).

Table 3.9(f)

Values of $D_L N$ for carbon monoxide

E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)
0.1295	0.50	0.612	0.43	1.855	0.46
0.2474	0.49	0.913	0.42	2.474	0.43
		1.231	0.39	3.10	0.40

Experimental:

All data taken from Wagner, *et al.*, J. Chem. Phys. 47, 3138 (1967).

Table 3.9(g)

Values of $D_L N$ for carbon dioxide

E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_L N$ ($10^{22} \text{cm}^{-1} \text{sec}^{-1}$)
0.2532	0.053 (a)	2.067	0.056 (a)	7.76	0.079 (b)
0.471	0.051 (a)	2.168	0.058 (a)	9.73	0.094 (b)
0.625	0.054 (a)	2.441	0.083 (b)	10.73	0.14 (b)
0.748	0.054 (a)	2.526	0.059 (a)	11.87	0.072 (b)
0.989	0.055 (a)	2.709	0.048 (b)	12.94	0.18 (b)
1.078	0.053 (a)	2.763	0.061 (a)	14.65	0.11 (b)
1.254	0.054 (a)	2.827	0.060 (a)	16.25	0.15 (b)
1.508	0.057 (a)	3.02	0.061 (a)	17.26	0.17 (b)
1.537	0.055 (a)	3.68	0.050 (b)	1000	2.32 (c)
1.766	0.055 (a)	5.32	0.054 (b)	1513	2.63 (c)
2.008	0.056 (a)	5.45	0.063 (b)	1995	2.43 (c)
		6.52	0.060 (b)	3162	2.74 (c)

Experimental:

(a) Wagner, *et al.*, J. Chem. Phys. 47, 3138 (1967).(b) Fink, *et al.*, Helv. Phys. Acta. 38, 717 (1965).

(c) Schlumbohm, Z. Physik 184, 492 (1965).

Table 3.10

Experimental values of $D_T N$ for helium

E/N (10^{-17}Vcm^2)	$D_T N$ ($10^{23} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_T N$ ($10^{23} \text{cm}^{-1} \text{sec}^{-1}$)	E/N (10^{-17}Vcm^2)	$D_T N$ ($10^{23} \text{cm}^{-1} \text{sec}^{-1}$)
0.00600	0.064	0.1500	0.104	3.00	0.34
0.00800	0.064	0.2000	0.115	4.00	0.40
0.01000	0.065	0.2500	0.125	5.00	0.46
0.01500	0.066	0.300	0.133	6.00	0.52
0.02000	0.067	0.400	0.149	8.00	0.67
0.02500	0.068	0.500	0.162	10.00	0.80
				12.00	0.88
0.0300	0.070	0.600	0.173		
0.0400	0.074	0.800	0.193	14.00	0.92
0.0500	0.078	1.000	0.209	16.00	0.94
0.0600	0.081	1.500	0.244	20.00	0.96
0.0800	0.087	2.000	0.277	23.00	0.97
0.1000	0.092	2.500	0.31	24.00	0.98
				26.00	1.05
				28.00	1.10

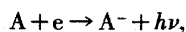
Experimental:

All data taken from Cavalleri, Phys. Rev. 179, 186 (1969).

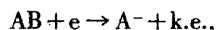
3.4. Attachment Coefficient

By definition, electronegative gases are those that form stable negative ions in an energy state lower than that of the ground state of the neutral atoms or molecules of the gas. The process of adding the extra electron to form the negative ion is known as electron attachment, and requires the removal not only of any kinetic energy that the electron may possess, but also of the energy difference between the original state of the neutral atom or molecule and the negative ion. The energy may be removed in a variety of ways, so giving rise to processes of

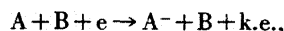
i) radiative attachment, represented by the equation



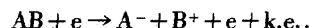
ii) dissociative attachment, represented by the equation



iii) three-body attachment, represented by the equation



iv) dissociation into ions, represented by the equation



Here A and B represent atoms, e an electron, A^- and B^+ negative and positive ions, respectively, and k.e., kinetic energy.

In general, when an electron swarm moves through an electronegative gas all four attachment processes are possible, although process (i) usually has a relatively small cross section and process (iv) is a high energy process so that these two processes are often negligible. The result of all these processes is that electrons are continually removed from the swarm to form negative ions.

When the value of E/N is sufficiently small, the mean energy of the electrons is so low that ionization by electrons in the high energy tail of the distribution is negligible. In these circumstances, a steady electron current, I_0 , released into a uniform field (by external illumination of a cathode of a uniform field electrode system by ultraviolet illumination, say) results, after travelling a distance, d , through the gas in the field direction, in an electron current

$$I_- = I_0 e^{-\eta d},$$

where η is an attachment coefficient representing all the possible attachment processes. Under these conditions, η can be determined [see e.g. 1317] by measuring the electron current as a function of d at a given value of N . Alternatively, a short duration pulse of electrons released from the cathode gives rise to an exponential

distribution of negative ions across the gap, which, after the disappearance of the electrons into the anode, gives an ion current of the form

$$I_0 e^{\eta W_i t},$$

where I_0 is a constant for a given gas, pressure, and electrode system and W_i is the negative ion drift velocity. Thus sampling the negative ion current arriving at a given distance from the cathode as a function of time [see e.g. 1383] gives values of η .

Values of η have also been determined [see e.g. 649] by measuring the transverse diffusion of a mixed swarm of electrons and negative ions at various distances from a point source.

These methods give values of η/N as a function of E/N which by carrying out experiments at sufficiently low E/N , can be extrapolated and used to obtain values of the attachment rate coefficient at thermal energies, which is the coefficient obtained directly from experiments on plasma decay using microwave methods.

Whether three-body attachment is a significant process in any given case may be determined by carrying out the measurements of the attachment coefficient over a range of gas pressures, because the three-body attachment coefficient depends on N^2 whereas the coefficients representing the other attachment processes are linearly dependent on N .

The situation becomes much more complicated at higher values of E/N , where in addition to electron attachment, ionization by electrons and detachment of electrons from negative ions in collisions with gas molecules both have to be taken into account and where ion conversion reactions may also be important. The resulting equation for the total current at a distance, d , in the direction of a uniform field from a cathode from which a steady initial electron current, I_0 , is released is given in section 3.7.c as equation (16). This equation is such that analysis of experimental data on I versus d to give significant values of the individual coefficients (including the attachment coefficient) is extremely difficult in many cases and the values obtained depend on the simplifying assumptions that are found to be necessary. Similar complications have, of course, to be taken into account in the analysis of experiments on the temporal development of pulses of electrons released from the cathode.

Of the gases for which data are included in this survey, only oxygen, carbon monoxide, nitric oxide, carbon dioxide, and air are electronegative and the values of η/N for these gases are given in the following graphs and tables.

3.4.a. Attachment Coefficient—Oxygen

At low values of E/N ($< 12 \times 10^{-17}$ V cm²), Chanin, Phelps, and Biondi [131, 1383], using a drift tube in which the negative ion current resulting from the release

of electrons from the cathode by a pulse of ultraviolet light was sampled as a function of time, established that η/N was a function of N as well as of E/N , indicating that in this region, a three-body attachment process is significant. Their results are shown in figure 4.1(a) as (η/N) , E/N (η on the graphs $\equiv \eta$ in text) together with another set of results obtained using a drift tube with electrical shutters [1269] covering approximately the same range. The range of gas number densities covered was extended in later experiments in which the development of an electron avalanche was measured oscillographically [4919] with the results shown in figure 4.1(b). To avoid confusion in the graphs, the three values obtained in [1006] that also show the dependence of η on N^2 are omitted from the figures but included in table 4.1 where the results of all these investigations are given as values of η/N^2 .

Measurements of attachment in oxygen have also been made by studying plasma decay by microwave methods. This procedure gives attachment rate coefficients at thermal energies. For comparison, rate coefficients at thermal energies can also be obtained from the drift-tube data using the procedure given below.

As discussed in section 2.1,

$$\eta W = a_3 N^2,$$

where a_3 is the three-body attachment rate coefficient. Thus by using the data given in figure 4.1, together with data for W such as that given in section 3.1 values of a_3 can be obtained as a function of E/N . Further, using data for D/μ such as that given in section 3.2, values of a_3 as a function of characteristic energy can be obtained. Extrapolation of the values of a_3 to thermal energy can then be made. Values of a_3 as a function of E/N can also be obtained from drift-tube measurements in gas mixtures. In this case

$$\eta W = \sum_X a_3(X) N(X) N(O_2),$$

where X indicates the various gases in the mixture and $N(X)$ the number density of molecules of gas X , so that a_3 for oxygen in different gases can be determined by measuring η for various small admixtures of O_2 in, say, N_2 or He.

Using values of W and D/μ given in their papers, Chanin et al. [1383] and Pack and Phelps [1662] used the above procedure to obtain the values of a_3 at thermal energies shown in table 4.2.

As can be seen from figure 4.2, the results in [1383] and [1662] for oxygen at low energies for $T = 300$ K are in general agreement with the recent data of McCorkle et al. [5046]. The discrepancy at higher energies, together with a similar discrepancy between results obtained with N_2 as the third body, suggests however, that the energy scale of [1383] and [1662] is in error. The results of recent microwave measurements [762,

1633, 4097, 5131, 5770, 5125] are also given in table 4.2, and it can be seen that values of a three-body attachment rate coefficient at 300 K in close agreement with the extrapolated drift-tube data are obtained, although in earlier microwave experiments [4095, 711, 4096] a two-body attachment rate coefficient of about $1.4 \times 10^{-15} \text{ cm}^3 \text{ s}^{-1}$ was reported.

As can be seen from figure 4.3, recent beam experiments [5047] and theoretical calculations show considerable structure, in the curves of three-body rate coefficient for the production of O_2^- as a function of electron energy, which was not revealed by the swarm data. The curves shown in figure 4.3 have been normalized at the peak of the swarm data given in [1383]. The envelope of these curves follows closely the swarm results shown in figure 4.2, for energies up to about 0.6 eV. At higher energies the swarm data are higher because of the effect of the much wider energy distribution in the swarm. The first peak of the beam data also shows a smaller dependence on temperature than the peak of the swarm data.

In the intermediate range of values of E/N between about 12×10^{-17} and $84 \times 10^{-17} \text{ V cm}^2$ a relatively large number of experimental investigations have been carried out using the various methods outlined in the introduction to this section. In this region of E/N , it is found that η/N is a function of E/N only, which indicates the predominance of two-body attachment over the range of values of N studied. Ionization is negligibly small at values of E/N up to about $56 \times 10^{-17} \text{ V cm}^2$, but above this value it becomes increasingly significant, and the coefficient measured by these methods is then $(\eta - \alpha)/N$ rather than η/N .

Many of the early data [1317, 1336, 2093] were expressed as values of h , the probability of attachment in a collision, which makes the recovery of accurate values of η/N difficult, because it involves the use of numerical values of other parameters such as W which were in some cases taken from different papers; e.g., values of W given in figure 1.28 were used in the case of [1317] and the values of k given in [195] in the case of [1336]. Moreover, temperatures to which the values of E/p given in [1317, 1433, 649, and 2093] correspond are not clearly stated and have been assumed to be 20°C . These factors possibly account for some of the scatter of the data shown in figures 4.4(a) and 4.4(b), but an analysis [1006] of the sources of error in the various experiments shows that most of the scatter is likely to be experimental in origin; for example, electron diffusion could have given rise to errors in some of the results given in [1433] and [1317], and the length of the pulse used in [1336] was such that it would be likely to lead to values of η/N which were too low.

Only the results of recent experimental work in which the results are given as η/N , E/N without ambiguity are given in table 4.3. The results of theoretical computations [2553] of η/N , using two slightly different sets of cross sections chosen to fit these and other (see

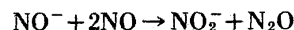
sec. 3.1.1 and sec. 3.7.d) swarm data, are given in figures 4.4(a) and 4.4(b) to help comparison of the results.

For values of E/N above $84 \times 10^{-17} \text{ V cm}^2$ the ionization coefficient increases rapidly, becoming greater than the attachment coefficient at about 10^{-15} V cm^2 . Many attempts have been made to determine the attachment coefficient from the spatial growth of ionization at constant values of E/N in this region. The appropriate equation for the analysis of the results is eq (16), but its use presents formidable problems. The experimental data in all cases show the effect of attachment through the departure of the $\ln I$, d graphs from linearity at low values of d , but the values obtained for the attachment coefficient depend on the assumptions made about which processes are significant; e.g., detachment and ion conversion are often assumed to be negligible. Furthermore, in oxygen as in air [2047], analysis of typical experimental data in which the ionization currents are measured to within 2 or 3 percent shows that the spread of attachment coefficients that will fit the experimental data is very large (typically ~ 100 percent). This situation is reflected in the wide range of values for η/N that have been obtained. These are shown in figure 4.5. The results of an investigation of the development of the current with time resulting from a short burst of electrons released from the cathode in a plane parallel gap [2370] are also shown in figure 4.5. Again there are considerable complications in the analysis and the results for η/N lie close to the lower limit of those obtained from spatial growth measurements in the region where they overlap. It is clear from consideration of these data that the value of the attachment coefficient in the region where ionization is significant is at present known only in order of magnitude because of the complexities in the reaction scheme for oxygen.

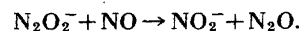
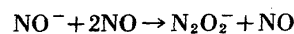
3.4.b. Attachment Coefficient—Nitric Oxide

The only published values of the attachment coefficient as a function of E/N for nitric oxide are those recently obtained by Parkes and Sugden [4943], whose data at 293 K are shown in figure 4.6 and tabulated in table 4.4. The earlier data of Bradbury [2099] for $25 \times 10^{16} < N < 100 \times 10^{16} \text{ cm}^{-3}$ were given as values of i , the probability of attachment, and require values of V to convert them to values of the attachment coefficient such as those shown in fig. 4.6) for comparison with the recent data. The data of Parkes and Sugden were obtained from measurements on the electron and ion pulses resulting from the release of a pulse of electrons from a cathode of a drift tube at relatively high number densities in the range $2.2 \times 10^{18} < N < 5.25 \times 10^{18} \text{ cm}^{-3}$. As can be seen from figure 4.6, all their results lie on a single curve of η/N^2 , E/N , showing that three-body attachment occurs in this case, which is in agreement with the earlier work of Bradbury [2099] and with recent studies of plasma decay using the microwave method [1608, 3229, 3610].

Extrapolation of the drift-tube data of Parkes and Sugden to thermal energies, however, gives a value of $(8 \pm 2) \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ for the thermal attachment rate coefficient which is much higher than the values 1.3 to $2.2 \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ obtained in the microwave experiments at low values of N . Further measurements [4943] of mass analyzed ion currents in NO, O₂, and NO/O₂ mixtures at low values of $N \sim 3 \times 10^{16} \text{ cm}^{-3}$ showed that the attachment rates obtained in the microwave experiments are too low because of the neglect of detachment. Detachment becomes less significant as the number density is increased because of the conversion of the NO⁻ ions from which electrons are readily detached to more stable ions in the processes



or



Both the microwave and drift-tube experiments show that the attachment coefficient decreases markedly with increasing temperature. The drift-tube experiments show, however, that the temperature dependence is larger at lower number densities over the range investigated, indicating that the situation is again complicated by detachment.

3.4.c. Attachment Coefficient—Carbon Monoxide

All the possible processes of attachment in carbon monoxide are dissociative processes with relatively high energy thresholds ($> 9.5 \text{ eV}$) (Hagstrum, Rev. Mod. Phys. **23**, 185, 1951; Chantry, Phys. Rev. **172**, 125, 1968). This, together with the fact that there is a very rapid detachment reaction between the O⁻ ions formed and CO to give CO₂ makes attachment unlikely to be significant under swarm conditions. The importance of this effect was illustrated (i) by mass spectrometric measurements [3352] (see also J. L. Moruzzi and A. V. Phelps, J. Chem. Phys. **45**, 4617, 1966) at relatively low gas number densities (3.3×10^{16} to $16.5 \times 10^{16} \text{ cm}^{-3}$) which showed negligible negative ion formation for $1.5 \times 10^{-17} < E/N < 300 \times 10^{-17} \text{ V cm}^2$, (ii) by measurements [2099, 2408] of the negative ion component of drifting electron swarms by means of a radio-frequency electron filter of the Bradbury type which gave no detectable attachment coefficient for $6.4 \times 10^{16} < N < 31.5 \times 10^{16} \text{ cm}^{-3}$ for E/N in the range from 14×10^{-17} to $118 \times 10^{-17} \text{ V cm}^2$, and (iii) by measurements of pre-breakdown ionization [4016] at relatively high gas number density (94×10^{16} to $321 \times 10^{16} \text{ cm}^{-3}$) which gave no measurable attachment coefficient for $122 \times 10^{-17} < E/N < 182 \times 10^{-17} \text{ V cm}^2$; earlier measurements [1254] seemed to show attachment in CO but the observed attachment has since been shown [5243] to be due to impurities.

3.4.d. Attachment Coefficient—Carbon Dioxide

The published results on values of attachment coefficients in carbon dioxide are shown in figure 4.7. No dependence of the coefficient η/N on N has been found, indicating that attachment occurs by a two-body process of associative detachment.

The experimental results shown in figure 4.7 for $E/N < 73 \times 10^{-17}$ V cm² were obtained [2408] for a range of values of N from 14×10^{16} to 28×10^{16} cm⁻³ by determining the negative ion component of a drifting swarm using a radio-frequency electron filter of the Bradbury type. Those for $E/N > 73 \times 10^{-17}$ V cm² were obtained either by analysis of the measured spatial growth of prebreakdown ionization [948, 3435, 4843] in the steady state for $78 \times 10^{16} < N < 386 \times 10^{16}$ cm⁻³ or by analysis of oscillographic observations of prebreakdown current pulses [1724]. Theoretical values [2553] obtained using cross sections obtained from consideration of swarm data are also shown.

The mean values given in figure 4.7 should not be taken as indicating more than approximate values of the attachment coefficient, because the nature of the analysis of the experimental data is such that a considerable spread (often greater than 100 percent) about the mean values given would be compatible with the experimental results.

3.4.e. Attachment Coefficient—Air

In air ionization becomes significant with respect to attachment at a value of E/N above about 70×10^{-17} V cm², so that it is convenient to consider the results obtained in two regions above and below this value of E/N .

The values obtained in the lower region of E/N are given in figure 4.8. All these results were obtained using various arrangements for the spatial analysis of an electron swarm moving through the gas, except for those in [3165] in which the temporal development of avalanches was measured. One of the sets of data [1317] was presented in the form of values of the probability of attachment, h , at an unspecified temperature, so that the conversion to η and the assumption of a temperature of

20 °C might have introduced some error in this case, but it seems likely that the spread in the other results arises from experimental factors such as those mentioned in the case of oxygen.

It is interesting to note that at very low E/N below about 10×10^{-17} V cm² the dependence of η/N on N observed by Hessenauer [3165] indicates the predominance of three-body attachment as in oxygen, the values for η /(partial pressure of oxygen) in air lying close to the values of η/p obtained in oxygen alone.

Thermal attachment rate coefficients have been obtained for air [762] in the range $10^{17} < N < 10^{18}$ cm⁻³ and for 4:1 nitrogen/oxygen mixtures [5770] in the range $3.54 \times 10^{16} < N < 3.54 \times 10^{17}$ cm⁻³ from microwave studies of the decay of ionization produced by pulsed electron beams. The results of Brodski and Zagik [762] were given in the form of a graph of $\nu_a/[O_2]$ versus p which tended to saturate at the higher pressures. Treating the low pressure part of the curve as linear, gave $k_1 = 3 \times 10^{-30}$ cm⁶ s⁻¹ and k_2 as 5×10^{-31} cm⁶ s⁻¹, where k_1 and k_2 are defined by

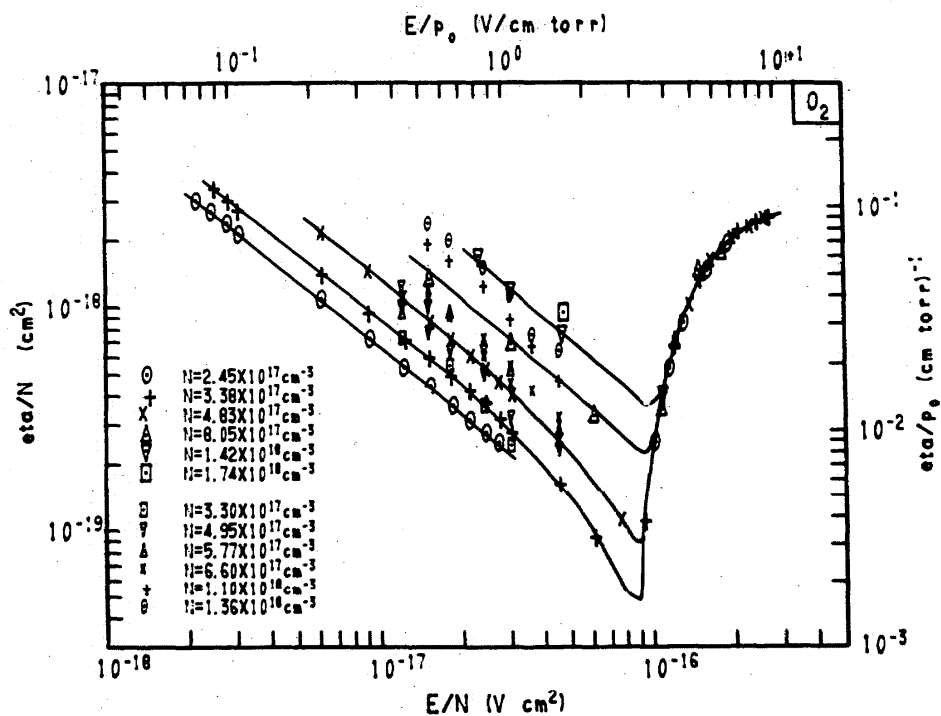
$$\nu_a/[O_2] = k_1[O_2] + k_2[N_2],$$

the brackets indicating concentrations. The results of Hirsh, Eisner, and Slevin [5770] showed that ν_a was linearly dependent on pressure over the range investigated and gave a value of $k_3 = 1.1 \times 10^{-31}$ cm⁶ s⁻¹, where k_3 is defined by

$$\nu_a = k_3([O_2] + [N_2])^2.$$

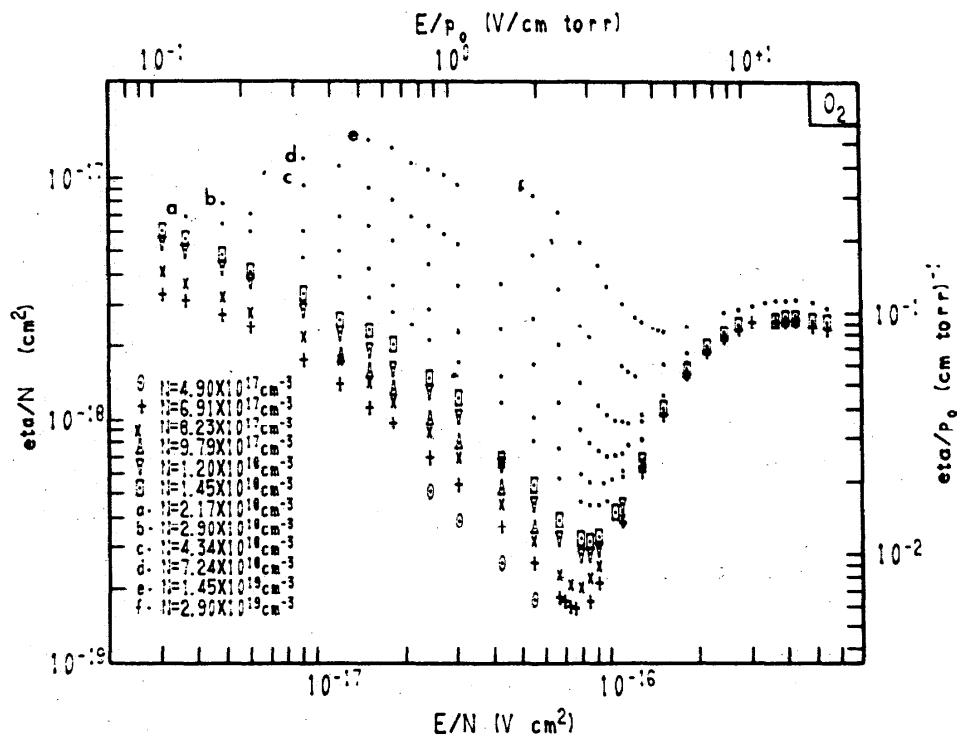
In air $k_3 = k_1 + 4k_2/25$, so that the value of k_3 from the results in [762] is 1.4×10^{-31} cm⁶ s⁻¹ in fair agreement with the value given in [5770].

The values for η/N at higher values of E/N when ionization becomes significant are shown in figure 4.9. Similar difficulties concerning the analysis of the experimental data arise as were discussed in the case of oxygen and these account for the lack of precise knowledge of the attachment coefficient in these conditions.



Experimental: Small symbols, Rees(1269);
Large symbols and lines, Chanin(1383).

FIGURE 4.1(a). Experimental values of η/N , E/N for oxygen at low values of E/N for $2.45 \times 10^{17} < N < 1.74 \times 10^{18} \text{ cm}^{-3}$.



Experimental: Grunberg(4919).

FIGURE 4.1(b). Experimental values of η/N , E/N for oxygen at low values of E/N for $4.90 \times 10^{17} < N < 2.9 \times 10^{19} \text{ cm}^{-3}$.

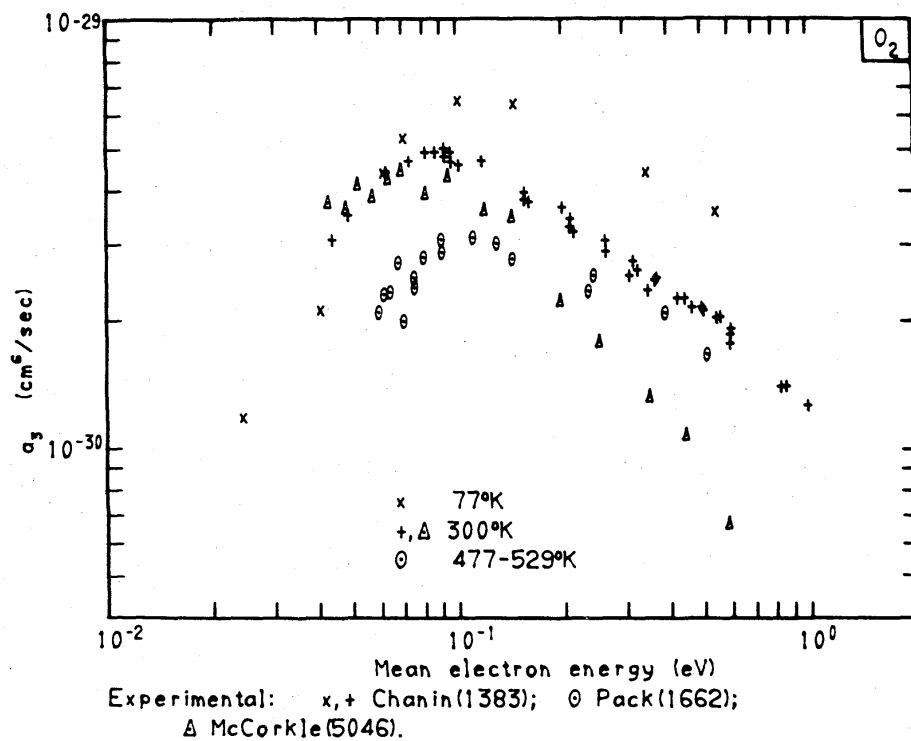


FIGURE 4.2. Three-body attachment rate coefficient for oxygen as a function of mean electron energy at various temperatures.

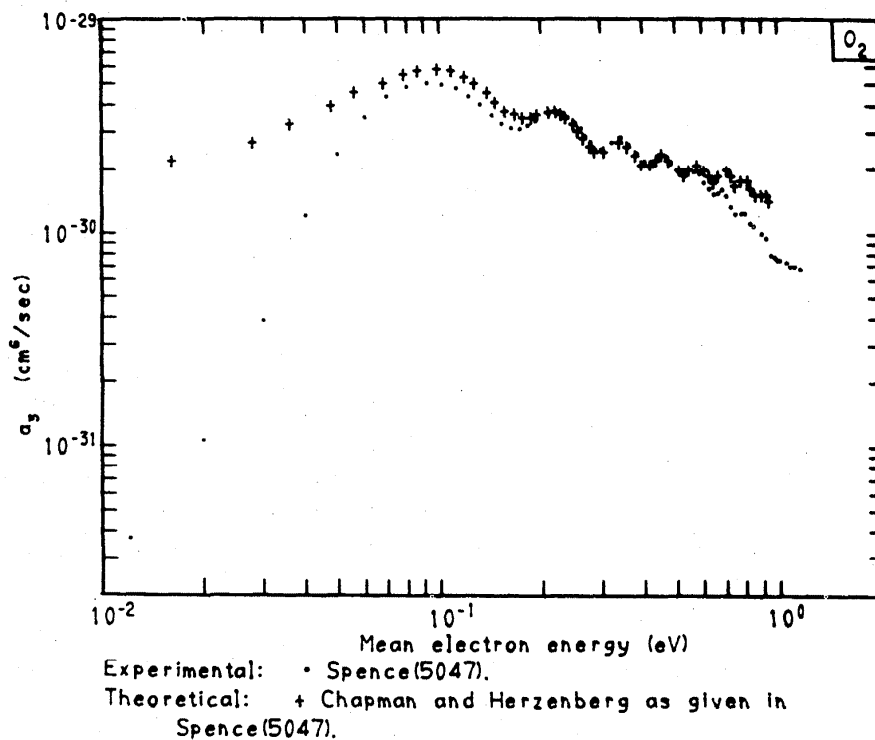
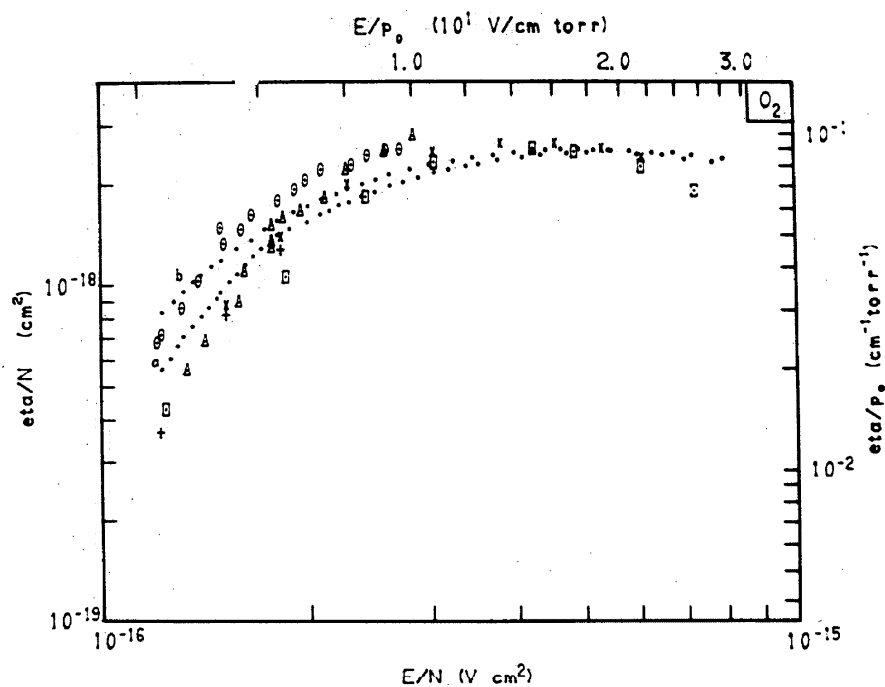
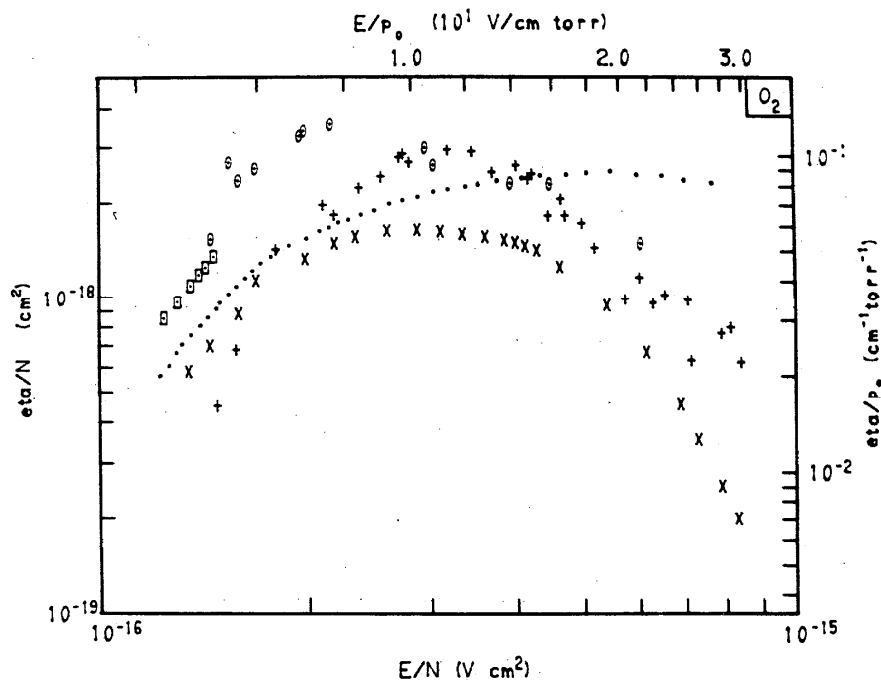


FIGURE 4.3. Comparison of the results of beam experiments on the three-body attachment rate coefficient as a function of electron energy for oxygen at 300 K with the maximum theoretically computed values. (The experimental beam results were normalized at the peak of the swarm data given in [1383] and the theoretical data to the second peak of the experimental beam results.)



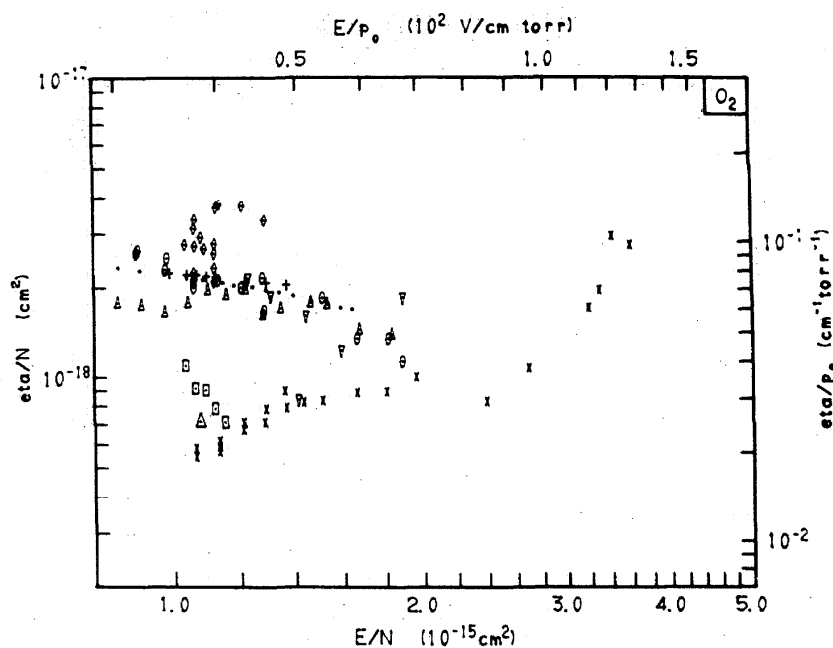
Experimental: θ Chanin(1383); + Rees(1269);
 x Huxley(649); $\square$ Chatterton(1006); Δ Doebling(727).
 Theoretical: $\cdot$ a, $\cdot$ b Hake(2553).

FIGURE 4.4(a). Values of η/N , E/N for oxygen for $12 \times 10^{-17} < E/N < 84 \times 10^{-17} \text{ V cm}^2$. (The experimental data for $E/N > 56 \times 10^{-17} \text{ V cm}^2$ represent values of $(\eta - \alpha)/N$.)



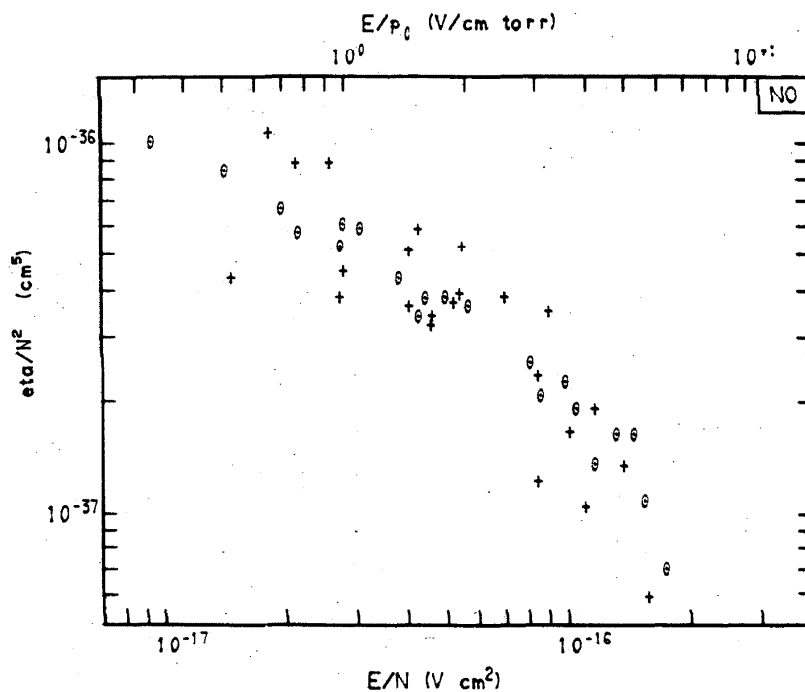
Experimental: θ Bradbury(1317); X Herrng(1336);
 + Kuffel(1433); $\square$ Cravath(2093).
 Theoretical: $\cdot$ Hake(2553).

FIGURE 4.4(b). Values of η/N , E/N for oxygen for $12 \times 10^{-17} < E/N < 84 \times 10^{-17} \text{ V cm}^2$. (The experimental data for $E/N > 56 \times 10^{-17} \text{ V cm}^2$ represent values of $(\eta - \alpha)/N$.)



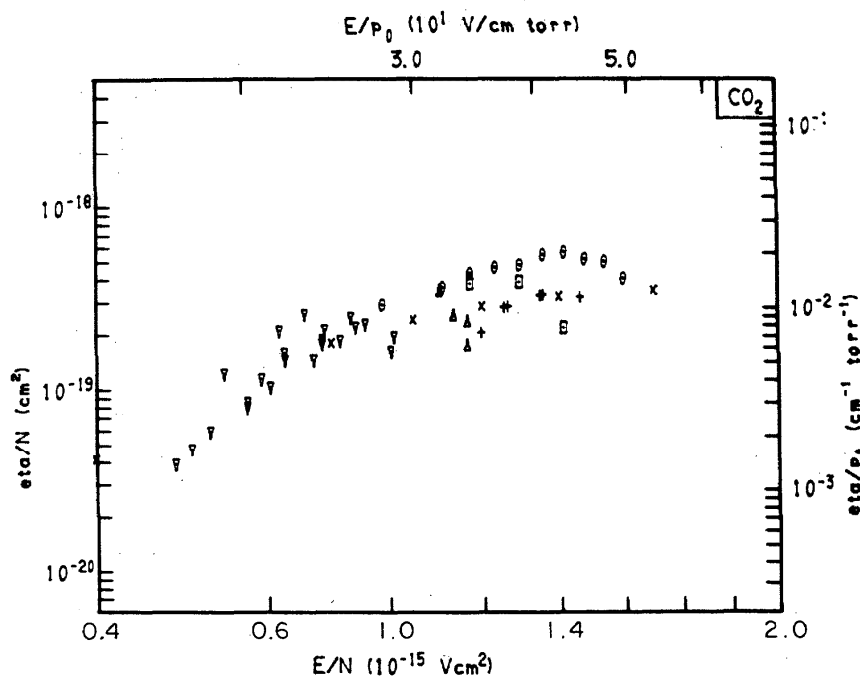
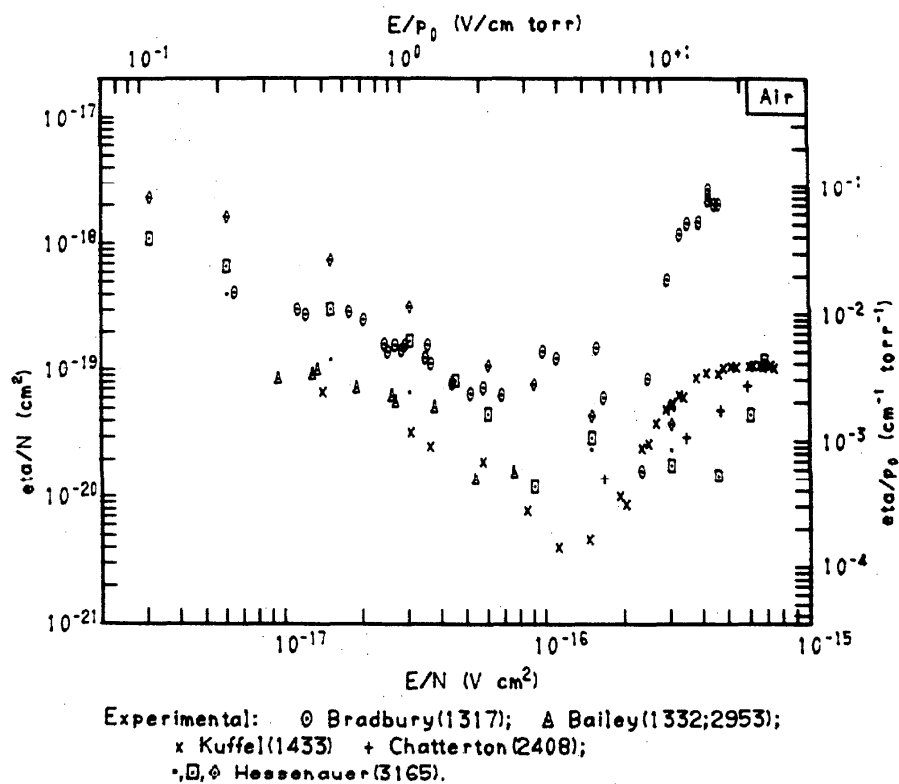
Experimental: $\diamond$ Sukhum(2555); $\times$ Frommhold(2370);
 $\square$ Dutton(573); Δ Freely(961); ∇ Schlumbohm(1445);
 $+$ Prasad(861); $\circ$ Harrison(791); $\triangle$ DeBitetto(4098).
 Theoretical: $\cdot$ Hake(2553).

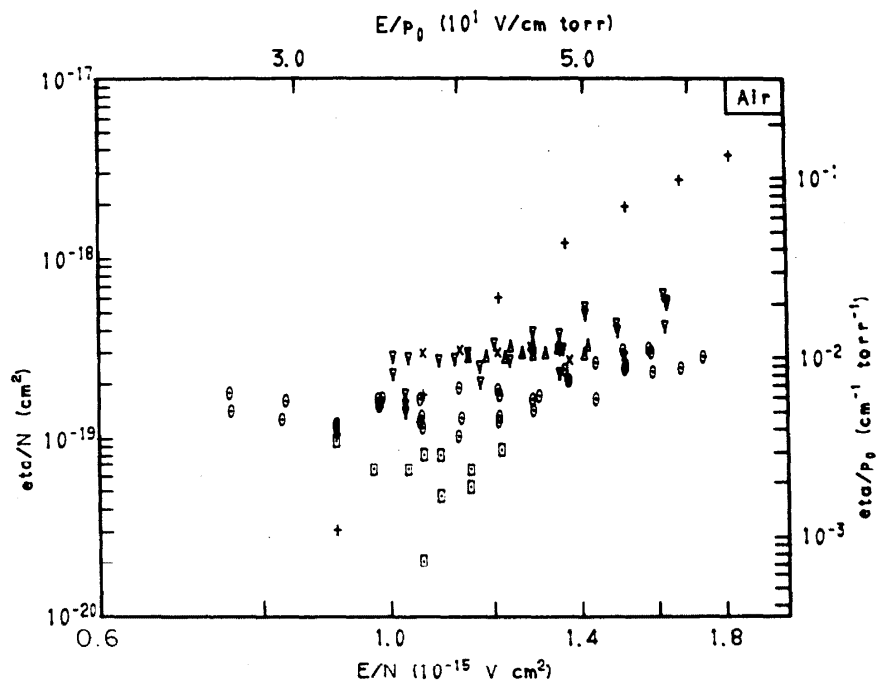
FIGURE 4.5. Values of η/N , E/N for oxygen at $E/N > 0.4 \times 10^{-17}$ V cm².



Experimental: $\circ$ Parkes(4943).
 Theoretical: $+$ Bradbury(2099).

FIGURE 4.6. Values of η/N^2 , E/N for nitric oxide. (The values of Bradbury [2099] were obtained from the published values of the probability of attachment using the assumption $W = W_M$ and the values of W_M given in fig. 1.32.)

FIGURE 4.7. Values of η/N , E/N for CO_2 .FIGURE 4.8. Experimental values of the attachment coefficient for air for $E/N < 68 \times 10^{-17} \text{ V cm}^2$.



Experimental: $\circ$ Harrison(791); Δ Ryzko(1911);
 $+$ Prasad(2047); ∇ Ryzko(2235); $\times$ Frommhold(2370);
 $\square$ Dutton(2388;2407).

Figure 4.9. Experimental values of attachment coefficient for air for $E/N > 84 \times 10^{-17} \text{ V cm}^2$.

Table 4.1
Experimental values of η/N^2 for oxygen for $E/N < 30 \times 10^{-17} \text{ Vcm}^2$

E/N (10^{-17} Vcm^2)	N (10^{17} cm^{-3})	η/N^2 (10^{-36} cm^5)	E/N (10^{-17} Vcm^2)	N (10^{17} cm^{-3})	η/N^2 (10^{-36} cm^5)
0.2154	2.45	12.2 (a)	1.224	4.83	2.28 (a)
0.2456	2.45	11.0 (a)	1.237	2.45	2.20 (a)
0.2501	3.38	10.1 (a)	1.250	3.38	2.13 (a)
0.2801	2.45	9.8 (a)		6.92	1.62 (d)
0.2819	3.38	8.9 (a)		8.24	1.72 (d)
				9.80	1.64 (d)
				11.99	1.59 (d)
0.303	6.92	4.8 (d)	1.515	14.49	1.61 (d)
	8.24	5.0 (d)		21.74	1.47 (d)
	11.99	4.5 (d)		28.99	1.45 (d)
	14.49	4.2 (d)		43.5	1.47 (d)
0.307	3.38	8.1 (a)		72.5	1.26 (d)
0.309	2.45	8.6 (a)		144.9	1.00 (d)
	6.92	4.5 (d)			
	8.24	4.5 (d)		4.95	1.56 (b)
0.364	11.99	4.2 (d)	1.516	4.95	2.02 (b)
	14.49	3.9 (d)		6.60	1.65 (b)
	21.74	3.2 (d)		6.60	1.83 (b)
				11.0	1.74 (b)
				13.6	1.75 (b)
0.485	6.92	3.9 (d)	1.523	8.05	1.70 (a)
	8.24	3.9 (d)	1.534	3.38	1.75 (a)
	11.99	3.5 (d)		2.45	1.80 (a)
	14.49	3.3 (d)	1.546	4.83	1.78 (a)
	21.74	2.99 (d)	1.694	10.62	0.64 (e)
	28.99	2.73 (d)			
	6.92	3.5 (d)		3.30	1.67 (b)
	8.24	3.4 (d)		4.95	1.29 (b)
0.606	11.99	3.1 (d)		4.95	1.74 (b)
	14.49	2.83 (d)		5.77	1.65 (b)
	21.74	2.81 (d)		6.60	1.38 (b)
	28.99	2.45 (d)		6.60	1.47 (b)
0.611	4.83	4.5 (a)		6.92	1.40 (d)
0.616	2.45	4.5 (a)		8.24	1.43 (d)
0.621	3.38	4.1 (a)	1.819	9.80	1.39 (d)
				11.0	1.46 (b)
	6.92	2.54 (d)		11.99	1.34 (d)
	8.24	2.65 (d)		13.6	1.47 (b)
	11.99	2.38 (d)		14.49	1.40 (d)
	14.49	2.28 (d)		21.74	1.30 (d)
0.909	21.74	2.16 (d)		28.99	1.24 (d)
	28.99	2.10 (d)		43.5	1.26 (d)
	43.5	2.16 (d)		72.5	1.13 (d)
	72.5	1.67 (d)		144.9	0.92 (d)
0.910	4.83	3.0 (a)	1.830	3.38	1.45 (a)
0.920	3.38	2.78 (a)	1.834	4.83	1.49 (a)
0.925	2.45	3.0 (a)	1.856	2.45	1.47 (a)
				21.74	1.15 (d)
				72.5	0.97 (d)
				144.9	0.79 (d)
	3.30	2.21 (b)	2.122		
	4.95	2.02 (b)		3.38	1.24 (a)
	4.95	2.49 (b)		2.45	1.27 (a)
	5.77	1.65 (b)	2.145	4.83	1.26 (a)
	6.92	2.01 (d)	2.154	14.2	1.18 (a)
	8.24	2.14 (d)	2.163		
1.212	9.80	1.89 (d)	2.314		
	11.99	1.87 (d)			
	14.49	1.80 (d)			
	21.74	1.79 (d)			
	28.99	1.73 (d)			
	43.5	1.61 (d)			
	72.5	1.55 (d)			

Table 4.1 (continued)

Experimental values of η/N^2 for oxygen for $E/N < 30 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	N (10^{17}cm^{-3})	η/N^2 (10^{-36}cm^5)		E/N (10^{-17}Vcm^2)	N (10^{17}cm^{-3})	η/N^2 (10^{-36}cm^5)
2.425	3.30	1.09 (b)		4.24	4.91	0.53 (d)
	4.91	1.04 (d)			6.92	0.53 (d)
	4.95	1.01 (b)			8.24	0.55 (d)
	4.95	1.29 (b)			9.80	0.55 (d)
	5.77	1.11 (b)			11.99	0.55 (d)
	6.60	0.93 (b)			14.49	0.48 (d)
	6.60	1.11 (b)			21.74	0.54 (d)
	6.92	1.01 (d)			28.99	0.52 (d)
	8.24	1.07 (d)			43.5	0.54 (d)
	9.80	1.05 (d)			72.5	0.51 (d)
	11.0	1.12 (b)		4.49	7.79	0.57 (e)
	11.99	1.08 (d)			4.95	0.47 (b)
	13.6	1.10 (b)			4.95	0.53 (b)
	14.49	1.03 (d)			5.77	0.50 (b)
	21.74	0.98 (d)			6.60	0.49 (b)
	28.99	0.99 (d)		4.55	10.44	0.34 (e)
	43.5	1.01 (d)			10.44	0.37 (e)
	72.5	0.88 (d)			11.0	0.42 (b)
	144.9	0.75 (d)			13.6	0.47 (b)
2.451	4.83	1.10 (a)				
2.456	2.45	1.14 (a)		4.56	4.83	0.53 (a)
2.459	3.38	1.09 (a)		4.57	3.38	0.48 (a)
2.728	72.5	0.81 (d)		4.68	14.2	0.54 (a)
	144.9	0.71 (d)		4.72	10.80	0.33 (e)
2.731	4.83	0.95 (a)			11.12	0.37 (e)
2.738	2.45	1.02 (a)		4.75	17.4	0.55 (a)
2.771	3.38	0.95 (a)		5.34	10.66	0.32 (e)
3.02	8.05	0.87 (a)			4.91	0.37 (d)
	14.2	0.83 (a)			6.92	0.37 (d)
3.03	3.30	0.74 (b)		5.45	8.24	0.39 (d)
	4.91	0.79 (d)			9.80	0.38 (d)
	4.95	0.65 (b)			11.99	0.38 (d)
	4.95	1.01 (b)			14.49	0.38 (d)
	5.77	0.92 (b)			21.74	0.38 (d)
	6.60	0.68 (b)			28.99	0.36 (d)
	6.60	0.83 (b)			43.5	0.39 (d)
	6.92	0.79 (d)			72.5	0.36 (d)
	8.24	0.85 (d)			144.9	0.33 (d)
	9.80	0.84 (d)			289.9	0.293 (d)
	11.0	0.80 (b)		5.97	5.21	0.39 (c)
	11.99	0.86 (d)			6.60	0.32 (c)
	13.6	0.83 (b)			8.05	0.41 (a)
	14.49	0.86 (d)			6.14	0.275 (a)
	21.74	0.80 (d)			3.38	0.253 (e)
	28.99	0.78 (d)		6.16	10.73	0.297 (e)
	43.5	0.83 (d)		6.50	10.76	0.297 (e)
	72.5	0.75 (d)			17.84	0.176 (c)
	144.9	0.65 (d)			6.92	0.267 (d)
3.05	3.38	0.82 (a)			8.24	0.279 (d)
3.06	4.83	0.85 (a)			11.99	0.267 (d)
3.14	10.62	0.44 (e)		6.67	14.49	0.269 (d)
3.21	9.98	0.37 (c)			21.74	0.267 (d)
3.38	10.58	0.48 (e)			28.99	0.262 (d)
3.64	6.60	0.64 (b)			43.5	0.271 (d)
	11.0	0.60 (b)			72.5	0.280 (d)
3.73	13.6	0.56 (b)			144.9	0.241 (d)
	10.62	0.47 (e)			289.9	0.252 (d)

Table 4.1 (continued)

Experimental values of η/N^2 for oxygen for $E/N < 30 \times 10^{-17} \text{ Vcm}^2$

E/N (10^{-17} Vcm^2)	N (10^{17} cm^{-3})	η/N^2 (10^{-36} cm^5)	E/N (10^{-17} Vcm^2)	N (10^{17} cm^{-3})	η/N^2 (10^{-36} cm^5)
6.81	10.62	0.263 (e)		6.92	0.55 (d)
6.97	6.92	0.259 (d)		8.24	0.49 (d)
7.06	14.34	0.181 (e)		11.99	0.38 (d)
	6.92	0.246 (d)		21.74	0.271 (d)
7.27	8.24	0.254 (d)	10.91	28.99	0.214 (d)
7.58	6.92	0.241 (d)		43.5	0.168 (d)
7.61	4.83	0.230 (a)		72.5	0.138 (d)
7.77	11.43	0.269 (e)		144.9	0.115 (d)
				289.9	0.104 (d)
	8.24	0.246 (d)			
	11.99	0.230 (d)	10.93	14.2	0.28 (a)
	14.49	0.228 (d)		43.5	0.182 (d)
	21.74	0.216 (d)	11.52	72.5	0.135 (d)
7.88	28.99	0.207 (d)		144.9	0.109 (d)
	43.5	0.209 (d)	11.62	2.45	2.20 (a)
	72.5	0.196 (d)	11.89	10.83	0.45 (e)
	144.9	0.169 (d)	12.01	3.38	2.01 (a)
	289.9	0.190 (d)	12.12	144.9	0.104 (d)
				289.9	0.092 (d)
8.42	11.33	0.232 (e)			
	6.92	0.259 (d)	12.19	8.05	0.89 (a)
	8.24	0.272 (d)		6.92	0.88 (d)
	11.99	0.233 (d)		8.24	0.81 (d)
8.49	14.49	0.221 (d)		14.49	0.47 (d)
	21.74	0.207 (d)	12.73	21.74	0.38 (d)
	28.99	0.193 (d)		28.99	0.286 (d)
	43.5	0.184 (d)		43.5	0.230 (d)
	144.9	0.150 (d)		72.5	0.146 (d)
				289.9	0.088 (d)
8.50	7.79	0.39 (e)			
	6.92	0.31 (d)	13.10	2.45	3.5 (a)
	8.24	0.31 (d)	13.31	14.34	0.40 (e)
	11.99	0.245 (d)	13.83	4.83	2.13 (a)
	14.49	0.228 (d)	13.94	289.9	0.082 (d)
9.09	21.74	0.207 (d)	14.10	10.34	0.67 (e)
	43.5	0.172 (d)	14.54	289.9	0.080 (d)
	72.5	0.159 (d)	14.88	8.05	1.85 (a)
	289.9	0.152 (d)	15.08	3.38	3.9 (a)
9.12	10.97	0.265 (e)		6.92	1.53 (d)
9.32	10.66	0.238 (e)		8.24	1.31 (d)
9.35	3.38	0.33 (a)		14.49	0.77 (d)
9.40	28.99	0.193 (d)		21.74	0.58 (d)
	21.74	0.216 (d)	15.15	28.99	0.44 (d)
	43.5	0.163 (d)		43.5	0.31 (d)
9.70	72.5	0.146 (d)		72.5	0.196 (d)
	289.9	1.24 (d)		144.9	0.117 (d)
				289.9	0.079 (d)
9.92	5.42	0.47 (e)			
10.00	28.99	0.197 (d)	15.82	5.42	1.67 (e)
10.19	2.45	1.02 (a)	16.00	2.45	6.0 (a)
	14.49	0.290 (d)	16.10	10.34	0.0108 (e)
10.31	43.5	0.163 (d)	16.50	4.83	3.4 (a)
	72.5	0.139 (d)		14.34	0.0092 (e)
	8.05	0.45 (a)	17.66	10.87	0.0126 (e)
10.75	10.58	0.35 (e)		10.87	0.0140 (e)
10.90			18.07	8.05	2.24 (a)

Table 4.1 (continued)
 Experimental values of η/N^2 for oxygen for $E/N < 30 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	N (10^{17}cm^{-3})	η/N^2 (10^{-36}cm^5)	E/N (10^{-17}Vcm^2)	N (10^{17}cm^{-3})	η/N^2 (10^{-36}cm^5)
18.19	6.92	2.25 (d)	22.63	11.12	0.0200 (e)
	8.24	1.91 (d)	23.09	4.83	4.7 (a)
	14.49	1.11 (d)	24.25	6.92	3.2 (d)
	21.74	0.78 (d)		8.24	2.75 (d)
	28.99	0.60 (d)		14.49	1.57 (d)
	43.5	0.41 (d)		21.74	1.00 (d)
	72.5	0.251 (d)		28.99	0.77 (d)
	144.9	0.130 (d)		72.5	0.31 (d)
	289.9	0.085 (d)		144.9	0.161 (d)
				289.9	0.097 (d)
18.36	17.84	0.0090 (e)			
19.10	2.45	7.9 (a)	24.31	3.38	7.2 (a)
19.49	7.79	0.0216 (e)	25.71	17.84	0.0141 (e)
19.80	4.83	4.3 (a)	25.88	4.83	5.2 (a)
20.85	3.38	6.5 (a)	27.09	3.38	7.5 (a)
21.07	11.08	0.0165 (e)			
21.22	6.92	2.76 (d)	27.28	6.92	3.4 (d)
	8.24	2.42 (d)		14.49	1.72 (d)
	14.49	1.37 (d)		21.74	1.12 (d)
	21.74	0.92 (d)		28.99	0.87 (d)
	28.99	0.71 (d)		43.5	0.63 (d)
	43.5	0.49 (d)		72.5	0.36 (d)
	72.5	0.30 (d)		144.9	0.186 (d)
	144.9	0.153 (d)		289.9	0.099 (d)
				10.73	0.0262 (e)
			28.33		

Experimental:

- (a) Chanin, *et al.*, Phys. Rev. **128**, 219 (1962).
- (b) Rees, Austr. J. Phys. **18**, 41 (1965).
- (c) Chatterton, *et al.*, J. Electron. Control **11**, 425 (1961).
- (d) Grunberg, Z. Naturforsch. **24a**, 1039 (1969).
- (e) Doehring, Z. Naturforsch. **7a**, 253 (1952).

Table 4.2
Experimental values of the thermal three-body attachment rate coefficient
for oxygen at various temperatures

T (°K)	a_3 ($10^{30} \text{ cm}^6/\text{sec}$)	T (°K)	a_3 ($10^{30} \text{ cm}^6/\text{sec}$)	T (°K)	a_3 ($10^{30} \text{ cm}^6/\text{sec}$)
77	<1 (a)	300	2.8 (a)	370	3.1 (a)
113	0.72 (b)		2.0 (c)	530	2.8 (c)
195	2 (a)		3.0 (d)		
200	1.5 (b)		2.3 (e)		
			2.4 (f)		
			2.12 (g)		
			2.2 (h)		

Experimental:

- (a) Chanin, *et al.*, Phys. Rev. **128**, 219 (1962).
 (b) Truby, Phys. Rev. A **6**, 671 (1972).
 (c) Pack and Phelps, J. Chem. Phys. **44**, 1870 (1966).
 (d) Brodski, *et al.*, Sov. Phys. Tech. Phys. **11**, 498 (1966).
 (e) Hackam, *et al.*, Proc. Phys. Soc. (London) **86**, 123 (1965).
 (f) Lennon, *et al.*, Proc. Phys. Soc. (London) **78**, 1543 (1961).
 (g) Hirsh, *et al.*, Phys. Rev. **178**, 175 (1969).
 (h) Warman, *et al.*, Chem. Phys. Lett. **12**, 211 (1971).

Table 4.3

Experimental values of η/N for oxygen for $E/N > 30 \times 10^{-17} \text{ Vcm}^2$

E/N (10^{-17} Vcm^2)	η/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	η/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	η/N (10^{-18} cm^2)
30.3	2.5 (a)	42.3	2.5 (c)	53.0	2.5 (a)
	1.28 (b)	42.4	1.39 (b)	54.5	1.06 (b)
33.3	0.23 (b)	45.5	2.6 (a)	60.6	2.4 (c)
36.4	1.36 (b)		0.71 (b)	72.1	2.2 (a)
37.9	2.6 (a)	48.5	2.5 (c)		1.9 (c)
39.4	1.46 (b)		1.42 (b)		

Experimental:

- (a) Huxley, *et al.*, Austr. J. Phys. **12**, 303 (1959).
 (b) Grunberg, Z. Naturforsch. **24a**, 1039 (1969).
 (c) Chatterton, *et al.*, J. Electron. Control **11**, 425 (1961).

Table 4.4

Experimental values of η/N^2 for nitric oxide

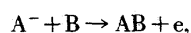
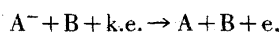
E/N (10^{-17} Vcm^2)	N (10^{-18} cm^{-3})	η/N^2 (10^{-37} cm^5)	E/N (10^{-17} Vcm^2)	N (10^{-18} cm^{-3})	η/N^2 (10^{-37} cm^5)
0.927	5.25	10.1	4.98	5.25	3.8
1.421	5.25	8.4	5.68	3.9	3.6
1.963	5.25	6.6	8.09	3.9	2.5
2.160	3.45	5.7	8.58	3.45	2.06
2.738	5.25	5.2	9.87	3.9	2.24
2.786	3.45	6.0	10.47	2.65	1.90
3.06	5.25	5.9	11.65	2.65	1.36
3.82	5.25	4.3	13.15	2.65	1.63
4.28	3.45	3.4	14.58	2.65	1.63
4.44	2.2	3.8	15.50	2.2	1.08
			17.43	2.2	0.70

Experimental:

All data taken from Parkes, *et al.*, J. Chem. Soc. Faraday Trans. II **68**, 600 (1972).

3.5. Detachment Coefficient

In section 3.4 a number of processes are listed by which, in an electronegative gas, negative ions can be produced in an electron swarm by collisions between electrons and the gas atoms or molecules. These processes of electron attachment are all reversible giving rise to corresponding processes of electron detachment in which electrons are removed from negative ions by providing the energy difference required. In swarm conditions, the only processes of detachment that are significant are those resulting from collisions of the negative ions with neutral gas particles and these processes can be represented by the equations



which are the reverse of three-body and dissociative attachment, respectively.

Of the gases considered in this review, values of the detachment rate coefficients and of the detachment coefficient have been obtained only (i) for oxygen negative ions, either in oxygen or in mixtures of oxygen with other gases, particularly air and (ii) for nitric oxide negative ions in mixtures of nitric oxide and oxygen. The data obtained are discussed in the next two subsections.

3.5.a. Detachment Coefficients—Oxygen

At low values of E/N for which ionization is negligible, the continuity equations for an electron swarm moving in an electronegative gas in which attachment and detachment are occurring may, neglecting diffusion, be written

$$\frac{\partial n}{\partial t} + W \frac{\partial n}{\partial x} = -an + dN_-$$

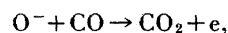
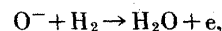
$$\frac{\partial N_-}{\partial t} + W_- \frac{\partial N_-}{\partial x} = an - dN_-$$

where n and N_- are the electron and negative ion number densities and W and W_- are the electron and negative ion drift velocities, respectively.

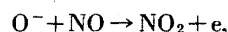
These equations were used by Pack and Phelps [1662] as the basis of the analysis of the wave-form of the current observed to result from the liberation of a pulse of electrons from the cathode of a drift tube. The value of E/N in the drift tube was maintained at a value sufficiently low ($\sim 3 \times 10^{-17}$ V cm²) for the negative ions to remain in thermal equilibrium with the gas molecules. In this way, values of the thermal detachment rate coefficient were obtained over the range $375 < T < 573$ K and for values of N from about 10^{17} to 10^{19} cm⁻³. These values are shown as a graph of d , T in figure 5.1 and given in table 5.1.

Detachment rate coefficients have also been determined for several mixtures of oxygen with small amounts

of other gases. These were obtained using flowing afterglow techniques [3042] and from measurements in drift tubes of mass identified ion currents [5700] and of the time-resolved electron component of pulses generated by photoelectric emission from the cathode [3352]. The values obtained from the drift tube measurements for the detachment rate coefficients for the associative detachment reactions



and



are shown as functions of E/N in figures 5.2, 5.3, and 5.4.

Using Wannier's expression (G. Wannier, Phys. Rev. **83**, 281, 1951; **89**, 795, 1952) for the mean energy of the negative ions, the range of values of mean energy corresponding to the range of experimental values of E/N was found to be about 0.04 to 0.7 eV. The results are thus in accord with the results of the afterglow studies [3042] which gave values for the associative detachment rates at thermal energy (0.039 eV at 300 K) of 7.6×10^{-10} cm³ s⁻¹ for H₂, 5.4×10^{-10} cm³ s⁻¹ for CO and 3.2×10^{-10} cm³ s⁻¹ for NO. (These values are updated values given graphically in [3352].)

Both drift tube and flowing afterglow studies show that the associative detachment rate for O⁻ in nitrogen is very low, an estimate $\sim 10^{-19}$ cm³ s⁻¹ being given in [3352].

Microwave measurements on a decaying afterglow in relative large admixtures (10 to 20 percent) of oxygen in nitrogen at a nitrogen pressure of about 9 Torr were made by Hackam and Lennon [1633]. The variation of decay rate with the oxygen content of the mixture was interpreted to give a value of $d = 8.4 \times 10^{-15}$ cm³ s⁻¹ at $T = 296$ K for the detachment rate coefficient of oxygen negative ions in collision with nitrogen molecules.

At higher values of E/N , when ionization becomes a significant process, attempts have been made to obtain values of the detachment coefficient both from measurements on pulsed [1444, 2370] and steady-state [2462, 2555, 4948] electron swarms. The analysis of the experimental data has, however, proved difficult, because of the large number of significant processes occurring and their relative magnitudes. The solution of the appropriate continuity equations in the steady state is given by equation (16). (See sec. 3.7.c.)

Analysis of the steady-state data has been carried out using this equation and various simplifying assumptions (e.g., that secondary ionization and ion conversion are negligible). With this assumption it was found that the detachment coefficient δ for $100 \times 10^{-17} < E/N < 140 \times 10^{-17}$ V cm² is small compared with the attachment coefficient η , values of δ/N in the range from zero to about 3×10^{-19} cm² having been obtained, depending on the conditions. Unfortunately, when

δ/η is small there is no significant difference in the best fit to the experimental data for a wide range of combinations of the coefficients which include the possibility of $\delta=0$. In a recent reanalysis [4947] of the steady-state data, it has been suggested that inclusion of ion conversion would lead to values of δ compatible with the much higher values ($\delta/N \sim 1.5 \times 10^{-17} \text{ cm}^2$) found by analysis [2370] of the pulse data. It should be pointed out, however, that the inclusion of ion conversion in the steady-state analysis enables an even wider range of combinations of coefficients to be used to give agreement with the experimental data and the possibility of $\delta=0$ is still not excluded.

That δ/p is finite but small in this region of E/N was definitely established by further experiments [2994, 4920, 4949] on the steady-state spatial growth of current resulting from an initial negative-ion current. The values obtained for δ/N were $\sim 3 \times 10^{-19} \text{ cm}^2$ at $p=20$ Torr, decreasing as the pressure increased to 60 Torr, but no detailed agreement has yet been obtained between different sets of data obtained using this procedure. Moreover none of the analyses used included either the effect of ion conversion reactions or of the presence of electrons mixed with the initial negative ion current. Comparison of these experiments is further complicated by the fact that the proportion of the ion species O^- , O_2^- , and O_3^- in the initial negative-ion current depends on the conditions in the sources which were different. To obtain definitive results on detachment coefficients in oxygen for $E/N > 99 \times 10^{-17} \text{ V cm}^2$ requires further work, preferably in a system in which the ion species can be identified.

3.5.b. Detachment Coefficient—Nitric Oxide

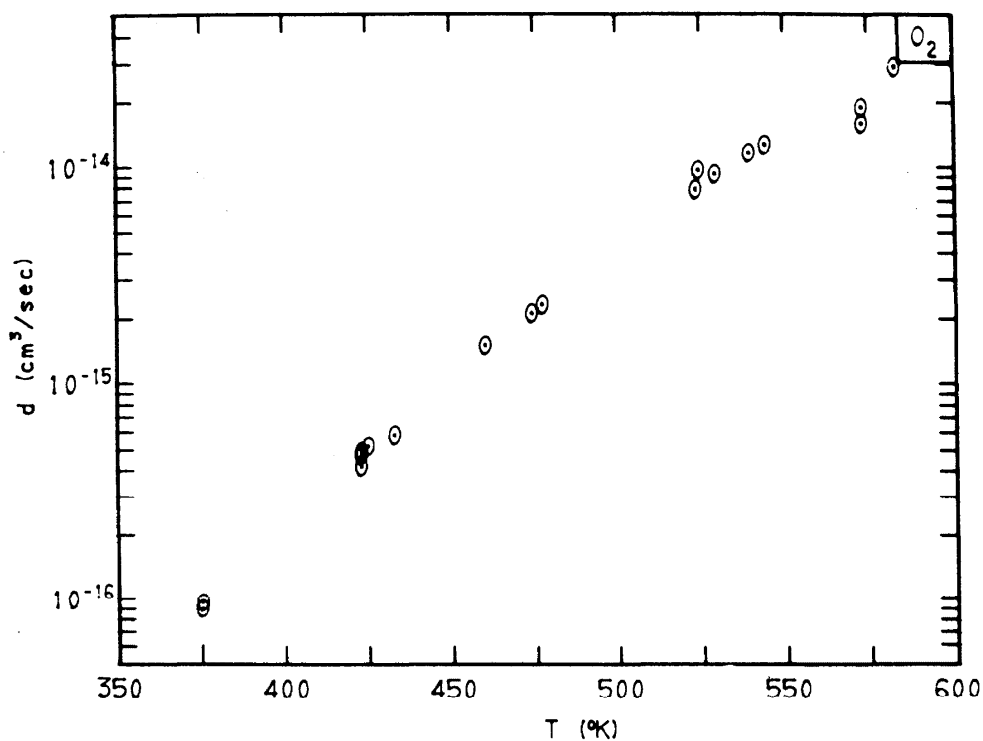
Parkes and Sugden [4943] obtained a value of

$(5 \pm 1) \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$ for the detachment coefficient of near-thermal NO^- ions from an analysis of experiments in which the mass identified ion current at the end of a drift tube was measured as a function of oxygen concentration in NO/O mixtures at $E/N = 3.1 \times 10^{-17} \text{ V cm}^2$.

3.5.c. Detachment Coefficient—Air

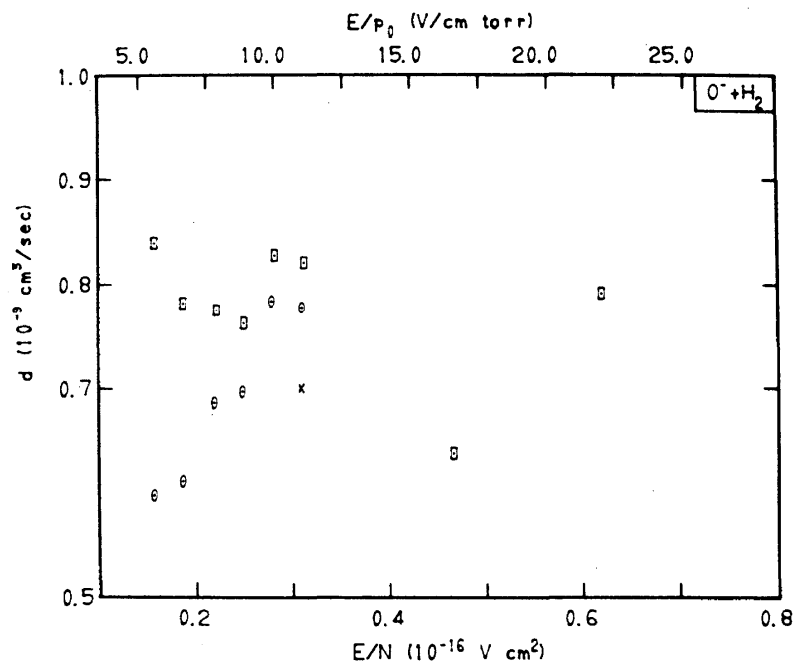
There are no values for the detachment coefficient at low values of E/N in air.

At values of $E/N > 85 \times 10^{-17} \text{ V cm}^2$ where ionization becomes significant, measurements have been made of both the spatial growth of an electron swarm in the steady state [791, 2047, 2407, 2919] and of the temporal growth under impulse conditions [2235, 2370, 4947] for $85 \times 10^{-17} < E/N < 140 \times 10^{-17} \text{ V cm}^2$. The same difficulties of analysis arise as in the case of oxygen so that the values of the detachment coefficient are not well defined. The values obtained are very dependent on the assumptions made in the analysis, and even for a given set of assumptions agreement between theory and experiment can be obtained for a wide range of combinations of values of the coefficients involved. Thus values of δ/N ranging from zero to $1.5 \times 10^{-17} \text{ cm}^2$ have been shown [791, 2047, 2407, 4947] to be compatible with the steady-state experimental results, while three sets of impulse data [2235, 2370, 4947] have been analyzed to give values of δ/N in the upper end of this range, but without any indication of the range of values of the coefficients which could be used to fit the experimental data. So far, no experiments on the development of an initial negative-ion current such as those carried out in oxygen have been reported for air.



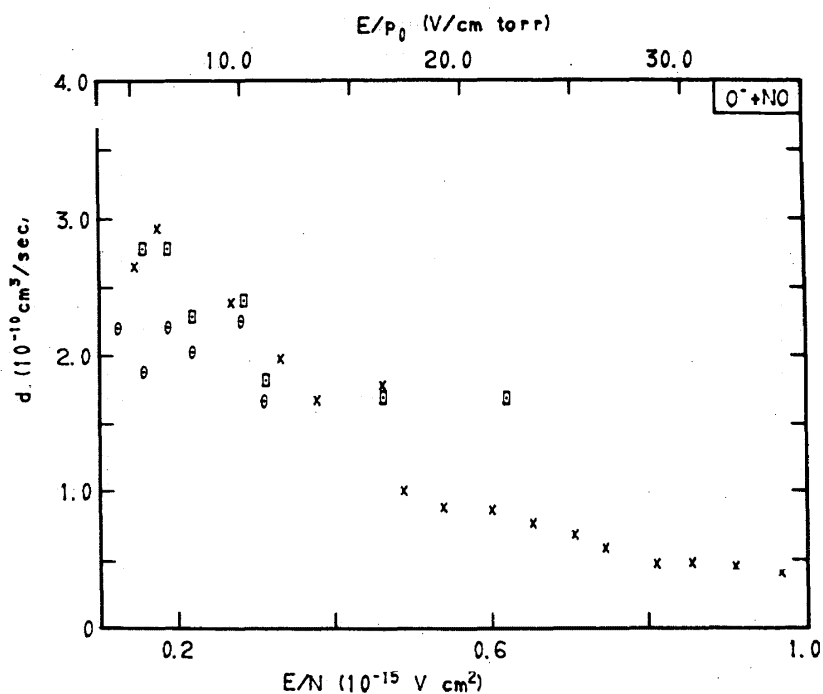
Experimental: O Pack(1662)

FIGURE 5.1. Thermal detachment rate coefficients for oxygen ions in oxygen as a function of temperature.



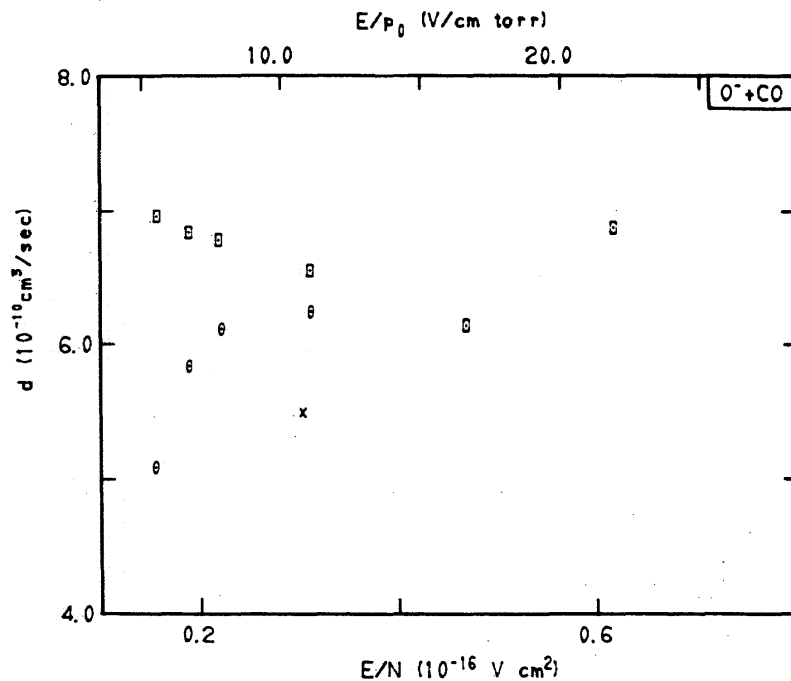
Experimental: □ Moruzzi(3352) using (η/N) from (649);
 O Moruzzi(3352) using (η/N) from (1383);
 x Moruzzi(5700) as corrected in (3352).

FIGURE 5.2. Associative detachment rate coefficient for the reaction $O^- + H_2 \rightarrow H_2O + e$ as a function of E/N .



Experimental: $\square$ Moruzzi(3352) using (η/N) from (649);
 $\circ$ Moruzzi(3352) using (η/N) from (1383);
 $\times$ Moruzzi(5700) as corrected in (3352).

FIGURE 5.3. Associative detachment rate coefficient for the reaction $\text{O}^- + \text{NO} \rightarrow \text{NO}_2 + e$ as a function of E/N .



Experimental: $\square$ Moruzzi(3352) using (η/N) from (649);
 $\circ$ Moruzzi(3352) using (η/N) from (1383);
 $\times$ Moruzzi(5700) as corrected in (3352).

FIGURE 5.4. Associative detachment rate coefficients for the reaction $\text{O}^- + \text{CO} \rightarrow \text{CO}_2 + e$ as a function of E/N .

Table 5.1

Thermal detachment coefficients for oxygen

Temperature (°K)	d ($10^{-15} \text{ cm}^3/\text{sec}$)	Temperature (°K)	d ($10^{-15} \text{ cm}^3/\text{sec}$)	Temperature (°K)	d ($10^{-15} \text{ cm}^3/\text{sec}$)
375.	0.090	433.	0.58	529.	9.3
	0.096	460.	1.5	539.	11.7
	0.42	474.	2.1	544.	12.7
423.	0.47	477.	2.3	573.	16.
	0.49	523.	7.9		19.
425.	0.52	524.	9.7	583.	29.

Experimental:

All data taken from Pack, *et al.*, J. Chem. Phys. **44**, 1870 (1966).

3.6. Excitation Coefficient

Some of the energy gained from the field by the electrons in an electron swarm is given up in inelastic collisions which lead to the excitation of the gas atoms or molecules. The number of excitations produced by n electrons moving a distance, dx , in the field direction is $n\epsilon dx$, where ϵ is the excitation coefficient.

If the excited states produce decay with the emission of radiation, measurements of the light emitted from the swarm, as a function of the distance from the cathode of a uniform field electrode system, may be used to determine ϵ_r , the excitation coefficient to radiative states. The light emitted by any given excited level at the currents ($i < 10^{-7}$ A) existing in electron swarms is of very low intensity. Thus, most measurements have been concerned with the determination of the total radiation emitted which leads to an average value of ϵ_r for all the radiating states involved. In some experiments [1761, 2229] however, the value of ϵ_r for excitation to states giving rise to particular radiations has been obtained by measuring the light emitted from the whole of a discharge in a narrow range of wavelengths.

In molecular gases, excitation may occur to repulsive states which leads to dissociation. Measurement of the number of dissociations produced by a given current can then be used to evaluate ϵ_d , the excitation coefficient to dissociating states, which is often termed, simply, the dissociation coefficient.

Of the gases with which this data survey is concerned, excitation coefficients have been obtained experimentally for H_2 , N_2 , O_2 and CO_2 and have been calculated theoretically for He and Ne. The values obtained are given in the succeeding subsections.

3.6.a. Excitation Coefficient—Helium

When an electron swarm moves through helium both metastable and radiating states are excited. There are no published experimental values of excitation coefficients for helium, but there have been a number [336, 3897, 4057, 5701, 4909] of theoretical calculations both of the total excitation coefficient and of the partial excitation coefficients to metastable states and to

radiating states. The values obtained are shown in figures 6.1 and 6.2 and given in table 6.1 (epsilon in the figures $\equiv \epsilon$ in the text).

All these values of ϵ are based on the published data [119, 318, 2010] (see also H. S. W. Massey and E. H. S. Burhop, *Electronic and Ionic Impact Phenomena*, Clarendon Press, Oxford, 1969, 2nd Ed., Vol. 1) on the cross sections involved. In [336, 4057, 5701, and 4909] analytical methods were used, based in one case [336] on a previously published [2010] energy distribution function, and in the others on distributions obtained for the purpose. In [3897] a Monte Carlo method was used.

3.6.b. Excitation Coefficient—Neon

The only available published information on excitation coefficients for electron swarms in neon are the values of the total excitation coefficient calculated by Hughes [4909] and the approximate values for the excitation coefficient to the $^3\text{P}_2$ level calculated in [2501] using expressions given in [3552]. These values are shown in figure 6.3, the values for the excitation coefficient also being tabulated in table 6.2.

3.6.c. Excitation Coefficient—Hydrogen

The radiation produced by an electron swarm moving through hydrogen lies mostly in the far ultraviolet (1000 to 1200 Å), resulting from the excitation of the $2^1\Pi_u$ state. A direct measurement of the excitation coefficient was made [354] by determining, at various distances from the cathode, the far-ultraviolet radiation emitted from a small length Δx of an electron avalanche (current $i < 10^{-8}$ A) established between parallel plates. These measurements gave values both of the excitation coefficient ϵ_r , which are shown in figure 6.4, and given in table 6.3, and of the ionization coefficient α . The values of α so obtained were in good agreement with those obtained using the Townsend method (see fig. 7.5).

In other experiments, the radiation emitted by the whole discharge, either axially from an avalanche with $i \leq 10^{-8}$ A [921] or radially from a self-sustaining Townsend discharge with $1 \mu\text{A} < i < 10 \mu\text{A}$ [3722], was measured together with the value of the current flowing.

both these experiments the emission at a constant value of E/N was found to depend on the gas pressure ($1 < p < 100$ Torr), because of the quenching of radiation in collisions of the second kind between excited and neutral gas molecules. When the results were corrected for this pressure effect they gave values of ϵ_r/α . In [921], absolute values of ϵ_r/α were obtained and converted into the values of ϵ_r/N shown in figure 6.4 (and given in table 6.3) by using the values of α/N given in [1356]. In [3722], however, relative values of ϵ_r/α were obtained, converted to relative values of ϵ_r/N using values of α/N given in [2150], and then normalized to give the values shown in figure 6.4. This normalization was carried out by adjusting the value of ϵ_r/N to agree with the absolute values of [354] and of [921] at low values of E/N .

Observations [1761] of the radiation from a self-sustained Townsend discharge, at currents up to 300 μ A, using interference filters gave values of ϵ_r for the excitation of states leading to the emission of H_α and H_β radiation and these are shown in figure 6.5.

Comparison of figures 6.5 and 6.4 shows that the number of molecules excited so as to give H_α and H_β radiation is much smaller than those excited to the $2^1\Pi_u$ state.

On the other hand, the dissociation coefficient ϵ_d , which is essentially a measure of the excitation coefficient to the $3\Sigma_u^+$ and $3\Sigma_g^+$ states is much larger than ϵ_r for the $2^1\Pi_u$ state, as can be seen from figures 6.6 and 6.4. Figure 6.6 and table 6.4 show the values of ϵ_d obtained from measurements of the number of molecules dissociated by a radiofrequency (5 MHz) discharge [354] and by a dc discharge [4950]. A maximum value of ϵ_d/E of 6.5×10^{-2} , which is close to that given in figure 6.6 was also obtained [4951] from measurements on the dissociation in a discharge at a frequency of 3000 MHz. Fair agreement between the experimental values of ϵ_d in figure 6.6 and those calculated [4950], on the assumption of a Maxwellian energy distribution, by the method given in [2109] can be obtained, provided the cross section for excitation to the $3\Sigma_u^+$ and $3\Sigma_g^+$ levels is assumed to be about a factor of 2.6 times greater than the theoretical value of about $0.2 \pi a_0^2$, obtained by a wave mechanical calculation [see 2109 and 4951].

Values of $(\epsilon_r + \epsilon_d)$ calculated [260] using cross sections obtained from numerical integration of the Boltzmann equation and comparison with swarm data are shown in figure 6.7 together with the experimental values obtained by the addition of the values of ϵ_r and ϵ_d given in figures 6.4 and 6.6.

It should be noted that the concept of a swarm coefficient ϵ/N at high values of E/N (say $> 4 \times 10^{-15}$ V cm^2 for hydrogen) is questionable, because equilibrium conditions are not established in the experiments.

3.6.d. Excitation Coefficient—Nitrogen

When an electron avalanche moves through nitrogen, ultraviolet radiation with wavelength in the range 3400

to 3800 Å, corresponding to the second positive group which involves $C^3\Pi_u \rightarrow B^3\Pi_g$ transitions, is emitted. Absolute measurements were made in [921] of the intensity of this radiation emitted axially from avalanches in which the current was $< 10^{-8}$ A. These showed that for 1 Torr $< p < 10$ Torr the intensity was independent of pressure at a given value of E/N . At higher pressures (10–100 Torr), however, quenching of the radiation in collisions of the second kind led to a decrease in emission with increasing pressure at a given E/N . For this region, the value of ϵ_r/α was obtained from the measured coefficient $(\epsilon_r/\alpha)_m$ say, using the relationship

$$\epsilon_r/\alpha = (\epsilon_r/\alpha)_m (1 + p/p_q),$$

where p is the pressure and p_q is a constant (found to be 60 Torr for nitrogen), called the “quenching pressure.” The pressure p_q is that at which one half of the excited molecules produced are destroyed by quenching. The values obtained for ϵ_r/α from this equation gave a single curve for ϵ_r/α , E/N , which was independent of pressure and in close agreement with the values obtained at low pressures. These values are shown in figure 6.8. Also shown in figure 6.8 are values of ϵ_r/α computed [921] assuming a Maxwellian energy distribution for the electrons, a maximum cross section for excitation of 0.88×10^{-16} cm^2 and an excitation potential of 11.04 eV.

The experimental values of ϵ_r/α were converted to the values of ϵ_r/N shown in figure 6.9 by using the values of α/N given in [2556]. These values of ϵ_r/N , which are tabulated in table 6.5, have recently been confirmed in other similar experiments [4993]. Also shown in figure 6.9 are values computed [218] using a self-consistent set of cross sections obtained by solving the Boltzmann equation.

3.6.e. Excitation Coefficient—Oxygen

Measurements [3268] of the intensity of radiation in the vacuum ultraviolet from electron avalanches, in which the current densities were less than 10^{-8} A/ cm^2 , in O_2 gave the values of ϵ_r/α shown in figure 6.10. The intensity of radiation was found to be dependent on pressure for $0.5 < p < 20$ Torr because of quenching, and the experimental results were interpreted using the equation given in section 3.6.d with the quenching pressure $p_q = 2.5$ Torr.

When the results for ϵ_r/α were published, there was doubt about the values of α/N in oxygen so that no values of ϵ_r/N were given. Reliable values of α/N have recently been obtained [4994], however, and those have here been used to convert the values of ϵ_r/α in figure 6.10 to give the values of ϵ_r/N for the range of E/N which the results overlap in figure 6.11.

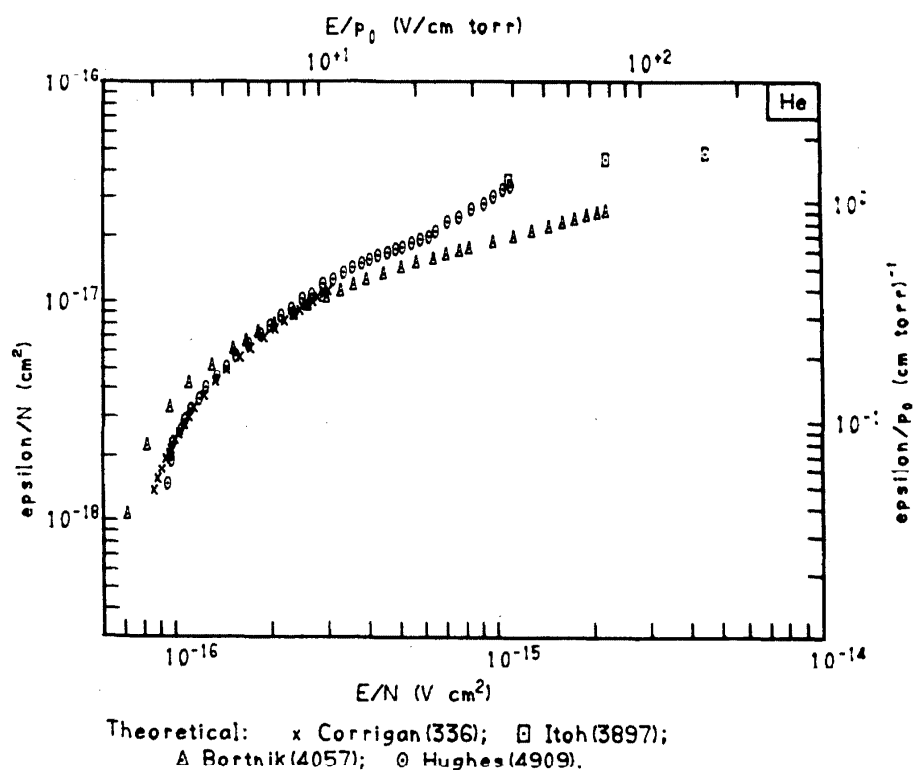
3.6.f. Excitation Coefficient—Carbon Dioxide

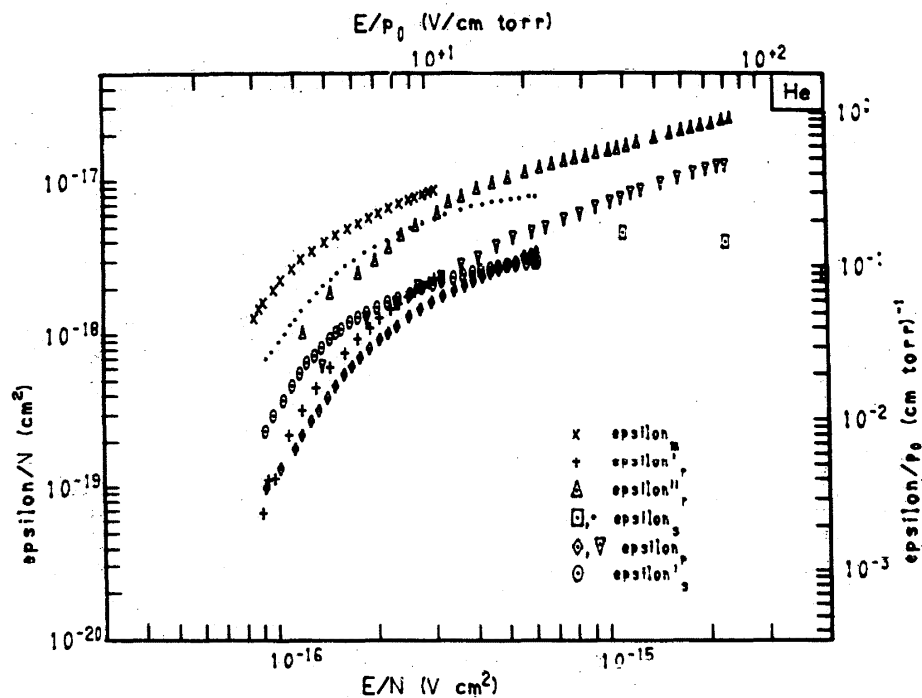
Measurements [4993] of the vacuum ultraviolet emission produced by electron avalanches in carbon dioxide showed that at a given value of E/N , the intensity

decreased with increasing pressure for $1 < p < 10$ Torr. A single curve of ϵ_r/α , E/N was obtained using the expression given in section 3.6.d and a quenching pressure of 3.6 Torr. From this curve, unspecified values of α were used to obtain the curve of ϵ_r/N , E/N shown in figure 6.12 and given in table 6.6(a).

Measurements of the increase in total pressure and of the pressure of the permanent gas products resulting from running a glow discharge in CO_2 for a given time

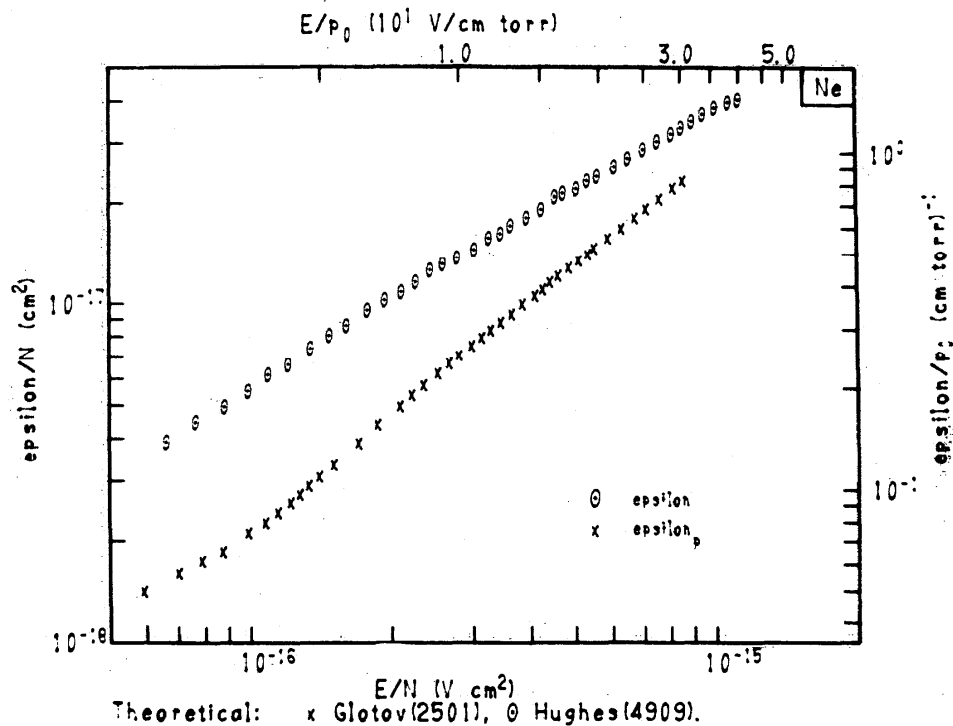
were used [5702] to obtain the values of the dissociation coefficient in CO_2 shown in figure 6.13 and given in table 6.6(b). A dissociative rate coefficient of $7 \times 10^{-15} \text{ cm}^3 \text{ s}^{-1}$ was obtained [5607] from measurements of the dependence of the fractional concentration of CO on flow velocity and position in a CO_2 laser for which the value of E/N on the axis was about $2.2 \times 10^{-16} \text{ V cm}^2$. This value is stated [5607] to be in good agreement with that calculated using the cross section in [2553] and with the data in [5702].





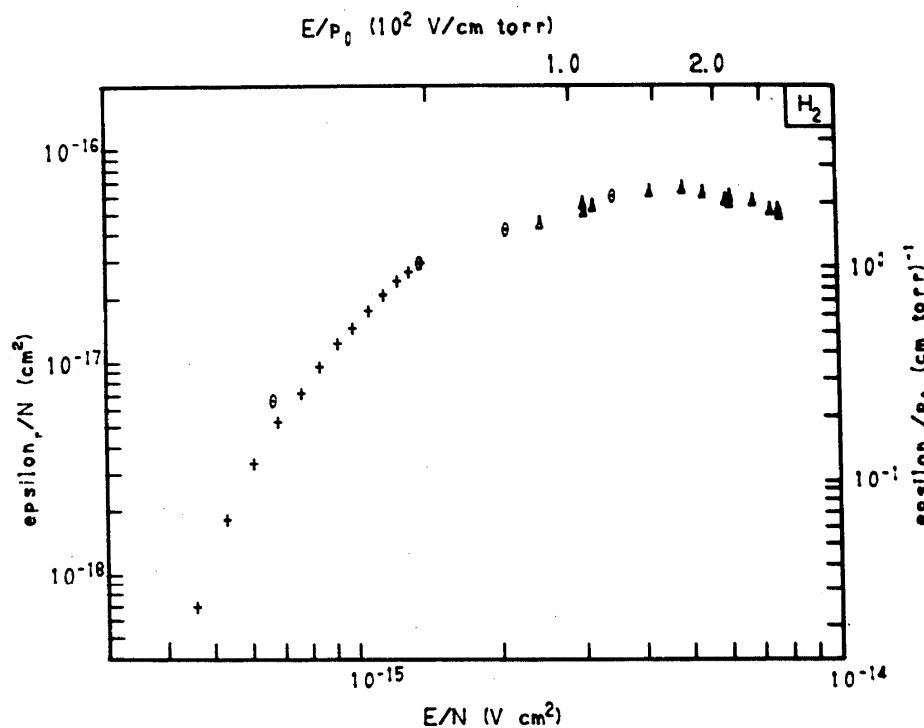
Theoretical: $\times$, + Corrigan(336); $\square$ Itoh(3897);
 Δ , ∇ Bortnik(4057); $\circ$, $\diamond$ Phelps(5701).

FIGURE 6.2. Theoretical values of partial excitation coefficients to various levels and groups of levels in helium. (ϵ , total excitation coefficient; ϵ_m , sum of excitation coefficients to the 2^3S , 2^1S and 2^3P levels; ϵ'_s , sum of the excitation coefficients to all states except the 2^3S , 2^1S and 2^3P levels; ϵ'' , sum of excitation coefficients to all states except the 2^3S and 2^1S levels; ϵ_p , excitation coefficient to the 2^1P level; ϵ_s , excitation to the 2^3S level; ϵ'_s , excitation to the 2^1S level.)



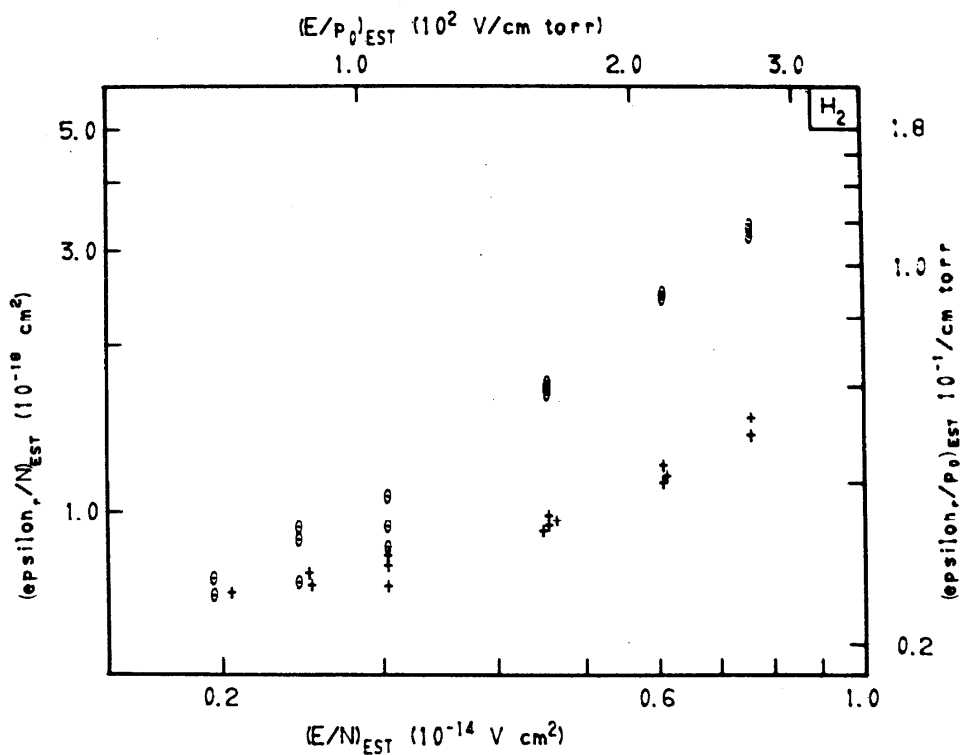
Theoretical: $\times$ Glotov(2501), $\circ$ Hughes(4909).

FIGURE 6.3. Theoretical values of the total excitation coefficient (ϵ) and the excitation coefficient to the 3P_2 level (ϵ_p) of neon. (A temperature of 0 °C was assumed for converting the value of E/p in [2501] to E/N .)



Experimental: $\circ$ Corrigan(354); $+$ Legler(921);
 Δ Nygaard(3722)

FIGURE 6.4. Experimental values of ϵ_r/N as a function of E/N for the $2^1\Pi_u$ state of hydrogen. (The temperature has been assumed to be 20 °C for the results given in [354] and [3722].)



Experimental: $\circ$ H_α ; $+$ H_β Nygaard(1761).

FIGURE 6.5. Experimental $(\epsilon_r/N)_{EST}$, $(E/N)_{EST}$ for excitations leading to the emission of H_α and H_β radiation. (A temperature of 20 °C was assumed.)

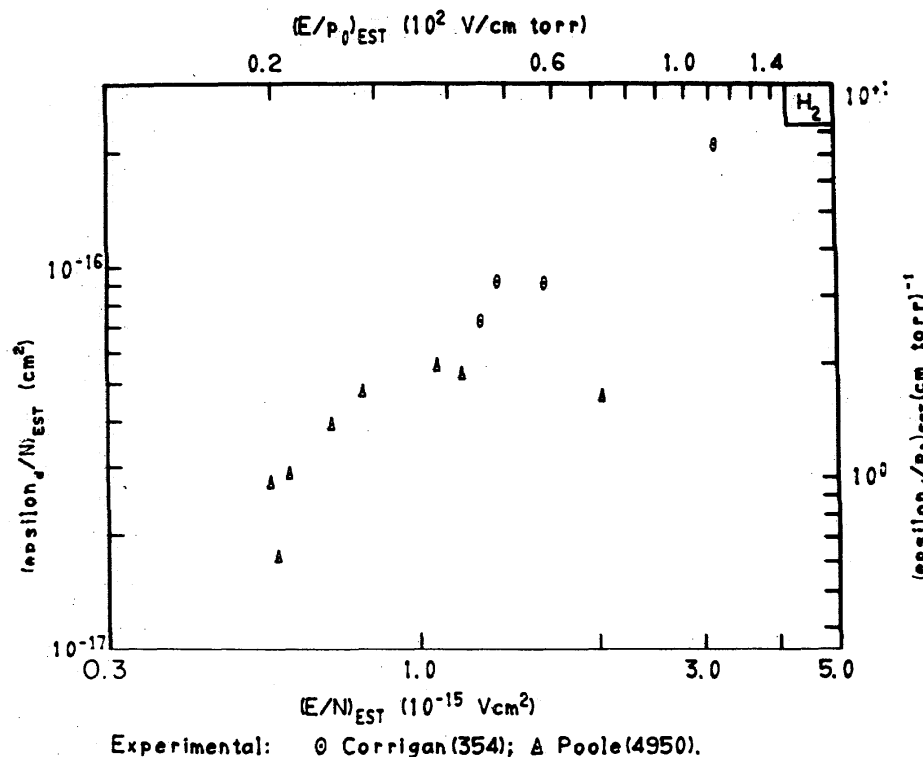


FIGURE 6.6. Experimental values of the excitation coefficient leading to dissociation (dissociation coefficient) $(\epsilon_d/N)_{EST}$ as a function of $(E/N)_{EST}$ (The temperature was assumed to be 20 °C.).

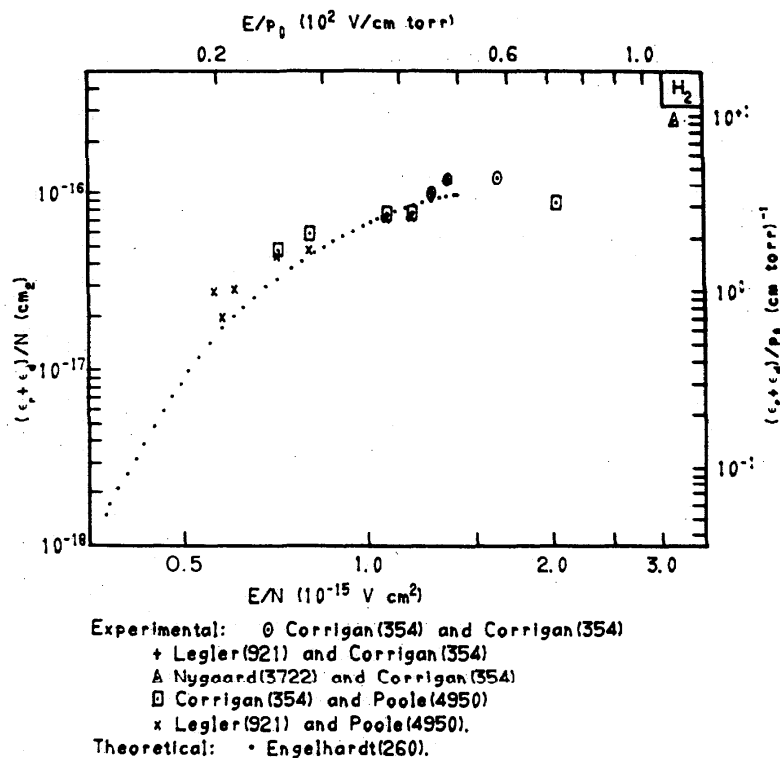
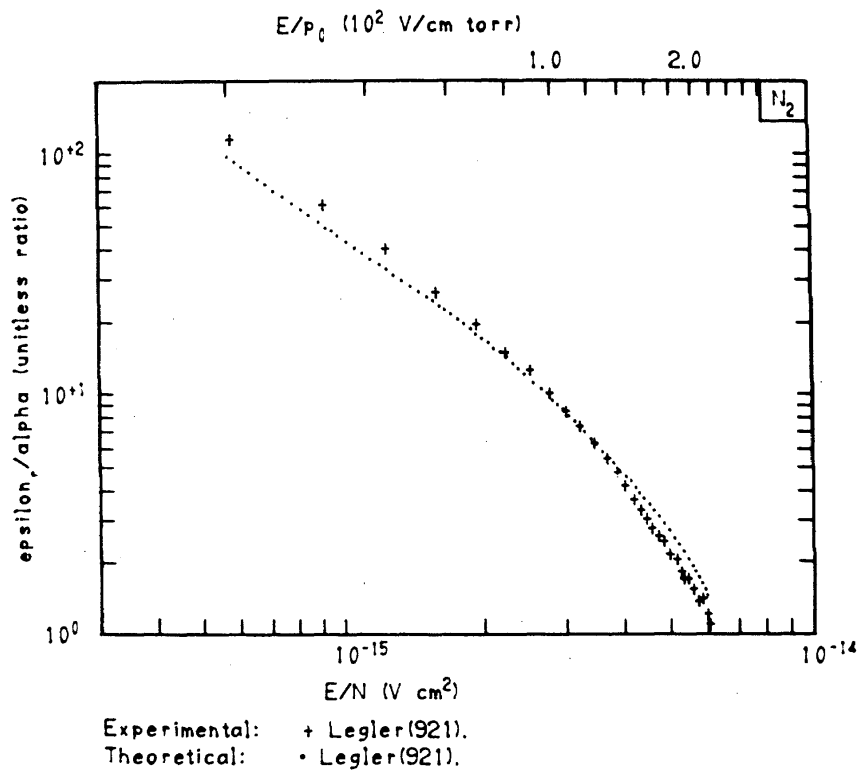
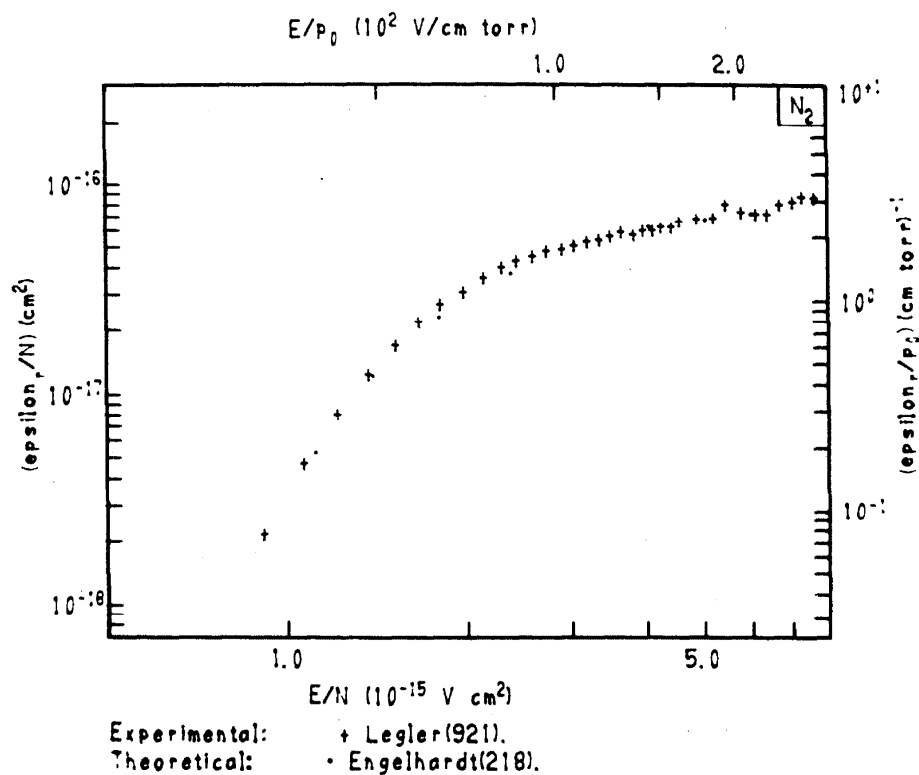


FIGURE 6.7. Comparison of theoretical and experimental values of the total excitation coefficient $(\epsilon_r + \epsilon_d)/N$ for hydrogen.

FIGURE 6.8. Comparison of experimental and theoretical values of ϵ_r/α for nitrogen.FIGURE 6.9. Comparison of experimental and theoretical values of ϵ_r/N , E/N for excitations to the $C^3\Pi_u$ state of nitrogen.

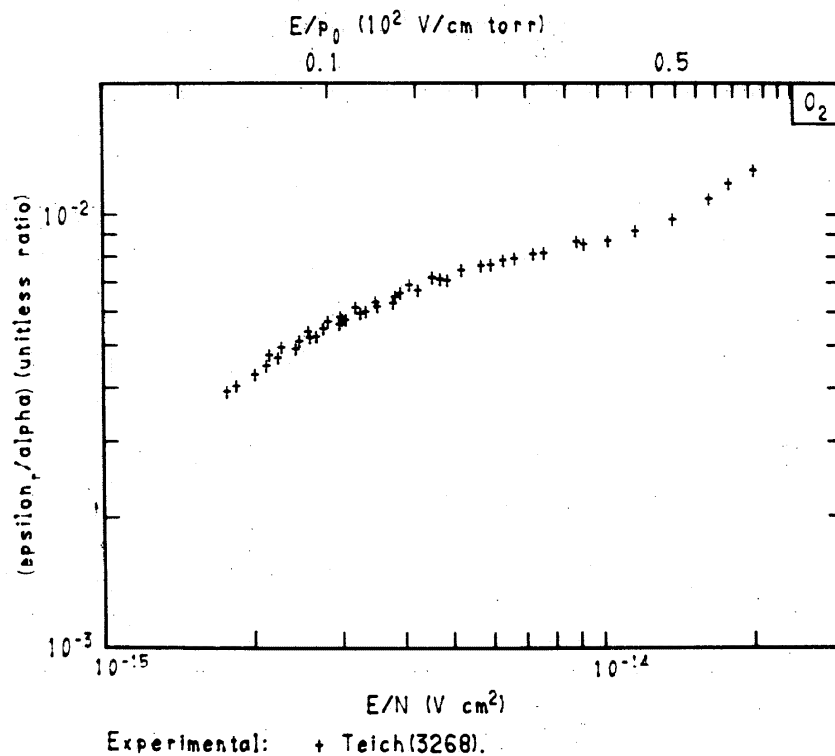


FIGURE 6.10. Experimental values of ϵ_r/α , E/N for excitation to states of oxygen leading to the emission of vacuum ultraviolet radiation.

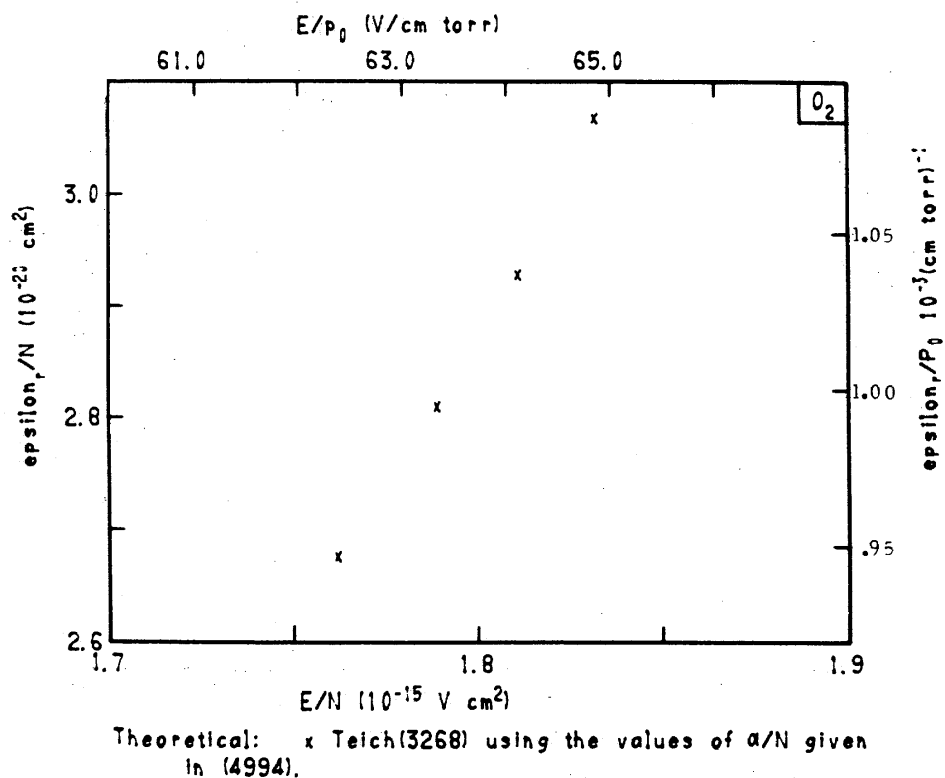


FIGURE 6.11. Values of ϵ_r/N , E/N for oxygen obtained from ϵ_r/α in [3268] and α/N in [4994].

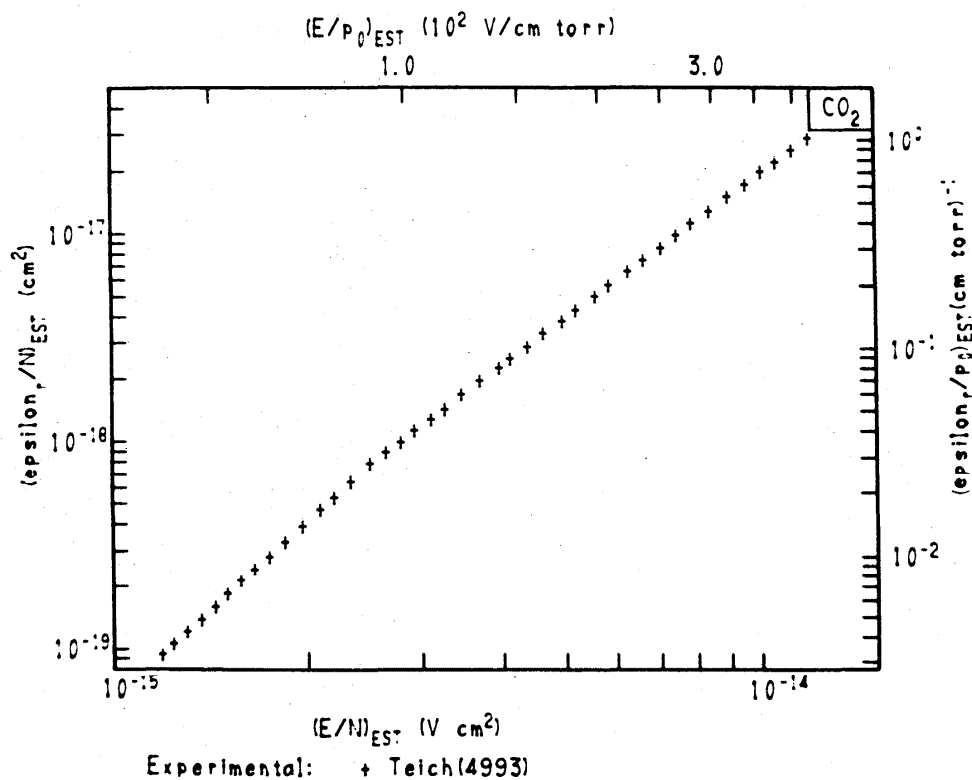


FIGURE 6.12. Experimental values of $(\epsilon_r/N)_{\text{EST}}$, $(E/N)_{\text{EST}}$ for the excitation of states of CO_2 leading to vacuum ultraviolet radiation. (A temperature of 20°C has been assumed for converting p to N .)

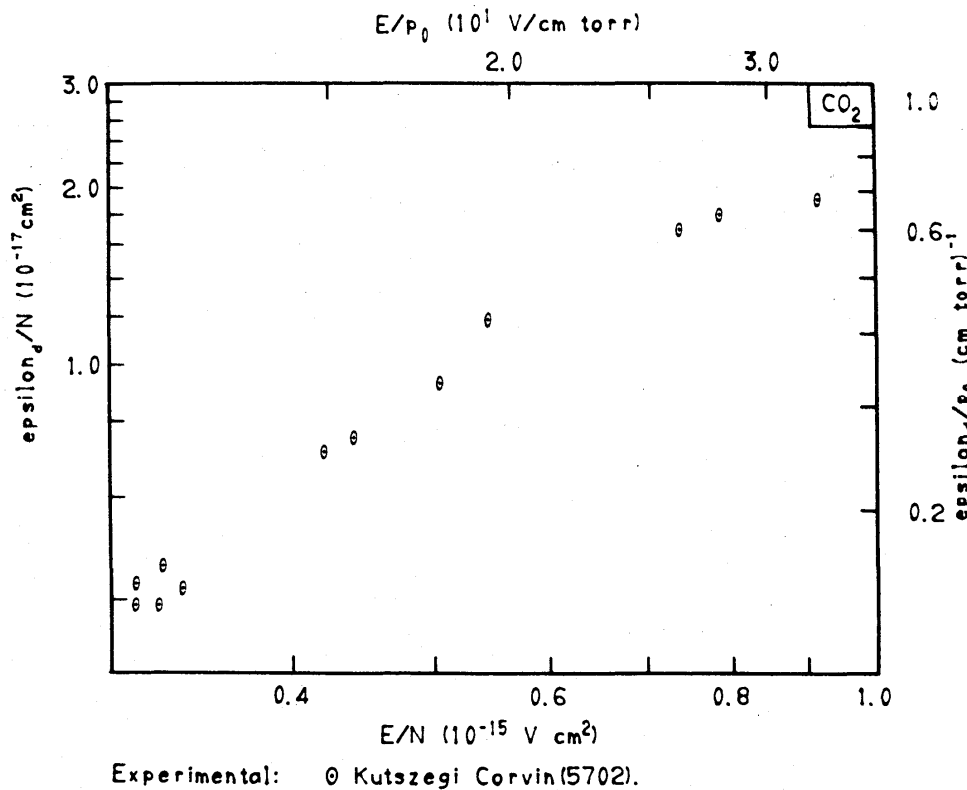


FIGURE 6.13. Experimental values of ϵ_d/N , E/N for carbon dioxide.

Table 6.1

Theoretical values of the total and partial excitation coefficients for helium *

E/N (10^{-17} V cm^2)	ϵ_m/N (10^{-18} cm^2)	ϵ'_S/N (10^{-18} cm^2)	ϵ'_r/N (10^{-18} cm^2)	ϵ''_r/N (10^{-18} cm^2)	ϵ_p/N (10^{-18} cm^2)	ϵ_S/N (10^{-18} cm^2)	ϵ/N (10^{-18} cm^2)
8.8	1.54 (1)		0.067 (1)				1.61 (1)
8.96							2.67 (2)
9.1		0.236 (3)					
9.2					0.098 (3)	0.70 (3)	
9.5							1.46 (4)
9.7		0.298 (3)					
9.8	2.05 (1)		0.112 (1)				2.16 (1)
10.1					0.132 (3)	0.90 (3)	
10.4		0.38 (3)					2.62 (4)
10.7	2.60 (1)		0.218 (1)				2.82 (1)
14.23				1.92 (2)	0.86 (2)		5.8 (2)
19.7							7.9 (4)
19.8						3.7 (3)	
20.0		1.53 (3)					
20.3	6.4 (1)		1.27 (1)				7.7 (1)
20.4					0.94 (3)		
27.40				5.7 (2)	2.14 (2)		10.0 (2)
28.8							12.1 (4)
29.9						5.7 (3)	
30.3	9.2 (1)		2.37 (1)				11.6 (1)
31.5		2.27 (3)					
31.7					1.82 (3)		
39.1					2.25 (3)		
40.4							15.6 (4)
41.4		2.64 (3)					
41.6				9.4 (2)	3.4 (2)		13.0 (2)
41.7						7.1 (3)	
49.6		2.86 (3)					
50.0					2.85 (3)		
51.0						7.6 (3)	
51.2							17.7 (4)
55.9				11.6 (2)	4.5 (2)		15.2 (2)
60.1						8.0 (3)	
60.2		2.98 (3)					
60.7					3.5 (3)		
61.8							19.9 (4)
70.1				13.4 (2)	5.3 (2)		16.6 (2)
70.9							23.1 (4)
79.7							25.5 (4)
83.8				14.9 (2)	6.3 (2)		17.8 (2)
91.8							27.8 (4)
98.6				16.0 (2)	7.1 (2)		18.9 (2)
101.9							32 (4)
110						4.6 (5)	36 (5)
112.2				17.1 (2)	7.9 (2)		19.8 (2)

Table 6.1 (continued)

Theoretical values of the total and partial excitation coefficients for helium*											
E/N	ϵ_m/N	ϵ'_s/N	ϵ'_r/N	ϵ''_r/N	ϵ_p/N	ϵ_s/N	ϵ/N				
(10^{-17}Vcm^2)	(10^{-18}cm^2)	(10^{-18}cm^2)	(10^{-18}cm^2)	(10^{-18}cm^2)	(10^{-18}cm^2)	(10^{-18}cm^2)	(10^{-18}cm^2)				
126.5				18.3	(2)	8.7	(2)		21.0	(2)	
140.2				19.9	(2)	9.4	(2)		21.7	(2)	
154.9				20.5	(2)	10.2	(2)		22.7	(2)	
169.2				21.7	(2)	10.9	(2)		23.4	(2)	
183.4				22.8	(2)	11.5	(2)		24.3	(2)	
197.6				23.7	(2)	12.1	(2)		25.0	(2)	
211.9				24.5	(2)	12.4	(2)		25.7	(2)	
219								4.0	(5)	45	(5)
226.6				25.4	(2)	12.7	(2)		26.4	(2)	
448								2.5	(5)	48	(5)

* ϵ , total excitation coefficient; ϵ_m , sum of excitation coefficients to the 2^3S , 2^1S and 2^3P levels; ϵ'_r , sum of the excitation coefficients to all states except the 2^3S , 2^1S and 2^3P levels; ϵ''_r , sum of excitation coefficients to all states except the 2^3S and 2^1S levels; ϵ_p , excitation coefficient to the 2^1P level; ϵ_s , excitation to the 2^3S level; ϵ'_s , excitation to the 2^1S level.

Theoretical:

- (1) Corrigan, *et al.*, Proc. Phys. Soc. London **72**, 786 (1958).
- (2) Bortnik, Sov. Phys. Tech. Phys. (Engl. trans.) **9**, 1496 (1965).
- (3) Phelps, Phys. Rev. **117**, 619 (1959).
- (4) Hughes, J. Phys. B **3**, 1544 (1970).
- (5) Itoh, *et al.*, J. Phys. Soc. Japan **15**, 1675 (1960).

Table 6.2

Theoretical values of the total excitation coefficient for neon

E/N (10^{-17}Vcm^2)	ϵ/N (10^{-17}cm^2)	E/N (10^{-17}Vcm^2)	ϵ/N (10^{-17}cm^2)	E/N (10^{-17}Vcm^2)	ϵ/N (10^{-17}cm^2)
6.59	0.39	30.3	1.45	70.2	2.87
7.63	0.44	32.5	1.57	72.5	2.96
8.80	0.49	34.5	1.62	75.1	3.0
9.85	0.55	36.3	1.71	78.4	3.1
10.89	0.61	39.3	1.81	80.7	3.2
12.07	0.66	42.3	1.92	83.0	3.2
13.50	0.74	45.4	2.08	84.7	3.3
14.81	0.80	47.2	2.14	86.7	3.4
16.12	0.86	50.3	2.21	89.3	3.5
17.82	0.96	53.1	2.34	92.1	3.6
19.39	1.03	55.7	2.41	94.6	3.7
21.10	1.08	58.3	2.48	97.5	3.7
22.67	1.16	60.7	2.55	100.1	3.8
24.37	1.26	62.5	2.63	103.5	3.9
25.94	1.31	64.9	2.71	106.9	4.0
27.91	1.37	67.3	2.79	110.5	4.0
				112.2	4.0

Theoretical:

All data taken from Hughes, J. Phys. B **3**, 1544 (1970).

Table 6.3

Experimental values of the excitation coefficient for the $2^1\Pi_u$ state of hydrogen*

E/N (10^{-15} Vcm^2)	ϵ_r/N (10^{-17} cm^2)		E/N (10^{-15} Vcm^2)	ϵ_r/N (10^{-17} cm^2)		E/N (10^{-15} Vcm^2)	ϵ_r/N (10^{-17} cm^2)	
0.460	0.069	(b)	1.142	2.06	(b)	3.47	6.1	(a)
0.532	0.181	(b)	1.220	2.39	(b)	4.14	6.5	(c)
0.608	0.34	(b)	1.293	2.63	(b)	4.84	6.7	(c)
0.666	0.66	(a)	1.356	2.92	(a)	5.33	6.4	(c)
0.686	0.53	(b)	1.369	2.90	(b)	5.94	5.9	(c)
0.766	0.71	(b)	2.062	4.2	(a)	6.05	5.7	(c)
0.837	0.95	(b)	2.430	4.6	(c)		6.1	(c)
0.915	1.23	(b)	2.995	5.6	(c)	6.76	5.8	(c)
0.987	1.44	(b)	3.02	5.2	(c)	7.31	5.3	(c)
1.064	1.75	(b)	3.13	5.5	(c)	7.61	5.3	(c)
						7.67	5.0	(c)

* a temperature of 20 C has been assumed for the results given in (a) and in (c).

Experimental:

- (a) Corrigan, *et al.*, Proc. Roy. Soc. London Ser. A 245, 335 (1958).
 (b) Legler, Z. Physik 173, 169 (1963).
 (c) Nygaard, J. Appl. Phys. 36, 743 (1965).

Table 6.4

Experimental values of the excitation coefficient leading to dissociation (dissociation coefficient) for hydrogen (a temperature of 20 C was assumed)

E/N (10^{-15} Vcm^2)	ϵ_d/N (10^{-17} cm^2)		E/N (10^{-15} Vcm^2)	ϵ_d/N (10^{-17} cm^2)		E/N (10^{-15} Vcm^2)	ϵ_d/N (10^{-17} cm^2)	
0.576	1.76	(b)	0.803	4.8	(b)	1.351	9.2	(a)
0.561	2.76	(b)	1.076	5.6	(b)	1.626	9.1	(a)
0.606	2.91	(b)	1.182	5.3	(b)	2.03	4.7	(b)
0.712	3.94	(b)	1.271	7.3	(a)	3.17	21.0	(a)

Experimental:

- (a) Corrigan, *et al.*, Proc. Roy. Soc. London Ser. A 245, 335 (1958).
 (b) Poole, Proc. Roy. Soc. London Ser. A 163, 424 (1937).

Table 6.5

Experimental values of the excitation coefficient
for excitation to the $C^3\Pi_u$ state of nitrogen

E/N (10^{-17} Vcm^2)	ϵ_r/N (10^{-17} cm^2)	E/N (10^{-17} Vcm^2)	ϵ_r/N (10^{-17} cm^2)	E/N (10^{-17} Vcm^2)	ϵ_r/N (10^{-17} cm^2)
76.1	0.067	257.6	4.5	441	6.2
90.8	0.212	271.5	4.8	455	6.7
106.3	0.47	288.7	4.8	486	6.8
121.4	0.80	303	5.0	519	6.9
136.8	1.23	318	5.3	544	8.0
151.7	1.68	334	5.4	576	7.4
166.1	2.18	348	5.6	610	7.3
180.3	2.68	364	5.9	638	7.3
196.7	3.1	380	5.7	669	8.0
212.7	3.6	396	6.0	702	8.3
228.0	4.0	410	6.0	727	8.8
241.3	4.3	424	6.2	763	8.7
				822	8.8

Experimental:

All data taken from Legler, Z. Physik 173, 169 (1963).

Table 6.6(a)

Experimental values of the excitation coefficient for carbon dioxide
for excitation to states leading to vacuum u.v. irradiation

E/N_{est} (10^{-17} Vcm^2)	$\epsilon_r/N_{\text{est}}$ (10^{-17} cm^2)	E/N_{est} (10^{-17} Vcm^2)	$\epsilon_r/N_{\text{est}}$ (10^{-17} cm^2)	E/N_{est} (10^{-17} Vcm^2)	$\epsilon_r/N_{\text{est}}$ (10^{-17} cm^2)
118.5	0.0092	249.2	0.079	555	0.50
123.5	0.0105	264.0	0.089	584	0.56
129.9	0.0120	277.7	0.100	625	0.66
136.6	0.0136	292.1	0.114	662	0.74
143.7	0.0158	309	0.128	702	0.85
149.8	0.0183	326	0.144	744	0.98
157.3	0.0211	345	0.169	783	1.12
165.4	0.0241	369	0.196	837	1.28
174.0	0.0279	394	0.226	895	1.50
184.3	0.033	411	0.251	950	1.71
195.7	0.039	436	0.286	1006	1.98
209.2	0.047	462	0.33	1058	2.19
219.6	0.053	494	0.38	1121	2.50
233.1	0.063	520	0.43	1190	2.85

Experimental:

All data taken from Teich, et al., 2nd Int. Conf. Gas Discharges, London, 335 (1972).

Table 6.6(b)

Experimental values of the dissociation coefficient for carbon dioxide

E/N (10^{-17} Vcm^2)	ϵ_d/N (10^{-17} cm^2)	E/N (10^{-17} Vcm^2)	ϵ_d/N (10^{-17} cm^2)	E/N (10^{-17} Vcm^2)	ϵ_d/N (10^{-17} cm^2)
31.2	0.42	33.6	0.42	54.5	1.19
	0.39	42.1	0.71	73.8	1.70
32.3	0.39	44.1	0.75	78.5	1.80
32.6	0.45	50.5	0.93	91.5	1.92

Experimental:

All data taken from Kutszegi Corvin, *et al.*, J. Chem. Phys. **50**, 2570 (1969).

3.7. Ionization Coefficient

When an electron swarm moves in an electric field that is sufficiently strong for ionization of gas atoms by electrons to become significant, there is a progressive amplification of the electron current as it moves in the field direction.

When the only significant process of ionization occurring is that of gas atoms by electrons and when electron removal processes are negligible, a current I_0 released into a gas from the cathode of an electrode system in which a uniform field E is established gives rise, in the steady state, to a current

$$I = I_0 e^{\alpha(d-d_0)} \quad (9)$$

at a distance d from the cathode. The inclusion of d_0 in eq (9) takes formal account of the fact that electrons emitted from the cathode attain a steady state, determined by the value of E/N in the gas, only after a number of collisions. The range over which eq (9) gives a satisfactory representation of the current has been discussed in detail by Folkard and Haydon [4907]. The value of d_0 decreases as N increases and thus at relatively low values of E/N , d_0 is often negligible with respect to d ; in these circumstances the current is given to a very good approximation by

$$I = I_0 e^{\alpha d} \quad (10)$$

Equations (9) and (10) define the coefficient α which is known as the Townsend primary ionization coefficient. Both experimentally and theoretically α/N is found to be a function of E/N so that α is constant for a given value of N and E/N . When primary ionization is the only significant process, α may clearly be determined from measurements of the steady-state ionization current as a function of d at constant E/N and N by using eq (9) or (10). This method of determining α is usually referred to as the Townsend method. The primary ionization coefficient may also be determined from measurements on the transient electron avalanche produced by a pulse of electrons released from the cathode of a uniform-field electrode system. If a number n_0 electrons is released

in the pulse, the number of electrons in the gap at any time t less than the transit time of the electrons across the gap is given by

$$n = n_0 \exp(\alpha W - t). \quad (11)$$

This equation may be used as the basis for obtaining values of αW by determining the electron component of the transient current in the gap from measurements either of the current in the external circuit of which the gap forms part, or of the light emission from the gap as a function of time. The total duration of the electron component of the avalanche gives the value of W and hence α may be obtained. If each avalanche is initiated by only one electron from the cathode, there is a measureable statistical fluctuation in the total number of electrons produced in an avalanche. Measurements of the probability $f(n)$ of such a single avalanche containing a total number n electrons may also be used to determine α , since

$$f(n) = (\bar{n})^{-1} \exp(-n/\bar{n}), \quad (12)$$

where $\bar{n} = \exp(\alpha d)$.

The interpretation of the Townsend primary ionization coefficient, defined by eq (9), in terms of ionization rate coefficients and of ionization coefficients per unit distance in the direction of the electric field depends on whether or not diffusion is significant. The effects of diffusion can often, in practice, be neglected and in this case the ionization coefficient α can be interpreted such that the number of new electrons produced on the average by n electrons in moving a distance dx in the field direction is $n\alpha dx$, i.e. α is a coefficient per unit drift distance as discussed in section 2. It should be remembered, however, that if diffusion is taken into account the number of new electrons produced on average by n electrons moving a distance dx in the field direction becomes (Crompton, J. Appl. Phys. **38**, 4093, 1967)

$$n\alpha [1 - \alpha(D/W)] dx = n\alpha_i dx, \quad (13)$$

say. Thus, strictly, the ionization frequency $\nu_i = \alpha_i W$,

and only when diffusion can be neglected can the approximation $\nu_i = \alpha W$ be used.

This can be important when relating data on ionization coefficients to the ionization cross section $Q_i(\epsilon)$ through the relationship

$$\nu_i = \int_{\epsilon_i}^{\infty} (e/150m)^{1/2} N Q_i(\epsilon) \epsilon^{1/2} F(\epsilon) d\epsilon, \quad (14)$$

where ν_i is the ionization frequency, ϵ_i the ionization energy, ϵ the electron energy and $F(\epsilon)$ the normalized electron energy distribution. It should also be borne in mind when comparing the results of measurements of steady-state growth of prebreakdown ionization currents (Townsend method) with those of pulse analysis and of high frequency breakdown measurements.

Where available, values of the Townsend primary ionization coefficient obtained by both steady state and pulse methods are given in the following subsections. Somewhat different problems are encountered in obtaining values of the coefficients for monatomic gases, for molecular gases which are not electronegative, and for electronegative gases. The data are thus grouped under these headings.

3.7.a. Ionization Coefficient—Monatomic Gases

In monatomic gases, eq (9) is not a good approximation to the prebreakdown ionization current, because secondary ionization, which results from a number of possible secondary ionization processes, is always significant compared with primary ionization. In these circumstances, the steady-state ionization may be expressed as

$$I = I_0 e^{\alpha d} / [1 - (\omega/\alpha) (e^{\alpha d} - 1)], \quad (15)$$

where ω/α is a coefficient representing the action of secondary ionization processes. Hence, by suitable analysis, values of α can be obtained from measurements of steady-state ionization currents as a function of d at a constant value of E/N . Care is needed in the use of eq (15), however. For a number of secondary ionization processes (such as the emission of electrons at the cathode by positive ions, for example) ω/α is, or can effectively be treated as, a constant at a given value of E/N but for others (such as emission of electrons from the cathode by metastable atoms) the dependence of ω/α on d has to be considered.

Moreover, monatomic gases have large excitation and ionization potentials relative to other gases, so that experimentally, it is necessary to use gas samples of the highest possible purity if significant data relating to the gas being investigated are to be obtained.

3.7.a(i). Ionization Coefficient—Helium

The problem of gas purity is acute in the case of helium which has an ionization potential (24.58 V)

higher than that of any other gas, and metastable levels with energies above 19.8 eV. As a result, any impurities present are not only preferentially ionized by direct ionization but are also ionized by Penning ionization which occurs in collisions between helium metastables and impurity atoms. The high cross section for Penning ionization means that very small concentrations of impurity ($\sim$ p.p.m.) can significantly affect the spatial growth of ionization. For this reason the values obtained for α in helium have decreased as vacuum techniques have improved.

The recent experimental results [646, 1173] given in figure 7.1(a) and table 7.1 were obtained using modern ultrahigh vacuum techniques and it is significant that there is good agreement between the two sets of data. There is, moreover, good agreement for $E/N > 17 \times 10^{-17}$ V cm² between these experimental data and the theoretically computed values, some of which are also shown in figure 7.1(a). An intercomparison of the available theoretical data obtained using both analytical methods [771, 1440, 2143, 5701, 4909] and Monte Carlo techniques [3897] is shown in figure 7.1(b).

It should be noted that recent experiments using He and Ne-He mixtures [3751] indicate that the generally accepted experimental values for helium may still be too large, not because of impurities but because of the operation of processes which do not conform to eq (13) where a constant value of ω/α is used to analyze the data.

Values of E/N below about 17×10^{-17} V cm² correspond to relatively large electrode separations and to high gas pressures so that experimental problems, especially those associated with the production of some litres of very high purity gas, become severe. Recent work [4910, 5077, 4049] in this range has shown that even for impurity contents as low as about 1 p.p.m., the values of α/N obtained are considerably higher than those expected for pure helium, because of the influence of Penning ionization. Earlier values are even higher [2406] probably because of relatively high levels of neon impurities or unreliable [646] because of the restricted range of parameters used. In these circumstances, the most reliable data available for α/N for $E/N < 1.7 \times 10^{-16}$ V cm² are probably those obtained by theoretical computations and shown in figure 7.1(b), although even these are subject to uncertainties due to the approximations adopted and uncertainties in the cross sections used [see 3753]. Table 7.1 gives a composite set of values of α/N in helium over the full range investigated.

At values of $E/N > 2.84 \times 10^{-15}$ V cm² which correspond to very low values of pressure and electrode separation, experiment [646] indicates that electrons do not experience sufficient collisions in moving from cathode to anode for steady-state conditions to be established and in these circumstances values of α have little significance.

3.7.a(ii). Ionization Coefficient—Neon

The experimental results of Kruithof and Penning [2355], which were themselves in fair agreement with the earlier results of Townsend and MacCallum [730] at an unspecified temperature, have been confirmed by three recent sets of measurements [751, 3751, 3612] in which ultrahigh vacuum techniques were used. The data of Kruithof and Penning together with the three recent sets of data are given in figure 7.2 and in table 7.2. (To avoid confusion two graphs—figs. 7.2(a) and (b)—are shown with one set of values [2355] common to both.) Measurements of the luminous flux from self-sustaining discharges in neon as a function of distance from the cathode gave values of α/N lower than those obtained by the Townsend method both for pure [2535, 5049] and impure [5771] neon. The reasons for this difference are not at present clear and the results for pure neon using this method seem to depend on the conditions since one set of data [2535] gave results of 20 to 30 percent below those given in figure 7.1, while another [5049] gave results about 16 percent below those in figure 7.1.

In attempting to obtain theoretically computed values of α/N in neon, one of the main difficulties is in the uncertainty associated with values of the excitation cross section as a function of energy. It has been shown in a number of investigations [2048, 5110, 4909] that the experimental values of this cross section obtained by Maier-Leibnitz (Z. Physik **95**, 499, 1935), gave calculated values of α/N in fair agreement with experiment. In recent more rigorous calculations, values of α/N were obtained using Monte Carlo calculations [3752] and using the energy distribution obtained from the solution of the Boltzmann equation [3753]. In this work it was found necessary to use excitation cross sections about 20 percent less than the Maier-Leibnitz experimental values in order to obtain the agreement between the theoretical and experimental values of α/N shown in figure 7.2.

The largest experimental uncertainty lies in the region of E/N below about 1.5×10^{-16} V cm² which is also the region in which there is discrepancy between theory and experiment as shown in figure 7.2(b).

Both theory [3752] and experiment [751] agree in indicating that above a value of E/N of about 2.82×10^{-15} V cm², steady-state conditions are not established so that the concept of a swarm coefficient α is not relevant.

3.7.a(iii). Ionization Coefficient—Argon

The most extensive investigations of argon to date are those of Kruithof and Penning [2139] and Kruithof [2237] with which the more recent data obtained over restricted ranges by Davies and Milne [2138] and by Willis [5241] are in good agreement as can be seen from figure 7.3. Kruithof's data are given numerically in table 7.3.

There has been no estimate of the upper limit of E/N for which equilibrium swarm conditions are likely to occur, but Kruithof [2237] remarked that although the region of the I, d curve satisfying eq (10) became shorter

and shorter as E/N increased, it was possible to determine α/N at values of E/N up to 4.7×10^{-14} V cm².

At low values of E/N , the experimental error in the values of α/N obtained by Kruithof and Penning increases considerably because of the relatively low pressures ($160 > p_0 > 1$ Torr) and small electrode separations ($d < 1.6$ cm) used. The uncertainty in the region of $E/N < 34 \times 10^{-17}$ V cm² is further increased by the recent work of Golden and Fisher [2333] at higher pressures (up to 700 Torr) and large electrode separations (up to 5 cm). They found that the I, d curves could not be represented by eq (15) but that values of α/N could be obtained, from measurements of I as a function of p_0 at a constant value of d ; the values of α/N obtained, however, depended on the distance used. The values of α/N at all values of d (0.6 to 1.2 cm) used for $34 \times 10^{-17} > E/N > 25 \times 10^{-17}$ V cm² were within 20 percent of the values of Kruithof and Penning. For $E/N < 25 \times 10^{-17}$ V cm², on the other hand, the values which were obtained for $d=4$ and 5 cm were lower than those of Kruithof and Penning by as much as a factor of 15 at $E/N = 14.1 \times 10^{-17}$ V cm².

Although the results of theoretical computations, which are also shown in figure 7.3, are close to the experimental values they were obtained by methods open to the general criticism of Thomas [3753] concerning similar computations for neon.

3.7.a(iv). Ionization Coefficient—Krypton and Xenon

The two available sets of data [2237, 5293] for krypton are shown in figure 7.4. There is good agreement for $E/N > 100 \times 10^{-17}$ V cm² but the recent results of Heylen [5293] fall increasingly below those obtained by Kruithof [2237] as E/N decreases, being about a factor of two lower at $E/N \sim 30 \times 10^{-17}$ V cm². Numerical values from both sets of data are given in table 7.4(a).

The only data available for xenon are the experimental values obtained by Kruithof [2237] shown in figure 7.4 and given in table 7.4(b).

3.7.b. Molecular Gases Which Are Not Electronegative

In molecular gases the secondary ionization coefficient ω/α is considerably smaller in relation to the primary ionization coefficient than is the case in monoatomic gases. Thus, in some circumstances, eq (9) can be used to determine α by the Townsend method but in other cases it has been shown [2150] that significant errors can be introduced if the full eq (15) is not used to analyze the data.

The relatively low value of ω/α also means that relatively large electron avalanches can develop in molecular gases at distances less than the distance d_s at which breakdown occurs, d_s being given by the value of d at which the denominator of eq (15) becomes zero. This simplifies the experimental problems associated with the measurement of the electron component of transient electron avalanches, so that for molecular

gases values of α have often been obtained both by the Townsend method and the methods based on measurements on transient avalanches (see sec. 3.7). The values for H_2 and N_2 using both methods are given and compared in the next two sections.

3.7.b(i). Ionization Coefficient—Hydrogen

The question of purity is not nearly as critical in hydrogen as in the inert gases [790]. As a result, there is good agreement between the values obtained in a large number of experimental determinations based on the measurement of steady-state currents [751, 782, 790, 1145, 1160, 1174, 1356, 1640, 2148, 2150], of pulsed avalanches [2154] and of radiation from self-sustaining discharges [5105] for values of $E/N \lesssim 3.4 \times 10^{-15} \text{ V cm}^2$. The data obtained in this range are so numerous that for clarity they are shown in two figures, 7.5(a) and 7.5(b), with one set of points [1356] common to both to aid the comparison. These data are also given in table 7.5.

By numerical solution of the Boltzmann equation, Engelhardt and Phelps obtained values for the appropriate collision cross sections which, as shown in figure 7.5, gave computed values of α/N in good agreement with the experimental data, although some of the cross sections used need modifying to take account of subsequent experimental data (Phelps, *Rev. Mod. Phys.* **40**, 399, 1968).

Until recently the situation for values of $E/N \sim 3.4 \times 10^{-15} \text{ V cm}^2$ was confused, there being a large number of observations, extending in some cases up to values of $E/N > 3 \times 10^{-14} \text{ V cm}^2$, with discrepancies between results of different investigations which increased as E/N increased. The situation has, however, been considerably clarified by the recent investigation of Folkard and Haydon [4907] in which special attention was paid to the effect of instrumental errors on the accurate determination of the ionization currents and to considerations of the range of experimental parameters over which a steady state is established. This work showed that significant values for α/N can be obtained only up to values of E/N about $7 \times 10^{-15} \text{ V cm}^2$. At higher values of E/N in the range $8 \times 10^{-15} \leq E/N \leq 14 \times 10^{-15} \text{ V cm}^2$, a quasi-steady state exists as shown by Monte Carlo calculations [5292]. When a steady-state analysis is used to analyze results obtained in this region the values of α/N obtained vary greatly, depending on the precision with which the ionization currents are measured. The region of $E/N \gtrsim 14 \times 10^{-15} \text{ V cm}^2$ is a completely non-steady-state region to which the concept of swarm coefficient α is not relevant.

Accordingly, the values obtained in various investigations up to $E/N = 7 \times 10^{-15} \text{ V cm}^2$ are shown in figure 7.6. In this region Jones and Llewellyn-Jones [2150] and Haydon and Stock [2148] have shown, respectively, that ignoring secondary ionization in the analysis of experimental data and including measurements at small distance when field distortion occurs, both give rise to high values of α/N . Other results [1160] show that the

purity of the gas sample plays a significant role in this region. These factors probably contribute to the remaining discrepancies. The numerical values given in table 7.5 for $E/N > 3.4 \times 10^{-15} \text{ V cm}^2$ are those of Folkard and Haydon [4907] since they are likely to be the most accurate obtained to date.

3.7.b(ii). Ionization Coefficient—Nitrogen

The results of recent investigations of steady-state ionization currents [1145, 1741, 1836, 1948, 2144, 2275, 5187] and pulsed avalanches [1836, 2154, 5228], at values of E/N below about $2.82 \times 10^{-15} \text{ V cm}^2$, using samples of nitrogen from cylinders (usually with impurity contents of about 0.1 percent) in conventional vacuum systems using backing and diffusion pumps, from which mercury was excluded, are in fairly good agreement with each other. In view of the large number of results available they are plotted for clarity on two figures—7.7(a) and 7.7(b)—with a set of data obtained from measurements of steady-state ionization for samples of higher purity [710] included in both figures to facilitate comparison. These values are also in fair agreement with the earlier data of Masch [2277, 2556] (if it is assumed that the temperature of Masch's measurements was 20°C) even though it is likely that Masch's results were for mercury-contaminated gas.

Engelhardt, Phelps, and Risk [218] were able to find a self-consistent set of collision cross sections which, as shown in figure 7.7(a), give computed values for the ionization coefficient in agreement with the experimental values shown.

It has been clear for some time that small amounts of impurities (<0.1 percent) significantly affected the results obtained in the range of E/N below $2.82 \times 10^{-15} \text{ V cm}^2$ in nitrogen. For example, Harrison [2102] using a degassed conventional vacuum system, obtained large (3 to 30 percent) differences in results for gas samples produced in different ways, the values in all cases being considerably greater (by two to six times) than those shown in figure 7.7. On the other hand, Heylen [710], using an ultrahigh vacuum system found that, although his values for α/N were higher than the others shown in figures 7.7(a) and (b) for values of $E/N < 1.3 \times 10^{-15} \text{ V cm}^2$, they lay much closer to them; for example, at $E/N \approx 1 \times 10^{-15} \text{ V cm}^2$, Heylen's value is about 40 percent higher, whereas Harrison's data lie between a factor of three and four higher. Recent investigations [5334, 5703] have shown that these differences result from the fact that in nitrogen the emission of electrons from the cathode by metastable nitrogen molecules (Haydon, *J. Phys. B* **6**, 227, 1973) can make an important contribution to the secondary ionization. This process gives rise to a secondary ionization ω/α which is dependent on electrode separation. Thus, primary and secondary ionization acting together can, in some circumstances, give an exponential spatial growth of current which if analyzed by means of eq (9), which assumes that primary ionization alone is operative,

gives erroneously high values of α/N . Thus the values of α/N shown in figure 7.7 which are obtained for slightly impure samples are more likely to be characteristic of nitrogen, because the impurities quench the metastables, thus markedly changing the secondary ionization, but having little effect on the primary ionization. This supposition is confirmed by the agreement of the values obtained from steady-state measurements on such samples with those obtained from pulse measurements [2154, 5228] which are not affected by secondary ionization.

A composite set of experimental results excluding the data of [2275] at the upper end and of [710] at the lower end of the ranges are given in table 7.6.

There are three recently published sets of data [3297, 2448, 5334] in mercury-free nitrogen for $E/N > 2.82 \times 10^{-16}$ V cm². These are compared with Bowls' [2275] earlier values in figure 7.8. The values are not very different from those obtained for mercury-contaminated nitrogen [789, 2277 and 2556]. The most recent investigation [5334] shows that all these data correspond to the steady-state region which was shown to extend to $E/N \approx 3.4 \times 10^{-14}$ V cm² in nitrogen.

3.7.c. Electronegative Gases

In electronegative gases, processes of electron attachment to and detachment from gas molecules have to be taken into account, and ion conversion reactions which convert negative ions of species A⁻, with a low cross section for collisional detachment, to species B⁻ with high cross section or vice versa can also be important. The equation for the spatial growth of pre-breakdown ionization taking into account these processes in addition to primary and secondary ionization processes, may be written as

$$I = I_0 \frac{\alpha \left(\frac{\lambda + \delta + \kappa}{\lambda_1 (\lambda_1 - \lambda_2)} \right) e^{\lambda_1 d} - \frac{\alpha (\lambda_2 + \delta + \kappa)}{\lambda_2 (\lambda_1 - \lambda_2)} e^{\lambda_2 d} + \frac{[(\eta + \eta^1)\kappa + \eta\delta]}{\lambda_1 \lambda_2}}{1 - \left(\frac{\omega}{\lambda_1 - \lambda_2} \right) \left[\left(\frac{\lambda_1 + \delta + \kappa}{\lambda_1} \right) (e^{\lambda_1 d} - 1) - \left(\frac{\lambda_2 + \delta + \kappa}{\lambda_2} \right) (e^{\lambda_2 d} - 1) \right]} \quad (16)$$

where η , η^1 are electron attachment coefficients to form A⁻ and B⁻ ions respectively, δ the electron detachment coefficient from ion species B⁻ (that from A⁻ being taken as negligibly small) and κ the ion conversion coefficient from A⁻ to B⁻ ions (all coefficients being per unit drift distance). The coefficients λ_1 and λ_2 are the roots of the equation

$$\lambda^2 - (\alpha - \eta - \eta^1 - \delta - \kappa) \lambda - [(\alpha - \eta - \eta^1) \times (\kappa + \delta) + \eta^1 \delta] = 0. \quad (17)$$

Analysis of experimental data in terms of eq (16) which contains seven unknown coefficients is difficult and this equation has not so far been used for the analysis of data in any published work. Analysis using the simplified equation obtained by assuming the formation

of only one ion species and neglecting secondary ionization has been carried out, but even in this case the combinations of values of the coefficients (α , η , δ) which will give agreement with the experimental data are in general subject to inherently large spreads (typically ~ 100 percent) even when the measurements of the ionization currents are accurate to within a few percent.

In practice, there is often a range of distance for which all but the first terms in the numerator and denominator of eq (16) are negligible. Over this range a graph of $\log I$, d is linear with a slope λ_1 . Under similar conditions, avalanches grow temporally with a time constant $\lambda_1 W$ so that λ_1 can be obtained both from steady state and pulse measurements.

The coefficient λ_1 , which can be thought of as an apparent or effective ionization coefficient, is a useful measure of the electrical properties of an electronegative gas and, unlike the true ionization coefficient α , can always be determined to within a few percent provided the ionization currents are measured accurately (to within a few percent) over a sufficiently wide range. Consequently, for the electronegative gases values of λ_1/N are given rather than values of α/N even though λ_1/N is often a function of N as well as of E/N . (In some cases the values presented have had to be recovered from the data given in the published papers by combining graphs and taking into account the analysis used.)

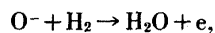
3.7.c(i). Apparent Ionization Coefficient—Oxygen

Recent investigations in oxygen have been confined to $E/N < 200 \times 10^{-17}$ V cm². The values of λ_1 obtained are shown in figure 7.9. For clarity two graphs, figures 7.9(a) and 7.9(b), have been drawn with the theoretical values given on each to aid comparison. It can be seen that the values obtained using the Townsend method

[573, 791, 861, 961, 2555, 2908] and those obtained using pulse [2154, 2387] methods show general agreement within rather a large scatter which is greatest at the lowest values of E/N . Some of this scatter may be accounted for by the fact that in a recent investigation [2908] in this region λ_1/N was shown to be dependent on N as well as E/N , a fact attributed to the occurrence of three-body processes which influence λ_1 .

Hake and Phelps [2553] were able to obtain a set of self-consistent cross sections which give theoretically computed values of λ_1/N which, as shown in figure 7.9, lie within the spread of the experimental data.

Use has recently been made [4994] of the fact that the addition of small percentages of hydrogen to oxygen gives a gas mixture which, because of the fast associative detachment reaction



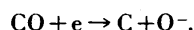
behaves as nonattaching gas, in which the true ionization coefficient can be accurately determined. Theoretical calculations showed that the energy distribution of the electrons remained unchanged in the mixture so that the values of α/N obtained for the mixture should thus correspond to the true ionization coefficient for O_2 . These values which lie within the large spread of values determined from the analysis of measurements in oxygen alone are shown in figure 7.9(c).

There is only one extensive published set of data [2556] extending beyond $E/N = 200 \times 10^{-17} \text{ V cm}^2$. These results, which were obtained for oxygen contaminated with mercury vapor from the manometer used, are shown in figure 7.10. The temperature corresponding to the values of E/p and λ_1/p given in the original paper is not clear and a value of 20°C has been assumed to obtain the data in figure 7.10. Frommhold [2154] gives two values for oxygen uncontaminated with mercury vapor at $E/N = 239 \times 10^{-17}$ and $326 \times 10^{-17} \text{ V cm}^2$, and these are also shown in figure 7.10.

No estimates have been made of the value of E/N above which steady-state swarm conditions cease to exist.

3.7.c(ii). Ionization Coefficient—Carbon Monoxide

The results of electron beam experiments have shown that negative ions are formed by electron collisions with gas atoms in carbon monoxide by processes such as



The processes are relatively high energy processes (thresholds $\sim 10 \text{ eV}$), however, and do not give rise to significant attachment in electron swarms for the range of values of E/N up to about $300 \times 10^{-17} \text{ V cm}^2$ for which determinations have been made. (See sec. 3.4.c.) Thus the coefficient determined from the linear section of the $\log I, d$ graphs in carbon monoxide represents a true ionization coefficient. The values obtained in the three published investigations [1254, 4016, 5243] are shown in figure 7.11 and tabulated in table 7.7.

There are no published values of α/N in carbon monoxide for values of $E/N > 275 \times 10^{-17} \text{ V cm}^2$.

3.7.c(iii). Apparent Ionization Coefficient—Nitric Oxide

There is no published information available on the ionization coefficient for nitric oxide.

3.7.c(iv). Apparent Ionization Coefficient—Carbon Dioxide

In contrast to the situation in carbon monoxide, there is no doubt that electron attachment occurs in low energy swarm conditions in CO_2 and there have been four recent determinations of apparent ionization coefficients in this gas, two [948, 3435] using the Townsend

method and two [1625, 1724] using pulse methods. The apparent ionization coefficient has been shown [3435] to be independent of gas number density for $35 \times 10^{16} < N < 1062 \times 10^{16} \text{ cm}^{-3}$, and the result of the four investigations are in good agreement as is shown in figure 7.12.

Hake and Phelps [2553] obtained a set of cross sections that gives theoretically computed values of λ_1/N which, as shown in figure 7.12, agrees with the experimental data.

There is one set [948] of recent data for λ_1/N in CO_2 for $E/N > 282 \times 10^{-17} \text{ V cm}^2$ and this is compared with the early values of the Townsend group [2271, 2593, 2815] and of Bishop [3490] in figure 7.13. The theoretical data of Hake and Phelps [2553] are included in this figure also to aid comparison. It can be seen that in the region where the results overlap, the data of the Townsend group lie considerably closer to the more recent results both of figure 7.12 and of [948] than those of Bishop. Table 7.8, which gives the numerical data, includes all the results shown in figure 7.12, together with the data of Townsend [2271] and of Bhalla and Craggs [948] for $E/N > 152 \times 10^{-17} \text{ V cm}^2$ and of Hurst [2815] for $E/N > 4000 \times 10^{-17} \text{ V cm}^2$.

No estimate has been made of the upper limit to the value of E/N at which steady-state conditions exist in CO_2 .

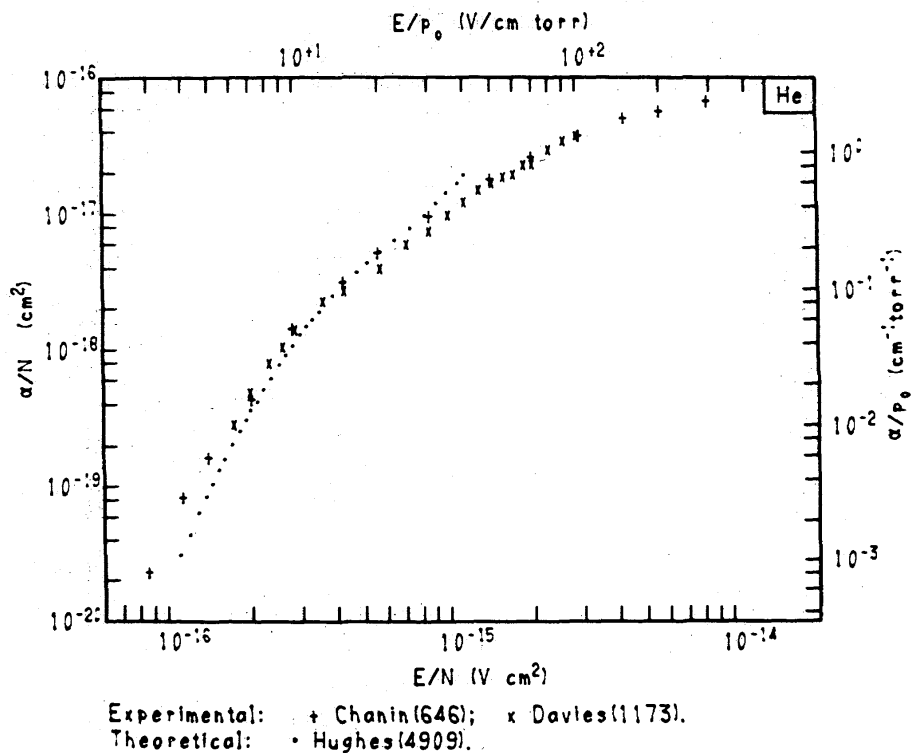
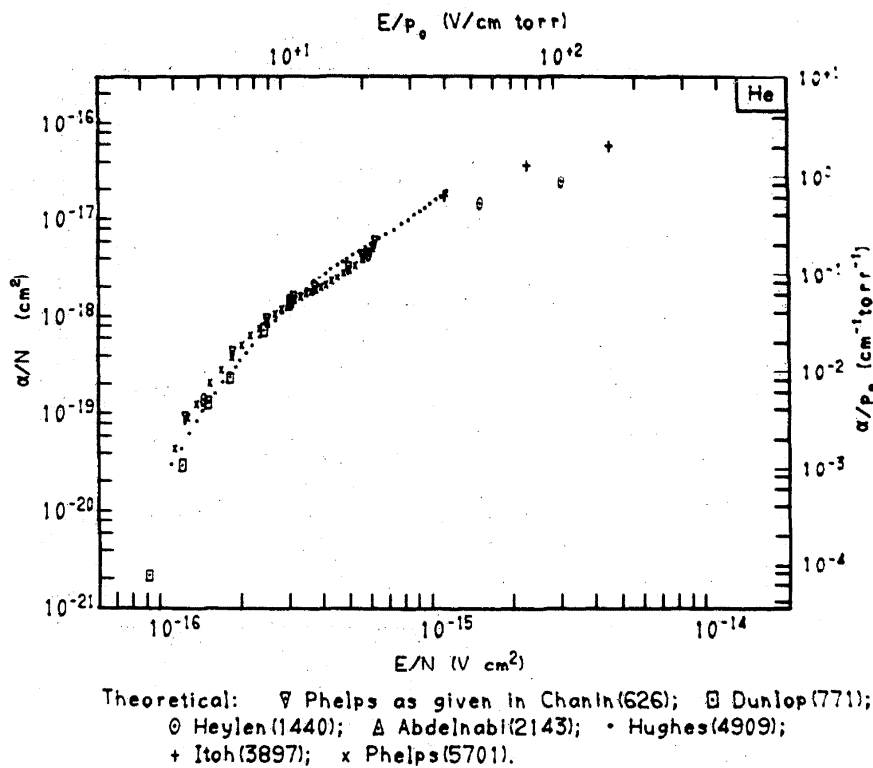
3.7.c(v). Apparent Ionization Coefficient—Air

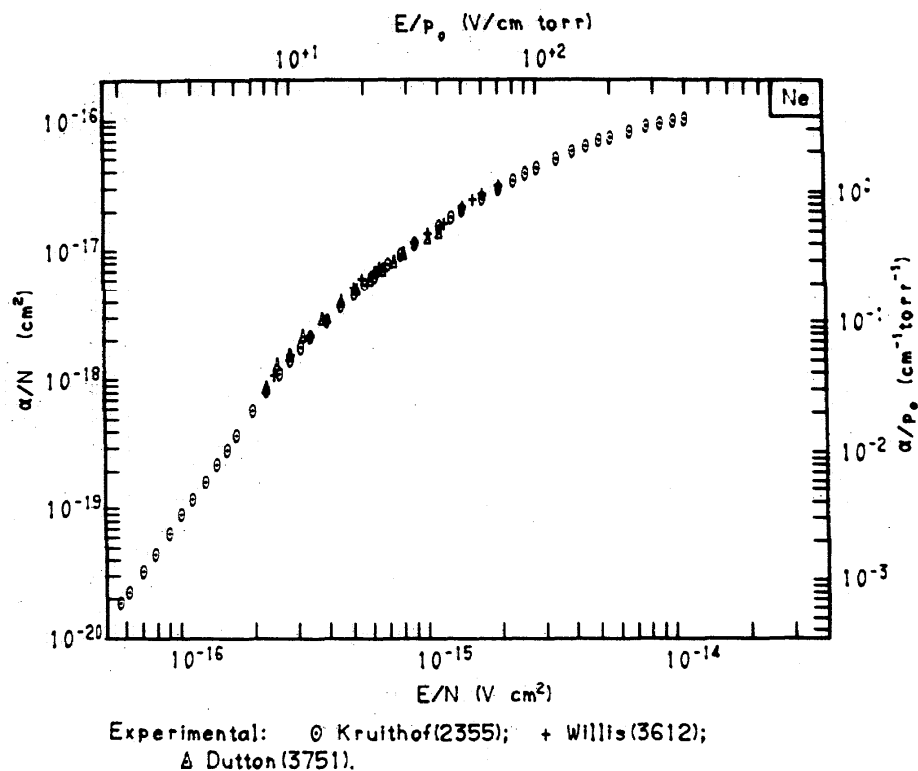
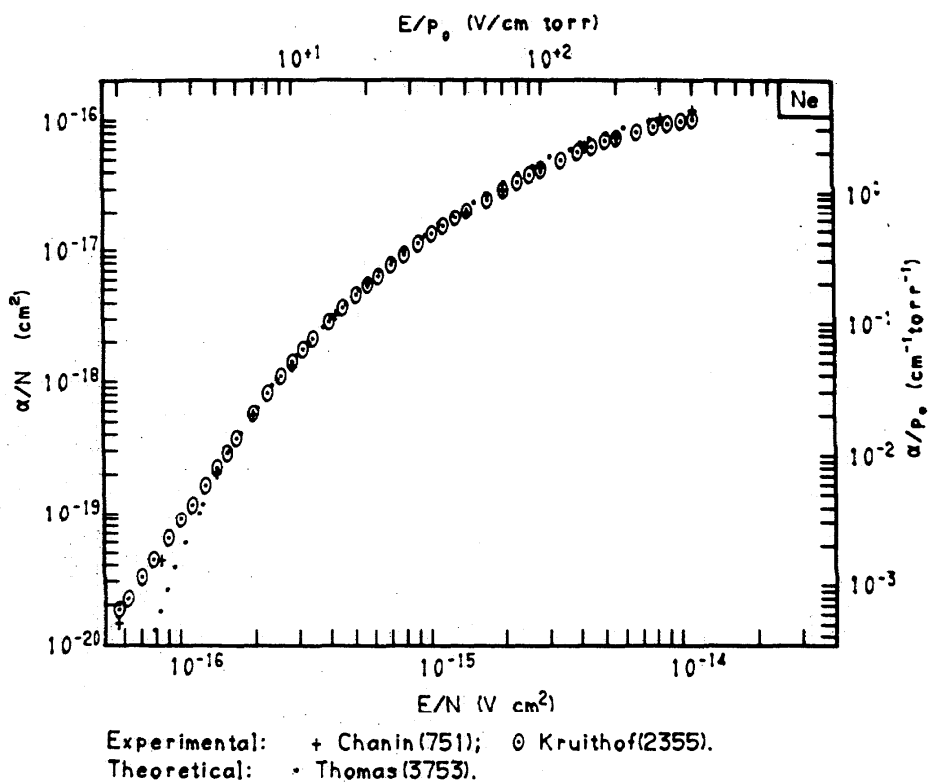
For values of $E/N < 1.7 \times 10^{-15} \text{ V cm}^2$, a number of sets of experimental data for λ_1/N in dry air uncontaminated with mercury vapor have been obtained. The values obtained using the Townsend method are given in figure 7.14.

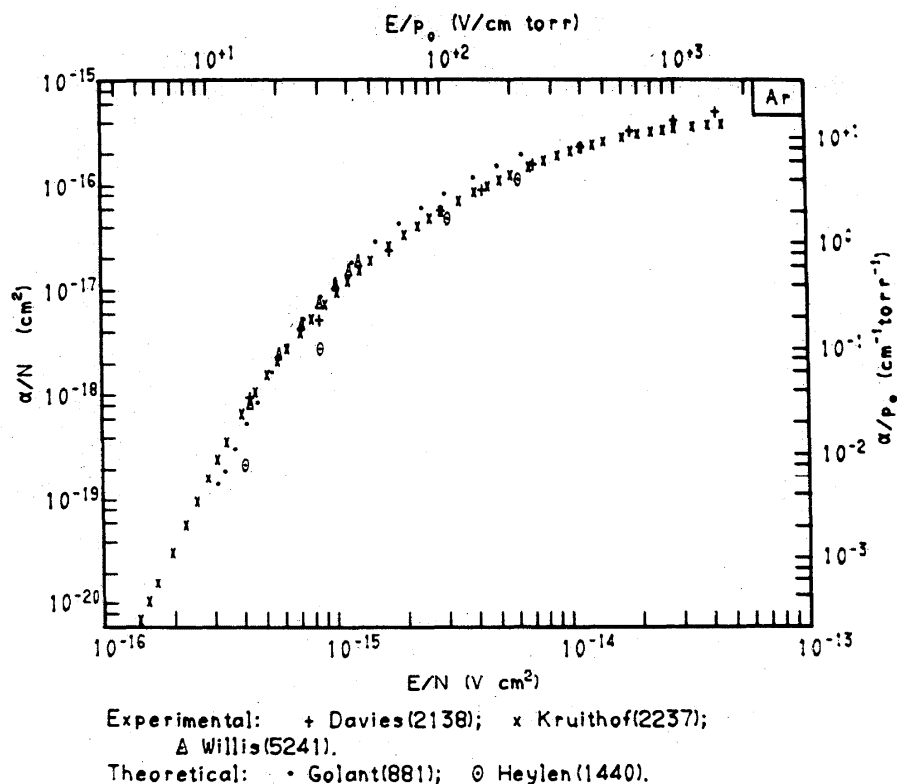
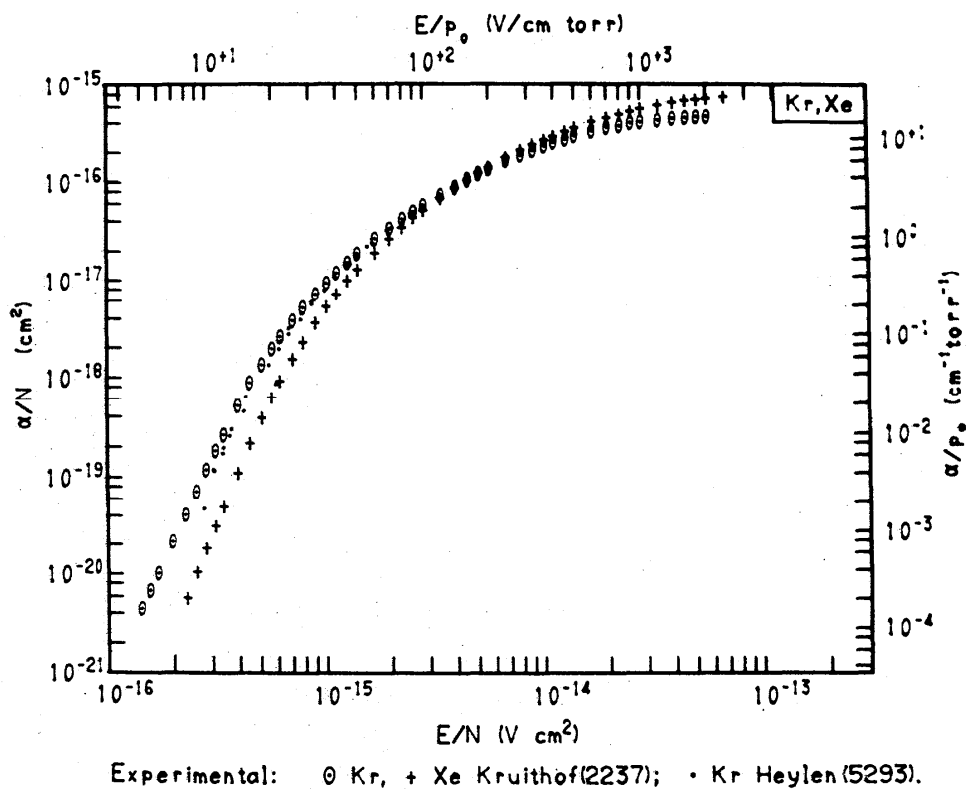
It can be seen that although the general trend of the various sets of results is the same, there is considerable scatter in the data. Some of this scatter, particularly at the lower values of E/N , is due to the variation of λ_1/N with N at a given value of E/N which has recently been shown [4015] to occur, and which is illustrated in figure 7.15.

The only results obtained using a pulsed avalanche method [1911] are between about a factor of two and ten lower than the results shown in figure 7.14. There are no published theoretical values of λ_1/N for air.

The two investigations [2274, 2556] of air which cover the widest range of E/N and which extend well beyond the value of $E/N = 1.7 \times 10^{-15} \text{ V cm}^2$ were both carried out using mercury-contaminated air, but give results, as can be seen from figure 7.16, lying close to those obtained in a recent investigation [5246] of mercury-free, dry air. Consequently the three sets of results are given in table 7.9. No estimate appears to have been made of the value of E/N above which steady-state swarm conditions cease to exist.

FIGURE 7.1(a) α/N , E/N for helium.FIGURE 7.1(b). Theoretical α/N , E/N for helium.

FIGURE 7.2(a). α/N , E/N for neon.FIGURE 7.2(b). α/N , E/N for neon.

FIGURE 7.3. α/N , E/N for argon.FIGURE 7.4. α/N , E/N for krypton and xenon.

J. DUTTON

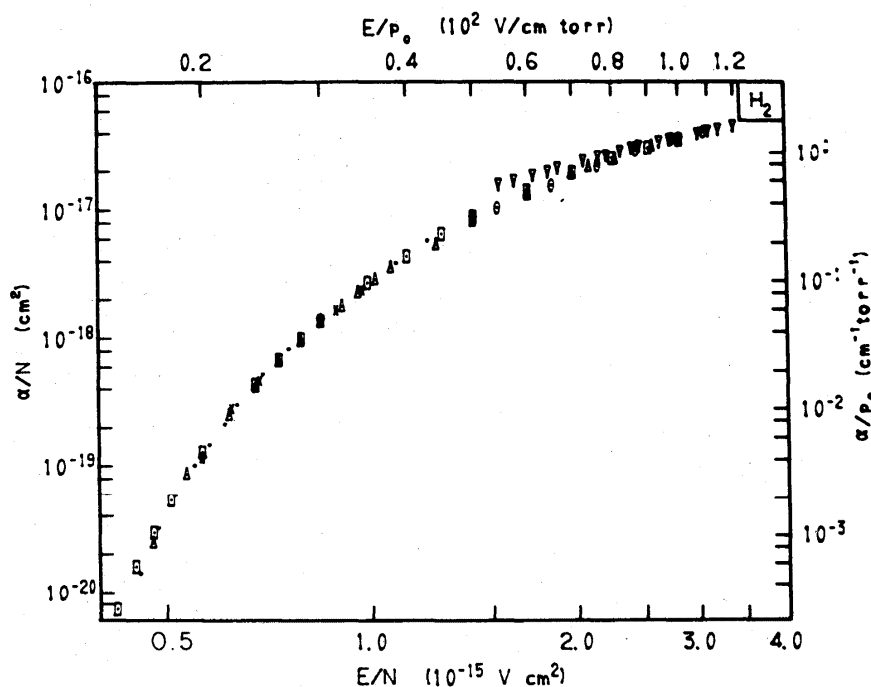


FIGURE 7.5(a). α/N , E/N for hydrogen for $E/N \leq 3.4 \times 10^{-15} \text{ V cm}^2$.

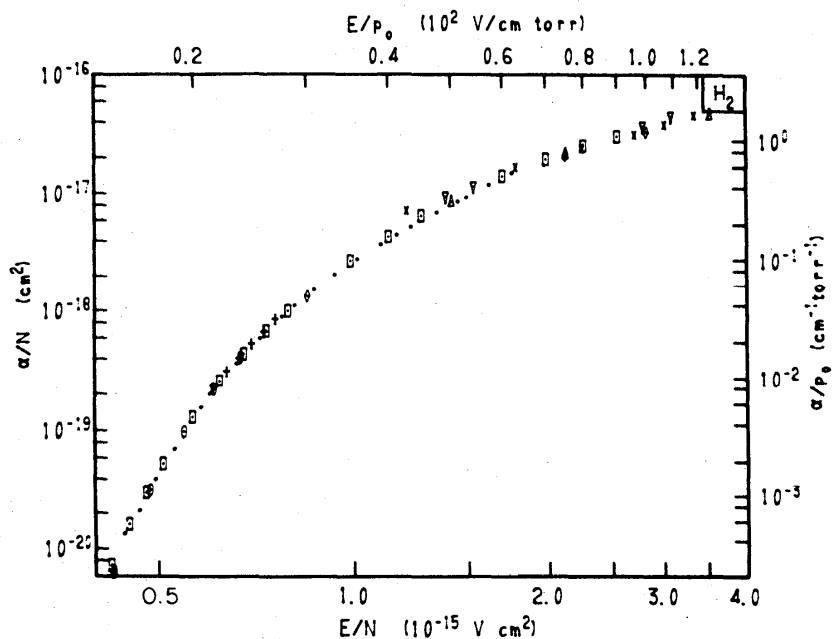
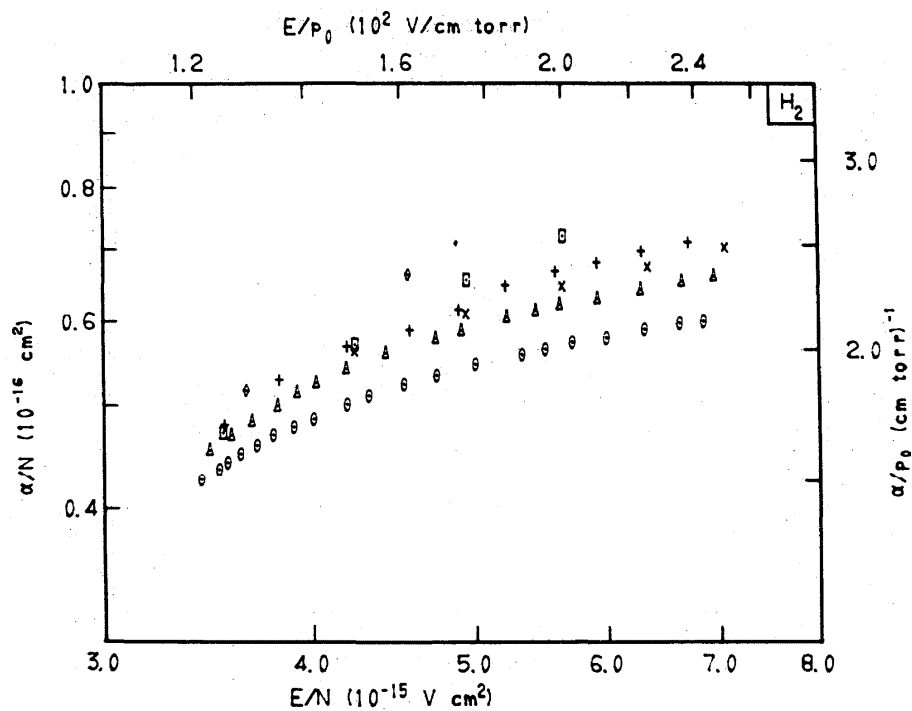
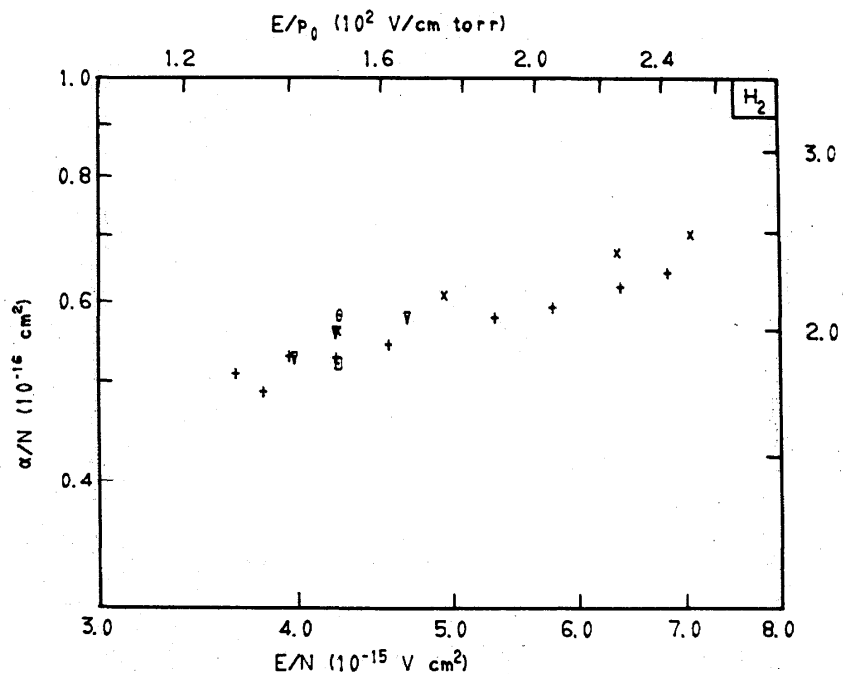


FIGURE 7.5(b). α/N , E/N for hydrogen for $E/N \leq 3.4 \times 10^{-15} \text{ V cm}^2$.



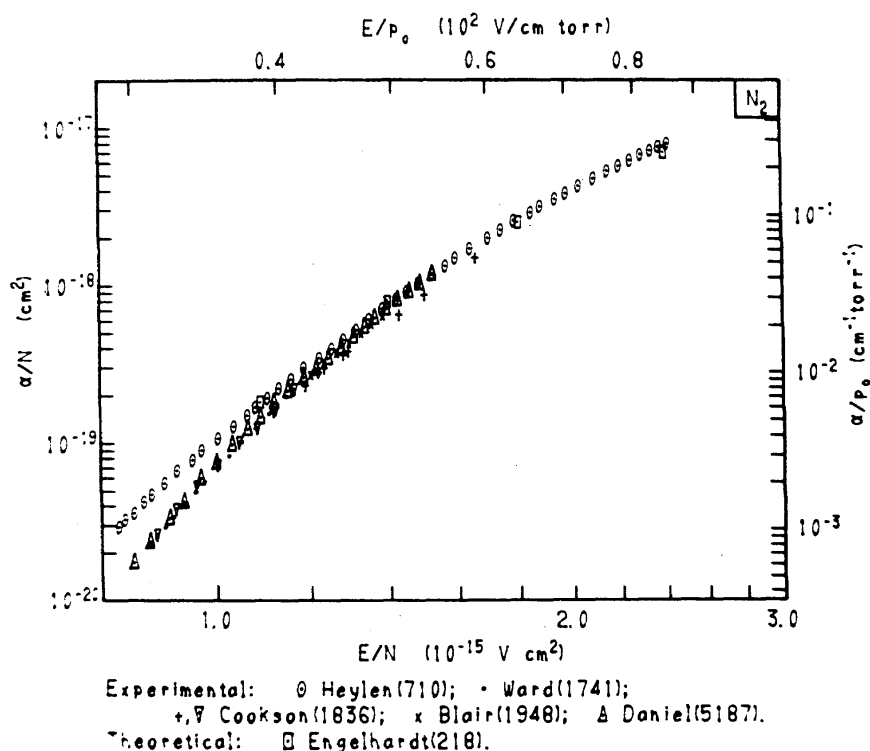
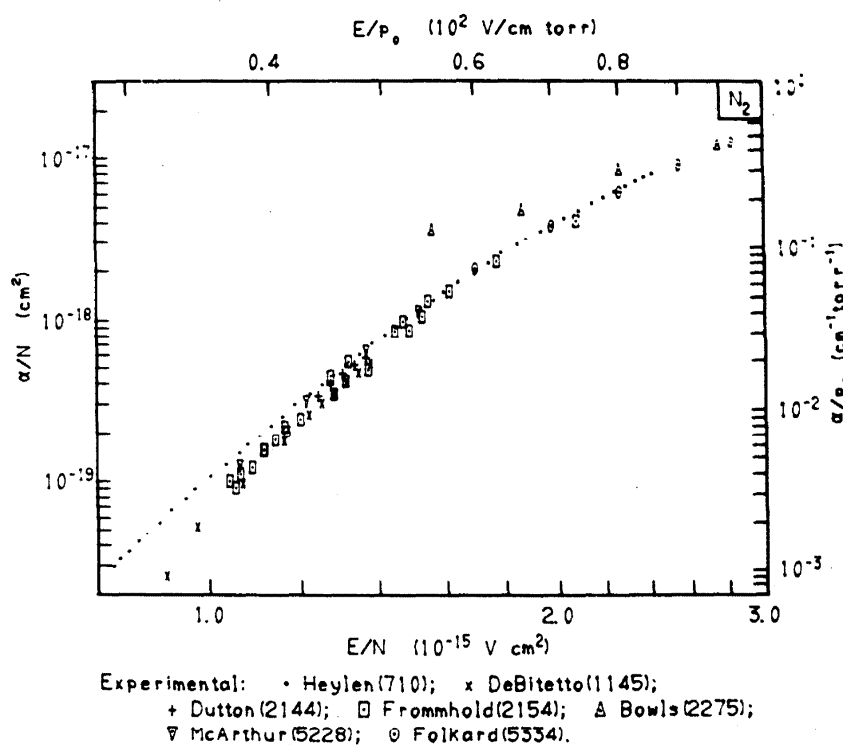
Experimental: $\circ$, Δ Fletcher(1160); $\diamond$ Blevin(790);
 $\square$ Rose(1356); $+$ Haydon(2040); $\times$ Folkard(4907).

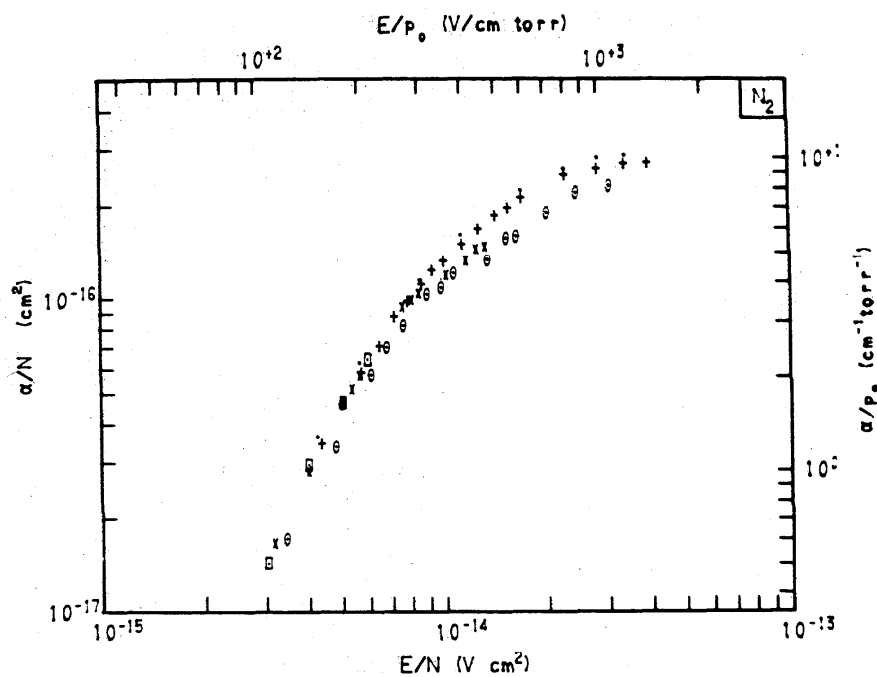
FIGURE 7.6(a). α/N , E/N for hydrogen for $E/N > 3.4 \times 10^{-15} \text{ V cm}^2$.



Experimental: $\circ$ Golden(2145); ∇ Haydon(2148); $+$ Jones(2150)
 Δ Chanin(751); $\times$ Folkard(4909); $\square$ Blasberg(5105);
 $\diamond$ Shallal(5225).

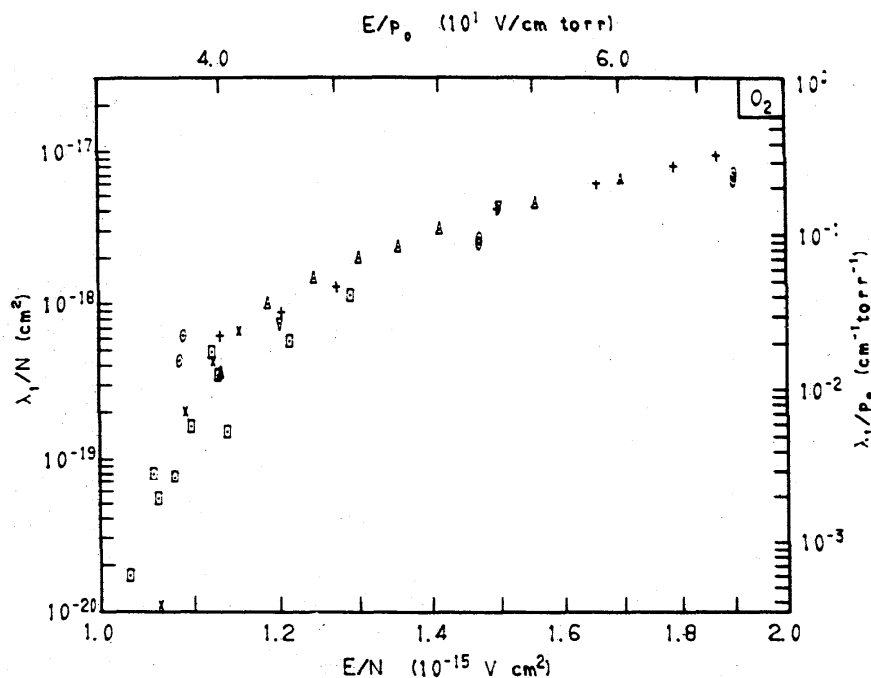
FIGURE 7.6(b). α/N , E/N for hydrogen for $E/N > 3.4 \times 10^{-15} \text{ V cm}^2$.

FIGURE 7.7(a). α/N , E/N for nitrogen for $E/N < 2.82 \times 10^{-15} \text{ V cm}^2$.FIGURE 7.7(b). α/N , E/N for nitrogen for $E/N < 2.82 \times 10^{-15} \text{ V cm}^2$. (The data of McArthur [5228] were obtained using the expression $\alpha/p_{20} = 6.36 \exp(-257 p_{20}/E)$ which is stated in the paper to fit the results to within ± 3 percent.)



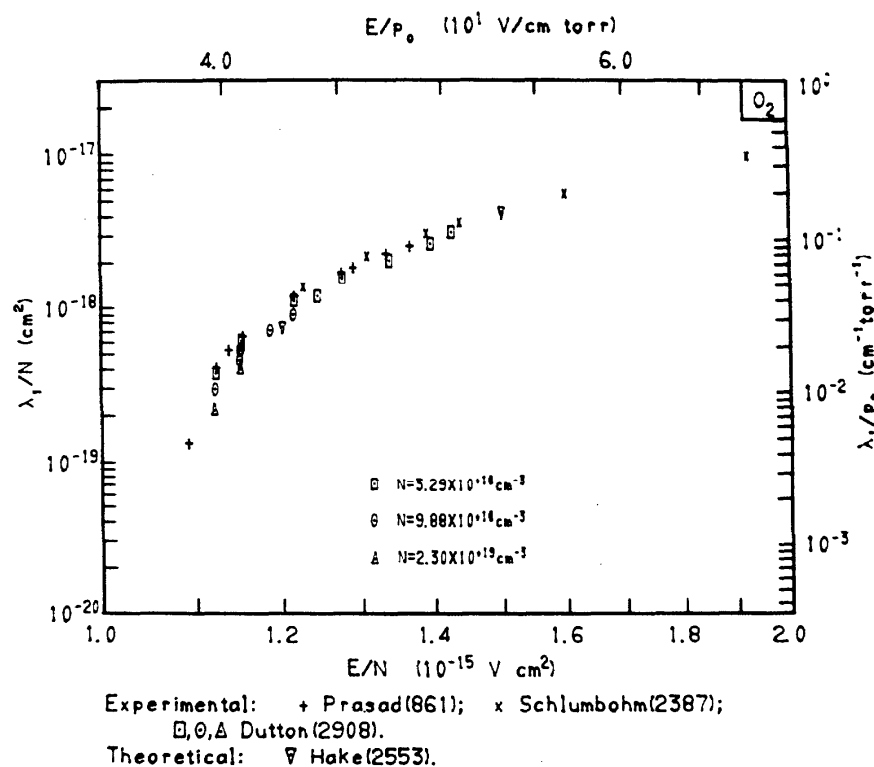
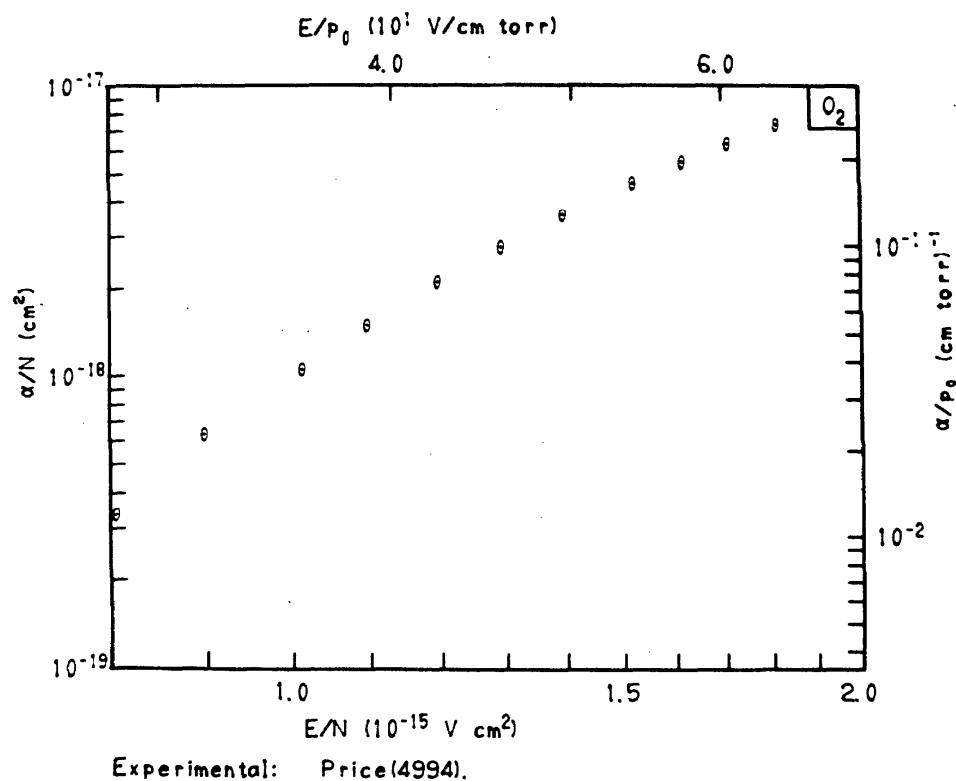
Experimental: $\circ$ Bawls(2275); $+$ Bagnall(2448);
 $\times$ Jones(3297); $\cdot$ Folkard(5334).
 Theoretical: $\square$ Engelhardt(218).

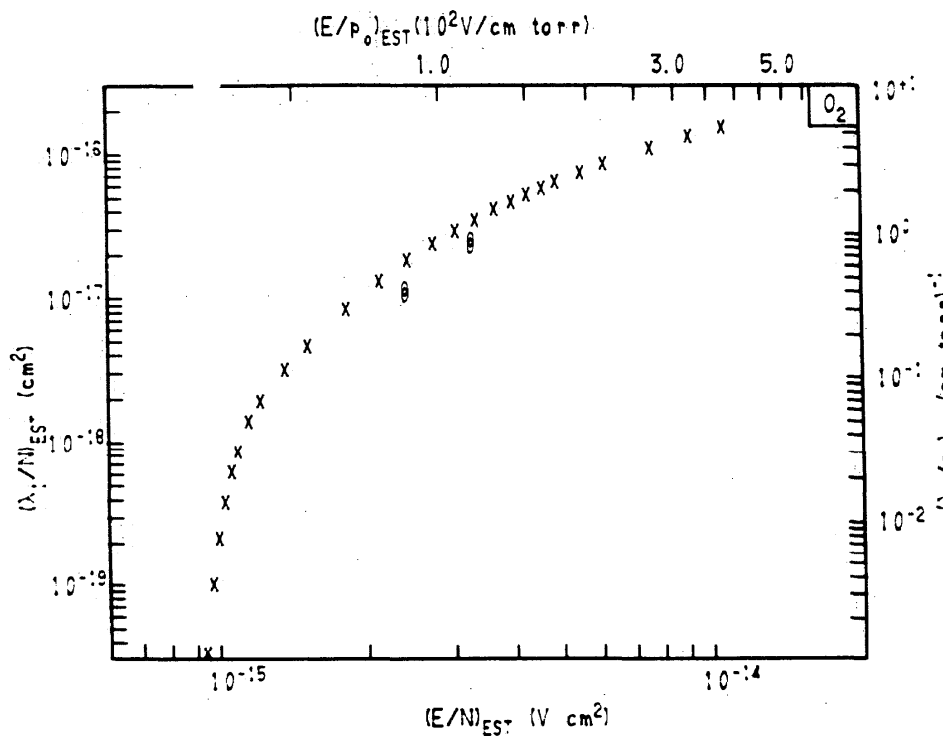
FIGURE 7.8. α/N , E/N for nitrogen for $E/N > 2.82 \times 10^{-15} \text{ V cm}^2$.



Experimental: $\times$ Dutton(573); $+$ Harrison(791);
 Δ Freely(961); $\circ$ Frommhold(2154); $\square$ Prasad(2555).
 Theoretical: ∇ Hake(2553).

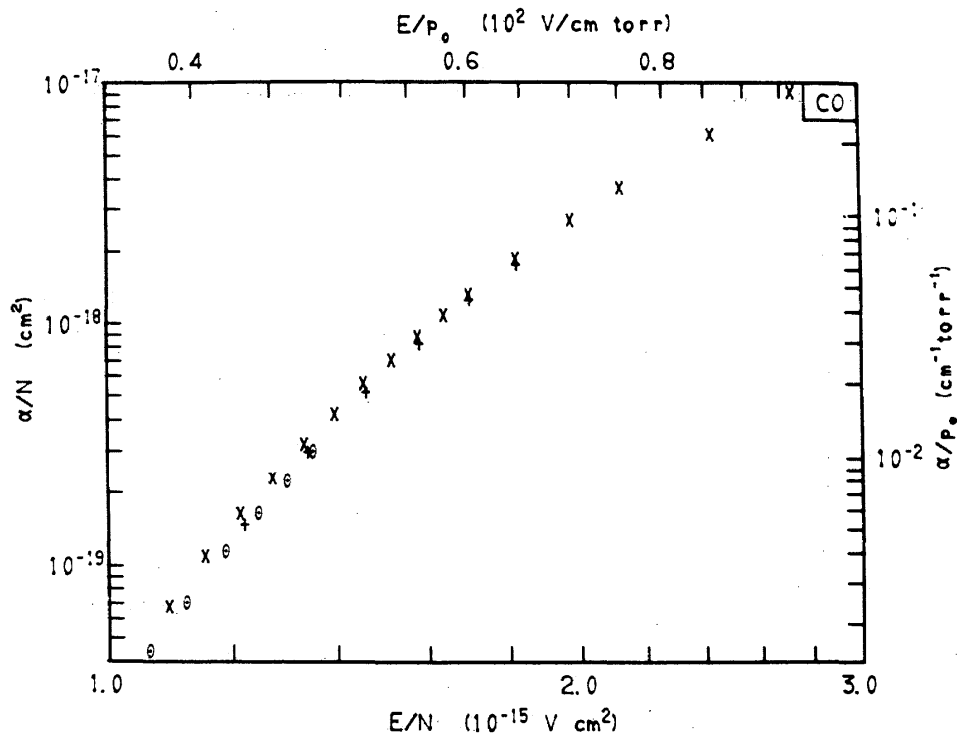
FIGURE 7.9(a). λ_1/N , E/N for oxygen for $E/N < 200 \times 10^{-17} \text{ V cm}^2$.

FIGURE 7.9(b). $\lambda_1/N, E/N$ for oxygen for $E/N < 200 \times 10^{-17} \text{ V cm}^2$.FIGURE 7.9(c). $\alpha/N, E/N$ for oxygen.



Experimental: θ Frommhold(2154); x Masch(2556).

FIGURE 7.10. $(\lambda_1/N)_{\text{Est}}$, $(E/N)_{\text{Est}}$ for mercury-contaminated oxygen including the region of $E/N > 200 \times 10^{-17} \text{ V cm}^2$. The two results of Frommhold [2154] are for λ_1/N for oxygen uncontaminated with mercury vapor.



Experimental: x Bhalla(1254); $+$ Davies(4016); θ Parr(5243).

FIGURE 7.11. α/N , E/N for carbon monoxide for $E/N < 300 \times 10^{-17} \text{ V cm}^2$.

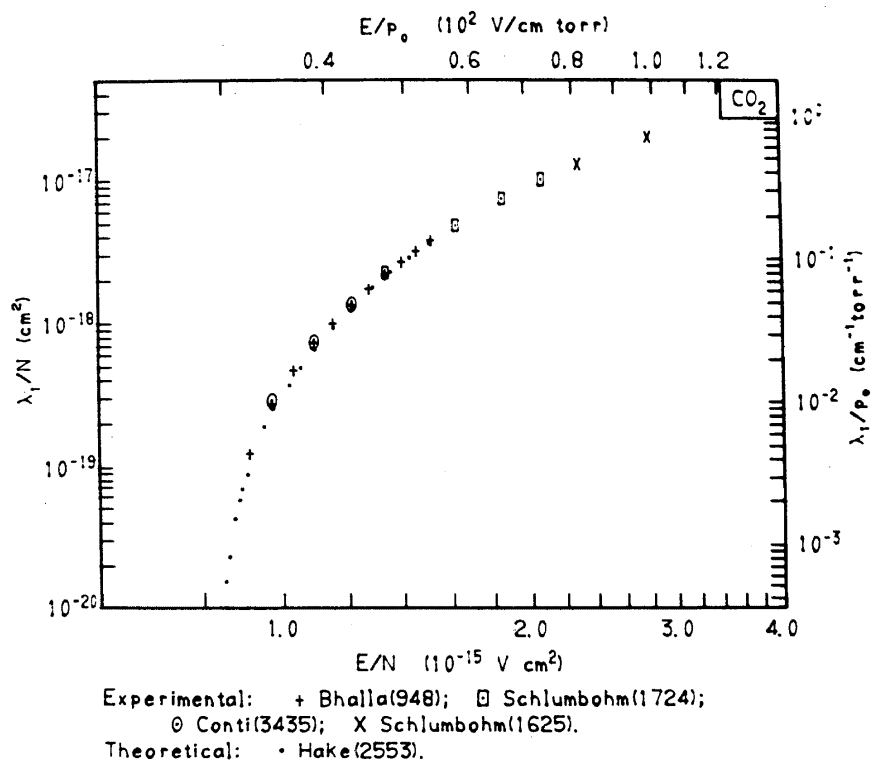


FIGURE 7.12. λ_1/N , E/N for carbon dioxide for values of E/N up to $300 \times 10^{-17} \text{ V cm}^2$.

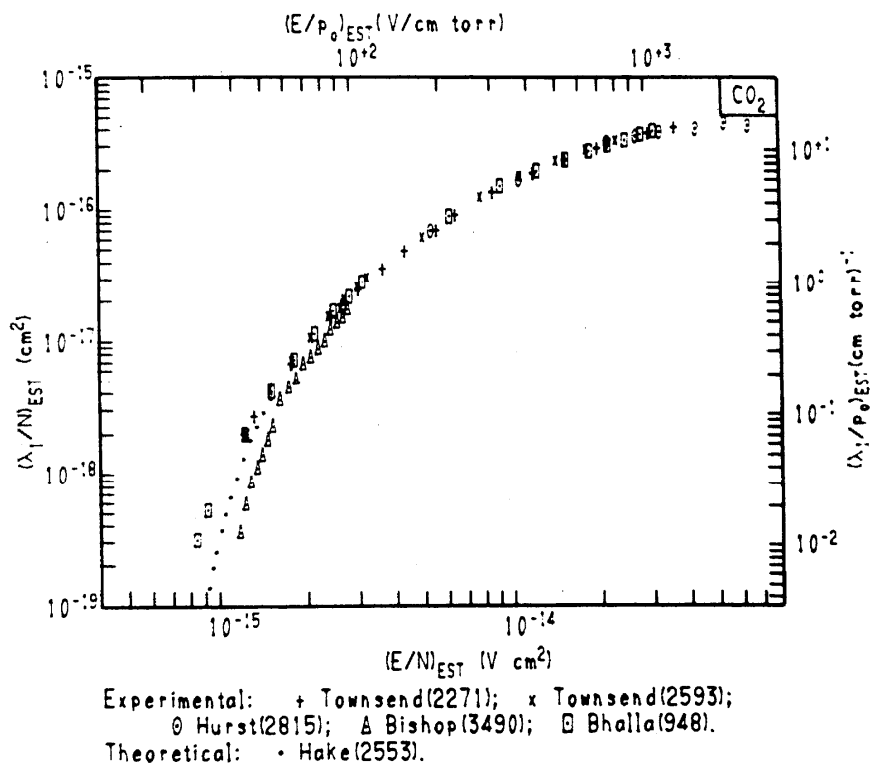
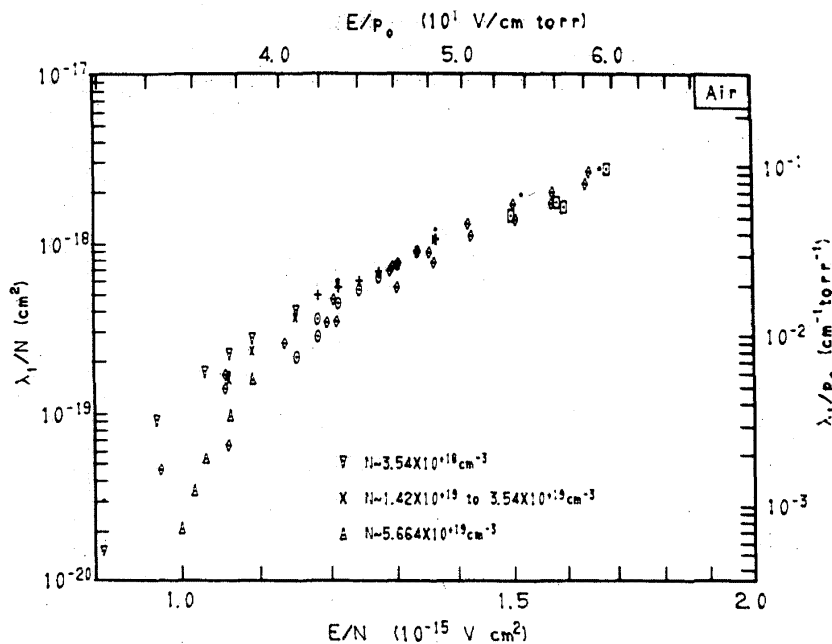
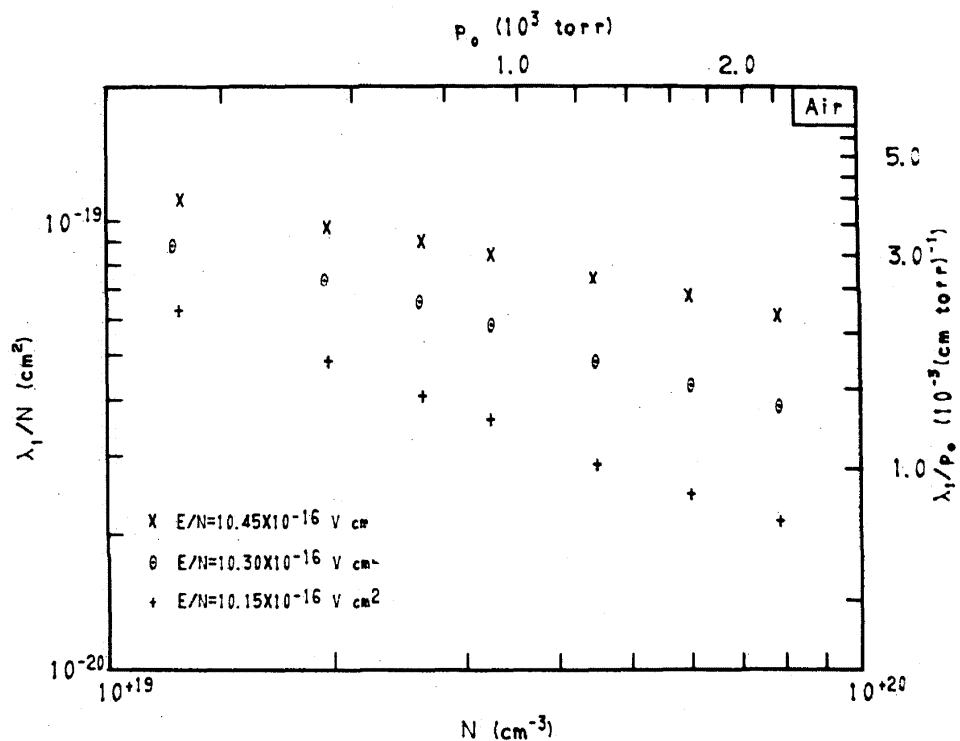
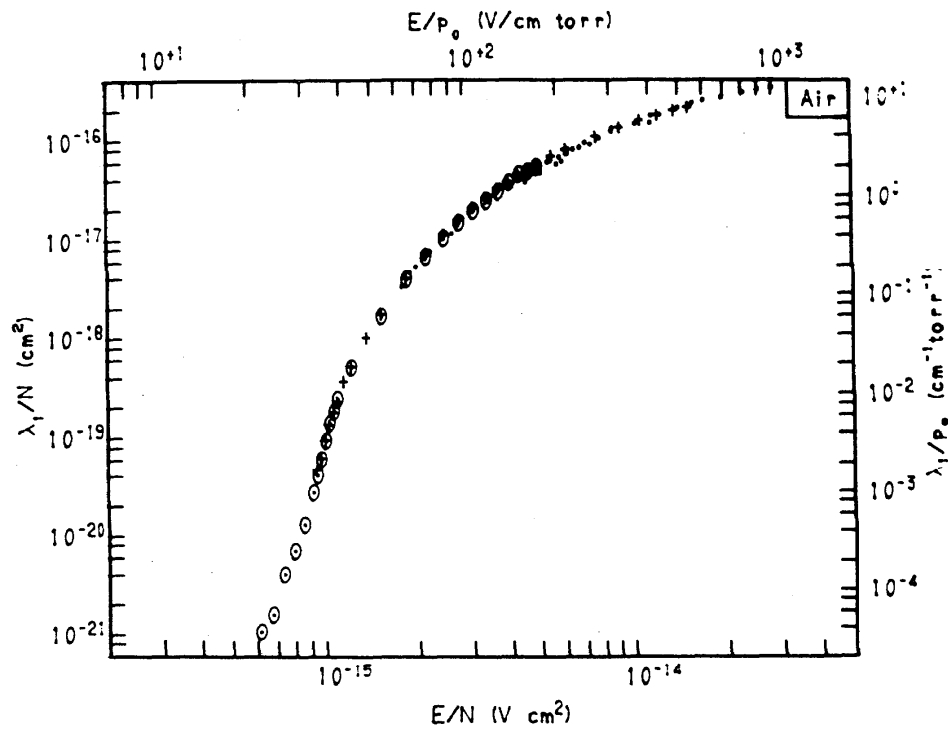


FIGURE 7.13. $(\lambda_1/N)_{\text{EST}}$, $(E/N)_{\text{EST}}$ for carbon dioxide for values of E/N up to about $7000 \times 10^{-17} \text{ V cm}^2$.

FIGURE 7.14. (λ_i/N) , (E/N) for air for $E/N < 1.7 \times 10^{-15} \text{ V cm}^2$.FIGURE 7.15. Variation of λ_i/N with N at given values of E/N for air.



Experimental: $\circ$ Sanders (2274); $+$ Masch (2556); $\bullet$ Rao (5246).

FIGURE 7.16. (λ_1/N) , (E/N) for air for values of E/N up to about $2000 \times 10^{-17} \text{ V cm}^2$. (The results of Sanders [2274] and Masch [2556] are for mercury-contaminated air while those of Rao [5246] are for mercury-free, dry air.)

Table 7.1

Values of α/N for helium

E/N (10^{-17} V cm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} V cm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} V cm^2)	α/N (10^{-18} cm^2)
9.09	0.00212 (1)	42.5	3.10 (a)	156.1	18.4 (b)
12.12	0.0303 (1)	42.6	2.67 (b)	170.1	19.2 (b)
15.16	0.127 (1)	56.1	5.1 (a)	184.2	22.3 (b)
17.34	0.277 (b)	57.1	4.0 (b)	196.2	25.7 (a)
19.87	0.48 (b)	71.2	5.9 (b)	198.3	22.8 (b)
19.95	0.43 (a)	85.1	9.4 (a)	226.5	29.2 (b)
23.12	0.79 (b)	85.3	7.4 (b)	254.7	34 (b)
26.01	1.05 (b)	99.3	9.6 (b)	282.8	37 (b)
27.97	1.41 (a)	113.8	12.1 (b)	288.2	37 (a)
28.54	1.39 (b)	127.9	14.8 (b)	420	50 (a)
35.8	2.26 (b)	140.9	17.5 (a)	563	56 (a)
		142.0	16.7 (b)	832	67 (a)

Experimental:

- (a) Chanin, *et al.*, Phys. Rev. **133**, A1005 (1964).
- (b) Davies, *et al.*, Proc. Phys. Soc. (London) **80**, 898 (1962).

Theoretical:

- (1) Dunlop, Nature **164**, 452 (1949).

Table 7.2

Values of α/N for neon

E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)
5.65	{0.0141 (a) 0.0184 (b)}	39.6	{2.87 (b) 2.74 (d)}	127.1	17.8 (b)
6.22	0.0223 (b)	42.4	3.2 (a)	141.2	{19.2 (a) 20.2 (b)}
7.06	0.032 (b)	45.0	4.0 (c)		{21.2 (d) 24.0 (d)}
7.91	0.044 (b)	45.2	3.7 (b)	155.4	
8.48	0.044 (a)		{4.5 (b) 5.1 (d)}	169.5	{24.9 (b) 26.3 (d)}
9.04	0.065 (b)	50.9	5.1 (c)		{29.1 (a) 29.3 (b)}
10.17	0.090 (b)	52.2	5.9 (d)	197.7	{30.8 (d) 34 (b)}
11.30	0.119 (b)	55.1	{5.6 (a) 5.4 (b)}		38 (b)
12.71	0.167 (b)	56.5	6.0 (c)	226.0	{43 (a) 42 (b)}
14.12	{0.205 (a) 0.226 (b)}	59.6	{6.4 (b) 6.8 (d)}	254.2	50 (b)
15.54	0.294 (b)	62.2	7.3 (d)	282.5	
16.95	0.38 (b)	65.0	7.2 (c)	339	56 (b)
19.77	{0.55 (a) 0.58 (b)}	67.2	7.8 (b)	396	61 (a)
22.29	0.88 (c)	70.6	8.3 (c)	424	63 (b)
22.60	0.82 (b)	74.2	{9.3 (b) 9.3 (d)}	452	69 (b)
24.01	1.05 (d)	79.1	9.5 (c)	509	{74 (a) 73 (b)}
24.86	1.32 (c)	81.4	10.6 (c)	565	
25.42	1.10 (b)		{11.2 (b) 11.3 (d)}	678	82 (b)
28.01	1.57 (c)	90.0	{13.3 (b) 13.3 (d)}	791	89 (b)
28.25	{1.31 (a) 1.42 (b) 1.53 (d)}	90.4		848	102 (a)
31.1	1.76 (b)	101.7		904	93 (b)
31.7	2.19 (c)	101.8		1017	98 (b)
33.9	{2.11 (b) 2.12 (d)	113.0	12.7 (c)	1130	{116 (a) 101 (b)}
37.9	2.94 (c)	113.3	15.3 (b)		
		118.6	13.8 (c)		
			15.8 (d)		

Experimental:

- (a) Chanin, et al., Phys. Rev. **132**, 2547 (1963).
 (b) Kruithof, et al., Physica **4**, 430 (1937).
 (c) Dutton, et al., J. Phys. B: Atom. Molec. Phys. **2**, 890 (1969).
 (d) Willis, et al., Brit. J. Appl. Phys. (J. Phys. D) **1**, 1219 (1968)

Table 7.3

Values of α/N for argon

E/N (10^{-17}Vcm^2)	α/N (10^{-17}cm^2)	E/N (10^{-17}Vcm^2)	α/N (10^{-17}cm^2)	E/N (10^{-17}Vcm^2)	α/N (10^{-17}cm^2)
14.12	0.00068	79.1	0.53	565	12.5
15.54	0.00104	90.4	0.74	678	14.9
16.95	0.00158	101.7	0.95	791	17.0
19.77	0.0032	113.0	1.22	904	19.0
22.60	0.0059	127.1	1.55	1017	20.7
25.42	0.0098	141.2	1.90	1130	22.3
28.25	0.016	169.5	2.62	1271	24.0
31.1	0.025	197.7	3.35	1412	25.6
33.9	0.036	226.0	4.1	1695	28.3
39.6	0.068	254.2	4.8	1977	30.5
45.2	0.109	282.5	5.6	2260	32
50.9	0.158	339	7.0	2542	34
56.5	0.216	396	8.5	2825	35
62.2	0.282	452	9.8	3390	36
70.6	0.40	509	11.2	3960	38
				4520	39

Experimental:

All data taken from Kruithof, *Physica* 7, 519 (1940).

Table 7.4(a)

Values of α/N for krypton

E/N (10^{-17}Vcm^2)	α/N (10^{-17}cm^2)	E/N (10^{-17}Vcm^2)	α/N (10^{-17}cm^2)	E/N (10^{-17}Vcm^2)	α/N (10^{-17}cm^2)
14.12	0.00044 (a)	62.4	0.233 (b)	226.0	4.2 (a)
15.54	0.00068 (a)	68.4	0.282 (b)	246.1	4.8 (b)
16.95	0.00103 (a)	70.6	0.316 (b)	249.7	4.7 (b)
19.77	0.00218 (a)	77.2	0.38 (a)	254.2	4.9 (a)
22.60	0.0041 (a)	78.3	0.38 (b)	282.5	5.7 (a)
25.42	0.0071 (a)	79.1	0.45 (b)	339	7.4 (a)
27.51	0.0046 (b)	87.1	0.51 (a)	396	9.0 (a)
28.25	0.0117 (a)	90.4	0.56 (b)	452	10.6 (a)
30.5	0.0118 (b)	99.0	0.59 (b)	509	12.2 (a)
30.8	0.0114 (b)	101.7	0.71 (a)	565	13.7 (a)
31.1	0.0182 (a)	110.9	0.79 (b)	678	16.4 (a)
33.7	0.0173 (b)	113.0	0.93 (a)	791	18.9 (a)
33.9	0.0268 (a)	124.2	1.08 (b)	904	21.2 (a)
33.9	0.0197 (b)	127.1	1.18 (a)	1017	23.3 (a)
36.4	0.0261 (b)	130.6	1.42 (b)	1130	25.3 (a)
36.9	0.030 (b)	139.2	1.51 (a)	1271	27.4 (a)
39.6	0.052 (a)	140.2	1.48 (b)	1412	29.5 (a)
42.3	0.046 (b)	141.2	1.78 (b)	1695	33 (a)
43.1	0.064 (b)	157.1	1.71 (b)	1977	36 (a)
45.2	0.089 (a)	166.4	1.87 (a)	2260	38 (a)
49.4	0.087 (b)	169.5	2.22 (b)	2542	40 (a)
50.9	0.137 (a)	193.2	2.39 (b)	2825	41 (a)
54.9	0.135 (b)	197.7	2.62 (a)	3390	44 (a)
56.5	0.195 (a)	203.1	3.2 (b)	4000	45 (a)
61.5	0.198 (b)		3.4 (a)	4520	46 (a)
62.2	0.262 (a)		3.4 (b)	5090	47 (a)
				5650	47 (a)

Experimental:

(a) Kruithof, *Physica* 7, 519 (1940).(b) Heylen, *Int. J. Electron.* 31, 19 (1971).

Table 7.4(b)

Values of α/N for xenon

E/N (10^{-17} Vcm^2)	α/N (10^{-17} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-17} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-17} cm^2)
22.60	0.00054	127.1	0.96	1017	26.7
25.42	0.00102	141.2	1.25	1130	29.4
28.25	0.00181	169.5	1.88	1271	33
31.1	0.0030	197.7	2.60	1412	35
33.9	0.0048	226.0	3.4	1695	41
39.6	0.0109	254.2	4.2	1977	45
45.2	0.0215	282.5	5.1	2260	49
50.9	0.038	339	6.8	2542	53
56.5	0.062	396	8.5	2825	56
62.2	0.092	452	10.3	3390	61
70.6	0.151	509	12.1	4000	65
79.1	0.227	565	14.0	4520	68
90.4	0.36	678	17.5	5090	70
101.7	0.51	791	20.8	5650	72
113.0	0.70	904	23.9	6780	74

Experimental:

All data taken from Kruithof, *Physica* 7, 519 (1940).

Table 7.5

Values of α/N for hydrogen

E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)
42.4	0.0073 (a)	62.0	0.255 (d)	79.0	0.98 (d)
42.7	0.0063 (b)	62.2	0.260 (a)	79.1	1.01 (a)
45.2	0.0161 (a)	62.5	0.282 (e)	84.1	1.37 (e)
45.8	0.0142 (c)		0.32 (g)	84.7	1.33 (m)
47.9	0.0254 (d)	63.7	0.30 (c)	84.8	1.40 (a)
					1.44 (f)
48.0	0.0299 (a)	64.0	0.31 (b)		
48.7	0.031 (b)	66.7	0.40 (g)	85.1	1.45 (c)
48.8	0.033 (c)	67.0	0.42 (b)	89.0	1.68 (e)
50.9	0.054 (a)	67.8	0.44 (a)	90.6	1.84 (d)
51.8	0.059 (c)	68.0	0.43 (d)	95.7	2.34 (d)
				97.2	2.49 (c)
53.7	0.089 (d)	68.4	0.47 (c)		
54.8	0.099 (b)	68.9	0.47 (e)	97.3	2.39 (e)
55.2	0.103 (c)	69.6	0.52 (c)	98.9	2.71 (a)
56.4	0.115 (e)	69.7	0.53 (g)	101.4	2.94 (d)
56.5	0.116 (f)	72.8	0.67 (g)	107.0	3.6 (d)
	0.130 (a)			109.1	3.9 (c)
		73.2	0.67 (d)		
56.6	0.134 (d)	73.5	0.68 (a)	113.0	4.4 (a)
57.9	0.150 (c)	75.8	0.85 (g)	120.4	7.1 (h)
60.6	0.227 (g)	76.0	0.84 (c)	121.0	5.8 (c)
60.9	0.221 (b)	78.7	0.99 (e)	124.5	5.5 (d)
61.1	0.215 (c)			127.1	6.6 (a)

Table 7.5 (continued)

Values of α/N for hydrogen

E/N (10^{-17}Vcm^2)	α/N (10^{-18}cm^2)	E/N (10^{-17}Vcm^2)	α/N (10^{-18}cm^2)	E/N (10^{-17}Vcm^2)	α/N (10^{-18}cm^2)
138.3	9.1 (i)	211.9	22.0 (l)		36 (a)
140.9	8.3 (d)		20.9 (m)	282.5	36 (f)
	8.8 (f)	214.6	22.1 (j)		36 (l)
141.2	8.9 (a)	215.4	25.5 (k)		33 (m)
	8.8 (l)	221.8	26.1 (k)	300	39 (k)
	8.3 (m)				39 (h)
		224.3	24.0 (h)		
152.7	10.41 (j)	226.0	25.1 (a)	306	41 (j)
152.9	11.22 (i)	228.0	25.4 (d)	308	44 (i)
154.0	15.5 (k)	232.3	28.2 (k)	311	41 (k)
162.4	16.6 (k)	240.8	29.8 (k)	321	42 (k)
	13.5 (d)			334	46 (h)
169.5	14.1 (a)	244.0	29.2 (j)		
		247.2	31 (k)	338	45 (k)
173.0	18.2 (k)	254.2	31 (a)	353	48 (l)
177.9	16.7 (h)	257.0	32 (d)	424	57 (l)
181.5	19.5 (k)	264.1	34 (k)	494	52 (m)
183.7	15.4 (j)				61 (l)
187.9	20.7 (k)	270.7	31 (h)		
		272.6	35 (k)	565	65 (l)
196.6	19.4 (d)	274.9	36 (j)		66 (m)
197.7	19.5 (a)	279.1	36 (i)	636	68 (l)
204.8	23.6 (k)			706	71 (l)
208.6	22.2 (d)			847	82 (m)

Experimental:

- (a) Rose, *Phys. Rev.* **104**, 273 (1956).
- (b) DeBitetto, *et al.*, *Phys. Rev.* **104**, 1213 (1956).
- (c) Hopwood, *et al.*, *Proc. Roy. Soc. (London) Ser. A* **235**, 334 (1956).
- (d) Barna, *et al.*, *J. Appl. Phys.* **35**, 2781 (1964).
- (e) Frommhold, *Z. Physik* **160**, 554 (1960).
- (f) Chanin, *et al.*, *Phys. Rev.* **132**, 2547 (1963).
- (g) Crompton, *et al.*, *Proc. Phys. Soc. (London) Ser. B* **69**, 2 (1956).
- (h) Jones, *et al.*, *Proc. Phys. Soc. (London)* **72**, 363 (1958).
- (i) Haydon, *et al.*, *Proc. Phys. Soc. (London)* **78**, 92 (1961).
- (j) Blevin, *et al.*, *Nature* **179**, 38 (1957).
- (k) Fletcher, *et al.*, in *Proc. of the Sixth International Conference on Ionization Phenomena in Gases*, Paris, 8-13 July, 1963, (P. Hubert and E. Cremieu-Alcan, Eds., Serma, Paris, 1963), Vol. 2, p. 217.
- (l) Folkard, *et al.*, *Austr. J. Phys.* **24**, 527 (1971).
- (m) Blasberg, *et al.*, *Physica* **54**, 468 (1971).

Table 7.b

Values of α/N for nitrogen

E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)	E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)
85.0	0.0181 (a)	121.2	0.31 (f)	165.8	1.50 (c)
87.5	0.0222 (b)	121.4	0.31 (a)	169.5	2.06 (j)
87.7	0.0242 (a)	122.1	0.258 (d)	169.7	2.00 (i)
88.7	0.0252 (b)		0.281 (c)	174.3	2.26 (i)
88.9	0.026 (c)	123.3	0.32 (g)	177.1	2.29 (e)
90.4	0.0293 (b)	123.5	0.299 (c)	179.0	2.60 (i)
91.2	0.034 (a)	124.3	0.33 (h)	184.7	2.93 (i)
91.5	0.0254 (d)	124.4	0.35 (a)	188.2	3.2 (i)
91.9	0.034 (b)	125.3	0.30 (d)	193.6	3.5 (i)
92.2	0.038 (c)	126.4	0.37 (g)	197.5	3.8 (i)
93.8	0.044 (a)		0.39 (h)	197.7	3.8 (j)
93.9	0.041 (b)	127.3	0.44 (e)	202.9	4.3 (i)
96.0	0.049 (b)	127.4	0.42 (a)	207.6	4.2 (e)
96.2	0.053 (c)	128.1	0.36 (c)	208.6	4.8 (i)
97.0	0.061 (a)	128.3	0.35 (d)	214.3	5.3 (i)
			0.34 (e)		
97.5	0.053 (d)			219.3	5.7 (i)
97.7	0.057 (b)	129.2	0.38 (c)	223.3	6.2 (i)
99.9	0.077 (a)	129.4	0.43 (g)	226.0	9.0 (j)
100.2	0.070 (b)	130.3	0.45 (h)	228.9	6.7 (i)
100.3	0.072 (c)	130.4	0.49 (a)	233.2	7.2 (i)
103.0	0.101 (a)	131.1	0.42 (e)	237.1	7.6 (i)
104.1	0.101 (e)	131.3	0.41 (d)	240.7	8.0 (i)
104.5	0.101 (c)	131.9	0.55 (e)	254.2	9.1 (j)
105.2	0.102 (b)	132.5	0.50 (g)	282.5	12.7 (j)
105.3	0.091 (e)	133.4	0.52 (h)	316	16.5 (k)
106.1	0.124 (f)	133.5	0.56 (a)	342	17.1 (l)
106.2	0.126 (a)	134.6	0.47 (d)	400	28.0 (k)
106.3	0.111 (e)	135.5	0.57 (g)	424	36 (j)
106.8	0.097 (d)	136.3	0.64 (a)	436	34 (m)
108.0	0.125 (b)		0.58 (h)	483	34.2 (l)
		136.4	0.64 (f)		
108.3	0.125 (c)			499	46 (m)
108.9	0.123 (e)	137.2	0.49 (e)	507	47 (k)
109.0	0.151 (a)	137.7	0.54 (d)	536	52 (k)
110.9	0.153 (b)	138.4	0.64 (g)	565	63 (j)
111.4	0.156 (e)	139.3	0.73 (a)	567	57 (k)
111.7	0.159 (c)	142.5	0.84 (a)	573	58 (m)
111.9	0.187 (a)	142.9	0.65 (c)	612	57 (l)
112.4	0.164 (c)	143.3	0.67 (c)	646	70 (m)
113.9	0.182 (e)	144.7	0.86 (e)	681	70 (l)
114.8	0.197 (b)	145.7	0.95 (a)	715	87 (m)
115.0	0.219 (a)	147.0	0.98 (e)	757	94 (k)
115.9	0.178 (d)	148.6	1.06 (a)	760	82 (l)
116.2	0.221 (e)	149.0	0.87 (e)	783	98 (m)
116.4	0.207 (e)	150.1	0.87 (c)	804	99 (k)
116.5	0.217 (c)	151.6	1.13 (f)	843	104 (k)
118.3	0.242 (b)	152.4	1.22 (a)	848	114 (j)
118.7	0.267 (a)	152.8	1.06 (e)	857	111 (m)
118.9	0.230 (c)	154.6	1.31 (e)	893	103 (l)
120.0	0.245 (e)	161.2	1.52 (e)	920	123 (m)
120.3	0.269 (g)	164.0	1.72 (i)	983	108 (l)

Table 7.6 (continued)

Values of α/N for nitrogen

E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)		E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)		E/N (10^{-17} Vcm^2)	α/N (10^{-18} cm^2)	
999	131	(m)	1326	148	(k)	2255	248	(m)
1017	119	(k)	1336	133	(l)	2260	260	(j)
1070	120	(l)	1414	185	(m)	2441	220	(l)
1130	160	(j)	1530	157	(l)	2812	260	(m)
1135	149	(m)	1550	195	(m)	2825	282	(j)
1159	132	(k)	1635	159	(l)	3050	231	(l)
1243	145	(k)	1692	211	(m)	3370	268	(m)
1267	167	(m)	1695	223	(j)	3390	288	(j)
			2013	189	(l)	3930	270	(m)

Experimental:

- (a) Daniel, *et al.*, J. Phys. B **3**, 363 (1970).
- (b) Ward, Nature **208**, 994 (1965).
- (c) Cookson, *et al.*, Brit. J. Appl. Phys. **17**, 891 (1966).
- (d) DeBitetto, *et al.*, Phys. Rev. **104**, 1213 (1956).
- (e) Frommhold, Z. Physik **160**, 554 (1960).
- (f) McArthur, *et al.*, 1st Int. Conf. Gas Discharges, London, 284 (1970).
- (g) Blair, Brit. J. Appl. Phys. **17**, 1051 (1966).
- (h) Dutton, *et al.*, Proc. Roy. Soc. (London) Ser. A **213**, 203 (1952).
- (i) Heylen, Nature **183**, 1545 (1959).
- (j) Folkard, *et al.*, J. Phys. B **6**, 214 (1973).
- (k) Jones, Brit. J. Appl. Phys. (J. Phys. D) **1**, 769 (1968).
- (l) Bowle, Phys. Rev. **53**, 293 (1938).
- (m) Bagnall, *et al.*, Austr. J. Phys. **18**, 227 (1965).

Table 7.7

Values of α/N for carbon monoxide

E/N (10^{-17} Vcm 2)	α/N (10^{-18} cm 2)	E/N (10^{-17} Vcm 2)	α/N (10^{-18} cm 2)	E/N (10^{-17} Vcm 2)	α/N (10^{-18} cm 2)
106.1	0.044 (c)	130.0	0.225 (c)	158.0	0.82 (b)
109.1	0.067 (a)	133.4	0.32 (a)	163.7	1.08 (a)
112.1	0.070 (c)	134.0	0.294 (b)	169.7	1.31 (a)
115.2	0.109 (a)	135.1	0.299 (c)	170.0	1.24 (b)
118.8	0.114 (c)	139.4	0.42 (a)	181.9	1.87 (a)
121.2	0.164 (a)	145.5	0.56 (a)	182.0	1.76 (b)
122.0	0.145 (b)	146.0	0.52 (b)	197.0	2.70 (a)
124.6	0.165 (c)	151.6	0.71 (a)	212.2	3.7 (a)
127.3	230 (a)	157.6	0.89 (a)	242.5	6.1 (a)
				272.8	9.2 (a)

Experimental:

- (a) Bhalla, *et al.*, Proc. Phys. Soc. (London) **78**, 438 (1961).
 (b) Davies, *et al.*, in *Contributed Papers of the Ninth International Conference on Phenomena in Ionized Gases*, Bucharest, 1-6 Sept. 1969, (Academy of the Socialist Republic of Romania, 3 Bis Gutenberg Ave., Bucharest, 1969), p. 46.
 (c) Parr, *et al.*, 10th Int. Conf. Phen. Ion. Gas. (1971), p. 8.

Table 7.8

Values of λ_1/N for carbon dioxide

E/N (10^{-17} Vcm 2)	λ_1/N (10^{-18} cm 2)	E/N (10^{-17} Vcm 2)	λ_1/N (10^{-18} cm 2)	E/N (10^{-17} Vcm 2)	λ_1/N (10^{-18} cm 2)
90.9	0.121 (a)	169.8	6.7 (a)	858	135 (e)
97.0	0.268 (a)	185.2	7.4 (c)	1140	181 (a)
	0.289 (b)	199.1	10.6 (a)	1188	190 (e)
103.1	0.47 (a)	206.7	10.2 (c)	1426	222 (a)
109.1	0.73 (a)	229.1	12.9 (d)	1518	236 (e)
	0.75 (b)	229.6	15.7 (a)	1714	258 (a)
115.2	1.00 (a)	260.0	20.3 (a)	1961	290 (e)
121.2	1.34 (a)	279.1	20.0 (d)	1991	285 (a)
	1.36 (b)	287.7	26.1 (a)	2275	310 (a)
127.3	1.72 (a)	298.6	24.2 (e)	2576	340 (a)
133.4	2.17 (a)	363	36. (e)	2825	360 (a)
133.5	2.25 (c)	430	49. (e)	2919	370 (e)
139.4	2.64 (a)	552	70. (e)	3580	420 (e)
145.5	3.2 (a)	570	83.8 (a)	4240	410 (f)
151.6	3.7 (a)	642	92 (e)	5300	460 (f)
162.3	4.8 (c)	857	141 (a)	6370	430 (f)

Experimental:

- (a) Bhalla, *et al.*, Proc. Phys. Soc. (London) **76**, 369 (1960).
 (b) Conti, *et al.*, in *Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases*, Vienna, 27 Aug.-2 Sept. 1967 (Springer-Verlag, Vienna, 1967), p. 23.
 (c) Schlumbohm, Z. Physik **166**, 192 (1962).
 (d) Schlumbohm, Z. Physik **184**, 492 (1965).
 (e) Townsend, Phil. Mag. **3**, 557 (1902).
 (f) Hurst, Phil. Mag. **11**, 535 (1906).

Table 7.9

Values of λ_1/N for air for $E/N > 140 \times 10^{-17} \text{Vcm}^2$

E/N (10^{-17}Vcm^2)	λ_1/N (10^{-18}cm^2)	E/N (10^{-17}Vcm^2)	λ_1/N (10^{-18}cm^2)	E/N (10^{-17}Vcm^2)	λ_1/N (10^{-18}cm^2)
141	1.57 (b)	311	22.8 (b)	565	73.5 (c)
141	1.61 (c)	311	24.0 (c)	566	57.1 (a)
150	1.47 (a)	316	19.7 (a)	579	68.7 (a)
160	1.64 (a)	333	23.2 (a)	589	61.9 (a)
168	2.76 (a)	336	25.2 (a)	612	72.8 (a)
170	3.59 (b)	339	28.5 (b)	612	80.9 (a)
170	3.67 (c)	339	29.7 (c)	643	81.5 (a)
176	3.30 (a)	341	24.4 (a)	675	83.6 (a)
182	3.94 (a)	355	27.9 (a)	701	94.4 (a)
188	4.52 (a)	367	34.9 (b)	706	98.9 (c)
191	4.04 (a)	367	34.8 (c)	731	90.0 (a)
197	5.36 (a)	378	32.4 (a)	780	105 (a)
198	6.33 (b)	379	28.0 (a)	847	123 (c)
198	6.64 (c)	395	41.7 (b)	859	121 (a)
205	5.19 (a)	395	39.6 (c)	865	131 (a)
210	6.67 (a)	397	31.6 (a)	915	135 (a)
217	7.23 (a)	404	34.5 (a)	989	144 (c)
222	7.64 (a)	420	39.6 (a)	1020	148 (a)
226	9.61 (b)	424	45.3 (b)	1130	164 (c)
226	10.3 (c)	424	45.2 (c)	1150	150 (a)
233	8.57 (a)	427	36.0 (a)	1160	178 (a)
240	10.1 (a)	427	41.9 (a)	1230	181 (a)
254	10.6 (a)	441	44.8 (a)	1270	184 (c)
254	13.9 (b)	448	37.4 (a)	1410	198 (c)
254	14.4 (c)	452	49.7 (b)	1410	215 (c)
261	11.6 (a)	452	51.7 (c)	1550	216 (a)
260	12.4 (a)	474	49.3 (a)	1570	237 (a)
269	13.9 (a)	476	47.3 (a)	1700	252 (a)
275	15.0 (a)	489	52.6 (a)	1950	270 (a)
282	16.5 (a)	490	49.3 (a)	2280	279 (a)
282	18.0 (a)	497	54.0 (a)	2550	318 (a)
282	19.2 (a)	501	46.7 (a)	2830	333 (a)
296	16.3 (a)	508	63.6 (c)		
296	18.8 (a)	530	59.4 (a)		
310	21.0 (a)	538	62.4 (a)		

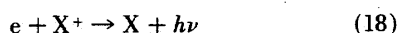
Experimental:

- (a) Rao, *et al.*, J. Phys. D 4, 494 (1971).
 (b) Sanders, Phys. Rev. 44, 1020 (1933).
 (c) Masch, Arch. Elektrotech. 26, 587 (1932).

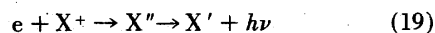
3.8. Recombination Coefficient

When both electrons and positive ions are present in a gas there is a finite probability that should an ion and electron come close to one another they will recombine to form a neutral atom or molecule. This process is different from all the others considered in this survey in that it involves the interaction of two charged particles rather than a charged and a neutral particle. Thus, although it is a process that can occur in electron avalanches, recombination is usually of greater importance in the study of more highly ionized gases where the probability of collision between electrons and ions is greater. In certain circumstances, e.g. in a highly ionized high-density plasma, it can be the dominant mechanism by which charged particles are removed from the gas.

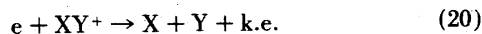
When an electron and an ion recombine, the energy of the system after collision is less than that before, and the difference has to be dissipated in some way. As in the case of electron attachment, the various possible recombination processes are designated by the way in which the excess energy is removed. Thus, radiative recombination may be represented by



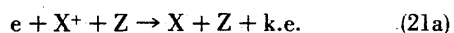
dielectronic recombination by



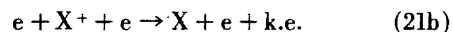
dissociative recombination by



and three-body recombination by



or by



In these equations e represents an electron, $h\nu$ a quantum of radiation, X , Y , and Z neutral atoms, X' and X'' singly and doubly excited electronic states of a neutral atom and k.e. indicates that the energy difference of the reaction is removed as kinetic energy of the particles concerned.

In a gas in which there exists a number density n of electrons and N_+ of positive ions, the rate of removal of the charged particle by the process denoted by eqs (18), (19), or (20) is given by

$$\frac{dn}{dt} = \frac{dN_+}{dt} = -r_2 n N_+, \quad (22)$$

where r_2 is a constant, called the two-body recombination (rate) coefficient. From eq (22) it can be seen that r_2 has the units of $\text{cm}^3 \text{s}^{-1}$.

The rate of removal of charged particles by three-body recombination is given by

$$\frac{dn}{dt} = \frac{dN_+}{dt} = -r_3 n N_+ N, \quad (23a)$$

or

$$\frac{dn}{dt} = \frac{dN_+}{dt} = -r_3 n^2 N, \quad (23b)$$

depending on whether the effective third body is an atom or an electron. In this case r_3 is the three-body recombination coefficient having units of $\text{cm}^6 \text{s}^{-1}$.

The values of the recombination coefficient are usually determined experimentally by measuring the decay of concentration of electrons with time for conditions in which recombination is the dominant loss mechanism. There are, however, numerous difficulties in the interpretation of the experimental data obtained.

First, diffusion always provides a competing mechanism for the decay of electron concentration and considerable care is needed to establish that the dominant loss mechanism is that of recombination [2169] (see also E. P. Gray and D. E. Kerr, *Ann. Phys. (New York)* **17**, 276, 1962) or to make corrections for the loss by diffusion [see, e.g., 3443].

Second, there is often more than one ion species present in the ionized gas, so that the measured decay rate depends on a number of different recombination processes proceeding at different rates. Thus, additional measurements of the rate of decrease of concentration of the ion species involved are most useful, but have only been carried out in a few recent experiments. Additional evidence from spectroscopic measurements can also be helpful in determining the processes occurring.

Third, even in a plasma decaying in a field-free region there are sometimes processes occurring which can give rise to the production of electrons which compete with the loss processes and must be taken into account. In helium, for example, the production of electrons in collisions between two 2^3S metastable atoms has an appreciable rate coefficient which can significantly affect the net rate of decay of electrons in an afterglow (A. W. Johnson and J. B. Gerardo, *Phys. Rev. A* **7**, 925, 1973).

Finally, at very low densities radiative recombination becomes the only significant recombination process and at very high densities collisional recombination is dominant; but for intermediate densities, both processes occur. In these circumstances the recombination is not simply the sum of the various processes because, as shown by Bates, Kingston, and McWhirter [444], the processes are coupled. Under these circumstances,

three-body recombination occurs to excited states, together with transitions between the bound states both by radiative and collisional processes and between the bound states and the continuum by collision processes. The term collisional radiative recombination is then used to describe the processes occurring. Although the processes themselves are complex, the experimental results are usually expressed as an equivalent two-body decay coefficient g .

The collisional radiative decay coefficient g_c can be expressed for optically thin plasmas in terms of a collisional radiative recombination coefficient r_c and a collisional radiative ionization coefficient i_c by the relationship

$$g_c = r_c - i_c (N/N_+), \quad (24)$$

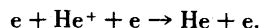
both r_c and i_c depending only on n and on the electron temperature T_e .

In general, the recombination coefficient depends on the temperature and density of the electrons, of the ions, and of the neutral gas particles, these temperatures and densities often themselves being interrelated. Graphs of r as a function of n and of r/n as a function of T_e can be a useful way of presenting data for comparison purposes if the other parameters of the experiments are similar. Probably the most satisfactory way of presenting the results, however, is in tabular form, when values of all the relevant parameters (if known) can be included.

3.8.a. Recombination Coefficient—Helium

There have been more than 30 experimental investigations of recombination in helium but most of them fall conveniently into two groups, viz.—those carried out at relatively high electron concentrations ($> 10^{11} \text{ cm}^{-3}$) mostly at low gas pressures ($\sim \text{m Torr}$), and those at lower electron concentrations ($< 10^{11} \text{ cm}^{-3}$) mostly at high gas pressures ($\geq 10 \text{ Torr}$).

(i) *High electron concentrations.* The results obtained in a number of experimental investigations of the recombination coefficient at high electron concentrations are shown in figure 8.1. These experimental results are compared with values computed theoretically by Bates and Kingston [2168] for quasi-equilibrium plasmas on the basis of the theory of collisional radiative recombination in which the three-body recombination process is



The results over the full range for which these computations have been made are given in table 8.1, but the theoretical values shown in figure 8.1 refer to a gas number density of 10^{14} cm^{-3} which is about the middle of the range used in the various experiments; the curves labelled (a) and (b) in this figure correspond respectively to plasmas which are optically thin and optically thick to the resonance lines. In the conditions under considera-

tion, the decay coefficient may be equated to the collisional radiative recombination coefficient, except for a correction for Penning ionization in the case of optically thick plasmas and this has been shown to be small [3381] except at low temperatures and high densities.

All the experimental results shown in figure 8.1 were obtained for static afterglows but for a wide range of conditions. Magnetic confinement was used in some [1639, 2331, 2447] but not in others [883, 1870, 1961, 3658] and the gas number densities ranged from about 8.8×10^{12} to $2.5 \times 10^{15} \text{ cm}^{-3}$. Other factors such as the size of the containing vessel may also have an influence, so that although there is considerable spread in the values obtained, the general agreement between theory and experiment supports the view that in this region collisional radiative recombination is the dominant decay process.

Numerical data corresponding to the experimental results on static afterglows shown in figure 8.1 are given in table 8.2, together with another set of data, obtained by Born [3464], for a range of gas temperatures at each value of n . These results were also shown to be consistent with the theory of collisional radiative recombination when the elevated gas and electron temperatures are taken into account. Values obtained from the study of plasma jets [759], which are appreciably different, are also included in this table.

Measurements on the early stages of afterglows have also been made by Chen [5745] to give r as a function of n and of T_e . These results are included in figure 8.2 where they are compared (i) with the other experimental results from table 8.2 for which T_e is available, and (ii) with the theoretically computed values of Bates and Kingston [2168] and of Mansbach and Kech [5761]. The latter were obtained using Monte Carlo trajectory methods and may be represented by the equation

$$r/n = 2 \times 10^{-27} (k T_e)^{-9/2},$$

with kT_e in electron volts. It can be seen that this expression gives a curve lying below and with a slightly smaller slope than a curve drawn through the majority of the experimental points. Recombination rates deduced from some spectroscopic observations [5327] of decaying afterglows appear to be in better agreement with those obtained from the theoretical computations of Mansbach and Kech than from those of Bates and Kingston.

Calculations of the collisional radiative recombination coefficient r_c as functions of n and T_e as independent variables have been made by Drawin and Emard [5732] and by Chen [5745]. The former used a method which is similar to that of Bates et al. but which did not assume hydrogenic energy levels. The latter used a simplified method based on the existence of a minimum in the total rate of de-excitation of atoms as a function of the energy level of the excited states. Results using this method are compared in figure 8.3 with the data for an optically thin plasma by Drawin and Emard.

The complete range of Drawin and Emard's data for both optically thin and optically thick plasmas is given in table 8.3. (Results for other assumed conditions are also given in their paper.)

The influence of the neutral particle density on the collisional radiative recombination coefficient through the reaction

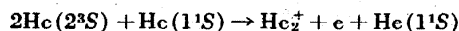
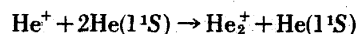


has also been investigated [1259, 5130, 5773, 5774, 3653]. The results for conditions in which $T_e = T_g$ and T_e is 250 and 300 K [1259, 5773, 5774] are shown in figure 8.4 and tabulated in table 8.4(a), together with theoretical values [1259, 5130] for higher values of $T_e = T_g$. As pointed out in [5130] the values at low electron densities are much higher than those calculated in [1259] for $n \rightarrow 0$, because even low densities of electrons are efficient in transporting excited electrons through the system of energy levels. In tables 8.4(b), (c), and (d) values of r as a function of N obtained [5773] with $T_g = 300$ K but different values of $T_e > T_g$ are given.

Measurements were made by Chen [4586] of rate of decay of the density of electrons and of atomic and molecular ions in an afterglow as a function of electron and atom temperatures for initial electron densities in the range 10^{12} to $3 \times 10^{14} \text{ cm}^{-3}$. Analysis of these results, taking into account processes of formation of molecular ions and the recombination of both atomic and molecular species but not the production of electrons in metastable collisions (see section 3.8.a(ii)) lead to electron temperature dependence of the dissociative recombination coefficient r_2 of the form

$$\begin{aligned} r_2 &= 1.9 \times 10^{-5} (T_e)^{-1.48} \text{ for } N \\ &= 6 \times 10^{16} \text{ cm}^{-3} \text{ and } T_g = 410 \text{ K} \\ \text{and} \\ r_2 &= 8.9 \times 10^{-9} (T_e)^{-0.36} \text{ for } N \\ &= 6 \times 10^{16} \text{ cm}^{-3} \text{ and } T_g = 1250 \text{ K.} \end{aligned}$$

(ii) *Low electron concentrations.* Experimental investigations of recombination at low electron concentrations involve the use of high gas pressures (~ 15 Torr) in order to reduce the influence of diffusion. The high pressure favors the formation of He_2^+ ions by the reactions



so that more than one recombination process is possible. Moreover, in some of the experiments, the results were found to depend on the way in which the plasma was produced initially, and in some the range of electron densities over which n^{-1} was found to be linearly dependent on t was not great enough (see E. P. Gray and D. E. Kerr, *Ann. Phys. (New York)* **17**, 276, 1959),

to be an adequate test of the assumption that the decay was recombination dominated, an assumption necessary to obtain the values quoted. These difficulties account, at least in part, for the fact that a wide range of values from 6×10^{-10} to $6.8 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$ have been obtained for the effective recombination coefficient r_e .

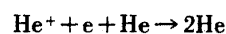
The value of the effective recombination coefficient ($\sim 10^{-8} \text{ cm}^3 \text{ s}^{-1}$) obtained from early measurements [249, 596, 802, 1642] was much higher than that for radiative recombination which is about $4.8 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$ at 250 K, decreasing to 4.31×10^{-13} and $2.69 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ at 10,000 and 20,000 K, respectively [715]. The relatively high experimental values led to the suggestion that the dominant recombination process was dissociative recombination of the He_2^+ ion and experimental results showing r_e to be independent of electron density [2456] and of gas pressure [249, 2456] lent support to this view.

A review of the available data on recombination coefficients and spectroscopic measurements in the light of theoretical considerations of dissociative and collisional radiative recombination by E. E. Ferguson, F. C. Fehsenfeld, and A. L. Schmeltekopf, (*Phys. Rev.* **138**, A381, 1965), however, led to the conclusion that the experimental results up to that time could be interpreted more satisfactorily on the basis of collisional radiative recombination. This seemed to be further confirmed by a subsequent experimental investigation by Berlande et al. [4374] which showed that the decay rate in carefully purified helium depended on both the electron concentration and the gas density. The results obtained over the ranges $10^9 < n < 5 \times 10^{11} \text{ cm}^{-3}$ and $3.5 \times 10^{17} < N < 3.5 \times 10^{18} \text{ cm}^{-3}$ could be represented by an effective recombination coefficient r_e given by

$$r_e = r_2 + kn + K'N$$

with $r_2 = 5 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$, $k = (2 \pm 0.7) \times 10^{-20} \text{ cm}^6 \text{ s}^{-1}$, and $K' = (2 \pm 0.5) \times 10^{-27} \text{ cm}^6 \text{ s}^{-1}$.

The above value for k compares with a value of 10×10^{-20} from the theoretical calculations of Bates et al. [444] and of 4×10^{-20} from the theoretical treatment of Deloche [3653]; the above value for K' compares with theoretical values of $2.5 \times 10^{-27} \text{ cm}^6 \text{ s}^{-1}$ obtained from the work of Pitaevski [2165] and of $1 \times 10^{-27} \text{ cm}^6 \text{ s}^{-1}$ from the values for the collisional radiative recombination coefficient for the process



calculated by Batcs and Kharc, which are given in table 8.4.

The conclusion that the recombination processes involved are collisional radiative recombination of He_2^+ with both electrons and helium atoms acting as third bodies is, however, invalidated by the recent work of Johnson and Gerardo [5570, 5488, 5730]. They have shown that the recombination coefficient determined in all previous work is actually, as already indicated by

the above nomenclature, an effective recombination coefficient. The reason is that no previous analysis had taken into account an intense source of free electrons which their measurements on the effect of a heating pulse on the electron and atomic metastable densities in a decaying plasma clearly showed to be present. This source is considered to be ionization in metastable-metastable collisions, and when it is taken into account an actual recombination coefficient is obtained which is as much as five times larger than the effective recombination coefficient. The actual recombination coefficient is given by

$$r = r_2 + KN$$

with $r_2 = 1.1 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$ and $K = 1.3 \times 10^{-26} \text{ cm}^6 \text{ s}^{-1}$ in the range $4.8 \times 10^{17} < N < 18 \times 10^{17} \text{ cm}^{-3}$. This work has also shown [5325] that under the conditions of the experiments dissociative recombination accounts for a large fraction of the total recombination.

The situation is, however, still not completely clarified, because although the results of Johnson and Gerardo [5570] for the effective recombination coefficient r_e range from about $3.4 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$ at 15 Torr to about $6.5 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$ at 65 Torr, in good agreement with the results of Berlande et al. [4374] for $n = 10^{11} \text{ cm}^{-3}$, they indicate no dependence on electron density in the range $4 \times 10^{10} < n < 5 \times 10^{11} \text{ cm}^{-3}$, being more in accord in this respect with other recent work [5772] in which the effective recombination coefficient at $N \sim 1.4 \times 10^{18} \text{ cm}^{-3}$ showed only a weak dependence on electron density for $10^{10} < n < 10^{12} \text{ cm}^{-3}$. Moreover, using a theoretical analysis which included a source term for the production of electrons during the afterglow, but an experimental tube of much smaller diameter than the above workers, Boulmer et al. [5343] obtained values of the recombination coefficient which showed no variation with N for $7 \times 10^{17} < N < 11 \times 10^{17} \text{ cm}^{-3}$. The values and their variation with temperature agree with those calculated by Mansbach and Kech [5761] on the basis of collisional radiative recombination.

At low temperatures where the dominant ion is He_2^+ there have been two recent investigations [5203, 5134] which show that the recombination coefficient is much higher than that at room temperature, the values of the effective recombination coefficient being $(3.4 \pm 1.4) \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$ for $T_g = T_e = 80 \text{ K}$ [5203] and $(5 \pm 0.75) \times 10^{-6}$ for $T_g = 4.2 \text{ K}$ and $T_e = 80 \text{ K}$ [5134]. The dependence on electron temperature in both cases was found to be of the form $T_e^{-\gamma}$ with γ somewhere within the range 0.8 to 1.6.

3.8.b. Recombination Coefficient—Neon

Most of the published data for recombination in neon have been obtained for relatively low electron densities ($< 10^{11} \text{ cm}^{-3}$) from measurements of plasma decay using microwave techniques. It can be seen from the results given in table 8.5 that when $T_e = T_+ = T_g = 300 \text{ K}$,

there is good agreement between the values obtained by a number of observers, especially if the appropriate corrections [see Frommhold, Biondi, and Mehr 3443] are made for diffusion and for the initial distribution of electrons in the plasma.

A number of observers [673, 1363, 2988] have shown that the recombination coefficient is independent of pressure indicating that the dominant recombination process is a two-body process. The suggestion [966] that, because of the relatively high value of the coefficient, this two-body process is dissociative recombination of electrons with Ne_2^+ ions has been confirmed experimentally by using a mass spectrometer to observe the decay of the Ne_2^+ ions simultaneously with that of the electrons [2988, 4044] and also by spectroscopic observations [1600].

The results of recent measurements [2988, 3443, 4044, 4585] on the temperature dependence of the recombination coefficient are shown in figure 8.5 and given in table 8.6. It can be seen that in contrast to the results of some earlier measurements [1363] in which no temperature dependence was apparent, the recombination coefficient varies as $T_e^{-\gamma}$ over the range $300 \text{ K} < T_e < 11,000 \text{ K}$.

When the electron temperature alone was increased by microwave heating the value of γ was found by Biondi et al. [3443, 4044] to be 0.43 over the whole range and by Hess [584] to be 0.4 in the range $900 < T_e < 2400 \text{ K}$, but 0.25 in the range $300 < T < 900 \text{ K}$.

When the electron, ion, and gas temperatures were all varied together over the range 295 to 503 K by heating the cavity, Kasner [2988] obtained a value of 0.42 for γ . Other experiments [4585] in which the gas was heated by means of a shock wave led to a similar value of γ for temperatures below 700 K but to a value of 1.5 for temperatures in the range from 700 to 3500 K. This change in value of γ at high temperatures can be satisfactorily accounted for theoretically (T. F. O'Malley, Phys. Rev. **185**, 101, 1969) if the recombination coefficient decreases rapidly for recombination to vibrationally excited states of the molecular ion. The experimental results of Chen [4586], at higher electron densities in plasmas produced by a condenser discharge, however, indicate no variation of the dissociative recombination coefficient with T_g in the range 420 to 1500 K for constant values of $T_e = 300 \text{ K}$.

Experimental measurements of the decay of electron density and of electron temperature in a decaying condenser discharge in which the electron density was about $5 \times 10^{12} \text{ cm}^{-3}$ led [5745] to much lower values of r than discussed above. These values are given in figure 8.6, where, bearing in mind the measured values of n and T_e , they can be seen to be consistent with the values calculated [5745] on the basis of collisional radiative recombination theory using the approximate method mentioned in section 3.8.a(i); these theoretical values are estimated in the original paper to be accurate within a factor of 4. A low value for r of $5.3 \times 10^{-14} \text{ cm}^3 \text{ s}^{-1}$ was also obtained [5775], at an even higher value

of $n \sim 5 \times 10^{17} \text{ cm}^{-3}$ but unmeasured temperature, from observations of the H_α and H_β lines from a decaying plasma in a small admixture of hydrogen in neon.

3.8.c. Recombination Coefficient—Argon

Measurements of the recombination coefficient for argon have been carried out at both relatively high and low electron densities and it is convenient to consider them in two groups corresponding to values of n greater and less than about 10^{11} cm^{-3} .

(i) *Large values of n ($> 10^{11} \text{ cm}^{-3}$).* Recombination coefficients at high electron concentrations have been measured for quasi-equilibrium plasmas in argon by a variety of techniques [1961, 3433, 3468, 5240, 5745, 5775, 5776, 5777, 5782] over a range of n up to about 10^{17} cm^{-3} and over a range of gas number densities from about $1.7 \times 10^{16} \text{ cm}^{-3}$ to 10^{20} cm^{-3} . In some of the measurements [1961, 3468, 5745] both n and T_e were determined but in most, either T_e or n was measured and equilibrium assumed. The results are shown as a graph of r , n in figure 8.7.

Also shown in this figure are the theoretical results obtained [1961, 5745] on the basis of collisional radiative recombination theory for the same values of n and T_e as those measured experimentally [1961, 5745]. The full range of the values calculated in [5201] and [5745] are shown in figure 8.8; r is given as a function of n at various values of T_e .

Some of the results obtained from a study of the radiation from a plasma jet [5702] at atmospheric pressure are also included in figure 8.7. For clarity of presentation, the data shown were chosen from the large number of values given so as to cover the range of parameters investigated. All the experimental results were found to fit the empirical equation

$$r = 1.28 \times 10^5 \times T^{-1.8} \times 10^{-(3410/T)} \times n^{-0.64}.$$

Theoretical values for the same range of parameters were also obtained in this study [5782] on the basis of both collisional radiative and dissociative recombination. The results are shown in figure 8.7, from which it can be seen that better agreement is obtained between theoretical and experimental results of this particular study on the assumption of dissociative recombination rather than collisional radiative recombination.

(ii) *Low values of n ($< 10^{11} \text{ cm}^{-3}$).* Although there have been a number of investigations [673, 711, 758, 811, 1363, 1642, 4065] of the recombination coefficient at relatively low values of n for values of $N \geq 3.5 \times 10^{17} \text{ cm}^{-3}$ for $T_e = T_g = T_\infty = 300 \text{ K}$, the value of r is not as well established as that for neon, values ranging from 2×10^{-7} to $1.1 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$ having been obtained. Bearing in mind difficulties with gas purity [1363] and with variation of r with the power of the exciting discharge [711, 1642, 2186] and the range of values of electron density over which measurements were made in a given experiment, the most reliable values seem likely

to be those in [673], [758], and [4065] which are given in table 8.7.

There have been no simultaneous observations of the decay of electron and ion densities using a mass spectrometer in argon in this range. Nevertheless, the observation [758] that the value of r for argon is at least a factor of 10^3 greater than that in argon-helium mixtures, together with spectroscopic observations (L. Frommhold and M. A. Biondi, Phys. Rev. **185**, 244, 1969) and mass spectrometer studies (G. E. Veatch and H. J. Oskam, Phys. Rev. A **1**, 1498, 1970) of the ion species formed in argon-helium mixtures, lead to the conclusion that the values of r in table 8.7 refer definitely to dissociative recombination of Ar_2^+ ions with electrons.

Investigations of the temperature dependence of the recombination coefficient in the range from room temperature up to a few thousand degrees Kelvin show similar results to those in neon, although the agreement between the absolute values obtained by various observers at slightly elevated temperatures is not as good as that in neon. As can be seen from figure 8.9 and table 8.8, the values of r vary as $T^{-\gamma}$ for the temperature ranges covered, but the value of γ depends on the method of heating used. When microwave heating was used [4065] to raise T_e to values in the range 300 to 10,000 K, while T_g and T_∞ remained at 300 K, γ was found to be 0.67. When, on the other hand, a shock wave was used [1873, 3438, 4585] to give conditions in which $T_e = T_g = T_\infty = T$, a similar value of γ was found for $T < 670 \text{ K}$ but γ became 1.5 for $670 < T < 3500 \text{ K}$. (Probe measurements [5184] in the Faraday dark space of a glow discharge gave a value for r of 1.2×10^{-8} at $T_e \approx 14,000 \text{ K}$ which lies between the extrapolation of the two curves in figure 8.9.)

The above change in value of γ can be explained (T. F. O'Malley, Phys. Rev. **185**, 101, 1969) if it is assumed that the coefficient representing recombination to vibrational states of the molecular ion decreases rapidly as the vibrational level increases. Whether such a decrease occurs depends on the details of the potential energy curves of the particular molecular ion and of the state of the neutral molecule formed initially [4586, 4065]. No information on the change of r_2 with T_g alone and thus on the change of r_2 with vibrational level is available in the conditions of the above experiments. Observations on the decay of plasmas of higher electron density (n_e in the range 10^{13} to 10^{16} are mentioned for He, but no details given for Ar) produced by a condenser discharge [4586] have, however, been interpreted as showing an increase, rather than a decrease of r_2 with increasing T_g for constant T_e in argon.

3.8.d. Recombination Coefficient—Krypton and Xenon

The only investigation of recombination coefficients in relatively pure samples of krypton (impurity content < 2.5 parts in 10^5) at room temperature, was that carried out by Oskam and Mittelstadt [673]; results are given in table 8.9.

Other investigations of afterglows in krypton containing xenon [254, 2445] at levels of 0.01 to 0.1 percent and 0.5 percent, respectively, gave electron-density-decay curves which had different slopes in the early and later afterglow. Spectroscopic observations [254] showed that the KrI spectrum predominated in the early afterglow and the XeI spectrum in the late afterglow. The different slopes of the electron-density-decay curves were thus interpreted as being due to recombination with krypton ions in the early afterglow and with xenon ions in the late afterglow. Analysis of the curves on this basis led to values of r for krypton of 6 to $12 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ [254] and not greater than $11 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ [2445], in fair agreement with the more accurate values in table 8.9.

The low value of $3 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ obtained for krypton from probe measurements of the decay of positive ion concentration following a dc discharge in krypton of unspecified purity [3188] is probably accounted for by complications arising from the interpretation of probe measurements in the presence of striations and the low pressure (1.5 Torr) used, at which diffusion is significant.

The variation of the recombination coefficient with temperature was measured [5128] over the range 800 to 2500 K in experiments in which the krypton was heated by means of shock waves so that $T_e = T_+ = T_g$. The results are given in figure 8.10 and in table 8.10, and it can be seen that r shows a $T^{-1.5}$ dependence as in argon over the same temperature range.

At a value of T_e of about 14,000 K, probe measurements in the Faraday dark space of a glow discharge gave a value of $1.4 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$ for r .

The situation in xenon is complicated by the fact that the cross section for momentum transfer is relatively large. Thus, at the pressures (≥ 10 Torr) required to make recombination the dominant loss mechanism, the assumption which is usually made (viz., that the collision frequency for momentum transfer is negligible with respect to the frequency of the microwave probing field) leads to measured values of the recombination coefficient, r_m say, which are related to the true values by the expression

$$r_m = [1 + (ap_0)^2] r.$$

In this expression $a = \nu_m/\omega_0$ and ν_m is the momentum transfer collision frequency at a pressure of 1 Torr and ω_0 the resonant frequency of the cavity in the absence of electrons.

The most reliable measurements in xenon are again those of Oskam and Mittelstadt [673], who observed the above predicted pressure dependence for samples of xenon containing less than 1 part in 10^5 krypton and obtained the value of r given in table 8.9 by extrapolating their results to zero pressure. Measurements [860] in samples containing not more than 1 percent krypton gave a similar pressure dependence and led to a value lying between 12 and $16 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ in good agree-

ment with the value in table 8.9. The late afterglow measurements in krypton containing 0.5 percent xenon, mentioned above [254], also gave values of r for xenon in the range 12 to $21 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$. An apparent pressure dependence was also found in other experiments but in one case [2130] it is not clear whether the values given are of r or of r_m and in the other [2445] only values of r_m are given.

There are no published measurements of the temperature dependence of r in Xe nor any mass spectrometer observation of the ion species involved in the plasma decay of Kr or Xe. From the spectroscopic observations [254] and the relatively high value of r however, it is assumed that dissociative recombination of Kr_2^+ and Xe_2^+ is the dominant recombination mechanism.

3.8.e. Recombination Coefficient—Hydrogen

Surprisingly few experimental investigations [239, 240, 2292, 1363, 5500] have been made of recombination coefficients in hydrogen at low electron densities ($< 10^{11} \text{ cm}^{-3}$) and there are marked differences between the results obtained. This probably arises from the fact that there are known to be four stable ion species H^+ , H_2^+ , H_3^+ , and H_5^+ and no measurements have been made, to date, in which a mass spectrometer has been used to identify the ions concerned.

At pressures of a few Torr where H_3^+ and H_5^+ ions might be expected to play a significant role, several observers [240, 2292, 5500] have obtained values of r which increase with pressure above 1 Torr. The absolute values obtained for r were, however, very different, ranging, for example, from about 0.12×10^{-6} to $2 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$ at about 3 Torr. Moreover, other measurements [1363] gave a pressure-independent value for r of $2.5 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$ for $3 < p < 12$ Torr. In one set of experiments [239], r was found to be too low to measure ($< 3 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$), the plasma decay being due entirely to diffusion and markedly dependent on very small amounts of impurities. In this case, the value of the ambipolar diffusion coefficient obtained indicated that the ion species being studied was probably H^+ . It seems likely that differing experimental conditions, including the purity of the gas samples, power of the exciting discharge, dimensions of the apparatus, and range of electron densities used give rise to different concentrations of the various possible ion species in different experiments, thus altering the plasma decay rate and making ion identification essential if significant values for r are to be obtained.

Theoretical computations of r for H_2^+ using Monte Carlo methods [1665] and perturbation theory [3471] lead to values of $3 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$ and $2.5 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$, respectively, at 300 K. The temperature dependence in the latter case is given by $r = 4.2 \times 10^{-8} T^{-1/2}$, with a value which does not exceed $3 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$. These computed values of r for H_2^+ are much lower than the experimental values which again suggests that the measured values refer to H_3^+ and/or H_5^+ .

For H^+ ions, theoretical calculations have been made of the radiative recombination coefficient r_2 [601, 1602, 2259, 2390], and of the collisional radiative recombination coefficient r_c [444, 731, 972, 994, 1639, 2147, 3381, 5781] over a wide range of conditions.

The values of the radiative recombination coefficient in table 8.11 are those given by Bates and Dalgarno [4656] based on the results of Seaton [601]. In addition to the total recombination coefficient r_2 , the partial radiative recombination coefficients $r_2(p)$ to each level of principal quantum number p are also given over a range of temperatures from 250 to 64,000 K.

Among the first calculations of r_c were those of d'Angelo [731] who developed a method in which he considered the reaction paths of individual electrons, on the basis that the processes occurring were three-body recombination of electrons into excited states, followed either by their transition to lower levels with the emission of radiation or by their ejection into the continuum in collisions with another electron. Collisional transitions between bound states were, however, neglected, so that at relatively low temperatures (~ 1000 K) where such collisions play a significant role, the values of r_c obtained are considerably lower than those calculated with the complete statistical theory of Bates et al. [444]. At relatively high temperatures when radiation processes dominate the two theories give results in good agreement.

The statistical theory, in which the populations of the states were obtained from the governing rate equations, was first developed [444] to calculate values of r_c for optically thin plasmas for a range of values of the electron temperature T_e and density n . These are given in table 8.12, together with the values of the collisional radiative ionization coefficient i_c which enable the collisional radiative decay coefficient g_c to be obtained using eq (24). Also given in table 8.12(a) are values of the collisional radiative recombination coefficient calculated using the general curves given by Bates and Kingston [994] for temperatures < 250 K.

The calculations were later extended [146] to several different cases of optically thick plasmas. Values of r_c for (a) a plasma optically thick towards the lines of the Lyman series, which is likely to correspond to most laboratory plasma, are given in table 8.13(a). The other cases considered were plasmas optically thick (b) to the Lyman continuum as well as to the Lyman series (c) to lines of all series and (d) to lines of all series and the Lyman continuum. The values obtained are given in table 8.13(b), (c) and (d), respectively.

Although these tables give r_c as a function of n and T_e , these are, in practice, not independent variables. Their relation to the values for the collisional radiative recombination coefficient were thus calculated by Bates and Kingston [2147] for two different plasma models. In the first, which corresponds to a magnetically confined plasma, the only energy flow was assumed to be to the boundary from the atom gas, the energy flow from the

electron and ion gas being zero. In the second, the energy flow to the boundary from the electron gas was assumed to be zero but that from the ion and atom gas was assumed to be such as to keep both T_i and T_e the same as that of the walls. The results (which are useful for comparison with experiment) are given in tables 8.14 to 8.17 for plasmas that are optically thin and optically thick to lines of the Lyman series.

Experimental results for recombination coefficients obtained for magnetically confined high density ($n > 10^{11} \text{ cm}^{-3}$) hydrogen plasmas [1389, 1639, 5778, 5779] are shown in figure 8.11, where they are compared with the theoretical curves for r_c for an optically thick magnetically confined plasma at 250 K obtained from table 8.14.

Taking into account the values of the initial atom density the experimental values of [1639] are in fair agreement with the theoretical curves. The experimental values of [1389] fall below the theoretical values, but the measured values of T_e are also different from those calculated for an optically thick plasma. Taking the measured values of n and T_e into account, the measured values lie between those calculated (using tables 8.12 and 8.13) for optically thin and optically thick plasmas.

The results obtained [5697] from the decay of spark channels lie closer to the calculated values for optically thin plasma as can be seen from figure 8.12.

In addition to the calculations of the collisional radiative recombination coefficients on the basis of the detailed statistical theory discussed above, a number of calculations [595, 640, 972, 1639, 1647, 5327, 5745, 5761, 5781] of the collisional recombination coefficient r_3 (to which r_c tends in the limit of high n) have been made. These calculations were carried out using various simplified models based, for example, on the simple Thomson formula [1614] or on the existence of a critical energy level in the atom, transitions across which determine the net rate of recombination. Several of these treatments [595, 640, 1614, 1639, 1647, 5761, 5781] led to analytical expressions which show that r_3 varies with electron temperature as $T_e^{-9/2}$. The actual values obtained for r_3 , which depend on the values used for the collision cross sections, are compared in figure 8.13 with those calculated from the detailed statistical theory [444] in which Gryzinski's cross sections (M. Gryzinski, Phys. Rev. **115**, 374, 1959) were used.

Comparison of some results of spectroscopic observations in helium with the recombination rates predicted by the two methods have led to the suggestion [5327] that Gryzinski's cross sections are too large.

Measured values of the recombination coefficient are also included in figure 8.13 for comparison.

3.8.f. Recombination Coefficient—Nitrogen

As noted above for hydrogen, the simultaneous existence of a number of different ion species in molecular gases at pressures of a few Torr greatly complicates the determination of recombination coefficients in these

gases. In nitrogen it is well known that the species N^+ , N_2^+ , N_3^+ and N_4^+ are all stable and that ion conversion reactions between species occur in collisions with N_2 molecules. Thus, as the pressure, temperature and conditions of excitation in the discharge change, so does the relative concentration of the various ion species. This largely accounts for the very wide range of pressure dependent values of r (from 0.1 to $2 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$) obtained [703, 728, 807, 811, 1363, 2372, 2394] in nitrogen at pressures ranging from 0.1 to 40 Torr.

The only extensive experiments in which the ions have been identified and their decay monitored simultaneously with that of the electrons are those of Biondi and his collaborators [1247, 1344, 1605, 2455, 2993]. In these experiments, the ion species N_2^+ was made the predominant one by carrying out the afterglow experiments in mixtures of neon and nitrogen in which the partial pressure of the neon ranged from 15 to 30 Torr and that of nitrogen from about 10^{-4} to 5×10^{-3} Torr. Two sets of measurements [1247 and 2993] for $T_e = T_i = T_g = 300 \text{ K}$, when the decay of N_2^+ ions was shown experimentally to follow closely the decay of electrons, gave a value for r for N_2^+ ions as $(2.7 \pm 0.3) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$, when the appropriate corrections were made for diffusion and the variation of the electron density distribution in the microwave cavity with time.

Measurements by Mahdavi et al. [5124] in a flowing afterglow, in a mixture of N_2 at pressures in the range 10^{-3} to 10^{-2} Torr with Ar at about 0.6 Torr, gave a value of $(2.2 \pm 0.4) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ for mass identified N_2^+ ions at 300 K, in good agreement with the above value obtained from static afterglow measurements.

At partial pressures of nitrogen greater than 10^{-2} Torr, in the static afterglow experiments, it was shown that the N_3^+ and N_4^+ ion species became predominant and that above about 0.1 Torr there were more N_4^+ than N_3^+ ions present. The recombination coefficient depended on the conditions, being between 1 and $2 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$, the value increasing as the ratio of N_4^+ to N_3^+ ions became larger. These values are not inconsistent with the pressure dependent value of $r \sim 10^{-6} \text{ cm}^3 \text{ s}^{-1}$ obtained at the higher pressures ($> \text{few Torr}$) in nitrogen by many observers [703, 807, 2372, 2394, 3166], suggesting that these values as would be expected, also refer to a mixture of N_3^+ and N_4^+ ions.

The concentration of different ion species also depends on the temperature so that to obtain significant results for the variation of recombination coefficient with temperature, it is necessary to determine the ion species involved by mass spectrometer. The results obtained for this variation under these conditions [2993, 4045, 4605, 4681] are shown in figure 8.14, both for the case in which the electron temperature T_e and the gas temperature T_g are varied simultaneously and for the case in which T_e alone is varied.

It can be seen that when both T_e and T_g were varied, r was found to be independent of temperature, whereas when T_e alone was varied r varied as $T^{-0.39}$. In the latter

experiment [4045], however, although the predominant ion species was N_2^+ the decay of this species did not follow the electron decay nearly as well as in the earlier work [2993, 1247] and the value of r at room temperature $(1.8 \pm 0.4) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ was considerably lower. The experimental conditions were very similar in all experiments and the discrepancy has not been explained. The two other results [4605, 4681] were obtained using a Langmuir probe, together with a mass spectrometer and do not agree well with those obtained by microwave methods.

Thus the variation of r_2 with temperature for N_2^+ ions is not well established at present.

There have been calculations [750] of the radiative and dielectronic recombination coefficient for atomic nitrogen and of the collisional radiative recombination coefficient for the reaction $Z^+ + e + N_2 \rightarrow Z + N_2$ for ions of mass 28 amu [4604]. The results of these calculations are given in tables 8.18 and 8.19, respectively.

The dissociative recombination coefficient for N_2^+ ions has also been calculated [1675] using a semiclassical method. The value of $2.0 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ so obtained is in surprisingly close agreement with the experimental value of $2.7 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$.

3.8.g. Recombination Coefficient—Oxygen

As in nitrogen and hydrogen, there are a number of stable species of ions in oxygen. Difficulties in the determination of the recombination coefficient arising from this source are, however, not as acute as in the case of nitrogen, because over a wide range of suitable experimental conditions the O_2^+ ion is the dominant species. The situation is, however, complicated by the occurrence of the process of three-body electron attachment to neutral gas molecules to form negative ions. As a result, the electron decay rate at low pressures in oxygen alone is pressure dependent, and measurements are thus usually carried out using a small concentration of O_2 (at a pressure of a few mTorr) in an inert buffer gas at a pressure of a few Torr. This ensures that the decay of electrons by recombination is much greater than that by attachment or by diffusion. To avoid the effects of negative ion accumulation, measurements are usually made in afterglows following single discharge pulses with relatively long delays ($\sim$ seconds) between pulses.

The four most recent results [3611, 3786, 4045, 5124] obtained for r_2 for O_2^+ ions at room temperature are in good agreement and are given in table 8.20. In two of these determinations [4045, 3611], the electron decay in a static afterglow was measured using the microwave method. The decay of the O_2^+ ions was also monitored simultaneously using a mass spectrometer and shown to follow closely the electron decay. In another [5124] a floating double probe and ion sampling arrangement were used to measure the decay of electrons and mass identified ions in a flowing afterglow in a mixture of oxygen and argon. Finally in [3786], the electron decay in a static afterglow was measured by a Langmuir probe

and although the ion species was not identified, conditions were chosen so that, on the basis of previous experiments in which a mass spectrometer was used, the O_2^+ ion was expected to be the dominant species. As indicated in the table, the buffer gases used were also different in the four experiments. In one of the microwave experiments [4045], r_2 was found to vary with the partial pressure of the oxygen in a mixture of O_2 and Ne. In this mixture the O_2^+ ions were formed by Penning ionization of O_2 in collision with metastable neon atoms and could thus have been in an electronically excited state. The effect of different mixtures was investigated and the dependence of r_2 on the oxygen partial pressure found to be least in the case of the O_2 -Ne-Kr mixture. In this case the O_2^+ ions were formed by the Penning ionization of Kr by Ne metastables followed by charge exchange between the Kr^+ and O_2 , so that the O_2^+ ions were formed in the ground electronic state and a low vibrational state.

The values given in table 8.20 are also in good agreement with the earlier values of 2.1×10^{-7} and $1.9 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ obtained by Mentzoni [3428] and by Asimov [1601] respectively, in experiments in which no identification was made of the species concerned. Both of these experiments were carried out in oxygen alone, Asimov et al. using low pressures (5×10^{-2} to 10^{-1} Torr) together with a confining magnetic field to reduce diffusion losses and Mentzoni using higher pressures from 0.15 to 4 Torr. In both cases the quoted value of r_2 was obtained from consideration of the observed variation of r with pressure.

Warke [1675] obtained theoretical values of the dissociative recombination coefficient for O_2^+ at room temperature on the basis of a semiclassical calculation and showed that with certain simplifying assumptions, a quantum treatment would lead to the same results. More detailed quantum calculations on the basis of Warke's approach but using different potential curves, were made by Chan [4020]. The results of both sets of calculations are also shown in table 8.20.

The results of experimental measurements of the temperature dependence of r_2 for O_2^+ are shown in figure 8.15. In one of these investigations [4045] T_e alone was varied, while in others [3611, 3428, and 3786] the temperature of the whole system was varied, thus keeping $T_e = T_i = T_g$. When T_e alone was varied for a mixture of O_2 and Ne, r_2 was found to vary as $T_e^{0.70}$ up to 1200 K and as $T_e^{-0.56}$ between 1200 K and 5000 K. It can be seen from the figure that this variation at the lower temperatures was similar to that obtained by Kasner and Biondi [3611] when T_e , T_i , and T_g were all varied together in an O_2 -Ne mixture. On the other hand, where T_e , T_i , and T_g were varied together in O_2 -He mixtures [3786] and in pure O_2 [3428], a less marked variation of r with temperature was observed.

The single values at elevated electron temperatures given by Sayers [4605, 4681] were obtained using a

Langmuir probe for O_2^+ ions identified using a mass spectrometer.

At temperatures ~ 200 K, for both O_2 -Ne-Kr mixtures [3611] and O_2 -He mixtures [3786] relatively high values of the effective recombination coefficient were obtained. Mass spectrometric studies in [3611] showed the existence of substantial quantities of O_4^+ ions under these conditions and the results were analyzed to give a value for r_2 of about $2.3 \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$ for O_4^+ ions and of $(3 \pm 0.3) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ for O_2^+ . Measurements of the decay of electron concentration by cylindrical Langmuir probes in static afterglows in krypton-oxygen mixtures at 180 K gave values for r_2 of $(1.8 \pm 0.6) \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$ for O_4^+ and of $(3.5 \pm 1.0) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ for O_2^+ in agreement with the microwave measurements.

The results of theoretical computations of the radiative recombination coefficient [1180] and the dielectronic recombination [750] for O^+ are given in table 8.21 from which it can be seen that the dielectronic coefficient is negligible with respect to the radiative one and that both are much smaller than the dissociative recombination coefficient for O_2^+ .

3.8.h. Recombination Coefficient—Carbon Monoxide

The only published value of the recombination coefficient for carbon monoxide at room temperatures is $(6.8 \pm 1.2) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ obtained by Mentzoni and Donohoe [3601]. From an experimental investigation of the decay of a static afterglow using a microwave method, the value obtained was independent of pressure for pressures of 0.55 and 0.92 Torr which, together with the high value of r , suggests that a dissociative recombination process is involved. No ion identification was made in the experiments but other experiments (H. E. Evans and P. P. Jennings, *Trans. Faraday Soc.* **61**, 2153, 1965) in this pressure range have shown that in active high frequency discharges CO_2^+ and CO^+ are the most numerous ions formed. For $p > 1.5$ Torr, the electron removal rate was found to be pressure dependent, possibly as a result of electron attachment. In an extension of the work to high temperatures [5783] a value of $r = 3.9 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$, independent of pressure for $0.2 < p < 2$ Torr, was obtained for $T = 775$ K.

3.8.i. Recombination Coefficient—Nitric Oxide

The three most recent investigations [1609, 3610, 5124] of recombination in nitric oxide, which are the only ones to include ion identification by mass spectrometer, gave results at room temperature that are in good agreement with each other. These results, together with the range of electron and gas number densities used, are given in table 8.22.

The data in [1609] and [3610] were obtained using mixtures of neon and nitric oxide in which the NO^+ ions were formed in the electronic ground state by photoionization. A long delay between successive pulses was also used to reduce complications resulting from the

accumulation of negative ions. The value of r_2 given by Weller and Biondi [3610] was obtained for relatively low partial pressure (5 to 41 mTorr) of NO when the decay of the NO^+ ions, which was shown to be the only species present, followed closely the electron decay. At higher partial pressures, the dimer ion $(\text{NO})_2^+$ became the dominant species and the measurement of electron decay under these conditions gave r_2 for $(\text{NO})_2^+ = (1.7 \pm 0.4) \times 10^{-6} \text{ cm}^3 \text{ s}^{-1}$. Gunton and Shaw's [1609] value of r_2 was also obtained at relatively low partial pressures (< 1 mTorr) of NO, because at higher partial pressures a pressure dependence of the coefficient was observed. On the basis of a mass spectrometer analysis at total pressures lower than that used in the experiment, this dependence is ascribed to the presence of significant amounts of NO_2^- ions at partial pressures of NO above 1 mTorr.

Mahdavi et al. [5124] used a flowing afterglow technique in which the electron decay in a mixture of NO and Ar was measured by means of a double probe and the ions identified by means of a quadrupole mass spectrometer.

Earlier investigations using a variety of methods [840, 920, 3229] but in which there was no identification of the ions involved gave less precise values of r_2 which are not inconsistent with the values given in table 8.22.

In both [3610] and [1609] the measurements were extended to temperatures within the range $180 < T < 450$ K and gave the values of r_2 shown in figure 8.16. The value of r_2 at $T = 200$ K given in [3610] was obtained under conditions when the decay of the NO^+ ion current to the walls followed the same form as that of electron decay during the early stages of the afterglow. Under most conditions, however, it was shown that the $(\text{NO})_2^+$ ion species was formed in considerable quantities at this temperature and this may account for the relatively high value for r_2 at $T = 186$ K obtained in [1609].

Also given in figure 8.16 is the value of r at 2900 K obtained from an investigation of the rate of decay of ionization behind a shock wave in air using microwave techniques [2491].

Analysis of the ionization occurring behind shock waves led to the conclusion [4789] that a dependence of recombination coefficient on temperature of the form

$r = 3 \times 10^{-3} T^{-3/2}$ would be consistent with the experimental observations. Data points obtained from this equation are also given in figure 8.16.

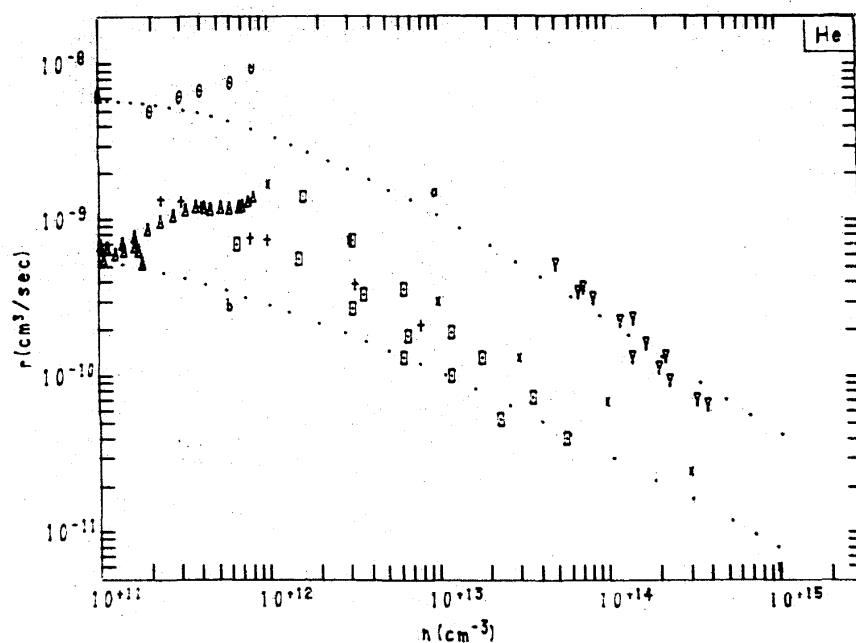
Bardsley [3462] has calculated the contribution of the $\text{B}^2\Pi$ and $\text{B}'^2\Delta$ states to the recombination coefficient for ions in their ground electronic and vibrational state and shown that this contribution varies as $T_e^{-1/2}$ as indicated in figure 8.16. The other theoretical curve shown in figure 8.16 is obtained from the expression $r_2 = 4.8 \times 10^{-8} (kT)^{-1/2} (1 - e^{-0.027/kT})$ deduced by Hansen [3476] on the assumption that excited species are present in equilibrium concentrations.

3.8.j. Recombination Coefficient—Carbon Dioxide

There is good agreement between the only published values of the recombination coefficient for CO_2 which are those obtained experimentally using static [2986] and flowing [5124] afterglow. In the experiments of Weller and Biondi [2986] the ions were produced in their ground electronic state by Penning ionization of CO_2 in a pulsed microwave discharge in a mixture of CO_2 and Ne. The nature and decay of the ions was obtained by a mass spectrometer, while the electron decay was determined by a microwave method. The apparatus was operated with long intervals between the pulses that created the ions, in order to prevent the accumulation of negative ions. Account was taken of the presence of O_2^+ ions which were observed significantly to affect the decay in the late afterglow (at times > 1 ms). At $T_g = T_e = T_i = 300$ K the value obtained for r for CO_2^+ was $(3.8 \pm 0.5) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$, being constant for mixtures in which the partial pressures of CO_2 ranged from 2×10^{-4} to 2×10^{-3} Torr and that of Ne ranged from 4 to 10 Torr.

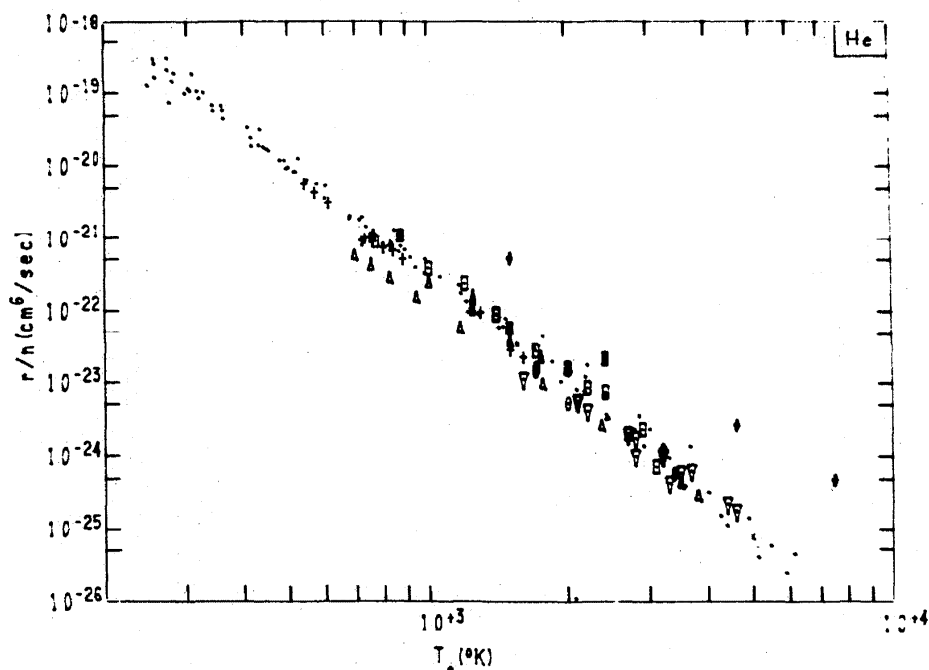
Despite some difficulty in observing the ion current decay at lower temperatures an approximate value of $r \approx 4 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ was given for $T = 210$ K.

In the experiments of Mahdavi et al. [5124] use of a floating double probe to measure the decay of the electron concentration in a flowing afterglow in a CO_2/Ar mixture at 300 K, led to a value of $(3.4 \pm 1.2) \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$ under conditions when CO_2^+ ions should have been the dominant species.



Experimental: Δ Mosburg(883); $\square$ Hinnov(1639) also includes Motley(2331) as given in (1639); $+$ Anisimov(1870); ∇ Gusinow(1961); $\circ$ Aleskovskii(2447); $\times$ Newton(3658).
Theoretical: $\cdot$ a, $\cdot$ b Bates(2168). (See text for details).

FIGURE 8.1. Recombination coefficient for helium as a function of electron density n for $n > 10^{11} \text{ cm}^{-3}$.



Experimental: $\circ$ Robben(759); $\square$ Hinnov(1639) including Motley and Kuckes; $+$ Anisimov(1870); ∇ Gusinow(1961); Δ Born(3464); $\times$ Newton(3658); $\diamond$ Chen(5745).
Theoretical: $\cdot$ Bates(2168); Δ Mansbach(5761).

FIGURE 8.2. Comparison of experimental and theoretical values for r/n , T_e for helium.

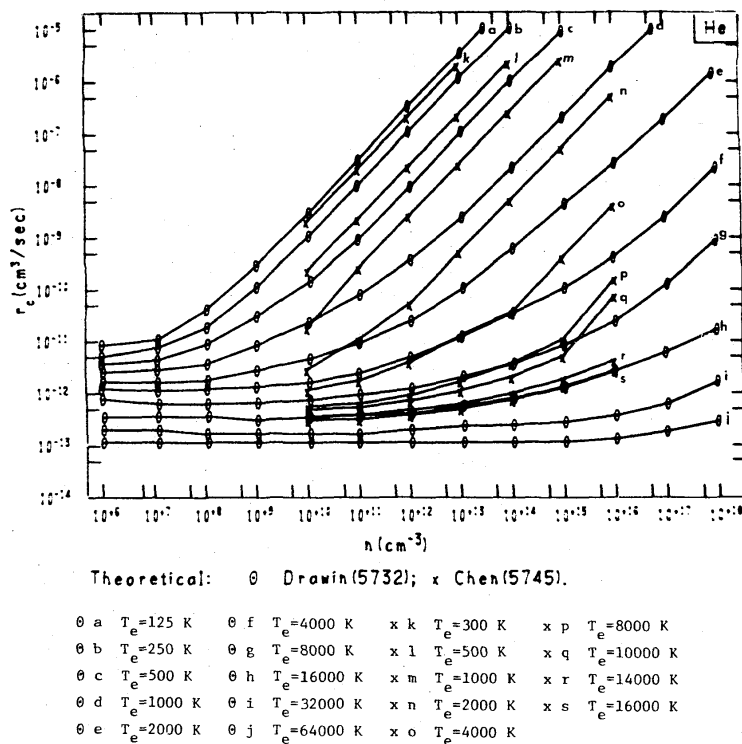
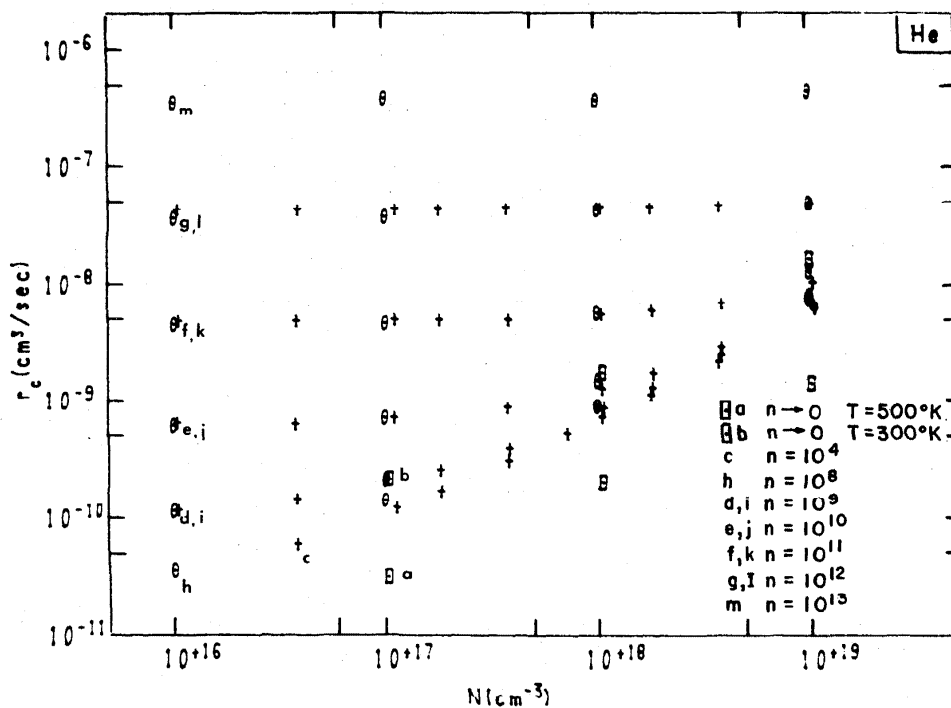


FIGURE 8.3. Recombination coefficient as a function of electron density for helium at various values of T_e in an optically thin plasma.



Theoretical: $\square$ Bates(1259); θ Collins(5773); $+$ Deloche(5774).

FIGURE 8.4. Theoretical values for the collisional radiative recombination coefficient for helium as a function of gas number density for various values of the electron concentration.

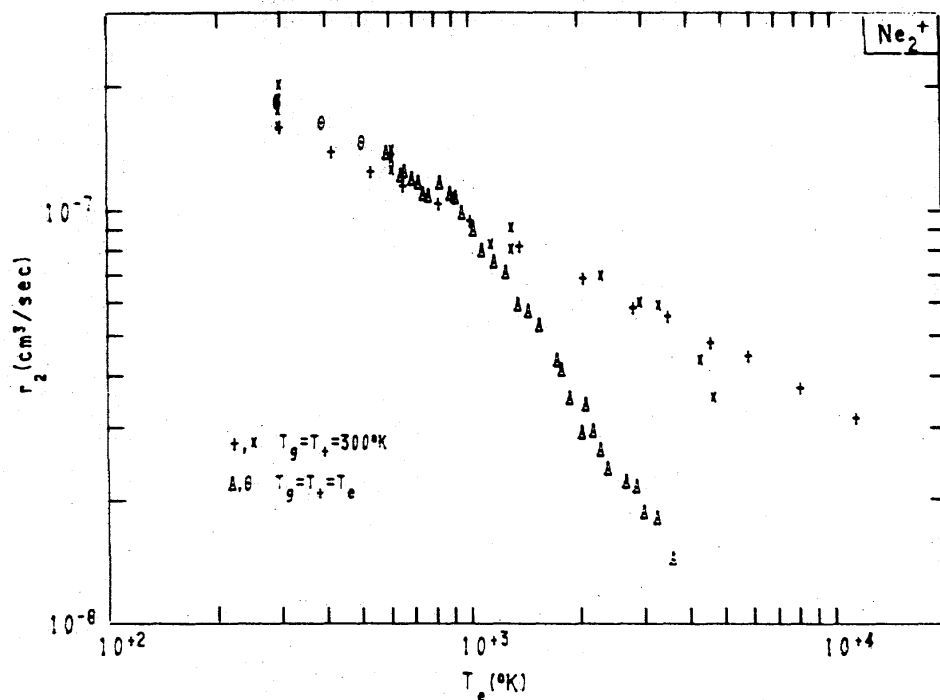
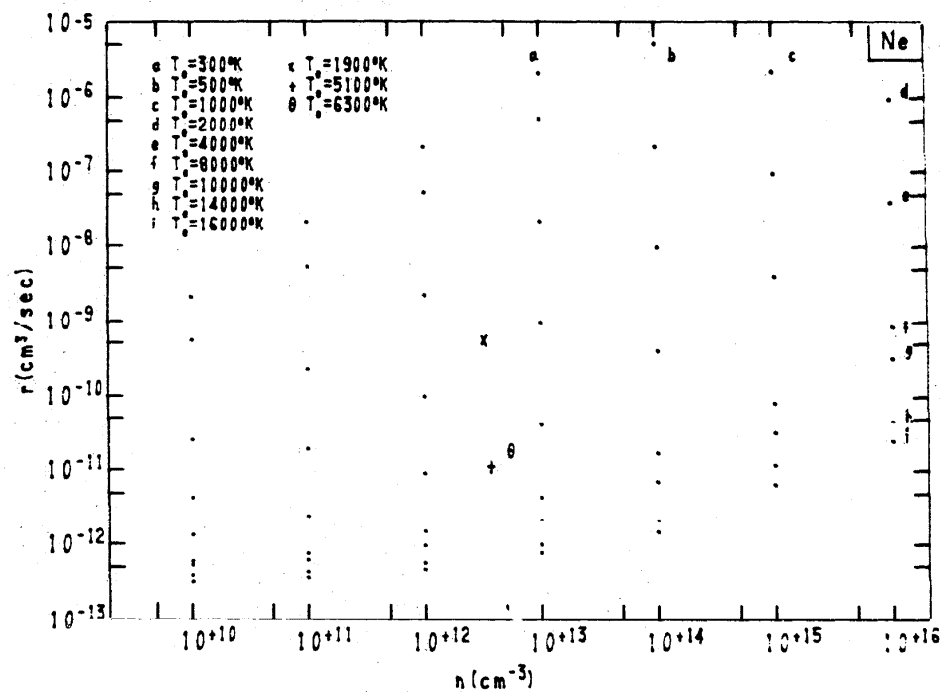
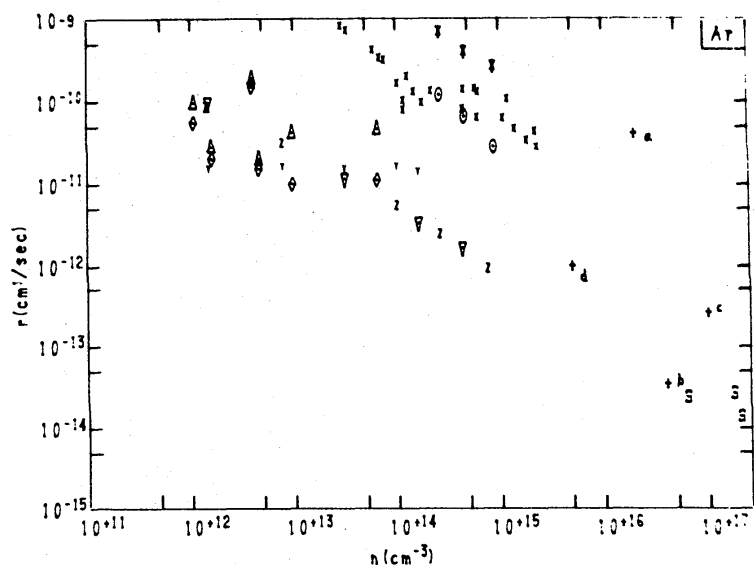


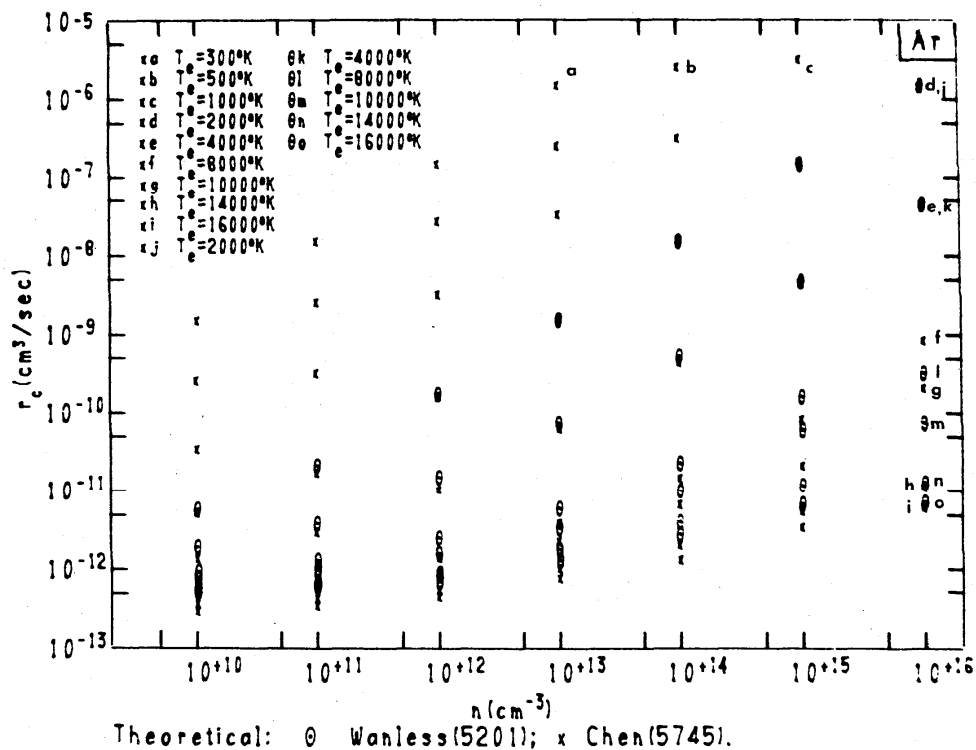
FIGURE 8.5. Temperature dependence of the dissociative recombination coefficient for neon.

FIGURE 8.6. Comparison of experimental values of r for plasmas of high electron concentration in neon, with approximate theoretical values calculated on the basis of collisional recombination theory.



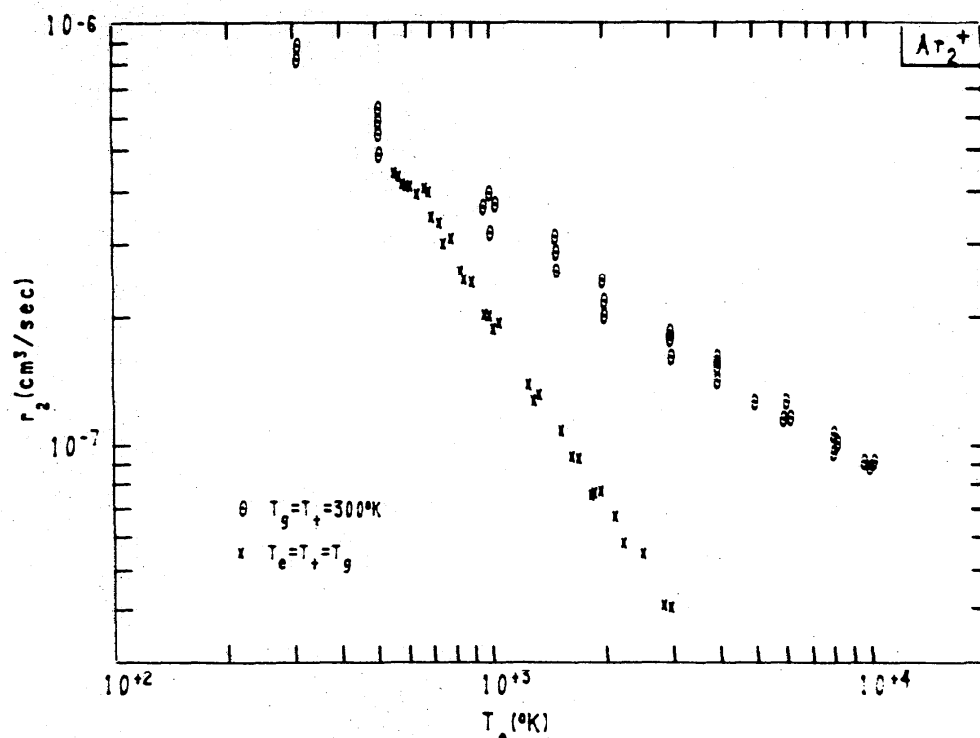
Experimental: θ Gusinow(1961); +a Aleksandrov(3433);
 x Funahashi(3468); +b Hughes(5240); Δ Chen(5745);
 +c Olsen(5775); $\square$ Mitin(5776); +d Van Trong(5777);
 ∇ Desai(5782).
 Theoretical: $\dagger$ Gusinow(1961); θ Chen(5745); γ Desai(5782)
 (Collisional radiative recombination coefficient);
 Z Desai(5782) (dissociative molecular-ion recombination
 coefficient).

FIGURE 8.7. Recombination coefficient as a function of electron concentration for quasi-equilibrium plasmas in argon. The theoretical values were calculated using different methods, but all on the basis of collisional radiative recombination except those of Desai (labeled Z) which were calculated on the basis of dissociative recombination.



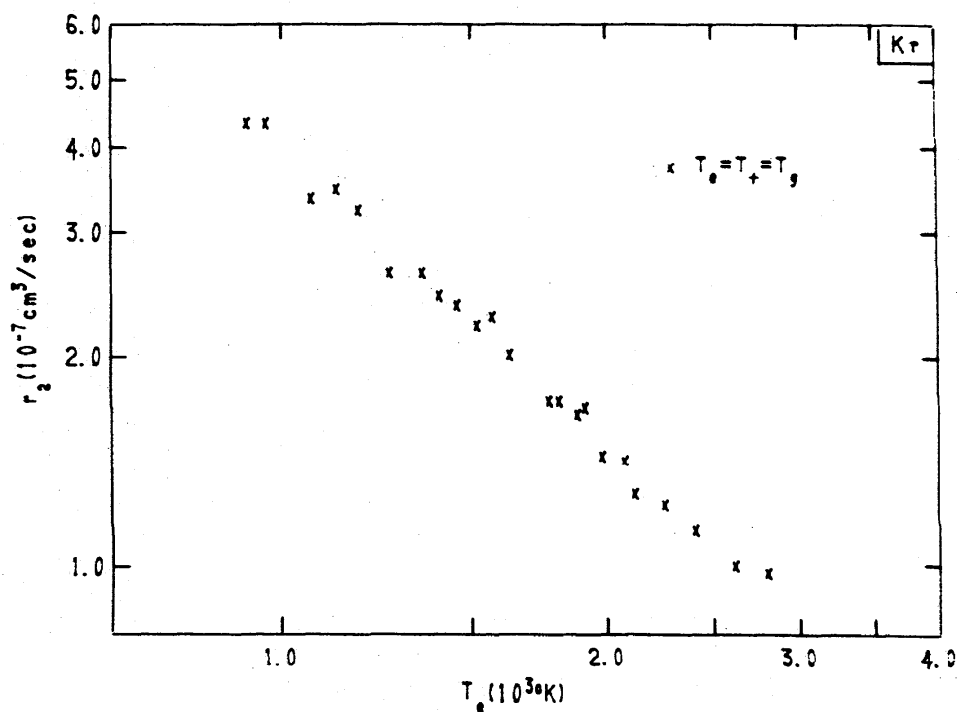
Theoretical: θ Wanless(5201); x Chen(5745).

FIGURE 8.8. Theoretical values of the collisional radiative recombination coefficient for argon as a function of n at various values of T_e .



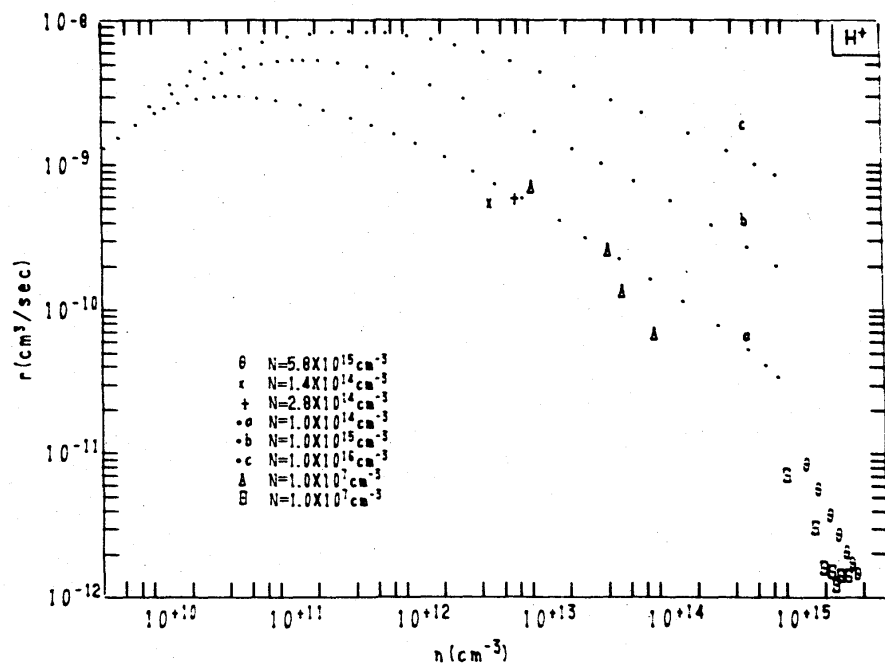
Experimental: $\circ$ Mehr(4065); $\times$ Cunningham(4585).

FIGURE 8.9. Temperature dependence of the dissociative recombination coefficient for argon.



Experimental: $\times$ Cunningham(5128).

FIGURE 8.10. Temperature dependence of the dissociative recombination coefficient for krypton.

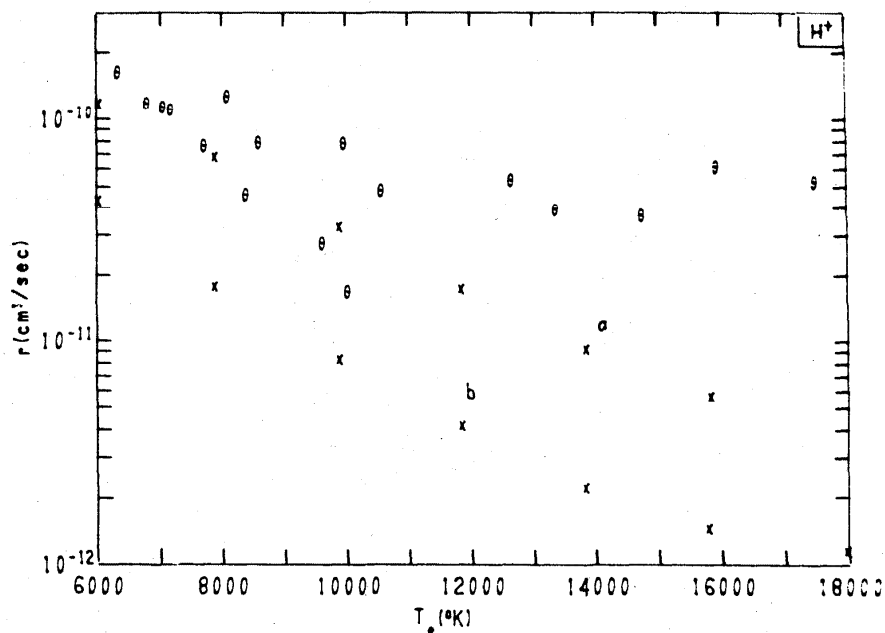


Experimental: θ Cooper(1389); $x, +$ Hinnov(1639).

Δ Brand(5778); $\square$ Irons(5779).

Theoretical: a, b, c Bates(2147).

FIGURE 8.11. Comparison of experimental recombination coefficients with computed values of r_c for optically thick, magnetically confined plasmas in hydrogen.



Experimental: θ Janssen(5697).

Theoretical: x Janssen(5697) using results of Bates(444) for an optically thin plasma. b Janssen(5697) using results of Bates(146) for an optically thick plasma.

FIGURE 8.12. Comparison of experimental recombination coefficients for decaying spark channels with computed values of r_c for atomic hydrogen in optically thin plasmas.

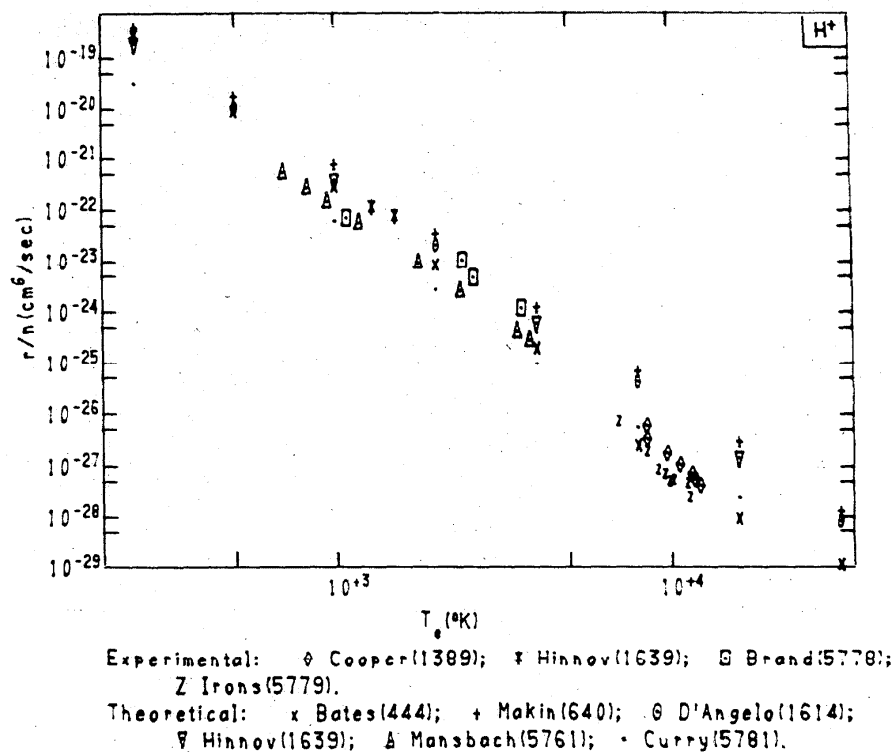


FIGURE 8.13. Comparison of the theoretical values of the collisional recombination coefficient r_3 for hydrogen using various theoretical models with experimental values of the recombination coefficient.

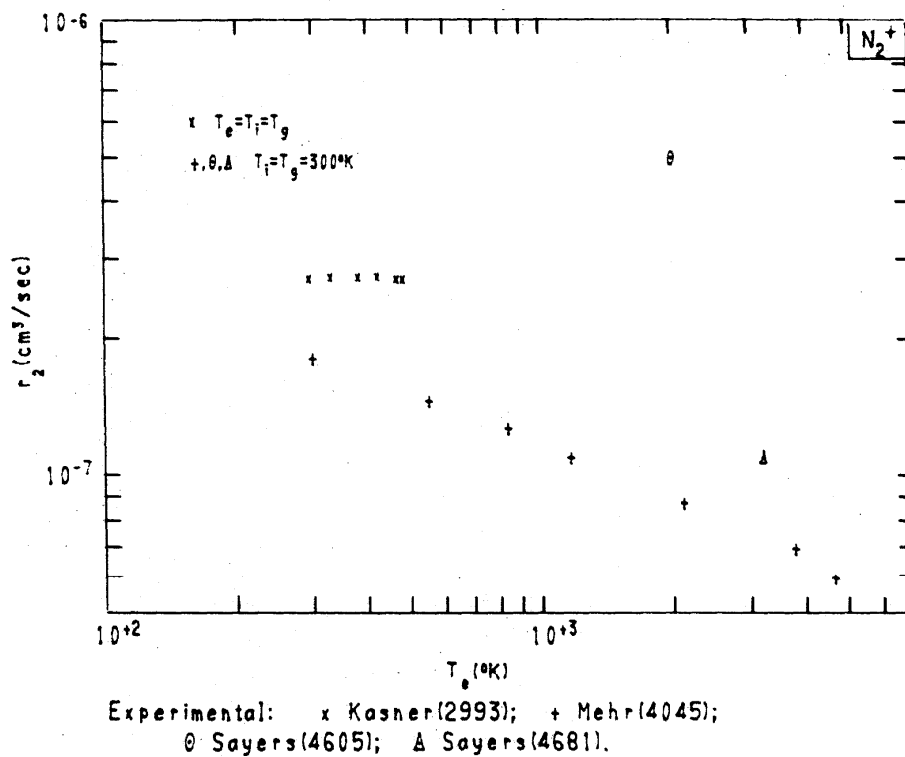


FIGURE 8.14. Temperature dependence of the dissociative recombination coefficient of N_2^+ ions.

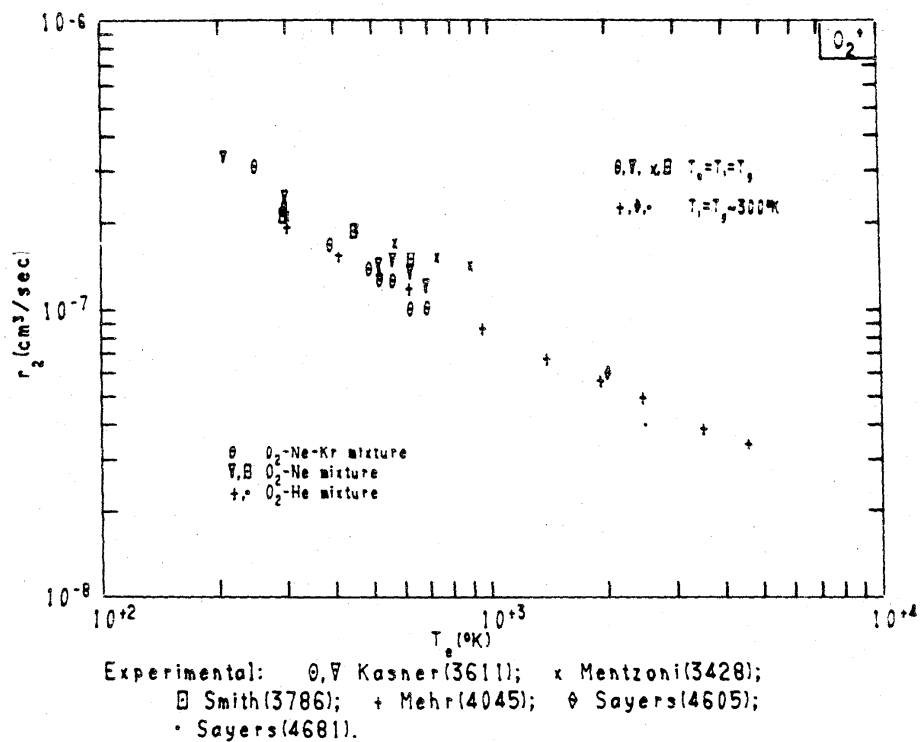


FIGURE 8.15. Experimental values of the dissociative recombination coefficient for O_2^+ as a function of temperature.

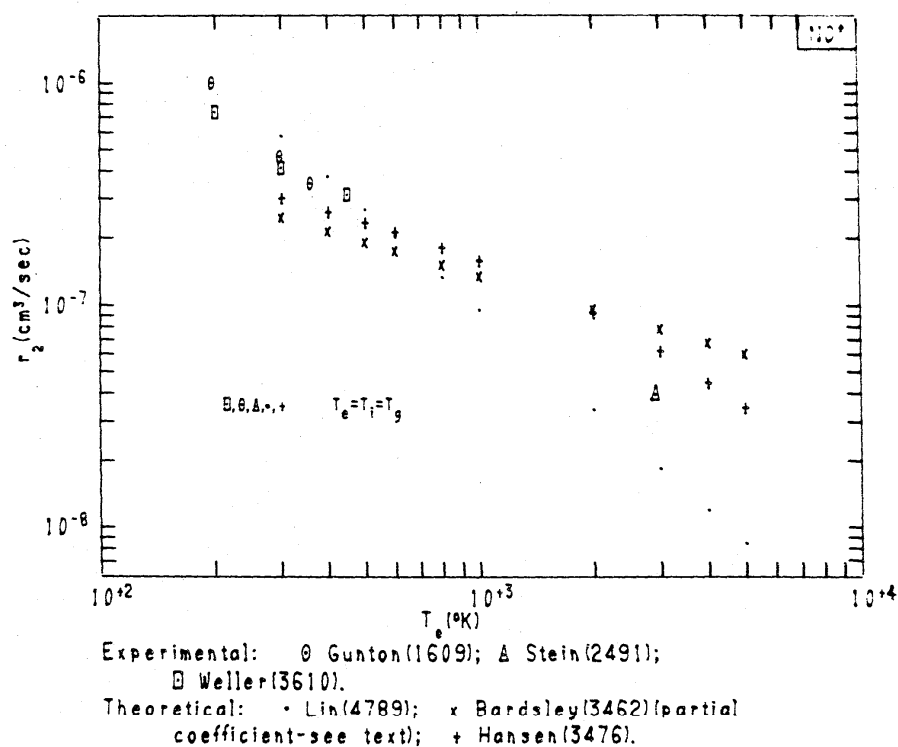


FIGURE 8.16. Temperature dependence of the recombination coefficient for NO^+ ions and electrons. (In [3462] r_2 is a partial recombination coefficient representing the contribution of only the $B^2\Pi$ and $B'^2\Delta$ states.)

Table 8.1

Theoretically computed values for the collisional radiative recombination coefficient for the reaction $\text{He}^+ + e + e \rightarrow \text{He} + e$ for magnetically confined plasmas (A) optically thin and (B) optically thick to resonance lines

N(cm ⁻³)	n(10 ¹¹ cm ⁻³)	T _e (°K)	T _e (°K)	T _i (°K)	T _i (°K)	r _c (10 ⁻¹⁰ cm ³ /sec)	r _c (10 ⁻¹⁰ cm ³ /sec)
		Case A	Case B	Case A	Case B	Case A	Case B
10 ¹²	.01		311		289		1.82
	.05	283	435	284	403	9.4	1.57
	.10	318	526	319	492	10.6	1.25
	.50	449	846	440	818	8.4	.64
	1.0	549	987	535	963	6.3	.51
	5.0	919	1770	915	1750	2.63	.213
	10.	1180	2200	1185	2200	1.70	.170
	50.	2180	3680	2190	3670	.59	.066
10 ¹³	100.	2850		2930		.35	
	.05	256	357	252	293	14.8	3.3
	.10	274	408	266	329	20.3	3.3
	.50	362	604	339	482	21.8	2.66
	1.0	432	722	399	591	18.8	1.95
	5.0	682	1170	653	1020	10.1	1.10
	10.0	895	1470	877	1320	6.6	.77
	50.0	1450	2410	1430	2234	2.97	.34
10 ¹⁴	100.	1850	3000	1820	2810	1.92	.226
	500.	3230	4580	3190	4500	0.64	.089
	1000.		5450		5323		.058
	.01		273		245		3.0
	.05		327		256		5.0
	.10	257	361	231	262	25.6	5.6
	.50	306	491	272	315	53.	5.9
	1.0	344	577	299	363	58	5.5
10 ¹⁵	5.0	496	873	429	577	45	3.7
	10.	600	1060	531	737	35	2.86
	50.	984	1690	924	1300	15.8	1.46
	100.	1240	2070	1180	1660	10.5	1.06
	500.	2170	3310	2100	3870	3.7	.47
	1000.	2780	4000	2700	3580	2.27	.32
	5000.	4900	5880	4800	5630	.70	.123
	10000.	6140		6130		.44	
10 ¹⁶	.05		303		230		5.8
	.10		342		238		6.7
	.50	281	455	236	260	72.	8.0
	1.0	321	519	252	272	85.	8.0
	5.0	442	735	311	331	90.	7.1
	10.	517	865	342	385	81.	6.3
	50.	774	1290	524	643	52.	4.3
	100	940	1550	670	843	39.	3.4
10 ¹⁷	500	1550	2410	1240	1560	16.0	1.81
	1000	1940	2910	1610	2040	10.4	1.32
	5000	3390	4390	2950	3640	3.3	.57
	10000		5130		4440		.39
	.01		166		224		3.7
	.05		247		232		6.5
	.10		276		235		7.4
	.50	258	415	229	244	80.	9.1
10 ¹⁸	1.0	298	501	236	248	97	9.4
	5.0	416	712	253	257	117	8.9
	10.0	481	833	261	264	115	8.3
	50.0	680	1210	311	323	92	6.6
	100.	800	1420	357	375	78	5.8
	500	1220	2090	586	630	47	3.9
	1000	1490	2450	782	835	35	3.2
	5000	2420	3580	1550	1690	15.5	1.95
10 ¹⁹	10000		4250		2220		1.53

Theoretical:

All data taken from Bates, et al., Proc. Roy. Soc. (London), 279A, 32 (1964).

Table 8.2

Experimental values of the recombination coefficient for helium for $n > 10^{11} \text{cm}^{-3}$

$n(10^{11} \text{cm}^{-3})$	$r(10^{-10} \text{cm}^3/\text{sec})$	$T_e(10^3 \text{K})$	$T_g(^{\circ}\text{K})$	$N(\text{cm}^{-3})$	
42	0.58	1.7			(a)
160	0.19	3.2			
190	3.0	1.7			
260	8.9	1.5			
660	3.4	2.0			
4400	2.5	3.4			
1.00	6.8				(b)
1.06	6.7				
1.06	5.6				
1.11	6.6				
1.14	5.6				
1.22	6.8				
1.26	6.0				
1.37	7.1				
1.42	6.4				
1.61	6.7				
1.63	7.8				
1.72	6.4				
1.78		0.610			
1.79	5.3				
1.81	5.2				
1.92	8.6				
2.31	9.6				
2.73	11				
3.2	11				
3.7	12				
3.8		0.750			
4.0	12				
4.2	12				
4.5	12				
5.2	12				
5.9	12				
6.6	12				
7.0	12				
7.5	13				
7.9		0.730			
8.1	14				
36	3.3	1.4			(c)
62	3.6	1.5			
66	1.8	1.7			
120	1.0	2.2			
180	1.3	2.2			
230	0.53	2.9		0.42×10^{14}	
360	0.73	2.7		3.5×10^{14}	
560	0.40	3.1		1.35×10^{14}	
6.5	7.0	0.87			(d)
15	5.6	1.0			
16	14	0.77			
31	$\begin{cases} 2.7 \\ 7.3 \end{cases}$	$\begin{cases} 1.4 \\ 1.2 \end{cases}$			
61	3.7	1.5			
62	1.3	2.4		0.18×10^{14}	
120	1.9	2.0			
180	1.3	2.4		1.4×10^{14}	

Table 8.2 (continued)

Experimental values of the recombination coefficient for helium for $n > 10^{11} \text{cm}^{-3}$

$n(10^{11} \text{cm}^{-3})$	$r(10^{-10} \text{cm}^3/\text{sec})$	$T_e(10^3 \text{K})$	$T_g(^{\circ}\text{K})$	$N(\text{cm}^{-3})$	
2.2		0.53	300	In range from 7×10^{14} to 7×10^{15}	(e)
2.3	13				
2.90		0.56			
3.1	13				
3.9		0.59			
4.0	12				
5.9		0.69			
7.7		0.76			
7.9	7.5				
9.4		0.81			
9.8	7.3				
28.1		1.2			
32	3.8		500	In range from 7×10^{14} to 7×10^{15}	(e)
53		1.4			
78	2.1				
2.2		0.73			
2.4	2.1				
2.9		0.72			
3.2	3.1				
3.9		0.75			
4.0	3.9				
5.7		0.79			
6.1	4.3				
7.6		0.83			
7.9	5.1			2.5×10^{17}	(f)
9.5		0.87			
10	4.9				
27		1.20			
32	2.9				
51		1.45			
79	1.7				
490	5.1	1.6			
660	3.4	2.1			
710	3.6	2.1			
820	3.1	2.2			
1200	2.2	2.7			
1400	$\begin{Bmatrix} 1.3 \\ 1.3 \\ 2.3 \end{Bmatrix}$	$\begin{Bmatrix} 2.8 \\ 3.2 \\ 2.8 \end{Bmatrix}$		$< 3.54 \times 10^{15}$	(g)
1700	1.6	3.2			
2000	1.1	3.5			
2200	1.3	3.7			
2300	0.92	3.3			
3300	0.70	4.4			
3900	0.65	4.6			
1.0	63				
2.0	50				
3.0	62				
4.0	68				
6.0	76				
8.0	96				

Table 8.2 (continued)

Experimental values of the recombination coefficient for helium for $n > 10^{11} \text{cm}^{-3}$				
$n(10^{11} \text{cm}^{-3})$	$r(10^{-10} \text{cm}^3/\text{sec})$	$T_e(10^3 \text{K})$	$T_g(^{\circ}\text{K})$	$N(\text{cm}^{-3})$
25	11		750	(h) $T_e = T_i = T_g$ for the lower electron densities and higher temperatures, but $T_e > T_g$ for higher electron densities and lower temperatures
	6.3		1000	
	4.0		1250	
	9.6		750	
50	5.8		1000	
	5.2		1250	
	2.9		1500	
	6.4		750	
	5.6		1000	
100	4.9		1250	
	3.8		1500	In range from 8.8×10^{15} to 7.1×10^{17} (i)
	2.3		1750	
200	3.1		1750	
	1.4		2000	
10	17	0.73		
30	7.3	1.2		
100	3.0	1.6		
300	1.3	2.4		
400		4.0 [†]		
1000	0.68	3.5		
3000	0.25	5.0 [†]		
10000	0.06	10		

[†] Measured values of T_e ; all other values of T_e given in this section of the table are calculated.

Theoretical:

- (a) Robben, *et al.*, Phys. Rev. 132, 2363 (1963). Data obtained using a plasma jet.
- (b) Mosburg, Phys. Rev. 152, 166 (1966).
- (c) Hinnov, *et al.*, Phys. Rev. 125, 795 (1962).
- (d) Motley and Kuches as given in (c).
- (e) Anisimov, *et al.*, Sov. Phys. Tech. Phys. (Eng. transl.) 10, 1554 (1964).
- (f) Gusinow, *et al.*, Phys. Rev. 149, 91 (1966).
- (g) Aleskovskii, *et al.*, Soviet Phys. JETP (Eng. transl.) 16, 887 (1963).
- (h) Born, Phys. Rev. 169, 155 (1968).
- (i) Newton, *et al.*, J. Phys. (B) 1, 669 (1968).

Table 8.3(a)

Theoretically computed values of the collisional-radiative recombination coefficient
for optically thin helium plasma

$n_e (10^{12} \text{ cm}^{-3})$	$T(^{\circ}\text{K})$	$r_c (10^{-10} \text{ cm}^3 \text{ sec}^{-1})$								
		125	250	500	1000	2000	4000	8000	16000	32000 64000
0.000001	0.083	.049	.035	.025	.016	.012	.0078	.0035	.0019	.0011
0.00001	0.11	.074	.043	.028	.016	.011	.0063	.0035	.0019	.0011
0.0001	0.40	.18	.089	.037	.018	.012	.0063	.0035	.0017	.0011
0.001	2.9	1.1	.30	.085	.027	.013	.0065	.0031	.0017	.0011
0.01	30.	10.	1.4	.23	.047	.016	.0074	.0035	.0017	.0011
0.1	300.	100.	9.1	.79	.093	.023	.0095	.0038	.0017	.0011
1.0	3300.	1100.	95.	3.7	.25	.049	.012	.0041	.0019	.0011
10.0	35000.	12000.	1100.	25.	1.1	.12	.021	.0058	.0023	.0011
100.0		100000.	10000.	220.	6.2	.35	.036	.0079	.0024	.0011
1000.0			91000.	2000.	45.	1.1	.081	.013	.0028	.0012
10000.0				19000.	270.	4.1	.24	.026	.0038	.0013
100000.0					1900.	25.	1.3	.060	.0063	.0018
1000000.0					14000.	220.	8.9	.17	.017	.0030

Table 8.3(b)

Theoretically computed values of the collisional-radiative recombination coefficient
for helium plasma optically opaque to the resonance lines

$n_e (10^{12} \text{ cm}^{-3})$	$T(^{\circ}\text{K})$	$r_c (10^{-10} \text{ cm}^3 \text{ sec}^{-1})$							
		250	500	1000	2000	4000	8000	16000	32000 64000
.000001	.050	.027	.022	.013	.0093	.0060	.0028	.0013	.00087
.00001	.068	.035	.025	.017	.0098	.0069	.0028	.0016	.00078
.0001	.17	.060	.035	.021	.010	.0066	.0031	.0015	.00087
.001	1.1	.23	.065	.030	.013	.0076	.0030	.0015	.00087
.01	11.	1.4	.19	.055	.017	.0085	.0026	.0016	.00081
.1	120.	11.	.71	.11	.025	.010	.0026	.0015	.00087
1.0	1300.	120.	3.5	.28	.047	.013	.0025	.0014	.00083
10.0	12000.	1100.	25.	.98	.10	.017	.0024	.0013	.00093
100.0	87000.	9800.	210.	5.0	.27	.021	.0020	.0012	.00093
1000.		91000.	2100.	33.	.78	.027	.0017	.0013	.00093
10000.			22000.	260.	3.4	.10	.0028	.0013	.00093
100000.				1800.	20.	.81	.011	.0018	.0011
1000000.				11000.	240.	9.5	.083	.0076	.0019

Theoretical:

All data taken from Drawin, *et al.*, Z. Physik 243, 326 (1971).

Table 8.4(a)

Theoretically computed values of the collisional-radiative
recombination coefficient for helium: $T_e = T_g = T$

		$r_c (10^{-11} \text{cm}^3 \text{-sec}^{-1})$											
		$T(^{\circ}\text{K})$	125	250	300	500	1000	2000	4000	8000	32000		
$n(10^{12} \text{cm}^{-3})$	$N(10^{16} \text{cm}^{-3})$												
$n \rightarrow 0$	10	147	20.8			3.1	0.70	0.22	0.088		(a)		
	100	1470	175.			20	2.70	0.50	0.14				
	1000	14200	1650.			144	10.6	1.38	0.238				
	10000	141000	16800.			980	42	2.14	0.44				
	100000		159000.			3900	113	4.3	0.58				
.0001	1.				3.61						(b)		
	10.				14.0								
	100.				90.								
	1000.				770								
.001	1.				11.5								
	10.				21.4								
	100.				90.								
	1000.				760								
.01	1.				61								
	10.				71								
	100.				149								
	1000.				820								
.1	1.				450								
	10.				450								
	100.				560								
	1000.				1290								
1.0	1.				3600								
	10.				3700								
	100.				4300								
	1000.				4900								
10.	1.				35000								
	10.				38000								
	100.				37000								
	1000.				45000								
.001	.000001						.56	.33	.15	.065	.015	(c)	
	.00001						.63	.33	.17	.059	.015		
	.0001						.87	.42	.17	.060	.014		
	.001						1.6	.53	.20	.059	.014		
	.01						3.2	.89	.22	.059	.011		
	.1						7.6	2.0	.30	.059	.0093		
	1.0						31.	7.8	.54	.060	.0085		
	10.						220.	27.0	1.3	.060	.0093		
	100.						2400.	130.0	4.0	.059	.011		
	1000.						29000.	710.0	16.0	.058	.015		
	.1	.000001						6.9	.91				
		.00001						6.9	.79				
.0001							6.9	.87					
.001							6.6	.96	.25				
.01							12.	1.1	.31				
.1							25.	2.1	.45				
1.0							71.	7.4	.91				
10.							300.	28.	2.0				
100.							2500.	120.	5.0				
1000.							29000.	710.	16.				

Table 8.4(a) continued

Theoretically computed values of the collisional-radiative
recombination coefficient for helium: $T_e = T_g = T$

		$r_c (10^{-11} \text{cm}^3 \text{-sec}^{-1})$								
		T(°K)	125	250	300	500	1000	2000	4000	8000 32000
$n(10^{12} \text{cm}^{-3})$	$N(10^{16} \text{cm}^{-3})$									
1.0	.000001						35	2.3	--	
	.00001						34	2.3	--	
	.0001						33	2.4	.28	
	.001						33	2.5	.33	
	.01						37	2.6	.49	
	.1						62	3.6	.78	
	1.						150	8.7	1.6	
	10.						420	30.	3.2	
	100.						2450	135.	6.5	
	1000.						29000	710.	16.0	
10.0	.000001						250	11.0	--	.012
	.00001						250	10.0	--	.012
	.0001						250	10.0	--	.012
	.001						250	10.0	--	.011
	.01						250	11.0	.87	.0093
	.1						260	12.0	1.0	.0083
	1.0						360	19.0	1.7	.0083
	10.						760	45.0	3.2	.0093
	100.						2500	150.	6.5	.074 .011
	1000.						29000	710.	16.0	.065 .015
.001	0.1						5.6	0.66	0.18	
	1.0						23.	1.9	.27	
	10.0						150.	7.7	.43	
	100.0						1800.	49.	.88	
	1000.0						20000.	340	2.6	
0.1	0.1						19.	1.3	0.27	
	1.0						60.	2.9	0.38	
	10.0						190.	9.1	0.54	
	100.0						1800.	52.	0.96	
	1000.0						20000.	340	2.6	
1.0	0.1						43.	3.2	0.43	
	1.0						120.	4.3	0.52	
	10.0						330	10.0	0.68	
	100.0						1900	54.	1.0	
	1000.0						20000	340.	2.6	
10.0	0.1						190	4.3	.95	
	1.0						280	5.4	1.1	
	10.0						510	12.	1.2	
	100.0						2000	55.	1.3	
	1000.0						20000	340	2.6	
.00000001	4.0				5.8					
	10.0				11.8					
	20.0				16.6					
	40.0				29.9					
	80.0				51.					
	100.0				71.					
	200.0				110					
	400.0				214					
	1000.0				610					

Table 8.4(a) continued

Theoretically computed values of the collisional-radiative
recombination coefficient for helium: $T_e = T_g = T$

		$r_c (10^{-11} \text{cm}^3 \text{-sec}^{-1})$								
		$T(^{\circ}\text{K})$	125	250	300	500	1000	2000	4000	8000 32000
$n(10^{12} \text{cm}^{-3})$	$N(10^{16} \text{cm}^{-3})$									
.001	1.0				11.4					
	4.0				14.2					
	20.0				25.1					
	40.0				38.					
	100.0				85.					
	200.0				130					
	400.0				243					
	1000.				660					
.01	1.0				63.					
	4.0				63.					
	10.0				69.					
	40.0				85.					
	100.				127.					
	200.				172					
	400.				284					
	1000.				660					
0.1	1.0				470					(e)
	4.0				470					
	10.0				480					
	20.0				480					
	40.				480					
	100.				540					
	200.				580					
	400.				680					
	1000.				1030					
1.0	1.0				4200					
	4.0				4200					
	12.6				4200					
	20.0				4100					
	40.0				4200					
	100.				4300					
	200.				4400					
	400.				4500					
	1000.				4800					

Theoretical:

- (a) Bates, *et al.*, Proc. Phys. Soc. (London) 85, 231 (1965).
- (b) Collins, Phys. Rev. 177, 254 (1968).
- (c) Drawin, *et al.*, Z. Physik 254, 202 (1972). (For plasma optically opaque to resonance lines).
- (d) Drawin, *et al.*, Z. Physik 254, 202 (1972). (For plasma optically opaque to resonance lines and resonance continuum).
- (e) Deloche, *et al.*, J. Phys. (Paris) 29, C3-27-30 (1968).

Table 8.4(b)

Theoretically computed values of the collisional-radiative recombination coefficient for helium: $T_g = 300^\circ\text{K}$, $T_e = 500^\circ\text{K}$

n (10^{12}cm^{-3})	N (10^{16}cm^{-3})	r_c ($10^{-11}\text{cm}^3\text{sec}^{-1}$)	n (10^{12}cm^{-3})	N (10^{16}cm^{-3})	r_c ($10^{-11}\text{cm}^3\text{sec}^{-1}$)
.0001	1.	1.59	0.1	1	54.
	10.	5.7		10	53.
	100.	43.		100	73.
	1000.	390		1000	299.
.001	1.	2.99	1.0	1	390
	10.	6.1		10	450
	100.	38.		100	420
	1000.	390.		1000	600
.01	1.	9.7	10.0	1	3600
	10.	11.9		10	3600
	100.	37.		100	3600
	1000.	350.0		1000	4100

Theoretical:

All data taken from Collins, Phys. Rev. 177, 254 (1968).

Table 8.4(c)

Theoretically computed values of the collisional-radiative recombination coefficient for helium: $T_g = 300^\circ\text{K}$, $T_e = 1000^\circ\text{K}$

n (10^{12}cm^{-3})	N (10^{16}cm^{-3})	r_c ($10^{-11}\text{cm}^3\text{sec}^{-1}$)	n (10^{12}cm^{-3})	N (10^{16}cm^{-3})	r_c ($10^{-11}\text{cm}^3\text{sec}^{-1}$)
.0001	1.	0.55	0.1	10	4.3
	10.	2.15		100	5.5
	100.	14.1		1000	49.
	1000.	132.			
.001	10	1.67	1.0	10	20.8
	100	12.8		100	19.6
	1000	122.		1000	31.
.0001	10	1.69	10.0	1	113
	100	6.9		10	880
	1000	112.		100	111
				1000	132

Theoretical:

All data taken from Collins, Phys. Rev. 177, 254 (1968).

Table 8.4(d)

Theoretically computed values of the collisional-radiative recombination coefficient for helium: $T_g = 300^\circ\text{K}$, $T_e = 2000^\circ\text{K}$

n (10^{12}cm^{-3})	N (10^{16}cm^{-3})	r_c ($10^{-11}\text{cm}^3\text{sec}^{-1}$)	n (10^{12}cm^{-3})	N (10^{16}cm^{-3})	r_c ($10^{-11}\text{cm}^3\text{sec}^{-1}$)
.0001	1	.222	.1	1	.71
	10	.74		10	.74
	100	5.0		100	.89
	1000	46.		1000	8.0
.001	1	.262	1.0	1	1.78
	10	.55		10	1.86
	100	4.2		100	1.81
	1000	42.		1000	2.96
.01	1	.400	10	1	7.1
	10	.52		10	7.3
	100	1.81		100	7.0
	1000	34.		1000	7.4

Theoretical:

All data taken from Collins, Phys. Rev. 177, 254 (1968).

Table 8.5

Experimental values of the dissociative recombination coefficient of electrons and Ne_2^+ ions for $T_e = T_+ = T_g = 300^\circ\text{K}$

	r_2 ($10^{-7}\text{cm}^3/\text{sec}$)	r_2^* corr ($10^{-7}\text{cm}^3/\text{sec}$)	n (10^{10}cm^{-3})	N (10^{17}cm^{-3})
(a)	2.0	1.7	0.11 to 1.2	6.4
(b)	2.2	1.8	0.028 to 2.0	5.5 to 11
(c)	3.4	1.9	0.05 to 0.22	8.3
(d)	2.1	1.7	0.066 to 0.59	4.8 to 9.7
(e)	2.0	2.0	1 to 10	23
(f)	2.4	1.8	0.18 to 1.2	5.8 to 6.4
(g)	1.8	1.8	0.023 to 0.44	2.6 to 9.7
(h)		1.7	0.012 to 0.29	6.4
(i)	1.75	1.75	0.058 to 0.96	1.9

* r_2 corr are the values by Frommhold, Biondi and Mehr (3443) when the original values (r_2) were corrected by them for diffusion and for the initial distribution of ions and electrons in the plasma.

Experimental:

- (a) Hess, Z. Naturforsch 20A, 451 (1965).
- (b) Oskam, et al., Phys. Rev. 132, 1445 (1963).
- (c) Biondi, Phys. Rev. 129, 1181 (1963).
- (d) Biondi, et al., Phys. Rev. 76, 1697 (1949).
- (e) Connor, et al., Phys. Rev. 140, A778 (1965).
- (f) Oskam, Philips Res. Rept. 13, 401 (1958).
- (g) Kasner, Scientific Paper 67-1E2-GASES-P3, Westinghouse Research Laboratories (1967).
- (h) Frommhold, et al., Phys. Rev. 165, 44 (1968).
- (i) Philbrick, et al., Phys. Rev. 181, 271 (1969).

Table 8.6(a)

Experimental values for the temperature variation of the dissociative recombination coefficient of electrons and Ne_2^+ ions: $T_e = T_g = T_+$

T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)		T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)		T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)	
290	18	(a)	880	11	(b)	1900	3.5	(b)
390	16	(a)	910	11	(b)	2000	2.9	(b)
500	15	(a)	950	9.9	(b)	2100	3.4	(b)
580	14	(b)	1000	9.0	(b)	2200	3.0	(b)
640	12	(b)	1100	8.0	(b)	2300	2.7	(b)
			1200	7.5	(b)	2400	2.4	(b)
660	12	(b)						
690	12	(b)	1300	7.1	(b)	2700	2.2	(b)
720	12	(b)		5.9	(b)	2800	2.2	(b)
740	11	(b)	1400	5.7	(b)	3000	1.9	(b)
770	11	(b)	1500	5.3	(b)	3200	1.8	(b)
830	12	(b)	1700	4.4	(b)	3600	1.4	(b)
			1800	4.1	(b)	4600	3.5	(b)

Experimental:

- (a) Kasner, Scientific Paper 67-1E2-GASES-P3, Westinghouse Research Laboratories (1967).
 (b) Cunningham, *et al.*, Phys. Rev. **185**, 98 (1969).

Table 8.6(b)

Experimental values for the temperature variation of the dissociative recombination coefficient of electrons and Ne_2^+ ions: $T_g = T_+ = 300^\circ\text{K}$

T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)		T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)		T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)	
300	20	(a)	650	11	(b)	2800	5.8	(b)
	19	(a)	820	10	(b)	2900	6.0	(a)
	17	(a)	900	11	(a)	3300	5.9	(a)
	16	(a)	1000	9.4	(b)	3500	5.5	(b)
	16	(b)	1100	8.3	(a)	4300	4.4	(a)
410	14	(b)				4600	4.8	(b)
530	12	(b)	1300	9.1	(a)	5800	4.4	(b)
	14	(a)	1400	8.1	(b)	8100	3.7	(b)
600	13	(a)	2100	6.8	(b)	11000	3.1	(b)
	12	(a)	2300	7.0	(a)			

Experimental:

- (a) Philbrick, *et al.*, Phys. Rev. **181**, 271 (1969).
 (b) Frommhold, *et al.*, Phys. Rev. **165**, 44 (1968).

Table 8.7

Experimental values of the dissociative recombination coefficient of electrons and Ar_2^+ ions for $T_e = T_+ = T_g = 300^\circ\text{K}$

Author	r_2 ($10^{-7} \text{ cm}^3/\text{sec}$)	n (10^8 cm^{-3})	N (10^{17} cm^{-3})
(a)	6.7	1.1 to 710	2.9 to 11
(b)	6	2.1 to 53	4.5
(c)	8.5	0.64 to 26	6.4

Experimental:

- (a) Oskam, *et al.*, Phys. Rev. **132**, 1445 (1963).
 (b) Biondi, Phys. Rev. **129**, 1181 (1963).
 (c) Mehr, *et al.*, Phys. Rev. **176**, 322 (1968).

Table 8.8(a)
Temperature dependence of the dissociative recombination coefficient
for Ar_2^+ : $T_g = T_+ = 300^\circ\text{K}$

T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)	T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)	T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)
310	85	1500	28	4000	15
				5000	13
510	56	2000	22	6000	12
960	37				
1000	36	3000	17	7800	10
				8000	11
				8200	10
				9600	9.1
				10000	9.1

Experimental:

All data taken from Mehr, *et al.*, Phys. Rev. **176**, 322, (1968).

Table 8.8(b)
Temperature dependence of the dissociative recombination coefficient
for Ar_2^+ : $T_e = T_+ = T_g$

T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)	T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)	T_e (°K)	r ($10^{-8} \text{ cm}^3/\text{sec}$)
550	44	790	31	1500	11
570	43	830	26	1600	9.4
580	41	850	25	1700	9.3
600	41	890	24	1800	7.5
610	41	960	20	1900	7.6
					7.7
640	39	990	20		
670	41	1000	19	2100	6.7
690	40	1100	19	2200	5.8
700	35			2500	5.5
730	34	1300	13	2800	4.1
750	30			3000	4.1

Experimental:

All data taken from Cunningham, *et al.*, Phys. Rev. **185**, 98 (1969).

Table 8.9
Experimental value for the dissociative recombination coefficient
for Kr_2^+ and Xe_2^+ for $T_e = T_+ = T_g = 300^\circ\text{K}$

Gas	r ($10^{-7} \text{ cm}^3/\text{sec}$)	n (10^8 cm^{-3})	N (10^{17} cm^{-3})
Kr	12	0.88 to 120	1.93 to 14.5
Xe	14	0.82 to 31	1.7 to 11.1

Experimental:

All data taken from Oskam, *et al.*, Phys. Rev. **132**, 1445 (1963).

Table 8.10

Temperature dependence of the dissociative
recombination coefficient for Kr_2^+ for $T_e = T_+ = T_g$

T_e (°K)	r ($10^{-6}\text{cm}^3/\text{sec}$)	T_e (°K)	r ($10^{-6}\text{cm}^3/\text{sec}$)	T_e (°K)	r ($10^{-6}\text{cm}^3/\text{sec}$)
930	.44	1450	.24	1970	.15
970	.44	1520	.22	2070	.14
1070	.34	1560	.23	2110	.13
1130	.35	1620	.20	2250	.12
1180	.33	1770	.17	2400	.11
1260	.27	1800	.17	2610	.10
1350	.27	1870	.17	2800	.10
1400	.25	1900	.17		

Experimental:

All data taken from Cunningham, *et al.*, J. Phys. B 5, 1773 (1972).

Table 8.11(a)

Theoretical radiative recombination coefficients for electrons with H^+ ions.
Partial radiative recombination coefficients $r_2(p)$ to individual
levels of principal quantum number p of the hydrogen atom.

Principal quantum No.	T_e (°K)	250	500	1000	2000	1000	8000	16000	32000	64000
		$r_2(p) (10^{-14} \text{cm}^3/\text{sec})$								
1	102	71.7	50.7	35.6	25.0	17.4	12.0	8.02	5.19	
2	56.6	39.8	27.9	19.4	13.2	8.80	5.63	3.42	1.95	
3	39.0	27.2	18.8	12.8	8.44	5.33	3.19	1.80	0.946	
4	29.5	20.4	14.0	9.23	5.86	3.53	2.00	1.06	0.533	
5	23.6	16.2	10.8	6.99	4.29	2.48	1.35	0.687	0.332	
6	19.6	13.3	8.70	5.48	3.26	1.82	0.953	0.471	0.222	
7	16.6	11.1	7.16	4.39	2.54	1.38	0.702	0.339	0.156	
8	14.3	9.46	5.99	3.59	2.02	1.07	0.534	0.253	0.114	
9	12.5	8.17	5.08	2.98	1.64	0.851	0.416	0.193	0.0866	
10	11.1	7.13	4.36	2.51	1.35	0.688	0.331	0.152	0.0672	
11	9.88	6.27	3.77	2.13	1.13	0.565	0.268	0.122	0.0533	
12	8.87	5.56	3.29	1.83	0.953	0.471	0.221	0.0989	0.0430	

Theoretical:

All data taken from Bates, *et al.*, in *Atomic and Molecular Processes* (Academic Press, New York, (1962), p. 245.

Table 8.11(b)

Theoretical radiative recombination coefficients for electrons with H^+ ions.
Total radiative recombination coefficient r_2 for electrons with H^+ ions.

Temperature °K	250	500	1000	2000	4000	8000	16000	32000	64000
$r_2 (10^{-13} \text{cm}^3/\text{sec})$	48.4	31.2	19.9	12.6	7.85	4.83	2.93	1.73	1.00

Theoretical:

All data taken from Bates, *et al.*, in *Atomic and Molecular Processes* (Academic Press, New York, (1962), p. 245.

Table 8.12(a)

Theoretical collisional-radiative decay coefficient g_c [see eq. (24)] for H^+ ions in optically thin plasmas.
Theoretical collisional-radiative recombination coefficient r_c .

T_e (°K)	50 ⁽¹⁾	100 ⁽¹⁾	150 ⁽¹⁾	200 ⁽¹⁾	250 ⁽²⁾	500 ⁽²⁾	1000 ⁽²⁾	2000 ⁽²⁾	4000 ⁽²⁾	8000 ⁽²⁾	16000 ⁽²⁾	32000 ⁽²⁾
n (cm ⁻³)	r_c (10 ⁻¹¹ cm ³ /sec)											
limit $n \rightarrow 0$					0.48	0.31	0.20	0.13	0.079	0.048	0.029	0.017
10 ²	2.3	1.1	0.80	0.65								
10 ³	3.4	1.4	0.90	0.70								
10 ⁴	6.8	1.9	1.1	0.82								
10 ⁵	21	3.5	1.7	1.1								
10 ⁶	95	9.5	3.4	1.9								
10 ⁷	700	39	10	4.5								
10 ⁸	6600	250	47	17	8.8	1.4	0.41	0.18	0.092	0.051	0.030	0.018
10 ⁹	65000	2300	350	96	40	3.8	0.75	0.25	0.10	0.053	0.030	0.018
10 ¹⁰		23000	3300	840	280	16	1.9	0.41	0.14	0.061	0.032	0.018
10 ¹¹			32000	8000	2700	100	6.9	0.91	0.22	0.081	0.034	0.018
10 ¹²					26000	900	39	2.9	0.44	0.12	0.043	0.020
10 ¹³					260000	8900	310	14	1.2	0.21	0.062	0.024
10 ¹⁴					2600000	88000	2900	98	5.1	0.51	0.10	0.031
10 ¹⁵						880000	29000	870	27	1.7	0.23	0.049
10 ¹⁶							290000	8500	230	8.4	0.50	0.073
10 ¹⁷								84000	2100	34	1.4	0.18
limit $n \rightarrow \infty$ *					2.6 ⁻⁸ _n	8.8 ⁻¹⁰ _n	2.9 ⁻¹¹ _n	8.4 ⁻¹³ _n	1.9 ⁻¹⁴ _n	2.4 ⁻¹⁶ _n	9.1 ⁻¹⁸ _n	1.1 ⁻¹⁸ _n

* In this row 2.6⁻⁸_n etc = 2.6 × 10⁻⁸_n etc and gives the value of r_c in units of (10⁻¹¹ cm³/sec).

Theoretical:

(1) Bates, et al., Proc. Phys. Soc. (London) **83**, 43 (1964).

(2) Bates, et al., Proc. Roy. Soc. (London) Ser A **257**, 297 (1962).

Table 8.12(b)

Theoretical collisional-radiative decay coefficient g_c [see eq. (24)] for H^+ ions in optically thin plasmas. Collisional-radiative ionization coefficient i_c .

$T_e (^{\circ}K)$ $n(\text{cm}^{-3})$	4000	8000	16000	32000
	$i_c (\text{cm}^3/\text{sec})$			
limit $n \rightarrow 0$	1.4×10^{-26}	9.7×10^{-18}	3.4×10^{-13}	8.2×10^{-11}
10^8	1.6×10^{-26}	1.1×10^{-17}	3.6×10^{-13}	8.4×10^{-11}
10^9	1.8×10^{-26}	1.2×10^{-17}	3.8×10^{-13}	8.8×10^{-11}
10^{10}	2.7×10^{-26}	1.4×10^{-17}	4.2×10^{-13}	9.2×10^{-11}
10^{11}	4.5×10^{-26}	1.9×10^{-17}	4.9×10^{-13}	1.0×10^{-10}
10^{12}	1.0×10^{-25}	3.0×10^{-17}	6.5×10^{-13}	1.2×10^{-10}
10^{13}	3.6×10^{-25}	6.8×10^{-17}	1.1×10^{-12}	1.7×10^{-10}
10^{14}	2.2×10^{-24}	2.3×10^{-16}	2.5×10^{-12}	2.9×10^{-10}
10^{15}	1.8×10^{-23}	1.3×10^{-15}	8.6×10^{-12}	6.9×10^{-10}
10^{16}	1.5×10^{-22}	5.9×10^{-15}	1.9×10^{-11}	1.0×10^{-9}
10^{17}	5.8×10^{-22}	1.0×10^{-14}	2.3×10^{-11}	1.1×10^{-9}
limit $n \rightarrow \infty$	8.3×10^{-22}	1.1×10^{-14}	2.3×10^{-11}	1.1×10^{-9}

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser A 267, 297 (1962).

Table 8.13(a)

Theoretical collisional-radiative recombination coefficients for H^+ ions in optically thick plasmas. Recombination coefficient r_c for plasma optically thick to the lines of the Lyman series.

$T_e (^{\circ}K)$ $n(\text{cm}^{-3})$	250	500	1000	3000	4000	8000	16000	32000	64000
	$r_c (10^{-12} \text{cm}^3/\text{sec})$								
limit $n \rightarrow 0$	4.8	3.1	2.0	1.3	0.79	0.46	0.19	0.093	0.055
10^8	79	12	3.7	1.7	0.89	0.48	0.19	0.093	0.055
10^9	380	34	6.4	2.2	1.0	0.50	0.18	0.092	0.055
10^{10}	2900	150	16	3.5	1.2	0.54	0.18	0.091	0.055
10^{11}	27000	1000	61	7.5	1.8	0.60	0.18	0.089	0.054
10^{12}	260000	8900	360	25	3.4	0.68	0.16	0.086	0.053
10^{13}		88000	3000	130	9.2	0.66	0.14	0.083	0.053
10^{14}			29000	940	35	0.81	0.14	0.083	0.053
10^{15}				8500	210	3.0	0.23	0.093	0.055
10^{16}					1900	25	1.1	0.20	0.079
limit $n \rightarrow \infty$ *	$2.6^{-7} n$	$8.8^{-9} n$	$2.9^{-10} n$	$8.4^{-12} n$	$1.9^{-13} n$	$2.4^{-15} n$	$9.1^{-17} n$	$1.1^{-17} n$	$2.7^{-18} n$

* In this row $2.6^{-7} n$ etc = $2.6 \times 10^{-7} n$ etc and gives the value of r_c in units of $(10^{-12} \text{cm}^3/\text{sec})$.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser A 270, 155 (1962).

Table 8.13(b)

Theoretical collisional-radiative recombination coefficients for H^+ ions in optically thick plasmas.
Recombination coefficient r_c for plasma optically thick to
the lines of the Lyman series and the Lyman continuum.

$T_e (^{\circ}K)$ $n(cm^{-3})$	250	500	1000	2000	4000	8000	16000	32000	64000
	$r_c (10^{-12} cm^3/sec)$								
limit $n \rightarrow 0$	3.8	2.4	1.5	0.94	0.54	0.29	0.07	0.013	0.003
10^8	78	11	3.2	1.3	0.64	0.31	0.07	0.013	0.003
10^9	380	33	5.9	1.8	0.75	0.33	0.06	0.012	0.003
10^{10}	2900	150	15	3.1	1.0	0.37	0.06	0.011	0.003
10^{11}	27000	1000	60	7.1	1.6	0.43	0.06	0.009	0.002
10^{12}	260000	8900	360	25	3.2	0.51	0.04	0.006	0.001
10^{13}		88000	3000	130	9.0	0.49	0.02	0.003	0.001
10^{14}			29000	940	35	0.64	0.02	0.003	0.001
10^{15}				8500	210	2.8	0.11	0.013	0.003
10^{16}					1900	2.5	1.0	0.12	0.027

Theoretical:

All data calculated from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 270, 155 (1962).

Table 8.13(c)

Theoretical collisional-radiative recombination coefficients for H^+ ions in optically thick plasmas.
Recombination coefficient r_c for plasma optically thick to lines of all series.

$T_e (^{\circ}K)$ $n(cm^{-3})$	250	500	1000	2000	4000	8000	16000	32000	64000
	$r_c (10^{-12} cm^3/sec)$								
limit $n \rightarrow 0$	3.4	2.1	1.3	0.75	0.40	0.19	0.12	0.081	0.052
10^8	29	3.0	1.3	0.75	0.40	0.19	0.12	0.081	0.052
10^9	260	11	1.6	0.76	0.40	0.19	0.12	0.081	0.052
10^{10}	2600	90	4.2	0.83	0.40	0.19	0.12	0.081	0.052
10^{11}	26000	880	30	1.6	0.42	0.19	0.12	0.081	0.052
10^{12}	260000	8800	290	9.2	0.60	0.19	0.12	0.081	0.052
10^{13}		88000	2900	85	2.3	0.21	0.12	0.081	0.052
10^{14}			29000	840	19	0.43	0.13	0.082	0.053
10^{15}				8400	190	2.6	0.21	0.092	0.055
10^{16}					1900	240	1.0	0.19	0.079

Theoretical:

All data calculated from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 270, 155 (1962).

Table 8.13(d)

Theoretical collisional-radiative recombination coefficients for H^+ ions in optically thick plasmas.
 Recombination coefficient r_c for plasma optically thick to
 lines of all series and the Lyman continuum.

T_e (°K)	250	500	1000	2000	4000	8000	16000	32000	64000
$n(\text{cm}^{-3})$	$r_c (10^{-12} \text{ cm}^3/\text{sec})$								
limit $n \rightarrow 0$	2.4	1.4	0.77	0.40	0.15	0.017	0.0024	0.00063	0.00024
10^8	28	2.3	0.80	0.40	0.15	0.017	0.0024	0.00063	0.00024
10^9	260	10	1.1	0.41	0.15	0.017	0.0024	0.00063	0.00024
10^{10}	2600	89	3.7	0.48	0.15	0.017	0.0024	0.00063	0.00024
10^{11}	26000	880	30	1.2	0.17	0.017	0.0024	0.00063	0.00024
10^{12}	260000	8800	290	8.8	0.34	0.019	0.0025	0.00064	0.00024
10^{13}		88000	2900	84	2.0	0.041	0.0033	0.00074	0.00027
10^{14}			29000	840	19	0.26	0.012	0.0017	0.00051
10^{15}				8400	190	2.4	0.093	0.012	0.0029
10^{16}					1900	2.4	0.91	0.11	0.027

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 270, 155 (1962).

Table 8.14 (a)

Collisional-radiative recombination coefficient r_c for a magnetically confined hydrogen plasma optically thick to the lines of the Lyman series at $T_g = 250^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^{-1}	10^0	10^1	10^2	10^3	10^4
10^9	T_e^*	1.2	0.64	0.39	0.29	0.26	0.25
	T_i^*	1.2	0.64	0.39	0.29	0.26	0.25
	r_c^*	0.49	1.7	7.9	22	32	35
10^{10}	T_e	1.5	0.83	0.51	0.36	0.3	0.27
	T_i	1.5	0.83	0.51	0.35	0.28	0.25
	r_c	0.63	2.7	13	55	140	220
10^{11}	T_e	1.9	1.1	0.71	0.5	0.40	0.35
	T_i	1.9	1.1	0.69	0.47	0.32	0.26
	r_c	0.93	4.5	24	98	270	520
10^{12}	T_e	2.4	1.5	1.0	0.74	0.59	0.50
	T_i	2.4	1.5	0.96	0.61	0.35	0.26
	r_c	1.4	7.8	38	150	410	890
10^{13}	T_e	3.2	2.1	1.4	1.1	0.92	0.77
	T_i	3.2	2.0	1.3	0.76	0.36	0.26
	r_c	2.1	11	56	180	450	1000
10^{14}	T_e	4.1	2.9	2.1	1.7	1.4	1.2
	T_i	4.1	2.8	1.	0.79	0.35	0.26
	r_c	3.0	16	70	220	460	1300
10^{15}	T_e	5.4	4.0	3.1	2.6	2.22	1.7
	T_i	5.3	3.7	2.0	0.76	0.33	0.26
	r_c	3.5	21	88	220	470	1700

* T_e in (10^3°K); T_i in (10^3°K); r_c in ($10^{-11} \text{cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.14(b)

Collisional-radiative recombination coefficient r_c for a magnetically confined hydrogen plasma optically thick to the lines of the Lyman series at $T_g = 1000^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^{-1}	10^0	10^1	10^2	10^3	10^4
10^9	T_e^*	1.6	1.1	1.0	1.0	1.0	1.0
	T_i^*	1.6	1.1	1.0	1.0	1.0	1.0
	r_c^*	0.32	0.56	0.65	0.65	0.65	0.65
10^{10}	T_e	1.8	1.2	1.0	1.0	1.0	1.0
	T_i	1.8	1.2	1.0	1.0	1.0	1.0
	r_c	0.46	1.1	1.5	1.6	1.6	1.6
10^{11}	T_e	2.1	1.4	1.1	1.0	1.0	1.0
	T_i	2.1	1.4	1.1	1.0	1.0	1.0
	r_c	0.69	2.3	4.8	5.9	6.0	6.0
10^{12}	T_e	2.6	1.7	1.3	1.1	1.0	1.0
	T_i	2.6	1.7	1.3	1.0	1.0	1.0
	r_c	1.1	4.6	13	28	32	35
10^{13}	T_e	3.4	2.3	1.6	1.3	1.2	1.1
	T_i	3.4	2.3	1.5	1.1	1.0	1.0
	r_c	1.8	8.1	32	76	120	170
10^{14}	T_e	4.3	3.1	2.2	1.9	1.7	1.4
	T_i	4.3	3.0	1.9	1.2	1.0	1.0
	r_c	2.5	12	55	130	230	510
10^{15}	T_e	5.6	4.1	3.2	2.8	2.4	1.9
	T_i	5.6	3.8	2.3	1.3	1.0	1.0
	r_c	3.2	18	76	160	320	1100

* T_e in (10^3 °K), T_i in (10^3 °K); r_c in (10^{-11} cm³/sec) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.14(c)

Collisional-radiative recombination coefficient r_c for a magnetically confined hydrogen plasma optically thick to the lines of the Lyman series at $T_g = 4000^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^{-1}	10^0	10^1	10^2	10^3	10^4
10^9	T_e^*	4.1	4.0	4.0	4.0	4.0	4.0
	T_i^*	4.1	4.0	4.0	4.0	4.0	4.0
	r_c	0.098	0.10	0.10	0.10	0.10	0.10
10^{10}	T_e	4.1	4.0	4.0	4.0	4.0	4.0
	T_i	4.1	4.0	4.0	4.0	4.0	4.0
	r_c	0.12	0.12	0.12	0.12	0.12	0.12
10^{11}	T_e	4.1	4.0	4.0	4.0	4.0	4.0
	T_i	4.1	4.0	4.0	4.0	4.0	4.0
	r_c	0.17	0.18	0.18	0.18	0.18	0.18
10^{12}	T_e	4.3	4.0	4.0	4.0	4.0	4.0
	T_i	4.3	4.0	4.0	4.0	4.0	4.0
	r_c	0.29	0.34	0.35	0.35	0.35	0.35
10^{13}	T_e	4.5	4.1	4.0	4.0	4.0	4.0
	T_i	4.5	4.1	4.0	4.0	4.0	4.0
	r_c	0.58	0.85	0.93	0.39	0.93	0.93
10^{14}	T_e	4.9	4.3	4.1	4.0	4.0	4.0
	T_i	4.9	4.2	4.0	4.0	4.0	4.0
	r_c	1.2	2.5	3.2	3.4	3.5	3.5
10^{15}	T_e	5.9	4.8	4.3	4.2	4.1	4.0
	T_i	5.9	4.7	4.1	4.0	4.0	4.0
	r_c	2.1	7.6	13	15	18	21

* T_e in (10^3°K); T_i in (10^3°K); r_c in ($10^{-11} \text{ cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.15(a)

Collisional-radiative recombination coefficient r_c for a magnetically confined optically thin hydrogen plasma at $T_g = 250^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^{-1}	10^0	10^1	10^2	10^3	10^4
10^8	T_e^*	0.36	0.27	0.25	0.25	0.25	0.25
	T_i^*	0.36	0.27	0.25	0.25	0.25	0.25
	r_c	3.3	6.8	8.5	8.7	8.7	8.7
10^9	T_e	0.51	0.36	0.26	0.25	0.25	0.25
	T_i	0.51	0.36	0.26	0.25	0.25	0.25
	r_c	3.5	10	32	38	40	40
10^{10}	T_e	0.72	0.48	0.35	0.28	0.26	0.25
	T_i	0.72	0.48	0.35	0.28	0.26	0.25
	r_c	4.9	18	71	180	260	280
10^{11}	T_e	0.98	0.67	0.48	0.37	0.31	0.29
	T_i	0.98	0.67	0.48	0.36	0.28	0.26
	r_c	7.4	29	120	430	930	1400
10^{12}	T_e	1.4	0.96	0.68	0.52	0.42	0.38
	T_i	1.4	0.95	0.67	0.47	0.32	0.26
	r_c	9.3	47	210	760	2100	3400
10^{13}	T_e	2.2	1.4	1.0	0.77	0.63	0.57
	T_i	2.2	1.4	0.95	0.60	0.34	0.26
	r_c	10	59	310	1100	2800	4900
10^{14}	T_e	3.3	2.2	1.6	1.2	1.1	0.88
	T_i	3.3	2.1	1.4	0.7	0.34	0.26
	r_c	11	69	320	1200	2100	5600
10^{15}	T_e	4.9	3.4	2.4	2.0	1.8	1.4
	T_i	4.8	3.1	1.8	0.72	0.32	0.26
	r_c	11	72	330	850	1700	6300

* T_e in (10^3 °K); T_i in (10^3 °K); r_c in ($10^{-11}\text{cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.15(b)

Collisional-radiative recombination coefficient r_c for a magnetically confined optically thin hydrogen plasma at $T_g = 1000^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^{-1}	10^0	10^1	10^2	10^3	10^4
10^8	T_e^*	1.0	1.0	1.0	1.0	1.0	1.0
	T_i^*	1.0	1.0	1.0	1.0	1.0	1.0
	r_c^*	0.42	0.42	0.42	0.42	0.42	0.42
10^9	T_e	1.0	1.0	1.0	1.0	1.0	1.0
	T_i	1.0	1.0	1.0	1.0	1.0	1.0
	r_c	0.72	0.76	0.76	0.76	0.76	0.76
10^{10}	T_e	1.1	1.0	1.0	1.0	1.0	1.0
	T_i	1.1	1.0	1.0	1.0	1.0	1.0
	r_c	1.6	1.9	1.9	1.9	1.9	1.9
10^{11}	T_e	1.2	1.0	1.0	1.0	1.0	1.0
	T_i	1.2	1.0	1.0	1.0	1.0	1.0
	r_c	3.5	6.0	6.9	6.9	6.9	6.9
10^{12}	T_e	1.6	1.2	1.0	1.0	1.0	1.0
	T_i	1.6	1.2	1.0	1.0	1.0	1.0
	r_c	6.3	17	32	38	38	38
10^{13}	T_e	2.3	1.6	1.2	1.1	1.0	1.0
	T_i	2.3	1.6	1.2	1.0	1.0	1.0
	r_c	7.9	35	100	190	250	270
10^{14}	T_e	3.4	2.3	1.7	1.4	1.3	1.2
	T_i	3.4	2.3	1.6	1.1	1.0	1.0
	r_c	9.5	52	220	550	830	1200
10^{15}	T_e	4.9	3.4	2.6	2.2	2.0	1.6
	T_i	4.9	3.3	2.0	1.2	1.0	1.0
	r_c	11	69	260	580	1000	3000

* T_e in (10^3°K); T_i in (10^3°K); r_c in ($10^{-11}\text{cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.15(c)

Collisional-radiative recombination coefficient r_c for a magnetically confined optically thin hydrogen plasma at $T_g = 4000^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^{-1}	10^0	10^1	10^2	10^3	10^4
10^8	T_e^*	4.0	4.0	4.0	4.0	4.0	4.0
	T_i^*	4.0	4.0	4.0	4.0	4.0	4.0
	r_c^*	0.096	0.096	0.096	0.096	0.096	0.096
10^9	T_e	4.0	4.0	4.0	4.0	4.0	4.0
	T_i	4.0	4.0	4.0	4.0	4.0	4.0
	r_c	0.11	0.11	0.11	0.11	0.11	0.11
10^{10}	T_e	4.0	4.0	4.0	4.0	4.0	4.0
	T_i	4.0	4.0	4.0	4.0	4.0	4.0
	r_c	0.14	0.14	0.14	0.14	0.14	0.14
10^{11}	T_e	4.0	4.0	4.0	4.0	4.0	4.0
	T_i	4.0	4.0	4.0	4.0	4.0	4.0
	r_c	0.22	0.22	0.22	0.22	0.22	0.22
10^{12}	T_e	4.0	4.0	4.0	4.0	4.0	4.0
	T_i	4.0	4.0	4.0	4.0	4.0	4.0
	r_c	0.44	0.45	0.45	0.45	0.45	0.45
10^{13}	T_e	4.1	4.0	4.0	4.0	4.0	4.0
	T_i	4.1	4.0	4.0	4.0	4.0	4.0
	r_c	1.1	1.2	1.2	1.2	1.2	1.2
10^{14}	T_e	4.6	4.1	4.0	4.0	4.0	4.0
	T_i	4.6	4.1	4.0	4.0	4.0	4.0
	r_c	3.1	4.8	5.1	5.1	5.1	5.1
10^{15}	T_e	5.8	4.7	4.2	4.1	4.0	4.0
	T_i	5.8	4.6	4.1	4.0	4.0	4.0
	r_c	5.8	13	23	26	26	26

* T_e in (10^3°K); T_i in (10^3°K); r_c in ($10^{-11}\text{cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, et al., Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.16(a)

Upper limit of the collisional-radiative recombination coefficient r_c for a hydrogen plasma optically thick to the Lyman series and not confined by a magnetic field: $T_g = T_i = 250^\circ\text{K}$

	N/n	10^0	10^1	10^2	10^3	10^4
$n(\text{cm}^{-3})$						
10^9	T_e^*	2.5	2.5	2.5	2.5	2.5
	r_c^*	3.6	3.6	3.6	3.6	3.6
10^{10}	T_e	2.8	2.8	2.8	2.8	2.7
	r_c	18	18	18	18	22
10^{11}	T_e	3.8	3.8	3.8	3.7	3.5
	r_c	36	36	36	38	52
10^{12}	T_e	5.7	5.7	5.6	5.5	5.0
	r_c	49	49	50	58	90
10^{13}	T_e	9.2	9.2	9.1	8.9	7.7
	r_c	44	44	46	51	100
10^{14}	T_e	16	15	15	14	12
	r_c	32	33	34	50	130
10^{15}	T_e	26	25	25	22	17
	r_c	24	25	28	50	180

* T_e in (10^2°K); r_c in ($10^{-10}\text{cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.16(b)

Upper limit of the collisional-radiative recombination coefficient r_c for a hydrogen plasma optically thick to the Lyman series and not confined by a magnetic field: $T_g = T_i = 1000^\circ\text{K}$

	N/n	10^0	10^1	10^2	10^3	10^4
$n(\text{cm}^{-3})$						
10^9	T_e^*	10	10	10	10	10
	r_c^*	0.065	0.065	0.065	0.065	0.065
10^{10}	T_e	10	10	10	10	10
	r_c	0.16	0.16	0.16	0.16	0.16
10^{11}	T_e	10	10	10	10	10
	r_c	0.60	0.60	0.60	0.60	0.60
10^{12}	T_e	10	10	10	10	10
	r_c	3.2	3.2	3.2	3.2	3.5
10^{13}	T_e	12	12	12	12	11
	r_c	10	10	10	12	17
10^{14}	T_e	18	18	18	16	14
	r_c	16	16	17	24	51
10^{15}	T_e	27	27	27	24	19
	r_c	16	16	18	32	110

* T_e in (10^2 °K); r_c in (10^{-10} cm³/sec) throughout this table.

Theoretical:

All data taken from Bates, et al., Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.17(a)

Upper limit of collisional-radiative recombination coefficient for an optically thin hydrogen plasma not confined by a magnetic field: $T_g - T_i = 250^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^0	10^1	10^2	10^3	10^4
10^8	T_e^*	2.5	2.5	2.5	2.5	2.5
	r_c^*	0.89	0.89	0.89	0.89	0.89
10^9	T_e	2.5	2.5	2.5	2.5	2.5
	r_c	4.1	4.1	4.1	4.1	4.1
10^{10}	T_e	2.5	2.5	2.5	2.5	2.5
	r_c	28	28	28	28	28
10^{11}	T_e	2.9	2.9	2.9	2.9	2.8
	r_c	140	140	140	140	150
10^{12}	T_e	3.9	3.9	3.9	3.9	3.7
	r_c	290	290	290	300	370
10^{13}	T_e	6.2	6.2	6.2	6.1	5.7
	r_c	320	320	320	330	490
10^{14}	T_e	11	11	11	10	8.8
	r_c	200	200	200	250	560
10^{15}	T_e	20	20	19	18	14
	r_c	100	100	110	180	630

* T_e in (10^2°K); r_c in ($10^{-10} \text{ cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.17(b)

Upper limit of collisional-radiative recombination coefficient for an optically thin hydrogen plasma not confined by a magnetic field: $T_g = T_i = 1000^\circ\text{K}$

$n(\text{cm}^{-3})$	N/n	10^0	10^1	10^2	10^3	10^4
10^8	T_e^*	10	10	10	10	10
	r_c^*	0.043	0.043	0.043	0.043	0.043
10^9	T_e	10	10	10	10	10
	r_c	0.078	0.078	0.078	0.078	0.078
10^{10}	T_e	10	10	10	10	10
	r_c	0.19	0.19	0.19	0.19	0.19
10^{11}	T_e	10	10	10	10	10
	r_c	0.69	0.69	0.69	0.69	0.69
10^{12}	T_e	10	10	10	10	10
	r_c	3.9	3.9	3.9	3.9	3.9
10^{13}	T_e	11	11	11	10	10
	r_c	24	24	24	25	28
10^{14}	T_e	13	13	13	13	12
	r_c	69	69	69	85	130
10^{15}	T_e	22	22	22	20	16
	r_c	65	65	65	100	320

* T_e in (10^2°K); r_c in ($10^{-10} \text{ cm}^3/\text{sec}$) throughout this table.

Theoretical:

All data taken from Bates, *et al.*, Proc. Roy. Soc. (London) Ser. A 279, 10 (1964).

Table 8.18
Theoretical values for dielectronic and radiative recombination coefficients for $N^+(3p)$

$T_e (^{\circ}K)$	$r(10^{-13} \text{ cm}^3/\text{sec})$ dielectronic	$r(10^{-13} \text{ cm}^3/\text{sec})$ radiative
1000	1.0	13
2000	1.6	7.8
3000	1.5	5.6

Theoretical:

All data taken from Bates, Planetary Space Sci. 9, 77 (1962).

Table 8.19
Theoretical values of the collisional-radiative recombination coefficient for the reaction
 $Z^+ + e + N_2 \rightarrow Z + N_2$ when the mass of the ion Z^+ is taken as 28 e.m.u. $T = T_e = T_i = T_g$

$N(\text{cm}^{-3})$	$T(^{\circ}K)$	125	250	500	1000	2000
		$r_c' (\text{cm}^3/\text{sec})$				
1.0×10^{17}		2.8×10^{-10}	7.7×10^{-11}	2.7×10^{-11}	1.2×10^{-11}	5.5×10^{-12}
1.0×10^{18}		2.7×10^{-9}	7.5×10^{-10}	2.6×10^{-10}	1.1×10^{-10}	5.3×10^{-11}
1.0×10^{19}		2.8×10^{-8}	7.6×10^{-9}	2.6×10^{-9}	1.1×10^{-9}	5.3×10^{-10}
1.0×10^{20}		2.8×10^{-7}	7.3×10^{-8}	2.6×10^{-8}	1.1×10^{-8}	5.3×10^{-9}
1.0×10^{21}			7.5×10^{-7}	2.6×10^{-7}	1.1×10^{-7}	5.1×10^{-8}

N is the number density of nitrogen molecules

r_c' is the collisional radiative recombination coefficient less the radiative recombination coefficient.

Theoretical:

All data taken from Bates, et al., Proc. Roy. Soc. (London) Ser A 320, 437 (1971).

Table 8.20

The dissociative recombination coefficient for O_2^+ ions for $T_e = T_i = T_g = 295^\circ K$

$r_2 (10^{-7} \text{ cm}^3 \text{ sec}^{-1})$	1.95 (a)	2.2 (b)	2.1 (c)	1.9±0.5 (d)	1.1 (1)	2.8 (2)
$n (10^{10} \text{ cm}^{-3})$	0.05 to 1	0.02 to 1	0.04 to 10	0.1 to 1		
$N(O_2) (10^{10} \text{ cm}^{-3})$	0.16 to 3.9×10^4	0.98 to 2.6×10^4	6.55×10^4	0.17 to 3.5×10^4		
$N(Ne) (\text{cm}^{-3})$	6.55×10^{17} 5.9×10^{17} 5.9×10^{17}	$\approx 3.28 \times 10^{17}$	-----	-----		
$N(Kr) (\text{cm}^{-3})$	----- 3.28×10^{16} -----	-----	-----	-----		
$N(He) (\text{cm}^{-3})$	-----	3.28×10^{16}	-----	6.55×10^{16}	-----	
$N(Ar) (\text{cm}^{-3})$	-----	-----	-----	3.18×10^{16}	-----	

Experimental:

- (a) Mehr, *et al.*, Phys. Rev., **181**, 264 (1969).
 (b) Kasner, *et al.*, Phys. Rev. **174**, 139 (1968).
 (c) Smith, *et al.*, Planetary Space Sci. **16**, 1177 (1968).
 (d) Mahdavi, *et al.*, J. Phys. B **4**, 1726 (1971).

Theoretical:

- (1) Warke, Phys. Rev. **144**, 120 (1966).
 (2) Chan, J. Chem. Phys. **49**, 2533 (1968).

Table 8.21

Theoretical values of the radiative and dielectronic recombination coefficient for electrons and O^+

$T_e (^\circ K)$	$r (10^{-12} \text{ cm}^3 \text{ sec}^{-1})$ dielectronic (2)	$r (10^{-12} \text{ cm}^3 \text{ sec}^{-1})$ radiative (1)
250		3.69
500		2.39
1000	0.0043	1.48
2000	0.025	0.89
3000	0.034	
4000		0.53
8000		0.33

Theoretical:

- (1) Massey, *et al.*, Rept. Progr. Phys. **9**, 62 (1943).
 (2) Bates, Planetary Space Sci. **9**, 77 (1962).

Table 8.22

Experimental values of the dissociative recombination coefficient r_2 for NO^+ ions at room temperature

$r_2 (10^{-7} \text{ cm}^3 \text{ sec}^{-1})$	$4.6^{+0.5}_{-1.3}$ (a)	$4.1^{+0.3}_{-0.2}$ (b)	3.4 ± 0.6 (c)
$T_e = T_i = T_g (^\circ K)$	298	300	300
$n (10^8 \text{ cm}^{-3})$	~ 35	~ 2 to 20	~ 10 to 100
$N(NO) (\text{cm}^{-3})$	1.9 to 3.9×10^{13}	1.9 to 16×10^{14}	0.28 to 2.8×10^{14}
$N(Ne) (\text{cm}^{-3})$	4.5 to 5.1×10^{18}	7.8 to 27×10^{16}	-----
$N(Ar) (\text{cm}^{-3})$	-----	-----	3.5×10^{16}

Experimental:

- (a) Gunton, *et al.*, Phys. Rev. **140**, A756 (1965).
 (b) Weller, *et al.*, Phys. Rev. **172**, 198 (1968).
 (c) Mahdavi, *et al.*, J. Phys. B **4**, 1726 (1971).

4. Scope and Arrangement of Bibliography

The intention is to include, in the Bibliography, references to published papers which contain original experimental or theoretical data on the eight properties of electron swarms listed in the first paragraph of the Introduction. By our definition of an electron swarm as a low number density electron gas in a neutral gas of considerably greater number density, papers concerned with very highly ionized gases are excluded. The scope has, however, been widened somewhat to include papers concerned with conductivity and energy distributions not only for swarms but also for the region lying between that of swarms and very highly ionized gases, in which interactions of electrons both with neutral and with charged particles are of importance; references to papers concerned with collisional radiative recombination are also included.

The Bibliography is arranged in two parts, sections 5.1 and 5.2; the first lists abbreviated references arranged by type of data, and the second lists full references in order of arbitrary file numbers.

In section 5.1, the abbreviated references consist of the arbitrary file number, the first author and the last two digits of the year of publication. In the case of references to experimental papers an indication of the principle of the method used to obtain the data is also given after the reference, where more than one well-recognized method is available. If only one method has been used, as for example in the case of D_T/μ where the Townsend method is the only one available to date, no method is indicated. Occasionally methods fall outside the broad categories allocated and in these cases also method is omitted. These abbreviated references are ordered as follows:

- A. Coefficient (e.g., drift velocity, diffusion coefficient, etc.)
- B. Experimental or theoretical
- C. Gas

Most of the listings under this third heading are of neutral gases in the following order:

- (1) monatomic gases in order of atomic number;
- (2) diatomic gases with molecules composed of two identical atoms in order of atomic number;
- (3) other molecular gases, except hydrocarbons, grouped in order of the atomic number of the first atom, and ordered within the group in order of the multiplicity of the first atom;
- (4) hydrocarbons grouped under the multiplicity of the carbon atom and ordered within groups in order of multiplicity of the hydrogen atom;
- (5) organic compounds of hydrogen and carbon plus one or more other atoms, the symbols for which do not extend to more than nine characters, which was the field width available;
- (6) organic compounds not containing hydrogen, the symbols for which do not extend to more than nine characters;
- (7) organic; organic compounds, the symbols for which exceed nine characters; in these cases the symbols of the compounds used are given after the titles in section 5.2;
- (8) air; and
- (9) mixture; in this case the mixtures used are indicated after the titles in section 5.2.

In the case of recombination coefficient, however, the nature of the ion involved is indicated in the following cases:

- (a) theoretical papers in which computations are made for a specific ion species, and (b) experimental papers if, and only if, a positive identification of the species is made mass spectrometrically. The ions are listed in order of increasing atomic number immediately following the parent gas (e.g., H_1^+ , H_2^+ , H_3^+ follow H_2 , in that order). Ionization is represented by a superscript which follows the symbol and indicates the degree of ionization (e.g., H^+ , H_2^+ , and H_2^{++} are indicated by H^{+1} , H_2^{+1} and by H_2^{+2} , respectively).
- D. File number.

5. Bibliography

5.1. Abbreviated References, Arranged by Type of Data

Drift Velocity (Experimental)

He	201	Townsend, 23	Magnetic Deflection
	530	Pack, 61	Time of Flight
	701	Stern, 63	
	888	Phelps, 60	Time of Flight
	1237	Hornbeck, 51	Time of Flight
	1387	Nielsen, 36	Time of Flight
	1434	Lowke, 62	Time of Flight
	1489	Crompton, 65	Time of Flight
	1489	Crompton, 65	Magnetic Deflection
	1623	Bowe, 60	Time of Flight
	1714	Levine, 62	Time of Flight
	1730	Levine, 65	Time of Flight
	2133	Wahlin, 26	Time of Flight
	2135	Loeb, 24	Time of Flight
	2164	Anderson, 64	
	2266	Levine, 67	Time of Flight
	2343	Meisel, 65	
	2433	Crompton, 67	Time of Flight
	2433	Crompton, 67	Magnetic Deflection
	3313	Wagner, 67	Time of Flight
	4023	Takayama, 62	
	4055	Grunberg, 69	Time of Flight
	4058	Grunberg, 68	Time of Flight
	4355	Crompton, 70	Time of Flight
	4913	Crompton, 71	Time of Flight
	5960	Errett, 51	
Ne	530	Pack, 61	Time of Flight
	1387	Nielsen, 36	Time of Flight
	1623	Bowe, 60	Time of Flight
	2164	Anderson, 64	
	2399	English, 53	Time of Flight
	4023	Takayama, 62	
	4862	Robertson, 72	Time of Flight
Ar	5780	Sugawara, 70	
	197	Townsend, 22	Magnetic Deflection
	199	Townsend, 22	Magnetic Deflection
	530	Pack, 61	Time of Flight
	732	Bortner, 57	Time of Flight
	1133	Caren, 63	Time of Flight
	1333	Colli, 52	Time of Flight
	1387	Nielsen, 36	Time of Flight
	1565	Christophorou, 65	Time of Flight
	1623	Bowe, 60	Time of Flight
	1752	Wahlin, 31	
	1811	Zelby, 66	

Drift Velocity (Experimental)—Continued

	2042	Kirshner, 52	Time of Flight
	2055	Klema, 50	Time of Flight
	2153	Jager, 62	
	2164	Anderson, 64	
	2261	Nagy, 60	Time of Flight
	2262	Herreng, 42	Time of Flight
	2264	Herreng, 43	Time of Flight
	2343	Meisel, 65	
	2357	Vorobev, 60	Time of Flight
	2399	English, 53	Time of Flight
	2542	Hudson, 46	Time of Flight
	2821	Colli, 51	
	2920	Levine, 64	Time of Flight
	3313	Wagner, 67	Time of Flight
	3480	Konjevic, 68	
	3896	Hearne, 68	
	4058	Grunberg, 68	Time of Flight
	4944	Brambring, 64	Time of Flight
	4971	Wagner, 64	Time of Flight
	5188	Allen, 70	Time of Flight
	5295	Carmichael, 72	
	5960	Errett, 51	
Kr	439	Pack, 62	Time of Flight
	1623	Bowe, 60	Time of Flight
	2399	English, 53	Time of Flight
Xe	439	Pack, 62	Time of Flight
	1623	Bowe, 60	Time of Flight
	2399	English, 53	Time of Flight
	4971	Wagner, 64	Time of Flight
Cs	764	Boeckner, 33	
	1154	Chanin, 64	Time of Flight
Hg	2401	Kerzar, 65	
	2481	Killian, 30	
H ₂	42	Bennett, 42	
	188	Bernstein, 62	
	195	Townsend, 21	Magnetic Deflection
	199	Townsend, 22	Magnetic Deflection
	530	Pack, 61	Time of Flight
	627	Breare, 63	Time of Flight
	1155	Lowke, 63	Time of Flight
	1314	Schlumbohm, 65	Time of Flight
	1381	Bradbury, 36	Time of Flight
	1625	Schlumbohm, 65	Time of Flight
	1964	Hurst, 66	Time of Flight
	1971	Haines, 15	Time of Flight
	2133	Wahlin, 26	Time of Flight
	2135	Loeb, 24	Time of Flight
	2137	Loeb, 22	Time of Flight
	2153	Jager, 62	
	2154	Frommhold, 60	Time of Flight

Drift Velocity (Experimental) — Continued

	2239	Bennett, 40	
	2278	Bailey, 30	
	2279	Bailey, 30	
	2285	Prasad, 67	Time of Flight
	2289	Hall, 55	Magnetic Deflection
	2399	English, 53	Time of Flight
	2489	Crompton, 57	Time of Flight
	2514	Bennett, 42	
	2594	Harris, 34	
	2952	Bradbury, 32	
	2992	Crompton, 68	Time of Flight
	3264	Craig, 53	
	3313	Wagner, 67	Time of Flight
	3382	Grunberg, 67	Time of Flight
	3384	Creaser, 67	Magnetic Deflection
	3459	Blevin, 67	Time of Flight
	3788	Breare, 64	Time of Flight
	4017	Gill, 49	
	4058	Grunberg, 68	Time of Flight
	4484	Crompton, 70	Time of Flight
	4913	Crompton, 71	Time of Flight
	4916	Robertson, 71	Time of Flight
	5103	Bartels, 72	Time of Flight
	5960	Errett, 51	
p - H ₂	2267	Crompton, 67	
	3659	Crompton, 68	Time of Flight
	4913	Crompton, 71	Time of Flight
	4916	Robertson, 71	Time of Flight
D ₂	42	Bennett, 42	
	188	Bernstein, 62	
	439	Pack, 62	Time of Flight
	1428	Hall, 55	Time of Flight
	2039	McIntosh, 66	Time of Flight
	2992	Crompton, 68	Time of Flight
	3384	Creaser, 67	Magnetic Deflection
	4913	Crompton, 71	Time of Flight
N ₂	195	Townsend, 21	Magnetic Deflection
	530	Pack, 61	Time of Flight
	725	Christophorou, 66	Time of Flight
	732	Bortner, 57	Time of Flight
	1155	Lowke, 63	Time of Flight
	1314	Schlumbohm, 65	Time of Flight
	1333	Colli, 52	Time of Flight
	1387	Nielsen, 36	Time of Flight
	1429	Jory, 65	Magnetic Deflection
	1434	Lowke, 62	Time of Flight
	1565	Christophorou, 65	Time of Flight
	1623	Bowe, 60	Time of Flight
	1625	Schlumbohm, 65	Time of Flight
	1716	Tholl, 64	
	1763	Fischer-Treuenfeld, 65	Time of Flight
	1833	Fink, 65	Time of Flight
	1971	Haines, 15	Time of Flight

Drift Velocity (Experimental) — Continued

	2046	Phelps, 59	Time of Flight
	2049	Wahlin, 24	Time of Flight
	2055	Klema, 50	Time of Flight
	2090	Loeb, 22	Time of Flight
	2135	Loeb, 24	Time of Flight
	2136	Wagner, 62	Time of Flight
	2154	Frommhold, 60	Time of Flight
	2164	Anderson, 64	
	2261	Nagy, 60	Time of Flight
	2285	Prasad, 67	Time of Flight
	2343	Meisel, 65	
	2504	Loeb, 21	Time of Flight
	2514	Bennett, 42	
	2529	Wagner, 63	Time of Flight
	2559	Christophorou, 65	Time of Flight
	2920	Levine, 64	Time of Flight
	2952	Bradbury, 32	
	3313	Wagner, 67	Time of Flight
	3382	Grunberg, 67	Time of Flight
	3460	Blevin, 67	Time of Flight
	3662	Hendrick, 68	Time of Flight
	4017	Gill, 49	
	4058	Grunberg, 68	Time of Flight
	4971	Wagner, 64	Time of Flight
	5188	Allen, 70	Time of Flight
	5960	Errett, 51	
O ₂	195	Townsend, 21	Magnetic Deflection
	727	Doehring, 51	Time of Flight
	1314	Schlumbohm, 65	Time of Flight
	1317	Bradbury, 33	
	1336	Herreng, 52	Time of Flight
	1412	Nielsen, 37	Time of Flight
	1625	Schlumbohm, 65	Time of Flight
	2089	Brose, 25	Magnetic Deflection
	2154	Frommhold, 60	Time of Flight
	2370	Frommhold, 64	Time of Flight
	5135	Nelson, 72	Time of Flight
Cl ₂	682	Bailey, 35	
Br ₂	683	Bailey, 37	Magnetic Deflection
I ₂	677	Healey, 38	Magnetic Deflection
HCl	2088	Bailey, 30	Magnetic Deflection
CO	200	Skinker, 23	Magnetic Deflection
	439	Pack, 62	Time of Flight
	1725	Wahlin, 25	Time of Flight
	3313	Wagner, 67	Time of Flight
NO	200	Skinker, 23	Magnetic Deflection
	2385	Bailey, 34	Magnetic Deflection
	4943	Parkes, 72	Time of Flight

Drift Velocity (Experimental)—Continued

H_2O	439	Pack, 62	Time of Flight
	1150	Lowke, 63	Time of Flight
	1964	Hurst, 66	Time of Flight
	2088	Bailey, 30	Magnetic Deflection
	2305	Ryzko, 65	Time of Flight
	5960	Errett, 51	
BF_3	3187	Bistline, 48	Time of Flight
CO_2	198	Skinker, 22	Magnetic Deflection
	439	Pack, 62	Time of Flight
	725	Christophorou, 66	Time of Flight
	732	Bortner, 57	Time of Flight
	1184	Hurst, 63	Time of Flight
	1314	Schlumbohm, 65	Time of Flight
	1625	Schlumbohm, 65	Time of Flight
	1833	Fink, 65	Time of Flight
	2041	Elford, 66	Time of Flight
	2154	Frommhold, 60	Time of Flight
	2399	English, 53	Time of Flight
	2559	Christophorou, 65	Time of Flight
	2920	Levine, 64	Time of Flight
	3313	Wagner, 67	Time of Flight
	4203	Lehning, 68	Time of Flight
NH_3	5188	Allen, 70	Time of Flight
	5206	Hurst, 63	Time of Flight
	5960	Errett, 51	
N_2O	439	Pack, 62	Time of Flight
	1412	Nielsen, 37	Time of Flight
	2088	Bailey, 30	Magnetic Deflection
SiH_4	159	Bailey, 32	Magnetic Deflection
	200	Skinker, 23	Magnetic Deflection
	439	Pack, 62	Time of Flight
	1412	Nielsen, 37	Time of Flight
SiD_4	1917	Cottrell, 65	Time of Flight
	5638	Pollock, 68	
PH_3	1917	Cottrell, 65	Time of Flight
	5638	Pollock, 68	
SF_6	5200	Cottrell, 68	Time of Flight
AsH_3	5119	Naidu, 72	Time of Flight
	5244	Dutton, 71	
CH_4	1917	Cottrell, 65	Time of Flight
	725	Christophorou, 66	Time of Flight
	732	Bortner, 57	Time of Flight
	973	Frommhold, 59	Time of Flight
	1314	Schlumbohm, 65	Time of Flight
	1716	Tholl, 64	

Drift Velocity (Experimental) — Continued

	1833	Fink, 65	Time of Flight
	1917	Cottrell, 65	Time of Flight
	1949	Cookson, 66	Time of Flight
	2156	Franke, 60	Time of Flight
	2269	Bowman, 67	Time of Flight
	2399	English, 53	Time of Flight
	2574	Frommhold, 60	Time of Flight
	3313	Wagner, 67	Time of Flight
	5638	Pollock, 68	
C_2H_2	1917	Cottrell, 65	Time of Flight
	2269	Bowman, 67	Time of Flight
	5200	Cottrell, 68	Time of Flight
C_2H_4	203	Bannon, 28	Magnetic Deflection
	488	Christophorou, 66	Time of Flight
	725	Christophorou, 66	Time of Flight
	732	Bortner, 57	Time of Flight
	1184	Hurst, 63	Time of Flight
	1917	Cottrell, 65	Time of Flight
	1964	Hurst, 66	Time of Flight
	2061	Hamilton, 66	Time of Flight
	2269	Bowman, 67	Time of Flight
	2559	Christophorou, 65	Time of Flight
	3313	Wagner, 67	Time of Flight
	3662	Hendrick, 68	Time of Flight
	5200	Cottrell, 68	Time of Flight
	5206	Hurst, 63	Time of Flight
C_2H_6	1917	Cottrell, 65	Time of Flight
	2269	Bowman, 67	Time of Flight
	3475	Huber, 68	Time of Flight
	4022	Huber, 69	
	5200	Cottrell, 68	Time of Flight
C_3H_6	732	Bortner, 57	Time of Flight
	2269	Bowman, 67	Time of Flight
C_3H_8	725	Christophorou, 66	Time of Flight
	1917	Cottrell, 65	Time of Flight
	2559	Christophorou, 65	Time of Flight
	4022	Huber, 69	
	5200	Cottrell, 68	Time of Flight
C_4H_8	2269	Bowman, 67	Time of Flight
C_4H_{10}	725	Christophorou, 66	Time of Flight
	2559	Christophorou, 65	Time of Flight
C_5H_{12}	725	Christophorou, 66	Time of Flight
	2284	McGee, 28	Magnetic Deflection
	2559	Christophorou, 65	Time of Flight
C_6H_6	725	Christophorou, 66	Time of Flight
	1314	Schlumbohm, 65	Time of Flight
	2559	Christophorou, 65	Time of Flight

Drift Velocity (Experimental) — Continued

CD ₄	1917 5638	Cottrell, 65 Pollock, 68	Time of Flight
CH ₃ Cl	1917 5200	Cottrell, 65 Cottrell, 68	Time of Flight Time of Flight
CH ₃ OH	5200	Cottrell, 68	Time of Flight
C ₂ D ₂	1917	Cottrell, 65	Time of Flight
C ₂ H ₆ O	1917	Cottrell, 65	Time of Flight
C ₂ H ₅ Cl	487	Christophorou, 66	Time of Flight
C ₂ H ₅ OH	1917	Cottrell, 65	Time of Flight
(C ₂ H ₅) ₂ O	1314 2154	Schlumbohm, 65 Frommhold, 60	Time of Flight Time of Flight
o -C ₆ H ₄ Cl ₂	487	Christophorou, 66	Time of Flight
m -C ₆ H ₄ Cl ₂	487	Christophorou, 66	Time of Flight
p -C ₆ H ₄ Cl ₂	487	Christophorou, 66	Time of Flight
C ₆ H ₅ Cl	487	Christophorou, 66	Time of Flight
C ₆ H ₅ Br	487	Christophorou, 66	Time of Flight
C ₆ D ₅ Br	487	Christophorou, 66	Time of Flight
CF ₄	5051	Naidu, 72	Time of Flight
CCl ₂ F ₂	5183	Naidu, 69	Time of Flight
C ₂ F ₆	5051	Naidu, 72	Time of Flight
C ₃ F ₈	5051 5123	Naidu, 72 Prasad, 70	Time of Flight Time of Flight
C ₄ F ₁₀	5051	Naidu, 72	Time of Flight
Organic	487 1314 1565 1917 5052	Christophorou, 66 Schlumbohm, 65 Christophorou, 65 Cottrell, 65 Naidu, 72	Time of Flight Time of Flight Time of Flight Time of Flight Time of Flight
Air	1317 1412 2054 2104 2263 2305	Bradbury, 33 Nielsen, 37 Huxley, 49 Townsend, 13 Loeb, 23 Ryzko, 65	Time of Flight Magnetic Deflection Magnetic Deflection Time of Flight Time of Flight

Drift Velocity (Experimental) — Continued

	2370	Frommhold, 64	Time of Flight
	2594	Harris, 34	
	2820	Deb, 56	
	3165	Hessenauer, 67	Time of Flight
Mixture	42	Bennett, 42	
	199	Townsend, 22	Magnetic Deflection
	488	Christophorou, 66	Time of Flight
	682	Bailey, 35	
	725	Christophorou, 66	Time of Flight
	732	Bortner, 57	Time of Flight
	1016	Hurst, 61	Time of Flight
	1150	Lowke, 63	Time of Flight
	1155	Lowke, 63	Time of Flight
	1333	Colli, 52	Time of Flight
	1379	Hurst, 63	Time of Flight
	1467	Nolan, 65	Time of Flight
	1716	Tholl, 64	
	2042	Kirshner, 52	Time of Flight
	2055	Klema, 50	Time of Flight
	2061	Hamilton, 66	Time of Flight
	2141	Bortner, 58	Time of Flight
	2151	Stevenson, 52	Time of Flight
	2155	Dibbern, 61	Time of Flight
	2261	Nagy, 60	Time of Flight
	2280	Bailey, 24	Magnetic Deflection
	2305	Ryzko, 65	Time of Flight
	2399	English, 53	Time of Flight
	2459	Fischer, 66	
	2514	Bennett, 42	
	2532	Compton, 66	Time of Flight
	2559	Christophorou, 65	Time of Flight
	2581	Blair, 62	Time of Flight
	2582	Dibbern, 62	Time of Flight
	2821	Colli, 51	
	2883	Townsend, 22	Magnetic Deflection
	2920	Levine, 64	Time of Flight
	2987	Colli, 53	Time of Flight
	3165	Hessenauer, 67	Time of Flight
	3550	Den Hartog, 49	Time of Flight
	3551	Den Hartog, 47	Time of Flight
	4018	Riemann, 44	
	4022	Huher, 69	
	4901	Fleming, 72	Time of Flight
	5960	Errett, 51	

Drift Velocity (Theoretical)

2187	Huxley, 51	3275	Dallenbach, 66
2198	Sodha, 59	3467	Cavalleri, 68
2201	Compton, 23	3592	Lucas, 67
2256	Huxley, 60	3655	Hassan, 68
2338	Huxley, 37	3711	Wilhelm, 67
2339	Gvosdover, 37	3720	Hertz, 24
2765	Bailey, 37		
2806	Pearson, 63	He	273 Allen, 37
2877	Ritchie, 67		397 Barbieri, 51

Drift Velocity (Theoretical) — Continued

	1062	Frost, 64		4450	Gibson, 70
	1440	Heylen, 63		3463	Bell, 68
	2202	Compton, 23			
	2812	Llewellyn-Jones, 44	D ₂	260	Engelhardt, 63
	3463	Bell, 68		4450	Gibson, 70
	3714	Elkomoss, 68			
	3897	Itoh, 60	N ₂	181	Frost, 62
	4053	Pfau, 69		218	Engelhardt, 64
	4057	Bortnik, 65		1043	Frost, 62
	4909	Hughes, 70		1447	Heylen, 62
	5490	Englert, 71		1950	Heylen, 66
				2202	Compton, 23
Ne	273	Allen, 37		3463	Bell, 68
	1440	Heylen, 63			
	1745	Salmon, 63	O ₂	1447	Heylen, 62
	2351	Druyvesteyn, 37		1950	Heylen, 66
	4053	Pfau, 69		2202	Compton, 23
	4909	Hughes, 70		2553	Hake, 67
Ar	273	Allen, 37	CO	1446	Wahlin, 30
	292	Engelhardt, 64		2553	Hake, 67
	397	Barbiere, 51			
	881	Golant, 59	CO ₂	2202	Compton, 23
	1062	Frost, 64		2553	Hake, 67
	1440	Heylen, 63			
	2202	Compton, 23	SiH ₄	5638	Pollock, 68
	2397	Uman, 60			
	4053	Pfau, 69	SiD ₄	5638	Pollock, 68
	5126	Sakai, 72			
			CH ₄	5638	Pollock, 68
Kr	1062	Frost, 64			
	4053	Pfau, 69	C ₂ H ₂	2142	Heylen, 62
Xe	1062	Frost, 64	C ₂ H ₄	2142	Heylen, 62
	4053	Pfau, 69			
			C ₂ H ₆	1950	Heylen, 66
Cs	3305	Postma, 68		2142	Heylen, 62
H ₂	181	Frost, 62	CD ₄	5638	Pollock, 68
	260	Engelhardt, 63			
	1043	Frost, 62	Air	1447	Heylen, 62
	1636	Heylen, 60		1950	Heylen, 66
	1638	Heylen, 60			
	1783	Heylen, 65	Mixture	292	Engelhardt, 64
	1950	Heylen, 66		2397	Uman, 60
	2202	Compton, 23		2921	Uman, 61
	2352	Kihara, 52		4909	Hughes, 70

(Diffusion Coefficient)/Mobility (Experimental)

He	201	Townsend, 23	Ar	197	Townsend, 22
	1036	Warren, 62		199	Townsend, 22
	1489	Crompton, 65			
	2433	Crompton, 67		1036	Warren, 62

(Diffusion Coefficient)/Mobility (Experimental) — Continued

H ₂	195	Townsend, 21	1036	Warren, 62
	199	Townsend, 22		
	352	Crompton, 52	NO	200 Skinker, 23
	689	Crompton, 63		2385 Bailey, 34
	736	Cochran, 62		
	841	Varnerin, 50	H ₂ O	1618 Crompton, 65
	1036	Warren, 62		2088 Bailey, 30
	1264	Lawson, 65		2463 Crompton, 66
	1437	Crompton, 66		
	1438	Crompton, 62	CO ₂	198 Skinker, 22
	1622	Crompton, 65		736 Cochran, 62
	1625	Schlumbohm, 65		1036 Warren, 62
	1778	Lawson, 65		1625 Schlumbohm, 65
	2110	Huxley, 55		2140 Rees, 64
	2992	Crompton, 68		5230 Virr, 70
	3303	Naidu, 68		
	5050	Virr, 72	NH ₃	2088 Bailey, 30
	5230	Virr, 70		2567 Bailey, 29
	5234	Kontoleon, 71		
p-H ₂	2267	Crompton, 67	N ₂ O	159 Bailey, 32
	3659	Crompton, 68		200 Skinker, 23
D ₂	1036	Warren, 62	SiH ₄	1917 Cottrell, 65
	1428	Hall, 55		2990 Cottrell, 67
	2039	McIntosh, 66		5638 Pollock, 68
	2992	Crompton, 68		
N ₂	195	Townsend, 21	SiD ₄	1917 Cottrell, 65
	352	Crompton, 52		2990 Cottrell, 67
	689	Crompton, 63		5638 Pollock, 68
	736	Cochran, 62	SF ₆	5119 Naidu, 72
	1036	Warren, 62		
	1429	Jory, 65	CH ₄	736 Cochran, 62
	1625	Schlumbohm, 65		1326 Schlumbohm, 65
	3303	Naidu, 68		1917 Cottrell, 65
	5230	Virr, 70		2990 Cottrell, 67
O ₂	195	Townsend, 21		3715 Brose, 35
	649	Huxley, 59		5638 Pollock, 68
	1269	Rees, 65	C ₂ H ₂	1917 Cottrell, 65
	1625	Schlumbohm, 65		
	2089	Brose, 25	C ₂ H ₄	203 Bannon, 28
Cl ₂	5226	Naidu, 70		736 Cochran, 62
			C ₂ H ₆	1917 Cottrell, 65
Br ₂	682	Bailey, 35		2990 Cottrell, 67
	683	Bailey, 37	C ₃ H ₆	736 Cochran, 62
I ₂				
	677	Healey, 38	C ₃ H ₈	2990 Cottrell, 67
HCl	2088	Bailey, 30	C ₅ H ₁₂	2284 McGee, 28
	2567	Bailey, 29		
CO	200	Skinker, 23	C ₆ H ₆	1625 Schlumbohm, 65

(Diffusion Coefficient)/Mobility (Experimental) — Continued

CD ₄	2990	Cottrell, 67	Organic	1625	Schlumbohm, 65
	5638	Pollock, 68		1917	Cottrell, 65
				5052	Naidu, 72
C ₂ D ₂	1917	Cottrell, 65			
C ₂ H ₆ O	1625	Schlumbohm, 65	Air	1332	Bailey, 25
	1917	Cottrell, 65		2054	Huxley, 49
	2990	Cottrell, 67		2097	Crompton, 53
				2104	Townsend, 13
CF ₄	5051	Naidu, 72		2813	Rees, 64
				5247	Rao, 71
CCl ₂ F ₂	5183	Naidu, 69			
C ₂ F ₆	5051	Naidu, 72	Mixture	199	Townsend, 22
				677	Healey, 38
C ₃ F ₈	5051	Naidu, 72		682	Bailey, 35
	5123	Prasad, 70		683	Bailey, 37
				2280	Bailey, 24
C ₄ F ₁₀	5051	Naidu, 72		4901	Fleming, 72

(Diffusion Coefficient)/Mobility (Theoretical)

	2766	Townsend, 37		4450	Gibson, 70
	3484	Liley, 67			
He			D ₂	260	Engelhardt, 63
	273	Allen, 37		4052	Lowke, 69
	1062	Frost, 64		4450	Gibson, 70
	2250	Orient, 66	N ₂	181	Frost, 62
	4052	Lowke, 69		218	Engelhardt, 64
	4053	Pfau, 69		1043	Frost, 62
Ne	4909	Hughes, 70		4052	Lowke, 69
	273	Allen, 37	O ₂	5233	Kline, 71
	2250	Orient, 66		2553	Hake, 67
	4053	Pfau, 69		4052	Lowke, 69
Ar	4909	Hughes, 70	CO	2553	Hake, 67
	273	Allen, 37		4052	Lowke, 69
	292	Engelhardt, 64	H ₂ O	4052	Lowke, 69
	1062	Frost, 64			
	4052	Lowke, 69			
Kr	4053	Pfau, 69	CO ₂	2553	Hake, 67
	1062	Frost, 64		4052	Lowke, 69
	4052	Lowke, 69	SiH ₄	5638	Pollock, 68
	4053	Pfau, 69			
Xe	1062	Frost, 64		5638	Pollock, 68
	4052	Lowke, 69	SiD ₄	5638	Pollock, 68
	4053	Pfau, 69			
H ₂			CH ₄	5638	Pollock, 68
	181	Frost, 62	CD ₄	5638	Pollock, 68
	260	Engelhardt, 63			
	1043	Frost, 62	Mixture	292	Engelhardt, 64
	4052	Lowke, 69		4909	Hughes, 70

Diffusion Coefficient (Experimental)			
He	3313	Wagner, 67	Longitudinal
	4048	Nelson, 69	
	4049	Cavalleri, 69	
Ne	1126	Cavalleri, 64	
	4048	Nelson, 69	
	4050	Cavalleri, 65	
Ar	3313	Wagner, 67	Longitudinal
	4048	Nelson, 69	
H ₂	188	Bernstein, 62	Longitudinal
	1964	Hurst, 66	
	2295	Madan, 57	
	3313	Wagner, 67	Longitudinal
	3788	Breare, 64	Longitudinal
	4048	Nelson, 69	
D ₂	188	Bernstein, 62	
N ₂	1833	Fink, 65	Longitudinal
	3313	Wagner, 67	Longitudinal
	4048	Nelson, 69	
O ₂	5135	Nelson, 72	
CO	3313	Wagner, 67	Longitudinal
	4048	Nelson, 69	
H ₂ O	1964	Hurst, 66	Longitudinal
CO ₂	1184	Hurst, 63	Longitudinal
	1833	Fink, 65	Longitudinal
	3313	Wagner, 67	Longitudinal
	4048	Nelson, 69	
	5206	Hurst, 63	Longitudinal
CH ₄	973	Frommhold, 59	Longitudinal
	1833	Fink, 65	Longitudinal
	2574	Frommhold, 60	Longitudinal
	3313	Wagner, 67	Longitudinal
	4048	Nelson, 69	
C ₂ H ₄	1184	Hurst, 63	Longitudinal
	1964	Hurst, 66	Longitudinal
	3313	Wagner, 67	Longitudinal
	4048	Nelson, 69	
	5206	Hurst, 63	Longitudinal

Diffusion Coefficient (Theoretical)

	2198	Sodha, 59		4063	Lloyd, 60
	2256	Huxley, 60			
	2437	Shkarofsky, 61	Ar	1440	Heylen, 63
	2877	Ritchie, 67		2438	Tan, 65
	3592	Lucas, 67		3798	Devoto, 67
	3711	Wilhelm, 67		4053	Pfau, 69
	3719	Townsend, 38			
	3785	Van Leeuwen, 67	Kr	4053	Pfau, 69
	3796	Devoto, 66			
	3800	Golant, 61	Xe	4053	Pfau, 69
	4194	Weijland, 68			
			H ₂	3463	Bell, 68
He	1440	Heylen, 63			
	3463	Bell, 68	N ₂	3463	Bell, 68
	4053	Pfau, 69			
	5490	Englert, 71	C ₂ H ₂	2142	Heylen, 62
			C ₂ H ₄	2142	Heylen, 62
Ne	1440	Heylen, 63			
	4053	Pfau, 69	C ₂ H ₆	2142	Heylen, 62

Attachment Coefficient (Experimental)

Br	4086	Razzak, 69	Drift Tube, Spatial
H ₂	1240	Whitmer, 56	Plasma Decay
	5500	Cronin, 70	Plasma Decay
O ₂	131	Chanin, 59	Drift Tube, Temporal
	573	Dutton, 63	Drift Tube, Spatial
	649	Huxley, 59	Drift Tube, Spatial
	711	Sexton, 60	Plasma Decay
	727	Doehring, 51	Drift Tube, Temporal
	762	Brodskii, 66	Plasma Decay
	791	Harrison, 53	Drift Tube, Spatial
	861	Prasad, 61	Drift Tube, Spatial
	961	Freely, 64	Drift Tube, Spatial
	1006	Chatterton, 61	Drift Tube, Spatial
	1269	Rees, 65	Drift Tube, Spatial
	1315	Geballe, 52	Drift Tube, Spatial
	1317	Bradbury, 33	Drift Tube, Spatial
	1322	Burch, 57	Drift Tube, Temporal
	1336	Herreng, 52	Drift Tube, Temporal
	1383	Chanin, 62	Drift Tube, Temporal
	1433	Kuffel, 59	Drift Tube, Temporal
	1444	Frommhold, 64	Drift Tube, Temporal
	1445	Schlumbohm, 59	Drift Tube, Temporal
	1565	Christophorou, 65	Drift Tube, Temporal
	1601	Anisimov, 64	Plasma Decay
	1633	Hackam, 65	Plasma Decay
	1662	Pack, 66	Drift Tube, Temporal
	1724	Schlumbohm, 62	Drift Tube, Temporal
	2093	Cravath, 29	Drift Tube, Spatial
	2370	Frommhold, 64	Drift Tube, Temporal
	2387	Schlumbohm, 60	Drift Tube, Temporal

Attachment Coefficient (Experimental) — Continued

	2461	Taylor, 66	Plasma Decay
	2462	Prasad, 66	Drift Tube, Spatial
	2492	De Bitetto, 58	Drift Tube, Spatial
	2555	Sukhum, 67	Drift Tube, Spatial
	2572	Dubs, 60	
	3154	Phelps, 61	Drift Tube, Temporal
	3495	Margenau, 46	Plasma Decay
	3647	Stein, 68	Plasma Decay
	4043	Thomas-Betts, 69	Drift Tube, Spatial
	4095	Biondi, 51	Plasma Decay
	4096	Holt, 59	Plasma Decay
	4919	Grunberg, 69	Drift Tube, Temporal
	5047	Spence, 72	
	5125	Warman, 71	Plasma Decay
	5131	Truby, 72	Plasma Decay
	5226	Naidu, 70	
	5227	Kinsman, 70	
	5770	Hirsh, 69	Plasma Decay
Cl ₂	682	Bailey, 35	Drift Tube, Spatial
	2099	Bradbury, 34	Drift Tube, Spatial
	2158	Wahlin, 22	
	2224	Bozin, 67	Drift Tube, Spatial
Br ₂	683	Bailey, 37	Drift Tube, Spatial
I ₂	541	Biondi, 58	Plasma Decay
	677	Healey, 38	Drift Tube, Spatial
	4108	Truby, 69	Plasma Decay
HCl	2088	Bailey, 30	Drift Tube, Spatial
	2099	Bradbury, 34	Drift Tube, Spatial
	2567	Bailey, 29	Drift Tube, Spatial
	3007	Christophorou, 68	Drift Tube, Temporal
HBr	3007	Christophorou, 68	Drift Tube, Temporal
HI	3007	Christophorou, 68	Drift Tube, Temporal
DCI	3007	Christophorou, 68	Drift Tube, Temporal
DBr	3007	Christophorou, 68	Drift Tube, Temporal
DI	3007	Christophorou, 68	Drift Tube, Temporal
CO	1254	Bhalla, 61	Drift Tube, Spatial
	2099	Bradbury, 34	Drift Tube, Spatial
	2158	Wahlin, 22	
	2408	Chatterton, 65	Drift Tube, Spatial
	4016	Davies, 69	Drift Tube, Spatial
	5243	Parr, 71	Drift Tube, Spatial
NO	658	Gunton, 63	Plasma Decay
	1181	Gunton, 61	Plasma Decay
	1608	Gunton, 65	Plasma Decay

Attachment Coefficient (Experimental) — Continued

	2099	Bradbury, 34	Drift Tube, Spatial
	2455	Biondi, 66	Plasma Decay
	3229	Mentzoni, 67	Plasma Decay
	3610	Weller, 68	Plasma Decay
H ₂ O	1433	Kuffel, 59	Drift Tube, Temporal
	1618	Crompton, 65	Drift Tube, Spatial
	1911	Ryzko, 66	Drift Tube, Temporal
	2088	Bailey, 30	Drift Tube, Spatial
	2093	Cravath, 29	Drift Tube, Spatial
	2100	Bradbury, 34	Drift Tube, Spatial
	2294	Prasad, 60	Drift Tube, Spatial
	2463	Crompton, 66	Drift Tube, Spatial
	5248	Parr, 72	Drift Tube, Temporal
H ₂ S	2100	Bradbury, 34	Drift Tube, Spatial
CO ₂	948	Bhalla, 60	Drift Tube, Spatial
	1724	Schlumbohm, 62	Drift Tube, Temporal
	2408	Chatterton, 65	Drift Tube, Spatial
	3435	Conti, 67	Drift Tube, Spatial
	4843	Baker, 70	Drift Tube, Spatial
	5290	Bouby, 71	Drift Tube, Spatial
NH ₃	2088	Bailey, 30	Drift Tube, Spatial
	2099	Bradbury, 34	Drift Tube, Spatial
	2158	Wahlin, 22	
	2273	Bailey, 28	Drift Tube, Spatial
	2567	Bailey, 29	Drift Tube, Spatial
	5248	Parr, 72	Drift Tube, Temporal
NO ₂	1649	Hasted, 65	Plasma Decay
N ₂ O	159	Bailey, 32	Drift Tube, Spatial
	2100	Bradbury, 34	Drift Tube, Spatial
	3680	Phelps, 68	Drift Tube, Temporal
SO ₂	1724	Schlumbohm, 62	Drift Tube, Temporal
	2100	Bradbury, 34	Drift Tube, Spatial
	5290	Bouby, 71	Drift Tube, Spatial
SF ₆	788	McAfee, 63	Drift Tube, Temporal
	1198	Compton, 66	Drift Tube, Temporal
	1260	Bhalla, 62	Drift Tube, Spatial
	1274	McAfee, 55	Drift Tube, Temporal
	1649	Hasted, 65	Plasma Decay
	1724	Schlumbohm, 62	Drift Tube, Temporal
	1923	Mahan, 66	Plasma Decay
	2544	Edelson, 64	Drift Tube, Temporal
	3662	Hendrick, 68	Drift Tube, Temporal
	4195	Chen, 68	
	5053	Boyd, 71	Drift Tube, Spatial
C ₂ H ₂	2158	Wahlin, 22	

Attachment Coefficient (Experimental) — Continued

C_2H_4	2158	Wahlin, 22	Drift Tube, Spatial
	5290	Bouby, 71	
C_2H_6	2158	Wahlin, 22	
C_5H_{12}	2284	McGee, 28	Drift Tube, Spatial
$C_{10}H_8$	4099	Chaney, 70	Drift Tube, Temporal
$CHCl_3$	4190	Blaunstein, 68	Drift Tube, Temporal
CH_2Cl_2	4190	Blaunstein, 68	Drift Tube, Temporal
CH_3Br	3600	Razzak, 68	Drift Tube, Spatial
C_2H_5Cl	2158	Wahlin, 22	Drift Tube, Temporal
	487	Christophorou, 66	
C_2HCl_3	4190	Blaunstein, 68	Drift Tube, Temporal
$C_2H_3Cl_3$	4190	Blaunstein, 68	Drift Tube, Temporal
<i>o</i> - $C_6H_4Cl_2$	487	Christophorou, 66	Drift Tube, Temporal
<i>m</i> - $C_6H_4Cl_2$	487	Christophorou, 66	Drift Tube, Temporal
<i>p</i> - $C_6H_4Cl_2$	487	Christophorou, 66	Drift Tube, Temporal
C_6H_5Cl	487	Christophorou, 66	Drift Tube, Temporal
C_6H_5Br	487	Christophorou, 66	Drift Tube, Temporal
C_6D_5Br	487	Christophorou, 66	Drift Tube, Temporal
$C_6H_5NO_2$	1198	Compton, 66	Drift Tube, Temporal
CF_4	3465	Bozin, 68	Drift Tube, Spatial
	5051	Naidu, 72	
CCl_4	4190	Blaunstein, 68	Drift Tube, Temporal
CCl_2F_2	791	Harrison, 53	Drift Tube, Spatial
	1724	Schlumbohm, 62	Drift Tube, Temporal
	2345	Moruzzi, 63	Drift Tube, Spatial
	2449	Moruzzi, 63	Drift Tube, Spatial
	5229	Boyd, 70	Drift Tube, Spatial
CF_3SF_5	791	Harrison, 53	Drift Tube, Spatial
C_2F_6	3465	Bozin, 68	Drift Tube, Spatial
	5051	Naidu, 72	
C_3F_8	2302	Moruzzi, 63	Drift Tube, Spatial
	5051	Naidu, 72	
	5123	Prasad, 70	

Attachment Coefficient (Experimental)—Continued

C_4F_{10}	3604	Razzak, 68	
	5051	Naidu, 72	
C_7F_{14}	1923	Mahan, 66	Plasma Decay
	4195	Chen, 68	
Organic	487	Christophorou, 66	Drift Tube, Temporal
	1565	Christophorou, 65	Drift Tube, Temporal
	3712	Blaunstein, 68	Drift Tube, Temporal
	4185	Christophorou, 69	Drift Tube, Temporal
	4187	Wentworth, 66	
	5052	Naidu, 72	
Air	762	Brodskii, 66	Plasma Decay
	791	Harrison, 53	Drift Tube, Spatial
	1317	Bradbury, 33	Drift Tube, Spatial
	1332	Bailey, 25	Drift Tube, Spatial
	1433	Kuffel, 59	Drift Tube, Temporal
	1444	Frommhold, 64	Drift Tube, Temporal
	1911	Ryzko, 66	Drift Tube, Temporal
	2047	Prasad, 59	Drift Tube, Spatial
	2149	Kuffel, 58	Drift Tube, Spatial
	2235	Ryzko, 67	Drift Tube, Temporal
	2370	Frommhold, 64	Drift Tube, Temporal
	2388	Dutton, 60	Drift Tube, Spatial
	2407	Dutton, 63	Drift Tube, Spatial
	2408	Chatterton, 65	Drift Tube, Spatial
	2488	Dutton, 63	Drift Tube, Spatial
	2528	Dutton, 63	Drift Tube, Spatial
	2953	Bailey, 32	Drift Tube, Spatial
	3165	Hessenauer, 67	Drift Tube, Temporal
Mixture	682	Bailey, 35	Drift Tube, Spatial
	738	Pack, 66	Drift Tube, Temporal
	956	Stockdale, 64	Drift Tube, Temporal
	961	Freely, 64	Drift Tube, Spatial
	1016	Hurst, 61	Drift Tube, Temporal
	1267	Hurst, 59	Drift Tube, Temporal
	1317	Bradbury, 33	Drift Tube, Spatial
	1379	Hurst, 63	
	1433	Kuffel, 59	Drift Tube, Temporal
	1633	Hackam, 65	Plasma Decay
	1649	Hasted, 65	Plasma Decay
	1923	Mahan, 66	Plasma Decay
	2093	Cravath, 29	Drift Tube, Spatial
	2099	Bradbury, 34	Drift Tube, Spatial
	2100	Bradbury, 34	Drift Tube, Spatial
	2141	Bortner, 58	Drift Tube, Temporal
	2294	Prasad, 60	Drift Tube, Spatial
	2407	Dutton, 63	Drift Tube, Spatial
	2408	Chatterton, 65	Drift Tube, Spatial
	2467	Ryzko, 66	Drift Tube, Temporal
	2532	Compton, 66	Drift Tube, Temporal
	2995	Stockdale, 67	Drift Tube, Temporal
	3165	Hessenauer, 67	Drift Tube, Temporal
	3229	Mentzoni, 67	Plasma Decay

Attachment Coefficient (Experimental) — Continued

3495	Margenau, 46	Plasma Decay
3595	Mahan, 67	Plasma Decay
3607	Truby, 68	Plasma Decay
3901	Lee, 63	Drift Tube, Temporal
4054	Christophorou, 70	Drift Tube, Temporal
4097	Lennon, 61	Plasma Decay
4192	Bouby, 65	Drift Tube, Spatial
4311	Ghosh, 67	Plasma Decay
5046	McCorkle, 72	Drift Tube, Temporal
5236	Davis, 69	
5237	Davis, 70	
5238	Christophorou, 69	
5770	Hirsh, 69	Plasma Decay

Attachment Coefficient (Theoretical)

H	1603	Dalgarno, 63		2553	Hake, 67
				4947	Wagner, 71
I ₂	4047	Shipsey, 70			
O	1180	Massey, 43	CO ₂	2553	Hake, 67
			Air	337	Tozer, 58
O ₂	337	Tozer, 58		4947	Wagner, 71
	1285	Bloch, 35			
	2086	Penning, 38	Mixture	2452	Golant, 57

Detachment Coefficient (Experimental)

O ₂	1444	Frommhold, 64	Drift Tube, Temporal
	1633	Hackam, 65	Plasma Decay
	1662	Pack, 66	Drift Tube, Temporal
	2370	Frommhold, 64	Drift Tube, Temporal
	2462	Prasad, 66	Drift Tube, Spatial
	2555	Sukhum, 67	Drift Tube, Spatial
	2994	Eccles, 67	Drift Tube, Spatial
	3154	Phelps, 61	Drift Tube, Temporal
	4920	Eccles, 70	Drift Tube, Spatial
	4949	Brabanec, 72	Drift Tube, Spatial
SF ₆	3472	Eccles, 67	Drift Tube, Spatial
Organic	4185	Christophorou, 69	Drift Tube, Temporal
Air	1444	Frommhold, 64	Drift Tube, Temporal
	2235	Ryzko, 67	Drift Tube, Temporal
	2370	Frommhold, 64	Drift Tube, Temporal
Mixture	1633	Hackam, 65	Plasma Decay
	3042	Fehsenfeld, 66	
	3352	Moruzzi, 68	Drift Tube, Temporal
	4311	Ghosh, 67	Plasma Decay
	4312	Malliaris, 68	
	4313	Johnston, 68	
	5700	Moruzzi, 66	

Detachment Coefficient (Theoretical)

H	3652	Dalgarno, 67	4948	Prasad, 65
O ₂	4947	Wagner, 71	Air	4947 Wagner, 71

Excitation Coefficient (Experimental)

C ⁺⁺	3656	Kunze, 68	Radiation
He	336	Corrigan, 58	
H ₂	354	Corrigan, 58	Radiation
	921	Legler, 63	Radiation
	1761	Nygaard, 65	Radiation
	2229	Nygaard, 63	Radiation
	2502	Lunt, 37	Radiation
	3111	Von Engel, 56	Radiation
	3240	Geballe, 44	Radiation
	3722	Nygaard, 65	Radiation
	4950	Pool, 37	
	4951	Shaw, 59	
N ₂	921	Legler, 63	Radiation
	4993	Teich, 72	
O ₂	3267	Teich, 67	
	3268	Teich, 67	
CO ₂	3267	Teich, 67	
	4993	Teich, 72	
	5607	Wiegand, 70	
	5702	Kutszegi, 69	
SF ₆	4993	Teich, 72	
Air	3267	Teich, 67	
Mixture	3267	Teich, 67	

Excitation Coefficient (Theoretical)

He	3897	Itoh, 60	Hg	3270	Cayless, 59
	4057	Bortnik, 65			
	4909	Hughes, 70	H ₂	260	Engelhardt, 63
	5701	Phelps, 59		2109	Lunt, 36
Ne	2501	Glotov, 38	D ₂	260	Engelhardt, 63
	3431	Kagan, 63	N ₂	218	Engelhardt, 64
	4909	Hughes, 70			
Ar	3432	Kagan, 63	Mixture	2501	Glotov, 38
				4909	Hughes, 70

Ionization Coefficient (Experimental)

He	619	Townsend, 34	Townsend
	646	Chanin, 64	Townsend
	900	Davies, 63	Townsend
	1160	Fletcher, 63	Townsend
	1173	Davies, 62	Townsend
	1250	Dutton, 63	Townsend
	2286	Dutton, 67	Townsend
	2406	Dutton, 65	Townsend
	2435	Burkley, 67	
	2496	Townsend, 31	
	2810	Townsend, 24	Townsend
	2816	Gill, 08	Townsend
	3171	Gill, 12	Townsend
	3272	Townsend, 28	
	4049	Cavalleri, 69	
	4910	Dutton, 71	Townsend
	5077	Dutton, 72	Townsend
	5771	Jones, 69	
C ⁺⁺	3656	Kunze, 68	
Ne	621	Davies, 60	Townsend
	730	Townsend, 28	Townsend
	751	Chanin, 63	Townsend
	1160	Fletcher, 63	Townsend
	2138	Davies, 59	Townsend
	2237	Kruithof, 40	Townsend
	2341	Glotov, 37	Townsend
	2355	Kruithof, 37	Townsend
	2403	Townsend, 28	Townsend
	2496	Townsend, 31	
	2535	De Hoog, 67	
	3272	Townsend, 28	
	3436	De Hoog, 67	
	3612	Willis, 68	Townsend
	3751	Dutton, 69	Townsend
	5049	Buursen, 72	
Ar	621	Davies, 60	Townsend
	624	Ayres, 23	Townsend
	2108	Engstrom, 40	
	2138	Davies, 59	Townsend
	2139	Kruithof, 36	Townsend
	2237	Kruithof, 40	Townsend
	2333	Golden, 61	Townsend
	2346	Huxford, 39	
	2355	Kruithof, 37	Townsend
	2435	Burkley, 67	
	2587	Golden, 62	Townsend
	2816	Gill, 08	Townsend
	3477	Heylen, 68	Townsend
	3478	Heylen, 68	Townsend
	3908	Scharfman, 64	A C Breakdown
	3956	Yamane, 60	

Ionization Coefficient (Experimental) — Continued

	5241	Willis, 71	Townsend
	5295	Carmichael, 72	
Br	4086	Razzak, 69	Townsend
Kr	2237	Kruithof, 40	Townsend
	5293	Heylen, 71	Townsend
Xe	2237	Kruithof, 40	Townsend
	2435	Burkley, 67	
Cs	5202	Davies, 70	Townsend
Hg	898	Smith, 63	Townsend
	1249	Davies, 62	Townsend
	1398	Davies, 65	Townsend
	2368	Grigorovici, 39	
	2481	Killian, 30	
	2930	Badareu, 44	Townsend
	3304	Overton, 68	Townsend
H ₂	354	Corrigan, 58	Townsend
	621	Davies, 60	Townsend
	624	Ayres, 23	Townsend
	751	Chanin, 63	Townsend
	779	Wilkes, 55	Townsend
	780	Rose, 56	Townsend
	782	Crompton, 56	Townsend
	784	Crompton, 55	Townsend
	790	Blevin, 57	Townsend
	841	Varnerin, 50	A C Breakdown
	1145	De Bitetto, 56	Townsend
	1160	Fletcher, 63	Townsend
	1174	Barna, 64	Townsend
	1249	Davies, 62	Townsend
	1356	Rose, 56	Townsend
	1640	Hopwood, 56	Townsend
	1712	Prowse, 64	A C Breakdown
	2040	Haydon, 66	Townsend
	2138	Davies, 59	Townsend
	2145	Golden, 65	Townsend
	2148	Haydon, 61	Townsend
	2150	Jones, 58	Townsend
	2154	Frommhold, 60	Pulse Analysis
	2247	Hale, 39	Townsend
	2248	Hale, 39	Townsend
	2271	Townsend, 02	Townsend
	2281	Townsend, 04	Townsend
	2295	Madan, 57	
	2367	Costa, 39	Townsend
	2474	Irish, 64	
	2483	Davies, 58	Townsend
	2561	Cottingham, 63	
	2569	Townsend, 03	Townsend
	2578	Davies, 60	Townsend

Ionization Coefficient (Experimental) — Continued

	2593	Townsend, 01	Townsend
	3490	Bishop, 11	Townsend
	4046	Tagashira, 69	Townsend
	4907	Folkard, 71	Townsend
	5105	Blasberg, 71	
	5225	Shallal, 71	Townsend
D_2	1174	Barna, 64	Townsend
	1186	Rork, 64	Townsend
	1249	Davies, 62	Townsend
	1356	Rose, 56	Townsend
	2533	Cottingham, 63	
	2817	Morgan, 67	Townsend
T_2	2465	Edelson, 66	Townsend
N_2	624	Ayres, 23	Townsend
	710	Heylen, 59	Townsend
	789	Posin, 36	Townsend
	1145	De Bitetto, 56	Townsend
	1728	Allen, 63	Townsend
	1741	Ward, 65	Townsend
	1836	Cookson, 66	Townsend
	1836	Cookson, 66	Pulse Analysis
	1948	Blair, 66	Townsend
	2102	Harrison, 57	Townsend
	2154	Frommhold, 60	Pulse Analysis
	2144	Dutton, 52	Townsend
	2275	Bowls, 38	Townsend
	2277	Masch, 32	Townsend
	2448	Bagnall, 65	Townsend
	2556	Masch, 32	Townsend
	2815	Hurst, 06	Townsend
	3908	Scharfman, 64	A C Breakdown
	3297	Jones, 68	Townsend
	4046	Tagashira, 69	Townsend
	4085	Daniel, 69	Townsend
	5102	Haydon, 72	Townsend
	5187	Daniel, 70	Townsend
	5228	McArthur, 70	Pulse Analysis
	5334	Folkard, 73	Townsend
O_2	573	Dutton, 63	Townsend
	791	Harrison, 5	Townsend
	861	Prasad, 61	Townsend
	961	Freely, 64	Townsend
	1315	Geballe, 52	Townsend
	1445	Schlumbohm, 5	Pulse Analysis
	1625	Schlumbohm, 6	Pulse Analysis
	1724	Schlumbohm, 6	Pulse Analysis
	2053	Frommhold, 58	Pulse Analysis
	2154	Frommhold, 60	Pulse Analysis
	2277	Masch, 32	Townsend
	2370	Frommhold, 64	Pulse Analysis
	2387	Schlumbohm, 6	Pulse Analysis

Ionization Coefficient (Experimental) — Continued

	2462	Prasad, 66	Townsend
	2492	De Bitetto, 5	Townsend
	2534	Skinner, 63	
	2555	Sukhum, 67	Townsend
	2556	Masch, 32	Townsend
	2908	Dutton, 67	Townsend
	3908	Scharfman, 64	A C Breakdown
	4043	Thomas-Betts, 69	Townsend
	4994	Price, 72	Townsend
	5226	Naidu, 70	
	5227	Kinsman, 70	
Cl ₂	2224	Bozin, 67	Townsend
HCl	2562	Townsend, 03	Townsend
CO	1254	Bhalla, 61	Townsend
	2271	Townsend, 02	Townsend
	4016	Davies, 69	Townsend
	5243	Parr, 71	Townsend
H ₂ O	1911	Ryzko, 66	Pulse Analysis
	2294	Prasad, 60	Townsend
	2562	Townsend, 03	Townsend
	2933	Bratescu, 50	Townsend
CO ₂	948	Bhalla, 60	Townsend
	1445	Schlumbohm, 59	Pulse Analysis
	1625	Schlumbohm, 65	Pulse Analysis
	1724	Schlumbohm, 62	Pulse Analysis
	2387	Schlumbohm, 60	Pulse Analysis
	2593	Townsend, 01	Townsend
	2809	Young, 50	
	2815	Hurst, 06	Townsend
	3435	Conti, 67	Townsend
	3490	Bishop, 11	Townsend
	4843	Baker, 70	Townsend
SO ₂	1724	Schlumbohm, 62	Pulse Analysis
	2881	Wheatley, 13	Townsend
SF ₆	1260	Bhalla, 62	Townsend
	1724	Schlumbohm, 62	Pulse Analysis
	2558	Hochberg, 42	Townsend
	2595	Hochberg, 46	Townsend
	5053	Boyd, 71	Townsend
CH ₄	973	Frommhold, 59	Pulse Analysis
	1161	Davies, 63	Townsend
	1335	Heylen, 63	Townsend
	1445	Schlumbohm, 59	Pulse Analysis
	1625	Schlumbohm, 65	Pulse Analysis
	1836	Cookson, 66	Townsend
	1836	Cookson, 66	Pulse Analysis
	1958	Cookson, 65	Townsend
	1958	Cookson, 65	Pulse Analysis

Ionization Coefficient (Experimental)—Continued

	2053	Frommhold, 58	Pulse Analysis
	2387	Schlumbohm, 60	Pulse Analysis
	2466	Cookson, 66	Townsend
	2466	Cookson, 66	Pulse Analysis
	2574	Frommhold, 60	Pulse Analysis
	2934	Giurgea, 55	Townsend
	3177	LeBlanc, 60	Townsend
	5228	McArthur, 70	Pulse Analysis
C_2H_2	1335	Heylen, 63	Townsend
C_2H_4	1335	Heylen, 63	Townsend
C_2H_6	3177	LeBlanc, 60	Townsend
C_3H_8	3177	LeBlanc, 60	Townsend
C_4H_{10}	3177	LeBlanc, 60	Townsend
C_5H_{12}	2595	Hochberg, 46	Townsend
	3177	LeBlanc, 60	Townsend
C_6H_5	2554	Swamy, 67	Townsend
C_6H_6	1625	Schlumbohm, 65	Pulse Analysis
	2931	Badareu, 41	Townsend
	2932	Badareu, 42	Townsend
	2935	Valeriu-Petrescu, 43	Townsend
C_6H_{12}	1445	Schlumbohm, 59	Pulse Analysis
	2387	Schlumbohm, 60	Pulse Analysis
	2935	Valeriu-Petrescu, 43	Townsend
C_6H_{14}	3177	Leblanc, 60	Townsend
C_7H_3	2932	Badareu, 42	Townsend
C_7H_8	2935	Valeriu-Petrescu, 43	Townsend
$CHCl_3$	2595	Hochberg, 46	Townsend
CH_3F	3600	Razzak, 68	Townsend
CH_3Cl	3178	Devins, 60	Townsend
	3600	Razzak, 68	Townsend
CH_3Br	3600	Razzak, 68	Townsend
CH_3OH	1445	Schlumbohm, 59	Pulse Analysis
	2387	Schlumbohm, 60	Pulse Analysis
C_2H_5Cl	2595	Hochberg, 46	Townsend
	3178	Devins, 60	Townsend
C_2H_5Br	2595	Hochberg, 46	Townsend

ELECTRON SWARM DATA

Ionization Coefficient (Experimental) — Continued

C_2H_5OH	2052	Schlumbohm, 58	Pulse Analysis
	2101	Frommhold, 56	Pulse Analysis
	2254	Raether, 37	Pulse Analysis
C_3H_7OH	2554	Swamy, 67	Townsend
C_4H_9OH	3922	Swamy, 66	Townsend
$(C_2H_5)_2O$	1625	Schlumbohm, 65	Pulse Analysis
	2500	Richter, 59	Pulse Analysis
CF_4	3465	Bozin, 68	Townsend
	5051	Naidu, 72	
CCl_4	2300	Howard, 58	Townsend
	2595	Hochberg, 46	Townsend
CCl_2F_2	791	Harrison, 53	Townsend
	1724	Schlumbohm, 62	Pulse Analysis
	2345	Moruzzi, 63	Townsend
	2449	Moruzzi, 6	Townsend
	5229	Boyd, 70	Townsend
CF_3SF_5	791	Harrison, 53	Townsend
C_2F_6	3465	Bozin, 68	Townsend
	5051	Naidu, 72	
C_3F_8	2302	Moruzzi, 6	Townsend
	5051	Naidu, 72	
	5123	Prasad, 70	
C_4F	3604	Razzak, 68	Townsend
	5051	Naidu, 72	
Organic	1445	Schlumbohm, 58	Pulse Analysis
	1625	Schlumbohm, 65	Pulse Analysis
	2052	Schlumbohm, 58	Pulse Analysis
	2387	Schlumbohm, 65	Pulse Analysis
	2547	Heylen, 61	Townsend
	2576	Heylen, 60	Townsend
	3178	Devins, 60	Townsend
Air	5052	Naidu, 72	Townsend
	791	Harrison, 53	
	1911	Ryzko, 66	
	2047	Prasad, 59	
	2050	Llewellyn-Jones, 58	
	2052	Schlumbohm, 58	
	2053	Frommhold, 58	
	2134	Sanders, 32	
	2271	Townsend, 6	
	2274	Sanders, 33	
	2277	Masch, 32	
	2281	Townsend, 04	

Ionization Coefficient (Experimental) — Continued

	2370	Frommhold, 64	Pulse Analysis
	2388	Dutton, 60	Townsend
	2402	Gill, 21	Townsend
	2407	Dutton, 63	Townsend
	2487	Dutton, 61	Townsend
	2488	Dutton, 63	Townsend
	2528	Dutton, 63	Townsend
	2556	Masch, 32	Townsend
	2557	Paavola, 29	Townsend
	2558	Hochberg, 42	Townsend
	2563	Townsend, 01	Townsend
	2569	Townsend, 03	Townsend
	2570	Davies, 65	Townsend
	2571	Davies, 66	Townsend
	2591	Dutton, 62	Townsend
	2595	Hochberg, 46	Townsend
	2767	Driver, 67	Pulse Analysis
	2881	Wheatley, 13	Townsend
	2882	Horton, 17	Townsend
	2919	Dutton, 67	Townsend
	3184	Dutton, 64	Townsend
	3273	Townsend, 00	Townsend
	3490	Bishop, 11	Townsend
	3553	Gill, 12	Townsend
	3908	Scharfman, 64	A C Breakdown
	4015	Daniel, 69	Townsend
	4085	Daniel, 69	Townsend
	5246	Rao, 71	Townsend
	5610	Bhiday, 70	
Mixture	961	Freely, 64	Townsend
	2247	Hale, 39	Townsend
	2275	Bowls, 38	Townsend
	2287	Penning, 34	
	2294	Prasad, 60	Townsend
	2341	Glotov, 37	Townsend
	2354	Chanin, 65	Townsend
	2355	Kruthof, 37	Townsend
	2407	Dutton, 63	Townsend
	2467	Ryzko, 66	Pulse Analysis
	2493	Chanin, 64	Townsend
	2554	Swamy, 67	Townsend
	2815	Hurst, 06	Townsend
	3171	Gill, 12	Townsend
	3477	Heylen, 68	Townsend
	3478	Heylen, 68	Townsend
	3751	Dutton, 69	Townsend
	3922	Swamy, 66	Townsend
	3956	Yamane, 60	
	4018	Riemann, 44	
	4994	Price, 72	Townsend
	5293	Heylen, 71	Townsend
	5577	Heylen, 71	Townsend

Ionization Coefficient (Theoretical)

	2352	Kihara, 52	Ne	1440	Heylen, 63
	2808	Crowe, 55		2048	Druyvesteyn, 36
	3592	Lucas, 67		2453	Golant, 57
He	674	Compton, 18		2501	Glotov, 38
	771	Dunlop, 49		3450	Zaitev, 39
	1440	Heylen, 63		3753	Thomas, 69
	2143	Abdelnabi, 53		4621	Lotz, 67
	2347	Compton, 16		4909	Hughes, 70
	3752	Thomas, 69		5110	DeHoog, 71
	3897	Itoh, 60	Ne ⁺¹	4621	Lotz, 67
	4057	Bortnik, 65	Ne ⁺²	4621	Lotz, 67
	4188	Lozanskii, 68	Ne ⁺³	4621	Lotz, 67
	4621	Lotz, 67	Ne ⁺⁴	4621	Lotz, 67
	5490	Englert, 71	Ne ⁺⁵	4621	Lotz, 67
	4909	Hughes, 70	Ne ⁺⁶	4621	Lotz, 67
	5701	Phelps, 59	Ne ⁺⁷	4621	Lotz, 67
He ⁺¹	4621	Lotz, 67	Ne ⁺⁸	4621	Lotz, 67
Li	4621	Lotz, 67	Ne ⁺⁹	4621	Lotz, 67
Li ⁺¹	4621	Lotz, 67	Na	4621	Lotz, 67
Li ⁺²	4621	Lotz, 67	Na ⁺¹	4621	Lotz, 67
Be	4621	Lotz, 67	Na ⁺²	4621	Lotz, 67
Be ⁺¹	4621	Lotz, 67	Na ⁺³	4621	Lotz, 67
Be ⁺²	4621	Lotz, 67	Na ⁺⁴	4621	Lotz, 67
Be ⁺³	4621	Lotz, 67	Na ⁺⁵	4621	Lotz, 67
B	4621	Lotz, 67	Na ⁺⁶	4621	Lotz, 67
B ⁺¹	4621	Lotz, 67	Na ⁺⁷	4621	Lotz, 67
B ⁺²	4621	Lotz, 67	Na ⁺⁸	4621	Lotz, 67
B ⁺³	4621	Lotz, 67	Na ⁺⁹	4621	Lotz, 67
B ⁺⁴	4621	Lotz, 67	Na ⁺¹⁰	4621	Lotz, 67
C	4621	Lotz, 67	Ar	881	Golant, 59
C ⁺¹	4621	Lotz, 67		1440	Heylen, 63
C ⁺²	4621	Lotz, 67		2276	Emeleus, 36
C ⁺³	4621	Lotz, 67		2452	Golant, 57
C ⁺⁴	4621	Lotz, 67			
C ⁺⁵	4621	Lotz, 67			

Ionization Coefficient (Theoretical) — Continued

Hg	3266	Klarfeld, 41		4621	Lotz, 67
	3270	Cayless, 59			
H	4621	Lotz, 67	O ⁺⁴	3485	Lotz, 68
				4621	Lotz, 67
H ₂	260	Engelhardt, 63	O ⁺⁵	3485	Lotz, 68
	767	Aamodt, 63		4621	Lotz, 67
	927	Lewis, 58			
	1638	Heylen, 60	O ⁺⁶	3485	Lotz, 68
	1723	Muller, 62		4621	Lotz, 67
	2034	Stuart, 60			
	2037	Gerjuoy, 60	O ⁺⁷	3485	Lotz, 68
	2087	Deas, 49		4621	Lotz, 67
	2276	Emeleus, 36			
	2519	Blevin, 57	O ⁺⁸	3485	Lotz, 68
	5235	Nasser, 71			
	5292	Folkard, 70	F	4621	Lotz, 67
D ₂	260	Engelhardt, 63	F ⁺¹	4621	Lotz, 67
N	4621	Lotz, 67	F ⁺²	4621	Lotz, 67
N ₂	218	Engelhardt, 64	F ⁺³	4621	Lotz, 67
	2087	Deas, 49			
	2276	Emeleus, 36	F ⁺⁴	4621	Lotz, 67
	2392	Compton, 16			
N ⁺¹	4621	Lotz, 67	F ⁺⁵	4621	Lotz, 67
N ⁺²	4621	Lotz, 67	F ⁺⁶	4621	Lotz, 67
N ⁺³	4621	Lotz, 67	F ⁺⁷	4621	Lotz, 67
N ⁺⁴	4621	Lotz, 67	F ⁺⁸	4621	Lotz, 67
N ⁺⁵	4621	Lotz, 67	HCl	2392	Compton, 16
N ⁺⁶	4621	Lotz, 67	H ₂ O	2392	Compton, 16
O	4621	Lotz, 67	CO ₂	2392	Compton, 16
				2553	Hake, 67
O ₂	2276	Emeleus, 36	C ₂ H ₄	1335	Heylen, 63
	2553	Hake, 67			
			Air	2087	Deas, 49
O ⁺¹	3485	Lotz, 68		2276	Emeleus, 36
	4621	Lotz, 67			
			Mixture	2096	Kruithof, 37
O ⁺²	3485	Lotz, 68		2501	Glotov, 38
	4621	Lotz, 67		3552	Moralew, 37
				4909	Hughes, 70
O ⁺³	3485	Lotz, 68		5239	Lozanskii, 71

Ionization Coefficient in Crossed Fields (Experimental)

H ₂	238	Fletcher, 66		
	2162	Haydon, 63		
	2468	Fletcher, 66		
	2539	Bernstein, 62		
	2580	Haydon, 62		Townsend
	2588	Haydon, 62		Townsend
N ₂	238	Fletcher, 66		
	2448	Bagnall, 65		
	2468	Fletcher, 66		
Air	2231	Bhiday, 67		
	5610	Bhiday, 70		

Ionization Coefficient in Crossed Fields (Theoretical)

			H ₂	2303	Blevin, 63
2450	Somerville, 52			2325	Blevin, 58

Ionization Coefficient, High Frequency (Experimental)

He	2184	Macdonald, 49	Air	2188	Herlin, 48
				2405	Herlin, 48
H ₂	2157	Macdonald, 49		3606	Taylor, 68

Ionization Coefficient, High Frequency (Theoretical)

He	2184	Macdonald, 49	H ₂	2157	Macdonald, 49
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Recombination Coefficient (Experimental)

He	249	Chen, 61		Radiation
	596	Biondi, 49		Plasma Decay, Microwave
	661	Mittelstadt, 63		Plasma Decay, Microwave
	673	Oskam, 63		Plasma Decay, Microwave
	759	Robben, 63		Radiation
	802	Johnson, 50		Plasma Decay, Microwave
	802	Johnson, 50		Radiation
	815	Newton, 66		Plasma Decay, Microwave
	1639	Hinnov, 62		Plasma Decay, Microwave
	1639	Hinnov, 62		Radiation
	1642	Sexton, 58		Plasma Decay, Microwave
	1671	Leycuras, 63		
	1713	Kuckes, 61		Plasma Decay, Microwave
	1870	Anisimov, 66		Plasma Decay, Microwave
	1901	Jeffries, 65		Radiation
	1961	Gusinow, 66		Plasma Decay
	1963	Stafford, 66		Plasma Decay, Microwave
	1963	Stafford, 66		Radiation
	2230	Craggs, 63		Radiation
	2329	Anderson, 62		Plasma Decay, Microwave
	2329	Anderson, 62		Radiation
	2330	Hinnov, 62		Plasma Decay, Microwave

Recombination Coefficient (Experimental) — Continued

	2330	Hinnov, 62	Radiation
	2331	Motley, 62	Plasma Decay, Microwave
	2447	Aleskovskii, 63	Plasma Decay, Probe
	2447	Aleskovskii, 63	Radiation
	2456	Thomas, 66	Plasma Decay, Microwave
	2549	Fugol, 66	Radiation
	3312	Robben, 67	Plasma Decay, Probe
	3312	Robben, 67	Plasma Decay, Radiation
	3464	Born, 68	Plasma Decay, Microwave
	3658	Newton, 68	Plasma Decay, Microwave
	3795	Davidson, 62	Plasma Decay, Microwave
	4193	Golant, 63	Plasma Decay, Microwave
	4374	Berlande, 70	Plasma Decay, Microwave
	4586	Chen, 69	Plasma Decay, Microwave
	5134	Delpech, 72	Plasma Decay, Microwave
	5184	Aizentson, 72	Plasma, Steady State
	5232	Donovan, 71	Plasma Decay, Microwave
	5325	Johnson, 72	Plasma Decay, Microwave
	5327	Stevefelt, 72	Radiation
	5327	Stevefelt, 72	Plasma Decay, Radiation
	5343	Boulmer, 73	Plasma Decay, Microwave
	5488	Johnson, 73	Plasma Decay, Microwave
	5501	Cronin, 70	Plasma Decay, Microwave
	5533	Baravian, 72	Radiation
	5570	Johnson, 72	Plasma Decay, Microwave
	5730	Johnson, 71	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Probe
	5772	Collins, 70	Plasma Decay, Microwave
	5772	Collins, 70	Plasma Decay, Radiation
He ⁺¹	883	Mosberg, 66	Plasma Decay, Microwave
	883	Mosberg, 66	Plasma Decay, Probe
	883	Mosberg, 66	Radiation
He ⁺²	883	Mosberg, 66	Plasma Decay, Microwave
	883	Mosberg, 66	Plasma Decay, Probe
	883	Mosberg, 66	Radiation
	1766	Hinnov, 66	Plasma Decay, Microwave
	1766	Hinnov, 66	Radiation
He ₃ ⁺¹	5203	Gerardo, 71	Plasma Decay, Microwave
C	5294	Dunn, 71	Plasma Decay, Microwave
	5294	Dunn, 71	Plasma Decay, Probe
Ne	217	Holt, 50	Plasma Decay, Microwave
	217	Holt, 50	Radiation
	584	Hess, 65	Plasma Decay, Microwave
	758	Biondi, 63	Plasma Decay, Microwave
	758	Biondi, 63	Radiation
	1363	Biondi, 49	Plasma Decay, Microwave
	1600	Connor, 65	Plasma Decay, Microwave
	2169	Oskam, 58	Plasma Decay, Microwave
	2409	Hess, 64	Plasma Decay, Microwave
	3443	Frommhold, 68	Plasma Decay, Microwave

Recombination Coefficient (Experimental) — Continued

	4585	Cunningham, 69	Plasma Decay, Probe
	4586	Chen, 69	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Probe
	5775	Olsen, 52	Plasma Decay, Radiation
Ne ₂ ⁺¹	661	Mittelstadt, 63	Plasma Decay, Microwave
	673	Oskam, 63	Plasma Decay, Microwave
	1344	Biondi, 64	Plasma Decay, Microwave
	2988	Kasner, 67	Plasma Decay, Microwave
	3483	Kasner, 68	Plasma Decay, Microwave
	4044	Philbrick, 69	Plasma Decay, Microwave
Ar	252	Redfield, 51	Plasma Decay, Microwave
	252	Redfield, 51	Radiation
	355	Kenty, 28	Plasma Decay, Probe
	711	Sexton, 60	Plasma Decay, Microwave
	758	Biondi, 63	Plasma Decay, Microwave
	758	Biondi, 63	Radiation
	811	Kretschmer, 62	Plasma Decay, Probe
	1363	Biondi, 49	Plasma Decay, Microwave
	1642	Sexton, 58	Plasma Decay, Microwave
	1786	Kozlov, 65	
	1872	Luhr, 31	
	1873	Fox, 66	Plasma Decay, Probe
	1961	Gusinow, 66	Plasma Decay
	2111	Luhr, 30	
	2230	Craggs, 63	Radiation
	2270	Marshall, 29	
	3277	Trong, 67	Radiation
	3433	Aleksandrov, 68	Radiation
	3438	Fox, 67	Plasma Decay, Probe
	3468	Funahashi, 68	Plasma Decay, Microwave
	4065	Mehr, 68	Plasma Decay, Microwave
	4585	Cunningham, 69	Plasma Decay, Probe
	4586	Chen, 69	Plasma Decay, Microwave
	5184	Aizentson, 72	Plasma, Steady State
	5240	Hughes, 71	Radiation
	5242	Novikova, 71	Plasma, Steady State
	5745	Chen, 69	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Probe
	5775	Olsen, 52	Plasma Decay, Radiation
	5776	Mitin, 69	Plasma Decay, Radiation
	5777	Van Trong, 67	Plasma Decay, Radiation
	5782	Desai, 69	Radiation
Ar ₂ ⁺¹	661	Mittelstadt, 63	Plasma Decay, Microwave
	673	Oskam, 63	Plasma Decay, Microwave
K	2546	D'Angelo, 61	Plasma, Steady State
Kr	3188	Popov, 60	
	5128	Cunningham, 72	Plasma Decay, Probe
	5184	Aizentson, 72	Plasma, Steady State
	5745	Chen, 69	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Probe

Recombination Coefficient (Experimental)—Continued

	5776	Mitin, 69	Plasma Decay, Radiation
Kr ₂ ⁺¹	673	Oskam, 63	Plasma Decay, Microwave
Xe	860	Chantry, 64	Plasma Decay, Microwave
	2130	Brodskii, 66	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Microwave
	5745	Chen, 69	Plasma Decay, Probe
	5776	Mitin, 69	Plasma Decay, Radiation
Xe ₂ ⁺¹	673	Oskam, 63	Plasma Decay, Microwave
Cs	638	Wada, 63	Plasma, Steady State
	697	Wada, 61	Plasma, Steady State
	737	Aleskovskii, 63	Plasma Decay, Probe
	737	Aleskovskii, 63	Radiation
	807	Kretschmer, 63	Plasma Decay, Probe
	811	Kretschmer, 62	Plasma Decay, Probe
	1620	Hammer, 66	
	1634	Harris, 65	
	1737	Dandurand, 51	Plasma Decay, Microwave
	2400	Mohler, 29	Radiation
	2457	Balfour, 66	Plasma Decay, Microwave
	2546	D'Angelo, 61	Plasma, Steady State
	2590	Knechtli, 62	Plasma, Steady State
	2950	Mohler, 37	Plasma Decay, Probe
	2950	Mohler, 37	Radiation
	2983	Harris, 68	Plasma Decay, Microwave
	3555	Mohler, 33	Radiation
	3609	Von Goeler, 68	
	3650	David, 67	Radiation
	5231	Archambault, 71	Plasma Decay, Microwave
Hg	807	Kretschmer, 63	Plasma Decay, Probe
	811	Kretschmer, 62	Plasma Decay, Probe
	1729	Dandurand, 51	Plasma Decay, Microwave
	1729	Dandurand, 51	Radiation
	2395	Biondi, 53	Plasma Decay, Microwave
	2518	Mohler, 37	Plasma Decay, Probe
	3613	Skrebov, 67	Radiation
	3654	Egorov, 67	Radiation
H	241	Fowler, 59	Radiation
	346	Craggs, 47	Radiation
	1639	Hinnov, 62	Plasma Decay, Microwave
	1639	Hinnov, 62	Radiation
	5778	Brand, 69	Plasma Decay, Microwave
	5778	Brand, 69	Plasma Decay, Radiation
	5779	Irons, 65	Plasma Decay, Radiation
H ₂	239	Persson, 55	Plasma Decay, Microwave
	240	Varnerin, 51	Plasma Decay, Microwave
	406	Craggs, 46	Radiation
	1240	Whitmer, 56	Plasma Decay, Microwave
	1363	Biondi, 49	Plasma Decay, Microwave

Recombination Coefficient (Experimental) — Continued

	1389	Cooper, 65	Radiation
	1872	Luhr, 31	
	2038	Litvak, 66	Radiation
	2106	Craggs, 47	Radiation
	2111	Luhr, 30	
	2230	Craggs, 63	Radiation
	2292	Richardson, 51	Plasma Decay, Microwave
	3278	Trong, 67	Radiation
	5500	Cronin, 70	Plasma Decay, Microwave
	5697	Janssen, 72	Radiation
N_2	703	Bialecke, 58	Plasma Decay, Microwave
	728	Mentzoni, 63	Plasma Decay, Microwave
	807	Kretschmer, 63	Plasma Decay, Probe
	811	Kretschmer, 62	Plasma Decay, Probe
	1363	Biondi, 49	Plasma Decay, Microwave
	1399	Faire, 59	Plasma Decay, Microwave
	1872	Luhr, 31	
	2111	Luhr, 30	
	2372	Van Lint, 64	Plasma Decay, Microwave
	2394	Hackam, 65	Plasma Decay, Microwave
	3166	Bryan, 57	
	4064	Bromer, 68	Plasma Decay, Microwave
	4605	Sayers, 58	Plasma Decay, Probe
N_2^{+1}	1247	Kasner, 65	Plasma Decay, Microwave
	1344	Biondi, 64	Plasma Decay, Microwave
	1605	Kasner, 61	Plasma Decay, Microwave
	2455	Biondi, 66	Plasma Decay, Microwave
	2993	Kasner, 67	Plasma Decay, Microwave
	4045	Mehr, 69	Plasma Decay, Microwave
	4681	Sayers, 56	Plasma Decay, Probe
	5124	Mahdavi, 71	Plasma Decay, Probe
N_4^{+1}	1247	Kasner, 65	Plasma Decay, Microwave
	1344	Biondi, 64	Plasma Decay, Microwave
O_2	807	Kretschmer, 63	Plasma Decay, Probe
	811	Kretschmer, 62	Plasma Decay, Probe
	1363	Biondi, 49	Plasma Decay, Microwave
	1601	Anisimov, 64	Plasma Decay, Microwave
	1872	Luhr, 31	
	2111	Luhr, 30	
	3428	Mentzoni, 65	Plasma Decay, Microwave
	3786	Smith, 68	Plasma Decay, Probe
	4605	Sayers, 58	Plasma Decay, Probe
	5699	Bragin, 70	Plasma Decay, Microwave
O_2^{+1}	1344	Biondi, 64	Plasma Decay, Microwave
	1605	Kasner, 61	Plasma Decay, Microwave
	3611	Kasner, 68	Plasma Decay, Microwave
	4045	Mehr, 69	Plasma Decay, Microwave
	5124	Mahdavi, 71	Plasma Decay, Probe
	5198	Plumb, 72	Plasma Decay, Probe
O_4^{+1}	3611	Kasner, 68	Plasma Decay, Microwave

Recombination Coefficient (Experimental) — Continued

	5198	Plumb, 72	Plasma Decay, Probe
CO	3601	Mentzoni, 68	Plasma Decay, Microwave
	5294	Dunn, 71	Plasma Decay, Microwave
	5294	Dunn, 71	Plasma Decay, Probe
	5783	Mentzoni, 69	Plasma Decay, Microwave
NO	658	Gunton, 63	Plasma Decay, Microwave
	840	Young, 66	
	920	Doering, 62	Plasma Decay, Probe
	920	Doering, 62	Plasma, Steady State
	1181	Gunton, 61	Plasma Decay, Microwave
	2491	Stein, 64	Plasma Decay, Microwave
	3229	Mentzoni, 67	Plasma Decay, Microwave
	4789	Lin, 63	
NO ⁺¹	1609	Gunton, 65	Plasma Decay, Microwave
	3610	Weller, 68	Plasma Decay, Microwave
	5124	Mahdavi, 71	Plasma Decay, Probe
H ₂ O	1450	Takeda, 60	Plasma Decay, Microwave
CO ₂ ⁺¹	2986	Weller, 67	Plasma Decay, Microwave
	5124	Mahdavi, 71	Plasma Decay, Probe
(NO) ₂ ⁺¹	3610	Weller, 68	Plasma Decay, Microwave
Air	1872	Luhr, 31	
	2111	Luhr, 30	
	5699	Bragin, 70	Plasma Decay, Microwave
Mixture	254	Richardson, 52	Plasma Decay, Microwave
	254	Richardson, 52	Radiation
	698	Faire, 58	Plasma Decay, Microwave
	705	Cool, 66	Radiation
	758	Biondi, 63	Plasma Decay, Microwave
	758	Biondi, 63	Radiation
	860	Chantry, 64	Plasma Decay, Microwave
	1605	Kasner, 61	Plasma Decay, Microwave
	2029	Morgulis, 66	Plasma Decay, Probe
	2169	Oskam, 58	Plasma Decay, Microwave
	2230	Craggs, 63	Radiation
	2268	Wilson, 67	Plasma Decay, Microwave
	2445	Lennon, 59	Plasma Decay, Microwave
	2550	Polushkin, 67	Plasma Decay, Probe
	2551	Morgulis, 66	Plasma Decay, Probe
	3229	Mentzoni, 67	Plasma Decay, Microwave
	3434	Braun, 67	Plasma, Steady State
	5291	Veatch, 70	Radiation

Recombination Coefficient (Theoretical)

	444	Bates, 62	C^{+3}	5288	Shore, 69
	595	Abramov, 66			
	640	Makin, 63	C^{+5}	4891	Bain, 71
	966	Bates, 50		5288	Shore, 69
	975	Aptekar, 56			
	1258	Burgess, 64	Ne	5289	Landini, 71
	1606	Makin, 64		5350	Tarter, 71
	1614	D'Angelo, 65		5745	Chen, 69
	1615	Collins, 65			
	1647	Gurevich, 64	Ne^{+9}	4891	Bain, 71
	2165	Pitaevskii, 62			
	2272	Biberman, 63	Na^{+1}	1259	Bates, 65
	2498	Thomson, 24			
	2885	Smirnov, 67	Mg	5289	Landini, 71
	2991	Veselovskii, 67		5350	Tarter, 71
	3311	Janev, 67			
	3469	Drawin, 68	Mg^{+1}	5288	Shore, 69
He	972	Byron, 62	Al^{+2}	5288	Shore, 69
	998	Kingston, 64			
	3437	Deloche, 67	Si	5289	Landini, 71
	3653	Deloche, 68			
	5130	Drawin, 72	Si^{+3}	5288	Shore, 69
	5732	Drawin, 71			
	5745	Chen, 69	S	5289	Landini, 71
	5761	Mansbach, 69			
	5773	Collins, 68	Ar	5201	Wanless, 71
	5774	Deloche, 68		5745	Chen, 69
He^{+1}	715	Burgess, 60	Ar^{+1}	3180	Dugan, 66
	1259	Bates, 65			
	2168	Bates, 64	K	5781	Curry, 70
	3381	Bates, 67			
	5288	Shore, 69	K^{+1}	1259	Bates, 65
He^{+2}	2259	Seaton, 60	Ca	5289	Landini, 71
He_2^{+1}	1675	Warke, 66	Ca^{+1}	5288	Shore, 69
Li^{+1}	1259	Bates, 65	Fe	5289	Landini, 71
Li^{+2}	5288	Shore, 69	Kr	5745	Chen, 69
Be^{+1}	5288	Shore, 69	Xe	5745	Chen, 69
Be^{+3}	5288	Shore, 69	Cs	737	Aleskovskii, 63
B^{+2}	5288	Shore, 69		3536	Abramov, 65
B^{+4}	5288	Shore, 69		3598	Norcross, 68
				5698	Shaw, 71
				5761	Mansbach, 69
				5781	Curry, 70
C	5289	Landini, 71			
	5350	Tarter, 71	Cs^{+1}	1805	Norcross, 66
C^{+1}	5288	Shore, 69		3180	Dugan, 66

Recombination Coefficient (Theoretical) — Continued

H	601	Seaton, 59	N_2^{+1}	1675	Warke, 66
	5350	Tarter, 71			
	5761	Mansbach, 69		5288	Shore, 69
	5781	Curry, 70			
H_2	4604	Bates, 71	N^{+6}	5288	Shore, 69
			O	5289	Landini, 71
H^{+1}	146	Bates, 62		5350	Tarter, 71
	731	D'Angelo, 61	O_2	750	Bates, 62
	972	Byron, 62	O^{+1}	243	Bates, 39
	994	Bates, 64		1180	Massey, 43
	998	Kingston, 64	O_2^{+1}	1675	Warke, 66
	1010	Stueckelberg, 30		4020	Chan, 68
	1602	Boardman, 64	O^{+5}	5288	Shore, 69
	2031	Bates, 61	O^{+7}	4891	Bain, 71
	2147	Bates, 64		5288	Shore, 69
	2259	Seaton, 60	NO^{+1}	3462	Bardsley, 68
	2390	Burgess, 58		3476	Hansen, 68
H_2^{+1}	2527	Stabler, 63	CO_2	4604	Bates, 71
	4656	Bates, 62	Mixture	3651	Dalidchik, 67
				4604	Bates, 71
N	5289	Landini, 71			
	5350	Tarter, 71			
N_2	750	Bates, 62			
	4604	Bates, 71			

Energy Distribution (Experimental)

He	1651	Roberts, 66	2469	Vorobeve, 66
	1800	Bocchieri, 66	2530	Crawford, 63
	1818	Kagan, 66	2807	VanGorcum, 36
	1822	Borodin, 66	2887	Afanaseva, 67
	1896	Bond, 65	3174	Twiddy, 61
	1899	Roberts, 65	3231	Kagan, 65
	2098	Borodin, 67	3235	Medicus, 56
	2163	Borodin, 65	3442	Franklin, 68
	2236	Vorobeve, 65	3496	Druyvesteyn, 30
	2469	Vorobeve, 66	3603	Rayment, 67
	3174	Twiddy, 61	3608	Vagner, 68
	3231	Kagan, 65	3679	Heymann, 68
	3442	Franklin, 68	3792	Pfau, 67
	3482	Koons, 68	3797	Fields, 63
	3504	Emeleus, 36	3951	Twiddy, 63
	3649	Borodin, 67		
	3792	Pfau, 67	Ar	1800
Ne				Bocchieri, 66
	1800	Bocchieri, 66		2568
	1818	Kagan, 66		Sloane, 33
	1896	Bond, 65		2886
	2236	Vorobeve, 65		Kagan, 67
				3169
				Haigh, 50
				3174
				Twiddy, 61
				3185
				Haigh, 52

Energy Distribution (Experimental)—Continued

	3442	Franklin, 68		3951	Twiddy, 63
	3470	Drouet, 68		5573	Kenny, 70
	3496	Druyvesteyn, 30			
	3679	Heymann, 68	N ₂	693	Noon, 66
	3792	Pfau, 67		2521	Thompson, 61
	3797	Fields, 63		2584	Swift, 62
	5191	Losee, 72		3185	Haigh, 52
Kr	1837	Barnes, 66		3299	Kilvington, 67
	3951	Twiddy, 63		3597	Noon, 68
			O ₂	2521	Thompson, 61
Xe	1837	Barnes, 66		3172	Boyd, 59
	3233	Wehner, 52			
	3790	Aisenberg, 64	CO	2494	Chao, 45
			Air	2442	Badareu, 58
Hg	2008	Vorobeve, 63		2494	Chao, 45
	2019	Vorobeve, 64		3169	Haigh, 50
	2469	Vorobeve, 66		3185	Haigh, 52
	3168	Kagan, 67			
	3231	Kagan, 65	Mixture	1837	Barnes, 66
	3288	Malyshev, 53		2028	Afanaseva, 66
	3306	Rayment, 68		2344	Vorobeve, 64
	3489	Langmuir, 24		2469	Vorobeve, 66
	3895	Guseva, 51		2887	Afanaseva, 67
	3951	Twiddy, 63		3168	Kagan, 67
	4189	Kagan, 51		3231	Kagan, 65
H ₂	2538	Boyd, 67		3599	Ostapchenko, 68
	3173	Boyd, 59		3614	Kagin, 68
	3175	Boyd, 60		3951	Twiddy, 63
	3716	Boyd, 54		4094	Malakhov, 67

Energy Distribution (Theoretical)

761	Parker, 63	2324	Druyvesteyn, 34
927	Lewis, 58	2337	Davydov, 36
1309	Ross, 67	2340	Davydov, 37
1388	Morse, 35	2348	Kagan, 61
1420	Holstein, 46	2352	Kihara, 52
1441	Druyvesteyn, 30	2369	Haseltine, 39
1789	Sodha, 66	2393	Compton, 16
1860	Townsend, 33	2436	Cahn, 49
2043	Bowe, 63	2437	Shkarofsky, 61
2044	Pidduck, 13	2515	Rose, 55
2094	Davydov, 35	2540	Kelly, 60
2152	Wu, 61	2583	Wu, 62
2185	Margcnau, 58	2006	Pearson, 63
2192	Margenau, 48	3156	Kagan, 64
2193	Margenau, 48	3170	Yarnold, 45
2196	Didlaukis, 33	3228	Brown, 66
2197	Druyvesteyn, 32	3235	Medicus, 56
2225	Townsend, 36	3242	Stenflo, 66
2297	Dreicer, 60	3261	Ulyanov, 64
2307	Cahn, 49	3263	Ornstein, 36
2323	Lichtenstein, 38	3276	Peyraud, 67

Energy Distribution (Theoretical) — Continued

	3281	Ornstein, 36		3534	Kagan, 62
	3284	Bayet, 56		3791	Rutscher, 67
	3285	Bayet, 56		3792	Pfau, 67
	3287	Ulyanov, 66			
	3290	Bayet, 55	Ar	292	Engelhardt, 64
	3308	Soshnikov, 68		397	Barbiere, 51
	3310	Stiller, 68		881	Golant, 59
	3314	Yen, 68		1440	Heylen, 63
	3430	Caldirola, 66		2452	Golant, 57
	3493	Kovrizhnykh, 60		2454	Kagan, 60
	3538	Didlaukis, 32		3157	Kagan, 62
	3592	Lucas, 67		3479	Jancel, 68
	3594	Lo Surdo, 68		3792	Pfau, 67
	3605	Sonin, 68			
	3615	Nastoyashchii, 68	Cs	3305	Postma, 68
	3648	De Hoog, 68			
	3661	Allouche, 68	Hg	2454	Kagan, 60
	3678	Peyraud, 68		3157	Kagan, 62
	3893	Gurevich, 57			
	3894	Gurevich, 57	H ₂	260	Engelhardt, 63
	3898	Jancel, 54		756	Baraff, 63
	3907	Sampson, 63		1043	Frost, 62
	3911	Sengupta, 61		1638	Heylen, 60
				1723	Muller, 62
He	397	Barbiere, 51		2037	Gerjuoy, 60
	1208	Reder, 54		2113	Allis, 52
	1297	Llewellyn-Jones, 36		2157	Macdonald, 49
	1440	Heylen, 63		2470	Hantzsche, 66
	2010	Smit, 36		2589	Stuart, 62
	2045	Dunlop, 51		3241	Hantzsche, 66
	2143	Abdelnabi, 53		3463	Bell, 68
	2184	Macdonald, 49		5224	Haydon, 70
	2194	Hartman, 48			
	3463	Bell, 68	D ₂	260	Engelhardt, 63
	3714	Elkomoss, 68			
	3792	Pfau, 67	N ₂	1043	Frost, 62
	4056	Postma, 70		3463	Bell, 68
	4057	Bortnik, 65		5233	Kline, 71
Ne	700	Salmon, 63	Air	1207	Carleton, 62
	1440	Heylen, 63			
	2194	Hartman, 48	Mixture	292	Engelhardt, 64
	2453	Golant, 57		3593	LoSurdo, 67
	2454	Kagan, 60		3902	Lyman, 66
	3157	Kagan, 62		3952	Uman, 64

Conductivity (Experimental)

	2928	Ionescu, 31	Ne	312	Phelps, 51
				3554	Clay, 37
He	306	Anderson, 55		3556	Clay, 37
	312	Phelps, 51		3679	Heymann, 68
	314	Gould, 54		3950	Tanaca, 64
	3179	Maksimov, 67		4021	Aleksandrov, 63
	3554	Clay, 37			
	5327	Stevelfelt, 72			

Conductivity (Experimental) — Continued

Ar	312 Phelps, 51	CO ₂	3383 Gupta, 67
	2018 Hamberger, 68		
	2819 Fedulov, 66	Air	2321 Childs, 32
	3167 Lin, 55		2926 Appleton, 32
	3245 Golubev, 64		3161 Lamb, 57
	3269 Maecker, 60		3164 Krinberg, 65
	3380 Rues, 67		3302 Morsell, 67
	3440 Ciampi, 67		3383 Gupta, 67
	3480 Konjevic, 68		3494 Carruthers, 62
	3679 Heymann, 68		3497 Szekely, 29
	3950 Tanaca, 64		3535 Finkelnburg, 50
Kr	312 Phelps, 51		3539 Karpenko, 52
Xe	312 Phelps, 51		3540 Appleton, 35
	2018 Hamberger, 68		3713 Cottureau, 68
	2472 Zauderer, 66		3721 Ionescu, 35
	3554 Clay, 37		3953 Valentin, 67
	3556 Clay, 37	Mixture	1473 Harris, 63
Cs	149 Mirlin, 62		2472 Zauderer, 66
	3202 Nighan, 67		2579 Dutt, 60
	3429 Mohler, 38		2818 Khozhatelev, 66
Hg	3309 Shimahara, 68		3164 Krinberg, 65
	3793 Prime, 52		3244 Morgulis, 66
H ₂	312 Phelps, 51		3245 Golubev, 64
	1559 Brasefield, 30		3291 Veyssiere, 67
	2018 Hamberger, 68		3302 Morsell, 67
	3466 Chiplonkar, 68		3427 Harris, 64
	3657 Bono, 67		3439 Bernard, 67
N ₂	312 Phelps, 51		3444 Fells, 67
	693 Noon, 66		3445 Goldenberg, 67
	2232 Formato, 60		3458 Ellington, 68
	2586 Formato, 62		3461 Brederlow, 68
	3181 Valentin, 67		3474 Garrison, 68
	3269 Maecker, 60		3481 Koyama, 67
	3282 Christmann, 67		3529 Goldenberg, 64
	3473 Fauchais, 68		3537 Donskoi, 63
O ₂	2586 Formato, 62		3539 Karpenko, 52
	3494 Carruthers, 62		3596 Maslennikov, 67
NO	1779 Mentzoni, 66		3677 Schwenn, 68
H ₂ O	305 Maecker, 55		3789 Akimov, 66
			3794 Croitoru, 63
			3892 Goldenberg, 64
			3900 Kerrebrock, 64
			3904 Mullaney, 61
			3955 Zukoski, 64
			3958 Zukoski, 65

Conductivity (Theoretical)

2183 Chambers, 52	2471 Delcroix, 66
2185 Margenau, 58	2473 Dolique, 66
2187 Huxley, 51	2524 Mallozzi, 63
2255 Huxley, 57	2540 Kelly, 60
2437 Shkarofsky, 61	2592 Croitoru, 62

Conductivity (Theoretical) — Continued

	2884	Schweitzer, 67		3441	De Barbieri, 68
	3155	Epstein, 60			
	3158	Cowling, 45	Ar	2452	Golant, 57
	3182	Bayet, 54		3307	Schweitzer, 67
	3237	De Barbieri, 67		3798	Devoto, 67
	3238	De Barbieri, 67		3799	Devoto, 67
	3274	Demetriades, 66			
	3286	Bayet, 56	Kr	3307	Schweitzer, 67
	3289	Johnson, 67			
	3298	Joyce, 67	Xe	3307	Schweitzer, 67
	3440	Ciampi, 67			
	3492	Schirmer, 58	Cs	1828	Eastlund, 66
	3602	Schweitzer, 67			
	3655	Hassan, 68	H ₂	3718	Stark, 01
	3660	Alievskii, 67			
	3717	Stark, 00	N ₂	3279	Manheimer-Timnat, 59
	3796	Devoto, 66		3718	Stark, 01
	3893	Gurevich, 57			
	3898	Jancel, 54	H ₂ O	2199	Molmud, 59
	3899	Kerrebrock, 64			
	3905	Nastoyashchii, 63	UF ₆	3300	Kudrin, 67
	3906	Pustovoi, 63			
	3907	Sampson, 63	Air	2199	Molmud, 59
	3911	Sengupta, 61		2203	Phelps, 60
	3954	Wilhelm, 66		3279	Manheimer-Timnat, 59
	3957	Yoshikawa, 62		3903	Margenau, 59
He	2199	Molmud, 59	Mixture	3227	Kasabov, 66
	3301	Devoto, 68		3491	Frost, 61
Ne	2453	Golant, 57		3646	Viegas, 68
	3307	Schweitzer, 67		3787	Shelton, 66
				3909	Schweitzer, 67
				3910	Schweitzer, 66

5.2. Full References, Listed in Order of Arbitrarily Assigned Filing Numbers

- 00042 Bennett, W. H., Thomas, L. H., Mobilities in Some Free Electron Gases, *Phys. Rev.*, **62**, 41-47 (1942).
Mixtures studied were $H_2 + N_2$, $H_2 + He$.
- 00131 Chanin, L. M., Phelps, A. V., Biondi, M. A., Measurement of the Attachment of Slow Electrons in Oxygen, *Phys. Rev. Letters*, **2**, 344-346 (1959).
- 00146 Bates, D. R., Kingston, A. E., McWhirter, R. W. P., Recombination between Electrons and Atomic Ions II. Optically Thick Plasmas, *Proc. Roy. Soc. London Ser. A*, **270**, 155-167 (1962).
- 00149 Mirlin, D. N., Pikus, G. E., Yurev, V. G., Determination of Electron Scattering Cross Section by the Electrical Conductivity of Weakly Ionized Gas, *Soviet Phys. Tech. Phys. English Transl.*, **7**, 559-561 (1962).
- 00159 Bailey, V. A., Rudd, J. B., The Behaviour of Electrons in Nitrous Oxide, *Phil. Mag.*, **14**, 1 033-1074 (1932).
- 00181 Frost, L. S., Phelps, A. V., Rotational Excitation and Momentum Transfer Cross Sections for Electrons in H_2 and N_2 from Transport Coefficients, *Phys. Rev.*, **127**, 1621-1633 (1932).
- 00188 Bernstein, M. J., Electron Drift and Diffusion Measurements in H_2 and D_2 with Crossed Electric and Strong Magnetic Fields, *Phys. Rev.*, **127**, 335-341 (1962).
- 00195 Townsend, J. S., Bailey, V. A., The Motion of Electrons in Gases, *Phil. Mag.*, **42**, 874-891 (1921).
- 00197 Townsend, J. S., Bailey, V. A., The Motion of Electrons in Argon, *Phil. Mag.*, **43**, 593-601 (1922).
- 00198 Skinker, M. F., The Motion of Electrons in Carbon Dioxide, *Phil. Mag.*, **44**, 994-999 (1922).
- 00199 Townsend, J. S., Bailey, V. A., The Motion of Electrons in Argon and Hydrogen, *Phil. Mag.*, **44**, 1033-1052 (1922).
Mixture studied was $Ar + H_2$.
- 00200 Skinker, M. F., White, J. V., The Motion of Electrons in Carbon Monoxide, Nitrous Oxide, and Nitric Oxide, *Phil. Mag.*, **46**, 630-637 (1923).
- 00201 Townsend, J. S., Bailey, V. A., Motion of Electrons in Helium, *Phil. Mag.*, **46**, 657-665 (1923).
- 00203 Bannon, J., Brose, H. L., The Motions of Electrons in Ethylene, *Phil. Mag.*, **6**, 817-824 (1928).
- 00217 Holt, R. B., Richardson, J. M., Howland, B., McClure, B. T., Recombination Spectrum and Electron Density Measurements in Neon Afterglows, *Phys. Rev.*, **77**, 239-241 (1950).
- 00218 Engelhardt, A. G., Phelps, A. V., Risk, C. G., Determination of Momentum Transfer and Inelastic Collision Cross Sections for Electrons in Nitrogen using Transport Coefficients, *Phys. Rev.*, **135**, A1566-1574 (1964).
- 00238 Fletcher, I., Haydon, S. C., Transport and Ionization Properties of Molecular Gases in a Transverse Magnetic Field *Australian J. Phys.*, **19**, 615-628 (1966).
- 00239 Persson, K. B., Brown, S. C., Electron Loss Process in the Hydrogen Afterglow, *Phys. Rev.*, **100**, 729-733 (1955).
- 00240 Varnerin, L. J., Electron Recombination and Collision Cross-Section Measurements in Hydrogen, *Phys. Rev.*, **84**, 563-566 (1951).
- 00241 Fowler, R. G., Atkinson, W. R., Electron Recombination in Atomic Hydrogen, *Phys. Rev.*, **113**, 1268-1269 (1959).
- 00243 Bates, D. R., Buckingham, R. A., Massey, H. S. W., Unwin, J. J., Dissociation, Recombination and Attachment Processes in the Upper Atmosphere II. The Rate of Recombination, *Proc. Roy. Soc. London Ser. A*, **170**, 322-340 (1939).
- 00249 Chen, C. L., Goldstein, L., Leiby, C. C., Electron Temperature Dependence of the Recombination Coefficient in Pure Helium, *Phys. Rev.*, **121**, 1391-1400 (1961).
- 00252 Redfield, A., Holt, R. B., Electron Removal in Argon Afterglows, *Phys. Rev.*, **82**, 874-876 (1951).
- 00254 Richardson, J. M., Electron Removal in Krypton Afterglows, *Phys. Rev.*, **88**, 895-900 (1952).
Mixture studied was $Kr + Xe$.
- 00260 Engelhardt, A. G., Phelps, A. V., Elastic and Inelastic Collision Cross Sections in Hydrogen and Deuterium from Transport Coefficients, *Phys. Rev.*, **131**, 2115-2128 (1963).
- 00273 Allen, H. W., Electron Temperatures and Mobilities in the Rare Gases, *Phys. Rev.*, **52**, 707-710 (1937).
- 00292 Engelhardt, A. G., Phelps, A. V., Transport Coefficients and Cross Sections in Argon and Hydrogen-Argon Mixtures, *Phys. Rev.*, **133**, A375-380 (1964).
Mixture studied was $Ar + H_2$.
- 00305 Maecker, H., Peters, T., Schenk, H., Ionen- und Atomquerschnitte im Plasma Verschiedener Gase, *Z. Phys.*, **140**, 119-138 (1955).
- 00306 Anderson, J. M., Goldstein, L., Interaction of Electromagnetic Waves of Radio-Frequency in Isothermal Plasmas Collision Cross Section of Helium Atoms and Ions for Electrons, *Phys. Rev.*, **100**, 1037-1046 (1955).
- 00312 Phelps, A. V., Fundingsland, O. T., Brown, S. C., Microwave Determination of the Probability of Collision of Slow Electrons in Gases, *Phys. Rev.*, **84**, 559-562 (1951).
- 00314 Gould, L., Brown, S. C., Microwave Determination of the Probability of Collision of Electrons in Helium, *Phys. Rev.*, **95**, 897-903 (1954).
- 00336 Corrigan, S. J. B., Von Engel, A., The Excitation of Helium by Electrons of Low Energy, *Proc. Phys. Soc. London*, **72**, 786-790 (1958).
- 00337 Tozer, B. A., Thorburn, R., Craggs, J. D., The Attachment of Slow Electrons in Air and Oxygen, *Proc. Phys. Soc. London*, **72**, 1081-1086 (1958).
- 00346 Craggs, J. D., Hopwood, W., Electron/Ion Recombination in Hydrogen Spark Discharges, *Proc. Phys. Soc. London*, **59**, 771-781 (1947).
- 00352 Crompton, R. W., Sutton, D. J., Experimental Investigation of the Diffusion of Slow Electrons in Nitrogen and Hydrogen *Proc. Roy. Soc. London Ser. A*, **215**, 467-480 (1952).
- 00354 Corrigan, S. J. B., Von Engel, A., Excitation and Dissociation of Hydrogen by an Electron Swarm, *Proc. Roy. Soc. London Ser. A*, **245**, 335-351 (1958).
- 00355 Kenty, C., Recombination of Argon and Ions and Electrons, *Phys. Rev.*, **32**, 624-635 (1928).
- 00358 Zanstra, H., Recombination and the Long Duration of the Balmer Spectrum, *Proc. Roy. Soc. London Ser. A*, **186**, 236-251 (1946).
- 00397 Barbieri, D., Energy Distribution, Drift Velocity, and Temperature of Slow Electrons in Helium and Argon, *Phys. Rev.*, **84**, 653-658 (1951).
- 00406 Craggs, J. D., Meek, J. M., The Emission of Light from Spark Discharges, *Proc. Roy. Soc. London Ser. A*, **186**, 241-260 (1946).
- 00439 Pack, J. L., Voshall, R. E., Phelps, A. V., Drift Velocities of Slow Electrons in Krypton, Xenon, Deuterium, Carbon Monoxide, Carbon Dioxide, Water Vapor, Nitrous Oxide, and Ammonia, *Phys. Rev.*, **127**, 2084-2089 (1962).

- 00484 Halse, D. R., Kingston, A. E., McWhirter, R. W. P., Recombination between Electrons and Atomic Ions. I. Optically Thin Plasmas, *Proc. Roy. Soc. London Ser. A*, **267**, 297-312 (1962).
- 00487 Christophorou, L. G., Compton, R. N., Hurst, G. S., Reinhardt, P. W., Dissociative Electron Capture by Benzene Derivatives, *J. Chem. Phys.*, **45**, 536-547 (1966).
- Organic compounds studied include $\text{o-C}_6\text{H}_4\text{CH}_3\text{Cl}$, $\text{m-C}_6\text{H}_4\text{CH}_3\text{Cl}$, $\text{p-C}_6\text{H}_4\text{CH}_3\text{Cl}$, $\text{o-C}_6\text{H}_4\text{CH}_3\text{Br}$, $\text{m-C}_6\text{H}_4\text{CH}_3\text{Br}$, $\text{p-C}_6\text{H}_4\text{CH}_3\text{Br}$.
- 00488 Christophorou, L. G., Hurst, G. S., Hendrick, W. G., Swarm Determination of the Cross Section for Momentum Transfer in Ethylene and in Ethylene Mixtures, *J. Chem. Phys.*, **45**, 1081-1085 (1966).
- Mixtures studied were $\text{C}_2\text{H}_4 + \text{H}_2\text{O}$.
- 00530 Pack, J. L., Phelps, A. V., Drift Velocities of Slow Electrons in Helium, Neon, Argon, Hydrogen and Nitrogen, *Phys. Rev.*, **121**, 798-806 (1961).
- 00541 Biondi, M. A., Dissociative Attachment of Electrons in Iodine. I. Microwave Determination of the Absolute Cross Section at 300°K, *Phys. Rev.*, **109**, 2005-2007 (1958).
- 00573 Dutton, J., Llewellyn-Jones, F., Morgan, G. B., Electron Attachment in Oxygen, *Nature*, **198**, 680-681 (1963).
- 00584 Hess, W., Rekombination und Diffusion in Abhängigkeit von der Elektronentemperatur eines Zerfallenden Neonplasmas, *Z. Naturforsch.*, **20A**, 451-457 (1965).
- 00595 Abramov, V. A., Smirnov, B. M., Electron-Ion Recombination in a Plasma, *Opt. Spectry. USSR English Transl.*, **21**, 9-12 (1966).
- 00596 Biondi, M. A., Brown, S. C., Measurements of Ambipolar Diffusion in Helium, *Phys. Rev.*, **75**, 1700-1705 (1949).
- 00601 Seaton, M. J., Radiative Recombination of Hydrogenic Ions, *Monthly Notices Roy. Astron. Soc.*, **119**, 81-89 (1959).
- 00619 Townsend, J. S., MacCallum, S. P., Ionization by Collision in Helium, *Phil. Mag.*, **17**, 678-698 (1934).
- 00621 Davies, D. E., Milne, J. G. C., Myatt, J., The Effect of Gas Purity, Tube Geometry, and Method of Calculation on the Ionization Coefficients in Hydrogen and in the Inert Gases, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 161-163 (1960).
- 00624 Ayres, T. L. R., The Ionization by Collision of Hydrogen, Nitrogen, and Argon, *Phil. Mag.*, **45**, 353-368 (1923).
- 00627 Breare, J. M., Von Engel, A., A New Method of Measuring Electron Drift Velocities, (In) *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 317-319 (1963).
- 00638 Wada, J. Y., Knechtli, R. C., Measurements of Electron-Ion Recombination in a Thermal Cesium Plasma, *Phys. Rev. Letters*, **10**, 513-516 (1963).
- 00640 Makin, B., Keck, J. C., Variational Theory of Three-Body Electron-Ion Recombination Rates, *Phys. Rev. Letters*, **11**, 281-283 (1963).
- 00646 Chanin, L. M., Rork, G. D., Experimental Determinations of the First Townsend Ionization Coefficient in Helium, *Phys. Rev.*, **133**, A1005-1009 (1964).
- 00649 Huxley, L. G. H., Crompton, R. W., Bagot, C. H., A New Method for Measuring the Attachment of Slow Electrons in Gases, *Australian J. Phys.*, **12**, 303-308 (1959).
- 00658 Gunton, R. C., Shaw, T. M., Electron Loss Processes in Nitric Oxide Plasmas, Final Report, Lockheed Missiles and Space Company, Sunnyvale, California, AFSWC-TDR-63-23, AD-410 463, 1963, 62 Pages.
- 00661 Mittelstadt, V. R., Oskam, H. J., Madson, J. M., The Mobility and Recombination of Positive Rare Gas Ions, Report Electrical Engineering Department, University of Minnesota, AF-AFOSR-62-103, AD-414 615, 1963, 29 Pages plus Figures.
- 00673 Oskam, H. J., Mittelstadt, V. R., Recombination Coefficient of Molecular Rare-Gas Ions, *Phys. Rev.*, **132**, 1445-1454 (1963).
- 00674 Compton, K. T., Benade, J. M., The Theory of Ionization by Collision. IV. Cases of Elastic and Partially Elastic Impact *Phys. Rev.*, **11**, 234-240 (1918).
- 00677 Healey, R. H., The Behaviour of Electrons in Iodine Vapour, *Phil. Mag.*, **26**, 940-953 (1938).
- Mixtures studied were $\text{I}_2 + \text{CO}_2$, $\text{I}_2 + \text{He}$.
- 00682 Bailey, V. A., Healey, R. H., The Behaviour of Electrons in Chlorine, *Phil. Mag.*, **19**, 725-746 (1935).
- Mixtures studied were $\text{Cl}_2 + \text{CO}_2$, $\text{Cl}_2 + \text{He}$.
- 00683 Bailey, J. E., Makinson, R. E. B., Somerville, J. M., The Behaviour of Electrons in Bromine, *Phil. Mag.*, **24**, 177-190 (1937).
- Mixtures studied were $\text{Br}_2 + \text{CO}_2$, $\text{Br}_2 + \text{He}$.
- 00689 Crompton, R. W., Elford, M. T., The Ratio of Drift Speed to Diffusion Coefficient for Quasi-Thermal Electrons in Nitrogen and Hydrogen, (In) *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 337-341 (1963).
- 00693 Noon, J. H., Holt, E. H., Radiation-Temperature Measurements of the Nitrogen Afterglow Plasma, *Phys. Rev.*, **150**, 121-123 (1966).
- 00697 Wada, J. Y., Knechtli, R. C., Generation and Application of Highly Ionized Quiescent Cesium Plasma in Steady State, *Proc. IRE*, **49**, 1926-1931 (1961).
- 00698 Faire, A. C., Fundingsland, O. T., Aden, A. L., Champion, K. S. W., Electron Recombination Coefficient Measurements in Nitrogen at Low Pressures, *J. Appl. Phys.*, **29**, 928-930 (1958).
- Mixtures studied were $\text{N}_2 + \text{He}$, $\text{N}_2 + \text{Ne}$, $\text{N}_2 + \text{Ar}$.
- 00700 Salmon, J., Une Nouvelle Theorie de la Mobilité des Ions et des Electrons, (In) *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 321-329 (1963).
- 00701 Stern, R. A., Drift Velocity of Electrons in Helium at High E/p, (In) *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 331-333 (1963).
- 00703 Bialecke, E. P., Dougal, A. A., Pressure and Temperature Variation of the Electron-Ion Recombination Coefficient in Nitrogen, *J. Geophys. Res.*, **63**, 539-546 (1958).
- 00705 Cool, T. A., Zukoski, E. E., Recombination, Ionization, and Nonequilibrium Electrical Conductivity in Seeded Plasmas, *Phys. Fluids*, **9**, 780-796 (1966).
- Mixtures studied were $\text{Ar} + \text{K}$.
- 00710 Heylen, A. E. D., Townsend's First Ionization Coefficient in Pure Nitrogen, *Nature*, **183**, 1545-1546 (1959).
- 00711 Sexton, M. C., Mulcahy, M. J., Lennon, J. J., Electron Removal Processes in the Afterglows of Microwave Discharges in Argon and Oxygen, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 94-98 (1960).
- 00715 Burgess, A., Seaton, M. J., Radiative Recombination of He, *Monthly Notices Roy. Astron. Soc.*, **121**, 471-473 (1960).

- 00725 Christophorou, L. G., Hurst, G. S., Hadjiantoniou, A., Interaction of Thermal Electrons with Polarizable and Polar Molecules, *J. Chem. Phys.*, **44**, 3506-3513 (1966).
Mixtures studied were C_2H_4 with the Nonpolar Gases C_3H_8 , C_4H_{10} , C_5H_{12} , C_6H_{14} , C_7H_{16} , C_8H_{18} , C_9H_{20} , $C_{10}H_{22}$, C_6H_6 , C_6H_{12} , p - $C_6H_4(CH_3)_2$, p - $C_6H_4Cl_2$, and C_2H_4 with the Polar Gases m - $C_6H_4(CH_3)_2$, o - $C_6H_4(CH_3)_2$, o - $C_6H_4CH_2Br$, m - $C_6H_4Cl_2$, C_6H_5Br , C_6H_5Cl , o - $C_6H_4Cl_2$.
- 00727 Doehring, A., Messung der Anlagerungswahrscheinlichkeit von Elektronen an Sauerstoff Mittels einer Laufzeitmethode, *Z. Naturforsch.*, **7A**, 253-270 (1952).
- 00728 Mentzoni, M. H., Effective Electron Recombination in Heated Nitrogen, *J. Geophys. Res.*, **68**, 4181-4186 (1963).
- 00730 Townsend, J. S., MacCallum, S. P., Electrical Properties of Neon, *Phil. Mag.*, **6**, 857-877 (1928).
- 00731 D'Angelo, N., Recombination of Ions and Electrons, *Phys. Rev.* **121**, 505-507 (1961).
- 00732 Bortner, T. E., Hurst, G. S., Stone, W. G., Drift Velocities of Electrons in some Commonly used Counting Gases, *Rev. Sci. Instr.*, **28**, 103-108 (1957).
Mixtures studied were $Ar+N_2$, $Ar+CH_4$, CH_4+O_2 , CH_4+CO_2 , and $Ar+Air$.
- 00736 Cochran, L. W., Forester, D. W., Diffusion of Slow Electrons in Gases, *Phys. Rev.*, **126**, 1785-1788 (1962).
- 00737 Aleskovskii, Yu. M., Investigation of Volume Recombination in a Cesium Plasma, *Soviet Phys. JETP English Transl.*, **17**, 570-575 (1963).
- 00738 Pack, J. L., Phelps, A. V., Electron Attachment and Detachment. II. Mixtures of O_2 and CO_2 and of O_2 and H_2O , *J. Chem. Phys.*, **45**, 4316-4329 (1966).
Mixtures studied were O_2+CO_2 and O_2+H_2O .
- 00750 Bates, D. R., Dielectric Recombination to Normal Nitrogen and Oxygen Ions, *Planetary Space Sci.*, **9**, 77-79 (1962).
- 00751 Chanin, L. M., Rork, G. D., Measurements of the First Townsend Ionization Coefficient in Neon and Hydrogen, *Phys. Rev.*, **132**, 2547-2553 (1963).
- 00756 Baraff, G. A., Buchsbaum, S. J., Anisotropic Electron Distribution and the dc and Microwave Avalanche Breakdown in Hydrogen, *Phys. Rev.*, **130**, 1007-1019 (1963).
- 00758 Biondi, M. A., Studies of the Mechanism of Electron-Ion Recombination. I., *Phys. Rev.*, **129**, 1181-1188 (1963).
Mixture studied was $He+Ar$.
- 00759 Robben, F., Kunkel, W. B., Talbot, L., Spectroscopic Study of Electron Recombination with Monatomic Ions in a Helium Plasma, *Phys. Rev.*, **132**, 2363-2371 (1963).
- 00761 Parker, J. H., Spatially Dependent Energy Distributions for Electrons Drifting through a Gas in a Uniform Electric Field *Phys. Rev.*, **132**, 2096-2108 (1963).
- 00762 Brodskii, V. B., Zagik, S. E., Measurement of the Capture Coefficient of Thermalized Electrons in Oxygen and in Air, *Soviet Phys. Tech. Phys. English Transl.*, **11**, 498-502 (1966).
- 00764 Boeckner, C., Mohler, F. L., Scattering of Electrons by Ions and the Mobility of Electrons in a Caesium Discharge, *J. Res. Nat. Bur. Std.*, **10**, 357-363 (1933).
- 00767 Aamodt, R. E., Model of an Electrical Discharge in a Neutral Gas, *Phys. Fluids*, **6**, 446-449 (1963).
- 00771 Dunlop, S. H., Townsend's Ionization Coefficient for Helium, *Nature*, **164**, 452 (1949).
- 00779 Wilkes, A., Hopwood, W., Peacock, N. J., Mechanism of Uniform Field Breakdown in Hydrogen, *Nature*, **176**, 837-839 (1955).
- 00780 Rose, D. J., De Bitetto, D. J., Fisher, L. H., The First Townsend Ionization Coefficient for Hydrogen, *Nature*, **177**, 945-946 (1956).
- 00782 Crompton, R. W., Dutton, J., Haydon, S. C., Precision Measurements of Ionization Coefficients in Uniform Static Fields, *Proc. Phys. Soc. London*, **69**, 2-13 (1956).
- 00784 Crompton, R. W., Dutton, J., Haydon, S. C., Growth of Pre-Breakdown Ionization Currents in Hydrogen, *Nature*, **176**, 1079 (1955).
- 00788 McAfee, K. D., Edelson, D., Identification and Mobility of Ions in a Townsend Discharge by Time-Resolved Mass Spectrometry, *Proc. Phys. Soc. London*, **81**, 382-384 (1963).
- 00789 Posin, D. Q., The Townsend Coefficients and Spark Discharge, *Phys. Rev.*, **50**, 650-658 (1936).
- 00790 Blevin, H. A., Haydon, S. C., Somerville, J. M., Effect of Gaseous Impurities on the First Townsend Coefficient in Hydrogen, *Nature*, **179**, 38-39 (1957).
- 00791 Harrison, M. A., Geballe, R., Simultaneous Measurement of Ionization and Attachment Coefficients, *Phys. Rev.*, **91**, 1-7 (1953).
- 00802 Johnson, R. A., Holt, R. R., McClure, R. T., Electron Removal in Helium Afterglows, *Phys. Rev.*, **80**, 376-379 (1950).
- 00807 Kretschmer, C. B., Petersen, H. L., Use of Langmuir Probes to Study Ion-Electron Recombination, *J. Appl. Phys.*, **34**, 3209-3217 (1963).
- 00811 Kretschmer, C. B., Kinetics of Recombination Processes, Aerojet-General Corporation, Azusa, California, Aerojet Report AN-671, AD-283 043, 1962, 24 Pages.
- 00815 Newton, A. A., Quinn, J. M. P., Sexton, M. C., Laser and Microwave Interferometric Study of Recombination in a Highly Ionised Helium Plasma, *Electron. Letters*, **2**, 157-158 (1966).
- 00840 Young, R. A., St. John, G., Recombination Coefficient of NO^+ with e , *Phys. Rev.*, **152**, 25-28 (1966).
- 00841 Varnerin, L. J., Brown, S. C., Microwave Determinations of Average Electron Energies and the First Townsend Coefficient in Hydrogen, *Phys. Rev.*, **79**, 946-957 (1950).
- 00860 Chantry, P. J., Afterglow Measurements in Xenon and Xenon-Water Vapour Mixtures, (In) *Atomic Collision Processes* M. R. C. McDowell, Editor, North-Holland Publishing Company, Amsterdam, Page 565-573 (1964).
Proceedings of the Third International Conference on the Physics of Electronic and Atomic Collisions (London, 22-26 July 1963).
Mixture studied was $Xe+H_2O$.
- 00861 Prasad, A. N., Craggs, J. D., Measurement of Townsend's Ionization Coefficients and Attachment Coefficients in Oxygen, *Proc. Phys. Soc. London*, **77**, 385-398 (1961).
- 00881 Golant, V. E., Coefficient of Ionization and Mobility of Electrons in Argon, *Soviet Phys. Tech. Phys. English Transl.* **4**, 680-682 (1959).
- 00883 Mosburg, E. R., Recombination of He^+ and He^{++} in the Afterglow of a Helium Discharge, *Phys. Rev.*, **152**, 166-176 (1966).
- 00888 Phelps, A. V., Pack, J. L., Frost, L. S., Drift Velocity of Electrons in Helium, *Phys. Rev.*, **117**, 470-474 (1960).
- 00898 Smith, D., Davies, D. E., Ionisation Phenomena in Mercury Vapour, (In) *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **2**, 213-216 (1963).

- 00900 Davies, D. K., Llewellyn-Jones, F., Morgan, C. G., Ionization in Helium, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **2**, 187-189 (1963).
- 00920 Doering, J. P., Mahan, B. H., Photoionization of Nitric Oxide, *J. Chem. Phys.*, **36**, 669-674 (1962).
- 00921 Legler, W., Anregung von UV-Strahlung in Stickstoff und Wasserstoff durch einen Elektronenschwarm, *Z. Phys.*, **173**, 169-183 (1963).
- 00927 Lewis, T. J., Electron Energy Distributions in Uniform Electric Fields and the Townsend Ionization Coefficient, *Proc. Roy. Soc. London Ser. A*, **244**, 166-185 (1958).
- 00948 Bhalla, M. S., Craggs, J. D., Measurement of Ionization and Attachment Coefficients in Carbon Dioxide in Uniform Fields, *Proc. Phys. Soc. London*, **76**, 369-377 (1960).
- 00936 Stockdale, J. A., Hurst, G. S., Swarm Measurement of Cross Sections for Dissociative Electron Capture in Heavy Water, Chlorobenzene, and Bromobenzene, *J. Chem. Phys.*, **41**, 255-261 (1964).
- Mixtures studied were $D_2O + Ar$, $C_6H_5Br + Ar$, $C_6H_5Br + N_2$, $C_6H_5Cl + Ar$, and $C_6H_5Cl + N_2$.
- 00961 Freely, J. B., Fisher, L. H., Ionization, Attachment, and Breakdown Studies in Oxygen, *Phys. Rev.*, **133**, A304-310 (1964).
- Mixture studied was $O_2 + He$.
- 00966 Bates, D. R., Dissociative Recombination, *Phys. Rev.*, **78**, 492-493 (1950).
- 00972 Byron, S., Stabler, R. C., Bortz, P. I., Electron-Ion Recombination by Collisional and Radiative Processes, *Phys. Rev. Letters*, **8**, 376-379 (1962).
- 00973 Frommhold, L., Eine Untersuchung der Elektronenkomponente von Elektronenlawinen, *Z. Phys.*, **156**, 144-158 (1959).
- 00975 Aptekar, I. L., Timan, B. L., On the Dependence of the Coefficient of Electron Recombination on Temperature and Pressure, *Soviet Phys. Tech. Phys. English Transl.*, **1**, 337-341 (1956).
- 00994 Bates, D. R., Kingston, A. E., Collisional-Radiative Recombination at Low Temperatures and Densities, *Proc. Phys. Soc. London*, **83**, 43-47 (1964).
- 00998 Kingston, A. E., Excitation and Ionization of Hydrogen Atoms by Electron Impact, *Phys. Rev.*, **135**, A1529-1536 (1964).
- 01006 Chatterton, P. A., Craggs, J. D., Measurements of Attachment Coefficients in Oxygen using an Electron Filter Technique, *J. Electron. Control*, **11**, 425-437 (1961).
- 01010 Stueckelberg, E. C., Morse, P. M., Computation of the Effective Cross Section for the Recombination of Electrons with Hydrogen Ions, *Phys. Rev.*, **36**, 16-23 (1930).
- 01016 Hurst, G. S., O'Kelly, L. B., Bortner, T. E., Dissociative Electron Capture in Water Vapor, *Phys. Rev.*, **123**, 1715-1718 (1961).
- Mixture studied was $Ar + H_2O$.
- 01036 Warren, R. W., Parker, J. H., Ratio of the Diffusion Coefficient to the Mobility Coefficient for Electrons in He, Ar, N_2 , H_2 , D_2 , CO, and CO_2 at Low Temperatures and Low E/p, *Phys. Rev.*, **128**, 2661-2671 (1962).
- 01043 Frost, L. S., Phelps, A. V., Molecular Excitation of Hydrogen and Nitrogen from Electron Swarm Experiments, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August - 1 September 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, **1**, 192, 1962.
- 01062 Frost, L. S., Phelps, A. V., Momentum-Transfer Cross Sections for Slow Electrons in He, Ar, Kr and Xe from Transport Coefficients, *Phys. Rev.*, **136**, A1538-1545 (1964).
- 01126 Cavalleri, G., Gatti, E., Principi, P., Experimental Measure of the Diffusion Coefficient for Thermal Electrons in Pure Neon, Using an Electron-Density Sampling Method, *Nuovo Cimento*, **31**, 318-324 (1964).
- 01133 Caren, R. P., Measurement of Electron Mobilities in Argon Gas in a Pure Vapor Cloud Chamber, *Phys. Rev.*, **131**, 1904-1906 (1963).
- 01145 De Bitetto, D. J., Fisher, L. H., Townsend Ionization Coefficients and Uniform Field Breakdown in Hydrogen and Nitrogen at High Pressures, *Phys. Rev.*, **104**, 1213-1220 (1956).
- 01150 Lowke, J. J., Rees, J. A., The Drift Velocities of Free and Attached Electrons in Water Vapour, *Australian J. Phys.*, **16**, 447-453 (1963).
- Mixture studied was $H_2O + H_2$.
- 01154 Chanin, L. M., Steen, R. D., Drift Velocities of Electrons in Cesium, *Phys. Rev.*, **136**, A138-141 (1964).
- 01155 Lowke, J. J., The Drift Velocity of Electrons in Hydrogen and Nitrogen, *Australian J. Phys.*, **16**, 115-135 (1963).
- Mixtures studied were $N_2 + H_2$ and $H_2O + N_2$.
- 01160 Fletcher, J., Davies, D. E., Ionization Coefficients in Hydrogen, Helium and Neon, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **2**, 217-222 (1963).
- 01161 Davies, D. K., Jones, E., Note on Ionization Coefficients in Methane, *Proc. Phys. Soc. London*, **82**, 537-542 (1963).
- 01173 Davies, D. K., Llewellyn-Jones, F., Morgan, C. G., Primary Ionization Coefficient of Helium, *Proc. Phys. Soc. London*, **80**, 898-908 (1962).
- 01174 Barna, S. F., Edelson, D., McAfee, K. B., First Townsend Ionization Coefficients in Hydrogen and Deuterium, *J. Appl. Phys.*, **35**, 2781-2782 (1964).
- 01180 Massey, H. S. W., Bates, D. R., The Properties of Neutral and Ionized Atomic Oxygen and Their Influence on the Upper Atmosphere, *Rept. Prog. Phys.*, **9**, 62-74 (1943).
- 01181 Gunton, R. C., Inn, E. C. Y., Rates of Electron Removal by Recombination, Attachment, and Ambipolar Diffusion in Nitric Oxide Plasmas, *J. Chem. Phys.*, **35**, 1896 (1961).
- 01184 Hurst, G. S., O'Kelly, L. B., Wagner, E. B., Stockdale, J. A., Time-of-Flight Investigations of Electron Transport in Gases, *J. Chem. Phys.*, **39**, 1341-1345 (1963).
- 01186 Rork, G. D., Chanin, L. M., First Townsend Ionization Coefficient Measurements in Deuterium, *J. Appl. Phys.*, **35**, 2801-2802 (1964).
- 01198 Compton, R. N., Christophorou, L. G., Hurst, G. S., Reinhardt, P. W., Nondissociative Electron Capture in Complex Molecules and Negative-Ion Lifetimes, *J. Chem. Phys.*, **45**, 4634-4639 (1966).
- 01207 Carleton, N. P., McGill, L. R., Electron Energy Distribution in Slightly Ionized Air Under the Influence of Electric and Magnetic Fields, *Phys. Rev.*, **126**, 2089-2099 (1962).
- 01208 Reder, F. H., Brown, S. C., Energy Distribution Function of Electrons in Pure Helium, *Phys. Rev.*, **95**, 885-887 (1954).
- 01237 Hornbeck, J. A., Microsecond Transient Currents in the Pulsed Townsend Discharge, *Phys. Rev.*, **83**, 374-379 (1951).
- 01240 Whitmer, R. F., Microwave Studies of the Electron Loss Processes in Gaseous Discharges, *Phys. Rev.*, **104**, 572-575 (1956).

- 01247 Kasner, W. H., Biondi, M. A., Electron-Ion Recombination in Nitrogen, *Phys. Rev.*, **137**, A517-329 (1965).
- 01249 Davies, D. E., Smith, D., Myatt, J., Townsend Ionization Coefficients and the Breakdown of Paschen's Law in Hydrogen, Deuterium, and Mercury Vapour, (In) *Proceedings of the Fifth International Conference on Ionization Phenomena in Gases* (Munich, Germany, 28 August-1 September 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, I, 678 (1962).
- 01250 Dutton, J., Llewellyn-Jones, F., Rees, D. E., Ionization Coefficients in Helium at High Pressures, *Nature*, **200**, 58-59 (1963).
- 01254 Bhalla, M. S., Craggs, J. D., Measurement of Ionization and Attachment Coefficients in Carbon Monoxide in Uniform Fields, *Proc. Phys. Soc. London*, **78**, 438-447 (1961).
- 01258 Burgess, A., Tables of Hydrogenic Photoionization Cross-Sections and Recombination Coefficients, *Mem. Roy. Astron. Soc.*, **69**, 1-26 (1964).
- 01259 Bates, D. R., Khare, S. P., Recombination of Positive Ions and Electrons in a Dense Neutral Gas, *Proc. Phys. Soc. London*, **28**, 231-243 (1965).
- 01260 Bhalla, M. S., Craggs, J. D., Measurement of Ionization and Attachment Coefficients in Sulphur Hexafluoride in Uniform Fields, *Proc. Phys. Soc. London*, **80**, 151-160 (1962).
- 01264 Lawson, P. A., Lucas, J., Electron Diffusion in Hydrogen at High Electric Fields and Low Gas Pressures, *Proc. Phys. Soc. London*, **85**, 177-183 (1965).
- 01267 Hurst, G. S., Bortner, T. E., Attachment of Low-Energy Electrons in Mixtures Containing Oxygen, *Phys. Rev.*, **114**, 116-120 (1959).
- Mixtures studied were $O_2 + N_2$ and $O_2 + C_2H_4$.
- 01269 Rees, J. A., The Behaviour of Free and Attached Electrons in Oxygen, *Australian J. Phys.*, **18**, 41-57 (1965).
- 01274 McAfee, K. E., Pulse Technique for Measurement of the Probability of Formation and Mobility of Negative Ions, *J. Chem. Phys.*, **23**, 1435-1440 (1955).
- 01285 Bloch, F., Bradbury, N. E., On the Mechanism of Unimolecular Electron Capture, *Phys. Rev.*, **48**, 689-695 (1935).
- 01297 Llewellyn-Jones, F., Electron Energies and Excitation in the Helium Positive Column, *Proc. Phys. Soc. London*, **48**, 513-526 (1936).
- 01309 Ross, D. W., Comments on the Solution to the Boltzmann Equation for a Weakly Ionized Plasma, *Phys. Rev.*, **158**, 244-249 (1967).
- 01314 Schlumbohm, H., Messung der Driftgeschwindigkeiten von Elektronen und Positiven Ionen in Gasen, *Z. Phys.*, **182**, 317-327 (1965).
- Organic compounds studied include CH_3OCH_3 and $(CH_3)_2CO$.
- 01315 Geballe, R., Harrison, M. A., Negative Ion Formation in Oxygen, *Phys. Rev.*, **85**, 372-373 (1952).
- 01317 Bradbury, N. E., Electron Attachment and Negative Ion Formation in Oxygen and Oxygen Mixtures, *Phys. Rev.*, **44**, 833-890 (1933).
- Mixtures studied were $O_2 + Ar$, $O_2 + He$, $O_2 + N_2$.
- 01322 Burch, D. S., Geballe, R., Ionic Drift Velocities and Electron Attachment Coefficients in Oxygen, *Phys. Rev.*, **106**, 136-137 (1957).
- 01326 Schlumbohm, H., Elektronenbewegung bei hohen E/N zur Bestimmung der Diffusionskoeffizienten von Elektronen um Ionen der Halogenwasserstoffmoleküle und der Elektronenionisation, *Ann. Physik*, **209**, 201-215 (1966).
- 01332 Bailey, V. A., On the Attachment of Electrons to CO_2 Molecules, *Phil. Mag.*, **50**, 825-848 (1925).
- 01333 Colli, L., Facchini, U., Drift Velocity of Electrons in Argon, *Rev. Sci. Instr.*, **23**, 39-43 (1952).
- Mixture studied was $Ar + N_2$.
- 01335 Heylen, A. E. D., Influence of Molecular Bonding on the Townsend Ionization Coefficients of Hydrocarbon Gases, *J. Chem. Phys.*, **38**, 765-771 (1963).
- 01336 Herreng, P., Mesure de la Mobilité des Electrons dans l'Oxygene et de leur Probabilité de Fixation sur les Molecules de Ce Gaz, *Cahiers Phys.*, **38**, 1-16 (1952).
- 01344 Biondi, M. A., Connor, T. R., Weller, C. S., Kasner, W. H., Recombination of Molecular Positive Ions with Electrons, Technical Report No. 3, Atomic and Plasma Physics Laboratory, University of Pittsburgh, AD-434 907, 1964, 21 Pages.
- 01356 Ross, D. J., Townsend Ionization Coefficient for Hydrogen and Deuterium, *Phys. Rev.*, **104**, 273-277 (1956).
- 01363 Biondi, M. A., Brown, S. C., Measurement of Electron-Ion Recombination, *Phys. Rev.*, **76**, 1697-1700 (1949).
- 01379 Hurst, G. S., Stockdale, J. A., O'Kelly, L. B., Interaction of Low-Energy Electrons with Water Vapor and with other Polar Molecules, *J. Chem. Phys.*, **38**, 2572-2578 (1963).
- Mixtures studied for drift velocity were $H_2O + N_2$, $H_2O + C_2H_4$, $H_2O + CH_4$, $H_2O + CO_2$.
- Mixture studied for Attachment Coefficient was $H_2O + CO_2$.
- 01381 Bradbury, N. E., Nielsen, R. A., Absolute Values of the Electron Mobility in Hydrogen, *Phys. Rev.*, **49**, 388-393 (1936).
- 01383 Chanin, L. M., Phelps, A. V., Biondi, M. A., Measurements of the Attachment of Low-Energy Electrons to Oxygen Molecules, *Phys. Rev.*, **122**, 219-230 (1962).
- 01387 Nielsen, R. A., Absolute Values of the Electron Drift Velocity in Nitrogen, Helium, Neon and Argon, *Phys. Rev.*, **50**, 950-954 (1936).
- 01396 Morse, P. M., Allis, W. P., Lamar, E. S., Velocity Distributions for Elastically Colliding Electrons, *Phys. Rev.*, **48**, 412-419 (1935).
- 01389 Cooper, W. S., Kunkel, W. B., Recombination of Ions and Electrons in a Highly Ionized Hydrogen Plasma, *Phys. Rev.*, **138**, A1022-1027 (1965).
- 01398 Davies, D. E., Smith, D., Primary Ionization Coefficients in Mercury Vapour at Low Pressures (<5 torr) Using a Pool Cathode, *Brit. J. Appl. Phys.*, **16**, 697-702 (1965).
- 01399 Faire, A. C., Champion, K. S. W., Measurements of Dissociative Recombination and Diffusion in Nitrogen at Low Pressures, *Phys. Rev.*, **113**, 1-6 (1959).
- 01412 Nielsen, R. A., Bradbury, N. E., Electron and Negative Ion Mobilities in Oxygen, Air, Nitrous Oxide and Ammonia, *Phys. Rev.*, **51**, 69-75 (1937).
- 01420 Holstein, T., Energy Distribution of Electrons in High Frequency Gas Discharges, *Phys. Rev.*, **50**, 367-384 (1946).
- 01428 Hall, E. I. M., The Diffusion of Slow Electrons in Deuterium, *Australian J. Phys.*, **10**, 33-35 (1957).
- 01429 Jory, R. L., Transport and Attachment of Low-Energy Electrons in Gases in Uniform Electric Fields, *Australian J. Phys.*, **14**, 211-215 (1961).
- 01435 Haffel, E., Mobility and Attachment Coefficients in Oxygen, *Ann. Physik*, **209**, 216-221 (1966).
- 01436 Haffel, E., Mobility and Attachment Coefficients in Nitrogen, *Ann. Physik*, **209**, 222-227 (1966).
- 01437 Haffel, E., Mobility and Attachment Coefficients in Argon, *Ann. Physik*, **209**, 228-233 (1966).
- 01438 Haffel, E., Mobility and Attachment Coefficients in Helium, *Ann. Physik*, **209**, 234-239 (1966).
- 01439 Haffel, E., Mobility and Attachment Coefficients in Neon, *Ann. Physik*, **209**, 240-245 (1966).
- 01440 Haffel, E., Mobility and Attachment Coefficients in Krypton, *Ann. Physik*, **209**, 246-251 (1966).
- 01441 Haffel, E., Mobility and Attachment Coefficients in Xenon, *Ann. Physik*, **209**, 252-257 (1966).
- 01442 Haffel, E., Mobility and Attachment Coefficients in Fluorine, *Ann. Physik*, **209**, 258-263 (1966).
- 01443 Haffel, E., Mobility and Attachment Coefficients in Chlorine, *Ann. Physik*, **209**, 264-269 (1966).
- 01444 Haffel, E., Mobility and Attachment Coefficients in Bromine, *Ann. Physik*, **209**, 270-275 (1966).
- 01445 Haffel, E., Mobility and Attachment Coefficients in Iodine, *Ann. Physik*, **209**, 276-281 (1966).
- 01446 Haffel, E., Mobility and Attachment Coefficients in Sulfur, *Ann. Physik*, **209**, 282-287 (1966).
- 01447 Haffel, E., Mobility and Attachment Coefficients in Selenium, *Ann. Physik*, **209**, 288-293 (1966).
- 01448 Haffel, E., Mobility and Attachment Coefficients in Tellurium, *Ann. Physik*, **209**, 294-299 (1966).
- 01449 Haffel, E., Mobility and Attachment Coefficients in Bismuth, *Ann. Physik*, **209**, 300-305 (1966).
- 01450 Haffel, E., Mobility and Attachment Coefficients in Antimony, *Ann. Physik*, **209**, 306-311 (1966).
- 01451 Haffel, E., Mobility and Attachment Coefficients in Arsenic, *Ann. Physik*, **209**, 312-317 (1966).
- 01452 Haffel, E., Mobility and Attachment Coefficients in Vanadium, *Ann. Physik*, **209**, 318-323 (1966).
- 01453 Haffel, E., Mobility and Attachment Coefficients in Niobium, *Ann. Physik*, **209**, 324-329 (1966).
- 01454 Haffel, E., Mobility and Attachment Coefficients in Tantalum, *Ann. Physik*, **209**, 330-335 (1966).
- 01455 Haffel, E., Mobility and Attachment Coefficients in Manganese, *Ann. Physik*, **209**, 336-341 (1966).
- 01456 Haffel, E., Mobility and Attachment Coefficients in Cobalt, *Ann. Physik*, **209**, 342-347 (1966).
- 01457 Haffel, E., Mobility and Attachment Coefficients in Nickel, *Ann. Physik*, **209**, 348-353 (1966).
- 01458 Haffel, E., Mobility and Attachment Coefficients in Zinc, *Ann. Physik*, **209**, 354-359 (1966).
- 01459 Haffel, E., Mobility and Attachment Coefficients in Cadmium, *Ann. Physik*, **209**, 360-365 (1966).
- 01460 Haffel, E., Mobility and Attachment Coefficients in Mercury, *Ann. Physik*, **209**, 366-371 (1966).
- 01461 Haffel, E., Mobility and Attachment Coefficients in Lead, *Ann. Physik*, **209**, 372-377 (1966).
- 01462 Haffel, E., Mobility and Attachment Coefficients in Thallium, *Ann. Physik*, **209**, 378-383 (1966).
- 01463 Haffel, E., Mobility and Attachment Coefficients in Bismuth, *Ann. Physik*, **209**, 384-389 (1966).
- 01464 Haffel, E., Mobility and Attachment Coefficients in Antimony, *Ann. Physik*, **209**, 390-395 (1966).
- 01465 Haffel, E., Mobility and Attachment Coefficients in Arsenic, *Ann. Physik*, **209**, 396-401 (1966).
- 01466 Haffel, E., Mobility and Attachment Coefficients in Vanadium, *Ann. Physik*, **209**, 402-407 (1966).
- 01467 Haffel, E., Mobility and Attachment Coefficients in Niobium, *Ann. Physik*, **209**, 408-413 (1966).
- 01468 Haffel, E., Mobility and Attachment Coefficients in Tantalum, *Ann. Physik*, **209**, 414-419 (1966).
- 01469 Haffel, E., Mobility and Attachment Coefficients in Manganese, *Ann. Physik*, **209**, 420-425 (1966).
- 01470 Haffel, E., Mobility and Attachment Coefficients in Cobalt, *Ann. Physik*, **209**, 426-431 (1966).
- 01471 Haffel, E., Mobility and Attachment Coefficients in Nickel, *Ann. Physik*, **209**, 432-437 (1966).
- 01472 Haffel, E., Mobility and Attachment Coefficients in Zinc, *Ann. Physik*, **209**, 438-443 (1966).
- 01473 Haffel, E., Mobility and Attachment Coefficients in Cadmium, *Ann. Physik*, **209**, 444-449 (1966).
- 01474 Haffel, E., Mobility and Attachment Coefficients in Mercury, *Ann. Physik*, **209**, 450-455 (1966).
- 01475 Haffel, E., Mobility and Attachment Coefficients in Lead, *Ann. Physik*, **209**, 456-461 (1966).
- 01476 Haffel, E., Mobility and Attachment Coefficients in Thallium, *Ann. Physik*, **209**, 462-467 (1966).
- 01477 Haffel, E., Mobility and Attachment Coefficients in Bismuth, *Ann. Physik*, **209**, 468-473 (1966).
- 01478 Haffel, E., Mobility and Attachment Coefficients in Antimony, *Ann. Physik*, **209**, 474-479 (1966).
- 01479 Haffel, E., Mobility and Attachment Coefficients in Arsenic, *Ann. Physik*, **209**, 480-485 (1966).
- 01480 Haffel, E., Mobility and Attachment Coefficients in Vanadium, *Ann. Physik*, **209**, 486-491 (1966).
- 01481 Haffel, E., Mobility and Attachment Coefficients in Niobium, *Ann. Physik*, **209**, 492-497 (1966).
- 01482 Haffel, E., Mobility and Attachment Coefficients in Tantalum, *Ann. Physik*, **209**, 498-503 (1966).
- 01483 Haffel, E., Mobility and Attachment Coefficients in Manganese, *Ann. Physik*, **209**, 504-509 (1966).
- 01484 Haffel, E., Mobility and Attachment Coefficients in Cobalt, *Ann. Physik*, **209**, 510-515 (1966).
- 01485 Haffel, E., Mobility and Attachment Coefficients in Nickel, *Ann. Physik*, **209**, 516-521 (1966).
- 01486 Haffel, E., Mobility and Attachment Coefficients in Zinc, *Ann. Physik*, **209**, 522-527 (1966).
- 01487 Haffel, E., Mobility and Attachment Coefficients in Cadmium, *Ann. Physik*, **209**, 528-533 (1966).
- 01488 Haffel, E., Mobility and Attachment Coefficients in Mercury, *Ann. Physik*, **209**, 534-539 (1966).
- 01489 Haffel, E., Mobility and Attachment Coefficients in Lead, *Ann. Physik*, **209**, 540-545 (1966).
- 01490 Haffel, E., Mobility and Attachment Coefficients in Thallium, *Ann. Physik*, **209**, 546-551 (1966).
- 01491 Haffel, E., Mobility and Attachment Coefficients in Bismuth, *Ann. Physik*, **209**, 552-557 (1966).
- 01492 Haffel, E., Mobility and Attachment Coefficients in Antimony, *Ann. Physik*, **209**, 558-563 (1966).
- 01493 Haffel, E., Mobility and Attachment Coefficients in Arsenic, *Ann. Physik*, **209**, 564-569 (1966).
- 01494 Haffel, E., Mobility and Attachment Coefficients in Vanadium, *Ann. Physik*, **209**, 570-575 (1966).
- 01495 Haffel, E., Mobility and Attachment Coefficients in Niobium, *Ann. Physik*, **209**, 576-581 (1966).
- 01496 Haffel, E., Mobility and Attachment Coefficients in Tantalum, *Ann. Physik*, **209**, 582-587 (1966).
- 01497 Haffel, E., Mobility and Attachment Coefficients in Manganese, *Ann. Physik*, **209**, 588-593 (1966).
- 01498 Haffel, E., Mobility and Attachment Coefficients in Cobalt, *Ann. Physik*, **209**, 594-599 (1966).
- 01499 Haffel, E., Mobility and Attachment Coefficients in Nickel, *Ann. Physik*, **209**, 600-605 (1966).
- 01500 Haffel, E., Mobility and Attachment Coefficients in Zinc, *Ann. Physik*, **209**, 606-611 (1966).

- Measurement of Electron Energy for Conditions Near Electrical Breakdown, (In) Proceedings of the Seventh International Conference on Phenomena in Ionized Gases (Belgrade, 22-27 August 1965) B. Perovic and D. Tosic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 86-91 (1966).
- 01438 Crompton, R. W., Jory, R. L., On the Swarm Method for Determining the Ratio of Electron Drift Velocity to Diffusion Coefficient, *Australian J. Phys.*, **15**, 451-469 (1962).
- 01440 Heylen, A. E. D., Lewis, T. J., Electron Energy Distribution Functions and Transport Coefficients for the Rare Gases, *Proc. Roy. Soc. London Ser. A*, **271**, 531-550 (1963).
- 01441 Druyvesteyn, M. J., De Invloed der Energieverliezen bij Elastische Botsingen in de Theorie der Electronendiffusie, *Physica*, **10**, 61-70 (1930).
- 01444 Frommhold, L., Simultaneous Determination of the Electron Attachment and Detachment Rates in Oxygen, (In) *Atomic Collision Processes*, M. R. C. McDowell, Editor, North-Holland Publishing Company, Amsterdam, Pages 556-564 (1964). Proceedings of the Third International Conference on the Physics of Electronic and Atomic Collisions (London, 22-26 July 1963).
- 01445 Schlumbohm, H., Elektronen-Stossionisierungskoeffizient (α) für Organische Dämpfe und Sauerstoff (aus der Tragerstatistik von Elektronenlawinen), *Z. Angew. Phys.*, **11**, 156-159 (1959).
- Organic compounds studied include $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, $\text{CO}(\text{CH}_3)_2$ and $\text{CH}_2(\text{OCH}_3)_2$.
- 01446 Wahlin, H. B., The Motion of Electrons in Carbon Monoxide, *Phys. Rev.*, **35**, 1568-1571 (1930).
- 01447 Heylen, A. E. D., Calculated Electron Mobility in Nitrogen, Oxygen and Air for $0.1 < E/p < 100 \text{ v cm}^{-1} \text{ mm Hg}^{-1}$, *Proc. Phys. Soc. London*, **79**, 284-292 (1962).
- 01450 Takeda, S., Dougal, A. A., Microwave Study of Afterglow Discharge in Water Vapor, *J. Appl. Phys.*, **31**, 412-416 (1960).
- 01467 Nolan, J. F., Phelps, A. V., Measurement of Cesium Excitation Cross Section near Threshold by a Swarm Technique *Phys. Rev.*, **140**, A792-799 (1965). Mixture Studied Was $\text{Ar} + \text{Cs}$.
- 01473 Harris, L. P., Electrical Conductivity of Cesium-Seeded Atmospheric Pressure Plasmas Near Thermal Equilibrium, *J. Appl. Phys.*, **34**, 2958-2965 (1963).
- Mixtures studied were $\text{N}_2 + \text{Cs}$, $\text{He} + \text{Cs}$, $\text{Ne} + \text{Cs}$, and $\text{Ar} + \text{Cs}$.
- 01489 Crompton, R. W., Jory, R. L., The Momentum-Transfer Cross Section for Low-Energy Electrons in Helium, (In Abstracts of) The Fourth International Conference on the Physics of Electronic and Atomic Collisions (Quebec, Canada, 2-6 August 1965) Science Bookcrafters, Inc., Hastings-on-Hudson, New York, Page 118, (1965).
- 01559 Brasefield, C. J., The Conductivity of a High Frequency Discharge in Hydrogen, *Phys. Rev.*, **35**, 1073-1079 (1930).
- 01565 Christophorou, L. G., Compton, R. N., Hurst, G. S., Reinhardt, P. W., Determination of Electron-Capture Cross Sections with Swarm-Beam Techniques, *J. Chem. Phys.*, **43**, 4273-4281 (1965).
- Organic compounds studied include $\text{o-C}_6\text{H}_4\text{CH}_3\text{Cl}$.
- 01600 Connor, T. R., Biondi, M. A., Dissociative Recombination in Neon-Spectral Line-Shape Studies, *Phys. Rev.*, **140**, A778-791 (1965).
- 01601 Anisimov, A. I., Vinogradov, N. I., Golant, V. E., Measurement of the Volume Removal Coefficient for Electrons in Plasma Decay in Oxygen, *Soviet Phys. Tech. Phys. English Transl.*, **8**, 850-851 (1964).
- 01602 Boardman, W. J., The Radiative Recombination Coefficients of the Hydrogen Atom, *Astrophys. J. Suppl. Ser.*, **9**, 185-192 (1964).
- 01603 Dalgarno, A., Kingston, A. E., Radiative Attachment of Electrons to Atomic Hydrogen, *Observatory*, **83**, 39-40 (1963).
- 01605 Kasner, W. H., Rogers, W. A., Biondi, M. A., Electron-Ion Recombination Coefficients in Nitrogen and in Oxygen, *Phys. Rev. Letters*, **7**, 321-323 (1961).
- 01606 Makin, B., Keck, J. C., Phase Space Calculation of Three Body Electron-Ion Recombination Rates, (In) *Atomic Collision Processes*, M. R. C. McDowell, Editor, North-Holland Publishing Company, Amsterdam, Page 510, (1964). Proceedings of the Third International Conference on the Physics of Electronic and Atomic Collisions (London, 22-26 July 1963).
- 01608 Gunton, R. C., Shaw, T. M., Ambipolar Diffusion and Electron Attachment in Nitric Oxide in the Temperature Range 196 to 358°K, *Phys. Rev.*, **140**, A748-754 (1965).
- 01609 Gunton, R. C., Shaw, T. M., Electron-Ion Recombination in Nitric Oxide in the Temperature Range 196 to 358°K, *Phys. Rev.*, **140**, A756-763 (1965).
- 01614 D'Angelo, N., Ion-Electron Recombination, *Phys. Rev.*, **140** A1488 (1965).
- 01615 Collins, C. B., Collisional-Dissociative Recombination of Electrons with Molecular Ions, *Phys. Rev.*, **140**, A1850-1857 (1965).
- 01618 Crompton, R. W., Rees, J. A., Jory, R. L., The Diffusion and Attachment of Electrons in Water Vapour, *Australian J. Phys.*, **18**, 541-551 (1965).
- 01620 Hammer, J. M., Aubrey, B. B., Ion Beam Measurements of Cesium Recombination Cross Sections, *Phys. Rev.*, **141**, 146-151 (1966).
- 01622 Crompton, R. W., Elford, M. T., Gascoigne, J., Precision Measurements of the Townsend Energy Ratio for Electron Swarms in Highly Uniform Electric Fields, *Australian J. Phys.*, **18**, 409-436 (1965).
- 01623 Bowe, J. C., Drift Velocity of Electrons in Nitrogen, Helium Neon, Argon, Krypton, and Xenon, *Phys. Rev.*, **117**, 1411-1415 (1960).
- 01625 Schlumbohm, H., Stossionisierungskoeffizient α , Mittlere Elektronenenergien und die Beweglichkeit von Elektronen in Gasen, *Z. Phys.*, **184**, 492-505 (1965).
- Organic compounds studied include $(\text{CH}_3)_2\text{CO}$ and $\text{CH}_2(\text{OCH}_3)_2$.
- 01633 Hackam, R., Lennon, J. J., Microwave Measurements of Temperature Dependence of Electron Density Decay Rates in Oxygen and Oxygen-Nitrogen Mixtures, *Proc. Phys. Soc. London*, **86**, 123-131 (1965).
- Mixture studied was $\text{O}_2 + \text{N}_2$.
- 01634 Harris, L. P., Ionization and Recombination in Cesium-Seeded Plasmas Near Thermal Equilibrium, *J. Appl. Phys.*, **36**, 1543-1553 (1965).
- 01636 Heylen, A. E. D., Calculated Electron Mobility in Hydrogen, *Proc. Phys. Soc. London*, **76**, 779-782 (1960).
- 01638 Heylen, A. E. D., Lewis, T. J., Theoretical Determinations of the Electron Energy Distributions, Drift Velocity and Townsend α Coefficient, (In) Proceedings of the Fourth International Conference on Ionization Phenomena in Gases (Uppsala 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, Pages 156-160 (1960).
- 01639 Hinnov, E., Hirschberg, J. G., Electron-Ion Recombination in Dense Plasmas, *Phys. Rev.*, **125**, 795-800 (1962).
- 01640 Hopwood, W., Peacock, N. J., Wilkes, A., A Study of Ionization Coefficients and Electrical Breakdown in

- Hydrogen Proc. Roy. Soc. London Ser. A, **235**, 334-348 (1956).
- 01642 Sexton, M. C., Craggs, J. D., Recombination in the Afterglows of Argon and Helium using Microwave Techniques, *J. Electron. Control*, **4**, 493-502 (1958).
- 01647 Gurevich, A. V., Pitaevskii, L. P., Recombination Coefficient in a Dense Low-Temperature Plasma, *Soviet Phys. JETP English Transl.*, **19**, 870-871 (1964).
- 01649 Hasted, J. B., Beg, S., Attachment of Electrons to Nitrogen Dioxide and Sulphur Hexafluoride, *Brit. J. Appl. Phys.*, **16** 1779-1786 (1965).
- Mixtures studied were $\text{NO}_2 + \text{Xe}$, $\text{NO}_2 + \text{N}_2$ and $\text{SF}_6 + \text{Xe}$.
- 01651 Roberts, T. D., Burch, D. S., Experimental Electron Energy Distributions for Townsend Discharges in Helium, *Phys. Rev.*, **142**, 100-104 (1966).
- 01662 Pack, J. L., Phelps, A. V., Electron Attachment and Detachment. I. Pure O_2 at Low Energy, *J. Chem. Phys.*, **44** 1870-1883 (1966).
- 01665 Wilkine, R. L., Monte Carlo Calculations of Cross Sections of Electron-Positive-Molecular-Ion Dissociative Recombination, *J. Chem. Phys.*, **44**, 1884-1888 (1966).
- 01671 Leczykas, Y., Decharges Luminescentes sous Pression et Coefficient de Recombinaison, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 457 (1963).
- 01675 Warke, C. S., Nonradiative Dissociative Electron Capture by Molecular Ions, *Phys. Rev.*, **144**, 120-126 (1966).
- 01712 Prowse, W. A., Nicholls, J. M., Ionization and Diffusion by Electrons in Hydrogen in Mixed Oscillatory and Unidirectional Fields, *Proc. Phys. Soc. London*, **84**, 545-555 (1964).
- 01713 Kuckes, A. F., Motley, R. W., Hinnov, E., Hirschberg, J. G., Recombination in a Helium Plasma, *Phys. Rev. Letters*, **6**, 337-339 (1961).
- 01714 Levine, J. L., Sanders, T. M., Anomalous Electron Mobility and Complex Negative Ion Formation in Low-Temperature Helium Vapor, *Phys. Rev. Letters*, **8**, 159-161 (1962).
- 01716 Tholl, H., Messung der Elektronendriftgeschwindigkeit N_2 und in CH_4 , *Z. Phys.*, **178**, 183-188 (1964).
- Mixture studied was $\text{N}_2 + \text{CH}_4$.
- 01723 Muller, K. G., Die Elektronenbewegung im Neutralgas bei Hohen Elektrischen Feldern, *Z. Phys.*, **169**, 432-455 (1962).
- 01724 Schlumbohm, H., Elektronenlawinen in Elektronegativen Gasen - Zur Gleichzeitigen Messung des Stossionisierungs- und Anlagerungskoeffizienten Alpha und Eta fur Elektronen und der Geschwindigkeiten Positiver und Negativer Ionen, *Z. Phys.*, **166**, 192-206 (1962).
- 01725 Wahlin, H. B., The Motion of Electrons in Carbon Monoxide, *Phys. Rev.*, **21**, 517-524 (1925).
- 01728 Allen, N. L., Townsend Ionization Coefficients in Compressed Nitrogen, *Brit. J. Appl. Phys.*, **14**, 589-590 (1963).
- 01729 Dandurand, P., Holt, R. B., Electron Removal in Mercury Afterglows, *Phys. Rev.*, **82**, 868-873 (1951).
- 01730 Levine, J. L., The Mobility of Electrons in Dense Helium Gas at Low Temperatures, Thesis, University of Minnesota, 1965, 133 Pages, University Microfilms, Inc., Ann Arbor, Michigan, Order No. 65-7893.
- 01737 Dandurand, P., Holt, R. B., Electron Density and Light Intensity Decay in Cesium Afterglows, *Phys. Rev.*, **82**, 278-279 (1951).
- 01741 Ward, B. W., Townsend's First Ionization Coefficient for Compressed Nitrogen, *Nature*, **208**, 994-995 (1965).
- 01745 Salmon, J., Mobilité et Temperature des Ions et Des Electrons dans un Gaz Faiblement Ionise, *J. Phys. (Paris)*, **24**, 1118-1126 (1963).
- 01752 Wahlin, H. B., The Motion of Electrons in Argon, *Phys. Rev.*, **37**, 260-262 (1931).
- 01761 Nygaard, K. J., Measurements of Balmer Line Radiation from Self-Sustained Townsend Discharges in Hydrogen, *Appl. Sci. Res. Sect. B*, **12**, 91-100 (1965).
- 01763 Fischer-Treuenfeld, W. F., Messung der Elektronendriftgeschwindigkeit in N_2 im E/p-Bereich von 0.05 bis 40 Volt/cm torr (Funkenkammerverfahren), *Z. Phys.*, **185**, 336-344 (1965).
- 01766 Hinnov, E., Measurement of Recombination-Rate Coefficient of He^{++} , *Phys. Rev.*, **147**, 197-200 (1966).
- 01778 Lawson, P. A., Lucas, J., Electron Diffusion in Hydrogen at High Electric Fields and Low Gas Pressures: II, *Brit. J. Appl. Phys.*, **16**, 1813-1820 (1965).
- 01779 Mentzoni, M. H., Donohoe, J., Electron Collision Frequency in Nitric Oxide, *Can. J. Phys.*, **44**, 693-703 (1966).
- 01783 Heylen, A. E. D., The Influence of a Crossed Magnetic Field on a Gaseous Townsend Discharge, *Brit. J. Appl. Phys.*, **16**, 1151-1159 (1965).
- 01786 Kozlov, G. I., Conductivity of Argon and Recombination Coefficient, *High Temp. USSR English Transl.*, **3**, 467-474 (1965).
- 01789 Sodha, M. S., Srivastava, H. K., Second-order Diffusion of Electrons in Ionized Gases, *Proc. Phys. Soc.*, **87**, 803-808 (1966).
- 01800 Bocchieri, P., Crosignani, E., Siragusa, G., Microwave Determination of the Distribution Functions of Electrons in Low Current Discharges, *Nuovo Cimento*, **42B**, 93-98 (1966).
- 01805 Norcross, D. W., Stone, P. M., Radiative Recombination in Cesium, *J. Quant. Spectry. Radiative Transfer*, **6**, 277-290 (1966).
- 01811 Zelby, L. W., Mehuron, W. O., Kalagher, R. J., Measurements of Electron Drift Velocity and Collision Frequency in an Argon Discharge, *Phys. Letters*, **21**, 522-524 (1966).
- 01818 Kagan, Yu. M., Milenin, V. M., Radial Dependence of the Electron Velocity Distribution Function in a Positive Column, *Soviet Phys. Tech. Phys. English Transl.*, **10**, 1470-1471 (1966).
- 01822 Borodin, V. S., Kagan, Yu. M., Investigation of Hollow-Cathode Discharge. I. Comparison of the Electrical Characteristics of a Hollow Cathode and a Positive Column, *Soviet Phys. Tech. Phys. English Transl.*, **11**, 131-134 (1966).
- 01828 Eastlund, B. J., Caesium Plasma Transport Properties, *J. Nucl. Energy P. C*, **8**, 31-41 (1966).
- 01833 Fink, X., Huber, P., Wanderungsgeschwindigkeit und Diffusionskonstante von Elektronen in Methan, *Helv. Phys. Acta*, **38**, 717-735 (1965).
- 01836 Cookson, A. H., Ward, B. W., Lewis, T. J., Townsend's First Ionization Coefficient for Methane and Nitrogen, *Brit. J. Appl. Phys.*, **17**, 891-903 (1966).
- 01837 Barnes, B. T., Electron Energy Distributions in Various Discharges, *J. Appl. Phys.*, **37**, 2679-2683 (1966).
- Mixtures studied were $\text{Ar} + \text{Hg}$ and $\text{Ne} + \text{Kr}$.
- 01860 Townsend, J. S., Distribution of Energies of Electrons in Gases, *Phil. Mag.*, **16**, 729-744 (1933).
- 01870 Anisimov, A. I., Budnikov, V. N., Vinogradov, N. I., Decay of a Helium Plasma in a Spherical Chamber, *Soviet Phys. Tech. Phys. English Transl.*, **10**, 1554 (1966).
- 01872 Luhr, O., Bradbury, N. E., Corrected Values for the Coefficient of Recombination of Gaseous Ions, *Phys. Rev.*, **37**, 998-1000 (1931).
- 01873 Fox, J. N., Hobson, R. M., Temperature Dependence of Dissociative Recombination Coefficients in Argon, *Phys. Rev. Letters*, **17**, 161-163 (1966).

- 01896 Bond, R. H., Directed Electron Velocity Distributions in Rare Gas Discharges using Guard Ring Probes. Thesis. California Institute of Technology, 1965, 103 Pages. University Microfilms Inc., Ann Arbor, Michigan, Order No. 65-11,698.
- 01899 Roberts, T. D., Experimental Determinations of Electron Energy Distributions for Townsend Discharges in Helium Gas, Thesis, Oregon State University, 1965, 82 Pages. University Microfilms Inc., Ann Arbor, Michigan, Order No. 65-9199.
- 01901 Jeffries, R. A., The Kinetics of Ionization of Strongly Shocked Helium, Thesis, University of Oklahoma, 1965, 108 Pages, University Microfilms Inc., Ann Arbor, Michigan, Order No. 65-9748.
- 01911 Ryzko, H., Ionization, Attachment and Drift Velocity of Electrons in Water Vapor and Dry Air, *Arkiv Fysik.*, **32**, 1-18 (1966).
- 01917 Cottrell, T. L., Walker, I. C., Drift Velocities of Slow Electrons in Polyatomic Gases, *Trans. Faraday Soc.*, **61**, 1585-1593 (1965).
- Organic Compounds Include $(CH_3)_2CO$.
- 01923 Mahan, B. H., Young, C. E., Gaseous Thermal Electron Reactions: Attachment to SF_6 and C_7F_{14} , *J. Chem. Phys.*, **44**, 2191-2196 (1966).
- Mixtures studied were $NO + He$, $He + NO + SF_6$ and $He + NO + C_7F_{14}$.
- 01948 Blair, D. T. A., The Use of Alpha-Particle Irradiation in Ionization Coefficient Measurements in Gases, *Brit. J. Appl. Phys.*, **17**, 1051-1060 (1966).
- 01949 Cookson, A. H., Positive-Ion and Electron Drift Velocities in Compressed Nitrogen and Methane, *Brit. J. Appl. Phys.*, **17**, 1069-1074 (1966).
- 01950 Heylen, A. E. D., Dargan, C. L., Calculated Magnetic Electron Drift Velocities in H_2 , N_2 , O_2 , Air and C_2H_6 for $10 < E/p < 10^4$ V m^{-1} torr $^{-1}$, *Brit. J. Appl. Phys.*, **17**, 1075-1082 (1966).
- 01958 Cookson, A. H., Ward, B. W., Townsend's First Ionisation Coefficient in Compressed Methane, *Electron. Letters*, **1**, 83-84 (1965).
- 01961 Gusinow, M. A., Gerardo, J. B., Verdeyen, J. T., Investigation of Electronic Recombination in Helium and Argon Afterglow Plasmas by Means of Laser Interferometric Measurements, *Phys. Rev.*, **149**, 91-96 (1966).
- 01963 Stafford, B., Durham, J., Schluter, H., Helium Recombination in a Radio-Frequency Discharge, *J. Chem. Phys.*, **45**, 670-678 (1966).
- 01964 Hurst, G. S., Parks, J. E., Time-of-Flight Determinations of Electron Diffusion Coefficients and Electron Drift Velocities in Ethylene, Water Vapor, and in Hydrogen, *J. Chem. Phys.*, **45**, 282-295 (1966).
- 01971 Haines, W. B., Ionic Mobilities in Hydrogen and Nitrogen, *Phil. Mag.*, **30**, 503-509 (1915).
- 02008 Vorobeva, N. A., Kagan, Yu. M., Milenin, V. M., Electron Velocity Distribution in the Positive Column of a Mercury Discharge. I., *Soviet Phys. Tech. Phys. English Transl.*, **8**, 423-426 (1963).
- 02010 Smit, J. A., Berechnung der Geschwindigkeitsverteilung der Elektronen bei Gasentladungen in Helium, *Physica*, **3**, 543-560 (1936).
- 02018 Hamberger, S. M., Electrical Conductivity of Highly Turbulent Plasma, *Friedman Letters*, **21**, 674-677 (1968).
- 02019 Vorobeva, N. A., Kagan, Yu. M., Lyagushchenko, R. I., Milenin, V. M., Velocity Distribution Function for the Electrons in the Positive Column of a Mercury Discharge. II. *Soviet Phys. Tech. Phys. English Transl.*, **9**, 114-116 (1964).
- 02028 Afanaseva, V. L., Lukin, A. V., Mustafin, K. S., Electron Energy Distribution in a Hollow-Cathode Discharge in a Helium-Neon Mixture, *Soviet Phys. Tech. Phys. English Transl.*, **11**, 389-395 (1966).
- Mixture studied was $He + Ne$.
- 02029 Morgulis, N. D., Polushkin, I. N., Recombination in High-Pressure Helium and Argon Afterglow with Cesium Impurities, *Soviet Phys. Tech. Phys. English Transl.*, **11**, 401-405 (1966).
- Mixtures studied were $He + Cs$ and $Ar + Cs$.
- 02031 Bates, D. R., Kingston, A. E., Recombination through Electron-Electron Collisions, *Nature*, **189**, 652-653 (1961).
- 02034 Stuart, G. W., Gerjuoy, E., Soluble Three-Dimensional Model for Townsend's α , *Phys. Rev.*, **119**, 892-899 (1960).
- 02037 Gerjuoy, E., Stuart, G. W., Ionization Growth in a Gas with a Constant Electric Field, *Phys. Fluids*, **3**, 1008-1015 (1960).
- 02038 Litvak, M. M., Edwards, D. F., Electron Recombination in Laser-Produced Hydrogen Plasma, *J. Appl. Phys.*, **37**, 4462 (1966).
- 02039 McIntosh, A. I., Electron Drift and Diffusion in Deuterium at 293 K, *Australian J. Phys.*, **19**, 805 (1966).
- 02040 Haydon, S. C., Stock, H. M. P., Non-Equilibrium Ionization Growth in Molecular Hydrogen, *Australian J. Phys.*, **19**, 795-803 (1966).
- 02041 Elford, M. T., The Drift Velocity of Electrons in Carbon Dioxide at 293°K, *Australian J. Phys.*, **19**, 629-634 (1966).
- 02042 Kirshner, J. M., Toffolo, D. S., Drift Velocity of Electrons in Argon and Argon Mixtures, *J. Appl. Phys.*, **23**, 594-598 (1952).
- Mixture studied was $Ar + N_2$.
- 02043 Bowe, J. C., Electron Velocity Distributions in Gases, *Am. J. Phys.*, **31**, 905-921 (1963).
- 02044 Pidduck, F. B., The Abnormal Kinetic Energy of an Ion in a Gas, *Proc. Roy. Soc. London Ser. A*, **88**, 296-302 (1913).
- 02045 Dunlop, S. H., Emeleus, K. G., The Electron Energy Distribution in Helium, *Brit. J. Appl. Phys.*, **2**, 163 (1951).
- 02046 Phelps, A. V., Pack, J. L., Electron Collision Frequencies in Nitrogen and in the Lower Ionosphere, *Phys. Rev. Letters*, **3**, 340-342 (1959).
- 02047 Prasad, A. N., Measurement of Ionization and Attachment Coefficients in Dry Air in Uniform Fields and the Mechanism of Breakdown, *Proc. Phys. Soc. London*, **74**, 33-40 (1959).
- 02048 Druyvesteyn, M. J., Calculation of Townsend's α for Ne, *Physica*, **3**, 65-74 (1936).
- 02049 Wahlin, H. B., The Motion of Electrons in Nitrogen, *Phys. Rev.*, **23**, 169-177 (1924).
- 02050 Llewellyn-Jones, F., Parker, A. B., Electrical Breakdown of Gases. I. Spark Mechanism in Air, *Proc. Roy. Soc. London Ser. A*, **213**, 185-202 (1952).
- 02052 Schlumbohm, H., Zur Statistik der Elektronenlawinen im Ebenen Feld. III., *Z. Phys.*, **151**, 563-576 (1958).
- Organic Gases Include $CO(CH_3)_2$ and $CH_2(OCH_3)_2$.
- 02053 Frommhold, L., Zur Statistik der Elektronenlawinen im Ebenen Feld. II., *Z. Phys.*, **150**, 172-181 (1958).
- 02054 Huxley, L. G. H., Zaazou, A. A., Experimental and Theoretical Studies of the Behaviour of Slow Electrons in Air. I., *Proc. Roy. Soc. London Ser. A*, **196**, 402-426 (1949).
- 02055 Klema, E. D., Allen, J. S., Drift Velocities of Electrons in Argon, Nitrogen, and Argon-Nitrogen Mixtures, *Phys. Rev.*, **77**, 661-665 (1950).
- Mixture studied was $Ar + N_2$.
- 02061 Hamilton, N., Stockdale, J. A., Interaction of Thermal Electrons with Polar Molecules, *Australian J. Phys.*, **19**, 813-822 (1966).

- Mixtures studied were of C_2H_4 with $NH_2CH_2CH_2NH_2$, CH_3COCH_3 , $NH_2CH_2CH_2OH$, $CH_3CH_2CH_2OH$, $(CH_3)_2CHOH$, C_6H_5Cl , $CH_3CH_2OCH_2CH_3$, $(C_2H_5)_2NH$, and $CH_3CH_2CH_3$.
- 02086 Penning, F. M., De Elementaire Processen bij de Doorslag Van Gassen Tusschen Vlakke Evenwijdige Platen, Ned. Tijdschr. Natuurk., **5**, 33-56 (1938).
- 02087 Deas, H. D., Emeleus, K. G., Determination of the Electron Energy Distribution in Gases from Townsend's Ionization Coefficient, Phil. Mag., **40**, 460-465 (1949).
- 02088 Bailey, V. A., Duncanson, W. E., On the Behaviour of Electrons Amongst the Molecules NH_3 , H_2O and HCl , Phil. Mag., **10**, 145-160 (1930).
- 02089 Brose, H. L., The Motions of Electrons in Oxygen, Phil. Mag. **50**, 536-546 (1925).
- 02090 Loeb, L. B., The Mobility of Electrons in Pure Nitrogen, Phys. Rev., **19**, 24-37 (1922).
- 02093 Cravath, A. M., The Rate of Formation of Negative Ions by Electron Attachment, Phys. Rev., **33**, 605-613 (1929).
- Mixture studied was $H_2O + O_2$.
- 02094 Davydov, B., Über die Geschwindigkeitsverteilung der Sich im Elektrischen Felde Bewegenden Elektronen, Physik Z. Sowjetunion, **8**, 59-70 (1935).
- 02096 Kruithof, A. A., Druyvesteyn, M. J., The Townsend Ionization Coefficient α and some Elementary Processes in Neon with Small Admixtures of Argon, Physica, **4**, 450-463 (1937).
- Mixture studied was $Ar + Ne$.
- 02097 Crompton, R. W., Huxley, L. G. H., Sutton, D. J., Experimental Studies of the Motions of Slow Electrons in Air with Application to the Ionosphere, Proc. Roy. Soc. London Ser. A, **218**, 507-519 (1953).
- 02098 Borodin, V. S., Kagan, Yu. M., Lyagushchenko, R. I., Investigation of a Hollow-Cathode Discharge. II., Soviet Phys. Tech. Phys. English Transl., **11**, 887-889 (1967).
- 02099 Bradbury, N. E., Formation of Negative Ions in Gases by Electron Attachment. Part I. NH_3 , CO , NO , HCl and Cl_2 , J. Chem. Phys., **2**, 827-834 (1934).
- Mixtures studied were $NH_3 + N_2$, $NH_3 + Ar$, $NH_3 + He$.
- 02100 Bradbury, N. E., Tatel, H. E., The Formation of Negative Ions in Gases. Part II. CO_2 , N_2O , SO_2 , H_2S , and H_2O , J. Chem. Phys., **2**, 835-840 (1934).
- Mixtures studied were $N_2O + Ar$, $N_2O + N_2$.
- 02101 Frommhold, L., Zur Statistik der Elektronenlawinen im Ebenen Feld, Z. Phys., **144**, 396-410 (1956).
- 02102 Harrison, M. A., Townsend's First Ionization Coefficient in Nitrogen, Phys. Rev., **105**, 360-368 (1957).
- 02104 Townsend, J. S., Tizard, H. T., The Motion of Electrons in Gases, Proc. Roy. Soc. London Ser. A, **88**, 336-347 (1913).
- 02106 Cragge, J. D., Hopwood, W., Ion Concentrations in Spark Channels in Hydrogen, Proc. Phys. Soc. London, **59**, 755-771 (1947).
- 02108 Engstrom, R. W., Huxford, W. S., Time-Lag Analysis of the Townsend Discharge in Argon with Activated Caesium Electrodes, Phys. Rev., **58**, 67-77 (1940).
- 02109 Lunt, R. W., Meek, C. A., Ionization, Excitation, and Chemical Reaction in Uniform Electric Fields. II. The Energy Balance and Energy Efficiencies for the Principal Electron Processes in Hydrogen, Proc. Roy. Soc. London Ser. A, **157**, 146-166 (1936).
- 02110 Huxley, L. G. H., Crompton, R. W., A Note on the Diffusion in a Gas of Electrons from a Small Source, Proc. Phys. Soc. London, **68**, 381-383 (1955).
- 02111 Luhr, O., The Recombination of Ions in Argon, Nitrogen, and Hydrogen, Phys. Rev., **36**, 24-34 (1930).
- 02113 Allis, W. P., Brown, S. C., High Frequency Electrical Breakdown of Gases, Phys. Rev., **87**, 419-424 (1952).
- 02130 Brodskii, V. B., Zagik, S. E., Lyutomskii, V. A., Determination of the Recombination Coefficient of Xenon in a Decaying Plasma, Soviet Phys. Tech. Phys. English Transl., **11**, 478-479 (1966).
- 02133 Wahlin, H. B., The Motion of Electrons in Hydrogen and Helium, Phys. Rev., **27**, 588-595 (1926).
- 02134 Sanders, F. H., The Value of the Townsend Coefficient for Ionization by Collision at Large Plate Distances and Near Atmospheric Pressure, Phys. Rev., **41**, 667-677 (1932).
- 02135 Loeb, L. B., The Mobilities of Electrons in Helium, Phys. Rev., **23**, 157-168 (1924).
- 02136 Wagner, K. H., Raether, H., Untersuchung von Elektronenlawinen mit Einem Mehrstufigen Bildverstärker, Z. Phys., **170**, 540-544 (1962).
- 02137 Loeb, L. B., The Mobilities of Electrons in Hydrogen, Phys. Rev., **20**, 397-404 (1922).
- 02138 Davies, D. E., Milne, J. G. C., First Ionization Coefficients in Hydrogen, Neon, and Argon, Brit. J. Appl. Phys., **10**, 301-306 (1959).
- 02139 Kruithof, A. A., Penning, F. M., Determination of the Townsend Ionization Coefficient α for Pure Argon, Physica, **3**, 515-533 (1936).
- 02140 Rees, J. A., Measurements of Townsend's Energy Factor k_1 for Electrons in Carbon Dioxide, Australian J. Phys., **17**, 462-471 (1964).
- 02141 Bortner, T. E., Hurst, G. S., An Apparatus for Measuring Electron Attachment: Results for Oxygen in Argon, Health Phys., **1**, 39-45 (1958).
- Mixture studied was $Ar + O_2$.
- 02142 Heylen, A. E. D., The Influence of the Townsend-Ramsauer Effect on the Electron Transport Coefficients of Molecular Gases, Proc. Phys. Soc. London, **80**, 1109-1116 (1962).
- 02143 Abdelnabi, I., Massey, H. S. W., Inelastic Collisions of Electrons in Helium and Townsend's Ionization Coefficient, Proc. Phys. Soc. London, **66**, 288-296 (1953).
- 02144 Dutton, J., Haydon, S. C., Llewellyn-Jones, F., Electrical Breakdown of Gases. II. Spark Mechanism in Nitrogen, Proc. Roy. Soc. London Ser. A, **213**, 203-214 (1952).
- 02145 Golden, D. E., Nakano, H., Fisher, L. H., First Townsend Ionization Coefficient in Hydrogen, Phys. Rev., **138**, A1613-1616 (1965).
- 02147 Bates, D. R., Kingston, A. E., Recombination and Energy Balance in a Decaying Plasma. I. $H-H^+-e$ Plasma, Proc. Roy. Soc. London Ser. A, **279**, 10-31 (1964).
- 02148 Haydon, S. C., Robertson, A. C., Pre-Breakdown Ionization in Hydrogen at Low Pressures, Proc. Phys. Soc. London, **78**, 92-102 (1961).
- 02149 Kuffel, E., A Note on the Cross Section for Electron Attachment in Air, Proc. Phys. Soc. London, **71**, 516-517 (1958).
- 02150 Jones, E., Llewellyn-Jones, F., The Experimental Determination of the Primary Ionization Coefficients at Low Gas Pressures, Proc. Phys. Soc. London, **72**, 363-368 (1958).
- 02151 Stevenson, A., Electron Velocities in Geiger Counter Gas Mixtures. Letter I, Rev. Sci. Instr., **23**, 93-94 (1952).
- Mixtures studied were $Ar + CH_3(CH_2)_2CH_3$ and $Ar + (C_2H_5)_2O$.
- 02152 Wu, C. S., The Effect of the Orientations of Electric and Magnetic Fields on the Electron Mean Energy and Drift Velocity in a Partially Ionized Gas, Proc. Roy. Soc. London Ser. A, **259**, 518-530 (1961).
- 02153 Jager, G., Otto, W., Driftgeschwindigkeiten von Ionen und Elektronen in Argon und Wasserstoff, Z. Phys., **169**, 517-525 (1962).
- 02154 Frommhold, L., Eine Untersuchung der Elektronen-

- komponente von Elektronenlawinen im Homogenen Feld II, *Z. Phys.*, **160** 554-567 (1960).
- 02155 Dibbern, U., Untersuchung der Elektronenkomponente von Einzellawinen mit Dem Photomultiplier, *Z. Phys.*, **163**, 582-593 (1961).
- Mixture studied was $N_2 + CH_4$.
- 02156 Franke, W., Über den Zeitlichen Verlauf der Lichtemission Einer Gasentladung, *Z. Phys.*, **158**, 96-110 (1960).
- 02157 MacDonald, A. D., Brown, S. C., High Frequency Gas Discharge Breakdown in Hydrogen, *Phys. Rev.*, **76**, 1634-1639 (1949).
- 02158 Wahlin, H. B., Behavior of Free Electrons toward Gas Molecules, *Phys. Rev.*, **19**, 173-186 (1922).
- 02162 Haydon, S. C., Robertson, A. G., The Effective Collision Frequency in Molecular Hydrogen in the Presence of a Transverse Magnetic Field, *Proc. Phys. Soc. London*, **82**, 343-352 (1963).
- 02163 Borodin, V. S., Kagan, Yu. M., A Study of the Hollow-Cathode Discharge, *Opt. Spectry. USSR English Transl.*, **18**, 546-547 (1965).
- 02164 Anderson, J. M., Hall Effect and Electron Drift Velocities in the Plasma of the Positive Column, *Phys. Fluids*, **7**, 1517-1526 (1964).
- 02165 Pitaevskii, L. P., Electron Recombination in a Monatomic Gas Soviet Phys. JETP English Transl., **15**, 919-921 (1962).
- 02168 Bates, D. R., Kingston, A. E., Recombination and Energy Balance in a Decaying Plasma. II. $He-He^+-e$ Plasma, *Proc. Roy. Soc. London Ser. A*, **279**, 32-38 (1964).
- 02169 Oskam, H. J., Microwave Investigation of Disintegrating Gaseous Discharge Plasmas, *Philips Res. Rept.*, **13**, 401-457 (1958).
- Mixture studied was $He + Ne$.
- 02183 Chambers, R. G., The Kinetic Formulation of Conduction Problems, *Proc. Phys. Soc. London*, **65**, 458-459 (1952).
- 02184 MacDonald, A. D., Brown, S. C., High Frequency Gas Discharge Breakdown in Helium, *Phys. Rev.*, **75**, 411-418 (1949).
- 02185 Margenau, H., Conductivity of Plasmas to Microwaves, *Phys. Rev.*, **109**, 6-9 (1958).
- 02187 Huxley, L. G. H., A General Formula for the Conductivity of a Gas Containing Free Electrons, *Proc. Phys. Soc. London*, **64**, 844-861 (1951).
- 02188 Herlin, M. A., Brown, S. C., Breakdown of a Gas at Microwave Frequencies, *Phys. Rev.*, **74**, 291-296 (1948).
- 02192 Margenau, H., Theory of High Frequency Gas Discharges. I. Methods for Calculating Electron Distribution Functions, *Phys. Rev.*, **73**, 297-308 (1948).
- 02193 Margenau, H., Hartman, L. M., Theory of High Frequency Gas Discharges. II. Harmonic Components of the Distribution Function, *Phys. Rev.*, **73**, 309-315 (1948).
- 02194 Hartman, L. M., Theory of High Frequency Gas Discharges. III. High Frequency Breakdown, *Phys. Rev.*, **73**, 316-325 (1948).
- 02196 Didlauskis, M., Über die Energieverteilung Diffundierender Langsamer Elektronen, *Z. Phys.*, **82**, 709-715 (1933).
- 02197 Druyvesteyn, M. J., Die Geschwindigkeitsverteilung der Elektronen in der Positiven Saule, *Physik Z.*, **33**, 856-863 (1932).
- 02198 Sodha, M. S., Transport Phenomena in Slightly Ionized Gases: Low Electric Fields, *Phys. Rev.*, **116**, 486-488 (1959).
- 02199 Molmud, P., Langevin Equation and the AC Conductivity of Non-Maxwellian Plasmas, *Phys. Rev.*, **114**, 29-32 (1959).
- 02201 Compton, K. T., On the Motions of Electrons in Gases, *Phys. Rev.*, **22**, 333-346 (1923).
- 02202 Compton, K. T., Mobilities of Electrons in Gases, *Phys. Rev.*, **22**, 432-444 (1923).
- 02203 Phelps, A. V., Propagation Constants for Electromagnetic Waves in Weakly Ionized, Dry Air, *J. Appl. Phys.*, **31**, 1723-1729 (1960).
- 02224 Bozin, S. E., Goodyear, C. C., Measurements of Ionization and Attachment Coefficients in Chlorine, *Brit. J. Appl. Phys.*, **18**, 49-57 (1967).
- 02225 Townsend, J. S., Distributions of Energies of Electrons, *Phil. Mag.*, **22**, 145-171 (1936).
- 02229 Nygaard, K. J., On the Emission of $H\alpha$ Radiation from Selfsustained Townsend Discharges in Hydrogen, (In) *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **2**, 191-195 (1963).
- 02230 Craggs, J. D., Afterglow Studies of Transient Discharges, (In) *Proceedings of the Sixth International Conference on Ionization Phenomena in Gases* (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 423-428 (1963).
- Mixture studied was $He + Ar$.
- 02231 Bhiday, M. R., Paithankar, A. S., Sharda, B. L., Collision Frequency in Air with Crossed Electric and Magnetic Fields, *Brit. J. Appl. Phys.*, **18**, 241-242 (1967).
- 02232 Formato, D., Gilardini, A., Microwave Determinations of Afterglow Temperatures and Electron Collision Frequencies in Nitrogen, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 99-104 (1960).
- 02235 Ryzko, H., Astrom, E., Electron Attachment-Detachment Processes in Dry Air, *J. Appl. Phys.*, **38**, 328-330 (1967).
- 02236 Vorobeva, N. A., Kagan, Yu. M., Milenin, V. M., Electron Distribution Function in the Positive Column of a Discharge in Neon and in Helium, *Soviet Phys. Tech. Phys. English Transl.*, **9**, 1598-1600 (1965).
- 02237 Kruithof, A. A., Townsend's Ionization Coefficients for Neon Argon. Krypton and Xenon, *Physica*, **7**, 519-540 (1940).
- 02239 Bennett, W. H., Mobilities in Hydrogen at High Current Densities, *Phys. Rev.*, **58**, 992-997 (1940).
- 02247 Hale, D. H., The Townsend Coefficients for Ionization by Collision in Pure and Contaminated Hydrogen as a Function of the Cathode Material, *Phys. Rev.*, **55**, 815-819 (1939).
- Mixture studied was $H_2 + Na + NaH$.
- 02248 Hale, D. H., The Townsend Ionization Coefficients for Ni and Al Cathodes in an Atmosphere of Hydrogen, *Phys. Rev.*, **56**, 1199-1202 (1939).
- 02250 Orient, O. J., Calculation of the Diffusion Coefficient to Mobility Ratio of Electrons for Noble Gases, *Acta Phys. Acad. Sci. Hung.*, **21**, 293-302 (1966).
- 02254 Raether, H., Untersuchung der Elektronenlawine mit der Nebelkammer, *Z. Phys.*, **107**, 91-110 (1937).
- 02255 Huxley, L. G. H., Free Path Formulae for the Electronic Conductivity of a Weakly Ionized Gas in the Presence of a Uniform and Constant Magnetic Field and a Sinusoidal Electric Field, *Australian J. Phys.*, **10**, 240-245 (1957).
- 02256 Huxley, L. G. H., Free Path Formulae for the Coefficient of Diffusion D and Velocity of Drift W of Ions and Electrons in Gases, *Australian J. Phys.*, **13**, 578-583 (1960).
- 02259 Seaton, M. J., H I, He I and He II Intensities in Planetary Nebulae, *Monthly Notices Roy. Astron. Soc.*, **120**, 326-337 (1960).

- 02261 Nagy, T., Nagy, L., Desi, S., Drift Velocities of Electrons in Argon, Nitrogen and Argon Mixtures, Nucl. Instr. Methods, **8**, 327-330 (1960).
Mixtures studied were Ar + N₂, Ar + CO₂, Ar + CH₄.
- 02262 Herreng, P., Application des Impulsions de Rayons X a la Mesure de la Mobilité des Electrons dans les Gaz, Compt. Rend., **215**, 79-83 (1942).
- 02263 Loeb, L. B., The Mobilities of Electrons in Air, Proc. Natl. Acad. Sci. U. S., **9**, 335-341 (1923).
- 02264 Herreng, P., Sur la Mobilité des Electrons Libres dans l'Argon, Compt. Rend., **217**, 75-77 (1943).
- 02266 Levine, J. L., Sanders, T. M., Mobility of Electrons in Low-Temperature Helium Gas, Phys. Rev., **154**, 138-149 (1967).
- 02267 Crompton, R. W., McIntosh, A. I., Electron Transport Coefficients in Gaseous Parahydrogen, Phys. Rev. Letters, **18**, 527-528 (1967).
- 02268 Wilson, L. N., Evans, E. W., Electron Recombination in Hydrocarbon-Oxygen Reactions Behind Shock Waves, J. Chem. Phys., **46**, 859-863 (1967).
Mixtures studied were Ar + O₂ + CH₄, Ar + O₂ + C₂H₆, Ar + O₂ + C₃H₈, Ar + O₂ + C₂H₄, Ar + O₂ + C₂H₂, Ar + O₂ + C₆H₆.
- 02269 Bowman, C. R., Gordon, D. E., Drift Velocities of Slow Electrons in Methane, Ethane, Ethylene, Propylene, Acetylene and i-Butene, J. Chem. Phys., **46**, 1878-1883 (1967).
- 02270 Marshall, I. C., The Recombination of Ions and of Ions and Electrons in Gases, Phys. Rev., **34**, 618-634 (1929).
- 02271 Townsend, J. S., The Conductivity Produced in Gases by the Aid of Ultra-Violet Light, Phil. Mag., **3**, 557-577 (1902).
- 02272 Biberman, L. M., Toropkin, Yu N., Ulyanov, K. N., The Theory of Multistage Ionization and Recombination, Soviet Phys. Tech. Phys. English Transl., **7**, 605-609 (1963).
- 02273 Bailey, V. A., McGee, J. D., On the Capture of Electrons by Molecules, Phil. Mag., **6**, 1073-1089 (1928).
- 02274 Sanders, F. H., Measurement of the Townsend Coefficients for Ionization by Collision, Phys. Rev., **44**, 1020-1024 (1933).
- 02275 Bowls, W. E., The Effect of Cathode Material on the Second Townsend Coefficient for Ionization by Collision in Pure and Contaminated N₂ Gas, Phys. Rev., **53**, 293-301 (1938).
Mixtures studied were N₂ + Hg and N₂ + Na.
- 02276 Emelius, K. G., Lunt, R. W., Meek, C. A., Ionization, Excitation, and Chemical Reaction in Uniform Electric Fields. I-The Townsend Coefficient of Ionization, Proc. Roy. Soc. London Ser. A, **156**, 394-411 (1936).
- 02277 Masch, K., Über Elektronenionisierung bei Geringen und Hohen Drucken, Z. Phys., **79**, 672-675 (1932).
- 02278 Bailey, V. A., The Behaviour of Electrons in Magnetic Fields Phil. Mag., **9**, 560-567 (1930).
- 02279 Bailey, V. A., The Behaviour of Electrons in Magnetic Fields Phil. Mag., **9**, 625-629 (1930).
- 02280 Bailey, V. A., The Motion of Electrons in Neon, Phil. Mag., **47**, 379-384 (1924).
Mixture studied was Ne + He.
- 02281 Townsend, J. S., Hurst, H. E., The Genesis of Ions by the Motion of Positive Ions, and a Theory of the Sparking Potential, Phil. Mag., **8**, 738-753 (1904).
- 02284 McGee, J. D., Jaeger, J. C., The Motion of Electrons in Pentane, Phil. Mag., **6**, 1107-1117 (1928).
- 02285 Prasad, A. N., Smeaton, C. P., Drift Velocities of Electrons in Nitrogen and Hydrogen, Brit. J. Appl. Phys., **18**, 371-373 (1967).
- 02286 Dutton, J., Rees, D. B., Ionization Processes in Helium at Pressures near to Atmospheric, Brit. J. Appl. Phys., **18**, 309-315 (1967).
- 02287 Penning, F. M., The Starting Potential of the Glow Discharge in Neon Argon Mixtures between Large Parallel Plates. II. Discussion of the Ionisation and Excitation by Electrons and Metastable Atoms, Physica, **1**, 1028-1044 (1934).
Mixture studied was Ne + Ar.
- 02289 Hall, B. I. H., Experimental Investigation of the Motions of Electrons in a Gas in the Presence of a Magnetic Field, Proc. Phys. Soc. London B, **68**, 334-341 (1955).
Discharge, Phys. Rev., **81**, 153-154 (1951).
- 02292 Richardson, J. M., Holt, R. B., Decay of the Hydrogen Discharge, Phys. Rev., **81**, 153-154 (1951).
- 02294 Prasad, A. N., Craggs, J. D., Measurement of Ionization and Attachment Coefficients in Humid Air in Uniform Fields and the Mechanism of Breakdown, Proc. Phys. Soc. London, **76**, 223 (1960).
Mixture studied was Air + H₂O.
- 02295 Madan, M. P., Gordon, E. I., Buchsbaum, S. J., Brown, S. C., Determination of the Coefficient of Diffusion and Frequency of Ionization in Microwave Discharges, Phys. Rev., **106**, 839 (1957).
- 02297 Dreicer, H., Electron Velocity Distributions in a Partially Ionized Gas, Phys. Rev., **117**, 343 (1960).
- 02300 Howard, P. R., Correlation between Ionizing and Attachment Coefficients and Breakdown Characteristics for Carbon Tetrafluoride, Nature, **181**, 645 (1958).
- 02302 Moruzzi, J. L., Craggs, J. D., Measurement of Ionization and Attachment Coefficients in C₃F₈, Proc. Phys. Soc. London **82**, 979 (1963).
- 02303 Blevin, H. A., Haydon, S. C., The Interpretation of Breakdown Characteristics of Electrodeless Discharges in Transverse Magnetic Fields, Proc. Phys. Soc. London, **81**, 490 (1963).
- 02305 Ryzko, H., Drift Velocity of Electrons and Ions in Dry and Humid Air and in Water Vapour, Proc. Phys. Soc. London, **85**, 1283 (1965).
Mixture studied was Air + H₂O.
- 02307 Cahn, J. H., Electron Velocity Distribution Function in High Frequency Alternating Fields Including Electronic Interactions, Phys. Rev., **75**, 838 (1949).
- 02321 Childs, E. C., Collisional Friction on Electrons Moving in Gases, Phil. Mag., **13**, 873-887 (1932).
- 02323 Lichtenstein, R., Die Geschwindigkeitsverteilung Elastisch Stossender Elektronen in Einem Gas Dessen Moleküle ihre Temperaturbewegung Ausführen, Physik Z., **39**, 646 (1938).
- 02324 Druyvesteyn, M. J., Bemerkungen zu Zwei Früheren Arbeiten über die Elektronendiffusion, Physica, **1**, 1003 (1934).
- 02325 Blevin, H. A., Haydon, S. C., The Townsend Ionization Coefficients in Crossed Electric and Magnetic Fields, Australian J. Phys., **11**, 18 (1958).
- 02329 Anderson, J. M., The Very Early Afterglow in Gaseous Helium, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August through 1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, **1**, 621 (1962).
- 02330 Hinnov, E., Hirschberg, J. G., Spectroscopic Measurements of Helium Afterglow, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August through 1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, **1**, 638 (1962).
- 02331 Motley, R. W., Kuckes, A. F., Recombination in a Helium Plasma, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28

- August through 1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, 1, 651 (1962).
- 02333 Golden, D. E., Fisher, L. H., Anomalies in Ionization Coefficients and in Uniform Field Breakdown in Argon for Low Values of E/P , *Phys. Rev.*, **123**, 1079 (1961).
- 02337 Davydov, B., Über die Geschwindigkeitsverteilung der Elektronen im Elektrischen Feld. II., *Physik Z. Sowjetunion*, **9**, 433 (1936).
- 02338 Huxley, L. G. H., Motions of Electrons in Magnetic Fields and Alternating Electric Fields, *Phil. Mag.*, **23**, 442 (1937).
- 02339 Gvosdover, S. D., The Mobility and Mean Free Path of Electrons in the Positive Column, *Physik Z. Sowjetunion*, **12**, 164 (1937).
- 02340 Davydov, B., On the Theory of the Motion of Electrons in Gases and Semi-Conductors, *Physik Z. Sowjetunion*, **12**, 269 (1937).
- 02341 Glotov, I. I., Über die Messung des Ionisations-koeffizienten in Reinem Neon und in Neon-Argon-Gemischen, *Physik Z. Sowjetunion*, **12**, 256 (1937).
- Mixture studied was Ne + Ar.
- 02343 Meisel, L. V., Method for the Determination of Electron Drift Velocity in Gases from the Characteristics of High-Pressure Gas Diodes, *J. Appl. Phys.*, **36**, 1389 (1965).
- 02344 Vorobeva, N. A., Kagan, Yu. M., Milenin, V. M., Electron Velocity Distribution Function in the Positive Column in a Mixture of Gases, *Soviet Phys. Tech. Phys. English Transl.*, **9**, 632 (1964).
- Mixtures studied were Hg + He, Hg + Ne, Hg + Ar, Hg + Xe.
- 02345 Moruzzi, J. L., Ionization and Attachment in Difluorodichloromethane, *Brit. J. Appl. Phys.*, **14**, 938 (1963).
- 02346 Huxford, W. S., Townsend Ionization Coefficients in Cs-Ag-O Photo-Tubes Filled with Argon, *Phys. Rev.*, **55**, 754 (1939).
- 02347 Compton, K. T., The Theory of Ionization by Collision. III. Case of Elastic Impact, *Phys. Rev.*, **7**, 509 (1916).
- 02348 Kagan, Yu. M., Lyagushchenko, R. I., Electron Energy Distribution Function in the Positive Column of a Neon Discharge, *Soviet Phys. Tech. Phys. English Transl.*, **6**, 321 (1961).
- 02351 Druyvesteyn, M. J., The Mobility of Electrons in Neon, *Physica*, **4**, 464 (1937).
- 02352 Kihara, T., The Mathematical Theory of Electrical Discharges in Gases, *Rev. Mod. Phys.*, **24**, 45 (1952).
- 02354 Chanin, L. M., Rork, G. D., Pressure-Dependent Breakdown, Potentials in Penning Mixtures, *J. Appl. Phys.*, **36**, 1515 (1965).
- Mixtures studied were He + H₂, Ne + H₂, Ne + Ar.
- 02355 Kruthof, A. A., Penning, F. M., Determination of the Townsend Ionization Coefficient Alpha for Mixtures of Neon and Argon, *Physica*, **4**, 430 (1937).
- Mixture studied was He + Ar.
- 02357 Vorobev, A. A., Ivanov, B. A., Komar, A. P., Korolev, V. A., The Influence of the Ramsauer-Townsend Effect upon Electron Mobility in Spectrally Pure Argon, *Soviet Phys. Tech. Phys. English Transl.*, **4**, 1148 (1960).
- 02367 Costa, H., Über die Nachlieferungselektronen durch Photoeffekt in einer Unselbständigen Wasserstoffentladung, *Z. Phys.*, **113**, 531 (1939).
- 02368 Grigorovici, R., Die Zündspannung von Reinem Quecksilberdampf, *Z. Phys.*, **111**, 596 (1939).
- 02369 Haseltine, W. R., The Mutual Interaction of Plasma Electrons, *J. Math. Phys.*, **18**, 174 (1939).
- 02370 Frommhold, L., Über Verzögerte Elektronen in Elektronenlawinen, Insbesondere in Sauerstoff und Luft, Durch Bildung und Zerfall Negativer Ionen O⁻, *Fortschr. Physik*, **12**, 597 (1964).
- 02372 Van Lint, V. A. J., Ionization Afterglow Measurements on Nitrogen, *IEEE Trans. Nucl. Sci.*, **11**, 266 (1964).
- 02385 Bailey, V. A., Somerville, J. M., The Behaviour of Electrons in Nitric Oxide, *Phil. Mag.*, **17**, 1169 (1934).
- 02387 Schlumbohm, H., A Method for Determining the Ionisation Coefficient by the Statistics of Avalanches, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 1B 127 (1960).
- Organic compounds studied were CO(CH₃)₂, CH₂(OCH₃)₂, and C₂H₅OC₂H₅.
- 02388 Dutton, J., Llewellyn-Jones, F., Palmer, R. W., Measurement of Ionization and Attachment Coefficients in Air, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 1B 137 (1960).
- 02390 Burgess, A., The Hydrogen Recombination Spectrum, *Monthly Notices Roy. Astron. Soc.*, **118**, 477 (1958).
- 02392 Compton, K. T., The Theory of Ionization by Collision. II. Case of Inelastic Impact, *Phys. Rev.*, **7**, 501 (1916).
- 02393 Compton, K. T., The Theory of Ionization by Collision. I. The Distribution of Velocities of the Electrons, *Phys. Rev.*, **7**, 489 (1916).
- 02394 Hackam, R., Temperature Dependence of Electron-Ion Recombination and Ion Mobilities in Nitrogen Afterglows, *Planetary Space Sci.*, **13**, 667 (1965).
- 02395 Biondi, M. A., Processes Involving Ions and Metastable Atoms in Mercury Afterglows, *Phys. Rev.*, **90**, 730 (1953).
- 02397 Uman, M. A., Warfield, G., Theoretical Study of the Electron Drift Velocity in Binary Gas Mixtures with Applications to A-CO₂ and A-N₂ Mixtures, *Phys. Rev.*, **120**, 1542 (1960).
- Mixtures studied were Ar + CO₂, Ar + N₂.
- 02399 English, W. N., Hanna, G. C., Grid Ionization Chamber Measurements of Electron Drift Velocities in Gas Mixtures, *Can. J. Phys.*, **31**, 768 (1953).
- Mixtures studied were Ar + N₂, Ar + CO₂, Ar + CH₄, Ne + CO₂, Xe + CO₂, Kr + CO₂, Kr + CH₄.
- 02400 Mohler, F. L., Recombination and Photo-Ionization, *Rev. Mod. Phys.*, **1**, 216 (1929).
- 02401 Kerzar, B., Weissglas, P., Determination of Electron Drift Velocity in a Gaseous Discharge from Doppler Shifts of Plasma Waves, *Electron. Letters*, **1**, 43 (1965).
- 02402 Gill, E. W. B., Comparison of Processes of Ionization which give Rise to Currents in Gases, *Phil. Mag.*, **42**, 852 (1921).
- 02403 Townsend, J. S., MacCallum, S. P., Electrical Properties of Monatomic Gases, *Phil. Mag.*, **5**, 695 (1928).
- 02405 Herlin, M. A., Brown, S. C., Microwave Breakdown of a Gas in a Cylindrical Cavity of Arbitrary Length, *Phys. Rev.*, **74**, 1650 (1948).
- 02406 Dutton, J., Llewellyn-Jones, F., Rees, D. B., Ionization in Helium at High Pressures, *Proc. Phys. Soc. London*, **85**, 909 (1965).
- 02407 Dutton, J., Harris, F. M., Llewellyn-Jones, F., The Determination of Attachment and Ionization Coefficients in Air, *Proc. Phys. Soc. London*, **81**, 52 (1963).
- Mixture studied was Air + H₂O.
- 02408 Chatterton, P. A., Craggs, J. D., Attachment Coefficient Measurements in Carbon Dioxide, Carbon Monoxide, Air and Helium-Oxygen Mixtures, *Proc. Phys. Soc. London*, **85**, 355 (1965).
- Mixture studied was He + O₂.
- 02409 Hess, W., Recombination and Diffusion in a Neon Plasma with Variable Electron Energy, *Phys. Letters*, **12**, 211 (1964).

- 02433 Crompton, R. W., Elford, M. T., Jory, R. L., The Momentum Transfer Cross Section for Electrons in Helium, *Australian J. Phys.*, **20**, 369-400 (1967).
- 02435 Burkley, C. J., Sexton, M. C., Ionization Rates in the Inert Gases, *Brit. J. Appl. Phys.*, **18**, 443 (1967).
- 02436 Cahn, J. H., Electronic Interaction in Electrical Discharges in Gases, *Phys. Rev.*, **75**, 293 (1949).
- 02437 Shkarofsky, I. P., Values of the Transport Coefficients in a Plasma for any Degree of Ionization Based on a Maxwellian Distribution, *Can. J. Phys.*, **39**, 1619 (1961).
- 02438 Tan, C. W., Soo, S. L., Bahadori, M. N., Transport Properties of Non-Equilibrium Ionized Argon Gas, *Z. Angew. Math. Phys.*, **16**, 255 (1965).
- 02442 Badareu, E., Popescu, I., Researches on the Double Cathode Effect, *J. Electron. Control*, **4**, 503 (1958).
- 02445 Lennon, J. J., Sexton, M. C., Recombination in Xenon and Krypton Afterglows, *J. Electron. Control*, **7**, 123 (1959).
- Mixture studied was Kr + Xe.
- 02447 Aleskovskii, Yu. M., Granovskii, V. L., Investigation of Volume Recombination in a Helium Plasma in a Magnetic Field, *Soviet Phys. JETP English Transl.*, **16**, 887 (1963).
- 02448 Bagnall, F. T., Haydon, S. C., Pre-Breakdown Ionization in Molecular Nitrogen in E X B Fields, *Australian J. Phys.*, **18**, 227 (1965).
- 02449 Moruzzi, J. L., Estimation of Initial Cathode Current in Townsend Electron Attachment Experiments, *Brit. J. Appl. Phys.*, **14**, 929 (1963).
- 02450 Somerville, J. M., Sparking Potentials in a Transverse Magnetic Field, *Proc. Phys. Soc. London*, **65**, 620 (1952).
- 02452 Golant, V. E., Formation of a Pulsed Discharge in Argon at Ultra-High Frequencies. I. Electron-Velocity Distribution Function, *Soviet Phys. Tech. Phys. English Transl.*, **2**, 684 (1957).
- Mixture studied was Ar + H₂.
- 02453 Golant, V. E., The Origin of a Pulse Discharge in Neon at Ultra-High Frequencies, *Soviet Phys. Tech. Phys. English Transl.*, **2**, 1370 (1957).
- 02454 Kagan, Yu. M., Mustafin, K. S., Electron Velocity Distribution Function in a Positive, Medium-High Pressure Discharge Column, *Soviet Phys. Tech. Phys. English Transl.*, **5**, 881 (1960).
- 02455 Biondi, M. A., Electron Loss Processes in Ionized Gases, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 19 (1966).
- 02456 Thomas, R. D., Hackam, R., Lennon, J. J., Electron-Ion Recombination in Helium Afterglows, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 27 (1966).
- 02457 Balfour, D., Harris, J. H., Loss Processes in Caesium Plasma (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 32 (1966).
- 02459 Fischer, J., Electron Drift Velocity Variations in Neon + 10 Per Cent Helium + Alcohol, using Spark Chamber Efficiency as a Sensitive Indicator, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 68 (1966).
- Mixture studied was Ne + He + C₂H₅O₄.
- 02461 Taylor, R. L., Narasinga Rao, K. V., Alteration of Electron Loss Processes in Oxygen Magnetoplasma by Microwave Absorption, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 74 (1966).
- 02462 Prasad, A. N., Electron Detachment in Oxygen, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 79 (1966).
- 02463 Crompton, R. W., Rees, J. A., Jory, R. L., The Diffusion and Attachment of Electrons in Water Vapour, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 83 (1966).
- 02465 Edelson, D., Barna, S. F., McAfee, K. B., Townsend Ionization Coefficients in Tritium, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 91 (1966).
- 02466 Cookson, A. H., Ward, B. W., Ionization and Positive Ion Mobility Measurements in Methane at Pressures above Atmospheric, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 95 (1966).
- 02467 Ryzko, H., Measurements of Ionization and Electron-Attachment Coefficients in Humid Air, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August, 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 97 (1966).
- Mixture studied was Air + H₂O.
- 02468 Fletcher, J., Haydon, S. C., Electron Impact Ionization of Molecular Gases in Crossed Electric and Strong Magnetic Fields, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade **1**, 100 (1966).
- 02469 Vorobeva, N. A., Kagan, Yu. M., Lyagushchenko, R. I., Milenin, V. M., Electron Energy Distribution in the Positive Discharged Column, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 330 (1966).
- Mixtures studied were Hg + He and Hg + Ne.
- 02470 Hantzsch, E., Wiczorek, L. W., Energy Distribution Functions in Anomalous Glow Discharges, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 339 (1966).
- 02471 Delcroix, J. L., Conductivite Electrique des Gaz Ionises, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August 1965) B. Perovic and D. Tasic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **2**, 56, (1966).
- 02472 Zauderer, B., Measurement of Electrical Transport Properties in a Shock Tube, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August 1965) B.

- Perovic and D. Tosic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **2**, 99 (1966).
Mixture studied was Xe + H.
- 02473 Dolique, J. H., Some Extensions of Ohm's Law in Gases, (In) Proceedings of the Seventh International Conference on Phenomena in Ionized Gases (Belgrade, 22-27 August 1965) B. Perovic and D. Tosic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **2**, 105 (1966).
- 02474 Irish, R. T., Bryant, G. H., Determination of Ionization Rate and Electron Temperature in a Mercury Discharge by a Microwave Method, *Proc. Phys. Soc. London*, **84**, 975 (1964).
- 02481 Killian, T. J., The Uniform Positive Column of an Electric Discharge in Mercury Vapor, *Phys. Rev.*, **35**, 1238 (1930).
- 02483 Davies, D. K., Dutton, J., Llewellyn-Jones, F., Secondary Ionization Processes in Hydrogen at High Gas Pressures, *Proc. Phys. Soc. London*, **72**, 1061 (1958).
- 02487 Dutton, J., Llewellyn-Jones, F., Palmer, R. W., Electrical Breakdown of Gases-Ionization Growth in Air at High Pressures, *Proc. Phys. Soc. London*, **78**, 569 (1961).
- 02488 Dutton, J., Harris, F. M., Llewellyn-Jones, F., The Influence of the Three-Body Process of Electron Attachment on the Growth of Ionization in Air at High Pressure, *Proc. Phys. Soc. London*, **82**, 581 (1963).
- 02489 Crompton, R. W., Hall, B. I. H., Macklin, W. C., A Source of Inaccuracy Encountered when Measuring Drift Velocities of Electrons in Gases by an Electrical Shutter Method, *Australian J. Phys.*, **10**, 366 (1957).
- 02491 Stein, R. P., Scheibe, M., Syverson, M. W., Shaw, T. M., Gunton, R. C., Recombination Coefficient of Electrons with NO^+ Ions in Shock-Heated Air, *Phys. Fluids*, **7**, 1641 (1964).
- 02492 De Bitetto, D. J., Fisher, L. H., Second Townsend Coefficient in Oxygen at High Pressures, *Phys. Rev.*, **111**, 390 (1958).
- 02493 Chanin, L. M., Rork, G. D., Primary Ionization Coefficient Measurements in Penning Mixtures, *Phys. Rev.*, **135**, A71 (1964).
Mixtures studied were $\text{H}_2 + \text{He}$ and $\text{H}_2 + \text{Ne}$.
- 02494 Chao, K. T., Tang, T. Y., Electron Temperatures in Electrical Discharges, *Phys. Rev.*, **68**, 30 (1945).
- 02496 Townsend, J. S., Uniform Columns in Electric Discharges, *Phil. Mag.*, **11**, 1112 (1931).
- 02498 Thomson, J. J., Recombination of Gaseous Ions, The Chemical Combination of Gases, and Monomolecular Reaction, *Phil. Mag.*, **47**, 337 (1924).
- 02500 Richter, K., Die Tragervermehrung einer Lawine mit Eigenraumladung, *Z. Phys.*, **157**, 130 (1959).
- 02501 Glotov, I. I., Calculation of the Coefficient of Volume Ionization for Pure Neon and Neon-Argon Mixtures, *Physik Z. Sowjetunion*, **13**, 84 (1938).
Mixture studied was Ne + Ar.
- 02502 Lunt, R. W., Meek, C. A., Smith, E. C. W., Ionization, Excitation, and Chemical Reaction in Uniform Electric Fields III. the Excitation of the Continuous Spectrum of Hydrogen, *Proc. Roy. Soc. London Ser. A*, **158**, 729 (1937).
- 02504 Loeb, L. B., The Mobilities of Electrons, *Proc. Natl. Acad. Sci. U. S.*, **7**, 307 (1921).
- 02514 Bennett, W. H., Mobilities in Nitrogen at High Current Densities, *Phys. Rev.*, **62**, 369 (1942).
Mixture studied was $\text{H}_2 + \text{N}_2$.
- 02515 Rose, D. J., Brown, S. C., High-Frequency Gas Discharge Plasma in Hydrogen, *Phys. Rev.*, **98**, 310 (1955).
- 02518 Mohler, F. L., Recombination in the Afterglow of a Mercury Discharge, *J. Res. Nat. Bur. Std.*, **19**, 559 (1937).
- 02519 Blevin, H. A., Haydon, S. C., The Theoretical Evaluation of the Townsend Coefficient α/p for Hydrogen, *Australian J. Phys.*, **10**, 590 (1957).
- 02521 Thompson, J. B., Electron Energy Distributions in Plasmas, IV. Oxygen and Nitrogen, *Proc. Roy. Soc. London*, **262**, 503 (1961).
- 02524 Mallozzi, P., Margenau, H., Absorption Coefficient and Microwave Conductivity of Isotropic Plasmas, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 277 (1963).
- 02527 Stabler, R. C., Upper Limit to Three Body Electron-Ion Recombination Rate, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **1**, 449 (1963).
- 02528 Dutton, J., Harris, F. M., Llewellyn-Jones, F., The Determination of Ionization and Attachment Coefficients from Measurements of Ionization Currents, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan Editors, Serma, Paris, **2**, 179 (1963).
- 02529 Wagner, K. H., The Development of an Electron Avalanche Investigated by Means of a High Gain Image Intensifier, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **2**, 309 (1963).
- 02530 Crawford, F. W., Garscadden, A., Palmer, R. S., A Double-Probe Technique for Determining Electron Velocity Distributions, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **4**, 53 (1963).
- 02532 Compton, R. N., Electron Capture Cross Sections and Negative Ion Lifetimes, Thesis, University of Tennessee, 1966, 180 pages University Microfilms Inc., Ann Arbor, Michigan, Order No. 66-8191.
Mixtures studied were of N_2 with $\text{C}_6\text{H}_5\text{Cl}$, $o\text{-C}_6\text{H}_4\text{Cl}_2$, $m\text{-C}_6\text{H}_4\text{Cl}_2$, $p\text{-C}_6\text{H}_4\text{Cl}_2$, $o\text{-C}_6\text{H}_4\text{CH}_2\text{Cl}$, $\text{C}_6\text{H}_5\text{Br}$, $\text{C}_6\text{D}_5\text{Br}$, $o\text{-C}_6\text{H}_4\text{CH}_2\text{Br}$ and $\text{C}_2\text{H}_5\text{Cl}$, and of Ar with O_2 and H_2O .
- 02533 Cottingham, W. B., Buchsbaum, S. J., Electron Ionization Frequency in Deuterium, (In) Proceedings of the Sixth International Conference on Ionization Phenomena in Gases (Paris, 8-13 July, 1963) P. Hubert and E. Cremieu-Alcan, Editors, Serma, Paris, **2**, 471 (1963).
- 02534 Skinner, J. G., Brady, J. J., Net Frequency of Ionization in Oxygen, *Phys. Rev.*, **129**, 2087 (1963).
- 02535 De Hoog, F. J., Kasdorp, J., A New Determination of Townsends Alpha for Neon, *Physica*, **34**, 63 (1967).
- 02538 Boyd, R. L. F., Boylett, F. D. A., Ion and Electron Energy Distribution in the Hydrogen Discharge, *Proc. Roy. Soc. London Ser. A*, **296**, 233 (1967).
- 02539 Bernstein, M. J., Townsend Ionization Coefficient for Hydrogen in a Transverse Strong Magnetic Field, *Phys. Rev.*, **127**, 342 (1962).
- 02540 Kelly, D. C., Microwave Conductivity of a Plasma in a Magnetic Field, *Phys. Rev.*, **119**, 27 (1960).
- 02542 Hudson, D. E., Method of Measuring the Drift Velocity of Electrons in Gases, A. E. C. Report, Manhattan District Declassified, 534 (1946) Technical Information Branch, Oak Ridge, Tenn.
- 02544 Edelson, D., McAfee, K. B., Improved Pulsed Townsend Discharge Experiment, *Rev. Sci. Instr.*, **35**, 187 (1964).
- 02546 D'Angelo, N., Rynn, N., Diffusion and Recombination of a Highly Ionized Cold Plasma in a Magnetic Field, *Phys. Fluids*, **4**, 1303 (1961).

- 02547 Heylen, A. E. D., Experimental Townsend Ionization Coefficients for Isopentane, *J. Chem. Phys.*, **35**, 169 (1961).
- Organic compounds studied include $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$.
- 02549 Fugol, I. Ya., Pakhomov, P. A., Shevchenko, Yu. F., Spectroscopic Study of Disintegrating Helium Plasmas at 20 Degrees K, *Opt. Spectry. USSR, English Transl.*, **21**, 404 (1966).
- 02550 Polushkin, I. N., Recombination in the Decay of a High-Pressure Helium Plasma Containing Potassium Vapor, *Soviet Phys. Tech. Phys. English Transl.*, **11**, 1558 (1967).
- Mixture studied was He + K.
- 02551 Morgulis, N. D., Polushkin, I. N., Charge Recombination during Deionization of Plasma in Helium and Argon at Elevated Pressures from Admixture of Cesium Vapor, (In) *Proceedings of the Seventh International Conference on Phenomena in Ionized Gases* (Belgrade, 22-27 August 1965) B. Perovic and D. Tosic, Editors, Gradevinska Knjiga Publishing House, Belgrade, **1**, 35 (1966).
- Mixtures studied were He + Cs and Ar + Cs.
- 02553 Hake, R. D., Phelps, A. V., Momentum-Transfer and Inelastic-Collision Cross Sections for Electrons in O_2 , CO and CO_2 , *Phys. Rev.*, **158**, 70 (1967).
- 02554 Swamy, M. N., Harrison, J. A., Townsend's Alpha in N-Propanol, Benzene and Hydrogen-N-Propanol Mixtures, *Brit. J. Appl. Phys.*, **18**, 781 (1967).
- Mixture studied was $\text{C}_3\text{H}_7\text{OH} + \text{H}_2$.
- 02555 Sukhum, N., Prasad, A. N., Craggs, J. D., Electron Attachment and Detachment in Oxygen, *Brit. J. Appl. Phys.*, **18**, 785 (1967).
- 02556 Masch, K., Über Elektronenionisierung von Stickstoff, Sauerstoff und Luft bei Geringen und Hohen Drucken, *Arch. Elektrotech.*, **26**, 587 (1932).
- 02557 Paavola, M., Der Dunkle Entladungsvorstrom in Luft von Atmospharendruck, *Arch. Elektrotech.*, **22**, 443 (1929).
- 02558 Hochberg, B. M., Sandberg, E. J., A Study of the Dielectric Strength of Gases, *Zh. Tekhn. Fiz.*, **12**, 65 (1942).
- 02559 Christophorou, L. G., Hadjiantoniou, A., Hurst, G. S., Interaction of Thermal Electrons with Polarizable and Polar Molecules, Report ORNL-TM-1262, Oak Ridge National Laboratory, Oak Ridge, Tennessee, 1965, 70 pages.
- Mixtures studied were C_2H_4 with C_3H_8 , C_5H_{12} , C_6H_{14} , C_7H_{16} , C_8H_{18} , C_9H_{20} , $\text{C}_{10}\text{H}_{22}$, C_6H_6 , C_6H_{12} , $p\text{-C}_6\text{H}_4(\text{CH}_3)_2$, $p\text{-C}_6\text{H}_4\text{Cl}_2$, $m\text{-C}_6\text{H}_4(\text{CH}_3)_2$, $o\text{-C}_6\text{H}_4(\text{CH}_3)_2$, $o\text{-C}_6\text{H}_4\text{CH}_2\text{Br}$, $m\text{-C}_6\text{H}_4\text{Cl}_2$, $\text{C}_6\text{H}_5\text{Br}$, $\text{C}_6\text{H}_5\text{Cl}$, $o\text{-C}_6\text{H}_4\text{Cl}_2$.
- 02561 Cottingham, W. B., Buchsbaum, S. J., Electron Ionization Frequency in Hydrogen, *Phys. Rev.*, **130**, 1002 (1963).
- 02562 Townsend, J. S., The Conductivity Produced in Gases by the Aid of Ultra-Violet Light, *Phil. Mag.*, **3**, 389 (1903).
- 02563 Townsend, J. S., The Conductivity Produced in Gases by the Motion of Negatively Charged Ions, *Phil. Mag.*, **1**, 198 (1901).
- 02567 Bailey, V. A., Higgs, A. J., On the Attachment of Electrons to the Molecules HCl and NH_3 , *Phil. Mag.*, **7**, 277 (1929).
- 02568 Sloane, R. H., Emeleus, K. G., An Effect of Positive Space Charge in Collector Analysis of Discharges, *Phys. Rev.*, **44** 333 (1933).
- 02569 Townsend, J. S., The Genesis of Ions by the Motion of Positive Ions in a Gas, and a Theory of the Sparking Potential, *Phil. Mag.*, **6**, 598 (1903).
- 02570 Davies, W. E. V. J., Dutton, J., Harris, F. M., Llewellyn-Jones, F., Electrical Breakdown of Air at High Voltages, *Nature*, **205**, 1092 (1965).
- 02571 Davies, W. E. V. J., Dutton, J., Harris, F. M., An Apparatus for the Investigation of Pre-Breakdown Ionization in Gases at High Voltages, High Gas Pressures and Large Electrode Separations, *J. Sci. Instr.*, **43**, 457 (1966).
- 02572 Dubs, C. W., Sen, H. K., Thermal Electron Attachment Coefficient of Oxygen Molecules near a Nuclear Burst, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 79 (1960).
- 02574 Frommhold, L., A Study of the Electron Component of Electron Avalanches, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden 17-21 August, 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 115 (1960).
- 02576 Heylen, A. E. D., Experimental Townsend Alpha Values of Molecular Gases, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August, 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 150 (1960).
- Organic compounds studied include $\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2$, $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$, $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$, $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$.
- 02578 Davies, D. K., Dutton, J., Llewellyn-Jones, F., Photon Processes in the Breakdown Mechanism, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August, 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **1**, 210 (1960).
- 02579 Dutt, T. L., Stainsby, A. G., Calculation of the Electronic Density and Collision Frequency in a Decaying Plasma, (In) *Proceedings of the Fourth International Conference on Ionization Phenomena in Gases* (Uppsala, Sweden, 17-21 August 1959) N. Robert Nilsson, Editor, North-Holland Publishing Company, Amsterdam, **2**, 755 (1960).
- Mixture studied was Kr + Xe.
- 02580 Haydon, S. C., Robertson, A. G., Ionization Coefficients for Hydrogen in Crossed Electric and Magnetic Fields, (In) *Proceedings of the Fifth International Conference on Ionization Phenomena in Gases* (Munich, 28 August-1 September 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, **1**, 75 (1962).
- 02581 Blair, D. T. A., McNaul, J., Tedford, D. J., Bruce, F. M., Avalanche Pulses in Nitrogen and Air, (In) *Proceedings of the Fifth International Conference on Ionization Phenomena in Gases* (Munich, 28 August-1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, **1**, 162 (1962).
- Mixture studied was Air + H_2O .
- 02582 Dibbern, U., Untersuchung Einzelner Elektronenlawinen mit dem Photomultiplier, (In) *Proceedings of the Fifth International Conference on Ionization Phenomena in Gases* (Munich, 28 August-1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, **1**, 174 (1962).
- Mixture studied was $\text{N}_2 + \text{CH}_4$.
- 02583 Wu, C. S., The Effect of Ionization and Recombination on the Electron Distribution Function of a Slightly Ionized Gas with Presence of Electric and Magnetic Fields, (In) *Proceedings of the Fifth International Conference on Ionization Phenomena in Gases* (Munich, 28 August-1 September 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, **1**, 214 (1962).
- 02584 Swift, J. D., Energy Distribution of Electrons in the Positive Column of a Nitrogen Discharge, (In) *Proceedings of the Fifth International Conference on*

- Ionization Phenomena in Gases (Munich, 28 August–1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, 1, 343 (1962).
- 02586 Formato, D., Gilardini, A., Microwave Determinations of the G Factor in Nitrogen and Oxygen, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August–1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, 1, 660 (1962).
- 02587 Golden, D. E., Fisher, L. H., Ionization Coefficients and Uniform Field Breakdown in Argon for Low Values of E/P, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August–1 September 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, 1, 731 (1962).
- 02588 Haydon, S. C., The Equivalent Pressure Concept for Ionization in a Transverse Magnetic Field, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August 1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam 1, 763 (1962).
- 02589 Stuart, G. W., Electron Distributions in H₂ Gas at High E/P, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August–1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, 1, 771 (1962).
- 02590 Knechtli, R. C., Wada, J. Y., Quiescent Cesium Plasmas in Steady State, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August–1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, 1, 786 (1962).
- 02591 Dutton, J., Harris, F. M., The Spatial Growth of Ionization in Air at High Pressure, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August–1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam, 2, 1607 (1962).
- 02592 Croitoru, Z., Conductivité Electrique des Gaz Faiblement Ionisés en Présence d'un Champ Magnétique, (In) Proceedings of the Fifth International Conference on Ionization Phenomena in Gases (Munich, 28 August–1 September, 1961) H. Maecker, Editor, North-Holland Publishing Company, Amsterdam 2, 1660 (1962).
- 02593 Townsend, J. S., Kirkby, P. J., Conductivity Produced in Hydrogen and Carbonic Acid Gas by the Motion of Negatively Charged Ions, *Phil. Mag.*, **1**, 630 (1901).
- 02594 Harris, S., Measurement of Ionic Mobilities in the Positive Column, *Phil. Mag.*, **17**, 131 (1934).
- 02595 Hochberg, B. M., Sandberg, E. J., Ionization of Gases and Their Breakdown Strength, *Compt. Rend. Acad. Sci. URSS*, **53** 511 (1946).
- 02765 Bailey, V. A., The Motions of Electrons in a Gas in the Presence of Variable Electric Fields and a Constant Magnetic Field, *Phil. Mag.*, **23**, 774–791 (1937).
- 02766 Townsend, J. S., The Equations of Motion of Electrons in Gases, *Phil. Mag.*, **23**, 481–486 (1937).
- 02767 Driver, C., Über die Elektronenlawine in Luft bei Grossen Elektrodenabständen (7.5...20cm) in Homogenen Feld bei Atmosphärendruck, *Z. Angew. Phys.*, **24**, 24–28 (1967).
- 02806 Pearson, G. A., Kunkel, W. B., Electron Velocity Distribution in a Slightly Ionized Gas with Crossed Electric and Strong Magnetic Fields, *Phys. Rev.*, **130**, 864–868 (1963).
- 02807 Van Gorpum, A. H., Determination of the Velocity Distribution of Electrons in a Low Pressure Discharge Tube, *Physica*, **3**, 207–218 (1936).
- 02808 Crowe, R. W., Bragg, J. K., Devins, J. C., A Semiempirical Expression for the First Townsend Coefficient of Molecular Gases, *J. Appl. Phys.*, **26**, 1121–1124 (1955).
- 02809 Young, D. R., Electric Breakdown in CO₂ from Low Pressures to the Liquid State, *J. Appl. Phys.*, **21**, 222–231 (1950).
- 02810 Townsend, J. S., McCallum, S. P., *Electrical Properties of Helium*, *Phil. Mag.*, **47**, 737–753 (1924).
- 02812 Llewellyn-Jones, F., The Mobilities of Electrons in Helium, *Proc. Phys. Soc. London A*, **56**, 239–248 (1944).
- 02813 Rees, J. A., Jory, R. L., The Diffusion of Electrons in Dry Carbon Dioxide Free Air, *Australian J. Phys.*, **17**, 307–314 (1964).
- 02815 Hurst, H. E., Genesis of Ions by Collision and Sparking–Potentials in Carbon Dioxide and Nitrogen, *Phil. Mag.*, **11**, 535–552 (1906).
- Mixture studied was N₂ + O₂.
- 02816 Gill, E. W. B., Pidduck, F. B., The Genesis of Ions by Collision of Positive and Negative Ions in a Gas. Experiments on Argon and Helium, *Phil. Mag.*, **16**, 280–290 (1908).
- 02817 Morgan, C. G., Powell, W. D., Williams, W. T., Primary Ionization Coefficient of Deuterium, *Brit. J. Appl. Phys.*, **18**, 939–943 (1967).
- 02818 Khozhatlev, M. B., Yarin, L. P., Method of Determining the Electrical Conductivity of Low-Temperature Plasma with a Double Probe, *High Temp. USSR English Transl.*, **4**, 576–569 (1966).
- Mixture studied was combustion products of C₃H₈ + Air.
- 02819 Fedulov, V. I., Efremova, G. D., Magnetic Method of Measuring the Electrical Conductivity of Ionized Gases, *High Temp. USSR English Transl.*, **4**, 580–584 (1966).
- 02820 Deb, S., Goswami, S. N., The Effect of a Uniform Magnetic Field on Electrodeless Discharge in a Tube and Measurement of Electronic Mobility, *Sci. Cult. Calcutta*, **22**, 283–285 (1956).
- 02821 Colli, L., Facchini, U., Velocità di Migrazione Degli Elettroni in Campo Elettrico in Argon, *Nuovo Cimento*, **8**, 891–892 (1951).
- Mixture studied was Ar + N₂.
- 02877 Ritchie, R. H., Turner, J. E., Electron Capture and Loss in Electron Swarm Experiments, *Z. Phys.*, **200**, 259–269 (1967).
- 02881 Wheatley, F. W., Ionization of Gases by Collision with Low Electric Forces, *Phil. Mag.*, **26**, 1034–1043 (1913).
- 02882 Horton, F., Application of Thermionic Currents to the Study of Ionization by Collision, *Phil. Mag.*, **34**, 461–477 (1917).
- 02883 Townsend, J. S., Bailey, V. A., The Abnormally Long Free Paths of Electrons in Argon, *Phil. Mag.*, **43**, 1127–1128 (1922).
- Mixture studied was Ar + H₂.
- 02884 Schweitzer, S., Mitchner, M., Electrical Conductivity of a Partially Ionized Gas on a Magnetic Field, *Phys. Fluids*, **10**, 799–806 (1967).
- 02885 Smirnov, B. M., Electron-Ion Recombination, *Soviet Phys. Tech. Phys. English Transl.*, **12**, 62–65 (1967).
- 02886 Kagan, Yu. M., Milenin, V. M., Mitrofanov, N. K., Electron Energy Distribution in the Positive Column of a Discharge in Argon, *Soviet Phys. Tech. Phys. English Transl.*, **11**, 1661–1662 (1967).
- 02887 Afanaseva, V. L., Lukin, A. V., Mustafin, K. S., Electron Energy Distribution in a Neon-Hydrogen Mixture in a Hollow Cathode Discharge, *Soviet Phys. Tech. Phys. English Transl.*, **12**, 233–235 (1967).
- Mixture studied was Ne + H₂.
- 02908 Dutton, J., Evans, N. B., Morgan, G. B., Secondary Ionization Coefficients in Oxygen at Pressures up to

- Atmospheric, Brit. J. Appl. Phys., **18**, 1287-1293 (1967).
- 02919 Dutton, J., Morris, W. T., The Mechanism of the Electrical Breakdown of Air in Uniform Fields at Voltages up to 400 kV, Brit. J. Appl. Phys., **18**, 1115-1120 (1967).
- 02920 Levine, N. E., Uman, M. A., Experimental Determination of the Drift Velocity of Low-Energy Electrons in Ar, N₂, CO₂, Several Ar-N₂ Mixtures and Several Ar-CO₂ Mixtures, J. Appl. Phys., **35**, 2618-2624 (1964).
Mixtures studied were Ar + N₂, Ar + CO₂.
- 02921 Uman, M. A., Comparison of Two Theoretical Approaches to Electron Behavior in Ar-CO₂, Ar-N₂, Ar-H₂ and Ar-CO Gas Mixtures, Phys. Rev., **123**, 399-403 (1961).
Mixtures studied were Ar + CO₂, Ar + N₂, Ar + H₂, Ar + CO.
- 02926 Appleton, E. V., Chapman, F. W., The Collisional Friction Experienced by Vibrating Electrons in Ionized Air, Proc. Phys. Soc. London, **44**, 246-254 (1932).
- 02928 Ionescu, T. V., Mihul, C., Sur la Constante Dielectrique et la Conductivite des Gaz Ionises, Compt. Rend., **192**, 343-345 (1931).
- 02930 Badareu, E., Gratescu, G. G., Determinations des Coefficients Townsend dans la Vapeur de Mercure, Bull. Soc. Roum. Phys., **45**, 9-25 (1944).
- 02931 Badareu, E., Valeriu, M., Die Townsend'schen Ionisierungszahlen im Benzoldampf, Bull. Soc. Roum. Phys., **42**, 3-22 (1941).
- 02932 Badareu, E., Valeriu, M., Die Townsendschen Ionisationskoeffizienten in Einigen Kohlen Wasserstoffdampfen Toluol und Benzol, Bull. Soc. Roum. Phys., **43**, 35-40 (1942).
- 02933 Bratescu, G. G., Potentialul Disruptiv in Vaporii de Apa Lucrarile Sesiunii Gen. Stiintifice Acad. RPR, **2**, 368-376 (1950).
- 02934 Ciurtea, M., Petrescu, C., Coeficientii de Ionizare in Metan Bul. Stiint. Sect. Stiint Mat Fiz, **7**, 153-160 (1955).
- 02935 Valeriu, M., Les Coefficients Townsend Dans la Vapeur de Cyclohexane, Bull. Soc. Roum. Phys., **43**, 3-7 (1943).
- 02950 Mohler, F. L., Recombination of Ions in the Afterglow of a Cesium Discharge, J. Res. Nat. Bur. Std., **19**, 447-456 (1937).
- 02952 Bradbury, N. E., Photoelectric Currents in Gases between Parallel Plates as a Function of the Potential Difference, Phys. Rev., **40**, 980-987 (1932).
- 02953 Bailey, V. A., Photoelectrons and Negative Ions, Nature, **129**, 166-167 (1932).
- 02983 Harris, J. H., Balfour, D., Microwave Cavity Studies of Ionization Decay in Caesium Vapour, Brit. J. Appl. Phys., **1**, 409-423 (1968).
- 02986 Weller, C. S., Biondi, M. A., Measurements of Dissociative Recombination of CO₂⁺ Ions with Electrons, Phys. Rev. Letters, **19**, 59-61 (1967).
- 02987 Colli, L., Deleonardis, M. T., A Measurement of the Drift Velocity of Electrons in the Electrical Field in Argon-Alcohol Mixtures, J. Appl. Phys., **24**, 255-257 (1953).
Mixture studied was Ar + C₂H₅OH.
- 02988 Kasner, W. H., Study of the Pressure and Temperature Dependence of Electron-Ion Recombination in Neon, Scientific Paper 67-1E2-Gases-P3, Westinghouse Research Laboratories, 1967, 15 pages.
- 02990 Cottrell, T. L., Walker, I. C., Townsend Energy Coefficients for Electron Swarms in Polyatomic Gases, Trans. Faraday Soc. **63**, 549-554 (1967).
- 02991 Veselovskii, I. S., Electron Recombination in a Weakly Ionized Gas, Soviet Phys. JETP English Transl., **25**, 687-690 (1967).
- 02992 Crompton, R. W., Elford, M. T., McIntosh, A. I., Electron Transport Coefficients in Hydrogen and Deuterium, Australian J. Phys., **21**, 43-63 (1968).
- 02993 Kasner, W. H., Study of the Temperature Dependence of Electron-Ion Recombination in Nitrogen, Phys. Rev., **164**, 194-200 (1967).
- 02994 Eccles, M. J., Craggs, J. D., Electron Detachment in Oxygen, Electron. Letters, **3**, 146-148 (1967).
- 02995 Stockdale, J. A., Christophorou, L. G., Hurst, G. S., Capture of Thermal Electrons by Oxygen, J. Chem. Phys., **47**, 3267-3270 (1967).
Mixtures studied were O₂ + C₂H₄ + H₂O and O₂ + C₂H₄.
- 03007 Christophorou, L. G., Compton, R. N., Dickson, H. W., Dissociative Electron Attachment to Hydrogen Halides and their Deuterated Analogs, J. Chem. Phys., **48**, 1949-1955 (1968).
- 03042 Fehsenfeld, F. C., Ferguson, E. E., Schmeltekopf, A. L., Thermal-Energy Associative-Detachment Reactions of Negative Ions, J. Chem. Phys., **45**, 1844-1845 (1966).
Mixtures studied were O⁻+H₂, O⁻+NO, O⁻+CO, O⁻+O, O⁻+N, O⁻+N₂, O⁻+O₂, Cl⁻+H, Cl⁻+O, Cl⁻+N, O₂⁻+O, O₂⁻+N, OH⁻+O, OH⁻+N, NO⁻+O₂.
- 03111 Von Engel, A., Internal Radiation and Nature of Discharges, Appl. Sci. Res. Sect. B, **5**, 34-42 (1956).
- 03154 Phelps, A. V., Pack, J. L., Collisional Detachment in Molecular Oxygen, Phys. Rev. Letters, **6**, 111-113 (1961).
- 03155 Epstein, M., Nonlinear Behavior of the Electrical Conductivity of a Slightly Ionized Gas, Phys. Fluids, **3**, 1016-1018 (1960).
- 03156 Kagan, Yu. M., Lyagushchenko, R. I., Electron Energy Distribution Function in the Positive Column of a Discharge, Soviet Phys. Tech. Phys. English Transl., **9**, 627-631 (1964).
- 03157 Kagan, Yu. M., Lyagushchenko, R. I., Energy Distribution of Electrons in a Positive Discharge Column, Soviet Phys. Tech. Phys. English Transl., **7**, 535-537 (1962).
- 03158 Cowling, T. G., The Electrical Conductivity of an Ionized Gas in a Magnetic Field, with Applications to the Solar Atmosphere and the Ionosphere, Proc. Roy. Soc. London Ser. A **183**, 453-479 (1945).
- 03161 Lamb, L., Lin, S. C., Electrical Conductivity of Thermally Ionized Air Produced in a Shock Tube, J. Appl. Phys., **28**, 754-757 (1957).
- 03164 Krinberg, I. A., Electrical Conductivity of Air in the Presence of an Impurity, J. Appl. Mech. Tech. Phys. USSR English Transl., **1**, 66-71 (1965).
Mixtures studied were Air + Cs, Air + Na, Air + Mg.
- 03165 Hessenauer, H., Anlagerungskoeffizienten und Driftgeschwindigkeiten von Elektronen in Luft, Z. Phys., **204**, 142-154 (1967).
Mixture studied was N₂ + O₂.
- 03166 Bryan, R. B., Holt, R. B., Oldenberg, O., Recombination and Afterglow in Nitrogen and Oxygen, Phys. Rev., **106**, 83-86 (1957).
- 03167 Lin, S. C., Reser, E. L., Kantrowitz, A., Electrical Conductivity of Highly Ionized Argon Produced by Shock Waves J. Appl. Phys., **26**, 95-109 (1955).
- 03168 Kagan, Yu. M., Milenin, V. M., Mitrofanov, N. K., Determination of the Electron Energy Distribution Function for a Noisy Plasma, Soviet Phys. Tech. Phys. English Transl. **12**, 87-89 (1967).
Mixtures studied were Hg + Ar, Hg + Kr.
- 03169 Haigh, C., Smith, T., A Determination of the Energy Distribution of Electrons in the Positive Columns of Gas Discharges, Phil. Mag., **41**, 557-567 (1950).
- 03170 Yarnold, G. D., The Energies of Uniformly Accelerated Particles in a Gas, Phil. Mag., **36**, 185-200 (1945).

- 03171 Gill, E. W. B., Pidduck, F. B., Ionization by Collision in Helium, *Phil. Mag.*, **23**, 837-849 (1912).
Mixture studied was He + Ar.
- 03172 Boyd, R. L. F., Thompson, J. B., The Operation of Langmuir Probes in Electronegative Plasmas, *Proc. Roy. Soc. London Ser. A*, **252**, 102-119 (1959).
- 03173 Boyd, R. L. F., Twiddy, N. D., Electron Energy Distributions in Plasmas. I., *Proc. Roy. Soc. London Ser. A*, **250**, 53-69 (1959).
- 03174 Twiddy, N. D., Electron Energy Distributions in Plasmas. III. The Cathode Regions in Helium, Neon and Argon, *Proc. Roy. Soc. London Ser. A*, **262**, 379-394 (1961).
- 03175 Boyd, R. L. F., Twiddy, N. D., Electron Energy Distributions in Plasmas. II. Hydrogen, *Proc. Roy. Soc. London Ser. A*, **259**, 145-158 (1960).
- 03177 LeBlanc, O. H., Devins, J. C., Townsend Ionization Constants in N-Alkanes, *Nature*, **188**, 219-220 (1960).
- 03178 Devins, J. C., LeBlanc, O. H., Ionization and Attachment Coefficients in N-Alkyl Chloride Gases, *Nature*, **187**, 409-410 (1960).
Organic compounds studied were $n\text{-C}_3\text{H}_7\text{Cl}$, $n\text{-C}_4\text{H}_9\text{Cl}$, $n\text{-C}_5\text{H}_{11}\text{Cl}$.
- 03179 Maksimov, A. I., Electron Density and Energy in a Microwave Helium Discharge, *Soviet Phys. Tech. Phys. English Transl.*, **11**, 1316-1321 (1967).
- 03180 Dugan, J. V., Calculation of Three-Body Collisional Recombination Coefficients for Cesium and Argon Atomic Ions with an Assessment of the Gryzinski Cross Sections, *J. Appl. Phys.*, **37**, 5011-5012 (1966).
- 03181 Valentin, P., Cottereau, M. J., Conductivite de l'Azote Ionise Par Choc, *Compt. Rend.*, **264**, 603-606 (1967).
- 03182 Bayet, M., Delcroix, J. L., Denisse, J. F., Theorie Cinetique des Plasmas Homogenes Faiblement Ionises. I., *J. Phys. Radium*, **15**, 795-803 (1954).
- 03184 Dutton, J., Harris, F. M., Llewellyn-Jones, F., The Three-Body Process of Electron Attachment in Air, (In) *Atomic Collision Processes*, M. R. C. McDowell, Editor, North-Holland Publishing Company, Amsterdam, Page 574, 1964, Proceedings of the Third International Conference on the Physics of Electronic and Atomic Collisions (London, 22-26 July 1963).
- 03185 Haigh, C., McCulloch, R. K., Further Determinations of the Energy Distribution of Electrons in the Positive Columns of Gas Discharges, *Proc. Leed Phil. Lit. Soc. Sci. Sect.*, **6**, 82-89 (1952).
- 03187 Bistline, J. A., Some Properties of BF_3 in Ionization Chambers, *Rev. Sci. Instr.*, **19**, 842-846 (1948).
- 03188 Popov, N. A., Afanaseva, E. A., Investigation of Deionization with Probes and by a Photoelectric Method, *Soviet Phys. Tech. Phys. English Transl.*, **4**, 764-769 (1960).
- 03202 Nighan, W. L., Low-Energy Electron Momentum Transfer Collisions in Cesium Plasmas, *Phys. Fluids*, **10**, 1085-1094 (1967).
- 03227 Kasabov, G. A., Nonequilibrium Conductivity of Argon-Cesium Plasma, *High Temp. USSR English Transl.*, **4**, 681-682 (1966).
Mixture studied was Ar + Cs.
- 03228 Brown, T. S., Rose, D. J., Plasma Diagnostics using Lasers-Relations between Scattered Spectrum and Electron-Velocity Distribution, *J. Appl. Phys.*, **37**, 2709-2714 (1966).
- 03229 Mentzoni, M. H., Donohoe, J., Electron Decay Following D-C. Discharge Ionization in NO and NO-Ne Mixtures, *Can. J. Phys.*, **45**, 1565-1578 (1967).
Mixture studied was NO + Ne.
- 03231 Kagan, Yu. M., The Velocity Distribution Function of Electrons in the Positive Column, *Beitr. Plasma Phys.*, **5**, 479-493 (1965).
Mixtures studied were Hg + Ne, Hg + He, Hg + Ar, Hg + Xe.
- 03233 Wehner, G., Medicus, G., Reliability of Probe Measurements in Hot Cathode Gas Diodes, *J. Appl. Phys.*, **23**, 1035-1046 (1952).
- 03235 Medicus, G., Simple Way to Obtain the Velocity Distribution of the Electrons in Gas Discharge Plasmas from Probe Curves, *J. Appl. Phys.*, **27**, 1242-1248 (1956).
- 03237 De Barbieri, O., Franchi, P., Orefice, A., Electromagnetic-Wave Propagation in a Weakly Ionized Plasma-III., *Nuovo, Cimento*, **48B**, 394-408 (1967).
- 03238 De Barbieri, O., Maroli, C., Orefice, A., Electromagnetic-Wave Propagation in a Weakly Ionized Plasma-II., *Nuovo Cimento*, **48B**, 378-393 (1967).
- 03240 Geballe, R., The Production of Photons Relative to Ionization by Collision in a Townsend Gap, *Phys. Rev.*, **66**, 316-320 (1944).
- 03241 Hantzsche, E., Energieverteilungsfunktionen im Fallraum von Glimmentladungen, *Beitr. Plasma Phys.*, **6**, 331-343 (1966).
- 03242 Stenflo, L., Instability of Velocity Distribution of Slightly Ionized Gases in Electric Fields, *J. Nucl. Sci. Technol.*, **3**, 189-193 (1966).
- 03244 Morgulis, N. D., Polushkin, I. N., Investigation of a Discharge Plasma in Helium and Argon with an Admixture of Cesium or Potassium Vapor. I., *High Temp. USSR English Transl.*, **4**, 699-704 (1966).
Mixtures studied were He + Cs, He + K, Ar + Cs, Ar + K.
- 03245 Golubev, V. S., Kasabov, G. A., Konakh, V. F., Study of a Steady-State Argon-Cesium Plasma with Nonequilibrium Conductivity, *High Temp. USSR English Transl.*, **2**, 455-459 (1964).
Mixture studied was Ar + Cs.
- 03261 Ulyanov, K. N., Stationary and Nonstationary Processes in Nonequilibrium Plasma, *High Temp. USSR English Transl.*, **2**, 758-766 (1964).
- 03263 Ornstein, L. S., Brinkman, H., Hamada, T., The Mechanism in the Positive Column of a Discharge, *Koninkl. Ned. Akad. Wetenschap. Proc. Ser. B*, **39**, 315-324 (1936).
- 03264 Craig, R. D., Craggs, J. D., Some Properties of Hydrogen Spark Channels, *Proc. Phys. Soc. London*, **66**, 500-511 (1953).
- 03266 Klarfeld, B., Characteristics of the Positive Column of Gaseous Discharge, *J. Phys. USSR*, **5**, 155-175 (1941).
- 03267 Teich, T. H., Emission Gasionisierender Strahlung aus Elektronenlawinen. II. Messungen in O_2 -He-Gemischen, Dampfen, CO_2 und Luft. Datenzusammenstellung, *Z. Phys.*, **199**, 395-410 (1967).
Mixture studied was He + O_2
- 03268 Teich, T. H., Emission Gasionisierender Strahlung aus Elektronenlawinen. I. Messanordnung und Messverfahren. Messungen in Sauerstoff, *Z. Phys.*, **199**, 378-394 (1967).
- 03269 Maecker, H., Messung und Auswertung von Bogencharakteristiken Ar, N_2 , *Z. Phys.*, **158**, 392-404 (1960).
- 03270 Cayless, M. A., Excitation and Ionization Rates of Mercury in Discharge Plasmas, *Brit. J. Appl. Phys.*, **10**, 186-190 (1959).
- 03272 Townsend, J. S., Theorie des Courants de haute Frequence dans les Gaz, *Compt. Rend.*, **186**, 55-58 (1928).
- 03273 Townsend, J. S., The Conductivity Produced in Gases by the Motion of Negatively-Charged Ions, *Nature*, **62**, 340-341 (1900).
- 03274 Demetriades, S. T., Argyropoulos, G. S., Ohm's Law in Multicomponent Nonisothermal Plasmas with

- Temperature and Pressure Gradients, *Phys. Fluids*, **9**, 2136-2149 (1966).
- 03275 Dallenbach, W., Einfluss eines Magnetfeldes auf die Durch ein Elektrisches Feld Bewirkte Driftgeschwindigkeit der Elektronen in einem Dichten Plasma, *Ann. Phys.*, **18**, 253-257 (1966).
- 03276 Peyraud, N., Theorie Cinetique des Plasmas-l'Operateur de Collisions Inelastiques dans le cas d'Excitations et de Desexcitations Electroniques, *J. Phys. (Paris)*, **28**, 147-151 (1967).
- 03277 Trong, N. V., Recombinaison Electronique dans un Plasma d'Argon, sous la Pression Atmospherique et au Voisinage de 10,000° K, *Compt. Rend.*, **264**, 217-219 (1967).
- 03278 Trong, N. V., Etude de la Recombinaison Electronique de l'Hydrogene dans une Striction Azimutale en Suivant l'Evolution du Profil des Raies Elargies par Effet Stark, *Compt. Rend.*, **264**, 770-773 (1967).
- 03279 Manheimer-Timnat, T., Low, W., Electron Density and Ionization Rate in Thermally Ionized Gases Produced by Medium Strength Shock Waves, *J. Fluid Mech.*, **6**, 449-461 (1959).
- 03281 Ornstein, L. S., Brinkman, H., Remark to the Paper-The Mechanism in the Positive Column of a Discharge, *Koninkl. Ned. Akad. Wetenschap, Proc. Ser. B*, **39**, 484-485 (1936).
- 03282 Christmann, H., Frie, W., Hertz, W., Experimentelle und Theoretische Untersuchungen des Leitwertabklingens in Stickstoffkaskadenbogen, *Z. Phys.*, **203**, 372-388 (1967).
- 03284 Bayet, M., Delcroix, J. L., Denisse, J. F., Theorie Cinetique des Plasmas Homogenes Faiblement Ionises. IV. Etude de l'Evolution de la Partie Isotrope de la Fonction de Distribution, *J. Phys. Radium*, **17**, 1005-1009 (1956).
- 03285 Bayet, M., Delcroix, J. L., Denisse, J. F., Theorie Cinetique des Plasmas Homogenes Faiblement Ionises. III. l'Operateur de Collision dans le cas du Gaz de Lorentz Imparfait, *J. Phys. Radium*, **17**, 923-930 (1956).
- 03286 Bayet, M., Theorie Cinetique du Gaz de Lorentz-cas des Molecules 'Maxwelliennes', *J. Phys. Radium*, **17**, 167-168 (1956).
- 03287 Ulyanov, K. N., The Electron Velocity Distribution in a Nonequilibrium Plasma, *High Temp. USSR English Transl.*, **4**, 309-317 (1966).
- 03288 Malyshev, G. M., Fedorov, V. L., Use of a Narrow-Band Amplifier in Oscillographic Investigation of the Electron Velocity Distribution Functions in an Electrical Discharge, *Dokl. Akad. Nauk. SSSR*, **92**, 269-271 (1953).
- 03289 Johnson, L. C., Electrical Conductivity of a Partially Ionized Gas, *Phys. Fluids*, **10**, 1080-1084 (1967).
- 03290 Bayet, M., Delcroix, J. L., Denisse, J. F., Theorie Cinetique des Plasmas Homogenes Faiblement Ionises. II., *J. Phys. Radium*, **16**, 274-280 (1955).
- 03291 Veyssiere, M., Manson, N., Conductivite Electrique des Produits de Detonation du Melange Propane-Oxygene-Azote, *Compt. Rend.*, **264**, 199-202 (1967).
Mixture studied was $C_3H_8 + O_2 + N_2$.
- 03297 Jones, J., Ionization Coefficients in Nitrogen, *Brit. J. Appl. Phys.*, **1**, 769-774 (1968).
- 03298 Joyce, G. R., Willett, J. E., High-Frequency Conductivity of a Partially Ionized Plasma in the Long-Wavelength Limit, *J. Appl. Phys.*, **38**, 3373-3384 (1967).
- 03299 Kilvington, A. I., Jones, R. P., Swift, J. D., Effect of AC Amplitude on the Measurement of Electron Energy Distribution Functions, *J. Sci. Instr.*, **44**, 517-520 (1967).
- 03300 Kudrin, L. P., Estimation of the Conductivity of a Low-temperature (U-F) Plasma of High Density, *Atomnaya Energiya*, **22**, 265-271 (1967).
- 03301 Devoto, R. S., Li, C. P., Transport Coefficients of Partially Ionized Helium, *J. Plasma Phys.*, **2**, 17-32 (1968).
- 03302 Morsell, A. L., Electrical Conductivity of Shock-Heated Air and Air-Plus-Teflon Mixtures, *Phys. Fluids*, **10**, 2171-2178 (1967).
Mixture studied was Air + Teflon.
- 03303 Naidu, M. S., Prasad, A. N., The Ratio of Diffusion Coefficient to Mobility for Electrons in Nitrogen and Hydrogen, *Brit. J. Appl. Phys.*, **1**, 763-768 (1968).
- 03304 Overton, G. D. N., Davies, D. E., Mechanisms of Electrical Breakdown in Mercury Vapour, *Brit. J. Appl. Phys.*, **1**, 881-888 (1968).
- 03305 Postma, A. J., Calculation of the Momentum Transfer Cross Section of Electron-Atom Collisions in Cesium Vapour from Drift Velocity Measurements, *Phys. Letters*, **26A**, 492-493 (1968).
- 03306 Rayment, S. W., Twiddy, N. D., Electron Energy Distributions in the Low-Pressure Mercury-Vapour Discharge-the Langmuir Paradox, *Proc. Roy. Soc. London Ser. A*, **340**, 87-98 (1968).
- 03307 Schweitzer, S., Mitchner, M., Convergence of Successive Approximations to the Scalar Electrical Conductivity of Some Weakly Ionized Real Gases, *AIAA J.*, **5**, 351-353 (1967).
- 03308 Soshnikov, V. N., Trekhov, E. S., Electron Distribution Functions in a Low-Temperature, Molecular Gas Discharge, *Soviet Phys. Tech. Phys. English Transl.*, **12**, 1026 (1968).
- 03309 Shimahara, H., Kiyama, S., Ogawa, K., Kato, K., Electrical DC Conductivity of a Turbulent Plasma, *J. Phys. Soc. Japan*, **24**, 604-610 (1968).
- 03310 Stiller, W., Calculation of the Electron Distribution Function of a Weakly Ionized Plasma in Time-Dependent Electric and Magnetic Fields, *Phys. Letters*, **27A**, 102-103 (1968).
- 03311 Janev, R. K., Danilovic, E. S., Radiative Attachment Coefficient of the Electrons to the $^2S_{1/2}$ Atoms, (In) Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases (Vienna, 27 August-2 September, 1967) Springer-Verlag Publishers, Vienna, Page 20, 1967.
- 03312 Robben, F., Stevefelt, J., Helium Recombination in the Afterglow of a Positive Column, (In) Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases (Vienna, 27 August-2 September 1967) Springer-Verlag Publishers, Vienna, Page 1, 1967.
- 03313 Wagner, E. B., Davis, F. J., Hurst, G. S., Time-of-Flight Investigations of Electron Transport in some Atomic and Molecular Gases, *J. Chem. Phys.*, **47**, 3138-3147 (1967).
- 03314 Yen, J. T., Kinetic Theory of Partially Ionized Gases, *Phys. Fluids*, **11**, 309-315 (1968).
- 03352 Moruzzi, J. L., Ekin, Jr., J. W., Phelps, A. V., Electron Production by Associative Detachment of O^- Ions with NO, CO, and H_2 , *J. Chem. Phys.*, **48**, 3070-3076 (1968).
Mixtures studied were $O_2 + H_2$, $O_2 + CO$, $O_2 + NO$.
- 03380 Bues, I., Patt, H. J., Richter, J., Über die Elektrische Leitfähigkeit und die Wärmeleitfähigkeit des Argons bei Hohen Temperaturen, *Z. Angew. Phys.*, **22**, 345-350 (1967).
- 03381 Bates, D. R., Bell, K. L., Kingston, A. E., Excited Atoms in Decaying Optically Thick Plasmas, *Proc. Phys. Soc. London*, **91**, 288-299 (1967).
- 03382 Grunberg, R., Bestimmung der Driftgeschwindigkeit von Elektronen in Wasserstoff und Stickstoff bei Hohem Druck, *Z. Phys.*, **204**, 12-16 (1967)

- 03383 Gupta, R. N., Mandal, S. K., Radio-Frequency Conductivity of Ionized Gases in Magnetic Field, *Indian J. Phys.*, **41**, 251-259 (1967).
- 03384 Creaser, R. P., Measurement of the Magnetic Drift Velocity and Magnetic Deflection Coefficient for Slow Electrons in Hydrogen and Deuterium at 293 K, *Australian J. Phys.*, **20**, 547-555 (1967).
- 03427 Harris, L. P., Electrical Conductivity of Potassium-Seeded Argon Plasmas Near Thermal Equilibrium, *J. Appl. Phys.*, **35**, 1993-1994 (1964).
Mixture studied was Ar + K.
- 03428 Mentzoni, M. H., Electron Removal During the Early Oxygen Afterglow, *J. Appl. Phys.*, **36**, 57-61 (1965).
- 03429 Mohler, F. L., Resistivity and Power Input in the Cesium Discharge at High Current Density, *J. Res. Nat. Bur. Std.*, **21**, 873-881 (1938).
- 03430 Caldirola, P., De Barbieri, O., Maroli, C., Electromagnetic Wave Propagation in a Weakly Ionized Plasma. I. *Nuovo Cimento*, **42**, 266-289 (1966).
- 03431 Kagan, Yu. M., Lyagushchenko, R. I., Khakhaev, A. D., The Excitation of Inert Gases in the Positive Column of a Discharge at Medium Pressures. I. Neon, *Opt. Spectry. USSR English Transl.*, **14**, 317-322 (1963).
- 03432 Kagan, Yu. M., Lyagushchenko, R. I., Khakhaev, A. D., The Excitation of Inert Gases in the Positive Column of a Discharge at Medium Pressures. II. Argon, *Opt. Spectry. USSR English Transl.*, **15**, 5-8 (1963).
- 03433 Aleksandrov, V. Ya., Gurevich, D. B., Podmoshenskii, I. V., Measurement of the Recombination Coefficient Alpha in the Dense Argon Plasma, *Opt. Spectry. USSR English Transl.*, **24**, 178-181 (1968).
- 03434 Braun, J., Larsson, E., Helium Recombination in a Steady State High Density Plasma, (In) *Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases* (Vienna, 27 August-2 September 1967) Springer-Verlag Publishers, Vienna, Page 2, 1967.
Mixtures studied were He + Cu, He + Ag, He + Pb.
- 03435 Conti, V. J., Williams, A. W., Measurement of Ionization and Attachment Coefficients in Carbon Dioxide, (In) *Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases* (Vienna, 27 August-2 September 1967) Springer-Verlag Publishers, Vienna, Page 23, 1967.
- 03436 De Hoog, F. J., Determining Townsend's Alpha in Neon by Measuring Excitation, (In) *Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases* (Vienna, 27 August-2 September 1967) Springer-Verlag Publishers, Vienna, Page 28, 1967.
- 03437 Deloche, R., Electron-Ion Recombination in a Dense Neutral Helium Plasma, (In) *Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases* (Vienna, 27 August-2 September 1967) Springer-Verlag Publishers, Vienna, Page 8, 1967.
- 03438 Fox, J. N., Hobson, R. M., The Variation of Dissociative Recombination Coefficients with Temperature, (In) *Contributed Papers of the Eighth International Conference on Phenomena in Ionized Gases* (Vienna, 27 August-2 September 1967) Springer-Verlag Publishers, Vienna, Page 5, 1967.
- 03439 Bernard, J., Labois, E., Ricateau, P., Zettwoog, P., Conductivite Electrique des Plasmas Argon, Cesium pour Conversion MHD, *Plasma Phys.*, **9**, 193-207 (1967).
Mixture studied was Ar + Cs.
- 03440 Ciampi, M., Talini, N., Determination of Average Plasma Conductivity by RF Probes, *J. Appl. Phys.*, **38**, 3771-3776 (1967).
- 03441 De Barbieri, O., Maroli, C., Negative Conductivity in a Neon Plasma, *Phys. Letters*, **26A**, 442-443 (1968).
- 03442 Franklin, J. L., Studniarz, S. A., Ghosh, P. K., Translational Energy Distribution of Electrons and Positive Ions in the Plasma of Microwave and High-Frequency Discharges of He, Ne, and Ar, *J. Appl. Phys.*, **39**, 2052-2061 (1968).
- 03443 Frommhold, L., Biondi, M. A., Mehr, F. J., Electron-Temperature Dependence of Electron-Ion Recombination in Neon, *Phys. Rev.*, **165**, 44-52 (1968).
- 03444 Fells, I., Gawen, J. C., Harker, J. H., An Investigation into the Electrical Conductivities of Propane-Air Flames Augmented with DC Electrical Power, *Combust. Flame*, **11**, 309-319 (1967).
Mixture studied was Air + C₂H₆.
- 03445 Goldenberg, S. A., Ievlev, V. N., Determination of the Electrical Conductivity of Argon Containing Alkali Metal Additions, *High Temp. USSR English Transl.*, **5**, 337-339 (1967).
Mixtures studied were Ar + K and Ar + Cs.
- 03450 Zaitsev, A., Calculation of the Townsend Ionization Coefficient, *Zh. Eksper. i Teor. Fiz.*, **9**, 469-474 (1939).
- 03458 Ellington, H. I., The Effect of the Bulk Gas on the Electrical Conductivity of Potassium-Seeded Non-Equilibrium Plasmas, *Brit. J. Appl. Phys.*, **1**, 189-192 (1968).
Mixtures studied were He + K, Ne + K, Ar + K.
- 03459 Blevin, H. A., Hasan, M. Z., The Measurement of Electron Drift Velocities, *Australian J. Phys.*, **20**, 735-739 (1967).
- 03460 Blevin, H. A., Hasan, M. Z., The Drift Velocity of Electrons in Nitrogen, *Australian J. Phys.*, **20**, 741 (1967).
- 03461 Brederlow, G., Hodgson, R. T., Electrical Conductivity in Seeded Noble Gas Plasmas in Crossed Electric and Magnetic Fields, *AIAA J.*, **6**, 1277-1284 (1968).
Mixtures studied were Ar + K, He + K.
- 03462 Bardsley, J. N., The Theory of Dissociative Recombination, *J. Phys. B*, **1**, 365-380 (1968).
- 03463 Bell, M. J., Kostin, M. D., Transport Coefficients and Energy Distributions of Electrons in Gases, *Phys. Rev.*, **169**, 150-155 (1968).
- 03464 Born, G. K., Recombination of Electrons and Molecular Helium Ions, *Phys. Rev.*, **169**, 155-164 (1968).
- 03465 Bozin, S. E., Goodyear, C. C., Growth of Ionization Currents in Carbon Tetrafluoride and Hexafluoride, *Brit. J. Appl. Phys.*, **1**, 327-343 (1968).
- 03466 Chiplonkar, V. T., Shah, R. G., Plasma Parameters by rf Conductivity Measurements, *Indian J. Pure Appl. Phys.*, **6**, 128-129 (1968).
- 03467 Cavalleri, G., Sesta, G., New Theory of Electron Drift Velocity in Gases, *Phys. Rev.*, **170**, 286-293 (1968).
- 03468 Funahashi, A., Takeda, S., Three-Body Electron-Ion Recombination in Argon Plasmas, *J. Phys. Soc. Japan*, **25**, 298-299 (1968).
- 03469 Drawin, H. W., Zur Formelmässigen Darstellung des Ionsierungsquerschnitts für den Atom-Atomstoss und über die Ion-Elektronen-Rekombination in Dichten Neutralgas, *Z. Phys.*, **211**, 404-417 (1968).
- 03470 Drouet, M. G., Sicha, M., Cloutier, G. G., Electron Energy Distribution in the Presence of Moving Striations, *Phys. Letters*, **27A**, 496-497 (1968).
- 03471 Dubrovskii, G. V., Obedkov, V. D., Decay of Molecular Hydrogen Ions through Collision with Thermal Electrons, *Soviet Astron. AJ English Transl.*, **11**, 305-307 (1967).
- 03472 Eccles, M. J., Prasad, A. N., Craggs, J. D., Electron Detachment in Sulphur Hexafluoride, *Electron. Letters*, **3**, 410-411 (1967).
- 03473 Fauchais, P., Moreau, M., Manson, N., Determination par la Mesure de la Conductivite Electrique, de la Temperature Moyenne d'un Jet Laminaire de Plasma d'Azote, *Compt. Rend.*, **266**, 1110-1113 (1968).

- 03474 Garrison, G. W., Electrical Conductivity of a Seeded Nitrogen Plasma, *ALAA J.*, **6**, 1264-1270 (1968).
Mixture studied was $N_2 + K$.
- 03475 Huber, B., Resonance Capture of Slow Electrons in Ethane, *Z. Naturforsch.*, **23A**, 1228-1229 (1968).
- 03476 Hansen, C. F., Temperature Dependence of the $NO^+ + e$ Dissociative-Recombination-Rate Coefficient, *Phys. Fluids*, **11**, 904-906 (1968).
- 03477 Heylen, A. E. D., Ionization Coefficients and Sparking Voltages in Argon and Argon-Ethane Mixtures, *Brit. J. Appl. Phys.*, **1**, 179-188 (1968).
Mixture studied was $Ar + C_2H_6$.
- 03478 Heylen, A. E. D., Ionization Coefficients and Sparking Voltages in Argon-Methane and Argon-Propane Mixtures, *Intern. J. Electron.*, **24**, 165-175 (1968).
Mixtures studied were $Ar + CH_4$, $Ar + C_3H_8$.
- 03479 Jancel, R., Calcul des Anisotropies et des Harmoniques de la Fonction de Distribution Electronique d'un Plasma Inhomogene Imparfaitement Lorentzien, *Compt. Rend.*, **266**, 153-156 (1968).
- 03480 Konjevic, N., Hearne, K. R., Edels, H., Determination of some Transport Properties of Argon from Transient Arc Behaviour, *Z. Phys.*, **214**, 109-126 (1968).
- 03481 Koyama, K., Sekiguchi, T., Measurement of Conductivity of Seeded Flame Plasmas, *Elec. Eng. Japan*, **87**, 12-19 (1967).
Mixtures studied were $C_3H_8 + O_2$ Flame with added K_2CO_3 , BaO , CaO , or SrO .
- 03482 Koons, H. C., Fiocco, G., Anisotropy of the Electron Velocity Distribution in a Reflex Discharge Measured by Continuous-Wave Laser Scattering, *Phys. Letters*, **26A**, 614-615 (1968).
- 03483 Kasner, W. H., Study of the Pressure and Temperature Dependence of Electron-Ion Recombination in Neon, *Phys. Rev.*, **167**, 148-151 (1968).
- 03484 Liley, B. S., Space Charge Effects in the Townsend-Huxley Swarm Technique, *Australian J. Phys.*, **20**, 527-545 (1967).
- 03485 Lotz, W., Electron-Impact Ionization Cross-Sections and Ionization Rate Coefficients for Atoms and Ions from Hydrogen to Calcium, *Z. Phys.*, **216**, 241-247 (1968).
- 03489 Langmuir, I., Mott-Smith, H., Studies of Electric Discharges in Gases at Low Pressures. Part IV. Data on Discharges in Mercury Vapor Obtained with Cylindrical Collectors, *Gen. Elec. Rev.*, **27**, 762-771 (1924).
- 03490 Bishop, E. S., An Absolute Determination of the Minimum Ionizing Energy of an Electron, and the Application of the Theory of Ionization by Collision to Mixtures of Gases, *Phys. Rev.*, **33**, 325-353 (1911).
- 03491 Frost, L. S., Conductivity of Seeded Atmospheric Pressure Plasmas, *J. Appl. Phys.*, **32**, 2029-2036 (1961).
Mixtures studied were $Ar + Cs$, $Ne + Cs$, $He + Cs$, Combustion Products of $CH_2 + 3/2 O_2 + Cs$, Combustion Products of $CH_2 + 3/2 O_2 + K$.
- 03492 Schirmer, H., Friedrich, J., Die Elektrische Leitfähigkeit eines Plasmas I, *Z. Phys.*, **151**, 1874-1868 (1958).
- 03493 Kovrizhnykh, L. M., Effect of Inelastic Collisions on the Velocity Distribution of Electrons, *Soviet Phys. JETP English Transl.*, **10**, 347-353 (1960).
- 03494 Carruthers, J. A., Conductivity of Electrons in Gases Weakly Ionized by X Rays, *Can. J. Phys.*, **40**, 1528-1536 (1962).
- 03495 Margenau, H., McMillan, F. L., Dearnley, I. H., Pearsall, C. S., Montgomery, C. G., Physical Processes in the Recovery of TR Tubes, *Phys. Rev.*, **70**, 349-357 (1946).
Mixtures studied were $Ar + H_2O$, $Ar + C_5H_{10}$, $Ar + H_2S$.
- 03496 Druyvesteyn, M. J., Der Niedervoltbogen, *Z. Phys.*, **64**, 781-798 (1930).
- 03497 Szekely, A., Experimentelle Untersuchungen über die Leitfähigkeit Ionisierter Luft bei Hochfrequenz, *Ann. Phys.*, **3**, 112-132 (1929).
- 03504 Emeleus, K. G., Greeves, F. D., Montgomery, F., Glow Discharges in Helium, *Proc. Roy. Irish Acad. Sect. A*, **43**, 35-47 (1936).
- 03529 Goldenberg, S. A., Ievlev, V. N., Leonteva, L. S., Measurement of the Electrical Conductivity of a High-Temperature Gas Jet, *High Temp. USSR English Transl.*, **2**, 616-621 (1964).
Mixture studied was Combustion Products of Gasoline, Air and Oxygen plus Added Potassium Salts.
- 03534 Kagan, Yu. M., Lyagushchenko, R. I., Electron Velocity Distribution, Excitation, and Ionization in the Positive Column of a Neon Discharge, *Soviet Phys. Tech. Phys. English Transl.*, **7**, 134-137 (1962).
- 03535 Finkelnburg, W., Segal, S. M., High Temperature Plasma Properties from High Current Arc Stream Measurements, *Phys. Rev.*, **80**, 258-260 (1950).
- 03536 Abramov, V. A., Electron-Ion Recombination in a Cesium Plasma, *High Temp. USSR English Transl.*, **3**, 18-21 (1965).
- 03537 Donskoi, K. V., Kunaev, Yu. A., Prokofev, I. A., Measurement of Electrical Conductivity in Gas Streams, *Soviet Phys. Tech. Phys. English Transl.*, **7**, 805-807 (1963).
Mixture studied was $C_2H_5OH + O_2 + K$.
- 03538 Didlauskis, M., Die Stationäre Geschwindigkeitsverteilung von in einem Elektrischen Feld Diffundierenden Elektronen, *Z. Phys.*, **74**, 624-630 (1932).
- 03539 Karpenko, A. S., Markevich, A. M., Ryabinin, Yu. N., The Electrical Conductivity of Gases at High Temperatures and Densities, *Zh. Exprim. i. Teor. Fiz.*, **23**, 468-476 (1952).
Mixtures studied were $N_2 + O_2$, $Ar + N_2 + O_2$.
- 03540 Appleton, E. V., Boohariwalla, D. B., The Influence of a Magnetic Field on the High-Frequency Conductivity of an Ionized Medium, *Proc. Phys. Soc. London*, **47**, 1074-1084 (1935).
- 03550 Den Hartog, H., Muller, F. A., Van Rooden, C. S. W., Electron Mobilities in Geiger-Muller Counters, *Physica*, **15** 581-587 (1949).
Mixture studied was $Ar + C_2H_5OH$.
- 03551 Den Hartog, H., Muller, F. A., Verster, N. F., Time Lags in Geiger-Muller Counters, *Physica*, **13**, 251-269 (1947).
Mixture studied was $Ar + C_2H_5OH$.
- 03552 Moralew, S., Die Bestimmung des Townsendschen Ionisations-koeffizienten Alpha Mit Berücksichtigung der Einzelnen Elementarprozesse, *Physik Z Sowjetunion*, **12**, 89-104 (1937).
Mixtures used were $Ar + N_2$, $Ar + Hg$, $Ar + Hg + N_2$.
- 03553 Gill, E. W. B., Note on Ionization Power of Negative Ions by Collisions at High Pressures, *Phil. Mag.*, **24**, 293-296 (1912).
- 03554 Clay, J., Ionization and Conductivity in Gases at High Pressures, *Koninkl. Ned. Akad. Wetenschap. Proc. Ser. B*, **40**, 824-836 (1937).
- 03555 Mohler, F. L., Recombination Radiation in the Cesium Positive Column, *J. Res. Nat. Bur. Std.*, **10**, 771-780 (1933).
- 03556 Clay, J., Van Kleef, C., Conductivity of Pure Gases at High Pressures, *Koninkl. Ned. Akad. Wetenschap. Proc. Ser. B*, **40**, 663-667 (1937).
- 03592 Lucas, J., The Spatial/Time Variation of the Energies of the Electrons in an Electron Swarm. Part I, *Intern. J. Electron.*, **22**, 529-547 (1967).
- 03593 Lo Surdo, C., Electron Velocity Distribution in a Partially Ionized Gas under the Action of Electric and Magnetic Field. I, *Nuovo Cimento*, **52B**, 429-454 (1967).
Mixture studied was $He + Cs$.

- 03594 Lo Surdo, C., Zampaglione, V., Electron Velocity Distribution in a Partially Ionized Gas Under the Action of Electric and Magnetic Fields. II, *Nuovo Cimento*, **55B**, 315-334 (1968).
- 03595 Mahan, B. H., Walker, I. C., Rate of Attachment of Gaseous Electrons to Nitrogen Dioxide, *J. Chem. Phys.*, **47**, 3780-3782 (1967).
Mixtures studied were NO + NO₂ + Ar, NO + NO₂ + He, NO + NO₂ + Ne, NO + NO₂ + Kr, NO + NO₂ + Xe, and NO + NO₂ + N₂.
- 03596 Maslennikov, N. M., Germanyuk, V. N., Experimental Study of the Electrical Conductivity of Plasma in a Model Magnetohydrodynamic Generator (MHDG), *High Temp. USSR English Transl.*, **5**, 506-510 (1967).
Mixture studied was Ar + K.
- 03597 Noon, J. H., Blaszk, P. R., Holt, E. H., Non-Maxwellian Form of the Electron Velocity Distribution in Nitrogen Plasmas, *J. Appl. Phys.*, **39**, 9-15 (1968).
- 03598 Norcross, D. W., Stone, P. M., Recombination, Radiative Energy Loss and Level Populations in Nonequilibrium Cesium Discharges, *J. Quant. Spectry. Radiative Transfer*, **8**, 655-685 (1968).
- 03599 Ostapchenko, E. P., Oreshak, O. N., Stepanov, V. A., The Effect of Striations on the Electron Energy Distribution in a Mercury-Krypton Mixture, *Soviet Phys. Tech. Phys. English Transl.*, **13**, 495-496 (1968).
Mixture studied was Hg + Kr.
- 03600 Razzak, S. A. A., Goodyear, C. C., Growth of Ionization Currents in the Methyl Halides, *Electron. Letters*, **4**, 54-55 (1968).
- 03601 Mentzoni, M. H., Donohoe, J., Electron Removal During the DC Discharge Afterglow of Carbon Monoxide, *Phys. Letters*, **26A**, 330-331 (1968).
- 03602 Schweitzer, S., On the Accuracy of Calculating the Scalar Electrical Conductivity at Very Low Ionization Levels, *AIAA J.*, **5**, 2086-2087 (1967).
- 03603 Rayment, S. W., Twiddy, N. D., Electron Energy Distributions in Moving Striations, *Nature*, **216**, 674-676 (1967).
- 03604 Razzak, S. A. A., Goodyear, C. C., Ionization and Attachment in Perfluorobutane, *Brit. J. Appl. Phys.*, **1**, 1215-1218 (1968).
- 03605 Sonin, E. B., Deviations from a Maxwellian Electron Energy Distribution in a Weakly Ionized Plasma, *Soviet Phys. JETP English Transl.*, **27**, 832-835 (1968).
- 03606 Taylor, W. C., Chown, J. B., Morita, T., Measurement of RF Ionization Rates in High-Temperature Air, *J. Appl. Phys.*, **39**, 191-194 (1968).
- 03607 Truby, F. K., Dissociative Electron Attachment in I₂ Vapor at 295°K, *Phys. Rev.*, **172**, 24-30 (1968).
Mixture studied was I₂ + He.
- 03608 Vagner, S. D., Virolainen, V. A., Kagan, Yu. M., Electron Energy Distribution in a High-Frequency Neon Discharge, *Soviet Phys. Tech. Phys. English Transl.*, **13**, 701-702 (1968).
- 03609 Von Goeler, S., Motley, R. W., Ellis, R., Measurement of Volume Recombination Coefficients in Highly Ionized Alkali Plasmas, *Phys. Rev.*, **172**, 162-165 (1968).
- 03610 Weller, C. S., Biondi, M. A., Recombination, Attachment, and Ambipolar Diffusion of Electrons in Photo-Ionized NO Afterglows, *Phys. Rev.*, **172**, 198-206 (1968).
- 03611 Kasner, W. H., Biondi, M. A., Temperature Dependence of the Electron-O₂⁺ Ion Recombination Coefficient, *Phys. Rev.*, **174**, 139-144 (1968).
- 03612 Willis, B. A., Morgan, C. G., Primary Ionization Coefficient of Neon, *Brit. J. Appl. Phys.*, **1**, 1219-1221 (1968).
- 03613 Skrebov, V. N., The Afterglow of a Pulsed Discharge in Mercury Vapor, *Opt. Spectry. USSR English Transl.*, **23**, 111-114 (1967).
- 03614 Kagan, Yu. M., Kolokolov, N. B., Milenin, V. M., Variation in Electron-Energy Distribution in Moving Striations, *Soviet Phys. Tech. Phys. English Transl.*, **13**, 1468-1469 (1968).
Mixtures studied were Hg + Ne, Hg + Kr.
- 03615 Nastoyashchii, A. F., Distribution Function of the Electrons in Inhomogeneous Weakly Ionized Plasma, *Soviet J. At. Energy English Transl.*, **25**, 1103-1109 (1968).
- 03646 Viegas, J. R., Kruger, C. H., Effect of Multispecies Ionization on Electrical Conductivity Calculations, *AIAA J.*, **6**, 1193-1195 (1968).
Mixtures studied were He + K, He + Cs, Ar + K, Ar + Cs.
- 03647 Stein, R. P., Gunton, R. C., Hackam, R., Lennon, J. J., Comments on a Paper by Hackam and Lennon, 'Microwave Measurements of Temperature Dependence of Electron Density Decay Rates in Oxygen and Oxygen-Nitrogen Mixtures', *J. Phys. B*, **1**, 128-129 (1968).
- 03648 De Hoog, F. J., A Random Walk to a Simple Stationary Electron Energy Distribution, *Physica*, **40**, 139-149 (1968).
- 03649 Borodin, V. S., Kagan, Yu. M., Excitation of Helium in a Hollow-Cathode Discharge, *Opt. Spectry. USSR English Transl.*, **23**, 108-110 (1967).
- 03650 David, J. P., Kaftandjian, V., Millet, J., Talin, B., Jarry, G., Influence de la Recombinaison Radiative Electron-Ion CS⁺ sur l'Intensite des Raies Spectrales du Cesium, *J. Phys. (Paris)*, **28**, 514-510 (1967).
- 03651 Dalidchik, F. I., Sayasov, Yu. S., Recombination of Electrons and Ions in Triple Collisions in a Dipole-Molecule Medium, *Soviet Phys. JETP English Transl.*, **25**, 1059-1061 (1967).
Mixture studied was K⁺ + H₂O.
- 03652 Dalgarno, A., Browne, J. C., The Associative Detachment of H and H⁻, *Astrophys. J.*, **149**, 231-232 (1967).
- 03653 Deloche, R., Etude des Mechanismes Collisionnels et Radiatifs dans de l'Helium Faiblement Ionise, *Compt. Rend.*, **266B**, 664-667 (1968).
- 03654 Egorov, V. S., Skrebov, V. N., Shukhtin, A. M., The Concentration of Excited Atoms in a Pulsed Discharge in Mercury Vapor, *Opt. Spectry. USSR English Transl.*, **22**, 4-6 (1967).
- 03655 Hassan, H. A., Drift Velocities and Thermal Flux Vectors in a Seeded Plasma with Magnetic Fields, *Phys. Fluids*, **11**, 106-111 (1968).
- 03656 Kunze, H. J., Gabriel, A. H., Griem, H. R., Measurement of Collisional Rate Coefficients for Helium-like Carbon Ions in a Plasma, *Phys. Rev.*, **165**, 267-276 (1968).
- 03657 Bono, R., Tondello, G., Conductivity and Temperature of Plasmas Produced by a Theta-Pinch Conical Discharge, *Ric. Sci.*, **37**, 923-928 (1967).
- 03658 Newton, A. A., Sexton, M. C., The Decay of a Highly Ionized Helium Plasma, *J. Phys. D*, **1**, 669-680 (1968).
- 03659 Crompton, R. W., McIntosh, A. I., Electron Drift and Diffusion in Parahydrogen at 77°K, *Australian J. Phys.*, **21**, 637-647 (1968).
- 03660 Alievskii, M. Ya., Radchenko, R. V., Maximum Conductivity of a Multicomponent Mixture of Ionized Gases, *High Temp. USSR English Transl.*, **4**, 711-717 (1967).
- 03661 Allouche, D., Jancel, R., Kahan, T., Application d'une Methode d'Approximations Successives au Calcul de la Fonction de Distribution Electronique d'un Plasma Lorentzien Homogene Soumis a un Champ Electrique Module en Frequence, *Compt. Rend.*, **267**, 392-395 (1968).
- 03662 Hendrick, W. G., Christophorou, L. G., Hurst, G. S., An Apparatus for Measurement of Electron Attachment and Electron Swarm Drift Velocities at High Temperatures,

- ORNL-TM-1444, Oak Ridge National Laboratory, Oak Ridge, Tennessee, (1968).
- 03677 Schwenn, R., Brederlow, G., Salvat, M., Electrical Conductivity of an Argon-Potassium Plasma at Low Current Densities as a Function of the Gas Temperature, *Plasma Phys.* **13**, 1077-1099 (1968).
Mixture studied was Ar + K.
- 03678 Peyraud, N., Action d'un Champ Electrique sur un Gaz Faiblement Ionise. I. Influence des Collisions Inelastiques sur la 'Queue' de la Fonction de Distribution Electronique Comportement Asymptotique, *J. Phys. (Paris)*, **29**, 201-211 (1968).
- 03679 Heymann, P., Der einfluss der Elektronenverteilungsfunktion auf Mikrowellenemission und Hochfrequenzleitfähigkeit Eines Schwach Ionisierten Entladungsplasmas, *Beitr. Plasma Phys.*, **8**, 233-246 (1968).
- 03680 Phelps, A. V., Voshall, R. E., Electron Attachment in N_2O , *J. Chem. Phys.*, **49**, 3246-3248 (1968).
- 03711 Wilhelm, V. J., Winkler, R., Zur Auswirkung Inelastischer Stosse auf die Elektronen-Transportkoeffizienten in Gasen und Plasmen Niedrigen Ionisationsgrades, *Ann. Phys.*, **20**, 296-302 (1967).
- 03712 Blaunstein, R. P., Christophorou, L. G., Electron Attachment to Halogenated Aliphatic Hydrocarbons, *J. Chem. Phys.*, **49**, 1526-1531 (1968).
Organic Compounds studied were CH_3Cl , CH_2Cl_2 , $CHCl_3$, CCl_4 , C_2Cl_6 , C_2H_5Cl , CHF_2Cl , CH_2F_2 , C_2F_5Cl , CH_3Br , CH_2Br_2 , C_2H_5Br , CF_3Br , CF_2Br_2 , CH_3I , C_2H_5I , CHF_3 , CF_4 , CF_3Cl , CF_2Cl_2 , $CFCl_3$.
- 03713 Cottureau, M. J., Valentin, P., Mesure de la Conductivite Electrique, Transverse a un Champ Magnetique, d'Air Ionise Par Choc, *Compt. Rend.*, **267**, 440-443 (1968).
- 03714 Elkomoss, S. G., Energy Distribution Function for Helium, *Phys. Rev.*, **172**, 153-162 (1968).
- 03715 Brose, H. L., Keyston, J. E., Collisions of Slow Electrons with Methane Molecules, *Phil. Mag.*, **20**, 902-913 (1935).
- 03716 Boyd, R. L. F., Twiddy, N. D., Electron Energy Distribution in the Striated Hydrogen Discharge, *Nature*, **173**, 633 (1954).
- 03717 Stark, J., Aenderung der Leitfähigkeit von Gasen durch einen Stetigen Elektrischen Strom, *Ann. Phys.*, **2**, 62-71 (1900).
- 03718 Stark, J., Berechnung der Leitfähigkeit Durchstromter Gase in der Positiven Lichtsaule, *Ann. Phys.*, **4**, 215-224 (1901).
- 03719 Townsend, J. S., Diffusion of Electrons in a Magnetic Field, *Phil. Mag.*, **25**, 459-470 (1938).
- 03720 Hertz, G., Über die Diffusion Langsamer Elektronen im Elektrischen Felde, *Z. Phys.*, **32**, 298-306 (1924).
- 03721 Ionescu, T. V., Mihul, C., Sur la Constante Dielectrique et la Conductibilite des Gaz Ionises, *J. Phys. Radium*, **6**, 35-48 (1935).
- 03722 Nygaard, K. J., Ultraviolet Light Emission from Townsend Discharges in Hydrogen, *J. Appl. Phys.*, **36**, 743-746 (1965).
- 03751 Dutton, J., Hughes, M. H., Tan, B. C., Ionization Coefficients in Helium, Neon and Helium-Neon Mixtures, *J. Phys. B*, **2**, 890-897 (1969).
Mixture studied was Ne + He.
- 03752 Thomas, R. W. L., Thomas, W. R. L., Monte Carlo Simulation of Electrical Discharges in Gases, *J. Phys. B*, **2**, 562-570 (1969).
- 03753 Thomas, W. R. L., The Determination of the Total Excitation Cross Section in Neon by Comparison of Theoretical and Experimental Values of Townsend's Primary Ionization Coefficient, *J. Phys. B*, **2**, 551-561 (1969).
- 03785 Van Leeuwen, J. M. J., Weijland, A., Non-Analytic Density Behaviour of the Diffusion Coefficient of a Lorentz Gas, *Physica*, **36**, 457-490 (1967).
- 03786 Smith, D., Goodall, C. V., The Dissociative Recombination Coefficient of O_2^+ Ions with Electrons in the Temperature Range 180-630 K, *Planetary Space Sci.*, **16**, 1177-1188 (1968).
- 03787 Shelton, S. V., Carlson, W. O., Variable Collision Cross-Section Effects on Electrical Conductivity, *AIAA J.*, **4**, 1676-1677 (1966).
Mixture studied was Ar + K.
- 03788 Breare, J. M., Von Engel, A., Locating Electron Swarms in Hydrogen by Far Ultra-Violet Signals, *Proc. Roy. Soc. London Ser. A*, **282**, 390-402 (1964).
- 03789 Akimov, A. V., Konenko, O. R., Measurement of Plasma Conductivity by a Radio-Frequency Method, *Soviet Phys. Tech. Phys. English Transl.*, **10**, 1126-1127 (1966).
Mixture studied was Ar + Hg.
- 03790 Aisenberg, S., Multiple Probe Measurements in High-Frequency Plasma Lasers, *J. Appl. Phys.*, **35**, 130-123 (1964).
- 03791 Rutscher, A., Zum Mechanismus der Positiven Neiderdrucksäule im Grenzfall Kleiner Elektronendichten. I. Allgemeine Übersicht, *Beitr. Plasma Phys.*, **7**, 43-56 (1967).
- 03792 Pfau, S., Zum Mechanismus der Positiven Niederdrucksäule im Grenzfall Kleiner Elektronendichten. II. Die Energieverteilung der Elektronen, *Beitr. Plasma Phys.*, **7**, 57-66 (1967).
- 03793 Prime, H. A., The Microwave Admittance of a Mercury-Vapor Discharge, *Australian J. Sci. Res.*, **5**, 607-617 (1952).
- 03794 Croitoru, Z., DeMontardy, A., Phenomenes de Contact, Tenseur de Conductivite et Temperature des Electrons dans un Gaz Ionise, *Rev. Gen. Elec.*, **72**, 429-438 (1963).
Mixtures studied were Ar + K, Ar + Na.
- 03795 Davidson, C. W., Farvis, W. E. J., Microwave Studies of a Pulsed Glow Discharge, *Phys. Rev.*, **127**, 1858-1864 (1962).
- 03796 Devoto, R. S., Transport Properties of Ionized Monatomic Gases, *Phys. Fluids*, **9**, 1230-1240 (1966).
- 03797 Fields, H., Bekefi, G., Brown, S. C., Microwave Emission from Non-Maxwellian Plasmas, *Phys. Rev.*, **129**, 506-515 (1963).
- 03798 Devoto, R. S., Transport Coefficients of Partially Ionized Argon, *Phys. Fluids*, **10**, 354-364 (1967).
- 03799 Devoto, R. S., Simplified Expressions for the Transport Properties of Ionized Monatomic Gases, *Phys. Fluids*, **10**, 2105-2112 (1967).
- 03800 Golant, V. E., Diffusion of Charged Particles Across a Magnetic Field in a Three-Component Plasma, *Soviet Phys. Tech. Phys. English Transl.*, **5**, 831-938 (1961).
- 03892 Goldenberg, S. A., Ievlev, V. N., Leonteva, Z. S., Determination of the Electrical Conductivity of High Temperature Combustion Products by an Induction Method, *High Temp. USSR English Transl.*, **2**, 48-53 (1964).
Mixture studied was Combustion Products of Gasoline, Air and Oxygen plus Added Potassium Salts.
- 03893 Gurevich, A. V., On the Effect of Radio Waves on the Properties of Plasma Ionosphere, *Soviet Phys. JETP English Transl.*, **3**, 895-904 (1957).
- 03894 Gurevich, A. V., Simplification of the Equations for the Distribution Function of Electrons in a Plasma, *Sov. Phys. JETP English Transl.*, **5**, 1006-1007 (1957).
- 03895 Guseva, L. G., A Study of the Fast Electrons in a Plasma, *Zh. Tekhn. Fiz.*, **21**, 427-437 (1951).
- 03896 Hearne, K. R., Konjevic, N., Electron Density

- Measurements during a Current Perturbation of a Wall Stabilized Argon Arc *Z. Phys.*, **208**, 65-72 (1968).
- 03897 Itoh, T., Musha, T., Monte Carlo Calculations of Motion of Electrons in Helium, *J. Phys. Soc. Japan*, **15**, 1675-1680 (1960).
- 03898 Jancel, R., Kahan, T., Theorie non Maxwellienne des Plasmas Homogenes et Anisotropes, *Nuovo Cimento*, **12**, 573-612 (1954).
- 03899 Kerrebrock, J. L., Nonequilibrium Ionization Due to Electron Heating. I. Theory, *AIAA J.*, **2**, 1072-1080 (1964).
- 03900 Kerrebrock, J. L., Hoffman, M. A., Nonequilibrium Ionization Due to Electron Heating. II. Experiments, *AIAA J.*, **2**, 1080-1087 (1964).
- Mixture studied was Ar + K.
- 03901 Lee, T. G., Electron Attachment Coefficients of some Hydrocarbon Flame Inhibitors, *J. Phys. Chem.*, **67**, 360-366 (1963).
- Mixtures studied were of N₂ with C₂H₅Br, C₂H₅I, C₂H₅Br₄, C₂F₄Cl₂, C₂F₆Cl₂, C₂H₄ClBr, CH₃I, CH₃Br, CH₃Cl, CH₂Br₂, CH₂ClBr, CH₂Cl₂, CHCl₃, CHF₃, CHFCl₂, CHF₂Cl, CCl₄, CF₄, CF₃Br, CF₃Cl, CF₃Br₂, CF₂Cl₂, CFC₂Cl₂, Fe(CO)₅, Pb(C₂H₅)₄, (CH₃C₅H₄)Mn(CO)₃, BF₃, O₂.
- 03902 Lyman, F. A., Maxwellization of Electrons in a Low-Voltage, Atmospheric Pressure Gas Discharge, *AIAA J.*, **4**, 1128-1130 (1966).
- Mixture studied was Ar + K.
- 03903 Margenau, H., Stillinger, D., Microwave Conductivity of Slightly Ionized Air, *J. Appl. Phys.*, **30**, 1385-1387 (1959).
- 03904 Mullaney, G. J., Kydd, P. H., Dibelius, N. R., Electrical Conductivity in Flame Gases with Large Concentrations of Potassium, *J. Appl. Phys.*, **32**, 668-671 (1961).
- Mixtures studied were Combustion Products of Propane + Oxygen Flames plus Potassium.
- 03905 Nastoyashchii, A. F., Complex Conductivity and the Stability of Ionized Gases, High Temp. USSR English Transl., **1**, 181-188 (1963).
- 03906 Pustovoi, V. I., Conductivity of Plasma Media in the Presence of a Drift, Soviet Phys. JETP English Transl., **16**, 1612-1617 (1963).
- 03907 Sampson, D. H., Enoch, J., Electron Distribution Function and Electrical Conductivity of a Slightly Ionized Gas, *Phys. Fluids*, **6**, 28-38 (1963).
- 03908 Scharfman, W., Morita, T., Focused Microwave Technique for Measurement of the Ionization Rate and Collision Frequency, *J. Appl. Phys.*, **35**, 2016-2020 (1964).
- 03909 Schweitzer, S., Tensor Electrical Conductivity of Atmospheric Cesium-Seeded Argon, *AIAA J.*, **5**, 844-847 (1967).
- Mixture studied was Ar + Cs.
- 03910 Schweitzer, S., Mitchner, M., Electrical Conductivity of Partially Ionized Gases, *AIAA J.*, **4**, 1012-1019 (1966).
- Mixture studied was Ar + Cs.
- 03911 Sengupta, D. L., The Electrical Conductivity of a Partially Ionized Gas, *Proc. I. R. E. Inst. Radio Engrs.*, **49**, 1872-1876 (1961).
- 03922 Swamy, M. N., Harrison, J. A., Pre-Breakdown Currents in Gas Mixtures, *Brit. J. Appl. Phys.*, **17**, 123-127 (1966).
- Mixture studied was H₂ + C₄H₉OH.
- 03950 Tanaka, H., Hagi, M., A Method of Measurement of Plasma Conductivity. II. Example of Measurement, *Japan. J. Appl. Phys.*, **3**, 338-341 (1964).
- 03951 Twiddy, N. D., Electron Energy Distributions in Plasmas. V. A Search for Evidence of a High Anomalous Rate of Energy Exchange between the Electrons of a Low-Pressure Discharge, *Proc. Roy. Soc. London Ser. A*, **275**, 338-356 (1963).
- Mixtures studied were Ne + Ar, Ne + Hg.
- 03952 Uman, M. A., Electron-Energy Distribution Function and Electron Average Energy in Ar-CO₂ and Ar-H₂ Mixtures, *Phys. Rev.*, **133**, A1266-1268 (1964).
- Mixtures studied were Ar + CO₂ and Ar + H₂.
- 03953 Valentin, P., Cottureau, M. J., Mesure de Conductivite Electrique d'Air Ionise par Choc, *Compt. Rend.*, **264**, 130-133 (1967).
- 03954 Wilhelm, H. E., Ohm's Law for Nonisothermal Plasma with Thermal Diffusion, *J. Appl. Phys.*, **37**, 2094-2098 (1966).
- 03955 Zukoski, E. E., Cool, T. A., Gibson, E. G., Experiments Concerning Nonequilibrium Conductivity in a Seeded Plasma, *AIAA J.*, **2**, 1410-1417 (1964).
- Mixture studied was Ar + K.
- 03956 Yamane, M., Ionization in Argon Mixtures by Electrons in an Electric Field, *J. Phys. Soc. Japan*, **15**, 1076-1086 (1960).
- Mixtures studied were of Ar with C₂H₄, C₃H₆, C₄H₈, CH₄, C₂H₆, C₃H₈, C₄H₁₀, C₅H₁₂, CH₃OH, C₂H₅OH, n-C₃H₇OH, Kr, Xe.
- 03957 Yoshikawa, S., Electrical Conductivity of a Turbulent Plasma *Phys. Fluids*, **5**, 1272-1277 (1962).
- 03958 Zukoski, E. E., Cool, T. A., Nonequilibrium Electrical Conductivity Measurements in Argon and Helium Seeded Plasmas *AIAA J.*, **3**, 370-371 (1965).
- Mixtures studied were Ar + K, He + K.
- 04015 Daniel, T. N., Dutton, J., Harris, F. M., The Influence of Processes Dependent on the Square of the Gas Number Density on the Spatial Growth of Ionization in Air, (In) *Contributed Papers of the Ninth International Conference on Phenomena in Ionized Gases* (Bucharest, 1-6 September 1969) Editor, Academy of the Socialist Republic of Romania, 3 Bis Butenberg Ave. Bucharest, Pg. 5, 1969.
- 04016 Davies, G. H. L., Williams, A. W., Ionization Growth in Carbon Monoxide, (In) *Contributed Papers of the Ninth International Conference on Phenomena in Ionized Gases* (Bucharest, 1-6 September 1969) Editor, Academy of the Socialist Republic of Romania, 3 Bis Butenberg Ave. Bucharest, Pg. 46 (1969).
- 04017 Gill, E. W. B., Von Engel, A., Starting Potentials of Electrodeless Discharges, *Proc. Roy. Soc. London Ser. A*, **197**, 107-124 (1949).
- 04018 Riemann, W., Untersuchung der Elektronenlawine mit der Nebelkammer, *Z. Phys.*, **122**, 216-229 (1944).
- Mixtures studied were of CO₂, O₂, N₂, H₂, Ar and Air with Various Cloud Chamber Vapor Mixtures of H₂O, C₂H₅OH and CH₃OH.
- 04020 Chan, F. T., Electron-Ion and Ion-Ion Dissociative Recombination of Oxygen. I. Electron-Ion Recombination, *J. Chem. Phys.*, **49**, 2533-2540 (1968).
- 04021 Aleksandrov, A. F., Kuzovnikov, A. A., High-Frequency Conductivity of the Positive Column in Neon, *Soviet Phys. Tech. Phys. English Transl.*, **8**, 410-411 (1963).
- 04022 Huber, B., Resonanzcinfang Langsamer Elektronen in Athan und Propan, *Z. Naturforsch.*, **24A**, 578-584 (1969).
- Mixture studied was C₂H₆ + CO₂.
- 04023 Takayama, K., Probe Measurements in Quiescent Plasmas, *Rev. Elec. Commun. Lab. Tokyo*, **10**, 100-116 (1962).
- 04043 Thomas-Betts, A., Davies, D. E., Ionization and Attachment in Oxygen at Low Pressures, *Brit. J. Appl. Phys.*, **2**, 213-219 (1969).
- 04044 Philbrick, J., Mehr, F. J., Biondi, M. A., Electron Temperature Dependence of Recombination of He₂⁺ Ions with Electrons, *Phys. Rev.*, **181**, 271-274 (1969).

- 04045 Mehr, F. J., Biondi, M. A., Electron Temperature Dependence of Recombination of O_2^+ and N_2^+ Ions with Electrons Phys. Rev., **181**, 264-271 (1969).
- 04046 Tagashira, H., Lucas, J., The Measurement of Ionization Coefficients in Hydrogen and Nitrogen, Brit. J. Appl. Phys., **2**, 867-880 (1969).
- 04047 Shipsey, E. J., Thermal-Energy Dissociative Attachment of I_2 ; Deduction of the Curve Crossing Point from Experimental Measurements, J. Chem. Phys., **52**, 2274-2277 (1970).
- 04048 Nelson, D. R., Davis, F. J., Determination of Diffusion Coefficients of Thermal Electrons with a Time-of-Flight Swarm Experiment, J. Chem. Phys., **51**, 2322-2335 (1969).
- 04049 Cavalleri, G., Measurements of Lateral Diffusion Coefficients and First Townsend Coefficients for Electrons in Helium by an Electron-Density Sampling Method, Phys. Rev. **179**, 186-202 (1969).
- 04050 Cavalleri, G., Gatti, E., Interlenghi, A. M., Improvements in the Measurement of Electron Diffusion Coefficients by the Light Chamber Technique, Nuovo Cimento, **40**, 450-453 (1965).
- 04052 Lowke, J. J., Parker, J. H., Theory of Electron Diffusion Parallel to Electric Fields. II. Application to Real Gases, Phys. Rev., **181**, 302-311 (1969).
- 04053 Pfau, S., Rutscher, A., Beweglichkeit und Diffusionskoeffizient der Elektronen, Ann. Phys., **22**, 166-179 (1969).
- 04054 Christophorou, L. G., Chaney, E. L., Electron Attachment to Gaseous Naphthalene, J. Chem. Phys., **52**, 2165-2166 (1970)
Mixtures studied were Naphthalene + C_2H_4 and Naphthalene + N_2 .
- 04055 Grunberg, R., Measurements of the Electron Drift Velocity in Helium Gas at High Pressure, Z. Naturforsch., **24A**, 1838-1839 (1969).
- 04056 Postma, A. J., Influence of Several Types of Inelastic Collision on the Electron Energy Distribution in Helium, Physica, **45**, 609-618 (1970).
- 04057 Bortnik, I. M., Voltage at which Electrical Discharge Appears in Helium for Moderate Values of (PD), Soviet Phys. Tech. Phys. English Transl., **9**, 1496-1502 (1965).
- 04058 Grunberg, R., Messungen der Elektronenbeweglichkeit bei Hohen Gasdrucken in Ar, He, N_2 und H_2 , Z. Naturforsch., **23A**, 1994-2004 (1968).
- 04063 Lloyd, J. L., On the Efficiency of the Neon Flash Tube, Proc. Phys. Soc. London, **75**, 387-394 (1960).
- 04064 Bromer, H. H., Elektron-Ion-Rekombination in Stickstoff, Abhandl Braunschweig Wiss Ges, **20**, 1-42 (1968).
- 04065 Mehr, F. J., Biondi, M. A., Electron-Temperature Dependence of Electron-Ion Recombination in Argon, Phys. Rev., **176**, 322-326 (1968).
- 04085 Daniel, T. N., Dutton, J., Harris, F. M., Similarity in Air and Nitrogen. II. Ionization, Attachment and Detachment Coefficients, Brit. J. Appl. Phys., **2**, 1559-1565 (1969).
- 04086 Razzak, S. A. A., Goodyear, C. C., Measurement of Ionization and Attachment Coefficients in Bromine, Brit. J. Appl. Phys., **2**, 1577-1581 (1969).
- 04094 Malakhov, V. B., Moskalenko, V. F., Ostapchenko, E. P., Excitation in a Neon-Krypton Mixture, Zh. Prikl. Spektroskopiya, **6**, 602-606 (1967).
Mixture studied was Ne + Kr.
- 04095 Biondi, M. A., Attachment of Thermal Electrons in Oxygen, Phys. Rev., **84**, 1072 (1951).
- 04096 Holt, E. H., Electron Loss Processes in the Oxygen Afterglow Bull. Am. Phys. Soc., **4**, 112-113 (1959).
- 04097 Lennon, J. J., Mulcahy, M. J., Microwave Measurement of Attachment in Oxygen-Nitrogen Mixtures, Proc. Phys. Soc. London, **78**, 1543-1545 (1961).
Mixture studied was $O_2 + N_2$.
- 04099 Chaney, E. L., Christophorou, L. G., Collins, P. M., Carter, J. G., Electron Attachment in the Field of the Ground and Excited Electronic States of the Azulene Molecule, J. Chem. Phys., **52**, 4413-4417 (1970).
- 04108 Truby, F. K., Temperature Dependence of Electron Attachment in I_2 Vapor, Phys. Rev., **188**, 508-512 (1969).
- 04185 Christophorou, L. G., Blaunstein, R. P., Nondissociative Electron Attachment to Aromatic Hydrocarbons, Radiation Res. **37**, 229-245 (1969).
Organic Compounds studied were Benzene, Naphthalene, Anthracene, Phenanthrene, Triphenylene, Chrysene, 1,2-Benzanthracene, Pyrene, Penylene.
- 04187 Wentworth, W. E., Chen, E., Lovelock, J. E., The Pulse-Sampling Technique for the Study of Electron-Attachment Phenomena, J. Phys. Chem., **70**, 445-458 (1966).
Organic Compounds studied were Naphthalene, Triphenylene, Phenanthrene, Chrysene, Benzo(c)-Phenanthrene, Anthracene, Pyrene, Benz(Alpha)Anthracene, Azulene.
- 04188 Lozanskii, E. D., Townsend's Ionization Coefficient of He with Allowance for Indirect Ionization, Akad. Nauk. SSR Dokl., **183**, 315-317 (1969) (Transl. **13**, 1134-1139 (1969)).
- 04189 Kagan, Yu. M., Fedorov, V. L., Malyshev, G. M., Cavallas, L. A., On Methodical Investigation of Electron Distribution Functions with Velocity in a Gas Discharge, Dokl. Akad. Nauk. SSSR, **76**, 215-217 (1951).
- 04190 Blaunstein, R. P., Christophorou, L. G., Electron Attachment to Organic Molecules, Thesis, University of Tennessee (1968) University Microfilms, Ann Arbor, Michigan, Order No. 69-1232.
- 04192 Bouby, L., Fiquet-Fayard, F., Abgrall, H., Attachment d'Electrons Thermiques sur Quelques Vapeurs Organiques, Compt. Rend., **261**, 4059-4062 (1965).
Mixtures studied were $CCl_4 + C_2H_4$, $CCl_4 + CO_2$, $CCl_4 + CH_3OH$, $SiCl_4 + C_2H_4$, $GeCl_4 + C_2H_4$, $SnCl_4 + C_2H_4$, $SiCH_3Cl_3 + C_2H_4$, $C_6H_5Cl + C_2H_4$, $CH_3(CO)_2CH_3 + C_2H_4$, $CH_3(CO)_2CH_3 + CO_2$, $CH_3(CO)_2CH_3 + CH_3OH$, $CH_3(CO)_2C_2H_5 + C_2H_4$, $CH_3COCH_2COCH_3 + CO_2$, $CH_3COCH_2COCH_3 + CH_3OH$, $CH_3CO(CH_2)_2COCH_3 + CH_3OH$, $CH_3OH + CH_2CHCOCH_3$, $CH_3OH + C(CH_3)_2CHCOCH_3$, $CH_3OH + CH_2CHCOOCH_3$, $CH_3OH + CCH_3COOCH_3$, $CH_3OH + COCH_3$, $CH_3OH + CH_2COCH_3$.
- 04193 Golant, V. E., Zhilinskii, A. P., Investigation of Diffusion Decay of a Plasma in a Magnetic Field. III. Soviet Phys. Tech. Phys. English Transl., **7**, 970-977 (1963).
- 04194 Weijland, A., Van Leeuwen, J. M. J., Non-Analytic Density Behaviour of the Diffusion Coefficient of a Lorentz Gas. II. Renormalization of the Divergencies, Physica, **38**, 35-47 (1968).
- 04195 Chen, E., George, R. D., Wentworth, W. E., Experimental Determination of Rate Constants for Thermal Electron Attachment to Gaseous SF_6 and C_7F_{14} , J. Chem. Phys., **49**, 1973-1974 (1968).
- 04203 Lehning, H., Resonance Capture of Very Slow Electrons in CO_2 , Phys. Letters, **28A**, 103-104 (1968).
- 04311 Ghosh, A. K., Carswell, A. L., Richard, C., Effect of Detachment Processes on Plasma Quenching by Electronegative Gases, Phys. Fluids, **10**, 1100-1106 (1967).
Mixtures studied were $N_2 + O_2$, $N_2 + SF_6$, $N_2 + H_2O$.

- 04312 Malliaris, A. C., Comments on the 'Effect of Detachment Processes on Plasma Quenching by Electronegative Gases', *Phys. Fluids*, **11**, 698-699 (1968).
Mixture studied was $N_2 + O_2$.
- 04313 Johnston, T. W., Ghosh, A. K., Reply to Comments by A. C. Malliaris, *Phys. Fluids*, **2**, 699-700 (1968).
Mixture studied was $O_2 + N_2$.
- 04355 Crompton, R. W., Elford, M. T., Robertson, A. G., The Momentum Transfer Cross Section for Electrons in Helium Derived from Drift Velocities at 77°K, *Australian J. Phys.*, **23**, 667-681 (1970).
- 04374 Berlande, J., Cheret, M., Deloche, R., Gonfalone, A., Mans, C., Pressure and Electron Density Dependence of the Electron-Ion Recombination Coefficient in Helium, *Phys. Rev. A*, **1**, 87-896 (1970).
- 04450 Gibson, D. K., The Cross Sections for Rotational Excitation of H_2 and D_2 by Low Energy Electrons, *Aust. J. Phys.*, **23**, 683-696 (1970).
- 04484 Crompton, R. W., Gibson, D. K., Robertson, A. G., Vibrational Excitation of H_2 by Low-Energy Electrons, *Phys. Rev. A*, **2**, 1386-1395 (1970).
- 04585 Cunningham, A. J., Hobson, R. M., Experimental Measurements of Dissociative Recombination in Vibrationally Excited Gases *Phys. Rev.*, **185**, 98-100 (1969).
- 04586 Chen, C. J., Temperature Dependence of Dissociative Recombination and Molecular Ion Formation in He, Ne, and Ar Plasmas, *Phys. Rev.*, **177**, 245-254 (1969).
- 04604 Bates, D. R., Malaviya, V., Young, N. A., Electron-Ion Recombination in Dense Molecular Gases, *Proc. Roy. Soc. London Ser. A*, **A320**, 437-458 (1971).
Mixtures studied were $Pb^+ + H_2 + N_2 + H_2O$.
- 04605 Sayers, J., Kerr, L. W., Ionic Reactions in Gases, (In) *Proceedings of the Third International Conference on Ionization Phenomena in Gases* (Venice, 1957) **1**, 908-911 (1958).
- 04621 Lotz, W., Electron-Impact Ionization Cross-Sections and Ionization Rate Coefficients for Atoms and Ions, *Astrophys. J. Suppl. Ser.*, **14**, 207-238 (1967).
- 04656 Bates, D. R., Dalgarno, A., Electronic Recombination, (In) *Atomic and Molecular Processes*, Academic Press, New York, 1962, Pages 245-271.
- 04681 Sayers, J., Recent Laboratory Studies of Recombination Cross-Sections, Solar Eclipses and the Ionosphere (Special Suppl. **6**, *J. Atmosph. Terr. Phys.*) 212-214 (1956).
- 04789 Lin, S. C., Teare, J. D., Rate of Ionization Behind Shock Waves in Air. II. Theoretical Interpretations, *Phys. Fluids*, **6**, 355-375 (1963).
- 04843 Baker, J. H., Williams, A. W., A New Apparatus for the Study of Pre-Breakdown Ionization Growth in Gases, (In) *Proceedings of the Conference on Gas Discharges* (I.E.E. London), **1**, 21-25 (1970).
- 04862 Robertson, A. G., The Momentum Transfer Cross Section for Low Energy Electrons in Neon, *J. Phys. B*, **5**, 648-664 (1972).
- 04891 Bain, R. A., Bardsley, J. N., Dielectronic Recombination of Highly Charged Ions, *Phys. Letters*, **37A**, 75-76 (1971).
- 04901 Fleming, I., Gray, D. R., Rees, J. A., The Drift and Diffusion of Electrons in Oxygen Containing Traces of Hydrogen, *J. Phys. D*, **5**, 291-296 (1972).
Mixture studied was $H_2 + O_2$.
- 04907 Folkard, M. A., Haydon, S. C., An Investigation of the Influence of Non-Equilibrium Effects on the Growth of Ionization in Hydrogen, *Australian J. Phys.*, **24**, 527-542 (1971).
- 04909 Hughes, M. H., Electron Energy Distribution Functions and Transport Coefficients in Helium and Neon, *J. Phys. B*, **3**, 1544-1551 (1970).
Mixture studied was He + Ne.
- 04910 Dutton, J., Powell, J. M., The Influence of Penning Ionization on the Experimental Determination of Ionization Coefficients in Helium at Low Values of E/N , *J. Phys. B*, **4**, 1506-1515 (1971).
- 04913 Crompton, R. W., Robertson, A. G., The Density Dependence of Electron Drift Velocities in Normal and Parahydrogen Deuterium and Helium at 77°K, *Australian J. Phys.*, **24**, 543-553 (1971).
- 04916 Robertson, A. G., Electron Drift Velocities in Normal and Parahydrogen and Deuterium, *Australian J. Phys.*, **24**, 445-449 (1971).
- 04919 Grunberg, K., Measurement of Attachment Coefficients of Electrons in Oxygen, *Z. Naturforsch.*, **24a**, 1039-1048 (1969).
- 04920 Eccles, M. J., O'Neill, B. C., Craggs, J. D., Electron Detachment in Oxygen, *J. Phys. B*, **3**, 1724-1731 (1970).
- 04943 Parkes, D. A., Sugden, T. M., Electron Attachment and Detachment in Nitric Oxide, *J. Chem. Soc. Faraday Trans. II*, **68**, 600-614 (1972).
- 04944 Brambring, J., Drift Velocity of Electrons in Argon, *Z. Phys.*, **179**, 539-543 (1964).
- 04947 Wagner, K. H., Ionization, Electron-Attachment, -Detachment, and Charge Transfer in Oxygen and Air, *Z. Phys.*, **241**, 258-270 (1971).
- 04948 Prasad, A. N., Craggs, J. D., Electron Detachment in Oxygen, *Electron. Letters*, **1**, 118-119 (1965).
- 04949 Brabanec, M. K., Williams, A. W., An Apparatus for the Study of Negative Ion Interactions under Swarm Conditions, (In) *Proceedings of Second International Conference on Gas Discharges* (I.E.E.) London, **1**, 332-334 (1972).
- 04950 Poole, H. G., Atomic Hydrogen. III. The Energy Efficiency of Atom Production in a Glow Discharge, *Proc. Roy. Soc. London Ser. A*, **163**, 424-454 (1937).
- 04951 Shaw, T. M., Dissociation of Hydrogen in a Microwave Discharge, *J. Chem. Phys.*, **30**, 1366-1367 (1959).
- 04971 Wagner, K. H., Mittlere Energien und Driftgeschwindigkeiten Elektronen in Stickstoff, Argon and Xenon, ermittelt aus Bildverstärkeraufnahmen von Elektronenlawinen, *Z. Phys.*, **178**, 64-81 (1964).
- 04993 Teich, T. H., Branston, D. W. Light Emission from Electron Avalanches in Electronegative Gases and Nitrogen, Second International Conference on Gas Discharges, (I.E.E. London,) **1**, 335-337 (1972).
- 04994 Price, D. A., Lucas, J., Moruzzi, J. L., Ionization in Oxygen-Hydrogen Mixtures, *J. Phys. D*, **5**, 1249-1259 (1972).
Mixture studied was $O_2 + H_2$.
- 05046 McCorkle, D. L., Christoprou, L. G., Anderson, V. E., Low Energy (<1eV) Electron Attachment to Molecules at Very High Gas Densities: O_2 , *J. Phys. B*, **5**, 1211-1220 (1972).
Mixture studied was $O_2 + N_2$.
- 05047 Spence, D., Schulz, G. J., Three-Body Attachment in O_2 using Electron Beams, *Phys. Rev. A*, **5**, 724-732 (1972).
- 05049 Buursen, C. G. J., De Hoog, F. J., Van Montfort, L. H., New Values of α/P_0 in Pure Neon Obtained with the Luminous Flux Method, *Physica*, **60**, 244-256 (1972).
- 05050 Virr, L. E., Lucas, J., Kontoleon, N., The Measurement of the Ratio of Diffusion Coefficient to Mobility for Electrons at Low E/p , *J. Phys. D*, **5**, 542-554 (1972).
- 05051 Naidu, M. S., Prasad, A. N., Mobility, Diffusion and Attachment of Electrons in Perfluoralkenes, *J. Phys. D*, **5**, 983-993 (1972).
- 05052 Naidu, M. S., Prasad, A. N., Craggs, J. D., Electron Transport, Attachment and Ionization in $c-C_4F_8$ and $iso-C_4F_8$, *J. Phys. D*, **5**, 741-746 (1972).
Organic compound studied was perfluorocyclobutane.

- 05053 Boyd, H. A., Crichton, H. A., Measurement of Ionisation and Attachment Coefficients in SF₆, *Proc. I.E.E.*, **118**, 1872-1877 (1971).
- 05077 Dutton, J., Powell, J. M., The Influence of Gas Sampling Techniques on Measurements of Pre-Breakdown Ionization in Helium, *J. Phys. B*, **5**, 1236-1240 (1972).
- 05102 Haydon, S. C., Williams, O. M., The Experimental Values of the Townsend First Ionization Coefficient in Nitrogen at Low Values of E/p, *J. Phys. D*, **5**, L79-L81 (1972).
- 05103 Bartels, A., Pressure Dependence of Electron Drift Velocity in Hydrogen at 77.8° K, *Phys. Rev. Letters*, **28**, 213-215 (1972).
- 05105 Blasberg, H. A. M., De Hoog, F. J., Measurements of the Primary Ionization Coefficient in H₂, *Physica*, **54**, 468-472 (1971).
- 05110 De Hoog, F. J., Holscher, J. G. A., A Calculation of the Primary Ionization Coefficient in Neon, *Physica*, **54**, 529-541 (1971).
- 05119 Naidu, M. S., Prasad, A. N., Diffusion and Drift of Electrons in SF₆, *J. Phys. D*, **5**, 1090-1095 (1972).
- 05123 Prasad, A. N., Naidu, M. S., Drift Diffusion and Attachment of Electrons in Perfluoropropane, (In) *Proceedings of the First International Conference on Gas Discharges* (London 15-18 September, 1970) *I.E.E.*, London, **1**, 416-420 (1970).
- 05124 Mahdavi, M. R., Hasted, J. B., Nakshbandi, M. M., Electron-Ion Recombination Measurements in the Flowing Afterglow, *J. Phys. B*, **4**, 1726-1737 (1971).
- 05125 Warman, J. M., Bansal, K. M., Fessenden, R. W., On the Pressure Dependence of Electron Attachment to O₂, *Chem. Phys. Letters*, **12**, 211-213 (1971).
- 05126 Sakai, Y., Tagashira, H., Sakamoto, S., The Variation of Steady State Electron Mean Energy between Parallel Plates in Argon, *J. Phys. B*, **5**, 1010-1016 (1972).
- 05128 Cunningham, A. J., Hobson, R. M., Dissociative Recombination at Elevated Temperatures I. Experimental Measurements in Krypton Afterglow, *J. Phys. B*, **5**, 1773-1783 (1972).
- 05130 Drawin, H. W., Emard, F., Influence of Atom-Atom Collisions on the Collisional-Radiative Ionization and Recombination Coefficients of Helium Plasmas, *Z. Phys.*, **254**, 202-217 (1972).
- 05131 Truby, F. K., Low-Temperature Measurements of Three-Body Electron-Attachment Coefficient in O₂, *Phys. Rev. A*, **6**, 671-676 (1972).
- 05134 Delpech, J.-F., Gauthier, J.-C., Electron-Ion Recombination in Cryogenic Plasmas, *Phys. Rev. A*, **6**, 1932-1939 (1972).
- 05135 Nelson, D. R., Davis, F. J., Thermal and Near-Thermal Electron Transport Coefficients in O₂ Determined with a Time-of-Flight Swarm Experiment Using a Drift-Dwell-Drift Technique, *J. Chem. Phys.*, **57**, 4079-4084 (1972).
- 05183 Naidu, M. S., Prasad, A. N., Diffusion and Drift of Electrons in Dichlorodifluoromethane, *Brit. J. Appl. Phys.*, **2**, 1431-1436 (1969).
- 05184 Aizentson, A. E., Volume Recombination in Steady State Discharges, *Soviet Phys. Tech. Phys. USSR English Transl.*, **16**, 2036-2040 (1972).
- 05187 Daniel, T. N., Harris, F. M., The Spatial Growth of Ionization Currents in Nitrogen at Voltages up to 500 kV, *J. Phys. B*, **3**, 363-368 (1970).
- 05188 Allen, N. L., Prew, B. A., Some Measurements of Electron Drift Velocities in Compressed Gases, *J. Phys. B*, **3**, 1113-1126 (1970).
- 05191 Losee, J. R., Burch, B. S., Experimental Electron Energy Distributions for Townsend Discharges in Argon Gas, *Phys. Rev. A*, **6**, 1652-1658 (1972).
- 05198 Plumb, I. C., Smith, D., Adams, N. G., Formation and Loss of O₂⁺ and O₄⁺ Ions in Krypton-Oxygen Afterglow Plasma, *J. Phys. B*, **5**, 1762-1772 (1972).
- 05200 Cottrell, T. L., Pollock, W. J., Walker, I. C., Electron Drift Velocities in Quadrupolar and Polar Gases, *Trans. Farad. Soc.*, **64**, 2260-2266 (1968).
- 05201 Wanless, D., Electron-Ion Recombination in Argon, *J. Phys. B*, **4**, 522-527 (1971).
- 05202 Davies, D. E., Garamoon, A. A. W. M., Surplice, N. A., Primary Ionization Processes in Caesium Vapour, *Phys. Letters*, **32A**, 106-107 (1970).
- 05203 Gerardo, J. B., Gusinow, M. A., Electronic Recombination of He⁺, *Phys. Rev. A*, **3**, 255-267 (1970).
- 05206 Hurst, G. S., O'Kelly, L. B., Wagner, E. B., Stockdale, J. A., Time-of-flight Investigations of Electron Transport in Gases, *J. Chem. Phys.*, **39**, 1341-1345 (1963).
- 05224 Haydon, S. C., McIntosh, A. I., Simpson, A. A., Electron Energy Distribution in a Townsend Discharge in Hydrogen, *J. Phys. B*, **3**, L110-L113 (1970).
- 05225 Shallal, M. A., Harrison, J. A., Pre-Breakdown Currents in Hydrogen in Uniform and Non-Uniform Fields, *J. Phys. D*, **4**, 1550-1559 (1971).
- 05226 Naidu, M. S., Prasad, A. N., Mobility Diffusion and Attachment of Electrons in Oxygen, *J. Phys. D*, **3**, 957-964 (1970).
- 05227 Kinsman, P. R., Rees, J. A., Ionization Attachment and Ion-Molecule Reactions in Oxygen, *Intern. J. Mass Spectrom. Ion Phys.*, **5**, 71-81 (1970).
- 05228 McArthur, W. T., Tedford, D. J., The Use of a Multi-Channel Pulse Height Analyser in the Determination of Townsend's α from Distributions of Avalanche Pulse Heights, (In) *Proceedings of the First International Conference on Gas Discharges* (London 15-18 September, 1970) *I.E.E.*, London, **1**, 284-288 (1970).
- 05229 Boyd, H. A., Crichton, G. C., Munk Nielsen, T., Determination of Ionization and Attachment Coefficients in CCl₂F₂, (In) *Proceedings of the First International Conference on Gas Discharges* (London, 15-18 September 1970) *I.E.E.*, London **1**, 426-430 (1970).
- 05230 Virr, L. E., Lucas, J., Kontoleon, N., Measurements of D/ μ for Electron Swarms in Gases at High E/p, (In) *Proceedings of the First International Conference on Gas Discharges* (London, 15-18 September 1970) *I.E.E.*, London, **1**, 530-533 (1970).
- 05231 Archambault, Y., Jeannot, J. C., Ronnsin, S., Sayer, B., Recombination and Diffusion Processes in a Cesium Afterglow, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, **1**, 8-23 (1971).
- 05232 Donovan, R. V., Sexton, M. C., Some Observed Effects on an Elevated, Time-Varying Neutral Gas Temperature on a Helium Afterglow, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, **1**, 17 (1971).
- 05233 Kline, L. B., Siambis, J. G., Characteristic Energy and Electron Energy Distribution for Electron Avalanches in Nitrogen, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, **1**, 53 (1971).
- 05234 Kontoleon, N., Lucas, J., Virr, L. E., Measurement of Discharge Parameters for Electron Swarms in Hydrogen, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, **1**, 54 (1971).
- 05235 Nasser, E., Parekh, H., Theoretical Determination of Townsend's First Ionization Coefficient, (In) *Proceedings of the Tenth International Conference on*

- Phenomena in Ionized Gases (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, 1, 50 (1971).
- 05236 Davis, F. J., Nelson, D. R., Pseudo-First-Order Attachment Rates of Thermal Electrons by SF₆ in Different Carrier Gases, *Chem. Phys. Letters*, **3**, 461-463 (1969).
- Mixtures studied were SF₆ with various carrier gases.
- 05237 Davis, F. J., Nelson, D. R., Attachment Rate of Thermal Electrons to SF₆, *Chem. Phys. Letters*, **6**, 277-278 (1970).
- Mixtures studied were SF₆ with Fourteen Different Carrier Gases.
- 05238 Christophorou, L. G., Chaney, E. L., Chistodoulides, A. A., Electron Attachment and 'Carrier Gas' Energy Distribution Functions, *Chem. Phys. Letters*, **3**, 363-366 (1969).
- Mixtures studied were P-Benzoquinone with C₂H₄, N₂ and Ar.
- 05239 Lozanskii, E. D., First Townsend Ionization Coefficient in a Mixture of Inert Gases, *Soviet Phys. Tech. Phys. USSR English Transl.*, **16**, 212-213 (1971).
- Mixture studied was He + Ar.
- 05240 Hughes, D. W., Shamin, M., Wooding, E. R., Diffusion and Recombination in a Dense Argon Plasma, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, 1, 26 (1971).
- 05241 Willis, B. A., Ionization Processes in Argon Discharges, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, 1, 48 (1971).
- 05242 Novikova, K. P., The Electron-Ion Recombination Coefficient Measurements in the Stable Argon Plasma, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, 1, 9 (1971).
- 05243 Parr, J. E., Moruzzi, J. L., Negative Ions in Carbon Monoxide, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, 1, 8 (1971).
- 05244 Dutton, J., Harris, F. M., Jones, G. J., Drift Velocities of Electrons in Sulphur Hexafluoride, (In) *Proceedings of the Tenth International Conference on Phenomena in Ionized Gases* (Oxford, 13-18 September 1971) R. N. Franklin, Editor, Donald Parsons and Co., Ltd., Oxford, 1, 60 (1971).
- 05246 Raja Rao, C., Govinda Raju, G. R., Growth of Ionization Currents in Dry Air at High Values of E/N, *J. Phys. D*, **4**, 494-503 (1971).
- 05247 Raja Rao, C., Govinda Raju, G. R., The Ratio of Diffusion Coefficient to Mobility for Slow Electrons in Dry Air, *J. Phys. D*, **4**, 769-772 (1971).
- 05248 Parr, J. E., Moruzzi, J. L., Electron Attachment in Water Vapour and Ammonia, *J. Phys. D*, **5**, 514-524 (1972).
- 05288 Shore, B. W., Dielectronic Recombination, *Astrophys. J.*, **158**, 1205-1218 (1969).
- 05289 Landini, M., Fossi, B. C., A Simple Formula for the Total Dielectronic Recombination Coefficient, *Solar Phys.*, **20**, 322-331 (1971).
- 05290 Bouby, L., Fiquet-Fayard, F., Bodere, C., Attachment d'Electrons Thermiques sur l'Anhydride Sulfureux en Phase Gazeuse, *Intern. J. Mass Spectrom. Ion Phys.*, **7**, 415-417 (1971).
- 05291 Veatch, G. E., Oskam, H. J., Recombination and Ion Conversion Processes in Helium-Neon Mixtures, *Phys. Rev. A*, **2**, 1422-1428 (1970).
- Mixture studied was He + Ne.
- 05292 Folkard, M. A., Haydon, S. C., Monte Carlo Investigations of Non-Equilibrium Effects at High Values of E/p in Swarm Experiments in Hydrogen, *Australian J. Phys.*, **23**, 847-861 (1970).
- 05293 Heylen, A. E. D., Ionization Coefficients and Sparking Voltages in Krypton and in Krypton-Olefin Gas Mixtures, *Intern. J. Electron.*, **31**, 19-25 (1971).
- Mixtures studied were Ethylene/Krypton and Propylene/Krypton.
- 05294 Dunn, M. G., Measurement of C⁺+e⁻+e⁻ and CO⁺+e⁻ Recombination in Carbon Monoxide Flows, *AIAA J.*, **9**, 2184-2191 (1971).
- 05295 Carmichael, C. H. H., Electron Drift Speed and Multiplication in a Pulsed Positive Column, *Intern. J. Electron.*, **32**, 49-56 (1972).
- 05325 Johnson, A. W., Gerardo, J. B., Dissociative Recombination of Electrons with He₂⁺, *Phys. Rev. Letters*, **28**, 1096-1098 (1972).
- 05327 Stevefelt, J., Robben, F., Spectroscopic Study of the Early Afterglow of a Recombining Helium Plasma, *Phys. Rev. A*, **5**, 1502-1515 (1972).
- 05334 Folkard, M. A., Haydon, S. C., Experimental Investigations of Ionization Growth in Nitrogen: I, *J. Phys. B*, **6**, 214-226 (1973).
- 05343 Boulmer, J., Davy, P., Delpech, J.-F., Gauthier, J.-C., Electronic Recombination of He₂⁺, *Phys. Rev. Letters*, **30**, 199-202 (1973).
- 05350 Tarter, C. B., Radiative Recombination Coefficients for Complex Ions, *Astrophys. J.*, **168**, 313-316 (1971).
- 05488 Johnson, A. W., Gerardo, J. B., Electronic-Recombination Coefficient of ³He₂⁺ Compared to ⁴He₂⁺, *Phys. Rev. A*, **7**, 1339-1341 (1973).
- 05490 Englert, G. W., Random Walk Theory of Elastic and Inelastic Time Dependent Collisional Processes in an Electric Field, *Z. Naturforsch.*, **26A**, 836-848 (1971).
- 05500 Cronin, J. C., Sexton, M. C., Mechanisms of Electron Loss in a Hydrogen Afterglow, *Proc. Roy. Irish Acad.*, **69A**, 1-9 (1970).
- 05501 Cronin, J. C., Sexton, M. C., Afterglow Studies in Helium, *Proc. Roy. Irish Acad.*, **70**, 13-25 (1970).
- 05533 Baravian, G., Benattar, R., Bretagne, J., Godart, J. L., Sultan, G., Electron-Ion Recombination in a Helium Plasma Produced by a Laser, *Z. Phys.*, **254**, 218-231 (1972).
- 05570 Johnson, A. W., Gerardo, J. B., Recombination and Ionization in a Molecular-Ion Dominated Helium Afterglow, *Phys. Rev. A*, **5**, 1410-1418 (1972).
- 05573 Kenny, T. E., Craggs, J. D., Electron Energy Distributions in Townsend Discharges in Hydrogen, *J. Phys. D*, **3**, 251-255 (1970).
- 05577 Heylen, A. E. D., Ionization Coefficients and Sparking Voltages in Argon-Long-Chain Hydrocarbon Gas Mixtures, *Intern. J. Electron.*, **30**, 121-132 (1971).
- Mixtures studied were n-Octane+Argon, Cis-Butene-2+Argon, and Benzene+Argon.
- 05607 Wiegand, W. J., Fowler, M. C., Benda, J. A., Carbon Monoxide Formation in CO₂ Lasers, *Appl. Phys. Letters*, **16**, 237-239 (1970).
- 05610 Bhiday, M. R., Paithankar, A. S., Sharda, B. L., Measurement of Townsend's First Ionization Coefficient for Air in Transverse Electric and Magnetic Fields, *J. Phys. D*, **3**, 943-950 (1970).
- 05638 Pollock, W. J., Momentum Transfer and Vibrational Cross-sections in Non-polar Gases, *J. Chem. Soc. Faraday Trans. II*, **64**, 2919-2926 (1968).
- 05697 Janssen, J. J., Craggs, J. D., Collisional-Radiative Decay of Spark Channels in Hydrogen, *J. Phys. B*, **5**, 89-94 (1972).

- 05698 Shaw, D. T., Wu, F. T., Effective Ionization Coefficients in a Non-L.T.E. Cesium Plasma, *Energy Conversion*, **11**, 113-117 (1971).
- 05699 Bragin, Yu. A., Vorontsov, S. S., Fedyashev, A. B., Measurement of the Effective Electron-Ion Recombination Coefficients in Air and in Oxygen, *Geomag. Aeron.*, **10**, 732-733 (1970).
- 05700 Moruzzi, J. L., Phelps, A. V., Survey of Negative-Ion-Molecule Reactions in O_2 , CO_2 , H_2O , CO and Mixtures of these Gases at High Pressures, *J. Chem. Phys.*, **45**, 4617-4627 (1966).
Mixtures studied were $O_2 + CO$ and $O_2 + H_2$.
- 05701 Phelps, A. V., Role of Molecular Ions, Metastable Molecules and Resonance Radiation in the Breakdown of Rare Gases, *Phys. Rev.*, **117**, 619-632 (1959).
- 05702 Kutszegi Corvin, K., Corrigan, S. J. B., Dissociation of Carbon Dioxide in the Positive Column of a Glow Discharge, *J. Chem. Phys.*, **50**, 2570-2574 (1969).
- 05730 Johnson, A. W., Gerardo, J. B., Electronic Recombination Coefficient of Molecular Helium Ions, *Phys. Rev. Letters*, **27**, 835-838 (1971).
- 05732 Drawin, H. W., Emard, F., Collisional-Radiative Volume Recombination and Ionization Coefficients for Quasi-Stationary Helium Plasmas, *Z. Phys.*, **243**, 236-340 (1971).
- 05745 Chen, C. J., Collisional-Radiative Electron-Ion Recombination Rate in Rare-Gas Plasmas, *J. Chem. Phys.*, **50**, 1560-1566 (1969).
- 05761 Mansbach, P., Keck, J. C., Monte Carlo Trajectory Calculations of Atomic Excitation and Ionization by Thermal Electrons, *Phys. Rev.*, **181**, 275-289 (1969).
- 05770 Hirsh, M. N., Eisner, P. N., Slevin, J. A., Ionization and Attachment in O_2 and Airlike N_2 : O_2 Mixtures Irradiated by 1.5 MeV Electrons, *Phys. Rev.*, **178**, 175-181 (1969).
Mixture Studied was $N_2 + O_2$.
- 05771 Jones, G. H., Morgan, C. G., Measurements of Primary Ionization Coefficients, (In) *Proceedings of the Ninth International Conference on Phenomena in Ionized Gases*, **1**, 139 (1969).
- 05772 Collins, C. B., Hicks, H. S., Wells, W. E., Direct Measurement of the Dependence on Electron Density of the Recombination-Rate Coefficient of He_2^+ with Electrons in a High-Pressure Helium Plasma, *Phys. Rev. A*, **2**, 797-806 (1970).
- 05773 Collins, C. B., Collisional-Radiative Recombination of Ions and Electrons in High-Pressure Plasmas in which the Electron Temperature Exceeds the Gas Temperature, *Phys. Rev.*, **177**, 254-257 (1968).
- 05774 Deloche, R., Gonfalone, A., Etude de la Recombinaison Electron-Ion dans un Plasma d'Helium, a Forte Densite de Neutres, *J. Phys. (Paris)*, **29**, C3-27-30 (1968).
- 05775 Olsen, H. N., Huxford, W. S., Dynamic Characteristics of the Plasma in Discharges through Rare Gases, *Phys. Rev.*, **87**, 922-930 (1952).
- 05776 Mitin, R. V., Pryadkin, K. K., Pulsed Plasma Heating in High-Pressure Electrodeless Discharge, *Soviet Phys. Tech. Phys.*, **13**, 1398-1400 (1969).
- 05777 Van Trong, N., Recombinaison Electronique dans un Plasma d'Argon sous la Pression Atmospherique et au Voisonage de 10000° K, *Compt. Rend.*, **264B**, 217-219 (1967).
- 05778 Brand, C. F., Heckenberg, N. R., Watson-Munro, C. N., Spectral and Microwave Studies of the Decay of a Highly Ionized Hydrogen Plasma, *Australian J. Phys.*, **22**, 337-344 (1969).
- 05779 Irons, F. E., Millar, D. D., A Spectroscopic Study of a Decaying Hydrogen Plasma, *Australian J. Phys.*, **18**, 23-39 (1965).
- 05780 Sugawara, M., Chen, C. J., Drift Velocities of Electrons in the Positive Column of a Neon Discharge, *J. Appl. Phys.*, **11**, 3442-3445 (1970).
- 05781 Curry, B. P., Collisional Radiative Recombination in Hydrogen Plasmas and in Alkali Plasmas, *Phys. Rev. A*, **1**, 166-176 (1970).
- 05782 Desai, S. V., Corcoran, W. H., Recombination of Electrons and Ions in an Atmospheric Argon Plasma, *J. Quant. Spectry. Radiative Transfer*, **9**, 1371-1386 (1969).
- 05783 Mentzoni, M. H., Donohoe, J., Electron Recombination and Diffusion in CO at Elevated Temperature, *Can. J. Phys.*, **47** 1789-1795 (1969).
- 05960 Errett, D. D., The Drift Velocity of Electrons in Gases, Thesis, Purdue University, West Lafayette, Indiana, 81 Pages 1951.
Mixtures studied were $Ar + CO_2$, $N_2 + H_2O$, $Ar + N_2$, and $Ar + H_2O$.

Aamodt, R. E.	00767					Bialecke, E. P.	00703				
Abdelnabi, I.	02143					Biberman, L. M.	02272				
Abgrall, H.	04192					Biondi, M. A.	00131	00541	00596	00758	01247
Abramov, V. A.	00595	03536					01344	01363	01383	01600	01605
Adams, N. G.	05198						02395	02455	02986	03443	03610
Aden, A. L.	00698						03611	04044	04045	04065	04095
Afanaseva, E. A.	03188					Bishop, E. S.	03490				
Afanaseva, V. L.	02028	02887				Bistline, J. A.	03187				
Aisenberg, S.	03790					Blair, D. T. A.	01948	02581			
Aizentson, A. E.	05184					Blasberg, H. A. M.	05105				
Akimov, A. V.	03789					Blaszuk, P. R.	03597				
Aleksandrov, A. F.	04021					Blaunstein, R. P.	03712	04185	04190		
Aleksandrov, V. Ya.	03433					Blevin, H. A.	00790	02303	02325	02519	03459
Aleskovskii, Yu. M.	00737	02447					03460				
Alievskii, M. Ya.	03660					Bloch, F.	01285				
Allen, H. W.	00273					Boardman, W. J.	01602				
Allen, J. S.	02055					Bocchieri, P.	01800				
Allen, N. L.	01728	05188				Bodere, C.	05290				
Allis, W. P.	01388	02113				Bockner, C.	00764				
Allouche, D.	03661					Bond, R. H.	01896				
Anderson, J. M.	00306	02164	02329			Bono, R.	03657				
Anderson, V. E.	05046					Boohariwalla, D. B.	03540				
Anisimov, A. I.	01601	01870				Born, G. K.	03464				
Appleton, E. V.	02926	03540				Borodin, V. S.	01822	02098	02163	03649	
Aptekar, I. L.	00975					Bortner, T. E.	00732	01016	01267	02141	
Archambault, Y.	05231					Bortnik, I. M.	04057				
Argyropoulos, G. S.	03274					Bortz, P. I.	00972				
Astrom, E.	02235					Bouby, L.	04192	05290			
Atkinson, W. R.	00241					Boulmer, J.	05343				
Aubrey, B. B.	01620					Bowe, J. C.	01623	02043			
Ayres, T. L. R.	00624					Bowls, W. E.	02275				
Badareu, E.	02442	02930	02931	02932		Bowman, C. R.	02269				
Bagnall, F. T.	02448					Boyd, H. A.	05053	05229			
Bagot, C. H.	00649					Boyd, R. L. F.	02538	03172	03173	03175	03716
Bahadori, M. N.	02438					Boylett, F. D. A.	02538				
Bailey, J. E.	00683					Bozin, S. E.	02224	03465			
Bailey, V. A.	00159	00195	00197	00199	00201	Brabanec, M. K.	04949				
	00682	01332	02088	02273	02278	Bradbury, N. E.	01285	01317	01381	01412	01872
	02279	02280	02385	02567	02765		02099	02100	02952		
	02883	02953				Brady, J. J.	02534				
Bain, R. A.	04891					Bragg, J. K.	02808				
Baker, J. H.	04843					Bragin, Yu. A.	05699				
Balfour, D.	02457	02983				Brambring, J.	04944				
Dannon, J.	00203					Brand, G. F.	05778				
Bansal, K. M.	05125					Branston, D. W.	04993				
Baraff, G. A.	00756					Brasefield, C. J.	01559				
Baravian, G.	05533					Bratescu, G. G.	02933				
Barbiere, D.	00397					Braun, J.	03434				
Bardsley, J											

Burkley, C. J.	02435					Croitoru, Z.	02592	03794						
Buursen, C. G. J.	05049					Crompton, R. W.	00352	00649	00689	00782	00784			
Byron, S.	00972						01437	01438	01489	01618	01622			
Cahn, J. H.	02307	02436					02097	02110	02267	02433	02463			
Caldirola, P.	03430						02489	02992	03659	04355	04484			
Caren, R. P.	01133						04913							
Carleton, N. P.	01207					Cronin, J. C.	05500	05501						
Carlson, W. O.	03787					Crosignani, E.	01800							
Carmichael, C. H. H.	05295					Crowe, R. W.	02808							
Carruthers, J. A.	03494					Cunningham, A. J.	04585	05128						
Carswell, A. I.	04311					Curry, B. P.	05781							
Carter, J. G.	04099					D'Angelo, N.	00731	01614	02546					
Cavalleri, G.	01126	03467	04049	04050		Dalgarno, A.	01603	03652	04656					
Cayless, M. A.	03270					Dalidchik, F. I.	03651							
Chambers, R. G.	02183					Dallenbach, W.	03275							
Champion, K. S. W.	00698	01399				Dandurand, P.	01729	01737						
Chan, F. T.	04020					Daniel, T. N.	04015	04085	05187					
Chaney, E. L.	04054	04099	05238			Danilovic, E. S.	03311							
Chanin, L. M.	00131	00646	00751	01154	01186	Dargan, C. L.	01950							
	01383	02354	02493			David, J. P.	03650							
Chantry, P. J.	00860					Davidson, C. W.	03795							
Chao, K. T.	02494					Davies, D. E.	00621	00898	01160	01249	01398			
Chapman, F. W.	02926						02138	03304	04043	05202				
Chatterton, P. A.	01006	02408				Davies, D. K.	00900	01161	01173	02483	02578			
Chen, C. J.	04586	05745	05780			Davies, G. H. L.	04016							
Chen, C. L.	00249					Davies, W. E. V. J.	02570	02571						
Chen, E.	04187	04195				Davis, F. J.	03313	04048	05135	05236	05237			
Cheret, M.	04374					Davy, P.	05343							
Childs, E. C.	02321					Davydov, B.	02094	02337	02340					
Chiplonkar, V. T.	03466					De Barbieri, O.	03237	03238	03430	03441				
Chistodoulides, A. A.	05238					De Bitetto, D. J.	00780	01145	02492					
Chown, J. B.	03606					De Hoog, F. J.	02535	03436	03648	05049	05105			
Christmann, H.	03282						05110							
Christophorou, L. G.	00487	00488	00725	01198	01565	DeMontardy, A.	03794							
	02559	02995	03007	03662	03712	Dearnley, I. H.	03495							
	04054	04099	04185	04190	05046	Deas, H. D.	02087							
	05238					Deb, S.	02820							
Ciampi, M.	03440					Delcroix, J. L.	02471	03182	03284	03285	03290			
Clay, J.	03554	03556				Deleonardis, M. T.	02987							
Cloutier, G. G.	03470					Deloche, R.	03437	03653	04374	05774				
Cochran, L. W.	00736					Delpach, J.-F.	05134	05343						
Colli, L.	01333	02821	02987			Demetriades, S. T.	03274							
Collins, C. B.	01615	05772	05773			Den Hartog, H.	03550	03551						
Collins, P. M.	04099					Denisse, J. F.	03182	03284	03285	03290				
Compton, K. T.	00674	02201	02202	02347	02392	Desai, S. V.	05782							
	02393					Desi, S.	02261							
Compton, R. N.	00487	01198	01565	02532	03007	Devins, J. C.	02808	03177	03178					
Connor, T. R.	01344	01600				Devoto, R. S.	03301	03796	03798	03799				
Conti, V. J.	03435					Dibbern, U.	02155	02582						
Cookson, A. H.	01836	01949	01958	02466		Dibelius, N. R.	03904							
Cool, T. A.	00705	03955	03958			Dickson, H. W.	03007							
Cooper, W. S.	01389					Didlaukis, M.	02196	03538						
Corcoran, W. H.	05782					Doehring, A.	00727							
Corrigan, S. J. B.	00336	00354	05702			Doering, J. P.	00920							
Costa, H.	02367					Dolique, J. H.	02473							
Cottureau, M. J.	03181	03713	03953			Donohoe, J.	01779	03229	03601	05783				
Cottingham, W. B.	02533	02561				Donovan, B. V.	05232							
Cottrell, T. L.	01917	02990	05200			Donskoi, K. V.	03537							
Cowling, T. G.	03158					Dougal, A. A.	00703	01450						
Craggs, J. D.	00337	00346	00406	00861	00948	Drawin, H. W.	03469	05130	05732					
	01006	01254	01260	01642	02106	Dreicer, H.	02297							
	02230	02294	02302	02408	02555	Driver, C.	02767							
	02994	03264	03472	04920	04948	Drouet, M. G.	03470							
	05052	05573	05697			Druryvesteyn, M. J.	01441	02048	02096	02197	02324			
Craig, R. D.	03264						02351	03496						
Cravath, A. M.	02093					Dubrovskii, G. V.	03471							
Crawford, F. W.	02530					Dubs, C. W.	02572							
Creaser, R. P.	03384					Dugan, J. V.	03180							
Crichton, G. C.	05229					Duncanson, W. E.	02088							
Crichton, H. A.	05053					Dunlop, S. H.	00771	02045						

Dunn, M. G.	05294					Frost, L. S.	00181	00888	01043	01062	03491
Durham, J.	01963					Fugol, I. Ya.	02549				
Dutt, T. L.	02579					Funahashi, A.	03468				
Dutton, J.	00573	00782	00784	01250	02144	Fundingsland, O. T.	00312	00698			
	02286	02388	02406	02407	02483	Gabriel, A. H.	03656				
	02487	02488	02528	02570	02571	Garamoon, A. A. W. M.	05202				
	02578	02591	02908	02919	03184	Garrison, G. W.	03474				
	03751	04015	04085	04910	05077	Garscadden, A.	02530				
	05244					Gascoigne, J.	01622				
Eastlund, B. J.	01828					Gatti, E.	01126	04050			
Eccles, M. J.	02994	03472	04920			Gauthier, J.-C.	05134	05343			
Edels, H.	03480					Cavallas, L. A.	04189				
Edelson, D.	00788	01174	02465	02544		Gawen, J. C.	03444				
Edwards, D. F.	02038					Geballe, R.	00791	01315	01322	03240	
Efremova, G. D.	02819					George, R. D.	04195				
Egorov, V. S.	03654					Gerardo, J. B.	01961	05203	05325	05488	05570
Eisner, P. N.	05770						05730				
Ekin, Jr., J. W.	03352					Gerjuoy, E.	02034	02037			
Elford, M. T.	00689	01622	02041	02433	02992	Germanyuk, V. N.	03596				
	04855					Choeh, A. K.	04311	04313			
Elkomoss, S. G.	03714					Ghosh, P. K.	03442				
Ellington, H. I.	03458					Gibson, D. K.	04450	04484			
Ellis, R.	03609					Gibson, E. G.	03955				
Emard, F.	05130	05732				Gilardini, A.	02232	02586			
Emeleus, K. G.	02045	02087	02276	02568	03504	Gill, E. W. B.	02402	02816	03171	03553	04017
Engelhardt, A. G.	00218	00260	00292			Giurgea, M.	02934				
Englert, G. W.	05490					Glotov, I. I.	02341	02501			
English, W. N.	02399					Godart, J. L.	05533				
Engstrom, R. W.	02108					Golant, V. E.	00881	01601	02452	02453	03800
Enoch, J.	03907						04193				
Epstein, M.	03155					Golden, D. E.	02145	02333	02587		
Errett, D. D.	05960					Goldenberg, S. A.	03445	03529	03892		
Evans, E. W.	02268					Goldstein, L.	00249	00306			
Evans, N. B.	02908					Golubev, V. S.	03245				
Facchini, U.	01333	02821				Gonfalone, A.	04374	05774			
Faire, A. C.	00698	01399				Goodall, C. V.	03786				
Farvis, W. E. J.	03795					Goodyear, C. C.	02224	03465	03600	03604	04086
Fauchais, P.	03473					Gordon, D. E.	02269				
Fedorov, V. L.	03288	04189				Gordon, E. I.	02295				
Fedulov, V. I.	02819					Goswami, S. N.	02820				
Fedyashev, A. B.	05699					Gould, L.	00314				
Fehsenfeld, F. C.	03042					Govinda Raju, G. R.	05246	05247			
Fells, I.	03444					Granovskii, V. L.	02447				
Ferguson, E. E.	03042					Gratescu, G. G.	02930				
Fessenden, R. W.	05125					Gray, D. R.	04901				
Fields, H.	03797					Greeves, F. D.	03504				
Fink, X.	01833					Griem, H. R.	03656				
Finkelnburg, W.	03535					Grigorovici, R.	02368				
Fiocco, G.	03482					Grunberg, R.	03382	04055	04058	04919	
Fiquet-Fayard, F.	04192	05290				Gunton, R. C.	00658	01181	01608	01609	02491
Fischer, J.	02459						03647				
Fischer-Treuenfeld, W.	01763					Gupta, R. N.	03383				
Fisher, L. H.	00780	00961	01145	02145	02333	Gurevich, A. V.	01647	03893	03894		
	02492	02587				Gurevich, D. B.	03433				
Fleming, I.	04901					Guseva, L. G.	03895				
Fletcher, J.	00238	01160	02468			Gusinow, M. A.	01961	05203			
Folkard, M. A.	04907	05292	05334			Gvosdover, S. D.	02339				
Forester, D. W.	00736					Hackam, R.	01633	02394	02456	03647	
Formato, D.	02232	02586				Hadjiantoniou, A.	00725	02559			
Fossi, B. C.	05289					Hagi, M.	03950				
Fowler, M. C.	05607					Haigh, C.	03169	03185			
Fowler, R. G.	00241					Haines, W. B.	01971				
Fox, J. N.	01873	03438				Hake, R. D.	02553				
Franchi, P.	03237					Hale, D. H.	02247	02248			
Franke, W.	02156					Hall, B. I. H.	01428	02289	02489		
Franklin, J. L.	03442					Hamada, T.	03263				
Freely, J. B.	00961					Hamberger, S. M.	02018				
Frie, W.	03282					Hamilton, N.	02061				
Friedrich, J.	03492					Hammer, J. M.	01620				
Frommhold, L.	00973	01444	02053	02101	02154	Hanna, C. C.	02399				
	02370	02574	03443			Hansen, C. F.	03476				

Hantzsche, E.	02470	03241				Irons, F. E.	05779				
Harker, J. H.	03444					Itoh, T.	03897				
Harris, F. M.	02407	02488	02528	02570	02571	Ivanov, B. A.	02357				
	02591	03184	04015	04085	05187	Jaeger, J. C.	02284				
	05244					Jager, G.	02153				
Harris, J. H.	02457	02983				Jancel, R.	03479	03661	03898		
Harris, L. P.	01473	01634	03427			Janev, R. K.	03311				
Harris, S.	02594					Janssen, J. J.	05697				
Harrison, J. A.	02554	03922	05225			Jarry, G.	03650				
Harrison, M. A.	00791	01315	02102			Jeannet, J. C.	05231				
Hartman, L. M.	02193	02194				Jeffries, R. A.	01901				
Hasan, M. Z.	03459	03460				Johnson, A. W.	05325	05488	05570	05730	
Haseltine, W. R.	02369					Johnson, L. C.	03289				
Hassan, H. A.	03655					Johnson, R. A.	00802				
Hasted, J. B.	01649	05124				Johnston, T. W.	04313				
Haydon, S. C.	00238	00782	00784	00790	02040	Jones, E.	01161	02150			
	02144	02148	02162	02303	02325	Jones, G. H.	05771				
	02448	02468	02519	02580	02588	Jones, G. J.	05244				
	04907	05102	05224	05292	05334	Jones, J.	03297				
	00677	00682				Jones, R. P.	03299				
Healey, R. H.	03480	03896				Jory, R. L.	01429	01438	01489	01618	02433
Hearne, K. R.	05778						02463	02813			
Heckenberg, N. R.	00488	03662				Joyce, G. R.	03298				
Hendrick, W. G.	02188	02405				Kaftandjian, V.	03650				
Herlin, M. A.	01336	02262	02264			Kagan, Yu. M.	01818	01822	02008	02019	02098
Herreng, P.	03720						02163	02236	02344	02348	02454
Hertz, G.	03282						02469	02886	03156	03157	03168
Hertz, W.	00584	02409					03231	03431	03432	03534	03608
Hess, W.	03165						03614	03649	04189		
Hessenauer, H.	00710	01335	01440	01447	01636	Kahan, T.	03661	03898			
Heylen, A. E. D.	01638	01783	01950	02142	02547	Kalagher, R. J.	01811				
	02576	03477	03478	05293	05577	Kantrowitz, A.	03167				
	03679					Karpenko, A. S.	03539				
Heymann, P.	05772					Kasabov, G. A.	03227	03245			
Hicks, H. S.	02567					Kasdorp, J.	02535				
Higgs, A. J.	01639	01713	01766	02330		Kasner, W. H.	01247	01344	01605	02988	02993
Hinnov, E.	01639	01713	02330				03483	03611			
Hirschberg, J. G.	05770					Kato, K.	03309				
Hirsh, M. N.	01873	03438	04585	05128		Keck, J. C.	00640	01606	05761		
Hobson, R. M.	02558	02595				Kelly, D. C.	02540				
Hochberg, B. M.	03461					Kenny, T. E.	05573				
Hodgson, R. T.	03900					Kenty, C.	00355				
Hoffman, M. A.	05110					Kerr, L. W.	04605				
Holscher, J. G. A.	01420					Kerrebrock, J. L.	03899	03900			
Holstein, T.	00693	03597	04096			Kerzar, B.	02401				
Holt, E. H.	00217	00252	00802	01729	01737	Keyston, J. E.	03715				
Holt, R. B.	02292	03166				Khakhaev, A. D.	03431	03432			
	00346	00779	01640	02106		Khare, S. P.	01259				
Hopwood, W.	01237					Khozhateliev, M. B.	02818				
Hornbeck, J. A.	02882					Kihara, T.	02352				
Horton, F.	02300					Killian, T. J.	02481				
Howard, P. R.	00217					Kilvington, A. I.	03299				
Howland, B.	03475	04022				Kingston, A. E.	00146	00444	00994	00998	01603
Huber, B.	01833						02031	02147	02168	03381	
Huber, P.	02542					Kinsman, P. R.	05227				
Hudson, D. E.	05240					Kirkby, P. J.	02593				
Hughes, D. W.	03751	04909				Kirshner, J. M.	02042				
Hughes, M. H.	01437					Kiyama, S.	03309				
Hurst, C. A.	00487	00488	00725	00732	00956	Klarfeld, B.	03266				
Hurst, G. S.	01016	01184	01198	01267	01379	Klema, E. D.	02055				
	01565	01964	02141	02559	02995	Kline, L. B.	05233				
	03313	03662	05206			Knechtli, R. C.	00638	00697	02590		
	02281	02815				Kolokolov, N. B.	03614				
Hurst, H. E.	02108	02346	05775			Komar, A. P.	02357				
Huxford, W. S.	00649	02054	02097	02110	02187	Konakh, V. F.	03245				
Huxley, L. G. H.	02255	02256	02338			Konenko, O. R.	03789				
	03445	03529	03892			Konjevic, N.	03480	03896			
Ievlev, V. N.	01181					Kontoleon, N.	05050	05230	05234		
Inn, E. C. Y.	04050					Koons, H. C.	03482				
Interlenghi, A. M.	02928	03721				Korolev, V. A.	02357				
Ionescu, T. V.	02474					Kostin, M. D.	03463				
Irish, R. T.											

Kovrizhnykh, L. M.	03493					Makinson, R. E. B.	00683				
Koyama, K.	03481					Maksimov, A. I.	03179				
Kozlov, G. I.	01786					Malakhov, V. B.	04094				
Kretschmer, C. B.	00807	00811				Malaviya, V.	04604				
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Kruger, C. H.	03646					Mallozzi, P.	02524				
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Kuckes, A. F.	01713	02331				Mandal, S. K.	03383				
Kudrin, L. P.	03300					Manheimer-Timnat, T.	03279				
Kuffel, E.	01433	02149				Mans, C.	04374				
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Kuzovnikov, A. A.	04021					Markevich, A. M.	03539				
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Larsson, E.	03434					McArthur, W. T.	05228				
Lawson, P. A.	01264	01778				McCallum, S. P.	02810				
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Lee, T. G.	03901					McCorkle, D. L.	05046				
Legler, W.	00921					McCulloch, R. K.	03185				
Lehning, H.	04203					McGee, J. D.	02273	02284			
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Lennon, J. J.	00711	01633	02445	02456	03647		05224				
	04097					McMillan, F. L.	03495				
Leonteva, Z. S.	03529	03892				McNaull, J.	02581				
Levine, J. L.	01714	01730	02266			McWhirter, R. W. P.	00146	00444			
Levine, N. E.	02920					Medicus, G.	03233	03235			
Lewis, T. J.	00927	01440	01638	01836		Meek, C. A.	02109	02276	02502		
Leycuras, Y.	01671					Meek, J. M.	00406				
Li, C. P.	03301					Megill, L. R.	01207				
Lichtenstein, R.	02323					Mehr, F. J.	03443	04044	04045	04065	
Liley, B. S.	01437	03484				Mehuron, W. O.	01811				
Lin, S. C.	03161	03167	04789			Meisel, L. V.	02343				
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Llewellyn-Jones, F.	00573	00900	01173	01250	01297		05783				
	02050	02144	02150	02388	02406	Mihul, C.	02928	03721			
	02407	02483	02487	02488	02528	Milenin, V. M.	01818	02008	02019	02236	02344
	02570	02578	02812	03184			02469	02886	03168	03614	
Lloyd, J. L.	04063					Millar, D. D.	05779				
Lo Surdo, C.	03593	03594				Millet, J.	03650				
Loeb, L. B.	02090	02135	02137	02263	02504	Milne, J. G. C.	00621	02138			

Mott-Smith, H.	03489					Philbrick, J.	04044				
Mulcahy, M. J.	00711	04097				Pidduck, F. B.	02044	02816	03171		
Mullaney, G. J.	03904					Pikus, G. E.	00149				
Muller, F. A.	03550	03551				Pitaevskii, L. P.	01647	02165			
Muller, K. G.	01723					Plumb, I. C.	05198				
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Mustafin, K. S.	02028	02454	02887			Poluehkin, I. N.	02029	02550	02551	03244	
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Nasser, E.	05235						02555	03303	03472	04948	05051
Nastoyashchii, A. F.	03615	03905					05052	05119	05123	05183	05226
Nelson, D. R.	04048	05135	05236	05237		Prew, B. A.	05188				
Newton, A. A.	00815	03658				Price, D. A.	04994				
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Nielsen, R. A.	01381	01387	01412			Principi, P.	01126				
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Norcross, D. W.	01805	03598				Pustovoit, V. I.	03906				
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Nygaard, K. J.	01761	02229	03722			Radchenko, R. V.	03660				
O'Kelly, L. B.	01016	01184	01379	05206		Raether, H.	02136	02254			
O'Neill, B. C.	04920					Raja Rao, C.	05246	05247			
Obedkov, V. D.	03471					Rayment, S. W.	03306	03603			
Ogawa, K.	03309					Razzak, S. A. A.	03600	03604	04086		
Oldenberg, O.	03166					Reuter, F. H.	01208				
Olsen, H. N.	05775					Redfield, A.	00252				
Orefice, A.	03237	03238				Rees, D. B.	01250	02286	02406		
Oreshak, O. N.	03599					Rees, J. A.	01150	01269	01618	02140	02463
Orient, O. J.	02250						02813	04901	05227		
Ornstein, L. S.	03263	03281				Reinhardt, P. W.	00487	01198	01565		
Oskam, H. J.	00661	00673	02169	05291		Reser, E. L.	03167				
Ostapchenko, E. P.	03599	04094				Ricateau, P.	03439				
Otto, W.	02153					Richard, C.	04311				
Overton, G. D. N.	03304					Richardson, J. M.	00217	00254	02292		
Paavola, M.	02557					Richter, J.	03380				
Pack, J. L.	00439	00530	00738	00888	01662	Richter, K.	02500				
	02046	03154				Riemann, W.	04018				
Paithankar, A. S.	02231	05610				Risk, C. G.	00218				
Pakhomov, P. A.	02549					Ritchie, R. H.	02877				
Palmer, R. S.	02530					Robben, F.	00759	03312	05327		
Palmer, R. W.	02388	02487				Roberts, T. D.	01651	01899			
Parekh, H.	05235					Robertson, A. G.	02148	02162	02580	04355	04484
Parker, A. B.	02050						04862	04913	04916		
Parker, J. H.	00761	01036	04052			Rogers, W. A.	01605				
Parkes, D. A.	04943					Ronnsin, S.	05231				
Parks, J. E.	01964					Rork, G. D.	00646	00751	01186	02354	02493
Parr, J. E.	05243	05248				Rose, D. J.	00780	01356	02515	03228	
Patt, H. J.	03380					Ross, D. W.	01309				
Peacock, N. J.	00779	01640				Rudd, J. B.	00159				
Pearsall, C. S.	03495					Rutscher, A.	03791	04053			
Pearson, G. A.	02806					Ryabinin, Yu. N.	03539				
Penning, F. M.	02086	02139	02287	02355		Rynn, N.	02546				
Persson, K. B.	00239					Ryzko, H.	01911	02235	02305	02467	
Peters, T.	00305					Sakai, Y.	05126				
Petersen, H. L.	00807					Sakamoto, S.	05126				
Petrescu, G.	02934					Salmon, J.	00700	01745			
Peyraud, N.	03276	03678				Salvat, M.	03677				
Pfau, S.	03792	04053				Sampson, D. H.	03907				
Phelps, A. V.	00131	00181	00218	00260	00292	Sandberg, E. J.	02558	02595			
	00312	00439	00530	00738	00888	Sanders, F. H.	02134	02274			
	01043	01062	01383	01467	01662	Sanders, T. M.	01714	02266			
	02046	02203	02553	03154	03352	Sayasov, Yu. S.	03651				
	03680	05700	05701			Sayer, B.	05231				

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Varnerin, L. J.	00240	00841				Wieczorek, L. W.	02470				
Veatch, G. E.	05291					Wiegand, W. J.	05607				
Verdeyen, J. T.	01961					Wilhelm, H. E.	03954				
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Veyssiere, M.	03291					Wilkins, R. L.	01665				
Viegas, J. R.	03646					Willett, J. E.	03298				
Vinogradov, N. I.	01601	01870				Williams, A. W.	03435	04016	04843	04949	
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Virr, L. E.	05050	05230	05234			Williams, W. T.	02817				
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	04017					Wilson, L. N.	02268				
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Vorobev, A. A.	02357					Wooding, E. R.	05240				
Vorobeva, N. A.	02008	02019	02236	02344	02469	Wu, C. S.	02152	02583			
Vorontsov, S. S.	05699					Wu, F. T.	05698				
Voshall, R. E.	00439	03680				Yamane, M.	03956				
Wada, J. Y.	00638	00697	02590			Yarin, L. P.	02818				
Wagner, E. B.	01184	03313	05206			Yarnold, G. D.	03170				
Wagner, K. H.	02136	02529	04947	04971		Yen, J. T.	03314				
Wahlin, H. B.	01446	01725	01752	02049	02133	Yoshikawa, S.	03957				
	02158					Young, C. E.	01923				
Walker, I. C.	01917	02990	03595	05200		Young, D. R.	02809				
Wanless, D.	05201					Young, N. A.	04604				
Ward, B. W.	01741	01836	01958	02466		Young, R. A.	00840				
Warfield, G.	02397					Yurev, V. G.	00149				
Warke, C. S.	01675					Zaazou, A. A.	02054				
Warman, J. M.	05125					Zagik, S. E.	00762	02130			
Warren, R. W.	01036					Zaitev, A.	03450				
Watson-Munro, C. N.	05778					Zampaglione, V.	03594				
Wehner, G.	03233					Zanstra, H.	00358				
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Weller, C. S.	01344	02936	03610			Zettwoog, P.	03439				
Wells, W. E.	05772					Zhilinskii, A. P.	04193				
Wentworth, W. E.	04187	04195				Zukoski, E. E.	00705	03955	03958		

7. List of Symbols and Units

Coefficient or Parameter	Symbol	Units	Coefficient or Parameter	Symbol	Units
1. (a) Drift velocity	W	cm s^{-1}	6. (a) Excitation coefficient	ϵ	cm^{-1}
(b) Magnetic drift velocity	W_M	cm s^{-1}	(b) Excitation coefficient to radiating states	ϵ_r	cm^{-1}
(c) Mobility	μ	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	(c) Excitation coefficient to dissociating states (dissociation coefficient)	ϵ_d	cm^{-1}
(d) Magnetic deflection coefficient	ψ		(d) Excitation rate coefficient	e_2	$\text{cm}^3 \text{s}^{-1}$
2. (a) (Drift velocity)/(Diffusion coefficient)	W/D	cm^{-1}	7. (a) Ionization coefficient	α	cm^{-1}
(b) (Diffusion coefficient)/(Mobility)	D/μ	V	(b) Apparent ionization coefficient	λ_i	cm^{-1}
(c) Characteristic energy	$\epsilon_k = e \frac{D}{\mu}$	eV	(c) Ionization rate coefficient	i	$\text{cm}^3 \text{s}^{-1}$
(d) Townsend energy factor	k		(d) Collisional radiative ionization rate coefficient	i_c	$\text{cm}^3 \text{s}^{-1}$
3. (a) Diffusion coefficient	D	$\text{cm}^2 \text{s}^{-1}$	(e) Ionization frequency	ν_i	s^{-1}
(b) Thermal equilibrium diffusion coefficient	D_{th}	$\text{cm}^2 \text{s}^{-1}$	8. (a) Two-body recombination rate coefficient	r_2	$\text{cm}^3 \text{s}^{-1}$
(c) Diffusion coefficient parallel to the electric field	D_L	$\text{cm}^2 \text{s}^{-1}$	(b) Three-body recombination rate coefficient	r_3	$\text{cm}^6 \text{s}^{-1}$
(d) Diffusion coefficient perpendicular to the electric field	D_T	$\text{cm}^2 \text{s}^{-1}$	(c) Decay rate coefficient	g	$\text{cm}^3 \text{s}^{-1}$
4. (a) Attachment coefficient	η	cm^{-1}	(d) Collisional radiative decay rate coefficient	g_c	$\text{cm}^3 \text{s}^{-1}$
(b) Two-body attachment coefficient	η_2	cm^{-1}	(e) Collisional radiative recombination rate coefficient	r_c	$\text{cm}^3 \text{s}^{-1}$
(c) Three-body attachment coefficient	η_3	cm^{-1}	(f) Effective recombination rate coefficient	r_e	$\text{cm}^3 \text{s}^{-1}$
(d) Attachment rate coefficient	a	$\text{cm}^3 \text{s}^{-1}$	9. Gas number density	N	cm^{-3}
(e) Two-body attachment rate coefficient	a_2	$\text{cm}^3 \text{s}^{-1}$	10. (a) (Electric field)/(Gas number density)	E/N	$\text{V cm}^2 (= 10^{17} \text{ Td})$
(f) Three-body attachment rate coefficient	a_3	$\text{cm}^6 \text{s}^{-1}$	(b) (Electric field)/(Gas pressure reduced to 0 °C)	E/p_0	$\text{V cm}^{-1} \text{ Torr}^{-1}$
(g) Attachment probability	h				
5. (a) Detachment coefficient	δ	cm^{-1}			
(b) Detachment rate coefficient	d	$\text{cm}^3 \text{s}^{-1}$			
(c) Two-body detachment rate coefficient	d_2	$\text{cm}^3 \text{s}^{-1}$			

$$(E/N (\text{V cm}^2) = 2.82 \times 10^{-17} E/p_0 (\text{V cm}^{-1} \text{ Torr}^{-1}))$$

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